

The Gartner Glossary of Information Technology Acronyms and Terms

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Numbers

1Base-5

One megabit per second (Mbps) transmission over a twisted-pair configuration.

10Base-x series

Any of several Institute of Electrical and Electronics Engineers (IEEE) specifications for baseband networks using the Ethernet protocol. It includes:

1. **10Base-2**: 10 megabit per second (Mbps) baseband network using thin Ethernet coaxial cable.
2. **10Base-5**: 10 Mbps baseband network using thick Ethernet coaxial cable.
3. **10Base-F**: Specification for baseband Ethernet over fiber-optic cabling.
4. **10Base-FB**: Part of the 10Base-F specification providing a synchronous signaling backbone that allows additional segments and repeaters to be connected to the network.
5. **10Base-FL**: Part of the 10Base-F specification designed to replace the Fiber-Optic Inter-Repeater Link (FOIRL) standard, providing Ethernet over fiber-optic cabling.
6. **10Base-FP**: Part of the 10Base-F specification that allows the organization of a number of end nodes into a star topology without the use of repeaters.
7. **10Base-T**: Standard enabling unshielded twisted-pair (UTP) cable to be used for Ethernet local-area networks (LANs).

10Broad-36

IEEE specification for broadband 10 Mbps networks running over thick coaxial cable.

100Base-T

IEEE specification for 100 Mbps networks using Fast Ethernet over copper cabling.

100Base-VG

LAN standard from IEEE's 802 committee for 100 Mbps transmission over UTP cable, promoted originally by Hewlett-Packard and AT&T, among others. It is an alternative to 100Base-T with Fast Ethernet.

100VG-AnyLAN

Method of LAN transmission that is based on 100Base-VG and builds on aspects of both token ring and Ethernet to run at 100 Mbps with resilience and high realization of potential. It is generally considered technically better than Fast Ethernet, but was less successful in the marketplace.

23B+D

An abbreviation for the Integrated Services Digital Network (ISDN Primary Rate Interface (PRI), composed of 23 bearer (B) channels and one data (D) channel.

24x7 (24 hours a day, seven days a week)

Continuous, round-the clock availability. A 24x7 system exhibits two characteristics: high availability (meaning that the service tolerates unscheduled outages) and continuous operations (meaning that it does not require scheduled downtime).

2B+D

An abbreviation for the ISDN Basic Rate Interface (BRI), composed of two bearer (B) channels and one data (D) channel.

2PC (two-phase commit)

A method for coordinating a single transaction across two or more database management systems (DBMSs) or other resource managers. 2PC guarantees the logical integrity of data by ensuring that transaction updates are either finalized in all participating databases or fully backed out of all of them (i.e., the update occurs "everywhere or nowhere"). 2PC is a required component of distributed databases, and is implemented in transaction management software that may be part of a DBMS, online transaction processing (OLTP) monitor or front-end application tool.

3270

IBM's interactive communications terminal standard used to communicate with an IBM mainframe or compatible system.

3380

An IBM direct-access storage device (DASD) family introduced in 1980.

3390

An IBM family of DASD products introduced in 1989.

3420

A family of IBM half-inch open-reel tape drives introduced in 1970, offering densities of up to 6,250 bits per inch.

3480

An IBM cartridge tape subsystem introduced in 1984.

3745

An IBM front-end processor introduced in 1988.

3780

A batch protocol used to communicate with an IBM mainframe or compatible system.

3880

A family of IBM DASD controllers introduced in 1980, supporting a wide range of devices (e.g., 3340, 3344, 3350, 3370, 3375 and 3380).

3990

An IBM DASD controller family introduced in 1987, supporting a wide range of devices (e.g., 3380, 3390, 9340 and RAMAC).

3-D visualization

The use of interactive graphics to represent and manipulate high-volume, multidimensional data as graphical objects with a wide range of characteristics (e.g., x/y/z axes, size, color, shape or movement).

3-D Web sites

A Web site that provides a virtual, three-dimensional "space" for visitors to explore (e.g., through the use of Virtual Reality Modeling Language). See VRML.

3G (third generation)

The term used to refer to the next generation of wireless communications technology, the "first generation" having been analog cellular, and the "second generation" being today's digital cellular networks. An initiative of the International Telecommunications Union (ITU) and regional standards bodies, 3G aims to provide universal, high-speed, high-bandwidth (up to 4 Mbps) wireless services supporting a variety of advanced applications. See UMTS, IMT-2000 and EDGE.

3GPP (3rd Generation Partnership Project)

A collaboration agreement, established in 1998 among a number of telecommunications standards bodies, to produce a series of technical specifications and standards for third-generation (3G) wireless communications. See 3G.

3GL (third-generation language)

A high-level programming language — such as FORTRAN, COBOL, BASIC, Pascal or C — that compiles to machine language.

3PL (third-party logistics)

A type of service provider that offers advanced logistics services, such as transportation management, inventory maintenance and logistical modeling.

3rd Generation Partnership Project (see 3GPP)

4GL (fourth-generation language)

A high-level language suitable for end-user or programmer data access and capable of reasonably complex data manipulation. It includes two categories of software development tools: application generators for production applications, and information generators for decision support applications. 4GLs are relatively nonprocedural and easier to use than 3GLs (e.g., COBOL and C), but are less powerful and more wasteful of computer resources. Gartner identifies three types of 4GLs:

- A procedural language integrated with a database management system. Examples include Computer Associates' CA-Ideal and Software AG's Natural.
- An information generator. Examples include Information Builders' Focus and Computer Associates' CA-Ramis.
- A code generator or nonprocedural tool. Examples include Oracle's SQL*Forms and Computer Associates' CA-Telon. These products are also known as lower-CASE tools.

802.11

A series of standards issued by the Institute of Electrical and Electronics Engineers (IEEE) for wireless LANs (WLANs). Various specifications cover transmission speeds from one megabit per second (Mbps) to 54 Mbps. The three main physical-layer standards are 802.11a, 802.11b and 802.11g (see separate entries).

802.11a

An IEEE standard for the physical layer of WLANs operating in the 5-gigahertz (GHz) radio band. It has eight radio channels. The maximum link rate is 54 Mbps per channel, but maximum user throughput will be about half this and the throughput is shared by all users of the same radio channel. Data rates fall off as the distance between the user and the radio access point increases. Frequency bands allowed for 802.11a (also called Wi-Fi5) differ in different parts of the world. See Wi-Fi5.

802.11b

An IEEE standard for the physical layer of WLANs operating at 2.4GHz. It has three radio channels. The maximum link rate is 11 Mbps per channel, but maximum user throughput will be about half this and the throughput is shared by all users of the same radio channel. Data rates for 802.11b fall off as the distance between the user and the radio access point increases. See Wi-Fi.

802.11g

An IEEE physical-layer standard for WLANs operating at 2.4GHz. It provides three available radio channels with a maximum link rate of up to 54 Mbps per channel. Support for complementary-code-keying modulation makes 802.11g backwardly compatible with 802.11b. The addition of other modulation schemes such as orthogonal frequency-division multiplexing (see OFDM) into the draft standard achieves higher link rates.

802.15.4

An IEEE draft standard for a low-data-rate, low-power-consumption wireless networking in the 2.4GHz radio band. With data rates of less than 220 Kbps over 75 meters, 802.15.4 (dubbed "ZigBee" by the vendor group that promotes it) is suitable for many automation and remote-control applications. (See ZigBee and ZigBee Alliance.)

802.x series

Any of several IEEE specifications covering network protocols. This series includes:

- 802.10: Standard for packet tagging for security within LANs, also used by some companies to tag packets for virtual LANs.
- 802.11: Standards for wireless LANs (see separate entry).
- 802.12: Standard that specifies the physical layer and the Media Access Control (MAC) sublayer of the Open Systems Interconnection model.
- 802.15: An IEEE working group addressing standardization of wireless personal-area networks
- 802.1AB: Draft standard for Gigabit Ethernet over UTP cable.
- 802.1Q: Draft standard for virtual LANs.
- 802.1Z: Draft standard for Gigabit Ethernet.
- 802.3: Broadband bus networking system that uses the CSMA/CD protocol. "Ethernet" has become the generic name, despite its being just one trademarked version of 802.3.
- 802.4: Standard that governs broadband bus and token bus; usually used in industrial applications.

- 802.5: Standard that governs token ring Networking systems.
- 802.6: Standard governing metropolitan-area networks.

80x86

A family of Intel microprocessors used in IBM and IBM-compatible PCs and workstations. It includes the 80286, 80386 and 80486. The Pentium processors are members of the same family, but the numerical designations have been dropped.

A

A2A (application-to-application)

An approach to enterprise application integration that provides visibility into internal systems, so that these systems can share information or business processes.

A3 (Adaptive Application Architecture)

A Gartner reference model for an application platform that can adapt to a variety of device types and networks (notably mobile devices and wireless data transmission), and that can support all types of data transactions. The A3 goes beyond the limitations of network- and device-specific wireless platforms to provide a more comprehensive model suitable to the new challenges of network computing (e.g., maintaining multiple, changing devices, unstable and mixed-media content, and multiple networks).

The A3 comprises three logical components:

- Link layer — the elements managing the link (i.e., the network transport), such as session management, synchronization, security, device discovery and location awareness.
- Transformation platform — the elements handling the transformation of the data stream, including filtering, selection, ordering and formatting.
- Profile management — the management of persistent data that supports the overall interaction.

Because it abstracts across multiple device types and media, the A3 is a key element in the evolution of the Supranet (see separate entry).

AA (application architecture)

The AA describes the layout of an application's deployment. This generally includes partitioned application logic and deployment to application server engines. AAs rely less on specific tool or language technology than on standardized middleware options, communications protocols, data gateways, and platform infrastructures such as Component Object Model (COM), JavaBeans and Common Object Request Broker Architecture (CORBA). The application architect is tasked with specifying an AA and supporting the deployment implementation.

AA (automated attendant)

A device, typically attached to a private branch exchange or voice mail system, that answers incoming calls.

AAC (ATM access concentrator)

A device used to concentrate a variety of services (e.g., frame relay, Internet Protocol and video) over a single asynchronous transfer mode (ATM) network access connection.

AAHP (American Association of Health Plans)

AAL (ATM adaptation layer)

The layer — where non-ATM data is converted to ATM format — serves as the “glue” that connects traditional packet and frame structures and multimedia data types with short, fixed-length ATM cells. It forms the top layer of the ATM version of the Broadband Integrated Services Digital Network (B-ISDN) protocol stack. AAL is divided into two sublayers: Convergence Sublayer, and Segmentation and Reassembly Sublayer.

A and B signaling

A type of in-band signaling used in T1 transmission, in which one bit, robbed from each of the 24 subchannels in every sixth frame, is used for carrying dial and control information.

AAR (automatic alternative routing)

Automatic switching of a call to other routes when first-choice routes are unavailable.

AARP (AppleTalk Address Resolution Protocol)

A protocol that maps network-assigned addresses to underlying data link addresses in an AppleTalk network.

ABAP (Advanced Business Application Programming)

The core tool for development in SAP's R/3 system.

ABC (activity-based costing)

An improved approach to understanding where and why costs are incurred within an enterprise. It provides the information for activity-based management, which focuses on the decisions and actions needed to reduce costs and increase revenue. ABC differs from traditional cost accounting in explicitly recognizing that not all cost objects place an equal demand on support resources.

ABEND (abnormal end)

A type of system error in which a task or program fails to execute properly (i.e., “abnormally ends”). The term is also used as the name for a type of error message that indicates such a failure has occurred.

ABI (application binary interface)

A set of specifications that enables an application written for one target operating system (OS) and hardware platform to run on a different OS and platform, where two hardware platforms share the same processor type. ABIs enable compatibility only among products built on the same microprocessor architectures.

ABM (activity-based management)

The use of activity-based costing (ABC) principles in the ongoing management of costs and resources. See ABC.

ABM (Asynchronous Balanced Mode)

A High-Level Data Link Control (HDLC) mode supporting peer-oriented, point-to-point communication between two stations. See HDLC.

ABR (automatic baud rate)

A process in which the receiving device in a network automatically detects, and adjusts for, the transmission rate of incoming data by examining the first character of the message to determine the data speed.

ABR (Available Bit Rate)

An asynchronous transfer mode (ATM) service category, defined by the ATM Forum and conceptually similar to a frame relay network, in which a minimal cell rate will be guaranteed and bursts can be supported if the network resources allow it.

ABSE (agent-based software engineering)

An object-oriented approach to software engineering, where the objects or components may have a range of additional capabilities, e.g., persistence, asynchrony, autonomy, code mobility and higher-level messaging protocols. The primary design rationale of ABSE is to advocate software engineering principles that better match the characteristics of the real world, i.e., having dynamic and persistent entities communicating and interacting in cooperative, and sometimes competitive, ways. ABSE targets the software engineering of complex distributed applications, e.g., Internet-based applications, e-commerce, e-business, telecommunications, personal assistants, office automation, information and travel brokerage.

abstract syntax

OSI term for a machine-independent data structure description.

Abstract Syntax Notation 1 (see ASN.1)

AC (alternating current)

A type of power supply in which the current periodically reverses its direction, as distinguished from a direct current (see DC).

ACA (automatic circuit assurance)

A PBX feature that assists the customer in identifying possible trunk malfunctions.

"A" carrier

A nonwireline cellular carrier.

Accelerated Graphics Port (see AGP)

access charge

The charge assessed to communications users for access to the local or regional exchange to send or receive calls. It also includes access to specialized telecommunications services and access to the customer's local access and transport area.

access code

The digit or digits that a user must dial to be connected to an outgoing trunk facility.

access control

System functions and administrative tasks that monitor system operation, perform user identification, and record system accesses and changes.

Access Control Facility 2 (see ACF2)

access control list (see ACL)

Access Device Messaging System (see ADMS)

access line

The connection to the customer's local telephone company for origination of local and long-distance calls. Also known as a local loop or trunk.

access method

1. The portion of a computer's operating system responsible for formatting data sets and their direction to specific storage devices. Examples from the mainframe world include Virtual Storage Access Method (VSAM) and Indexed Sequential Access Method (ISAM).
2. In local-area networks, the technique or program code used to arbitrate the use of the communications medium by granting access selectively to individual stations. Examples are Carrier Sense Multiple Access with Collision Detection (CSMA-CD) and token passing.

access point

The basic building block of a roaming wireless LAN infrastructure. Access points attach to a wired backbone and provide wireless connectivity to all devices within range. As devices move out of the range of one access point, they move into the range of another.

access router

An access device with built-in basic routing protocol support, specifically designed to allow remote LAN access to corporate backbone networks. Access routers are not designed to replace backbone routers or to build backbone networks. They usually have limited protocols, few ports and low speed.

accounting rate

The charge per traffic unit — a unit of time or information content — covering communications between zones controlled by different telecommunications authorities. Accounting rates are used to establish international tariffs.

accounts payable

The process of managing payments to suppliers or vendors.

accounts receivable

The process of managing payments from customers.

Accredited Standards Committee (see ASC)

ACD (automatic call distributor)

A specialized phone system that handles many incoming calls. ACD is used for a variety of order-taking functions, such as calls to help desks or dispatching of service technicians. It is designed to distribute a large volume of incoming calls uniformly to a number of operators or agents, e.g., for airline reservations.

ACE (Advanced CMOS-ECL)

A high-end processor technology from Hitachi Data Systems.

ACE (Advanced Computing Environment)

A consortium founded in April 1991 whose goal was to establish at least source code compatibility between Santa Cruz Operation (SCO) and System V Release 4 (SVR4) Unix systems on either reduced instruction set computer (RISC) or Intel processors, as well as source compatibility of Windows NT systems on RISC or Intel systems. The primary founding members were Digital Equipment, Compaq Computer, MIPS Computer Systems, SCO and Microsoft.

A+ certification

A service technician credential issued by the Computing Technology Industry Association (CompTIA).

ACF (Advanced Communications Function)

A family of IBM communications programs that handle tasks such as resource sharing and distribution of functions. They include ACF/Virtual Telecommunications Access Method (ACF/VTAM) and ACF/Network Control Program (ACF/NCP). IBM eventually dropped the "ACF/" prefix from many of these program names; for example ACF/VTAM and ACF/NCP are now known simply as "VTAM" and "NCP."

ACF2 (Access Control Facility 2)

A host-based security subsystem from Computer Associates; also known as CA-ACF2.

ACF/NCP (see NCP)

ACF/VTAM (see VTAM)

ACH (automated clearinghouse)

A type of funds transfer network that processes debit and credit transactions between accounts from participating financial institutions.

ACID (atomicity, consistency, isolation, durability)

Four well-established tests for verifying the integrity of business transactions in a data-processing environment.

ACL (access control list)

Manages users and their access to files and directories. Access control requires linking users with content. User information is stored in a directory, and content is referenced in ACLs.

ACM (Association for Computing Machinery)

An educational and scientific computing society, founded in 1947, whose membership today includes more than 80,000 computing professionals and students worldwide.

ACMS (Application Control and Management System)

A transaction-processing monitor from Compaq (now part of Hewlett-Packard); originally a product of Digital Equipment, which Compaq acquired in 1998.

acoustic coupler

A device that allows a telephone handset to be used for access to the switched telephone network for data transmission. Digital signals are modulated as sound waves, and data rates are typically limited to a range of 300 bits per second (bps) to 1,200 bps.

ACP (array control processor)

A type of processor used in storage systems.

ACPI (Advanced Configuration and Power Interface)

A standard developed by Intel, Microsoft and Toshiba to improve PC power management and plug-and-play capabilities.

A-CPR (ambulatory computer-based patient record)

A computer-based patient record (CPR) that is used exclusively in the ambulatory/outpatient care delivery environments of healthcare.

ACR (Allowed Cell Rate)

ATM Forum-defined parameter for asynchronous transport mode (ATM) traffic management.

ACR (attenuation-to-crosstalk ratio)

A measure of signal quality in network cabling.

Acrobat

A product from Adobe Systems used to display documents in digital form as they appeared in their original, paper format. As an Internet plug-in, it is often used display documents, such as brochures, on the Web.

ACS (Affiliated Computer Services)

A business process and IT outsourcing provider headquartered in Dallas, Texas.

ACSE (Association Control Service Element)

An ISO standard specifying how application processes on two remote machines must initiate and terminate a communications session. This session is called an association.

ACSLs (Automated Cartridge System Library Software)

A Unix-based tape-library-sharing system from Storage Technology.

ACT I (application, channel, technology and industry)

A Gartner acronym representing the "survival locations" for integrated document management (IDM) vendors. A survival location is a market segment where a vendor can develop sustainable competitive advantage. A critical mass of sustainable competitive advantage is necessary for a vendor to thrive in the long term in any market.

active data dictionary

A facility for storing dynamically accessible and modifiable information relating to midrange-system data definitions and descriptions.

Active Data Warehouse (see ADW)

Active Directory

The “directory service” portion of the Windows 2000 operating system. Active Directory manages the identities and relationships of the distributed resources that make up a network environment. It stores information about network-based entities (e.g., applications, files, printers and people) and provides a consistent way to name, describe, locate, access, manage and secure information about these resources. It is the central authority that manages the identities and brokers the relationships between these distributed resources, enabling them to work together.

active-matrix liquid crystal display (see AMLCD)

Active Server Pages (see ASP)

ActiveX

An application programming interface (API) that enhances Microsoft’s OLE protocol. Often compared to Java, ActiveX facilitates various Internet applications, and therefore extends and enhances the functionality of Microsoft’s Internet Explorer browser. Like Java, ActiveX enables the development of interactive content. When an ActiveX-aware browser encounters a Web page that includes an unfamiliar feature, it automatically installs the appropriate applications so the feature can be used.

ActiveX Data Objects (see ADO)

activity-based costing (see ABC)

activity-based management (see ABM)

actuator

The mechanism that positions the disk read-write head over the selected track.

ACU (automatic calling unit)

A unit, which may or may not be integrated within a modem, that automatically dials calls based on digits supplied by the attached business machine.

AD (application development)

The function of creating applications for an enterprise. The term refers not simply to programming, but to the larger overall process of defining application requirements, planning the application structure, developing the code, monitoring development progress and testing results.

A/D (analog-to-digital)

The term used to describe a type of converter used to bridge analog and digital circuitry. A/D converters can be either stand-alone microcomponents, or included in the functionality embedded in certain advanced processor types.

Adaptive Application Architecture (see A3)

adaptive control

A control strategy that automatically changes the type or influence of control parameters to improve overall control system performance.

adaptive differential pulse code modulation (see ADPCM)

adaptive equalization

Equalization that is adjusted during transmission to compensate for line characteristics.

adaptive routing

Routing that automatically adjusts to network changes, such shifts in network traffic, to find the most efficient path for transmission.

ADC (analog-to-digital converter)

Component used to bridge the digital and analog circuitry contained in certain advanced integrated circuits. Also known as an A/D converter (see A/D).

ADC (automated data collection)

The automated conversion of disparate types of information into computer records. ADC devices and technologies include bar-code systems, optical character recognition, radio frequency identification devices and magnetic media recording.

address

1. Coded representation of the origin or destination of data. Each terminal on a communications line, for example, has a unique address.
2. In software, a location in memory that can be specifically referred to in a program.

Address Resolution Protocol (see ARP)

address translation

The process of changing the identifier associated with an item of data, or an instruction, to the actual location in main storage where it is held.

address verification service (see AVS)

ADDS (automated drug dispensing system)

An automated system designed to dispense drugs at the point of care — including ATM-like equipment found in patient care areas — to allow caregivers to dispense drugs as needed, improving workflow, timeliness, administrative accuracy and clinical effectiveness.

ADE (application development environment)

An product that offers a range of tools or features (e.g., for programming, interface development and testing) to provide a complete “environment” for developing applications.

ADF (Application Development Facility)

An IBM program for developing Information Management System (IMS) applications. See IMS.

ADF (automated document factory)

Gartner’s term for an architecture and set of processes to manage the creation and delivery of mission-critical, high-volume digital documents. The ADF applies factory production concepts to the document production — raw materials, including data and

preparation instructions, enter the ADF, where they are transformed into digital documents and prepared for delivery.

Adjunct Switch Application Interface (see ASAI)

ADLC (Asynchronous Data Link Control)

An IBM Systems Network Architecture (SNA)-compatible protocol for asynchronous connection of remote terminals.

ADMF (Asynchronous Data Mover Facility)

A Hitachi mainframe feature designed to enhance system performance in data moves between central and expanded storage.

admission, discharge and transfer system (see ADT)

ADMS (Access Device Messaging System)

A proprietary standard developed by Visa Interactive, a PC and Internet banking outsourcing organization acquired by Integrion in 1997. In 1998, Integrion announced it would "sunset" ADMS and the Visa Interactive platform by migrating its customers to the Integrion platform and its GOLD standard. See Integrion and GOLD.

ADO (ActiveX Data Object)

A high-level data access object model introduced by Microsoft in 1996.

Adobe Type Manager (see ATM)

ADP (Automatic Data Processing)

A check-processing and payroll services company based in Roseland, New Jersey.

ADPCM (adaptive differential pulse code modulation)

A speech-coding method that calculates the difference between two consecutive speech samples in standard pulse code modulation (PCM) coded telecommunications voice signals. This calculation is encoded using an adaptive filter and, as a result, allows analog voice signals to be carried on a 32 Kbps digital channel in half the space PCM uses.

ADSI (Active Directory Services Interface)

ADSI (Analog Display Services Interface)

ADSL (asymmetric digital subscriber line)

ADSL allows cable TV, video, telephony, and other multimedia services to be sent over voice-grade twisted-pair cable carrying from 1.5 megabits per second (Mbps) to 8 Mbps downstream, and from 16 kilobits per second (Kbps) to 640 Kbps upstream (hence the designation "asymmetric"), over distances ranging from two to six kilometers without the use of repeaters. ADSL uses adaptive digital filtering, which adjusts to compensate for noise and other problems on the line.

ADSM (ADSTAR Distributed Storage Manager)

An IBM software product that enables a user to manage storage, data access and backup across multi-vendor enterprisewide networks.

ADSP (Apple Datastream Protocol)

A transport mechanism for communication between Apple Macintosh systems and Digital Equipment Virtual Address Extension (VAX) systems.

ADSTAR (Automated Document Storage and Retrieval)

ADSTAR Distributed Storage Manager (see ADSM)

ADT (abstract data type)

ADT (Access Developer's Toolkit)

ADT (admission, discharge and transfer system)

A software system that records admissions to, discharges from and transfers within a hospital, and maintains the hospital census.

AD technology profile

A Gartner method of assessing an enterprise and its methodology. It includes:

- Development process profile: Life cycle coverage, ease of use, change management, methodologies supported, project management, information model, templates/componentware.
- Development technology profile: Workgroup support, development platform, repository, object-oriented (OO) component concepts, technical quality, openness, integration.
- Target environment profile: Execution platforms, human interface, database management systems, middleware supported, portability, communications protocols, execution technology, reliability.
- Application capabilities profile: Application topology, application types, complexity supported, transaction volume, security, usability support, type of users supported.
- (Built-in) execution environment profile: Application servers, middleware, database gateways, wrapper facilities, workflow engine, rules engine, dynamic repartitioning.

Advanced Business Application Programming (see ABAP)

Advanced CMOS-ECL (see ACE)

Advanced Communications Function (see ACF)

Advanced Communications Function/Virtual Terminal Access Method (see ACF/VTAM)

Advanced Computing Environment (see ACE)

Advanced Configuration and Power Interface (see ACPI)

Advanced Function Printing (see AFP)

Advanced Intelligent Network (see AIN)

Advanced Intelligent Tape (see AIT)

Advanced Interactive Executive (see AIX)

Advanced Metal Evaporated (see AME)

Advanced Micro Devices (see AMD)

Advanced Mobile Phone Service (see AMPS)

advanced payer revised DRG (see APR-DRG)

Advanced Peer-to-Peer Networking (see APPN)

Advanced Planner and Optimizer (see APO)

Advanced Program-to-Program Communications (see APPC)

Advanced Research Projects Agency Network (see ARPANET)

advanced shipment notice (see ASN)

advanced technology

A technology that is still immature but promises to deliver significant value, or that has some technical maturity but still has relatively few users. Among current examples: artificial intelligence, agents, speech and handwriting recognition, virtual reality and 3-D visualization, smart cards, real-time collaboration, enhanced user authentication, data mining, and knowledge management.

Advanced Technology Attachment (see ATA)

advanced technology group (see ATG)

Advanced Visual Systems (see AVS)

ADW (Active Data Warehouse)

A relational database from i2 Technologies. The ADW serves as an operational data store for the data used in systems such as i2's Supply Chain Planner tool.

AEC (architecture, engineering and construction)

AEP (AppleTalk Echo Protocol)

AFC (antiferromagnetically coupled)

A type of storage media that uses an advanced magnetic coating expected to dramatically increase hard disk drive (HDD) capacity. IBM, which shipped the first AVC-based storage products in 2001, predicts that advances in the new material will quadruple HDD data density (to 100 gigabits per square inch) by 2003.

affiliate

An enterprise that sells products of other manufacturers or retailers (i.e., sponsoring merchants) on its Web site. Users select products at the affiliate Web site, but the sale is actually transacted at the sponsoring merchant's site. Affiliates are similar in concept to industry-based manufacturing representatives that sell multiple manufacturers' product lines.

Affiliated Computer Services (see ACS)

AFP (Advanced Function Printing)

An IBM all-points-addressable enterprisewide print architecture.

AFP (AppleTalk Filing Protocol)

A standard way to display the filing system of a server to Apple Macintosh clients.

AFPDS (Advanced Function Printing Data Stream)

agent

Software that acts as an intermediary for a person by performing an activity. Agents can “learn” an individual’s preferences and act in the person’s best interest. For example, an agent for a purchasing manager could learn corporate specifications, determine when inventory is low, search the Internet for the lowest-cost supplier, and even negotiate and complete transactions.

agent-based software engineering (see ABSE)

agent profile

A basic agent technology that enables the user to be selective about what reaches the desktop. It describes users’ information interests in words, concepts and relationships, not in a simulation of artificial intelligence. When invoked on a user’s behalf, agent profiles can be applied to real-time broadcasts, office applications including e-mail and distributed filing systems such as Microsoft Exchange. Technically, an agent profile is an inquiry that is reused against new, incoming information to determine if, when and to whom information should be delivered.

aggregator

Enables buyers within a market to select among competitors by aggregating information about the market and its suppliers and providing this information via a Web site. Aggregators may provide decision-support applications that integrate supplier information with third-party information and with user requirements or preferences to allow users to differentiate the services and features of various competitors. Content aggregators aggregate information and match it to user preferences. These preferences may be declared actively (i.e., the user explicitly specifies it) or passively (e.g., the software discerns preferences from patterns of user behavior or interest) and are used to filter aggregated content and deliver only the content that matches the user’s preferences.

AGP (Accelerated Graphics Port)

An Intel technology for desktop systems. It increases system performance by offloading graphic requirements from the system bus to a bus dedicated to video processing.

AGVS (automated guided vehicle system)

Aho Weinberger Kernighan (see AWK)

AHP (analytical hierarchy process)

A process that uses hierarchical decomposition to deal with complex information in multicriterion decision making, such as information technology vendor and product evaluation. It consists of three steps:

1. Developing the hierarchy of attributes germane to the selection of the IT vendor.
2. Identifying the relative importance of the attributes.
3. Scoring the alternatives' relative performance on each element of the hierarchy.

Developed by Thomas Saaty while he was teaching at the University of Pennsylvania's Wharton School of Business, the AHP is recognized as the leading theory in multicriterion decision making.

AI (artificial intelligence)

A wide-ranging discipline of computer science that at its core seeks to make computers behave more like humans. The term was coined by John McCarthy of the Massachusetts Institute of Technology in 1956. AI attempts to resolve problems by "reasoning," similar to the process used by the human mind. AI involves the capability of a machine to learn (to remember results produced on a previous trial and to modify the operation accordingly in subsequent trials) or to reason (to analyze the results produced in similar operations and select the most favorable outcome). Today, applications of artificial intelligence include voice recognition, robotics, neural networks and expert systems (i.e., systems that can make decisions an expert would otherwise have to make to, for example, forecast financial performance, or diagnose illnesses).

AIAG (Automotive Industry Action Group)

An association of automotive companies which has defined a number of standards, including the E-5 standard for sending e-commerce transactions over the Automotive Network Exchange, a private-IP-based commerce network. The group's EDI standards work includes implementation guides for X12 transactions, and the contribution of content to the EDIINT AS2 specification.

AIDC (automatic identification data capture)

AIF (Application Integration Feature)

AIIM (Association for Information and Image Management)

A Maryland-based organization dedicated to promoting development of systems that store, retrieve and manage document images.

AIM (AOL Instant Messenger)

The instant messaging (IM) technology offered in America Online's (AOL's) Internet service. AIM is among the most widely used IM services, with more than 60 million registered users worldwide.

AIML (Astronomical Instrument Markup Language)

AIN (Advanced Intelligent Network)

A U.S. variant of intelligent network (IN) technology evolving from earlier research work and specifications carried out by Bellcore (now Telcordia Technologies).

AIR (architecture implementation review)

Designed to enable an alignment between information technology (IT) strategy and implementation, this review, which involves IS managers and end users, assures that tactical initiatives such as product acquisitions support the enterprise information architecture and, therefore, the business and technology objectives. The

AIR can be a periodic, formal review or it can be ongoing (e.g., each time the enterprise distributes a request for proposal for IT equipment).

AIT (Advanced Intelligent Tape)

The first 8-millimeter helical-scan tape drive designed and manufactured by Sony's tape streaming division using advanced metal evaporated media. Sony has differentiated its AIT drive from other 8-millimeter tape drives with a unique media feature on the tape cartridge called memory in cassette (MIC).

AIX (Advanced Interactive Executive)

A Unix-based operating system from IBM.

A/L (Archive Link)

An interface that enables enterprises to link traditional storage archives with SAP R/3. A/L is the product of a partnership between SAP and iXOS Software.

A-law encoding

Encoding according to ITU Recommendation G711, used with European 30-channel pulse code modulation (PCM) systems that comply with ITU Recommendation G732. It employs nonuniform quantifying to obtain the desired compression characteristic.

alert

A message that an application or operating system may display to notify the user of certain conditions.

algorithm

A process for calculations involving manipulation of numbers.

ALL-IN-1

A VAX/VMS-based office information system that was a dominant player in corporate messaging in the late 1980s, from Digital Equipment (acquired in 1998 by Compaq Computer, which in turn was acquired by Hewlett-Packard in 2002).

Allowed Cell Rate (see ACR)

all points addressable (see APA)

Alpha

A 64-bit reduced instruction set computing (RISC) microprocessor from Compaq (now part of Hewlett-Packard), which acquired the technology with its 1998 purchase of Digital Equipment.

alternate buffer

Two input/output buffers that are alternated to provide continuous throughput.

alternate mark inversion (see AMI)

alternate voice/data circuits (see AVD circuit)

alternating current (see AC)

alternative routing

Method of routing whereby a secondary communications path is used if the usual one is not available.

ALU (arithmetic logic unit)

The core element of a central processing unit (CPU) which actually carries out the arithmetic computations.

ALUW (application logical unit of work)

A design container that developers define for allocation across client/server environments. It is useful in designing enterprise client/server applications of a scope and complexity beyond traditional applications.

AM (amplitude modulation)

A form of modulation in which the amplitude of the carrier is varied in accordance with the instantaneous value of the modulating signal.

AM (asset management)

A system of practices intended to address shortcomings, inefficiencies, waste and unavoidable failures in managing technology and information technology equipment. It involves five major areas:

1. Requisition
2. Procurement
3. Deployment
4. Maintenance
5. Retirement strategies

At its core is an integrated data repository that contains:

- Asset tracking — technical information about the equipment or software
- Portfolio information — acquisition and financial details
- A contracts database — summarizing key software and maintenance contracts terms and conditions

AMA (automatic message accounting)

An automatic recording system that documents all the necessary billing data of subscriber-dialed long-distance calls.

ambient noise

Communications interference present in a signal path at all times.

ambulatory computer-based patient record (see A-CPR)

ambulatory suite (see AS)

AMD (Advanced Micro Devices)

A microprocessor manufacturer headquartered in Sunnyvale, California.

AME (Advanced Metal Evaporated)

A media formulation manufactured by Sony for use in its consumer and computer product lines.

American Management Systems (see AMS)

American Medical Informatics Association (see AMIA)

American National Standards Institute (see ANSI)

American Standard Code for Information Interchange (see ASCII)

AMF (annual maintenance fee)

AMHE (automated materials-handling equipment)

AMI (alternate mark inversion)

A digital signaling method in which the signal carrying the binary value alternates between positive and negative polarities; zero and one values are represented by the signal amplitude at either polarity, while no-value "spaces" are at zero amplitude. Also called bipolar.

AMIA (American Medical Informatics Association)

A not-for-profit organization dedicated to the development and application of medical informatics in the support of patient care, teaching, research and healthcare administration. The AMIA serves as an authoritative body in the field of medical informatics and represents the United States in the informational arena of medical systems and informatics in international forums.

AMIS (Audio Messaging Interchange Specification)

An enhanced key system feature for voice/call processing that enables enterprise locations to transfer and forward voice messages between systems. It is a voice processing standard that specifies the procedures to network voice processing systems, regardless of who manufactures the system.

AML (Astronomical Markup Language)

AMLCD (active-matrix liquid crystal display)

A key flat panel display (FPD) technology.

AMO (application management outsourcing)

The ongoing maintenance, management, conversion, enhancement and support of an application portfolio by an external company. AMO, a subset of application outsourcing (see separate entry), includes changes that generally take less than some predefined time to implement (e.g., 10 days or 30 days). Examples of maintenance include regulatory changes, software upgrades, new release installations and "fix it if it breaks" troubleshooting. AMO may involve the transfer of people and application software to the vendor.

amplifier

An electronic component that boosts the strength or amplitude of a transmitted, usually analog, signal; functionally equivalent to a repeater in digital transmissions.

amplitude

A departure of the value of a wave or alternating current from its average value.

amplitude modulation (see AM)

AMPS (Advanced Mobile Phone Service)

The term used by AT&T's Bell Laboratories (prior to divestiture in 1984) to refer to its cellular technology.

AMR (automated medical record)

AMS (American Management Systems)

A business and IT consulting firm headquartered in Fairfax, Virginia.

AMS (Application Management Specification)

A set of application-programming interfaces that enable the consistent definition of applications for deployment and management, from IBM's Tivoli subsidiary.

AMS (automation management system)

A subsystem of the warehouse management system. It controls automated material-handling equipment such as carousels, pick to light, in-line scales, and conveyors. The AMS is designed to provide a standardized interface between the warehouse management system and the automated material-handling equipment.

analog

Information presented in the form of a continuously varying signal. A transmission is conveyed by modulating (varying) the frequency, amplitude or phase of a carrier.

analog loopback

A technique for testing transmission equipment and devices which isolates faults to the analog signal receiving or transmitting circuitry, where a device such as a modem echoes back a received (test) signal that is then compared with the original signal.

analog signal

A signal in the form of a continuous wavelike pattern representing a physical quantity, such as voltage, which reflects variations in some quantity, such as loudness of the human voice. Analog signaling is generic to the public switched telephone network (PSTN), as well as to certain other audio frequency and radio frequency facilities such as broadcasting. A digital baseband signal generated by a business machine must be converted to analog form in order to be transmitted over an analog facility, e.g., a voice grade telephone line. This process is performed by a modem.

analog-to-digital (see A/D)

analog-to-digital converter (see ADC)

analog transmission

Transmission of a continuously variable signal as opposed to a discretely variable signal. Also called analog signaling.

analytical hierarchy process (see AHP)

anchor

A fragment of information in a hypertext document to which a link is attached, enabling connection to another document.

ANDF (Architecture-Neutral Distribution Format)

The Open Group's format enabling distribution of a single version of an application to computers with different hardware and software architectures.

ANI (automatic number identification)

A series of digits, in either analog or digital form, which tells a user the originating number of the incoming phone call. Caller ID is the most familiar form of ANI.

angle of arrival (see AOA)

ANOVA (analysis of variance)

ANSI (American National Standards Institute)

In the United States, ANSI serves as a quasi-national standards organization. It provides "area charters" for groups that establish standards in specific fields. These groups include the Institute of Electrical and Electronics Engineers (IEEE) and the Electronic Industries Association (EIA). ANSI is unique among the world's standards groups as a nongovernmental body granted the sole vote for the United States in the International Organization for Standardization (ISO). This status is part of the reason ANSI bends over backward to limit its role to that of facilitator, or catalyst, in the production of standards.

ANSI X series

Any of a number of standards administered by ANSI, including:

1. ANSI X3B11: The evolving standards for smaller-format optical storage subsystems.
2. ANSI X9.9: U.S. electronic data interchange (EDI) standard for message authentication in banking.
3. ANSI X.12: U.S. EDI standard.

antenna

A device used to transmit or receive radio waves. The physical design of the antenna determines the frequency range of transmission and reception.

antiferromagnetically coupled (see AFC)

ANX (Automotive Network Exchange)

Established by the Automotive Industry Action Group (AIAG) to offer extranet-based applications to suppliers of Chrysler, Ford and General Motors.

AOA (angle of arrival)

Based on triangulation, a method of processing cellular phone signals, AOA allows the physical position of switched-on wireless devices to be located. The position is determined by comparing the AOA of the existing reverse control channel at various cell sites.

AOL (America Online)

AOL Instant Messenger (see AIM)

AOR (architecture objectives review)

Part of enterprise information management focused on achieving an alignment between business and information technology strategies. This alignment was explored in some detail throughout the 1980s. In fact, more than 45 different methods are now being used throughout the industry.

AP (see access point)

AP (see accounts payable)

AP (see agent profile)

APA (all points addressable)

A monitor or printer that can place a dot or mark at any x/y coordinate within the limit of its resolution.

Apache

An open-source Web server.

APACS (Association for Payments Clearing Services)

The organization that manages U.K. payment systems.

aperture

The diameter of a parabolic antenna.

APG (Ambulatory Patient Group)

API (application programming interface)

A set of calling conventions that defines how a service is invoked through software. An API is an interface that enables programs written by users or third parties to communicate with certain vendor-supplied software.

APICS (American Productivity and Inventory Control Society)

APM (Advanced Power Management)

APO (Advanced Planner and Optimizer)

A supply-chain-planning suite from SAP.

APPC (Advanced Program-to-Program Communications)

The programming interface to LU 6.2, IBM's protocol for peer relationship program-to-program communications under Systems Network Architecture (SNA).

AppleShare

Apple software for the Apple Macintosh environment, which enables the sharing of files and network services.

applet

A small program that runs within an application. Applets are commonly used to make otherwise static Web pages more interactive. Examples include animated graphics, games, configurable bar charts and scrolling messages. Applets also play an important role in network computers (NCs). They increase an NC's independence from the server because they do not have to communicate with the operating system (resident on the server) to function once the applet has been received by the NC.

AppleTalk Address Resolution Protocol (see AARP)

AppleTalk Filing Protocol (see AFP)

AppleTalk Remote Access (see ARA)

AppleTalk Update-Based Routing Protocol (see AURP)

appliance

See computing appliance.

Applicability Statement 1 (see AS1)

Applicability Statement 2 (see AS2)

applicant tracking system

An application used to track resumes and data on job applicants. Typically, these applications include matching features, which rank candidates by matching them to criteria specified in the requisition for an open position.

application

A specific use for a computer, such as for accounts payable or payroll. The term is commonly used in place of the terms "application program," "software" or "program." Examples of programs and software include pre-packaged productivity software (such as spreadsheets and word processors) and larger, customized packages designed for multiple users (such as e-mail and workgroup applications).

application architecture (see AA)

application binary interface (see ABI)

application, channel, technology and industry (see ACT I)

Application Control and Management System (see ACMS)

application database

The initial point of capture for much of the data that enters an enterprise. Application databases are accessed and updated by end users through online transaction processing (OLTP) applications. These databases tend to contain very detailed and up-to-date data; however, because OLTP applications normally support specific business processes, application databases tend to be process-specific, rather than enterprisewide, in nature. They may support some degree of data analysis, but this tends to be secondary to OLTP activity. An application database is often a source for refinement of data into one or several of the other database implementation styles (e.g., a data warehouse or operational data store).

Application Development Facility (see ADF)

application generator

A tool or toolset that is more sophisticated than simple query/report facilities, and that enables reasonably complex interactive or batch applications to be created at a higher level more easily than the compiler languages mentioned under “traditional processing.” Code that is generated should be compiled rather than interpreted for better performance.

application hosting

A service in which a vendor will house shared or dedicated servers and applications for an enterprise at the provider’s controlled facilities. The vendor is responsible for all day-to-day operations and maintenance of the application. Application hosting is typically based on service arrangements with vendors that provide the hardware, software and networking infrastructure that enable an enterprise to run applications externally by connecting electronically using a browser. A vendor may offer the services directly or, more commonly, through an arrangement between a vendor and an application service provider (ASP).

application integration

The process of 1) keeping redundant copies of data (in independently designed applications) consistent, or 2) enabling end-users to access data and functionality from independently designed applications on a single user interface.

application layer

The top of the seven-layer Open Systems Interconnection (OSI) model, generally regarded as offering an interface to, and largely defined by, the network user. See OSI.

application logical unit of work (see ALUW)

application management outsourcing (see AMO)

Application Management Specification (see AMS)

application portfolio analysis

A tool to divide current and proposed applications into three categories — utility, enhancement and frontier — based on the degree to which they contribute to the enterprise’s performance. The utility category is essential but does not enhance the enterprise’s performance (e.g., payroll); the enhancement category contains applications that improve the enterprise’s performance based on the use of established technology (e.g., documentation automation); and the frontier category is aimed at greatly improving enterprise performance (e.g., through aggressive use of rules-based decision support) but usually entails substantial risk. The management issues for each category are, respectively, cost, opportunity identification and innovation. The planning process should consider the best balance among the three categories to gain optimal future performance and the appropriate value from the application of IT.

application program

Software programs in a system are either application programs or supervisory programs, also called system software. Application programs contain instructions

that transfer control to the system software to perform input/output and other routine operations, working through the application programming interface (API).

application programming interface (see API)

application response measurement (see ARM)

application development (see AD)

application development environment (see ADE)

application server

1. A hardware server designated to run applications (but not a database).
2. A single runtime executable that contains modules of application logic.
3. Systems software used to host the business logic tier of applications — i.e., to manage, optimize, monitor, secure, scale and make its functionality accessible.

For three-tier applications, the application server is the fundamental enabling technology. It enables the management of the business logic of an application in the middle tier, and it enables access to the middle tier from the user interface tier of the application. In a service-oriented architecture, an application server is the host for the application services and also plays the role of a fundamental enabling technology. Transaction processing monitors, object request brokers and object transaction monitors are native application server products.

Application Server Evaluation Model (see ASEM)

application service provider (see ASP)

application sharing

The ability of two or more participants to have equal and simultaneous control over the content of a document inside an application (e.g., a word processing document, spreadsheet or conference slide) over a wide-area network, local-area network or modem connection. Enables users in different locations to work together on the same documents, with shared control and editing capabilities. A component of data conferencing.

applications outsourcing

An outsourcing arrangement for a wide variety of application services including new development, legacy systems maintenance, offshore programming, management of packaged applications and staff augmentation. While this form of outsourcing generally involves a transfer of staff, the use of the term has recently broadened to include arrangements where this is not the case, as in staff augmentation. It does not include system integration activities.

application-specific integrated circuit (see ASIC)

application-specific standard product (see ASSP)

applications portfolio analysis (see APA)

Application System/400 (see AS/400)

APPN (Advanced Peer-to-Peer Networking)

IBM's APPN is an extension of Systems Network Architecture (SNA) Low-Entry Networking (LEN, also known as Physical Unit 2.1), which provides dynamic multipath routing among nodes.

APRC (Amdahl Peer-to-Peer Remote Copy)

APR-DRG (advanced payer revised DRG)

An enhanced diagnosis-related group (DRG) scheme that provides more granular, appropriate groupings.

APS (advanced planning and scheduling)

A subcomponent of supply chain planning, contextually describing manufacturing planning and scheduling.

AQ (Advanced Queuing)

AQL (acceptable quality level)

AR (see accounts receivable)

ARA (AppleTalk Remote Access)

Protocol enabling remote Macintosh users to have direct access to information on a network, via a dial-up connection.

ARC (additional resource charge)

Archie

An early Internet search tool, used to locate document files via File Transfer Protocol. The name was derived from "archive."

architecture

1. In reference to computers, software or networks, the overall design of a computing system and the logical and physical interrelationships between its components. The architecture specifies the hardware, software, access methods and protocols used throughout the system.
2. A framework and set of guidelines to build new systems. IT architecture is a series of principles, guidelines or rules used by an enterprise to direct the process of acquiring, building, modifying and interfacing IT resources throughout the enterprise. These resources can include equipment, software, communications, development methodologies, modeling tools and organizational structures.

Architecture for Voice, Video and Integrated Data (AVVID)

architecture implementation review (see AIR)

Architecture-Neutral Distribution Format (see ANDF)

architecture objectives review (see AOR)

Architecture of Integrated Information Systems (see ARIS)

Archive Link (see A/L)

ARCnet (Attached Resource Computer Network)

A 2.5 megabits per second (Mbps) baseband, token-bus local-area network (LAN) in bus or star topologies developed by Datapoint in the late 1970s. Versions can support coax, twisted pair, or fiber cabling.

Arden Syntax

The Arden Syntax for Medical Logic Modules (MLMs) is a de facto standard in healthcare for coding clinical information (e.g., to generate clinical alerts, suggest interpretations or diagnoses, and guide compliance with protocols).

Ariba

A business-to-business (B2B) e-commerce platform and network services provider. Among the early players in Internet-based procurement, Ariba is one of the largest B2B e-marketplace technology providers, which also include Commerce One and RosettaNet. It offers an XML-centric transactional infrastructure based on its Commerce XML (cXML) specification. (See cXML, Commerce One and RosettaNet.)

ARIS (Architecture of Integrated Information Systems)

Suite of software tools developed by IDS Scheer that allows the user to design business processes and to measure their impact.

arithmetic logic unit (see ALU)

ARM (Application Response Measurement)

A set of application programming interfaces that enables enterprises to accurately measure application transaction response times across a distributed computing infrastructure. The ARM standard, approved by the Open Group, enables organizations to extend their enterprise management tools directly to applications, thus creating comprehensive end-to-end management capabilities that include measuring application availability, application performance, application usage and end-to-end transaction response time.

ARP (Address Resolution Protocol)

An Internet protocol used to map an IP address to a media access control (MAC) address.

ARPANET (Advanced Research Projects Agency Network)

The ARPANET, the forerunner of the Internet, was a pioneering long-haul network funded by the U.S. Department of Defense's Advanced Research Projects Agency (ARPA). It served as the test bed for many areas of internetworking technology development and testing, and acted as the central backbone during the development of the Internet. The ARPANET was built using packet-switching computers interconnected by leased lines.

ARQ (automatic repeat request)

An error control technique that requires retransmission of a data block which contains detected errors.

array control processor (see ACP)

ARS (automatic route selection)

Provides automatic routing of outgoing calls over alternative customer facilities based on the dialed long-distance number.

artificial intelligence (see AI)

artificial life/emergent computation

Produces complex behavior from the interaction of many simple behaviors (as in societies of ants and bees). Uses include simulation and planning.

ARTS (Association for Retail Technology Standards)

ARU (audio response unit)

A device that provides prerecorded spoken responses to digital inquiries from a telephone caller once the connection is established. Also called voice response unit (VRU).

ARUP (AppleTalk Update-Based Routing Protocol)

Encapsulates AppleTalk traffic in the header of a foreign protocol, enabling the connection of separate AppleTalk networks through a different network.

AS (ambulatory suite)

An application suite consisting of practice management, contract management, and ambulatory computer-based patient record (A-CPR) application components.

AS (autonomous system)

An administrative domain. All members of an AS that share route information can handle traffic to and from any destination.

AS1 (Applicability Statement 1)

An Internet Engineering Task Force (IETF) draft specification for the secure exchange of electronic data interchange (EDI) data over the Internet. Based on Simple Mail Transfer Protocol (SMTP), AS1 supports security features such as digital signatures, encryption and digitally signed return receipts. AS1 was last tested in November 1999.

AS2 (Applicability Statement 2)

A second IETF draft specification (after AS1 — see separate entry) to address security and interoperability issues in the exchange of EDI data over the Internet. Based on Hypertext Transport Protocol (HTTP), AS2 expands on AS1 to include support for additional security features, such as digital signing via Secure Multipurpose Internet Messaging Extensions (SMIME) and Open Specification for Pretty Good Privacy (OpenPGP).

AS/400 (Application System/400)

IBM's midrange system that replaced the IBM System/36 and System/38 product families.

ASA (average speed of answer)

A standard quantitative method for measuring the speed at which call center calls are answered.

ASAI (Adjunct Switch Application Interface)

An AT&T product that provides a detailed set of technical specifications for having an outside computer control an AT&T private branch exchange (PBX).

ASC (Accredited Standards Committee)

An organization, certified by the American National Standards Institute, that produces standard communication protocols for electronic data interchange.

Ascend Tunnel Management Protocol (see ATMP)

ASCII (American Standard Code for Information Interchange)

A standard table of seven-bit designations for digital representation of uppercase and lowercase Roman letters, numbers and special control characters in teletype, computer and word processor systems. Some IBM systems use similar code called Extended Binary-Coded Decimal Interchange Code (EBCDIC). Since most computer systems use a full byte to send an ASCII character, many hardware and software companies have made their own nonstandard and mutually incompatible extensions of the official ASCII 128-character set into a 256-character set.

ASCOM (asynchronous communication)

ASD (automated supply dispensing)

An extension of supply chain management capabilities to automated and monitored dispensing of tangible supplies.

ASEM (Application Server Evaluation Model)

A Gartner server-selection tool that seeks to provide selection guidance using a standard set of selection criteria that remain consistent for all application types. An enhanced version of the Gartner System and Server Evaluation (SSEM) model.

ASI (application-specific interface)

ASIC (application-specific integrated circuit)

A chip on which the pattern of connections has been set up exclusively for a specific function.

ASM (abnormal situation management)

The process of and systems supporting the handling of a deviation to the normal procedures involved with the control and management of a manufacturing or other production process.

ASN (advanced shipment notice)

An electronic data interchange (EDI) message sent from the shipper to the receiver prior to the departure of the shipment from the shipper's facility. The message includes complete information about the shipment and its contents. In today's environment, this message is more often an "as shipped notice" sent after the departure of the shipment.

ASN (autonomous system number)

A number assigned to a local network, registered into the carrier's routing community and placed under the umbrella of an administrative domain called an autonomous system (see AS).

ASN.1 (Abstract Syntax Notation 1)

A specification language adopted by the International Telecommunications Union (ITU) for the Open Systems Interconnection (OSI) Reference Model, described in ITU Recommendations X.409, X.208, and X.209.

ASO (automated system operations)

Often referred to as “lights-out operations.” This is a combination of hardware and software that allows a computer installation to run unattended — that is, without the need for a human operator to be physically located at the site of the installation.

ASP (Active Server Pages)

A technology introduced by the Mesa Group in 1997 and now owned by Microsoft (which acquired Mesa Group in 1998). ASP automatically senses whether the user’s browser supports ActiveX. If it does, the applet is downloaded; if not, the ASP runs the applet on the server and broadcasts the result to the client, essentially switching dynamically between Stage 0 (time sharing) and Stage 2 (applet-based clients).

ASP (application service provider)

An ASP is a company that provides the use of applications — from basic e-mail to enterprise applications — and associated services across a network to multiple customers. Gartner defines the ASP market as the delivery of standardized application software via a network — usually, but not necessarily, the Internet — through an outsourcing contract predicated on usage-based transaction pricing. The ASP may host the applications on the customer’s site. Most ASPs, however, house the applications in their own data centers, where the ASP is responsible for maintaining the applications and all associated hardware, software and network services to link the applications to the customer base. Despite the early promise of the ASP proposition, most ASP services to date have been limited to e-mail and enterprise resource planning.

ASP (average selling price)

ASR (automatic speech recognition)

ASRS or AS/RS (automated storage and retrieval system)

asset management (see AM)

Association Control Service Element (see ACSE)

Association for Computing Machinery (see ACM)

associativity

The ability to link computer-aided design (CAD) data and models together in a manner that allows design changes to be reflected automatically. Unidirectional or downstream associativity permits model changes to automatically change downstream data such as drafting, analysis or computer-aided manufacturing (CAM) data. Bidirectional associativity allows downstream changes, such as in drawings, to change the model.

ASSP (application-specific standard product)

An integrated circuit (IC) dedicated to a specific application market and sold to more than one user. A type of embedded programmable logic, ASSPs combine digital,

mixed-signal and analog products. When sold to a single user, Gartner defines such ICs as “application-specific integrated circuits” (see ASIC).

ASTD (American Society for Training and Development)

ASTM (American Society for Testing and Materials)

asymmetric cryptography

See public-key cryptography.

asymmetric digital subscriber line (see ADSL)

asynchronous

Characterized by not having a constant time interval between successive bits, characters or events. Transmission generally uses one start and one stop bit for character element synchronization (often called start-stop transmission).

Asynchronous Balanced Mode (see ABM)

asynchronous data link control (see ADLC)

Asynchronous Data Mover Facility (see ADMF)

asynchronous transfer mode (see ATM)

asynchronous transmission

A process in which each information character, and sometimes each word or small block, is individually synchronized, usually by the use of start and stop elements.

AT

Control language for dial-up modems developed by modem pioneer Hayes Microcomputer Products. Use of these commands is one of the characteristics that makes a modem “Hayes compatible.”

AT (advanced technology)

ATA (Advanced Technology Attachment)

A disk drive interface standard, also known as Integrated Drive Electronics (see IDE).

ATE (automatic test equipment)

ATG (advanced technology group)

The role of the ATG is to provide a continuing stream of technology opportunities to the enterprise. It typically takes the lead in prototype and pilot projects.

ATL (automated tape library)

ATM (Adobe Type Manager)

A program that enables the user to view type of any size with the highest resolution the user’s monitor can provide.

ATM (asynchronous transfer mode)

A wide-area network (WAN) technology, a transfer mode for switching and transmission that efficiently and flexibly organizes information into cells; it is asynchronous in the sense that the recurrence of cells depends on the required or instantaneous bit rate. Thus, empty cells do not go by when data is waiting. ATM's powerful flexibility lies in its ability to provide a high-capacity, low-latency switching fabric for all types of information, including data, video, image and voice, that is protocol-, speed- and distance-independent. Using statistical multiplexing, cells from many different sources are multiplexed onto a single physical circuit. The fixed-length fields in the cell, which include routing information used by the network, ensure that faster processing speeds are enabled using simple hardware circuits. The greatest benefit of ATM is its ability to provide support for a wide range of communications services while providing transport independence from those services.

ATM (automated teller machine)

A public banking machine that is usually hooked up to a central computer through leased local lines and a multiplexed data network.

ATM access concentrator (see AAC)

ATM adaptation layer (see AAL)

ATM Forum

An international body, mostly composed of networking vendors, that sets standards for — and promotes the use of — asynchronous transfer mode networking.

ATMI (application transaction manager interface)

ATMP (Ascend Tunnel Management Protocol)

A Layer 3 tunneling protocol developed by Ascend Communications (later acquired by Lucent Technologies). ATMP performs Internet Protocol (IP) encapsulation at the communications server, thus enabling remote users to connect into their networks over the Internet.

ATO (assemble to order)

A strategy allowing a product or service to be made to specific order, where a large number of products can be assembled in various forms from common components. This requires sophisticated planning processes to anticipate changing demand for internal components or accessories while focusing on mass customization of the final products to individual customers.

atomicity, consistency, isolation, durability (see ACID)

ATP (available to promise)

The uncommitted portion of a company's inventory or planned production. This figure is frequently calculated from the master production schedule and is maintained as a tool for order promising.

ATS (alternative trading system)

attachment unit interface (see AUI)

attenuation

A decrease in the magnitude of the current, voltage or power of a signal in transmission between points because of the transmission medium. Attenuation is usually expressed in decibels.

attenuation-to-crosstalk ratio (see ACR)

auction

An electronic market that can exist in a business-to-business and business-to-consumer context. Sellers offer products or services to buyers through a Web site with a structured process for price setting and fulfillment. Web auctions may follow English, Dutch, reverse-bid or sealed-bid processes.

Audio Messaging Interchange Specification (see AMIS)

audio response unit (see ARU)

Audiotex

A communications system that enables a host computer to pass data to a voice mail computer — a store-and-forward mechanism for digitized voice — where the data is interpreted and delivered over the telephone.

AUI (autonomous unit interface or attachment unit interface)

Most commonly used in reference to the 15-pin D-type connector and cables used to connect single- and multiple-channel equipment to an Ethernet transceiver.

AUNet (Asian UUNet)

AURP (AppleTalk Update-Based Routing Protocol)

authentication

The use of passwords, tokens (such as smart cards), digital certificates or biometrics (more commonly fingerprint, hand geometry and voice biometrics) to verify the identity of a user and better ensure against fraud.

authentication service

A mechanism, analogous to the use of passwords on time-sharing systems, for the secure authentication of the identity of network clients by servers and vice versa, without presuming the operating system integrity of either (e.g., Kerberos).

authorization

A process ensuring that correctly authenticated users can access only those resources for which the owner has given them approval.

auto baud rate (see ABR)

AUTODIN (Automatic Digital Network)

auto-identification technologies

Technologies used to identify physical objects (including humans) automatically and transparently. They allow an object to declare its identity to an auto-identification reading device. Examples include:

- Bar code scanners
- Magnetic strips and chip cards
- Optical character recognition (see OCR)
- Biometrics (for example, machine recognition of a unique human voice)
- Radio frequency identification (see RFID)

Auto Industry Action Group (see AIAG)

automated attendant (see AA)

automated backup

Delivers the most basic form of storage availability — recoverable data. Most enterprises are struggling with the implementation of this function. It is a conceptually complex and labor-intensive process. Backup design must address multiple elements (e.g., hardware, network, file system and application) across heterogeneous platforms and geographically dispersed sites. Labor intensive, departmental processes are replaced with automated, enterprise-level solutions to increase availability.

Automated Cartridge System Library Software (see ACSLS)

automated clearinghouse (see ACH)

automated data collection (see ADC)

automated document factory (see ADF)

automated drug dispensing system (see ADDS)

automated supply dispensing (see ASD)

automated system operations (see ASO)

automated teller machine (see ATM)

Automated Work Distributor (see AWD)

automatic alternative routing (see AAR)

automatic call distributor (see ACD)

automatic calling unit (see ACU)

automatic circuit assurance (see ACA)

Automatic Data Processing (see ADP)

automatic error correction

A transmission system feature that automatically detects and corrects a proportion of errors in a received signal. It performs fault detection and isolation and reconfigures

the system, dynamically invoking redundant components without the need to bring the system down.

automatic message accounting (see AMA)

automatic message-switching center

In a communications network, the location at which data is automatically routed according to its destination.

automatic number identification (see ANI)

automatic repeat request (see ARQ)

automatic restart

Also known as “warm recovery,” this is the resumption of operation after a system failure with minimal loss of work or processes (as opposed to a “cold” restart, which requires a complete reload of the system with no processes surviving).

automatic route selection (see ARS)

automation management system (see AMS)

Automotive Network Exchange (see ANX)

autonegotiation

Automatic adjustment to transmission type or speed.

autonomous system (see AS)

autonomous system number (see ASN)

autonomous unit interface (see AUI)

autopartitioning

A 10Base-T feature that causes an Ethernet hub to stop forwarding packets if it detects too many sequential collisions.

autosensing

Automatic adjustment to differing operating conditions or to transmission type or speed.

autotopology

A feature of network management systems that automates the creation of a graphical network configuration map.

autovectorizing

Software used in technical document control systems to convert certain bitmapped data to geometrical values.

AV (audiovisual)

Available Bit Rate (see ABR)

available to promise (see ATP)

availability

The assurance that an enterprise's IT infrastructure has suitable recoverability and protection from system failures, natural disasters or malicious attacks.

avatar

Computer representation of users in a computer-generated 3-D world, used primarily in chat and entertainment Web sites. Potential business applications include customer support, training or sales, where avatars in an enterprise's Web site may assist potential customers through text or audio links.

AVD circuits (alternate voice/data circuits)

Circuits that have been conditioned to handle both voice and data traffic.

average inventory

In an inventory system, this is the sum of one-half the lot sizes plus the reserve stock in formula calculations.

average speed of answer (see ASA)

AVL (automatic vehicle locator)

AVS (address verification service)

A fraud detection method in which the address provided by the buyer at the time of purchase is matched against the address registered with the card issuer. The result of this check (match/mismatch) is sent back to the merchant, but a mismatch does not result in automatic transaction denial. Since there are valid reasons for a mismatch, merchants are left to consider a variety of factors (e.g., size of the transaction) to determine whether to accept the purchase.

AVS (Advanced Visual Systems)

A vendor of data visualization technology, headquartered in Waltham, Massachusetts.

AVVID (Architecture for Voice, Video and Integrated Data)

An architectural approach for integrating enterprise voice, video and data traffic over IP-based networks, introduced by Cisco Systems in September 1999.

AWD (Automated Work Distributor)

A suite of imaging, telephony and work management software tools developed by DST Systems.

AWK (Aho Weinberger Kernighan)

A Unix text parser.

AWT (Abstract Windowing Toolkit)

B

B1

A class of computer system security defined by the U.S. Department of Defense. In a military context, B1 systems offer a higher degree of security than C2 systems by enforcing the concept of information sensitivity classifications (e.g., unclassified, confidential, secret and top secret) with corresponding user clearance requirements. B1 security has little or no relevance to commercial applications.

B2B (business-to-business)

A form of e-commerce conducted among businesses, typically because of formal, contractual arrangements. B2B functions include:

- Sophisticated Web authorization and control for delivery of sensitive price, contract and content information for each partner
- Catalogs that provide custom views based on access controls and parametric searches
- Order entry functions such as standardized "ship to" locations, dynamic order recalculation and payment options

B2C (business-to-consumer)

A form of e-commerce conducted between businesses and consumers. B2C commerce includes both formal relationships (e.g., customers with subscription-based services or content) and ad hoc relationships (formed in real time to enable a new user to buy, sell or access information).

B2E (business-to-employee)

The use and leverage of e-business approaches and Internet technologies to deliver a comprehensive set of services to an enterprise's employees and managers. B2E is the automated delivery of enterprise relationship management (ERM), but goes beyond ERM's support of human capital management (HCM) to include workforce management and workforce optimization. The full term is sometimes presented as "business-to-enterprise." See ERM and HCM.

B2G (business-to-government)

backbone

A high-speed line or series of lines that forms the fastest (measured in bandwidth) path through a network. It often acts as a metanetwork.

backbone network

A high-speed transmission facility, or an arrangement of such facilities, designed to interconnect lower-speed distribution channels or clusters of dispersed user devices.

backbone router

A router designed to be used to construct backbone networks using leased lines. Backbone routers typically do not have any built-in digital dial-up wide-area network interfaces.

back end

The server side of a client/server system.

background task

A task performed by a system during the time when its primary application is idle.

backhaul

The terrestrial link between an earth station and a switching or data center.

back-off algorithm

The formula used to determine when a second attempt will be made to access a network after an initial failed access attempt.

backplane

The physical connection between the interface cards and the data and power distribution buses in a network device such as a router, hub or switch.

backup server

A software or hardware system that copies or "shadows" the contents of a server, providing redundancy.

balanced-to-ground

In a two-wire circuit, describes the state where the impedance-to-ground on one wire equals the impedance-to-ground on the other wire.

balanced scorecard

A performance measurement and management approach that recognizes that financial measures by themselves are not sufficient and that an enterprise needs a more holistic, balanced set of measures which reflects the different drivers that contribute to superior performance and the achievement of the enterprise's strategic goals. The balanced scorecard is driven by the premise that there is a cause-and-effect link between learning, internal efficiencies and business processes, customers, and financial results.

Balun

Balanced/unbalanced. In the IBM cabling system, this refers to an impedance-matching device used to connect balanced twisted-pair cabling with unbalanced coaxial cable.

BAM (business activity monitoring)

A Gartner term that defines the concept of providing real-time access to critical business performance indicators to improve the speed and effectiveness of business operations. At its broadest level, BAM is the convergence of operational business intelligence (BI) and real-time application integration. A goal of the zero-latency enterprise (ZLE), BAM aims to improve business performance through advances in IT.

Compared with traditional business event monitoring and reporting, BAM requires a higher degree of organizational and architectural planning and investment due to:

- Its focus on real-time data, as well as real-time access to data.
- The use of information from multiple application systems, and other internal and external sources (unlike traditional monitoring, which uses a single source).
- The delivery of information through alerts and graphical displays (dashboards), which are customized and optimized for different users across the enterprise.

See application integration, BI and ZLE.

band

1. A range of frequencies between two defined limits.
2. In wide-area telephone service (WATS), the specific geographical area in which the customer is entitled to call.

bandpass filter

A circuit designed to allow a single band of frequencies to pass.

bandwidth

1. The range of frequencies that can pass over a given transmission channel. The bandwidth determines the rate at which information can be transmitted through the circuit: the greater the bandwidth, the more information that can be sent in a given amount of time. Bandwidth is typically measured in bits per second. Increasing bandwidth potential has become a high priority for network planners due to the growth of multimedia, including videoconferencing, and the increased use of the Internet.
2. The signal-to-noise ratio (SNR) of a modulated signal is tested within a range of frequencies on either side of the carrier frequency. This range is also called a bandwidth. In the case of videodiscs, it is often 15 kilohertz (KHz) on either side.

Bandwidth On Demand Interoperability Group (see BONDING)

BANI (Bell Atlantic Network Integration)

Bank Global Trust Enterprise

See Identrus.

Banking Industry Technology Secretariat (see BITS)

banner

An advertisement that appears on a Web site. The banner ad format combines graphic and textual content to induce the site user to "click through" for further information on an advertised product or service.

BAPI (Biometric API)

A specification developed by I/O Software for application at the device level. Although primarily focused on fingerprinting, the specification can be applied to other biometrics such as voice and eye. This group is defining a standard software protocol and application program interface (API) for communication between software and biometric devices.

BAPI (Business Application Programming Interface)

A set of documented, server-side interfaces to one or more R/3 processes, from SAP. BAPI packages multiple internal functions to enable programmatic access to such higher-order tasks as checking customer numbers, providing product descriptions, selecting products, creating quotations or creating orders.

barrel nut connector (see BNC)

BASDA (Business Application Software Developers' Association)

baseband

Transmitting a signal in its original, unmodulated form. A baseband signal can be analog (e.g., originating from a telephone) or digital (e.g., originating from a computer).

Baseline Privacy Interface Plus (see BPI+)

base station

Within a mobile radio system, a fixed radio station providing communication with mobile stations and, where applicable, with other base stations and the public telephone network.

BASIC (Beginner's All-Purpose Symbolic Instruction Code)

A high-level algebraic programming language developed at Dartmouth College in the 1960s and widely taught to beginning programmers. It is simple to use but lacks speed.

Basic Encoding Rules (see BER)

basic input/output system (see BIOS)

basic phone

A voice-centric device designed to provide only voice functions and limited contact management, as distinguished from an enhanced phone or smart phone (see separate entries).

Basic Rate Access (see BRA)

Basic Rate Interface (see BRI)

Basic Telecommunications Access Method (see BTAM)

BASIS

SAP's proprietary middleware between the SAP graphical user interface and the R/3 application servers.

Batch Data Interface (see BDI)

batch processing

The processing of application programs and their data individually, with one being completed before the next is started. It is a planned processing procedure typically used for purposes such as preparing payrolls and maintaining inventory records.

battery backup

Provision of sufficient battery power to enable the nonvolatile storage of data and graceful system shutdown in the event of a power interruption.

baud

A unit of signaling speed. The speed in bauds is the number of discrete changes per second in some aspect of a signal (e.g., voltage in a wire). Transmission speeds are now more commonly measured bits per second (bps), rather than bauds. The two terms were roughly synonymous until modems began to exceed 2,400 bps, after which they diverged more widely as modem speed has increased. Modems now use coding techniques to transmit more than one bit per baud, making their true baud ratings irrelevant.

BBBOnLine

The Better Business Bureau's Internet program. BBBOnLine's Privacy Seal identifies Internet companies that use appropriate processes and operations to protect the confidentiality of consumer information. License fees are based on a company's annual revenue. Vendors must reapply for licensing each year; during this process, their operations are re-evaluated to ensure that they meet all requirements for displaying the program emblem.

BBNS (Broadband Network Services)

IBM's architecture for implementing high-speed multimedia networks. It is designed to be complementary to existing asynchronous transfer mode (ATM) standards and the implementers' agreements developed by the ATM Forum.

BBP (Business-to-Business Procurement)

A catalog-based procurement product from SAP.

BBS (bulletin board system or service)

A network-based system or service for communicating and sharing information in the form of posted messages.

BCC (block check character)

A control character appended to blocks in character-oriented protocols. Used for determining if the block was received in error in longitudinal and cyclic redundancy checking.

BCD (binary-coded decimal)

A numeric notation in which each of decimal digit is represented by a binary numeral. For example, in BCD notation, the number 23 is represented as "0010 0011" (as compared to the representation "10111" in the pure binary numeration system).

bCentral

Microsoft's Web technology for running businesses. bCentral offers users business tools that include Web sites, business-class e-mail, and sales and customer management tools.

B channel (bearer channel)

One of two 64 kilobit per second data channels in the Integrated Services Digital Network (ISDN) Basic Rate Interface (see BRI).

BCM (business continuity manager)

The person responsible for business recovery planning (BRP) and business continuity planning (BCP).

BCP (business continuity planning)

A broad disaster recovery approach whereby enterprises plan for recovery of the entire business process. This includes a plan for workspaces, telephones, workstations, servers, applications, network connections and any other resources required in the business process.

BCS (Balanced Cluster Service)

A load-balancing product from Cubix (Carson City, Nevada).

BCV (business continuance volume)

BDC (Backup Domain Controller)

BDI (Batch Data Interface)

An interface used within SAP's R/3 enterprise resource planning system to enable developers to load data into R/3 on a batch basis.

BDLC (Burroughs Data Link Control)

A bit-oriented protocol similar to High-level Data Link Control (HDLC). (Burroughs is now part of Unisys.)

BDLS (bidirectional loop switching)

The ability of fiber rings to be recovered in either of two directions, typically by using two pairs of fiber in the ring.

beacon

In token-ring and fiber distributed data interface (FDDI) networks, a frame indicating a serious problem, and containing the address of any stations that are down.

beam width

Angular width of an antenna radiation pattern, or beam, within which the radiation exceeds some proportion of the maximum.

bearer channel (see B channel)

bearer service

A network technology or vendor that provides wireless transmission. Examples of such technologies are Mobitex, Radio Data Link Access Protocol (RDLAP), Reflex, cellular digital packet data (CDPD), Global System for Mobile Communications (GSM) and digital cellular/personal communications services (PCS) packet services, including general packet radio service (GPRS), enhanced data rates for GSM environments (EDGE) and Wideband Code Division Multiple Access (W-CDMA). The vendors are too numerous to mention, but consist of the world's prominent vendors of GSM, time division multiple access (TDMA), code division multiple access (CDMA), packet radio and paging.

BECN (Backward Explicit Congestion Notification)

Beginner's All-Purpose Symbolic Instruction Code (see BASIC)

Bell Communications Research (see Bellcore)

Bellcore (Bell Communications Research)

A jointly owned, financed and centrally staffed organization of the regional holding companies (RHCs) formed after the AT&T divestiture in 1984, charged with establishing network standards and interfaces. Bellcore changed its name to Telcordia Technologies in 1999.

Bell operating company (see BOC)

benchmarking

The comparison between vendor performance and designated benchmark organizations or indexes. An index is a publicly available indicator for a factor that is associated with a pricing element.

- Internal benchmarking is the continuous process of measuring a company's products, services and practices to determine whether the best possible job is performed with the resources at hand. This can include comparing similar functions of different operating units in an organization or comparing the operations of a specific division from one year to the next.
- External benchmarking is the continuous process of measuring a company's products, services and practices and comparing them with those of another company. This can include comparisons with industry peers, functional leaders, or best-in-class performers.

bend loss

Attenuation in fiber cabling resulting from a bend being at too great an angle or from distortion within the cable.

bend radius

The radius of the smallest circle possible when bending a cable, if transmission is not to be faulty.

benefits administration

A function of human resources responsible for providing and managing competitive employee benefits, e.g., health and welfare, deferred compensation, and retirement/pension programs. Services to the enterprise include regulatory compliance and employee communication related to benefits.

BER (bit error rate)

BER (Basic Encoding Rules)

The Open Systems Interconnection (OSI) language for transfer syntax description.

Berkeley 4.2Bsd

The Unix version developed by the University of California at Berkeley, which dominated technical computing applications until it was merged with AT&T's System V Unix operating system through an agreement between AT&T and Sun Microsystems. The merger was completed in System V release 4 (SVR4).

Berkeley Internet Name Daemon (see BIND)

Berkeley Software Distribution (see BSD)

best-in-class

The superior product within a category of hardware or software. It does not necessarily mean best product overall, however. For example, the best-in-class product in a low-priced category may be inferior to the best product on the market, which could sell for much more. See best-of-breed.

best-of-breed

The best product of its type. Enterprises often purchase software from different vendors to obtain the best-of-breed offering for each application area. For example, enterprises may purchase a human-resource package from one vendor and an accounting package from another. Although enterprise resource planning (ERP) vendors offer numerous enterprise applications and claim that their integrated system is a superior solution, all modules in an ERP system are rarely best-of-breed. See best-in-class.

best practice

A group of tasks that optimizes the efficiency (cost and risk) or effectiveness (service level) of the business discipline or process to which it contributes. It must be implementable, replicable, transferable and adaptable across industries.

beta test

The stage at which a new product is tested under actual usage conditions.

BETRS (basic exchange telephone radio system)

BGP (Border Gateway Protocol)

The interdomain routing protocol implemented in Transmission Control Protocol/Internet Protocol (TCP/IP) networks.

BGP-4 (Border Gateway Protocol-4)

A Border Gateway Protocol-based networking redundancy service. It enables an enterprise to route Information Protocol (IP) traffic destined for the same IP address via different network connections. In a BGP-4 environment, when a transmission comes from an Internet service provider's network, it will look for the primary router that connects to the enterprise's location. If that router becomes unavailable, the transmission will automatically be redirected to the backup router without interrupting the original transmission.

BI (business intelligence)

An interactive process for exploring and analyzing structured, domain-specific information (often stored in data warehouses) to discern trends or patterns, thereby deriving insights and drawing conclusions. The BI process includes communicating findings and effecting change. Domains include customers, suppliers, products, services and competitors.

BIA (business impact analysis)

BIA identifies the cost (financial and nonfinancial) of a set of business processes that are not functioning correctly. It is one of the early steps required for preparing a disaster recovery plan (DRP) or a business continuity plan (BCP). Without this step, it is difficult for the business continuity manager to balance the cost of the DRP or BCP against the potential loss.

BiCMOS (bipolar complementary metal-oxide semiconductor)

bid

1. An attempt to gain control over a line in order to transmit data, usually associated with the contention style of sharing a single line among several terminals.
2. A vendor's proposal to win a contract.

bidirectional loop switching (see BDLS)

big-endian

A method of storing or transmitting data where the most significant bit or byte is presented first. See little-endian.

BI IDA (business intelligence information delivery architecture)

A key element of business intelligence (BI) implementation planning deals with the means and mechanisms for providing the needed data and information to users. Information delivery via desktop databases, intermediate servers, client/server database applications, or Web server interaction presents a challenge to most enterprises and many users. The BI IDA provides a framework and implementation model for evaluating and prioritizing users' needs as well as matching those needs to the evolving delivery approaches and technologies.

bill of lading (see BOL)

bill-of-material processor

Computer applications supplied by many manufacturers for maintaining, updating and retrieving bill-of-material information on direct-access files.

BiMOS (bipolar metal oxide semiconductor)

An integrated circuit technology.

binary code

Representation of quantities expressed in the base-2 number system.

binary-coded decimal (see BCD)

binary digit (see bit)

binary large object (see BLOB)

Binary Synchronous Communications (see BSC)

binary table (see byte)

BIND (Berkeley Internet Name Daemon)

A public domain code used for name resolution.

bindery

A database used by a workstation or network operating system to store internal data such as user or node definitions.

biometric API (see BAPI)

biometric authentication

A form of user authentication based on a physical (e.g., fingerprint, iris, face or hand) or behavioral (e.g., signature or voice) characteristic. Because it is based on something the person "is," biometric authentication can provide a higher level of security than something a person "knows" (e.g., password, PIN or mother's maiden name) or something a person "has" (e.g., a card, key or hardware token). Biometrics can be used to verify a person's claimed identity or to identify a user from a database of candidates. To use a biometric system, the user must enroll by providing several samples of the biometric, from which the system creates a template for that user. At the time of verification or authentication, the live sample of the biometric is compared against the stored template.

biometrics

The technologies that analyze and measure biological and behavioral characteristics of individuals. These technologies allow for both verification and authentication, and recognition and identification. All biometric systems follow similar operational procedures, including enrollment and matching phases. Systems require users to enroll to generate a template for later comparison and matching.

BIOS (basic input/output system)

The part of an operating system that links the specific hardware devices to the software. It obtains the buffers required to send information from a program to the hardware/desktop receiving the information.

biphase coding

A bipolar coding scheme developed for Ethernet use where the clocking information is embedded into, and recovered from, the synchronous data stream without the need for separate clocking leads.

bipolar

1. The predominant signaling method used for digital transmission services, such as digital data service (DDS) and T1, in which the signal carrying the binary value successfully alternates between positive and negative polarities. Zero and one values are represented by the signal amplitude at either polarity, while no-value "spaces" are at zero amplitude.
2. A type of integrated circuit that uses both positively and negatively charged currents, characterized by high operational speed and cost. Also called alternate mark inversion.

bipolar metal-oxide semiconductor (see BiMOS)

bipolar violation

Modified bipolar signaling in which a control code is inserted into the original digital format.

BIS (Boot Integrity Services)

A secure version of PXE.

B-ISDN (Broadband ISDN)

A high-speed (greater than ISDN primary rate), asynchronous, time division multiplexed transmission facility, or an arrangement of such facilities, designed to provide a wide range of audio, video and data applications in the same network.

bisync (see BSC)

bit (binary digit)

The minimum unit of binary information stored in a computer system. A bit can have only two states, on or off, which are commonly represented as ones and zeros. The combination of ones and zeros determines which information is entered into and processed by the computer.

BitBLT (bit block transfer)

Bitbus

A serial, low-speed network developed by Intel in 1983 for sensor and factory-level controlling devices.

bit interleaving

A method of time-division multiplexing where inputs are sampled so that the sequence and number of bits are maintained, facilitating synchronization.

bitmapped

Composed of a vast number of picture elements, or pixels, which are displayed in a pattern that forms an image on a video screen.

BITS (Banking Industry Technology Secretariat)

The technology arm of the Bankers Roundtable, whose membership comprises the top 125 bank holding companies in the United States. BITS' mission is to help banks develop electronic-banking and e-commerce initiatives, and to address and resolve critical industry issues (e.g., competitive disintermediation and reduction of infrastructure costs).

bits per inch (see bpi)

bits per second (see bps)

BIW (Business Information Warehouse)

A component of SAP's R/3 system.

BizTags

Microsoft "bridges" that translate between different Extensible Markup Language (XML) schemas.

BizTalk

A Microsoft Extensible Markup Language (XML) schema.

BlackBerry

A two-way wireless device developed by Research in Motion (see RIM). It allows users to check e-mail and voice mail (translated into text) and to page other users via a wireless network service. The device has a miniature keyboard used to type messages, which are delivered using the Short Message Service (SMS) protocol. BlackBerry users must subscribe to a wireless service that allows for data transmission. See SMS.

blanking interval

The area in a video signal that falls between frames. It is often used to accommodate data such as synchronizing information.

BLAST (Bell Labs Layered Space-Time)

An emerging wireless communications technique, developed by Bell Labs, that uses multi-element antennas at both the transmitter and receiver to permit transmission rates significantly higher than those possible using conventional techniques. A laboratory prototype has already demonstrated spectral efficiencies of 20 to 40 bps per hertz of bandwidth.

BLEC (building local-exchange carrier)

A type of local-exchange carrier (LEC) that offers voice and data communications service over wiring it has installed itself within a building. BLECs typically install fiber wiring in vertical risers within the building, and connect it externally to their own point of presence or to another LEC, usually over a broadband circuit of at least DS-1 size. In addition to basic connectivity, some BLECs offer application and Internet services, typically to small and midsize businesses. BLECs are among the competitive LECs (CLECs) that have arisen in the wake of telecommunications deregulation. See CLEC.

BLERT (block error rate test)

A test conducted by transmitting a known blocked bit pattern, comparing the pattern received with the pattern transmitted, and counting the number of blocks containing errored bits.

BLOB (binary large object)

A generic term used to describe the handling and storage of long strings of data by database management systems. A BLOB is a category of data, characterized by large size (including media formats such as audio and video), which can place extreme demands on storage systems and network bandwidth.

block

A group of bits or bytes treated as a unit.

block check character (see BCC)

block down conversion

The conversion of a full satellite band to a lower frequency (e.g., from SHF to UHF or VHF).

block error rate

In data communications testing, the ratio between the total number of blocks transmitted in a given message and the number of blocks in that message received in error; a measure of the quality of a data transmission.

block error rate test (see BLERT)

blocking

The inability to connect two lines in a network because all possible paths between them are already in use.

Blog

Slang for "weblog" - a whimsical truncation of the term. This slang term may also be used as other parts of speech (as in "to blog" and "blogging"). See **Weblog**.

Blowfish

A public-domain algorithm developed by Bruce Schneier.

Blue Book

Recommendations approved in 1988 by Comite Consultatif International Telegraphique et Telephonique (CCITT), now known as the International Telecommunications Union (ITU).

Bluetooth

A wireless networking technology with a range of about 10 meters and a raw data transmission rate of 1 Mbps. Bluetooth supports ad hoc networking of up to 80 devices within a 10-meter radius (supporting voice and data).

BMAN (broadband metropolitan-area network)

BMIT (business management of IT)

A framework developed by Gartner to help managers focus on the key issues affecting each area or stage of an IT strategy and how these issues impact the business or organizational side of the enterprise. Each stage requires measurement of IT costs, operations, efficiency and effectiveness. The six-stage management process focuses on risk and investment criteria across four major BMIT management practices: people, money, infrastructure and strategic-process architecture.

BNC (barrel nut connector)

A type of coaxial cable connector with a plug that is turned after insertion to tighten the connection.

BOB (see best-of-breed)

BOB (Business Object Broker)

SAP's message broker for R/3.

BOC (Bell operating company)

Any of the 22 original companies (or their successors) that were created from the breakup of AT&T in 1983. They were reorganized into seven Bell regional holding companies (RHCs). (There are now five.) The divestiture distributed the right to provide local telephone service in a given geographic area. Before this, companies had existed as subsidiaries of AT&T and were called the "Bell System." The breakup was designed to create competition at both the local and long-distance service levels. As a group, companies that offer local telephone service are legally referred to as "local exchange carriers."

BOCs are not allowed to manufacture equipment and were initially not allowed to provide long-distance service. The Telecommunications Act of 1996 now permits them to engage in long-distance business under certain circumstances.

BOI (business object interface)

Application programming interfaces (APIs) published by Baan to permit access to functions within its enterprise resource planning (ERP) system.

BOL (Books Online)

A bookstore on the Internet created as a joint venture of German publisher Bertelsmann and French publisher Havas. It is designed to be pan-European with a Web site available in various local languages.

BOL (bill of lading)

A contract of carriage between a shipper and carrier to consign a load for delivery to another party.

BOM (bill of materials)

A structured list of the raw materials, parts and assemblies that constitute a product to be manufactured, typically used as part of a manufacturing resource planning (MRP II) system.

BONDING (Bandwidth On Demand Interoperability Group)

A networking specification for a standardized inverse multiplexing protocol.

bookmark

Pointer to an Internet address kept within a Web client (browser).

Books Online (see BOL)

Boolean

A term that originated in the in the realm of mathematics, and that is now commonly known for its application to search engine logic. A Boolean search allows for the inclusion or exclusion of documents containing certain keywords, using Boolean operators such as "and," "not" and "or." For example:

- A search structured "Bush and China" would return all documents that refer to both "Bush" and "China," but not those that contain only one term or the other.
- "Bush not China" would return all documents referring to "Bush," except for those that also contain references to "China."
- "Bush or China" would return all documents that reference either "Bush" or "China," as well as those that reference both.

Some Boolean search engines allow keywords and operators to be nested using parentheses. For example, a search structured as "Bush and (Japan or China)" would return all documents that reference both Bush and Japan, as well as those that reference both Bush and China.

boot

To start a computer system. A boot from a power-off condition is called a "cold boot," while merely reinitializing the system is called a "warm boot."

BOOTP (Bootstrap Protocol)

A protocol defined by the Internet Engineering Task Force (IETF) and used to remotely boot systems that communicate with the Transmission Control Protocol/Internet Protocol (TCP/IP) suite. IP address, configuration, and software load are a few of the things that can be accomplished with BOOTP.

BOP (beginning of partition)

Border Gateway Protocol (see BGP)

Border Gateway Protocol-4 (see BGP-4)

BOT (beginning of tape)

BOT (build-operate-transfer)

A process used in several Asia/Pacific countries that allows foreign companies to build a telecommunications network in the country, operate it for a period and then transfer ownership to the government.

bottleneck

The operation with the least capacity in a total system with no alternative routings; the total system can be effectively scheduled by simply scheduling the limiting operation.

bounce

The return of an undeliverable e-mail.

BPA (business process automation)

The automation of complex business processes and functions beyond conventional data manipulation and record-keeping activities, usually through the use of advanced technologies. It focuses on "run the business" as opposed to "count the business" types of automation efforts and often deals with event-driven, mission-critical, core processes. BPA usually supports an enterprise's knowledge workers in satisfying the needs of its many constituencies.

BPI+ (Baseline Privacy Interface Plus)

A specification related to Version 1.1 of the Data-Over-Cable Service Interface Specification (DOCSIS) standard. It strengthens cable modem service protection by incorporating digital-certificate-based authentication of cable modems and customers. It also supports filtering services, such as blocking User Data Protocol (UDP) port 137, which supports Network Basic Input/Output System (NetBIOS) and server message block (SMB) file-sharing protocols.

bpi (bits per inch)

A measurement used to calculate the number of bits stored in a linear inch of a track on a disk, tape or other recording surface.

BPM (business process management)

A general term describing a set of services and tools that provide for explicit process management, including process analysis, definition, execution, monitoring and administration. Ideally, BPM should include support for both human and application-level interactions. The workflow market has been a significant source of BPM, although forms of BPM are now emerging from many other sources, such as collaborative applications, integration brokers, Web integration servers, development tools, rules engines and e-commerce offerings.

BPM (business process modeling)

A process that links business strategy to IT systems development to ensure business value. It combines process/workflow, functional, organizational and data/resource

views with underlying metrics such as costs, cycle times and responsibilities to provide a foundation for analyzing value chains, activity-based costs, bottlenecks, critical paths and inefficiencies.

BPO (business process outsourcing)

The delegation of one or more IT-intensive business processes to an external provider that, in turn, owns, administers and manages the selected processes, based on defined and measurable performance metrics.

BPP (bill presentment and payment)

BPR (business process re-engineering)

The fundamental analysis and radical redesign of business processes and management systems to accomplish change or performance improvement. It uses objective, quantitative methods and tools to analyze, redesign and transform business processes, including supporting organization structures, information systems, job responsibilities and performance standards.

BPR analytical techniques

Mathematical, graphical, logical and managerial algorithms for describing and modeling business processes, information systems or management decision-making systems.

BPR methodology

An integrated set of management policies, project management procedures, and modeling, analysis, design and testing techniques for analyzing existing business processes and systems; designing new processes and systems; testing, simulating and prototyping new designs prior to implementation; and managing the implementation process.

BPR tools

Combinations of techniques and software products that allow electronic capture, analysis, testing, simulation, reconfiguration and persistent memory of business and systems models

bps (bits per second)

The basic unit of measurement for serial data transmission capacity. Represented as Kbps for thousands of bits per second; Mbps for millions of bits per second; Gbps for billions of bits per second; and Tbps for trillions of bits per second. The numbers are not exact: Kbps, for example, is really 1,024 bps.0

BPSK (binary phase-shift keying)

BR (business routing)

A Gartner category of call routing used for sales or service functions. The general objective is to determine, as transparently as possible, the caller's needs, business value and relationship, and ultimately to automate the call's routing to a resource. The process is usually driven by multiple databases, with each factor considered becoming a decision or value point.

BRA (Basic Rate Access)

The name used in Canada and Europe for Basic Rate Interface (see BRI).

brand service company

Similar to the in-sourcing model, a company that is built to provide services to a large organization or a group of business-oriented companies. Services provided (which may include non-IT services and business processes) are carefully compared against the market, and the services company leverages external service providers (ESPs), selectively outsourcing part of their services.

break

An interruption to a transmission; frequently a provision permitting a controlled terminal to interrupt the controlling computer.

breakout box

A device that allows access to individual points on a physical interface connector (e.g., EIA RS-232-C) for testing and monitoring.

breakout panel

A breakout box mounted as a component in some larger device.

BRI (Basic Rate Interface)

The Integrated Services Digital Network (ISDN) BRI — known as Basic Rate Access (BRA) in Canada and Europe — consists of two 64 kilobit per second (Kbps) data or voice channels, which are designated as B (bearer) channels. The interface also has a 16-Kbps signaling or packet data channel designated as the D (delta) channel. The interface is, therefore, often referred to as 2B+D.

brick and mortar

Refers to a traditional company with a physical presence and non-Web channels as the sales outlet for its products or services.

bridge

A relatively simple device that passes data without examining it. Bridges interconnect networks, or network segments, running the same protocols. Operating at the MAC Layer, they are protocol-independent; the decision as to whether to forward a signal depends only on the address. There are various special types of bridges: encapsulating, translating, and source routing are the main categories.

bridge/router

Strictly, this provides bridging at Level 2 and routing at Level 3 of the OSI stack, but this very precise meaning has been largely superseded by a looser categorization that includes any device that combines the functions of bridges and routers.

broadband

Channels carried on coaxial or fiber-optic cables that have a wider bandwidth than conventional telephone lines, giving them the ability to carry video, voice and data simultaneously. Cable modems and digital subscriber line (DSL) technologies are examples of broadband connectivity.

Broadband ISDN (see B-ISDN)

Broadband Network Services (see BBNS)

broadcast

1. Delivery of a transmission to two or more stations at the same time, such as over a bus-type local network or by satellite.
2. Protocol mechanism whereby group and universal addressing is supported.

broadcast storm

Excessive one-to-many or many-to-many transmissions, especially troublesome on Ethernet networks.

broker

The middleware that mediates communication between applications (including external legacy applications and packaged applications) and enables them to share information.

brownout

In response to heavy demand, main system voltages are sometimes lowered, leading to brownouts in which power is not lost but reduced.

browser

A software program (also known as a "Web browser") used to locate and display information on the Internet or an intranet. Examples include Microsoft's Internet Explorer and Netscape Communications' Navigator. Most browsers can display graphics, photographs and text; multimedia information (such as sound and video) may require additional software, known as "plug-ins."

browsing

The near-random search for content on the Internet.

BRP (business recovery planning)

Planning for the recovery of an enterprise's systems in the event of a major outage.

BRU (business requirement unit)

A unit of measure developed to compensate IS organizations for performing up to business requirements. The unit cost to deliver server environments to the enterprise can then be calculated by accumulating all the delivered BRUs and dividing them by the total cost to operate the environment. BRUs are determined based on IS functions delivered by the IS organization, weighted by importance, scaled for size and added together to provide a total.

BS7799

The British Standard 7799, Section 7, which offers guidelines for password policy, including:

- Passwords should be for individual users in order to maintain accountability.
- Passwords must be kept confidential.
- Passwords should not be kept on paper unless they can be securely stored.
- Passwords must be changed whenever there is a chance that the password or the system could be compromised.
- Passwords should not contain aspects of a date, family or company identifiers, telephone numbers, the user ID or other system identifiers, more than two

consecutive identical characters or be all numbers or all letters.

- Passwords should be changed approximately every 30 days (more frequently for privileged accounts) and cannot be recycled.
- Passwords should not be included in a macro or function key to automate log-in.
- Passwords should be stored only as encrypted hash files.

BSA (Business Software Alliance)

A software industry coalition whose stated missions include industry education and copyright enforcement.

BSC (Binary Synchronous Communications)

A half-duplex, character-oriented data communications protocol originated by IBM in 1964. It includes control characters and procedures for controlling the establishment of a valid connection and the transfer of data. Also called Bisync. Although still in use, it has largely been replaced by IBM's more efficient protocol, Synchronous Data Link Control (SDLC), which is under Systems Network Architecture (SNA).

BSD (Berkeley Software Distribution)

A version of Unix developed at the University of California.

BSI (British Standards Institute)

BSP (business service provider)

A domain of enterprise application outsourcing best suited for confined processes with a few, well-defined interfaces to other business processes of the enterprise. BSP is the extension of the application service provider (ASP) model into business process management. A BSP manages and operates standardized business processes on behalf of its customers, delivering its service across a network to multiple customers using a "pay as you go" payment model. See ASP.

BSS (business solution server)

A platform for development and execution of complete line-of-business (LOB) applications. A unit of BSS development abstraction is a business process, not a program. Architecturally, a BSS is a framework of business processes. Technologically, BSSs may rest on a variety of technologies, exposed to developers via business application programming interfaces.

BTAM (Basic Telecommunications Access Method)

IBM's lowest-level input/output (I/O) macro-routine support for providing communications programs on a host computer.

BTU (British thermal unit)

BU (business unit)

budgeting

The process of building plans or forecasts of revenue and expenditures for a budget entity (such as a business unit or a product).

buffer

A storage device used to compensate for a difference in rate of data flow, or time of occurrence of events, when transmitting data from one device to another.

bug

An unexpected problem with software or hardware. Typical problems are often the result of external interference with the program's performance that was not anticipated by the developer. Minor bugs can cause small problems like frozen screens or unexplained error messages that do not significantly effect usage. Major bugs may not only affect software and hardware, but could also have unintended effects on connected devices or integrated software and may damage data files.

building local-exchange carrier (see BLEC)

build-operate-transfer (see BOT)

bulletin board system or service (see BBS)

bundling

Packaging multiple features and products together for a single price.

Burroughs Data Link Control (see BDLC)

burst

In data communications, a sequence of signals counted as one unit in accordance with a specific criterion or measure.

bursty

Data that is transmitted in uneven spurts.

bus

1. Physical transmission path or channel. Typically an electrical connection with one or more conductors, wherein all attached devices receive all transmissions at the same time.
2. Local network topology, such as that used in Ethernet, where all network nodes listen to all transmissions, selecting certain ones based on address identification. It involves some type of contention-control mechanism for accessing the bus transmission medium.

business activity monitoring (see BAM)

Business Application Programming Interface (see BAPI)

business continuity manager (see BCM)

business continuity planning (see BCP)

business impact analysis (see BIA)

business intelligence (see BI)

business intelligence information delivery architecture (see BI IDA)

business management of IT (see BMIT)

Business Object Broker (see BOB)

business object interface (see BOI)

business portal (see enterprise portal)

business process

An event-driven, end-to-end processing path that starts with a customer request and ends with a result for the customer. Business processes often cross departmental and even organizational boundaries.

business process automation (see BPA)

business process management (see BPM)

business process modeling (see BPM)

business process outsourcing (see BPO)

business process re-engineering (see BPR)

business recovery planning (see BRP)

business requirement unit (see BRU)

business routing (see BR)

business service marketplace

A type of e-marketplace that offers content or services for specific business processes, such as financing, logistics, marketing and requests for quotation (RFQs). Business service marketplaces aggregate the offerings of business service providers (BSPs), and will proliferate as outsourcing grows.

Business service marketplaces are one of three types of public e-marketplaces (along with integration service marketplaces and commodity marketplaces) that Gartner has identified in the evolving public e-marketplace landscape. (See BSP, integration service marketplace and commodity marketplace.)

business service provider (see BSP)

Business Software Alliance (see BSA)

Business Solution Server (see BSS)

business-to-business (see B2B)

Business-to-Business Procurement (see BBP)

business-to-consumer (see B2C)

business-to-employee (see B2E)

business-to-enterprise— see "B2E (business-to-employee)"

Business Workflow (see BW)

bus topology

An equal-access network design in which all devices are connected to a single linking cable with two distinct ends.

buy side

A process enabling companies to purchase products that include requisitioning, product catalogs, approvals, user identification, purchase order creation, payment processing and integration with other systems.

BW (Business Workflow)

The workflow component introduced by SAP with release 3.0 of R/3. SAP Business Workflow provides the infrastructure and tools to allow business processes to be managed, automated and analyzed.

bypass

Any of several configurations of alternative transmission arrangements whose purpose is to avoid the local telephone company switched network. "Service" bypass involves using the telephone company's own (cheaper) facilities, while "facilities" bypass involves privately owned fiber or radio transmission.

byte (binary table)

A group of eight bits handled as a logical unit. In text files, a byte is equivalent to a single character such as a letter, number or punctuation mark.

byte code

The intermediate code compiled and executed by a virtual machine (VM). Byte code can be used unchanged on any platform on which the VM operates. See VM and Java.

C

C

The programming language created by Dennis Ritchie at the former Bell Laboratories in 1972. C provides very precise control of the computer's operation.

C++

An extension to the C language defined by Bjarne Stroustrup at Bell Laboratories in 1986. As a superset of C, it provides additional features for data abstraction and object-oriented programming. C++ can be used to develop programs for almost all computers. Together, C and C++ are among the most common programming languages in use today.

C#

An object-oriented programming language from Microsoft based on C++. C# (pronounced "C sharp") has elements from Visual Basic and Java (e.g., automatic garbage collection), whereas C++ does not. C# supports Extensible Markup Language (XML) and Simple Object Access Protocol (SOAP) and has access to the .NET class library. See .NET.

CA (certification authority)

Also known as a "certificate authority," this is an internal or third-party entity that creates, signs and revokes digital certificates that bind public keys to user identities. A repository or directory stores digital certificates and certificate revocation lists (CRLs) to allow users to obtain the public keys of other users and determine revocation status. Typically, the repository is a traditional X.500 directory or a database that supports Lightweight Directory Access Protocol (LDAP).

CA (Computer Associates)

CA (continuous availability)

cable

An assembly of one or more conductors within a protective sheath.

cable modem

The fastest of the current Internet connection technologies. Cable modems are devices that enable very high-speed data access via a cable TV network. Fast cable modems can transfer a megabyte of information in less than one second.

CAC (connection admission control)

A function in asynchronous transfer mode (ATM) networks that checks whether network resources are available to support the quality of service and traffic parameters of an incoming connection. Also known as "call admission control."

cache

Used as a noun or a verb, this term refers to the temporary storage of instructions or data for quick access by a computer system. For example, data is often cached near a computer's central processing unit to replicate information from main memory or storage in a way that facilitates quicker access, using fewer resources than the original source. In an Internet context, cache refers to the storage of commonly accessed Web pages or graphics locally on a user's PC or a Web server. This helps to minimize download time and preserve bandwidth for frequently accessed Web sites, and to reduce the load on a Web server.

caching server

A device that efficiently stores frequently requested data from protocols such as Hypertext Transport Protocol (HTTP) and File Transfer Protocol (FTP) and has the ability to "prefetch" additional data at preset intervals. A network caching server can "listen" on the network and intercept protocol requests on its associated Transmission Control Protocol (TCP) ports. This alleviates the need to have the enterprise's browser configured to "know" where the server sits on the network.

CAD (computer-aided design)

High-speed workstations or personal computers that use specialized software and input devices such as graphic tablets and scanners for specialized use in architectural, electrical and mechanical design. With few exceptions, CAD systems rely extensively on graphics.

CADAM (Computer-Aided Design and Manufacturing)

A computer-aided design program by IBM consisting of three-dimensional construction, modification, analysis and display (geometrical representation) of mechanical parts, including a control system for automated machine tools to manufacture the parts.

CAD/CAM (computer-aided design/computer-aided manufacturing)

CADD (computer-aided design and drafting)

Interactive graphic programs that automate the methodologies of drafting and design layouts. A few programs are successful enough so that it is difficult to justify designing layouts manually. Applications include integrated circuits and printed circuit boards.

CADDY (Computer-Aided Dossier and Data Supply)

An electronic dossier interchange and archiving format used in registering agrochemical products, such as pesticides .

CAE (computer-aided engineering)

An area of automated manufacturing and design technology for building end products that had its roots in finite element methods, but today it includes all types of performance systems, e.g., heat transfer, structural, electromagnetic, aeronautics and acoustic analysis. Major improvements have been in the architecture, mechanical, electronic and electrical-engineering disciplines.

CAFM (Computer-aided facilities management)

CAGR (compound annual growth rate)

Average yearly growth rate during a specified period.

CAI (Common Air Interface)

An interface for digital cordless communications, adopted by the British Department of Trade and Industry.

CA-IDMS (Computer Associates Integrated Data Management System)

See IDMS.

CAL (Client Access License)

A Microsoft software license program.

call

In networking, a call is any demand to set up a connection. In telecommunications, it is a unit of telephone traffic.

call accounting system

A device that tracks outgoing calls and records data for reporting. See call detail recording (CDR) and station message detail recording (SMDR).

call admission control (see CAC)

call center

A group or department where employees receive and make high volumes of telephone calls. Call centers can have internal customers (e.g., help desks) or external customers (e.g., customer service and support centers). The call center uses a variety of technologies to improve the management and servicing of the call (see call center suite). A center that use both phone- and non-phone-based communication channels (e.g., e-mail or the Web) is known as a "contact center" (see separate entry).

call center suite

A product that offers a suite of integrated components to support a call center (e.g., a help desk or a customer service and support center). In the past, integrating these components often required the services of an independent integrator; however, as call center functions move to open software platforms, many vendors now offer bundled suites of call center functionality.

An *all-in-one* call center suite provides a complete set of call center functions as a single platform, within the control of a single administrative view. Components include:

- An open computing platform (usually Windows NT or Unix)
- Telephone switch functionality and computer-telephony integration (CTI)
- Intelligent routing, based on business rules or agent skills
- Automatic call distribution (ACD), interactive voice response (IVR) and voice mail functions
- Outbound (e.g., predictive) dialing
- Application integration interfaces and tools
- "Cradle to grave" contact reporting, and component administration

A *multifunction* call center suite differs from an all-in-one suite in that it does not require switching integration. Instead, the switch functionality can be on a separate platform, controlled via CTI links and administered separately.

See call center, contact center and contact center suite.

call detail recording (see CDR)

caller ID

A telephone service that records the telephone numbers of incoming calls; it is a form of automatic number identification (ANI). Caller ID systems can be integrated with customer databases to streamline call management processes. This integration gives the agent receiving a call instantaneous access to relevant information about the caller. For example, when a customer calls, that customer's name immediately appears on the agent's computer screen. The screen might include information about the product a customer purchased and the purchase date. The system could also display the client's previous call history, information about other products the customer owns and price promotions on products that might also be appealing to that caller, based on a profile in the database.

call information logging

Automatic recording of information on chargeable calls made on a private branch exchange (PBX) system, including the extension number, exchange line number, time, call duration and digits dialed. This can be used for call accounting or billing.

CallPath

An IBM proprietary standard that covers switch and application interfaces and client/server architectures.

call processing

The sequence of operations performed by a switching system from the acceptance of an incoming call through the final disposition of the call.

call record

All recorded data pertaining to a single call.

CALS (Continuous Acquisition and Life Cycle Support)

A joint project of industry and the U.S. Department of Defense to exchange technical-support information in digital form. (The acronym originally stood for "Computer-Aided Acquisition and Logistics Systems," and is now sometimes expanded as "Commerce at Light Speed.") It has become a common set of programs for integrating electronic commerce initiatives, intended to enhance the development of pro forma and de facto standards (particularly for graphics exchanges) and to drive new methods for concurrent manufacturing in the automotive, aerospace, electronics and heavy-equipment industries. CALS is a useful way for manufacturing enterprises to combine a number of productivity-enhancing initiatives under one umbrella.

CAM (computer-aided manufacturing)

The manufacturing of goods controlled and automated via computer and robot. Frequently used in conjunction with computer-aided design (CAD).

CAMA (centralized automatic message accounting)

An automatic message accounting system that is located at an exchange but that serves various adjacent exchanges.

campaign management system (see CMS)

Canadian Information Processing Society (see CIPS)

CANDA (computer-assisted new drug application)

CAP (carrierless amplitude phase modulation)

A multilevel, multiphase encoding method for transmitting data over twisted pair lines. It is a superset of the legacy protocol used by analog modems. It is simple to implement in silicon, uses less power than discrete multitone (DMT), is supported by Paradyne and a number of original equipment manufacturer (OEM) providers, and is currently in use in several Internet service providers' digital subscriber line (DSL) services.

CAP (competitive access provider)

A U.S. provider of bypass services for telecommunications.

capabilities exchange (see CEX)

Capability Maturity Model (see CMM)

capacity requirements planning (see CRP)

CAPE (concurrent art-to-product environment)

CAPE represents the third wave of design. It requires a wide variety of synergistic applications to work together, including visualization, rapid prototyping, analysis, materials selection, machining and cost estimation. Key to CAPE are application frameworks, data management and product geometry exchanges, so that any person who is involved in product design and approval can participate in the process.

Seven elements comprise the technological foundation of the CAPE system architecture:

- Hardware independence
- Software architecture
- Framework incorporation
- Application integration
- Data exchange
- Data management
- Enterprise pricing policies

Beyond these base elements, CAPE systems include technological components targeted to specific application areas. These application-specific elements are grouped into three markets — mechanical design, process plant design and electronics design — which account for the majority of industrial design activity.

CAPP (computer-aided process planning)

Application software used to develop work instructions and product manufacturing or assembly steps in discrete manufacturing operations.

CAR (committed access rate)

A metric used Internet quality of service (QOS) agreements to classify and limit customer traffic and manage excess traffic according to the network policy.

card

A removable board that carries the necessary circuits for a particular computer function; such cards (or boards) are designed to fit expansion slots provided by computer manufacturers.

card cage

A frame for holding circuit cards in a midsize computer system. Also referred to as a card chassis.

care delivery organization (see CDO)

Carlisle Adams, Stafford Tavares (see CAST)

carrier alarm

An alarm condition that occurs when excessive zeros or all zeros are transmitted.

carrier detect (see CD)

carrier frequency

The frequency of a carrier wave, measured in cycles per second, or Hertz, that is modulated to transmit signals.

carrierless amplitude phase modulation (see CAP)

carrier loss

An alarm condition that occurs when 32 consecutive zeros are transmitted.

Carrier Sense Multiple Access with Collision Detection (see CSMA/CD)

carrier signaling

Any of the signaling techniques used in multichannel carrier transmission. The most commonly used techniques are in-band signaling, out-of-band signaling and separate channel signaling.

carrier system

The means of obtaining a number of channels over a single path by modulating each channel on a different carrier frequency and demodulating at the receiving point to restore the signals to their original frequency.

cartridge

In optical technology, an enclosure, generally of plastic, in which an optical medium is kept for protection; also called a "cassette." Some vendors captivate their media in the cartridges (this mode is called "spin in"), providing a window for the light beams; others remove the medium from the cartridge inside the drive.

Cartridge Memory (see CM)

cascade control

A control strategy that uses the output of one controller as the set point for another.

cascading faults

Network faults (outages) that generate other faults.

cascading hubs

A hierarchy of hubs enabling many local-area network (LAN) segments to be connected to a backbone efficiently but without great expense.

CASE (computer-aided software engineering)

An umbrella term for a collection of application development tools designed to increase programmer productivity. They include technologies such as application generators and PC-based workstations that provide graphics-oriented automation of the front end of the development process.

CASE analysis and design tools

Graphical, interactive tools for the analysis and design phases of application software development.

case-based reasoning (see CBR)

cash concentration and disbursement plus addenda (see CCD+)

CASE Data Interchange Format (see CDIF)

CAST (Carlisle Adams, Stafford Tavares)

A Canadian-designed telecommunications algorithm, named after its designers. CAST is used by Nortel Networks and Entrust Technologies.

CAT (Communications Authority of Thailand)

The exclusive operator of Thailand's international telecommunications services to the rest of the world. Set up in 1976, it has challenged the Telephone Organization of Thailand's (TOT's) dominance in the domestic services sector and, apart from international telecommunications services, provides data communications, mobile and satellite services. It also has some regulatory and licensing powers and operates Thailand's postal service.

catalog

Data sets grouped and placed permanently in a storage device for use when required.

catalog content management

Processes, services and applications used to allow electronic catalog creation and updating in an e-commerce environment.

Catalog Interchange Format (see CIF)

Category (cabling)

A series of five grades of unshielded twisted-pair (UTP) cabling defined by the Electronic Industries Association/Telecommunication Industry Association (EIA/TIA)-586 standard. It includes:

Category 1: Used for voice communications; not normally used for data.

Category 2: Used for data transmission up to 4 megabits per second (Mbps).

Category 3: Used in 10Base-T networks.

Category 4: Used in token ring networks transmitting up to 16 Mbps.

Category 5: Used in local-area networks transmitting at up to 100 Mbps.

CATI (computer-aided telephone interviewing)

CATIA (Computer-Aided Three-Dimensional Interactive Application)

A computer-aided design and product lifecycle management product from IBM and Dassault Systemes.

CATV (community antenna television)

1. Commonly known as "cable TV." Television signals are received at a selected site by and retransmitted to subscribers via a cable network. Additional channels, not normally available in that area, can also be transmitted.
2. Data communications based on radio frequency (RF) transmission, generally using 75-ohm coaxial cable as the transmission medium. CATV offers multiple frequency-divided channels, allowing mixed transmissions to be carried simultaneously.

CAU (controlled access unit)

An intelligent hub on a token ring network.

CAV (constant angular velocity)

One of two standards for rotating storage media, in which the disk spins at the same rate at all times. The other, constant linear velocity (CLV), spins the disk more slowly on the inside tracks where the circumference is smaller.

CB (Component Broker)

An object request broker (ORB)-based message broker from IBM, focused on solving the problem of application integration. It will enable enterprises to build applications that draw on the enterprise applications' resources.

C-band

The portion of the electromagnetic spectrum, approximately 4 gigahertz (GHz) to 6 GHz, used primarily for satellite and microwave transmission.

CBC (cipher block chaining)

A Data Encryption Standard (DES) mode of operation.

CBD (component-based development)

A set of reuse-enabling technologies, tools and techniques that allow application development (AD) organizations to go through the entire AD process (i.e., analysis design, construction and assembly) or through any particular stage via the use of

predefined component-enabling technologies (such as AD patterns, frameworks, design templates) tools and application building blocks.

CBDS (Connectionless Broadband Data Service)

A European metropolitan-area networking term similar in many respects to Switched Multimegabit Data Service.

CBL (Common Business Language)

Commerce One's Extensible Markup Language (XML) schema development effort.

CBP (constraint-based programming)

A technique that looks for a workable solution by reducing the search space (i.e., possibilities) by processing the necessary conditions to be satisfied. Configuration engines, planning and scheduling systems are among its most successful applications.

CBQ (class-based queuing)

A mechanism that defines various levels of service for access to the Internet and can be used with or without Resource Reservation Protocol (RSVP). CBQ divides traffic into queues and assigns each a specific amount of network bandwidth.

CBR (case-based reasoning)

An artificial-intelligence problem-solving technique that catalogs experience into "cases" and correlates the current problem to an experience. CBR is used in many areas, including pattern recognition, diagnosis, troubleshooting and planning. These systems are easy to maintain in comparison to rule-based expert systems.

CBR (constant bit rate)

An asynchronous transfer mode (ATM) service category, defined by the ATM Forum, that guarantees a constant bandwidth with low delay, jitter and cell loss. Circuit emulation is a typical application.

CBR (content-based retrieval)

Search methodology for retrieving information based on words or phrases in the text.

CBT (computer-based training)

A training method that uses a mixture of software, disks and manuals to convey information. Trainees work at their own pace, often over an extended period.

CBX (computerized branch exchange)

CC (competence center)

A permanent center of expertise in an enterprise that supports implementation, enhancement and maintenance of common business processes and systems built around SAP's R/3.

CC (concurrent copy)

CC (control code)

A multibit code reserved for controlling hardware, such as printers.

CCA (Compatible Communications Architecture)

A protocol for transmission of asynchronous data over X.25 lines.

CCBP (consumer-centric bill provider)

CCBPs offer bill management services that bring all of a consumer's bills online so that they can be viewed, paid and managed.

CCD (charge-coupled device)

A semiconductor device capable of both photo-detection and memory, which converts light to electronic impulses. One- and two-dimensional CCD arrays are used in scanners to perform the first stage in converting an image into digital data. They are particularly attractive because they can step the signals from each detector across the array in response to a clock signal, permitting each scan line to be read through a single electrical connection.

CCD+ (Cash Concentration and Disbursement plus addenda)

One of the primary message formats necessary for enterprise-initiated payments to traverse the U.S. national banks' clearinghouse system. The format is limited to a single addendum record (one invoice, one payment), and many banks can process it.

CCIR (Comite Consultatif International des Radio Communications)

Abbreviation of the French name for the International Radio Communications Consultative Committee, now part of the International Telecommunication Union (ITU).

CCIS (Common Channel Interoffice Signaling)

An electronic means of signaling between any two switching systems independent of the voice path.

CCITT (Comite Consultatif International Telegraphique et Telephonique)

Abbreviation of the French name for the International Telegraph and Telephone Consultative Committee. In March 1993, the name was changed to the International Telecommunication Union (ITU) Telecommunications Standards Sector.

CCITT Group III

Also now called ITU Group III, this is the original standard for compression and decompression of facsimile transmissions.

CCITT Group IV

Also now called ITU Group IV, this is an optimized standard for the transmission of black-and-white office documents. Neither Group III nor Group IV handles color. Both are required to reduce bandwidth and storage demand.

CCL (common-carrier line)

A type of access charge paid by interexchange carriers (IXCs) to local-exchange carriers (LECs). A usage-sensitive portion is included, even though the actual costs to the LECs are not usage-sensitive.

CCM (clinical context manager)

CCM (Customer Contact Manager)

A computer-telephony integration (CTI) system from Hewlett-Packard.

CCMS (Computing Center Management System)

A management framework developed by SAP that provides an event console, a set of management services and sets of application programming interfaces to be integrated with third-party management tools.

CCMS (converged communications management system)

CCN (common control node)

ccNUMA (cache-coherent nonuniform memory access)

A category of high-performance computing systems.

c-commerce (collaborative commerce)

Collaborative, electronically enabled business interactions among an enterprise's internal personnel, business partners and customers throughout a trading community. The trading community could be an industry, industry segment, supply chain or supply chain segment.

C conditioning

A type of line conditioning that controls attenuation, distortion and delay distortion so they lie within specific limits.

CCOW (Clinical Context Object Workgroup)

A group that defines standards for collaboration among visual (GUI-based) applications on clinical workstations. Originally an independent consortium, CCOW is now technical committee of the Health Level Seven (HL7) standards organization.

CCP (Certified Computer Professional)

A technology credential issued by the Institute for Certification of Computing Professionals (ICCP). To receive CCP certification, applicants must have four years of industry experience and pass a series of core and specialty exams.

CCR (customer-controlled routing)

CCS7 (Common Channel Signaling No. 7)

See SS7.

CCS (hundred call seconds)

A metric used in calculating call center enquiry volume or efficiency.

CCSA (common control switching arrangement)

An automatic switching arrangement in which the control equipment necessary for the establishment of connections is shared, being associated with a given call only during the period required to accomplish the control function.

CCSA access

The provision of inward and outward service between the private branch exchange (PBX) and the Common Control Switching Arrangement (CCSA) network.

CCTA (Central Computer and Telecommunications Agency)

A former U.S. agency, which was subsumed by the Office of Government and Commerce in April 2001.

CCTO (client configure-to-order)

CD (carrier detect)

A signal indicating that a connection has been made. Also known as a received line signal detector (RLSD) signal.

CD (compact disc)

The trademarked name for the laser-read digital audio disc, 12 centimeters in diameter, developed jointly by Philips and Sony.

CDA (Compound Document Architecture)

A proprietary method for describing and exchanging document data, developed by Digital Equipment (later acquired by Compaq, now part of Hewlett-Packard). The data exchanged may include rich text, synthetic graphics, scanned images, voice, relational and spreadsheet-like tables, and full-motion video.

CDBS (Connectionless Broadband Data Service)

European high-speed, packet switched wide-area networking standard, similar to Switched Multimegabit Data Service (SMDS).

CDDI (Copper Distributed Data Interface)

A method, favored by the American National Standards Institute (ANSI), of transmitting Fiber Distributed Data Interface (FDDI) signals over copper media. CDDI runs on both shielded and unshielded twisted-pair cabling.

CDE (Common Desktop Environment)

The first user interface specification (based on Motif) from the now-defunct Common Open Software Environment (COSE) group, a consortium of major vendors that was dedicated to standardizing Unix. CDE later became a specification of the Open Group.

CDE (Cooperative Development Environment)

CDER (Center for Drug Evaluation and Research)

CDF (Channel Definition Format)

A format introduced by Microsoft in 1997 in an effort at standardizing the marketplace for push delivery.

CDG (CDMA Development Group)

An international consortium of companies that focuses on the adoption and evolution of code division multiple access (CDMA) wireless systems.

CD-I (compact disc-interactive)

CDIA (Certified Document Imaging Architect)

A type of technology certification issued by the Computing Technology Industry Association (CompTIA).

CDIF (CASE Data Interchange Format)

The primary interchange standard for computer-aided software engineering (CASE) tools.

CDLA (Computer Dealers and Lessors Association)

A trade association of North American-based computer dealers and lessors. The principal purpose of the organization is to promote professional integrity among its members. It also promotes alternate instruments for computer financing.

CDM (Common Data Model)

A metadata rendering of the data elements used within i2 Technologies products.

CDMA (code division multiple access)

A digital wireless technology used in radio communication for transmission between a mobile phone and a radio base station. CDMA was developed by Qualcomm, and commercially introduced in 1995. It enables the simultaneous transmission and reception of several messages, each of which has a coded identity to distinguish it from the other messages.

CDMA2000

An updated version of CDMA technology, developed by Qualcomm. It doubles the voice capacity of cdmaOne systems and also supports high-speed data services. See cdmaOne.

cdmaOne

A standard based on code division multiple access (see CDMA), offering increased voice capacity over analog systems and data speeds from 14.4 kilobits per second (kbps) to 64 kbps. cdmaOne was developed by Qualcomm and endorsed by the International Standards Organization (IS-95A/B).

CDO (care delivery organization)

A category of enterprises that use healthcare information systems. CDOs are organizations, such as hospitals and physician practices, whose primary mission is to deliver healthcare-related services.

CDO (Collaboration Data Object)

An event-based architecture, formerly known as Active Messaging.

CDPD (cellular digital packet data)

Cellular data that is transmitted over a cellular network. In early deployments, packet data moved at 19.2 Kbps over ever-changing unused intervals in the voice channels. Modern deployments use dedicated data channels. CDPD is an IP-based network with RC4 encryption that allows cellular networks to offer remote and mobile computing.

CDR (call detail recording)

A means of capturing telephone system information on calls made, including who made the call, where it went and what time of day it was made, for processing into meaningful management reports. With such information, it is easier to spot exceptions to regular calling patterns such as out-of-hours calling, international calls, significant variances from previous reporting periods and call destinations that do not reflect normal calling patterns for the enterprise. Also known as station message detail recording (SMDR).

CDR (clinical data repository)

A database for storage of clinical information in a computer-based patient record (CPR).

CDR (continuous data recognition)

CD-R (compact disc-recordable)

A standard and technology that enables users to write on and read from a compact disc.

CD-ROM (compact disc read-only memory)

A version of the standard compact disc intended to store general-purpose digital data. CD-ROMs can store a wide variety of data such as music, video and graphics. CDs are often used by software companies to deliver programs like word processing or spreadsheet applications, because they store significantly more data (650 megabytes) than a typical floppy disk.

CD-RW (compact disc rewritable)

A drive that permits the user to back up data to compact discs (CDs).

CDS (cell directory service)

A core Distributed Computing Environment (DCE) server component provided for applications to locate resources. It functions primarily as a naming service, which provides a mapping between a logical name and a physical address.

CDSS (clinical decision support system)

An application that enhances decision making by health caregivers by providing context-sensitive advice relating to clinical situations.

CDV (cell delay variation)

One of three negotiated quality of service (QOS) parameters defined by the ATM Forum.

CE (concurrent engineering)

A collaborative, team-based approach for designing products that combines multiple departments and disciplines into a project team.

CE (customer engineer)

CE (see Windows CE)

CEC (central electronics complex)

The term generically used to refer to the central processing unit (CPU), and includes the power unit, service units, console and other units, but not any peripherals. Within a sysplex or coupled complex it is any serial-numbered processor (which may be made up of multiple engines). Within a parallel sysplex complex, it is a single parallel processor that can house multiple engines. (Today this is an uncommon configuration.)

CEFACT (Center for Facilitation of Procedures and Practices for Administration, Commerce and Transport)

A body that is under the auspices of the United Nations Economic Commission for Europe and is responsible for the United Nations rules for Electronic Data Interchange for Administration, Commerce and Transport (UN/EDIFACT).

cell

1. A block of fixed length identified by a label at Layer 1 of the Open Systems Interconnection (OSI) reference model. It is the fundamental building block of asynchronous transfer mode (ATM) and Broadband Integrated Services Digital Network (B-ISDN).
2. The storage position of one unit of information such as a character, bit, or word.
3. Also used to describe a precise location in an electronic spreadsheet where specific row and column coordinates intersect.
4. In data transmission, a fixed number of bytes of data sent together, as opposed to a frame, which is of variable length.
5. In mobile communications, the basic geographic unit of a cellular system. Also, the basis for the generic industry term "cellular."

cell controller

A supervisory computer used to sequence and coordinate multiple machines and operations.

cell delay variation (see CDV)

cell directory service (see CDS)

cell loss ratio (see CLR)

cell of origin

A type of wireless location service that requires no modification to the handset or the cellular network. This method is fast (about three seconds), but its accuracy depends on the radius of cells — which can range from 100 meters in urban areas to several kilometers in rural areas. See "location services."

cell relay

A transmission mode that utilizes fixed-length cells as the bearer mechanism, as with asynchronous transfer mode (ATM), which uses 48 bytes of payload plus five overhead bytes as the standard cell size.

cell splitting

A means of increasing the capacity of a cellular system by subdividing or splitting cells into two or more smaller cells.

cell transfer delay (see CTD)

cellular

Wireless services in the 800 megahertz (MHz) radio spectrum.

cellular digital packet data (see CDPD)

Cellular Multi-Processing (see CMP)

cellular radio

Technology employing low-power radio transmission as an alternative to local loops for accessing the switched telephone network. It differs from older forms of mobile

telephony in that service is provided through a large number of areas or cells that are served by a low-power transmitter in each cell, rather than through a single high-power transmitter for the entire region. Because any given frequency can be reused in each cell, the number of subscribers that can be served is multiplied dramatically. Users can be stationary or mobile.

CEM (contract electronics manufacturer)

CEMS (Contrat d'Entreprise Maintenance et Service)

IBM's Extended Service Agreement (ESA) offering in France.

CEN (Comite Europeen de Normalisation)

French name of the European Committee for Standardization, a Brussels-based standards body.

CENELEC (Comite Europeen de Normalisation Electrique)

French name of the European Committee for Electrotechnical Standardization, a Brussels-based standards organization founded in 1973.

Center for Facilitation of Procedures and Practices for Administration, Commerce and Transport (see CEFACT)

center of excellence

A central clearinghouse for knowledge capital that is used across all business transformation projects.

central electronics complex (see CEC)

centralized automatic message accounting(see CAMA)

centralized dispatching

Organization of the dispatching function into one central location. This often involves the use of data collection devices for communication between the centralized dispatching function, which usually reports to the production control department, and the shop manufacturing departments.

central office (see CO)

central processing unit (see CPU)

Centrex

Centrex switching equipment is located in the central office (exchange) and enables subscribers to access facilities normally provided by a separate private branch exchange (PBX). It can be located in a central office or on a large customer's premises.

CEPS (Common Electronic Purse Standard)

A standard endorsed by Visa and American Express for electronic purse or stored-value applications on smart cards.

CEPT (European Conference of Postal and Telecommunications Administrations)

Formed in 1959, CEPT underwent a major reform during the early 1990s. CEPT's involvement in standardization and operational activities has been transferred to other bodies, leaving CEPT to deal solely with the regulatory issues affecting postal and telecommunications sectors across Western and Eastern Europe.

CERDIP (ceramic dual in-line package)

CERN (Conseil Européen pour la Recherche Nucleiare)

In English, the European Laboratory for Particle Physics.

CERT (Computer Emergency Response Team)

A group formed in 1998 by the U.S. Defense Advanced Research Projects Agency — and coordinated through Carnegie Mellon University's Software Engineering Institute (SEI) — to research and report on Internet-related security problems. SEI's CERT Coordination Center publishes security information and advisory bulletins through its Web site at www.cert.org.

CERT (computer emergency response team)

See CIRT.

certificate authority (see CA)

certificate chaining

Recognition by one certificate of the root key of a different certification authority.

certificate management system (see CMS)

certificate revocation list (see CRL)

certification (see accredited certification)

certification authority (see CA)

certification practice statement (see CPS)

Certified Computer Professional (see CCP)

Certified Document Imaging Architect (see CDIA)

Certified Network Professional (see CNP)

CES (Consumer Electronics Show)

CEX (capabilities exchange)

A method whereby applications "understand" device attributes such as screen size, color vs. monochrome, resolution, amount of storage and bandwidth available.

CF (coupling facility)

CFCC (coupling facility control code)

CFI (Computer-Aided Design Framework Initiative)

A nonprofit organization formed to develop framework standards to facilitate the integration of CAD tools, particularly in the electronics design arena.

CFML (ColdFusion Markup Language)

The server-side declarative script used in Allaire's ColdFusion Web development product.

CGA (Color Graphics Adapter)

The first color video adapter, introduced by IBM in 1981. It displays four colors simultaneously.

CGI (Common Gateway Interface)

A data-passing specification used when a Web server must send or receive data from an application such as a database. A CGI script passes the request from the Web server to a database, gets the output and returns it to the Web client.

CHAID (chi-squared automated induction)

Challenge Handshake Authentication Protocol (see CHAP)

change integration

Occurs at the back end of the change management process to coordinate and schedule requested changes, with the goal of reducing downtime and risk.

change management

Automated support for development, rollout and maintenance of system components (i.e., intelligent regeneration, package versioning, state control, library control, configuration management, turnover management and distributed impact sensitivity reporting).

channel

A channel is defined by a combination of three factors: A target device, for example, a mobile phone, personal digital assistant (PDA) or a PC; an interaction paradigm, for example, a graphical user interface (GUI), e-mail, voice response or video; and business practices, that is, channel-specific elements of business that may include payment services, shipment processes, taxation, customer support and authentication

channel analytics

A superset of Web analytics (see separate entry), channel analytics are not restricted to Web channels, but include direct mail, the customer contact center, mass media, store or branch locations, and all other distribution or customer-contact channels. The different elements of business — for example, payment and shipment processes, and customer support and authentication — need to be measured and analyzed. Channel analytics examine costs, usage, efficiency, integrity, integration with other systems and the value of each channel, separately and in relation to each other.

channel assembly

A sales channel initiative aimed at offloading much of the system assembly task from the initial manufacturers to an intermediate dealer or distributor. The reseller or distributor then assembles the system to the buyer's specifications.

channel bank

The equipment typically used in a telephone central office that performs multiplexing of lower-speed, digital channels into a higher-speed composite channel. The channel bank also detects and transmits signaling information for each channel and transmits framing information so that time slots allocated to each channel can be identified by the receiver.

channel capacity

An expression of the maximum data traffic that can be handled by the channel.

Channel Definition Format (see CDF)

channel integration

Strategies aimed at consolidating — either physically or logically — customer information and its use to provide an all-encompassing view of the customer.

channel service unit (see CSU)

CHAP (Challenge Handshake Authentication Protocol)

A security protocol used to provide challenges and responses with encryption that allows access between data communications systems prior to and during data transmission. CHAP uses challenges to verify that a user is allowed access to a system. See Password Authentication Protocol.

charge-coupled device (see CCD)

chat

A keyboard conference between two or more users on a local-area network (LAN), on the Internet or via a bulletin board system (BBS). See Internet Relay Chat (IRC) and instant messaging.

chat room

The “location” in which an interactive keyboard discussion on a topic, hosted on the Internet or on a bulletin board system (BBS), takes place. Chat rooms can be accessed from major Internet service providers (ISPs), individual Web sites and the Internet Relay Chat (IRC) system — the Net’s traditional means of conferencing by computer.

check digit

A number added to each character in a coded system which allows for the detection of errors; through the use of a mathematical formula, recording errors such as number reversals can be noted. See parity bit.

checksum

A value calculated from a block of data, used to detect errors in transmitted data.

CHES (Comprehensive Health Enhancement Support System)

An ongoing research project at the University of Wisconsin focused on providing seriously ill patients with free computers loaded with educational software, access to databases with information about their diagnosed illnesses and access to support groups.

chief information officer (see CIO)

chief knowledge officer (see CKO)

chief technology officer (see CTO)

CHIME (College of Healthcare Information Management Executives)

An organization formed in 1992 to advocate more effective use of information management in healthcare, and to meet the professional-development needs of healthcare CIOs.

CHIN (community health information network)

An encompassing term to describe any community-based network open to (and perhaps required for) all healthcare organizations.

chip

An integrated circuit that is the foundation of computer processing and data storage. It holds the logic circuitry that responds to and processes the basic instructions that run all types of computers. Chips are used in everything from watches and calculators to personal computers (PCs) and high-performance computers.

CHIP (client host interface processor)

chip cards

Smart cards and memory cards. A smart card includes embedded microcontroller silicon. A memory card includes embedded silicon memory and possibly other functions, such as cryptography, but no microprocessor.

CHIPCo (Clearing House Interbank Payments Company)

A private company responsible for the CHIPS payment-clearing system (see CHIPS).

CHIPS (Clearing House Interbank Payments System)

A clearing system that processes very large payments (valued at an average of \$1.4 trillion a day) in more than 27 countries.

CHMIS (community health management information system)

choke packet

A packet sent to a transmitter which informs it that congestion exists and requests that it reduce its sending rate.

CHRP (Common Hardware Reference Platform)

An effort subsequently abandoned by Apple Computer, IBM and others to establish a multivendor system architecture for the PowerPC processor.

cHTML (Compact Hypertext Markup Language)

A proprietary version of Hypertext Markup Language (HTML) adopted by NTT DoCoMo for i-mode. See i-mode and HTML.

CI (competitive intelligence)

Analysis of an enterprise's marketplace to understand what is happening, what will happen and what it means to the enterprise. CI business goals may be offensive (to

position the company in the marketplace, plot a course for the future and allocate resources) or defensive (to react to unfolding or potential market developments).

CI (Computer Interconnect)

The local-area network (LAN) used in a Virtual Address Extension (VAX) cluster.

CIC (circuit identification code)

An end-point identifier (ISDN term).

CICS (Customer Information Control System)

IBM's strategic general-purpose subsystem for implementing transaction-processing applications. CICS invokes customer-written application programs in response to transactions entered at teleprocessing terminals, and provides the services needed by those applications to retrieve and update data in files and respond to the terminal that invoked them.

CICS/SM (Customer Information Control System/System Manager for AIX)

The Unix-environment CICS management capability.

CICS/TS (Customer Information Control System/Transaction Server)

CIDR (classless interdomain routing)

The successor to class-oriented domains for Internet routing, it allows for better allocations of Internet addresses. It combines a number of Class C Internet Protocol (IP) addresses to reduce the burden on routing tables in the Internet.

CIF (Catalog Interchange Format)

A lightweight specification from Ariba used to communicate catalog information between selling and buying organizations. CIF is a public-domain specification used by corporations to exchange contract information about product and service offerings.

CIF (Common Intermediate Format)

A videoconferencing data rate specification. See QCIF.

CIF (customer information file)

A system that consolidates customer account information and combines it with basic demographic information to create a current snapshot of a customer relationship. CIFs are often a central component of integrated banking application packages and are primarily used to support operational activities with current, as opposed to historical, data. See MCIF.

CIFS (Common Internet File System)

A remote file system access protocol that allows groups of users to work together and share documents via the Internet or their corporate intranets. CIFS is an open, cross-platform technology based on the native file-sharing protocols built into Microsoft Windows and other operating systems, and is supported on numerous platforms, including Unix. Microsoft submitted a preliminary draft of the CIFS 1.0 protocol specification to the Internet Engineering Task Force in December 1997.

CIL (Component Integration Labs)

CIM (Common Information Model)

A modeling schema that describes managed system, hardware, and software objects. CIM is a component of the Web-Based Enterprise Management (WBEM) initiative, an emerging Web-oriented systems management standard controlled by the Distributed Management Task Force (DMTF).

CIM (computer-integrated manufacturing)

The integration of manufacturing operations by integrating human systems, information systems and manufacturing systems. The goal of such systems is to combine electronically the systems and functions necessary to manufacture products more effectively.

CIM (customer information management)

The systematic support of business strategy through customer knowledge.

CIMA (customer information management and application)

A six-step process starting with a business plan followed by a technology plan. The six steps are:

1. planning
2. information acquisition
3. information compilation, storage and maintenance
4. information analysis
5. information application
6. information distribution

CIO (chief information officer)

The person responsible for planning, choosing, buying and installing a company's computer and information-processing operation. Originally called data-processing managers, then management information system (MIS) directors, CIOs develop the information technology (IT) vision for the company. They oversee the development of corporate standards, technology architecture, technology evaluation and transfer; sponsor the business technology planning process; manage client relations; align IT with the business; and develop IT financial management systems. They also oversee plans to reinvest in the IT infrastructure, as well as in business and technology professionals. They are responsible for leading the development of an IT governance framework that will define the working relationships and sharing of IT components among various IT groups within the corporation.

CIP (Channel Interface Processor)

CIP (Classical Internet Protocol)

cipher block chaining (see CBC)

cipher feedback (see CFB)

CIPS (Canadian Information Processing Society)

A Canadian organization that provides information technology certification.

CIR (committed information rate)

In a frame relay network, the minimum speed to be maintained between nodes.

CIRC (Cross-Interleaved Reed-Solomon Code)

circuit

1. A continuous electrical connection between any two points.
2. The means of two-way communication between two or more points.
3. A group of electrical/electronic components connected to perform a specific function. See channel.

circuit board

A flat card with connections for electronic components; part of an electronic system.

circuit grade

The data-carrying capability of a circuit; the grades of circuit are broadband, voice, subvoice, and telegraph.

circuit identification code (see CIC)

circuit switching

Temporary direct connection of one or more channels between two or more points to provide the user with exclusive use of an open channel with which to exchange information. A discrete circuit path is set up between the incoming and outgoing lines, in contrast to message switching and packet-switching, in which no such physical path is established. Also called line switching.

CIRP (cyber incident response plan)

Also known as a "computer incident response plan," this is formulated by an enterprise to respond to potentially catastrophic, computer-related incidents, such as viruses or hacker attacks. The CIRP should include steps to determine whether the incident originated from a malicious source — and, if so, to contain the threat and isolate the enterprise from the attacker. (See cyber incident and CIRT.)

CIRT (cyber incident response team)

Also known as a "computer incident response team," this group is responsible for responding to security breaches, viruses and other potentially catastrophic incidents in enterprises that face significant security risks. In addition to technical specialists capable of dealing with specific threats, it should include experts who can guide enterprise executives on appropriate communication in the wake of such incidents. The CIRT normally operates in conjunction with other enterprise groups, such as site security, public-relations and disaster recovery teams. (See cyber incident and CIRP.)

CIS (clinical information system)

CIS (customer information system)

CISC (complex instruction set computer)

A computer in which individual instructions may perform many operations and take many cycles to execute, in contrast with reduced instruction set computer (RISC).

Examples include IBM S/370 and Hewlett-Packard's (formerly Digital Equipment's and, later, Compaq's) VAX.

CISO (chief information security officer)

CISSP (Certified Information Systems Security Professional)

A certification program administered by the International Information Systems Security Certification Consortium (ISC2).

CKO (chief knowledge officer)

Articulates and champions the knowledge management (KM) vision, provides the enterprise level leadership to implement and sustain KM, and is the ultimate focal point for knowledge creation, sharing and application.

CL (coupling link)

CLA (Corporate License Agreement)

A licensing agreement option under Novell's "Customer Connections" program. It offers an alternative for midsize organizations that cannot qualify for Novell's Master License Agreement (MLA).

cladding

In fiber-optic cable, a colored, low refractive index material that surrounds the core and provides optical insulation and protection to the core.

clamshell

A computer system that weighs less than 3 pounds and opens lengthwise to expose a keyboard and screen.

class

A specification that defines the operations and the data attributes for a set of objects.

CLASS (Custom Local Area Signaling Services)

A bundle of telephone features introduced by U.S. telephone companies in the late 1980s to offer users more control over incoming calls. CLASS features include caller ID, anonymous call rejection, automatic callback, automatic recall, caller ID blocking, distinctive ringing, call waiting, selective call rejection, call trace, selective call acceptance and selective call forwarding.

class-based queuing (see CBQ)

Classical IP over ATM

A specification for running Internet Protocol over asynchronous transfer mode (IP over ATM) as defined in RFC 1577. This differs from traditional IP, whose Address Resolution Protocol (ARP) uses broadcasts to learn remote addresses.

classless interdomain routing (see CIDR)

class of exchange

The ranking assigned to a switching point in the telephone network determined by its switching functions, interrelationships with other exchanges, and transmission requirements. Also called class of office.

class of service (see COS)

1. In X.25 networks, refers to the speed at which the data circuit-terminating equipment (DCE) and the data terminal equipment (DTE) communicate.
2. In telephony, the categorization of telephone subscribers according to specific types of telephone usage.

Clearing House Interbank Payments Company (see CHIPCo)

Clearing House Interbank Payments System (see CHIPS)

cleartext operation

An option permitting a user to bypass a device's crypto functions, enabling communication with nonsecure terminals and enabling data to be sent or stored in clear mode.

clear to send (see CTX)

CLEC (competitive local-exchange carrier)

Provides competition in the local telephone services market. CLECs include local-services resellers or aggregators, which buy local services in volume at wholesale prices and resells them to the market, as well as "hybrid" resellers — carriers that build portions of the local network band and buy the remaining service components.

CLI (call-level interface)

CLI (command-line interface)

CLI (Common Language Infrastructure)

A platform-independent development system from Microsoft that enables programs written in different programming languages to run on different types of hardware. CLI is part of Microsoft's .NET platform and is expected to become an ECMA standard. The CLI includes the Common Type System (CTS) and Common Language Specification (CLS). No matter which programming language they are written in, CLI applications are compiled into Intermediate Language (IL), which is further compiled into the target machine language by the Common Language Runtime (CLR) software. See CLS, CTS, .NET and IL.

C library

A group of tested application-level routines in C language used by programmers to develop applications.

clicks and bricks

The combining of e-business channels and network-based processes with selective investment in physical locations to control local markets, distribution channels, and critical labor accessibility.

clickstream analysis

A form of Web analytics (see separate entry), clickstream analysis is the tracking and analysis of visits to Web sites. Although there are other ways to collect this data, clickstream analysis typically uses the Web server log files to monitor and measure Web site activity. This analysis can be used to report user behavior on a specific Web site, such as routing, stickiness (a user's tendency to remain at the Web site), where

users come from and where they go from the site. It can also be used for more aggregate measurements, such as the number of hits (visits), page views, and unique and repeat visitors, which are of value in understanding how the Web site operates from a technical, user experience and business perspective.

click-through

Term applied to the act of clicking with a mouse button on a Web page advertisement, which brings the user to the advertiser's site. See CTR (click-through rate).

click-through rate (see CTR)

click-wrapped

License terms for downloaded software that are accepted by the user by clicking the mouse on a button labeled "I accept" or similar words. See shrink-wrapped.

client

A system or a program that requests the activity of one or more other systems or programs, called servers, to accomplish specific tasks. In a client/server environment, the workstation is usually the client.

client appliance

A type of computing appliance that provides end-user access to applications. Examples include network computers and certain handheld computers. See computing appliance.

client library (see CTLIB)

client/server

The splitting of an application into tasks performed on separate computers connected over a network. In most cases, the "client" is a desktop computing device (e.g., a PC) or a program "served" by another networked computing device (i.e., the "server"). Gartner has defined five styles of client/server computing, based on how presentation, application logic and data management functions are partitioned between the client and server device — see separate definitions for "distributed presentation," "remote presentation," "distributed function," "remote data management" and "distributed data management."

Client/Server Interface (see C/SI)

Clinical Context Object Workgroup (see CCOW)

clinical data repository (see CDR)

clinical decision support system (see CDSS)

Clinical Resource Management System (see CRMS)

clipping

1. Loss of parts of words or of syllables in the operation of voice-actuated devices.
2. Distortion of a signal that has reached the limit of its modulation parameter (e.g., amplitude) and can no longer be effectively modulated.

CLLM (Consolidated Link Layer Management)

CLNS (Connectionless Network Service)

An Open Systems Interconnection (OSI) Network Layer service that does not require a circuit to be established before transmitting.

clock, clocking

Repetitive, regularly timed signals used to control synchronous processes.

closed-loop MRP (closed-loop material requirements planning)

A system built around MRP that also includes production planning, master production schedule, and capacity requirements planning. Once the planning phase is complete and the plans have been accepted as realistic and attainable, the execution functions come into play. These include the shop floor control functions of input/output measurement, detailed scheduling and dispatching, as well as anticipated delay reports from both the shop and vendors, purchasing follow-up and control, and other functions. The term "closed loop" implies that not only is each of these elements included in the overall system, but also that there is feedback from the execution functions so the planning can be kept valid at all times.

closed user group (see CUG)

CLR (cell loss ratio)

One of three negotiated quality of service (QOS) parameters defined by the Asynchronous Transfer Mode (ATM) Forum.

CLR (Common Language Runtime)

Software for Microsoft's .NET platform that executes the Intermediate Language (IL) code or compiles it into machine language. The CLR contains object-oriented and security services that all .NET applications can utilize. See CLI and .NET.

CLS (Common Language Specification)

A subset of Microsoft's Common Type System (CTS) that defines minimum compliance. See CTS.

cluster

Two or more terminals connected to a single point or node.

cluster controller

A device that handles the remote communications processing for multiple (usually dumb) terminals or workstations.

clustered system

An architecture that ties together uniprocessor, symmetric multiprocessing (SMP) and/or massively parallel processing (MPP) systems with all nodes sharing access to disks. Also called a shared-disk system.

clustering

The capability to define resources on one or more interconnected midrange systems as transparently available to users and applications from within the specified group of loosely coupled systems in a local-or metropolitan-area network.

CLV (constant linear velocity)

One of two standards for rotating storage media, in which the disk spins at a higher speed on the outside tracks than on the inside tracks (where the circumference is smaller), so that all data moves past the head at the same rate. The other standard is constant angular velocity (CAV), in which the disk spins at the same rate at all times.

CM (Cartridge Memory)

A Linear Tape Open (LTO) tape format feature. It consists of an embedded, noncontacting, passive radio frequency interface that accesses 4 kilobytes of nonvolatile memory. It stores metadata and data without consuming drive resources.

CM (configuration management)

A function that enables impact/dependency analysis of application components. There are two views of configuration management. The first is a microcosmic view that looks at a single device or system and the knobs, dials, switches, buffers and settings individually for the device or system. The second is a macrocosmic view that looks at these settings and components as well as the relationships and the dependencies among them.

CM (content management)

A broad term referring to applications and processes to manage Web content, document content and e-commerce-focused content.

CMC (common messaging calls)

A set of calls developed by the X.400 API Association (XAPIA) for use on top of any existing messaging system. Programmers developing applications using these calls may request services of whatever messaging system is accepting the call. The number of calls is limited to the most popular messaging and directory lookup functions. CMC is similar to simple Messaging Application Programming Interface (MAPI) in its breadth of services, but provides greater portability of applications.

CMDE (Comparex Multiplatform Data Exchange)

CMIP (Common Management Information Protocol)

The Open Systems Interconnection (OSI) protocol for the exchange of network management information.

CMIP over LLC (see CMOL)

CMIP over TCP/IP (see CMOT)

CMIS (Common Management Information Standard)

An Open Systems Interconnection (OSI) standard that defines the network monitoring and control function.

CML (Chemical Markup Language)

CMM (Capability Maturity Model)

A model from the Software Engineering Institute (SEI) used to assess the maturity of an organization's application development (AD) capabilities and processes, and to provide a road map for improvement. The CMM is a highly popular approach to AD process management and improvement. Five levels are defined in the model:

- Level 1: Ad Hoc — No development methodology and few, if any, controls are in place. Progress, if attainable, may not be recognized due to lack of measurement.
- Level 2: Repeatable — A set of tasks and processes has been defined well enough to forecast project results with reasonable accuracy. However, no method for forecasting improvements or making trade-offs has been implemented.
- Level 3: Defined — The development process is implemented and understood. Measurements are taken, and the process is predictable enough to forecast the effect of implementing new technologies.
- Level 4: Managed — Significant quantitative and qualitative improvements are possible because the process is managed and evolutionary. Each technology implementation is part of an overall architecture.
- Level 5: Optimized — A theoretical level in the development organization where the environment drives the process. Effort can be channeled into improving the process rather than executing it.

CMMS (computerized maintenance management system)

Application software used to provide for work and materials management of maintenance activities in a manufacturing organization. See EAM.

CMNS (Connection Mode Network Service)

Extends local X.25 switching to Ethernet, token ring or Fiber Distributed Data Interface (FDDI) networks.

CMO (chief marketing officer)

CMOL (CMIP over LLC)

A proprietary network management draft developed jointly by 3Com and IBM that specifies using Common Management Information Protocol (CMIP) over Logical Link Control (LLC) to provide network management of devices on mixed-media local-area networks (LANs).

CMOS (complementary metal-oxide semiconductor)

A semiconductor technology that uses less power and generates less heat (enabling higher circuit density), but is typically slower than bipolar technologies.

CMOT (CMIP over TCP/IP)

Use of the Common Management Information Protocol (CMIP) to manage gateways in a Transmission Control Protocol/Internet Protocol (TCP/IP) network.

CMP (Cellular Multi-Processing)

A heterogeneous server technology from Unisys that can run any combination of operating systems.

CMS (campaign management system)

An application used by marketers to design multichannel marketing campaigns and track the effect of those campaigns, by customer segment, over time.

CMS (certificate management system)

A system that manages digital certificates in a public-key infrastructure (PKI) security implementation. See "digital certificate" and PKI.

CMS (Conversational Monitor System)

A single-user, interactive operating system that was implemented for, and together with, IBM's Virtual Machine (VM) environment.

CMS (course management system)

A platform for delivering distributed learning (DL) courseware. See DL and courseware.

CMV (controlled medical vocabulary)

An approved list of terms coded in a fashion that facilitates the use of the computer. Controlled vocabularies are essential if clinical applications are to function as intended. Widely used systems include the American College of Radiology (ACR) Code, Current Procedural Terminology (CPT), Diagnostic and Statistical Manual of Mental Disorders (DSM-IV) and the International Classification of Diseases, Ninth Revision (ICD-9).

CMYK

Refers to the traditional four-color printing process incorporating four inks (cyan, magenta, yellow, and black). The majority of printed color paper and photographic documents incorporate the CMYK process.

CNC (computer numerical control)

CNE (Certified NetWare Engineer)

CNI (Coalition for Networked Information)

CNLP (Connectionless Network Protocol)

The Open Systems Interconnection (OSI) version of the Internet Protocol (IP).

CNP (Certified Network Professional)

An information technology certification by the Network Professional Association (NPA) requiring two years of experience, two vendor certifications and a passing grade on the core fundamentals exam.

CNS (converged network services)

The delivery of voice, data, video and other forms of network services with the following characteristics:

1. Services are usually from a customer premises-based access concentrator, owned by a network service provider (NSP) that converts traffic to an asynchronous transfer mode (ATM) or Internet Protocol (IP) stream
2. Service is delivered via one means of access, transmitted via one facility, with one switching infrastructure

CO (central office)

The telephone company's centralized switching facility, where subscriber loops terminate. The CO handles a specific geographic area, identified in the United States by the first three digits of the local telephone number.

coaxial cable

Cable consisting of an outer conductor surrounding an inner conductor, with a layer of insulating material in between. Such cable can carry a much higher bandwidth than a wire pair.

COBIT (Control Objectives for Information and Related Technology)

An auditing standard developed by the Information Security Audit and Control Association for assessing information security risk.

COBOL (Common Business Oriented Language)

COCOT (customer-owned, coin-operated telephone)

CODASYL (Conference on Data Systems Languages)

Codd's Rule Zero

Ted Codd, whose theoretical work on relational databases stimulated today's plethora of relational products, defined a fundamental "Rule Zero" for classifying relational database management systems (RDBMSs). The intent of this rule was to help enterprises focus on the requirement for a consistent integrity layer in the RDBMSs they evaluated: "For any system that is advertised as, or claimed to be, a relational database management system, that system must be able to manage databases entirely through its relational capabilities, no matter what additional capabilities the system may support."

codec (coder/decoder)

A device used to convert analog signals, such as speech, music, or television, to digital form for transmission over a digital medium, and back again to the original analog form. One is required at each end of the channel.

code division multiple access (see CDMA)

coder/decoder (see codec)

COL (Component Object Library)

The object library in Microsoft's Component Object Model (COM).

COLD (computer output to laserdisc)

Microfiche replacement system. COLD systems offer economies as a replacement medium when rapid or frequent access to archived documents is necessary. Typically, a 12-inch optical-disc platter holds approximately 1.4 million 8.5-by-11-inch pages of information, equal to 7,000 fiche masters.

ColdFusion Markup Language (see CFML)

Collaboration Data Object (see CDO)

collaborative commerce (see c-commerce)

collaborative planning, forecasting and replenishment (see CPFR)

collaborative product commerce (see CPC)

College of Healthcare Information Management Executives (see CHIME)

collision

Overlapping transmissions that interfere with one another. This occurs when two or more devices attempt to transmit at or about the same instant.

Color Graphics Adapter (see CGA)

COLS (commercial online service)

A company that provides its subscribers with access to a data base, via a modem and telephone line, for a fee.

COM (communications port)

A port that allows an application to access a modem.

COM (Component Object Model)

A component architecture in Microsoft's desktop **operating** system. A distributed version of COM (i.e., DCOM — the Distributed Component Object Model) enables the development of applications in which components are distributed over several computers (e.g., a client and one or more servers).

COM (computer output to microfilm or microfiche)

A system in which digital data is converted into an image on dry processed microfilm.

COM+

A successor to Microsoft Transaction Server (MTS) and Microsoft's Component Object Model (COM).

Comdex (Computer Dealers' Exposition)

Comite Consultatif International de Radio Communications (see CCIR)

Comite Consultatif International Telegraphique et Telephonique (see CCITT)

Commerce One

An e-marketplace company based in Pleasanton, California. Originally founded in 1994 as DistriVision, the company changed its name to Commerce One in 1997 and went public in 1999. Commerce One's e-marketplace technology uses an XML-centric transactional infrastructure, based on the XML Common Business Library specification (see xCBL).

commerce service provider (see CSP)

Commerce XML (see cXML)

commercial off-the-shelf (see COTS)

commercial online service (see COLS)

commercial parallel processing (see CPP)

committed access rate (see CAR)

committed information rate (see CIR)

commodity marketplace

A type of e-marketplace that enables suppliers to mitigate risk and enables investors to speculate on commodity values. The presence of commodity marketplaces will be limited to industries with fungible products, such as energy, agribusiness or bulk commodity chemicals. The commodity marketplace is one of three types of emerging e-marketplaces identified by Gartner; the other two are business service marketplaces and integration service marketplaces (see separate entries).

Common Air Interface (see CAI)

common carrier

An organization in the business of providing regulated telephone, telegraph, telex and data communications services.

common-carrier line (see CCL)

Common Channel Interoffice Signaling (see CCIS)

Common Channel Signaling No. 7 (CCS7)

See SS7.

common control switching arrangement (see CCSA)

Common Data Model (see CDM)

Common Desktop Environment (see CDE)

Common Electronic Purse Standard (see CEPS)

Common Gateway Interface (see CGI)

Common Hardware Reference Platform (see CHRP)

Common Information Model (see CIM)

Common Intermediate Format (see CIF)

Common Internet File System (see CIFS)

Common Language Infrastructure (see CLI)

Common Language Runtime (see CLR)

Common Language Specification (see CLS)

Common Management Information Protocol (see CMIP)

Common Management Information Standard (see CMIS)

common messaging calls (see CMC)

Common Object Request Broker Architecture (see CORBA)

Common Open Software Environment (see COSE)

Common Programming Interface for Communications (see CPI-C)

Common Type System (see CTS)

COMMS (customer-oriented maintenance management system)

communication

Transmission of information between points of transmission and reception, without alteration of sequence or structure of the content. May include data, voice, video or other content. Also "communications."

communication controller

A dedicated computer with special processing capabilities for organizing and checking data and handling information traffic to and from many remote terminals or computers, including functions such as message switching.

communication protocol

The exchange of a special sequence of control characters between a computer and a remote terminal to establish synchronous communications.

communication satellite

An earth satellite designed to act as a telecommunications radio relay. A communication satellite is usually positioned in geosynchronous orbit 35,800 kilometers (23,000 miles) above the equator so that it appears from earth to be stationary in space, but may also be in low earth orbit (LEO) 150-200 miles from earth.

Communications Authority of Thailand (see CAT)

Communications Oriented Production Information and Control System (see COPICS)

community

A constantly changing group of people collaborating and sharing their ideas over an electronic network (e.g., the Internet). Communities optimize their collective power by affiliation around a common interest, by the compression of the time between member interactions (i.e., communicating in real time), and by asynchronous "postings" that potentially reach more participants and permit more reflection time than real-time interactions.

community antenna television (see CATV)

community health information network (see CHIN)

community of practice

People associated and interlinked in a communication or knowledge network because of their shared interest or shared responsibility for a subject area. Examples are people who hold similar job functions (project managers, department managers, team leaders or customer service agents); all the people on a project team; and

people interested in specific technologies (e-commerce or network management). Communities continually emerge and dissolve, and their membership, processes and knowledge continually change and evolve.

compact disc (see CD)

compact disc read-only memory (see CD-ROM)

Compact Hypertext Markup Language (see cHTML)

compandor

The combination of a compressor at one point in a communications path, for reducing the volume range of signals, followed by an expander at another point, for restoring the original volume range. It is designed to improve the ratio of the signal to the interference entering the path between the compressor and expander.

COMPARE (Compliance Progress and Readiness)

A scale introduced by Gartner in April 1997 to assist business units, enterprises or business partners in measuring their progress toward year 2000 compliance.

COMPARE Operational Readiness Evaluation (see CORE)

comparison-shopping agent

Comparison-shopping agents are pieces of software code that automatically crawl from Web site to Web site gathering information to assist consumers in making purchasing decisions — e.g., locating a particular product at the lowest price.

compatibility

The characteristic of computer hardware or software by which one machine or program may accept and process data prepared by another without conversion or code modification.

Compatible Communications Architecture (see CCA)

competency center

An organizational structure used to coordinate IT skills with an enterprise. Competency centers provide expertise for project or program support, acting both as repositories of knowledge and resource pools for multiple business areas. Skills-based competency centers, the most common type in an information services organization, are used for application development, software language skills, data management, Internet development and network design. Within the enterprise, it is increasingly common to find competency centers (or shared services) for travel, finance and human resources. Repository-based competencies act exclusively as sources of information.

competence center (see CC)

competitive access provider (see CAP)

competitive intelligence (see CI)

competitive local-exchange carrier (see CLEC)

complementary metal-oxide semiconductor (see CMOS)

complex instruction set computer (see CISC)

Compliance Progress and Readiness (see COMPARE)

compliance unit (see CU)

component

Technically, a dynamically bindable package of functionality that is managed as a unit and accessed through documented interfaces that can be discovered at runtime. Pragmatically, components tend to fall into two major groups: technical components, which perform a technology-specific task that is application-independent (e.g., a graphical user interface control), and business components, which encapsulate a piece of business functionality.

component-based development (see CBD)

Component Broker (see CB)

Component Object Library (see COL)

Component Object Model (see COM)

component supplier management (see CSM)

composite application

An emerging applications architecture in which functionality and data from multiple applications are exploited to present an integrated user interface. A composite application is one that has the appearance of a single application (from the point of view of the end user) but is, in fact, composed of multiple, independently designed applications. They defragment the end user's view into data, providing, for example, a "customer-centric" perspective in which all relevant information on a customer is accessible from a single user interface. The enabling software required for a composite application is essentially a hub for connections between back-end (often legacy) systems and systems designed to interact with end users (e.g., Windows clients, Web servers and integrated computer-telephony systems).

compound document

Any document containing more than one data type, which may include rich text, vector graphics and raster images.

Compound Document Architecture (see CDA)

Comprehensive Health Enhancement Support System (see CHESS)

comprehensive production architecture (see CPA)

Compressed Serial Link Internet Protocol (see CSLIP)

compression

1. The application of any of several techniques that reduce the number of bits required to represent information in data transmission or storage, therefore reducing bandwidth or storage requirements.

compressor

An electronic device that compresses the volume range of a signal.

CompTIA (Computing Technology Industry Association)

An industry trade group whose members are primarily computer resellers and vendors. CompTIA offers certification programs such as A+ and Certified Document Imaging Architect.

Computer-Aided Acquisition and Logistics Systems (see CALS)

computer-aided design (see CAD)

computer-aided design and drafting (see CADD)

Computer-Aided Design and Manufacturing (see CADAM)

Computer-Aided Design Framework Initiative (see CFI)

Computer-Aided Dossier and Data Supply (see CADDY)

computer-aided engineering (see CAE)

computer-aided manufacturing (see CAM)

computer-aided process planning (see CAPP)

computer-aided software engineering (see CASE)

Computer-Aided Three-Dimensional Interactive Application (see CATIA)

computer-based patient record (see CPR)

computer-based training (see CBT)

Computer Dealers and Lessors Association (see CDLA)

Computer Emergency Response Team (see CERT)

computer incident response plan (see CIRP)

computer incident response team (see CIRT)

computer-integrated manufacturing (see CIM)

Computer Interconnect (see CI)

computerized maintenance management system (see CMMS)

computer network

An interconnection of two or more computer systems, terminals or communications facilities.

computer output to laserdisc (see COLD)

computer output to microfilm or microfiche (see COM)

Computer Supported Telephony Architecture (see CSTA)

computer-telephony integration (see CTI)

computer-to-PBX interface (see CPI)

computing appliance

A computing device that provides pre-defined services, and that has its underlying operating (OS) software hidden beneath an application-specific interface. Computing appliances offer reduced complexity (e.g., installation, administration and maintenance) and faster deployment by hiding the operating software and embedding the application within the device. A computing appliance may be based on a general-purpose OS (e.g., Windows, Solaris or Linux) if the OS's complexity is hidden and the ability to load arbitrary services is removed.

Computing appliances can provide one or more services; however, they not general-purpose devices in that they are not flexible in the services they provide.

Administrators do not need platform expertise — just limited application and appliance-specific expertise.

Gartner has defined four fundamental types of computing appliances — see separate entries for server appliance, storage appliance, network appliance and client appliance.

Computing Center Management System (see CCMS)

Computing Technology Industry Association (see CompTIA)

COM-R (computer output to microfiche replacement)

COMTI (COM Transaction Integrator)

A feature within Microsoft SNA Server that allows applications developers to extend and expose Customer Information Control System (CICS) and Information Management System (IMS) transaction programs with Component Object Model (COM) components.

concentrator

A device that merges many low-speed asynchronous channels into one or more high-speed synchronous channels to achieve economies of data transmission.

concurrent art-to-product environment (see CAPE)

concurrent backup

A system-level facility to allow a database or disk file to be backed up to another disk or to magnetic tape while it is still open for application access.

concurrent database restore

A system-level facility to allow a database or portion thereof to be restored while the database is still open for application access.

concurrent engineering (see CE)

concurrent use

A way to measure the usage of software licenses. Software can be licensed in one of the following ways: individual (it cannot be shared with other users); site (a limited number of people can use the software); concurrent use (usage is limited to a maximum number of users at a particular time). The software vendor usually sets the guidelines on the type of software license. Monitoring is facilitated by the vendor or by the user. When software usage is measured by the user, the vendor can require a user to supply detailed reports or can rely on the user's pledge. Enterprises can avoid liability and save money by accurately measuring the number of users.

conditioning

A procedure that restricts transmission impairments of a circuit to certain specified limits.

conductor

1. Any equipment, such as a wire or cable, that can carry an electric current.
2. One wire of a multiwire cable.

configuration management (see CM)

connection

The establishment of a communication path for the transfer of information as per agreed conventions.

connection admission control (see CAC)

connectionless

The interconnection model in which communication takes place without first establishing a connection.

Connectionless Broadband Data Service (see CBDS)

Connectionless Network Service (see CLNS)

connectionless service

In a connectionless service, no fixed path is set up between sender and recipient. Every unit of data that is exchanged is self-contained in that it contains all the necessary control and address information to ensure correct delivery, e.g., packet switching.

Connection Mode Network Service (see CMNS)

Connection-Oriented Network Protocol (see CONP)

connection-oriented service

A service in which a connection (real or virtual) is set up and maintained for the duration of the communication. See circuit switching.

connect time

The amount of time that a circuit, typically in a circuit switched environment, is in use. See holding time.

CONP (Connection-Oriented Network Protocol)

An Open Systems Interconnection (OSI) protocol enabling connection-oriented operation to upper-layer protocols.

console

The part of a computer used for communicating between the user and the system. A cathode ray tube (CRT) terminal with mouse and keyboard is the most common type.

consolidated security administration

See EUA (enterprise user administration).

consolidated service desk (see CSD)

consolidated user administration

See EUA (enterprise user administration).

constant angular velocity (see CAV)

constant bit rate (see CBR)

constant linear velocity (see CLV)

constraint-based programming (see CBP)

consulting

1. Management consulting — Tier 1 (i.e., executive) assistance regarding setting or execution of corporate business strategy, or Tier 2 (i.e., business) assistance regarding business processes (e.g., re-engineering) or change management.
2. IS consulting — includes systems architecture design or development and IS organizational planning.
3. Application or technical consulting — includes application project management and development, technology assessment, and product tuning.

consumer-centric bill provider (see CCBP)

contact center

Also called a "customer contact center," this includes all customer contact channels, including telephone, interactive voice response (IVR), speech recognition, e-mail, Web and fax. This is an inbound and outbound service-based environment in which customer service representatives handle all types of contacts regarding sales, customer service and support (CSS), marketing and other functions.

contact center suite

This contains all the components of a call center suite (see separate entry) with additional functions to support non-phone-based inquiry channels such as Web and e-mail communication. This added functionality includes:

- Web contact functions (e.g., Web chat and collaborative browsing)
- E-mail response management system (ERMS) functions
- Unified messaging
- Tools for integration with front- and back-office applications, or with applications that support customer relationship management (CRM) strategies

In this definition, the CRM applications that support marketing, sales, customer service and support, or field service and dispatch are not considered part of the suite. However, contact center suite functionality is often included with CRM applications and sold as a bundled CRM suite.

(See call center, contact center, ERMS and CRM.)

contact chips

Chips used in smart cards that are inserted into a reader to conduct transactions or pass information from the card to the reader. Contactless chips can perform a function by being passed near a reader. Contactless chips are often preferred for use in applications where speed is essential.

contact database

A database containing names, addresses and other information on sales contacts , used for contact tracking and management purposes.

contactless card

A card that uses radio frequencies and needs no physical contact to be read by readers or terminals. Contactless cards are simply waved near the reader to record transactions or to identify the user. Systems are either passive, with the readers generating the frequency, or active, in which the card activates the reader.

content

in commercial publishing, content refers to individual documents that can be graphic, textual or illustrative in nature, or amalgamations of documents that can be combined into individual articles or complete publications. On the Internet, the term refers to the content of Web sites.

content aggregation

The presentation of content from multiple sources at a single Web location for review by the customer.

content-based retrieval (see CBR)**content filtering**

See "filtering."

content management (see CM)

content provider

An enterprise with information-based (i.e., content) products; it also includes services to access and manage the content.

contention

A method of line control in which the terminals request to transmit. If the channel in question is free, transmission proceeds; if it is not free, the terminal must wait until it becomes free.

context management engine

A core enabling technology for effective portals, this application collects, analyzes and distributes personalization and customization information.

context

Refers to meanings that are clear to the sender or receiver (e.g., application or person), either because they are stated elsewhere in the message or because they have been predefined (e.g., the number "30" means "30 pounds of flour" in one message and "30 cases of orange soda" in another message). Sender and receiver may have different interpretations of meaning (i.e., different context). For example, "customer" could be the party that pays the freight bill in a billing application, but the receiver of the item (which might not be the payer) in a warehousing application.

contingent workforce

An on-demand workforce staffing strategy using contract workers, agency staffing, independent contractors and people who will work directly for an enterprise on an on-call basis.

continuity check

In common-channel signaling, a test performed to check that a path exists for speech or data transmission.

Continuous Acquisition and Life Cycle Support (see CALS)

continuous improvement

A manufacturing methodology used to improve overall quality by continuously increasing precision in parts specification.

continuous moves

Normally used in conjunction with private truck fleets in process industries to keep the trucks moving with different loads and driver crews on regular routes with just-in-time materials. This concept is being adopted by public fleets to utilize their assets more effectively.

continuous operations

Those characteristics of a data-processing system that reduce or eliminate the need for planned downtime, such as scheduled maintenance. One element of 24-hour-a-day, seven-day-a-week operation.

continuous process improvement (see CPI)

continuous production

A production system in which the productive units are organized and sequenced according to the steps to produce the product. The routing of the jobs is fixed, and setups are seldom changed.

continuous quality improvement (see CQI)

Continuous Variable Slope Delta Modulation (see CVSD)

Contrat d'Enterprise Maintenance et Service (see CEMS)

control character

A character inserted into a data stream to signal the receiving station to perform a function or to identify the structure of the message. Newer protocols are moving away from character-oriented control procedures toward bit-oriented control procedures.

control code (see CC)

controlled access unit (see CAU)

controlled authoring

Methods for limiting the vocabulary and the complexity of text. Controlled authoring is usually used to help with machine translation, but can also be used any time that writers must be compelled to use simple sentences and a limited vocabulary. It is impractical and counterproductive to impose controlled-authoring methods on most business users. It is most widely used for producing complex technical documents.

controlled medical vocabulary (see CMV)

Control Objectives for Information and Related Technology (see COBIT)

Control Program for Microcomputers (see CP/M)

converged network services (see CNS)

Conversational Monitor System (see CMS)

cookie

A file placed on a computer's hard drive by a Web site that the computer user has visited. A code in this file uniquely identifies, or "registers," that user and can be accessed for a number of marketing and site-tracking purposes.

COOL (C++ Object Oriented Language)

A Microsoft application development (AD) tool initiative and language; a predecessor to C# (see separate entry).

cooperative processing

The splitting of an application into tasks performed on separate computers. Physical connectivity can occur via a direct channel connection, a local-area network (LAN), a peer-to-peer communication link or a master/slave link. The application software can exist in a distributed processing environment, but this is not a requirement.

coordination mechanics

A term coined by Coordination Technology's founder, Anatole Holt. It generally refers to a class of workflow that is heuristic in nature; i.e., a higher form of workflow concentrating on human behavior.

COPICS (Communications Oriented Production Information and Control System)

A mainframe material requirements planning product from IBM.

COPPA (Children's Online Privacy Protection Act)

Copper Distributed Data Interface (see CDDI)

COPS (Common Open Policy Service)

CORBA (Common Object Request Broker Architecture)

An Object Management Group (OMG) interoperability standard for object-oriented applications communicating over heterogeneous networks.

Cordless Telephone version 2 (see CT2)

Cordless Telephone version 3 (see CT3)

CORE (COMPARE Operational Readiness Evaluation)

Gartner's CORE risk assessment and reporting steps are used to define business operational risks, to report risks to management, investors, regulators and customers in a consistent form, and to determine when contingency and recovery strategies are required.

core banking systems

The back-end data processing applications for processing all transactions that have occurred during the day and posting updated data on account balances to the mainframe. Core systems typically include deposit account and CD account processing, loan and credit processing, interfaces to the general ledger and reporting tools.

Corporate License Agreement (see CLA)

corporate portal (see enterprise portal)

corporate property management software (see CPMS)

corporate trade exchange (see CTX)

corporate trade payment (see CTP)

correspondence-tracking application (CTA)

COS (class of service)

An Internet service provider (ISP) offering that prioritizes which traffic is delivered before other traffic. With COS, when an ISP's network is not congested, all traffic is treated equally. When the network is congested, however, traffic that has been

designated as a higher priority will be delivered first, while lower priority traffic will be held in a queue until the higher-class traffic has been transmitted.

COSE (Common Open Software Environment)

A now-defunct vendor consortium, which was formed to promote interoperability and portability across Unix platforms. The group's first project was the Common Desktop Environment (CDE) specification

cost per click (see CPC)

cost per thousand (see CPM)

COTS (commercial off-the-shelf)

Descriptive term for software that can be purchased from an external supplier, as opposed to that which is developed within the enterprise.

country code

In direct distance dialing, a code characterizing a particular country. Codes corresponding to the world numbering plan start with a single digit that identifies a particular geographical area. This can be followed by one or two extra digits.

course management system (see CMS)

courseware

Technology-enabled content for an education or training course. Courseware technology includes the content and functionality required to view and navigate a learning session. Related functionality, such as course registration and student or course tracking, is not included in courseware. These related functions are provided by courseware administration technology, such as a learning management system (see LMS).

CP (central processor)

CPA (Canadian Payment Association)

CPA (comprehensive production architecture)

Enables simultaneous and cost-effective production and archiving in print and Internet formats by integrating five key processes: document creation and retrieval, assembly and formatting, internal archiving, prepress and printing and Internet publishing.

CPAF (Cost Plus Award Fee)

CPBX (cordless PBX)

CPC (collaborative product commerce)

An e-business strategy for exploiting new Web-based commerce opportunities across product development and product life cycle processes. CPC opportunities include both inbound (business-to-business) and outbound (business-to-consumer) commerce such as collaborative product development, customer driven design, collaborative product and component sourcing, manufacturing/supply-chain collaboration, and product maintenance self-service portals.

CPC (cost per click)

An online advertising metric which refers to advertising spending divided by the number of recipients who click on the message (e.g., a banner ad on a Web site or a link embedded in e-mail message).

CPE (customer premises equipment)

Any telephone apparatus — including telephone handsets, private branch exchange (PBX) switching equipment, key and hybrid telephone systems, and add-on devices — that is physically located on a customer's property, as opposed to being housed in the telephone company's central office or elsewhere in the network.

CPEMH (computer-aided patient-entered medical history)

A system that presents a series of questions, thereby collecting data that traditionally would require a receptionist, a "traditional clipboard," a nurse and a doctor. These systems use knowledge management and branching logic to tailor each session to the individual patient. Once the data is collected, it can be parsed, translated and highlighted. The clinician then has access to data prior to seeing the patient and can better direct the encounter.

CPEX (customer profile exchange)

CPFR (collaborative planning forecasting and replenishment)

A collection of business practices that leverage the Internet and electronic data interchange (EDI) to radically reduce inventories and expenses while improving customer service. CPFR is a cross between continuous replenishment and vendor-managed inventory.

CPG (consumer packaged goods)

CPI (computer-to-PBX interface)

A gateway providing 24 digital pulse code modulation (PCM) channels at an aggregate rate of 1.544 megabits per second (Mbps). Developed by Nortel Networks (formerly Northern Telecom). See DMI.

CPI (continuous process improvement)

A methodology for improving production, driven by formal metrics and measurement programs, including methodology upgrades.

CPI-C (Common Programming Interface for Communications)

A superset of IBM communications technologies that contains bits of Advanced Program-to-Program Communications/Virtual Machine (APPC/VM), Transparent Services Access Facility (TSAF) and Server/Requester Programming Interface (SRPI). It provides a high-level interface to APPC.

CPM (cent per minute)

CPM (cost per thousand)

A marketing metric which refers to the cost to reach every thousand people with an advertising message. In Internet advertising, CPM is commonly used to measure the cost per thousand "impressions" (i.e., views) of an advertiser's message (as distinguished from the number of users who click on the ad, the basis of the "cost per click" or "CPC" metric). Banner ads on the Web are typically sold on a CPM or CPC basis. (See CPC.)

CPM (critical path method or critical path management)

The series of activities and tasks in a project that do not have built-in slack time. Any task in the critical path that takes longer than expected lengthens the total time of the project.

CP/M (Control Program for Microcomputers)

The first major operating system for personal computers, introduced by Digital Research in the late 1970s. In 1981, it was largely supplanted by PC-DOS from IBM, licensed from Microsoft.

CPMS (corporate property management software)

Software whose basic functionality is to produce basic rent rolls and feed to general ledger; to track simple information about leased or owned property (e.g., renewal dates, term dates, amount of square footage and basic breakdown of divisions or departments in space for space allocation); and to capture data regarding parties to agreement and basic contract information.

CPP (commercial parallel processing)

The use of parallel-processing systems for complex commercial applications.

CPR (computer-based patient record)

A system that contains electronically maintained information about an individual's health status and care. It focuses on tasks directly related to patient care, unlike other healthcare information systems that support providers' and payers' operational processes (which may, however, serve as source or feeder systems for the CPR). The CPR completely replaces the paper medical chart and thus must meet all clinical, legal and administrative requirements.

CPRI (Computer-Based Patient Record Institute)

cps (characters per second)

CPS (certification practice statement)

A document defining all the operational practices that will be used to maintain the required level of public-key infrastructure (PKI) security. To prove that issued certificates are valid, an enterprise must demonstrate (usually through an audit) adherence to its CPS. The Internet Engineering Task Force's (IETF's) request for comment (RFC) 2527 contains draft guidelines for the format and content of a CPS.

CPT (Current Procedural Terminology)

A widely used controlled medical vocabulary authored by the American Medical Association, this system describes medical and surgical procedures in a hierarchical format with six major sections and more than 7,300 codes. A series of two-digit modifiers are used to increase specificity, allowing the reporting of a procedure under specific circumstances.

CPT-4 (Current Procedural Terminology, 4th Edition)

CPU (central processing unit)

The component of a computer system that controls the interpretation and execution of instructions. The CPU of a PC consists of a single microprocessor, while the CPU of a more powerful mainframe consists of multiple processing devices, and in some cases, hundreds of them. The term "processor" is often used to refer to a CPU.

CQI (continuous quality improvement)

A methodology for to continuous improvement of the quality of an enterprise's products, services or internal processes.

CR (computed radiography)

CRC (camera-ready copy)

CRC (cyclic redundancy check)

An error detection technique using a polynomial to generate a series of two 8-bit block check characters that represent the entire block of data. These block check characters are incorporated into the transmission frame and then checked at the receiving end.

create, retrieve, update, delete (see CRUD)

CREDIT (Compliance Rating for EMU-Dependent IT Systems)

critical-path method or management (see CPM)

critical-path scheduling

A project-planning and monitoring system used to check progress toward the completion of a project by scheduling events, activities, milestones, etc.

critical ratio

A dispatching rule that calculates a priority index number by dividing the time to due date remaining by the expected elapsed time to finish the job. Typically ratios of less than 1.0 are behind, ratios greater than 1.0 are ahead, and a ratio of 1.0 is on schedule.

critical success factor (see CSF)

CRL (certificate revocation list)

A "hot lists" that identifies certificates that have been withdrawn, canceled or compromised or that should not be trusted because of other identified reasons. CRLs should be replicated to all subscribing servers to a specific root certification authority.

CRM (customer relationship management)

A business strategy designed to optimize profitability, revenue and customer satisfaction by organizing the enterprise around customer segments, foster customer-centric behavior and implementing customer-centric processes. The application domains of CRM include technology-enabled selling (TES), customer service and support (CSS) and technology-enabled marketing (TEM).

CRM analytics

A set of analyses that support customer relationship management (CRM) on both an individual-customer and an aggregate level, including the real-time monitoring of day-to-day operations. A significant aspect of performing true CRM analytics (as opposed to product or channel analytics) is the capability to integrate data and analyses across various distribution channels and business units, creating a holistic understanding of relationships.

CRMS (Clinical Resource Management System)

A suite of six integrated decision support products from HBO & Company (HBOC) Payor Solutions Group that provides an information infrastructure for health plan medical management reporting and analysis.

CRMS (Customer Relationship Management Services)

An IBM business unit.

cross-assembler program

Symbolically coded input running on one computer that produces machine language output for another computer. See cross-compiler.

crossbar switch

A switch having multiple vertical and horizontal paths and a means for interconnecting any one of the vertical paths within any one of the horizontal paths.

cross-certification

A process by which two enterprises are able to recognize and trust each other's digital certificates.

cross-compiler

A program that translates instructions from a high-level language on one computer to the machine language of another computer — the one on which the program is to be run.

cross-docking

The planning of warehouse "put away" assignments so that inventory can be moved from one shipment to another on a dock without movement to a rack or warehouse location. Although this type of inventory movement may violate lot and code date movement parameters, cross-dock planning is used frequently to minimize labor costs and handling in warehouses and distribution centers.

cross-media publishing (see multichannel publishing)

cross-modulation

Interference caused by two or more carriers interacting in a transmission system.

crossover cable

A cable with wires that cross over so the terminating ends of the cable have opposite wire assignments. See straight-through cable.

crosspoint

1. A switching element that can be mechanical or electronic.
2. A two-state semiconductor switching device having a low transmission system impedance in one state and a very high one in the other.

cross-posting

The act of posting an Internet message to multiple topic groups at the same time.

cross section

Signal transmission capacity of a transmission system, usually measured in the number of two-way voice channels.

Cross System Product (see CSP)

crosstalk

Interference or an unwanted signal from one transmission circuit detected on another, usually parallel, circuit.

crosstalk attenuation

The extent to which a communications system resists crosstalk.

CRP (capacity requirements planning)

The process of specifying the level of resources (facilities, equipment and labor force size) that best supports the enterprise's competitive strategy for production.

CRS (call reporting system)

CRS (Customer Relationship Solutions)

Former name of the IBM business unit now known as Customer Relationship Management Services.

CRT (cathode ray tube)

CRTC (Canadian Radio-Television and Telecommunications Commission)

CRU (customer replaceable unit)

CRUD (create, retrieve, update, delete)

Guidelines for defining how different people or communities within an organization deal with data elements owned by the organization.

CS (computer science)

C&S (calendar and scheduling)

C/S (see client/server)

CS 1 (Capability Set 1)

CSA (Calendar and Scheduling API)

CSA (client services applications)

CSA (consolidated security administration)

See EUA (enterprise user administration).

CSD (consolidated service desk)

The hub where the needs of support groups, distributors, suppliers and customers are consolidated, and where network and systems management (NSM) tools are integrated. The CSD has become the integration point for multiple management disciplines, the single point of contact for providing multiple IS services to the end users, and the source of automation for multiple workflow processes.

C-SET

A chip card electronic-payment implementation supported by French consortium Europay. It is not compliant with the international Secure Electronic Transaction (SET) standard, so it requires software at the merchant site to convert SET transactions to C-SET, allowing for interoperability between the two standards but adding cost and time to the transaction itself. Europay France, formed in January of 1996, includes: Credit Mutuel, Credit Agricole, Credit Industrial et Commercial, Banques Populaires, the Post, Europay France and Europay International.

CSF (critical success factor)

A methodology, management tool or design technique that enables the effective development and deployment of a project or process.

CSI (Computer Security Institute)

C/SI (Client/Server Interface)

An interface published by Sybase and delivered with release 4.0 of its SQL Server product. Open Client, which resides on the client, is an application programming interface (API) that allows applications or third-party products to access SQL Server or Open Server. Open Server is a set of protocols that allows communications with other servers, and can provide gateways to critical data that is not managed by the native SQL server.

CSLIP (Compressed SLIP)

Extension of Serial Link Internet Protocol (SLIP) that allows just the header information to be sent on a SLIP connection.

CSM (component supplier management)

CSM was originally covered as an adjunct product data management (PDM) technology. CSM centers around the desire to increase design reuse of proprietary and standard part components and to lower the cost of procuring components by improving the ability to manage more effectively supplier issues. CSM systems contain four major elements, part classification and retrieval, component libraries, Web component cataloging, and component/supplier process management.

CSMA/CD (Carrier Sense Multiple Access with Collision Detection)

A local-area network (LAN) access technique in which multiple stations connected to the same channel can sense transmission activity on that channel and defer the initiation of transmission while the channel is active. Sometimes called contention access.

CSO (cable service operator)

CSP (cable service provider)

CSP (commerce service provider)

A provider specializing in Web-enabled e-commerce services, as well as offering specific software or outsourcing support for these services. CSPs provide payment processing bundled with front-end commerce functions, including hosted catalogs, site wallets and shopping.

CSP (Cross System Product)

An IBM fourth-generation language (4GL).

CSR (continuous speech recognition)

CSR (customer service representative)

An employee who communicates with customers in a call or contact center. See "call center," "contact center" and "CSS (customer service and support)."

CSRC (Comparex Synchronous Remote Copy)

CSS (cascading style sheets)

CSS (customer service and support)

Once known as the "complaint department," CSS is responsible for retaining and extending customer relationships once a product or service is sold. Due to the increasing complexity of customer interactions, customer service organizations need a complex technological infrastructure that is flexible, extensible and scalable and that integrates front-office applications with back-end processes and data. The components of CSS include

- Call management — The core functionality of CSS applications. This component is used to log all incoming telephone calls and transactions and to manage the transaction from initiation through closure.
- Internet-based customer service suites — Also known as e-service suites, these applications and tools empower customers, partners and prospects for self-service and interactions with the enterprise via the Web, an intranet or an extranet.
- Field service and dispatch (FS/D) systems — FS/D has become an essential element of complete CSS suites and is an important element of customer relationship management (CRM). FS/D software is evolving from solely back-office functionality to an enterprise system that tightly couples the back office with the front-office servicing systems.
- Contact centers — Traditional call centers handle voice-only customer contact, whereas contact centers include all types of channels of customer contact, including voice (e.g., telephone, IVR, speech recognition and voice verification), the Web, fax, video kiosks and e-mail. This is an inbound and outbound service-based environment in which agents handle all types of contacts regarding sales, customer service, marketing, telemarketing, collections and other functions.

CSTA (Computer Supported Telephony Architecture)

A European Computer Manufacturers Association standard for linking computers to telephone systems.

CSU (channel service unit)

A device found on digital links that transfers data faster than a modem (in a range from 56 kilobits per second to 1.5 megabits per second) but does not permit dial-up functions. It also performs certain line-conditioning and equalization functions, and responds to loop-back commands sent from a central office. A CSU is the link between digital lines from the central office and devices such as channel banks or data communications devices.

CT (computed tomography)

CT2 (Cordless Telephone version 2)

A standard for digital mobile telephony, using 4 megahertz (MHz) of channel frequency between 864.1 MHz and 868.1 MHz. CAI (Common Air Interface) enables CT2 manufacturers' handsets to communicate with other manufacturers' base stations.

CT3 (Cordless Telephone version 3)

A technology for digital cordless telephony developed by Ericsson. Many of the principles of CT3 are being incorporated into Digital European Cordless Telephone (DECT).

CTA (correspondence-tracking application)

A blend of integrated document and output management (IDOM) technologies that facilitates high-volume document routing and response, and a strong example of an overlooked document-enabled vertical application (DEVA).

CTD (cell transfer delay)

One of three negotiated quality of service (QOS) parameters defined by the Asynchronous Transfer Mode (ATM) Forum.

CTD (Centura Team Developer)

CTG (Computer Task Group)

CTI (computer-telephony integration)

The intelligent linking of computers with switches, enabling coordinated voice and data transfers to the desktop.

CTLIB (Client Library)

Sybase's application programming interface (API) set required for users to exploit the System 10 version of its SQL Server database management system.

CTO (chief technology officer)

The CTO has overall responsibility for managing the physical and personnel technology infrastructure including technology deployment, network and systems management, integration testing, and developing technical operations personnel. CTOs also manage client relations to ensure that service objective expectations are developed and managed in the operations areas.

CTP (capable-to-promise)

A system that allows an enterprise to commit orders against available capacity, as well as inventory. These systems are evolving to include multiple sites, as well as the entire distribution network.

CTP (corporate trade payment)

An electronic-commerce standard format that was established in the early 1980s using bank-developed transaction sets. It was retired in 1996.

CTR (click-through rate)

The number of times a Web page advertisement is clicked, compared to the number of times it is displayed. Royalties are often based on CTR. This term is also used in reference a Web site's ability to persuade a visitor to "click through" to another site.

CTS (Common Type System)

A component of Microsoft's Common Language Infrastructure (CLI), CTS defines the different programming types and operations supported by Common Language Runtime (CLR), the .NET runtime engine. See CLI and .NET.

CTS (Component Transaction Server)

CTX (clear to send)

A control circuit that indicates to the data terminal equipment that data can or cannot be transmitted.

CTX (corporate trade exchange)

An electronic-commerce standard format that allows for inclusion of 9,999 "addenda" records in addition to the primary financial records (i.e., amount being moved, bank routing number and checking-account number).

CU (compliance unit)

A logical partition of applications and data to be upgraded together.

CUA (consolidated user administration)

See EUA (enterprise user administration).

CUG (closed user group)

A set-up that restricts access to and from one or more terminals to other members of the user group (found on packet switched systems, e-mail, etc.).

cumulative lead time

The longest length of time involved to accomplish the activity in question. For any item planned through material requirements planning, it is found by reviewing each bill-of-material path below the item, and whichever path adds up to the greatest number defines cumulative material lead time.

CUOD (Capacity Upgrade On Demand)

current loop

A transmission technique that recognizes current flows rather than voltage levels. It was traditionally used in teletypewriter networks incorporating batteries as the transmission power source.

Current Procedural Terminology (see CPT)

custom configuration

A computer system built to order (i.e., to the customer's specifications).

customer contact center

See contact center.

Customer Information Control System (see CICS)

Customer Information Control System/System Manager for AIX (see CICS/SM)

customer information file (see CIF)

customer information management (see CIM)

customer information management and application (see CIMA)

customer premises equipment (see CPE)

customer relationship management (see CRM)

customer service and support (see CSS)

customer service representative (see CSR)

customized network management

Tools to allow assignment of levels of network management functions and capabilities to selected nodes throughout the network. With this, the degree of centralization vs. decentralization of network management can be varied depending on the environment.

custom key set

Specialized multibutton telephones designed expressly for a particular private branch exchange (PBX). Unlike the locking buttons on normal key telephones, the buttons on a custom key set are used to communicate with the system and are typically nonlocking buttons. Custom key set buttons can be arranged to activate specific features such as speed dialing and executive override, as well as to select lines.

Custom Local Area Signaling Services (see CLASS)

CVC2 (Card Verification Code)

MasterCard's card verification method.

CVD (chemical vapor deposition)

CVO (Credentialing Verification Organization)

CVP (Content Vectoring Protocol)

CVSD (Continuous Variable Slope Delta Modulation)

A speech-encoding and digitizing technique that uses one bit to describe the change in the slope of the curve between two samples, rather than the absolute change between the samples. Sampling is usually done at 32 kilobits per second (Kbps).

CVV2 (Card Verification Value)

Visa's card verification method.

CWF (Common Working File)

CXI (Common X Interface)

cXML (Commerce XML)

An XML-based specification for managing inventory control, purchase orders, purchase documentation and bidding procedures in Ariba marketplaces.

cyber incident

Any real or perceived event that could be considered a threat, encompasses some form of technology and has the potential to cause significant business disruption at a level that impacts the enterprise's infrastructure, customers or reputation. See CIRT.

cyber incident response plan (see CIRP)

cyber incident response team (see CIRT)

cyber library

An electronic version of a physical library that is implemented on behalf of workers for information self-service.

cyberbrochure

Slang term for a corporate Web site that offers nothing but marketing and public-relations information.

cyclic redundancy check (see CRC)

D

DA (data administration)

See DBA (database administration).

DA (data administrator)

See DBA (database administrator).

D/A (digital-to-analog)

The term used to describe a type of converter used to bridge digital and analog circuitry. D/A converters can either be stand-alone microcomponents, or included in the functionality embedded in advanced integrated circuits.

DAA (data access arrangement)

An interface device once used to interconnect customer-owned data terminal equipment (DTE) to the telephone network. It is now generally integrated into such equipment.

DAA (distributed applications architecture)

An architecture in which the computational and data-processing requirements of an application are performed by two or more computer systems. DAAs include client/server, peer-to-peer and other cooperative-processing approaches.

DAC (digital-to-analog converter)

Component used to bridge the digital and analog circuitry contained in certain advanced integrated circuits. Also known as a "D/A converter."

DAC (Design Automation Conference)

Annual electronics industry conference devoted to electronic design automation (EDA) methodologies and tools.

DAC (dual-attached concentrator)

In Fiber Distributed Data Interface (FDDI), a concentrator that can be attached to both rings and can also be dual-homed from the master ports of other FDDI concentrators.

DACS (digital access and cross-connect system)

An AT&T term for a digital cross-connect system (see DCS).

DAE (Disk Array Enclosure)

A type of disk array module offered in EMC's CLARiiON storage systems.

DAE (Distributed Automation Edition)

IBM software for writing factory-floor applications independent of the distributed data, LAN protocol or terminal type.

daemon

A Unix process initiated during system boot and activated automatically to perform a particular task.

daisy-chaining

The connection of multiple devices in a serial fashion. An advantage of daisy-chaining is savings in transmission facilities. A disadvantage is that if a device malfunctions, all of the devices daisy-chained behind it are disabled.

DAL (Data Access Language)

From Apple Computer, a database interface — itself a superset of Structured Query Language (SQL) — that enables access to Macintoshes or non-Apple computers.

DAMA (Demand Assigned Multiple Access)

Allocation of communication satellite time to earth stations as the need arises.

DAMOS (Drug Application Methodology with Optical Storage)

A European registered pharmacy standard.

DAO (Data Access Objects)

The programming interface for the Microsoft Jet database engine; a precursor to ActiveX Data Objects (see ADO).

DAP (Directory Access Protocol)

A protocol for working among X.500 Directory Service Agents.

DAP (Distributed Application Platform)

An application framework introduced in 1997 by Visigenic (later acquired by Borland International).

DAPP (data analysis and provider profiling)

DAPP vendors are those that provide healthcare value-added analytic applications to support analysis of administrative data for the purposes of network management, actuarial and underwriting functions, medical management and performance measurement — including Health Plan Employer Data and Information Set (HEDIS) reporting.

dark fiber

Fiber-optic cable deployments that are not yet being used to carry network traffic. (The word “dark” refers to the fact that no light is passing through the optical fibers.)

DARPA (Defense Advanced Research Projects Agency)

The U.S. government department that developed the Transmission Control Protocol/Internet Protocol (TCP/IP) ARPANET protocol architecture.

DAS (direct-attached storage)

A broad category of storage technology that includes:

- Redundant array of independent disks (RAID) storage that is internal and host-attached (both the RAID controller and software technologies and the hard disk drives are housed within the server)

- Host-attached, external RAID storage (the RAID controller and software technologies are housed within the server, and the hard disk drives are housed separately from the server in a high-availability RAID enclosure)
- RAID controller-based storage (a separate enclosure — incorporating both the RAID controller and software technologies and the hard disk drives — is attached directly to the server).

DAS (dual-attached station)

In Fiber Distributed Data Interface (FDDI), a device that is attached to both the primary and secondary rings.

DASD (direct-access storage device)

Generic nomenclature for a storage peripheral that can respond directly to random requests for information; usually denotes a disk drive.

DASS (Digital Access Signaling System)

The original British Telecom (BT) Integrated Services Distributed Network (ISDN) signaling system developed for single-line and multiline integrated digital access to the public network, but used in the BT ISDN pilot service for single-line access only.

DASSII

A message-based signaling system developed by BT and its suppliers for multiline integrated digital access to the public network; an updated version of Digital Access Signaling System (see DASS).

DAT (digital audiotape)

A magnetic tape that stores audio data converted to digital form.

DAT (dynamic address translation)

The change of a logical storage address to an actual storage address.

data

Raw facts and figures that a computer processes into usable information. Data such as cash receipts mean little until processed into information such as an open receivable balance.

data access arrangement (see DAA)

Data Access Language (see DAL)

Data Access Objects (see DAO)

data acquisition

The process of converting a physical phenomenon (such as temperature or pressure) into an electrical signal and measuring that signal to extract information. A transducer converts the physical phenomenon into an electrical signal. The signal is then measured using signal conditioning and data acquisition hardware.

data administration

See DBA (database administration).

data administrator

See DBA (database administrator).

data analysis and provider profiling (see DAPP)

data archive

A static, independent copy of a related set of files for intermediate or long-term storage, usually to satisfy financial-reporting, audit, regulatory and ad hoc retrieval requirements. The primary files can be left intact or deleted when the archive is created.

Database 2 (see DB2)

database administration/administrator (see DBA)

Database/Data Communication (see DB/DC)

Database Library (see DBLIB)

database management system (see DBMS)

database marketing

The process of analyzing database-captured customer, product and purchasing information in order to allocate marketing resources to those activities that have the greatest impact and return on profitable customer relationships.

database partitioning

The practice of separating a database into portions that may reside on more than one disk volume or more than one system.

data center

The department in an enterprise that houses and maintains back-end information technology (IT) systems and data stores — its mainframes, servers and databases. In the days of large, centralized IT operations, this department and all the systems resided in one physical place, hence the name data center. With today's more distributed computing methods, single data center sites are still common, but are becoming less so. The term continues to be used to refer to the department that has responsibility for these systems, no matter how dispersed they are.

Datacenter Server (see DCS)

data circuit

A communications facility that enables transmission of data in either direction, in either analog or digital form.

data communications

1. The transmission, reception and validation of data.
2. Data transfer between a data source and a data destination via one or more data links, according to appropriate protocols, by means of an electromagnetic transmission system.

data communications equipment (see DCE)

data compression

A technique that saves storage space by eliminating gaps, empty fields, redundancies or unnecessary information to shorten the length of records or blocks.

data concentration

Collection of data at an intermediate point from several low- and medium-speed lines for retransmission across high-speed lines.

data conversion

The process of changing data from one form of representation to another.

Data Country Code (see DCC)

data definition language (see DDL)

data dictionary

A repository of information about data that supplies the meaning of the data, its relationship to other data, its origin, its usage and its format. The dictionary assists management, database administrators, systems analysts and application programmers in effectively planning, controlling and evaluating the collection, storage and use of data. A data dictionary manages data categories such as aliases, data elements, data records, data structure, data stores, data models, data flows, data relationships, processes, functions, dynamics, size, frequency, resource consumption and other user-defined attributes.

Data Encryption Standard (see DES)

Data Exchange Interface (see DXI)

Data Extract (see DXT)

Data Facility Data Set Services (see DFDSS)

Data Facility Hierarchical Storage Manager (see DFHSM)

Data Facility Product (see DFP)

Data Facility Sort (see DFSORT)

Data Facility Storage Management Subsystem (see DFSMS)

datagram

A packet of data that can be transmitted over a packet-switched system in a connectionless mode. This is a type of "fire and forget" mechanism that has no concept of a transaction or acknowledgment.

Datagram Delivery Protocol (see DDP)

data integrity

The assurance that data has not been lost or corrupted in the course of being stored or communicated. Aside from hackers or deliberate destruction, the main threats to data integrity include network transmission errors such as dropped packets, storage hardware errors such as bad disk sectors, and application errors such as database

replication errors. Ways of ensuring data integrity include redundancy and parity checking for data storage and transmission, and two-phase commit and template filtering for data entry and database management.

Data Interchange Format (see DIF)

Data Interpretation System (see DIS)

Data Language/1 (see DL/1)

data line interface (see DLI)

data link

Any serial data communications transmission path, generally between two adjacent nodes or devices and without any intermediate switching nodes.

data link control

Procedures to ensure that both the sending and receiving devices agree on synchronization, error detection and recovery methods, and initialization and operation methods for point-to-point or multipoint configurations.

Data Link Control Layer

Layer 2 of the Systems Network Architecture (SNA) architectural model, which is responsible for the transmission of data over a particular physical link. It corresponds approximately to the Data Link Layer of the Open Systems Interconnection (OSI) model.

Data Link Layer

The layer in the Open Systems Interconnection (OSI) model concerned with transmission of data between adjacent network nodes. It is the second layer in the OSI model, between the Physical and the Network Layers.

Data Link Switching (see DLS)

data logging

Recording data about events in the time sequence in which they occur.

data manipulation language (see DML)

data mart A data mart provides access to a limited number of data sources and a data model to aid in decision making for a specific business area or application — in contrast to the enterprisewide, strategic focus of the data warehouse.

Data Migration Interface (see DMI)

data mining

The process of discovering meaningful correlations, patterns and trends by sifting through large amounts of data stored in repositories. Data mining employs pattern recognition technologies, as well as statistical and mathematical techniques.

data network identification code (see DNIC)

Data-Over-Cable Service Interface Specification (see DOCSIS)

data over voice (see DOV)

Data Packet Network (see DPN)

data PBX

A switch that allows a user on an attached circuit to select from other circuits, usually one at a time and on a contention basis, for the purpose of establishing a through connection. A data private branch exchange (PBX) is distinguished from a PBX in that only digital transmission, and not analog, is supported.

data rate

The speed at which a channel carries data, measured in bits per second (bps).

data service unit (see DSU)

data synchronization

A form of embedded middleware that allows applications to update data on two systems so that the data sets are identical. These services can run via a variety of different transports but typically require some application-specific knowledge of the context and notion of the data being synchronized.

data terminal equipment (see DTE)

Data Universal Numbering System (see DUNS)

data VPN (data virtual private network)

A virtual private network (VPN) devoted to data, rather than voice, communications. See VPN.

data warehouse

A storage architecture designed to hold data extracted from transaction systems, operational data stores and external sources. The warehouse then combines that data in an aggregate, summary form suitable for enterprisewide data analysis and reporting for predefined business needs. The five components of a data warehouse are production data sources, data extraction and conversion, the data warehouse database management system, data warehouse administration and business intelligence (BI) tools.

date windowing

A software technique used to convert a two-digit year value to a four-digit one, based predefined assumptions of the correct year implied based on the size of the value. For example, "87" might be interpreted as "1987," while "08" may be interpreted as "2008."

dB (decibel)

The unit for measuring the relative strength of a signal parameter such as power or voltage. The number of decibels is 10 times the logarithm (base 10) of the ratio of the power of two signals or the ratio of the power of one signal to a reference level.

DB (database)

An electronic filing system organized by fields, records and files. A field is a single piece of information, a record is a set of fields and a file is a collection of records.

DB2 (Database 2)

IBM's relational database management system (RDBMS) offering, originally built for Multiple Virtual Storage (MVS) systems. It uses Structured Query Language (SQL) as its data manipulation and definition language. IBM has released versions of DB2 for several other operating systems, such as Windows and Unix.

DBA (database administration)

The role of a database administrator, i.e., managing the data contained in a database. Also known as "data administration." See DBA (database administrator).

DBA (database administrator)

The person responsible for managing data, namely data set placement, database performance, and data recovery and integrity at a physical level. Also known as a "data administrator."

DBCS (double-byte character set)

A type of character set large enough to support languages with high numbers of characters (notably Asian languages such as Kanji).

DB/DC (Database/Data Communication)

This acronym refers to IBM's collection of transaction managers and database managers, e.g., Customer Information Control System (CICS), Information Management System (IMS), Database 2 (DB2) and SQL/DS.

DBLIB (Database Library)

The native application programming interface (API) for Sybase's SQL Server.

DBM (see database marketing)

DBMS (database management system)

A system that enables end users or application programmers to share data. It provides a systematic method of creating, updating, retrieving and storing information in a database. DBMSs are generally also responsible for data integrity, data access control, and automated rollback, restart and recovery.

DBS (direct broadcast satellite)

A wireless technology for broadcasting by transmitting a compressed high-speed video signal via satellite.

DC (direct current)

Type of power supply in which the electric current flows in only one direction, as distinguished from an alternating current (see AC).

DC (see distributed computing)

DCA (distributed component architecture)

An architecture in which the various logical functions of an integration broker (such as message capture, transformation, routing and delivery) are implemented as separate executables that can be placed on different machines.

DCC (Data Country Code)

One of two formats from the Asynchronous Transfer Mode (ATM) Forum for address usage. Also see ICD.

DCD (Document Content Description)

A World Wide Web Consortium (W3C) specification for expressing Extensible Markup Language (XML) data types and value constraints.

DCE (data communications equipment)

Equipment that performs the functions required to connect data terminal equipment (DTE) to the data circuit. In a communications link, it is either part of the network, an access point to the network, a network node, or equipment at which a network circuit terminates. In the case of an RS-232-C connection, modems are usually regarded as DCE while user devices are considered to be data terminal equipment (DTE). In an ITU X.25 connection, the network access and packet-switching node is viewed as the DCE.

DCE (Distributed Computing Environment)

A vendor-neutral networking and distributed computing environment created by the Open Software Foundation (now The Open Group) to support distributed applications, by integrating the appropriate technology or combination of technologies into a single environment while addressing interoperability, standards and security. DCE integrates remote procedure calls, presentation services, a naming directory, security, threads (sequential flows of control similar to tasks), time services (to synchronize clocks) and a distributed file system.

DCE (distributed computing environment)

Generic term for a computing environment in which data and applications are distributed among disparate computers or systems, but are connected and integrated such that they function as a single environment.

DCF (Document Composition Facility)

Program used in IBM's Professional Office System (PROFS) for creating text. Also known as SCRIPT/VS.

DCM (document component management)

DCM applies the principles of managing entire documents to managing parts of documents. Every significant unit of information is identifiable as a separate entity and is subject to the same core library services (check in/check out, version control, security) that integrated document management (IDM) provides on complete documents or sections of documents. Structural markup — i.e., using Standard Generalized Markup Language (SGML) or Extensible Markup Language (XML) — within the document identifies the components of information. Unlike IDM, information about the content — metadata or attributes — is captured on the inside and embedded in the content rather than stored on the outside as a separate index of information. This is essential when pointing to information elements required for selective reuse, updating, associating data with related information, storing translations, validity checking and other component-level processes.

DCOM (Distributed Component Object Model)

Microsoft's Component Object Model (COM) distributed via a remote procedure call (RPC).

D conditioning

Conditioning that controls the harmonic distortion and the signal-to-noise ratio of a signal so that they lie within specified limits.

DCS (Datacenter Server)

A Microsoft product focused on the highest end of the Windows 2000 server market. It provides support for up to 32-way systems, as much as 64 gigabytes of physical memory, and four-node high-availability clustering.

DCS (Defense Communications System)

A long-haul voice, data and record communications system maintained by the U.S. Department of Defense.

DCS-1800 (Digital Cellular Service 1800)

A GSM-based standard for digital cellular communications in the 1.8-GHz frequency range, introduced in Europe and Asia in the mid-1990s.

DCS (digital cross-connect system)

A computerized facility allowing DS1 lines (1.544 Mbps) to be remapped electronically at the DS0 level (64 Kbps). This means that DS0 channels can be individually rerouted and reconfigured into different DS1 lines.

DCS (distributed control system)

A form of direct digital control for process automation, distributing specialty-purpose controllers across a common communication network throughout the plant. In DCSs, measurement, control and communications are distributed in function and location. By partitioning and distributing control functions, local controllers throughout the plant remain in control of the process if central control room consoles are lost. Likewise, if one local control station fails, other local controllers continue to operate. DCSs are usually deployed in fault-tolerant modes using redundant system configurations to achieve high measures of system availability.

DCT (Discrete Cosine Transform)

A video compression algorithm similar to Fast Fourier Transform (FFT), but easier to compute.

DDA (Distributed Data Architecture)

Groupe Bull's scheme for data interoperability.

DDBMS (distributed database management system)

A DBMS that enables end users or application programmers to view a collection of physically separate databases as one logical single-system image. The concept that is most fundamental to the DDBMS is location transparency, meaning the user should not be conscious of the actual location of data.

DDC (Display Data Channel)

To simplify installation, the Video Extended Standards Association (VESA) developed the DDC standards to allow "plug and play" connection. The monitor and the system unit communicate to agree on the configuration options that both sides can support.

DDCMP (Digital Data Communications Message Protocol)

A synchronous protocol developed by Digital Equipment (later acquired by Compaq Computer, which merged with Hewlett-Packard in 2002).

DDD (direct distance dialing)

A telephone exchange service that enables telephone users to place long-distance calls without operator assistance.

DDE (Dynamic Data Exchange)

A Microsoft technology for sharing data between two applications or files. When data is used in one file, with DDE it can be automatically updated in the other. Both applications must be in use, and DDE does not work over a network. Object Linking and Embedding (see OLE) is a more sophisticated development.

DDI (Digital Document Interchange)

The format for encoding revisable form text, graphics and image information in Compound Document Architecture (see CDA).

DDL (data definition language)

A language used to describe the data model for a database, i.e., the names and access paths for the data and how they are interrelated. In some software products, the DDL describes the logical, not the physical, data. Other products use it to describe both.

DDM (see distributed data management)

DDN (Defense Data Network)

A U.S. Defense Department network used for switching data-processing systems; part of the Defense Communications System (see DCS).

DDN (Digital Data Network)

A network service established in China in 1994, connecting Beijing, Tianjin, Shanghai and other provincial capitals. Each PTA in China has a department regulating DDN service in its region.

DDNS (Dynamic Domain Name System)

A technology that automatically updates the logical names of dynamically assigned Internet Protocol (IP) addresses in the DNS server of an IP network. Specifications defining a standard approach to DDNS were defined in RFC 2136, introduced in 1997. See DNS (Domain Name System).

DDP (Datagram Delivery Protocol)

AppleTalk network layer protocol.

DDP (distributed data processing)

The use of computer systems or intelligent terminals within an enterprise to perform data processing or storage functions. The devices may or may not be interconnected via transmission facilities.

DDS (digital data service)

A dedicated digital data transmission service that operates over interconnected, point-to-point digital private lines.

DDS (Digital Data Storage)

A format for digital storage drives, originally introduced in 1988 by Hewlett-Packard and Sony.

deadlock

Occurs in a network when two users each try to access a file the other has locked.

debit/credit benchmark

A performance measurement once used for online transaction processing (OLTP) throughput. The debit/credit transaction, sometimes called ET-1, simulates an automated teller machine (ATM) function. It has been replaced by TPC-A and TPC-B, defined by the Transaction Processing Performance Council (TPC).

deblocking

Separating blocked data into individual logical records.

decentralized

The description for an application that is distributed, but originates with the IS organization, has central application development and is centrally supported.

decibel (see dB)

decision support system (see DSS)

DECnet

A family of peer-to-peer, Ethernet-based network products from Compaq (formerly Digital Equipment), now part of Hewlett-Packard.

DEComni

A version of the Manufacturing Message Service protocol (see MMS) from Compaq (formerly Digital Equipment), now part of Hewlett-Packard. Used for control-level device communications.

DECS (Domino Enterprise Connection Services)

Technology that provides real-time integration between Domino applications, from IBM's Lotus subsidiary.

DECT (Digital Enhanced Cordless Telecommunications)

An interface specification for European digital mobile telephony. DECT employs 10 carrier frequencies between 1.88 gigahertz (GHz) and 1.9GHz, and has a transmission speed of 144 kilobits per second. It is typically used for short-range communications and wireless-local-loop applications.

DECwrite

A compound document editor originally developed by Digital Equipment (later acquired by Compaq Computer, now merged with Hewlett-Packard) based on the editor engine from Frame Technologies.

dedicated circuit

A communications line dedicated for private enterprise or personal use.

deep user interface

A design approach for user interaction that generates a new window for each step of a given task, most appropriate if the subtasks represent new environments and applications requiring different user-interface controls.

Defense Advanced Research Projects Agency (see DARPA)

Defense Communications System (see DCS)

Defense Data Network (see DDN)

Defense Message System (see DMS)

delay

In communications, the wait time between two events, such as the time when a signal is sent until the time it is received.

delay announcement

An automatic call distribution (ACD) function that plays a recorded message when calls are delayed and placed in queue.

delay distortion

The change in a signal from the transmitting end to the receiving end resulting from the tendency of some frequency components within a channel to take longer to be propagated than others.

delimiter

In data communications, a character that separates and organizes elements of data.

delta modulation

A method of representing a speech waveform (or other analog signal) in which successive bits represent increments of the waveform. The increment size is not necessarily constant.

Demand Assigned Multiple Access (see DAMA)

demand planning

The process of forecasting and managing the demand for products and services by end users, as well as by intermediate supply chain members.

demand management

The process of matching future demand with (sometimes constrained) projected available supply.

demilitarized zone (see DMZ)

demodulation

The process of retrieving data from a modulated carrier wave; the reverse of modulation.

DEN (Directory-Enabled Networks)

An initiative formed by Microsoft and Cisco to define a directory schema foundation for common network objects as well as for the use of Lightweight Directory Access Protocol (LDAP) as a query protocol. Microsoft and Cisco turned over control of DEN to the Distributed Management Task Force (DMTF), which has incorporated the specification under its Common Information Model.

denial of service (see DoS)

dense wave division multiplexing (see DWDM)

dependent demand

Demand is considered dependent when it is directly related to or derived from the demand for other items or end products. Such demand is thus calculated and does not have to be forecast. A given inventory item may have both dependent and independent demand at any given time.

Dependent Logical Unit Requester (see DLUR)

deployment planning and scheduling

Specifications for deploying available supplies to satisfy demand requirements.

derivation equipment

Equipment used to produce narrowband facilities from a wider band facility. Commonly used on alternate voice/data (AVD) circuits to derive telegraph-grade lines from unused portions of a voice circuit.

DES (Data Encryption Standard)

A security specification developed by IBM in 1977. Still in use today, it is available at no charge from many online bulletin boards and is based on a single key encryption algorithm. If user A wants to send an encrypted file to user B, user A would first encrypt it with a secret key. User B would then decrypt the file using the same key. Recipients must prearrange for possession of the appropriate key for decryption to take place.

Design Automation Conference (see DAC)

design for logistics (see DFL)

design for manufacturability (see DFM)

design synthesis

The set of technological capabilities for modeling, simulating and analyzing product designs. Also known as conceptual design or top-down design.

desktop alternative

A computer system that meets all the criteria for a notebook PC but is designed to replicate the functionality of a desktop. It weighs 6 pounds or more. The screen can be as large as 15 inches or 16 inches with Super Video Graphics Array (SVGA) resolution or higher. Target markets include engineers and end users who want to travel with minimal weight.

Desktop Management Interface (see DMI)

Desktop Management Task Force (see DMTF)

desktop services

Acquisition, maintenance and ongoing management of PC hardware and software assets.

desktop videoconferencing

The ability of two or more geographically separated users to meet via their personal computers. A typical system involves a PC, a monitor-mounted camera, a network connection, and special software that facilitates the exchange of video and audio.

despreading

In spread spectrum technology, the correlation by which the data is rejoined.

destination field

A field in a message header that contains the address of the station to which a message is being directed.

destination point code/origination point code (see DPC/OPC)

Deutsche Institut fur Normung (see DIN)

DEVA (document-enabled vertical application)

A Gartner concept that applies integrated document and output management (IDOM) technologies in specific industries for support of vertical (or sometimes horizontal) processes. Examples of industries and related processes include insurance (claims processing), engineering (technical document management), pharmaceuticals (new drug application), financial services (retirement processing) and cross-industry applications (call-center support).

DFDSS (Data Facility Data Set Services)

An IBM software product that provides copy, dump-restore and direct access storage device (DASD) space management functions. It was later renamed DFSMSdss.

DFHSM (Data Facility Hierarchical Storage Manager)

An IBM product that provides automatic space and availability management within a storage hierarchy. It was later renamed DFSMSHsm.

DFL (design for logistics)

A product design approach that considers logistical issues, such the manner in which the product will be stored or transported.

DFM (design for manufacturability)

An approach for designing products in ways that optimize their manufacture.

DFP (Data Facility Product)

Originally called MVS/DFP, it was the MVS component that contained the access methods and handled catalog management. This facility was later renamed DFSMSdfp.

Dfs (Distributed file system)

Microsoft's implementation of a distributed file system that allows multiple servers to provide file shares under the hierarchy of a common logical share. Multiple servers can sponsor identical file shares to provide redundancy and differing levels of response time.

DFS (Distributed File System)

A Distributed Computing Environment standard that provides users with a common file system across different operating systems. When files are moved, DFS tracks the new location by storing its address in a database. DFS also eases backup difficulties by automating regular backup routines across each cell through a backup server.

DFSMS (Data Facility Storage Management Subsystem)

The conceptual repackaging of IBM's Data Facility family of products and Resource Access Control Facility (RACF). It is intended to simplify the management and use of external storage resources by providing a device-independent means of requesting services by data set.

DFSMSdfp (DFSMS Data Facility Product)

See DFP.

DFSMSdss (DFSMS Data Set Services)

See DFDSS.

DFSMShsm (DFSMS Hierarchical Storage Manager)

See DFHSM.

DFSORT (Data Facility Sort)

An IBM products that provides high-performance rearrangement of data.

DHCP (Dynamic Host Configuration Protocol)

An Internet Engineering Task Force (IETF) Transmission Control Protocol/Internet Protocol (TCP/IP) specification for allocating Internet Protocol (IP) addresses and other configuration information based on network adapter addresses. It enables address pooling and allocation and simplifies TCP/IP installation and administration.

DHTML (Dynamic HTML)

Netscape and Microsoft technology solutions that offer client-side mechanisms for enhancing the capabilities of the Web browser and HTML documents. DHTML is often confused with dynamically generated HTML, but the terms are not synonymous — see HTML.

diagnostic programs

Software routines used to check equipment malfunctions and to pinpoint faulty components.

Dialed Number Identification Service (see DNIS)

dial-up

The process of, or the equipment or facilities involved in, establishing a temporary network connection via the public switched telephone network.

DICOM (Digital Imaging and Communications in Medicine)

A communication standard developed by the American College of Radiology/National Electrical Manufacturers Association and used in digital radiology systems (e.g., computed-radiography, computed-tomography and magnetic resonance imaging systems).

DIF (Data Interchange Format)

A file format developed for VisiCalc, the first electronic spreadsheet. Still used today as a means for transferring files to and from spreadsheets.

DIF (Directory Interoperability Forum)

Coalition formed in 1999 by IBM, Novell, Oracle, Lotus Development and Isocor to promote the use of Lightweight Directory Access Protocol (LDAP) by developers, and to accelerate directory interoperability standards.

differential phase-shift keying (DPSK)

See phase modulation.

Diffie-Hellman

The first widely deployed public-key algorithm. Held primarily by Cylink, it passed into the public domain in September 1997.

DiffServ (Differentiated Services)

A protocol that helps support network quality of service. DiffServ enables data packet fields to carry information about the specific level of service a packet should receive on the network.

digital

A form of representation in which distinct objects, or digits, represent something in the real world, so that counting and other operations can be performed precisely. Data represented digitally can be manipulated to produce a calculation, a sort or some other computation. In digital electronic computers, two electrical states correspond to the 1s and the 0s of binary numbers, which are manipulated by computer programs.

digital access and cross-connect system (see DACS)

Digital Access Signaling System (see DASS)

digital audiotape (see DAT)

Digital Cellular Service 1800 (see DCS-1800)

digital certificate

An encrypted code issued to both buyers and sellers to verify their identities through the exchange of their public keys.

digital cross-connect system (see DCS)

Digital Data Communications Message Protocol (see DDCMP)

Digital Data Network (see DDN)

digital data service (see DDS)

Digital Data Storage (see DDS)

digital dial tone

A Gartner term describing the combination of XML and Internet transport protocols — such as HTTP, SMTP and FTP — to create a ubiquitous capability to exchange structured information. The metaphor relies on the contrast between previous business to business (B2B) message exchanges — such as electronic data exchange (EDI) using X12, EDI for Administration, Commerce and Transportation (EDIFACT)

and various vertical-industry-specific standards — and the unstructured communications media of telephone and fax. The former required considerable investment and the interposition of a private network — or substantial bilateral negotiation — to achieve interoperability, while the latter are ubiquitous and broadly interoperable. With the Internet and ubiquitous software to implement Internet transport protocols, business partners can reduce the cost, delay and risk of implementing B2B messaging.

Internet transport protocols by themselves do not constitute a complete digital-dial-tone solution. For secure application messaging to occur over the Internet, communicating systems must be programmed to additional specifications in key areas such as security, routing and access control. Ubiquity will not be achieved until integration products agree on standards for meeting these requirements.

digital divide

The gap in opportunities experienced by those with limited accessibility to technology, especially the Internet. This includes, but is not limited to, accessibility challenges in the following areas:

- Cultural (e.g., membership of a community that prohibits or restricts access to technology)
- Physical (e.g., having a disability that make it difficult or impossible to use a computer)
- Economic (e.g., being unable to afford a computer)
- Educational (e.g., not knowing how to use a computer)

Digital Document Interchange (see DDI)

Digital Enhanced Cordless Telecommunications(see DECT)

Digital Imaging and Communications in Medicine (see DICOM)

digital linear tape (see DLT)

digital loopback

A technique for testing the digital processing circuitry of a communications device. It can be initiated locally or remotely via a telecommunications circuit; the device being tested will echo back a received test message (after first decoding and then re-encoding it) the results of which are compared with the original message.

digital mock-up (see DMU)

digital modem

A system component that enables communication over digital access facilities with a remotely located system connected to the public network over analog facilities.

Digital Multiplexed Interface (see DMI)

digital network

A network incorporating both digital switching and digital transmission.

Digital Private Network Signaling System (see DPNSS)

digital rights management (see DRM)

digital signal

A discrete or discontinuous signal; one whose various states are discrete intervals apart.

Digital Signal Level 0-4 (see DS-0 through DS-4)

digital signal processor (see DSP)

digital signature

A core function of a public key infrastructure (PKI). A digital signature can prove identity because it is created with the private key portion (which only the key holder should access) of a public/private key pair. Anyone with the sender's widely published public key can decrypt the signature and, by doing so, receive the assurance that the data must have come from the sender (nonrepudiation of the sender) and that the data has not changed (integrity). The data that is encrypted with the private key is not the entire message, but a short, fixed-length block of data that is computed from the message using a so-called "hash" function.

Digital Signature Algorithm (see DSA)

digital simultaneous voice and data (see DSVD)

digital speech interpolation (see DSI)

Digital Standard Relational Interface (see DSRI)

digital subscriber line (see DSL)

digital switching

The process of establishing and maintaining a connection under stored program control where binary-encoded information is routed between an input and an output port.

Digital Table Interchange Format (see DTIF)

digital-to-analog converter (see DAC)

digital versatile disc (see DVD)

digital versatile disc read-only memory (see DVD-ROM)

digitize

To convert or express an analog form in a digital format.

digitizer

A purely graphical input device — i.e., a computer-aided design/computer-aided manufacturing (CAD/CAM) system digitizer used in computer-aided engineering (CAE) for converting locations into storable electronic impulses — with a surface on

which a location or a point is selected and then automatically converted into a digital X-Y coordinate suitable for transmission to a computer.

DIMM (dual in-line memory module)

A memory packaging and form factor that allows for a wider data path to memory, higher memory capacities in systems, and potentially easier upgrading than traditional single in-line memory modules (SIMMs).

DIN (Deutsche Institut fur Normung)

1. German national standards organization.
2. A German standard for cable connectors.

DIN connector

Multipin connector format used in some Macintosh and IBM PC computers and in some network processor panels.

DIP (document image processing)

A technology used to scan, digitize and store documents (e.g., checks or invoices) as images.

DIP (dual in-line package)

A method of packaging electronic components for mounting on printed circuit boards.

direct-access storage device (see DASD)

direct-attached storage (see DAS)

direct broadcast satellite (see DBS)

direct dial

The ability to place calls directly to an extension, bypassing the switchboard operator.

direct distance dialing (see DDD)

direct memory access (see DMA)

directed speech recognition

A system that uses a script-like dialog instead of complete, free-form natural language. For each question asked, there are a limited number of valid responses. With this approach, accuracy rates may go up dramatically on less-expensive hardware.

Directory Access Protocol (see DAP)

Directory-Enabled Networks (see DEN)

Directory Interoperability Forum (see DIF)

Directory Management Option (see DMO)

directory services

Middleware that locates the correct and full network address for a mail addressee from a partial name or address. A directory service provides a naming service and extends the capabilities to include intelligent searching and location of resources in the directory structure.

directory system agent (see DSA)

direct sequence spread spectrum (see DSSS)

DirSync

Microsoft's proposal, announced in March 1999, for a directory synchronization protocol.

"dirty" protocols

Many Internet Protocol (IP) applications assume that direct IP connectivity exists between hosts. In today's Internet or extranets, this is often not true. The problems of limited IP address space have caused many enterprises to use private Request for Comment (RFC) 1918 addresses. These addresses cannot be routed and, for enterprises to connect to the Internet or to communicate in an extranet, address translations or application proxies must be used. For applications that exchange their IP addresses between the client and the server, these "dirty" IP address are not valid when one or both of the end systems exist on an RFC 1918 network. In addition, without using special techniques, applications like File Transfer Protocol (FTP) will not work when an enterprise uses private RFC 1918 addresses.

DirXML (Directory XML)

DIS (Data Interpretation System)

A marketing-oriented decision support system (DSS) product from IBM.

DIS (draft international standard)

DISA (Data Interchange Standards Association)

DISA (Defense Information Systems Agency)

DISA (direct inward system access)

disaster recovery (see DR)

disaster recovery planning (see DRP)

Discrete Cosine Transform (see DCT)

discrete-event simulation

A modeling technique where the traffic is represented as sequences of messages, packets or frames.

discrete manufacturing

The production of a discrete category of goods (e.g., automobiles, aircraft, computers or component assemblies).

discrete multitone (see DMT)

discretionary security controls

An operating-system security rating of C2 or higher based on U.S. Department of Defense trusted computer system evaluation criteria.

discussion database

A database designed specifically for the capture, exchange and storage of ideas (e.g., Lotus Notes).

Disk Array Enclosure (see DAE)

disk drive

A computer device that reads from, and stores information on, a floppy or hard disk.

diskless workstation

A network-attached computer that has no disk drives, depending instead on file servers for all data storage.

disk mirroring

The duplication of disks and controllers so that two access paths exist in case a failure occurs on one of them.

Disk Operating System (see DOS)

DISN (Defense Information Systems Network)

DISOSS (Distributed Office Support System)

DISP (Directory Information Shadowing Protocol)

Display Data Channel (see DDC)

distance learning (see DL)

distance vector protocol

Any routing protocol derived from the Bellman-Ford algorithm, such as DECnet or Routing Information Protocol (RIP). The routers using such protocols create a network map by communicating in a periodic and progressive sequence.

distortion

The unwanted change in a waveform that occurs between points in a transmission system.

distributed applications architecture (see DAA)

Distributed Applications Platform (see DAP)

Distributed Automation Edition (see DAE)

distributed component architecture (see DCA)

Distributed Component Object Model (see DCOM)

distributed computing

A form of computing in which data and applications are distributed among disparate computers or systems, but are connected and integrated by means of network services and interoperability standards such that they function as a single environment. See DCE (distributed computing environment).

Distributed Computing Environment (see DCE)

distributed computing environment (see DCE)

distributed control system (see DCS)

Distributed Data Architecture (see DDA)

distributed database

A database whose objects (tables, views, columns and files) reside on more than one system in a network, and can be accessed or updated from any system in the network.

distributed database management system (see DDBMS)

distributed data management

A form of client/server computing in which some portion of the application data executes on two or more computers.

distributed data processing (see DDP)

Distributed File System (see DFS)

distributed function

A form of client/server computing in which some of the application program logic executes on one computer, possibly with a database, and the rest of the application resides on another computer, possibly along with presentation services.

Distributed interNet Applications (see DNA)

distributed learning (see DL)

distributed lock manager (see DLM)

Distributed Management Environment (see DME)

Distributed Management Task Force (see DMTF)

distributed memory parallel processors (see DMPP)

Distributed Object Management Facility (see DOMF)

distributed output management (see DOM)

distributed parallel processing (see DPP)

distributed presentation

A form of client/server computing in which some of the presentation handling executes on one computer and the rest of the presentation, along with the remainder of the application and the database, executes on another. See client/server.

Distributed Print Architecture (see DPA)

distributed processing

The ability of an application to run on one or more nodes of a multiplatform network. The user need not be aware of the physical location of the data or the application software. The application can operate using cooperative processing, but this condition is not necessary.

Distributed Queue Dual Bus (see DQDB)

Distributed Relational Data Architecture (see DRDA)

distributed request (see DR)

distributed resource management (see DRM)

Distributed System Object Model (see DSOM)

distributed systems

Refers to computer systems in multiple locations throughout an organization working in a cooperative fashion. The system at each location serves the needs of that location but also is able to receive information from, and supply information to, other systems within the network.

Distributed Systems License Option (see DSLO)

distributed systems management (see DSM)

Distributed Systems Network (see DSN)

distributed unit of work (see DUOW)

distribution and logistics

A category of product distribution management and warehousing functions that includes physically moving parts, components, spares and products between two or more locations. The processes involved include rating and routing of both inbound and outbound freight, tracking and tracing of shipments, freight bill payment and auditing, import and export compliance and documentation, and load optimization. Advanced concepts utilize logistics modeling, financial optimization and third-party logistics management tools.

distribution channel

The selling channels supported by an enterprise. These may include retail sales, distribution partner (e.g., wholesale) sales, original equipment manufacturer (OEM) sales, Internet exchange or marketplace sales, and Internet auction or reverse auction sales.

distribution management system (see DMS)

distribution planning

The ability to assess where products and services should be deployed and determine the stock-keeping unit/location-level replenishment plan.

distribution requirements planning (see DRP)

distribution service

Middleware that ensures a mail package is delivered to the specified addressee.

divestiture

The U.S. Justice Department's legal action against AT&T leading to the terms of the Modified Final Judgment. AT&T agreed to the 1982 Consent Decree, and in 1984 it relinquished ownership and control of its 22 wholly owned Bell operating companies. They were separated into seven independent Regional Holding Companies (RHCs).

DL (distributed learning)

The distribution of the learning environment across boundaries of time and space (also known as "distance learning"). DL includes e-learning (the network-enabled delivery of digital learning content), but also includes other types of learning that don't involve computers. For example, correspondence courses and classes taught via videotape or closed-circuit television are forms of DL. See e-learning.

DL/1 (Data Language/1)

The language in which the application programmer specifies requirements to IBM's Information Management System/Database Manager (IMS/DB).

DLEC (data local-exchange carrier)

DLI (data line interface)

The point at which a data line is connected to a telephone system.

DLL (Dynamic Link Library)

A Windows feature.

DLM (distributed lock manager)

An architecture critical to maintaining locking consistency across multiple processing nodes and maintaining coherency between the multiple database buffer pools that exist on those different nodes.

DLS or DLSw (Data Link Switching)

An IBM-developed technique for carrying Systems Network Architecture (SNA) and NetBIOS over Transmission Control Protocol/Internet Protocol (TCP/IP). DLS "tunnels" or encapsulates SNA traffic using the transport services of TCP/IP. Upon entry into a router-based intranet, SNA traffic is placed into TCP segments, which are then sent across the network as IP datagrams. Intermediate routers along the session path "see" only standard IP datagrams and addresses, with no need to recognize SNA information flowing through them. At the destination router, TCP/IP headers are stripped off, and the original SNA traffic is sent in "native" mode to an SNA network or destination device using SNA addresses.

DLT (digital linear tape)

A linear recording tape drive designed by Digital Equipment and sold to Quantum in 1993. The technology became a de facto standard for backup and archive applications in 1997.

DLUR (Dependent Logical Unit Requester)

An IBM technology that performs a "gateway" type of conversion to get Systems Network Architecture (SNA) traffic onto Advanced Peer-to-Peer Networking/High-Performance Routing (APPN/HPR) sessions.

DM (see data mart)

DMA (direct memory access)

A facility that permits data transfers directly into or out of main storage without passing through the processor's general registers; either performed independently of the processor or on a cycle-stealing basis.

DMA (Document Management Alliance)

DME (Distributed Management Environment)

A mix of network management specifications introduced by the Open Software Foundation (OSF) in the early 1990s.

DMI (Data Migration Interface)

An interface published by SAP for the R/3 enterprise resource planning (ERP) system. DMI supports the batch process required to load data into an R/3 database as part of a migration to R/3 from another application.

DMI (Desktop Management Interface)

An architecture specification defined by the Distributed Management Task Force (DMTF), designed to give vendors and users a common vendor- and protocol-neutral framework to manage desktop systems, servers, peripherals and components.

DMI (Digital Multiplexed Interface)

A gateway providing 23 digital channels plus one signaling channel at an aggregate rate of 1.544 megabits per second (Mbps). Developed by AT&T. See CPI.

DML (data manipulation language)

The language by which an application programmer invokes functions from a database management system (DBMS) to access and manipulate data.

DMO (Directory Management Option)

An enterprise user administration tool offered as an option for Computer Associates' Unicenter TNG product.

DMP (distributed message passing)

DMPP (distributed memory parallel processor)

A massively parallel processing system without shared-memory capabilities; also known as "loosely coupled."

DMS (Defense Message System)

A protocol for communications over the Defense Information Systems Network (DISN). Connectivity between DMS and other systems is handled through gateways called Multifunction Interpreters (MFIs).

DMS (distribution management system)

Another term for warehouse management system (WMS). Some people use the term DMS because the system is used to control the activities within a distribution center. The terms DMS and WMS can be used interchangeably.

DMT (discrete multitone)

A frequency division method for transmitting data over twisted pair lines using 256 frequencies in the 64 kilohertz (KHz) to 1.1 megahertz (MHz) range. It is a more sophisticated signaling method that has more robust error correction and bandwidth management features. DMT is the official standard for generic digital subscriber line (xDSL) signaling.

DMTF (Distributed Management Task Force)

An industry organization promoting the development, adoption and unification of management standards and initiatives for desktop, enterprise and Internet environments. Formerly known as the Desktop Management Task Force.

DMU (digital mock-up)

A 3-D application used to simulate mechanical designs.

DMX (Document Manager for Microsoft Exchange)

A software component developed by Eastman Software that adds document management services to Microsoft Exchange.

DMZ (demilitarized zone)

Technical jargon for the protected subnet between an intranet and the public Internet.

DNA (Digital Network Architecture)

A network architecture from Compaq (formerly Digital Equipment), now part of Hewlett-Packard.

DNA (Distributed interNet Applications)

A Microsoft application development framework.

DNAs (Distributed interNet Applications for Financial Services)

An application development framework from Microsoft that is 75 percent industry-neutral Component Object Model (COM) object libraries, and 25 percent financial-services extensions for support of existing industry standards — e.g., Open Financial Exchange (OFX) for banking, Olife and Objx for insurance, and Federal Internet Exchange (FIX) and Society for Worldwide Interbank Financial Telecommunications (SWIFT) for securities — as well as financial-services-specific logic to address specific integration issues and standard definitions for various transaction participants.

DNA-M (Distributed interNet Applications for Manufacturing)

A Microsoft standard for integrated plant system (IPS) integration with enterprise and supply chain planning.

DNC (digital numeric control)

Software used to control machine tools or other devices that use pre-programmed motion control to remove material from a stock piece as part of the manufacturing process. It is usually stored in a general-purpose computer and distributed to multiple devices on demand.

DNIC (data network identification code)

A four-digit number assigned to public data networks.

DNIS (Dialed Number Identification Service)

DNIS is a feature of 800 and 900 lines. The long-distance carrier transmits the DNIS of the incoming call to identify the numbers the person is calling.

DNS (Digital Nervous System)

A comprehensive architecture for Windows-based client and server computing.

DNS (Domain Name System)

The system that serves as the map between logical names and network addresses in an Internet Protocol (IP) network.

DOCSIS (Data-Over-Cable Service Interface Specification)

A specification managed by the MCNS partners (Comcast Cable Communications, Cox Communications, Tele-Communications and Time Warner Cable) for residential cable modems.

document component management (see DCM)

Document Composition Facility (see DCF)

Document Content Description (see DCD)

document-enabled vertical application (see DEVA)

document image processing (see DIP)

document imaging

The conversion (i.e., digitizing) of paper documents into electronic documents by using a scanner or similar device, and the subsequent storage and management of those electronic images.

document management

A function in which applications or middleware perform data management tasks tailored for typical unstructured documents (including compound documents). It may also be used to manage the flow of documents through their life cycles.

Document Manager for Microsoft Exchange (see DMX)

Document Object Model (see DOM)

document rendering

The ability to control the appearance of a document.

document type definition (see DTD)

DOM (distributed output management)

Middleware that drives the output process and supports the automated creation and delivery of documents. Key DOM capabilities include the ability to deliver documents to hard-copy output devices, such as printers or fax machines, as well as electronically via e-mail or Web servers.

DOM (Document Object Model)

A language-neutral interface from the World Wide Web Consortium (W3C), designed to enable programs and scripts to dynamically access and update the content, structure and style of documents.

domain

1. A group of nodes on a network forming an administrative entity.
2. On the Internet, a part of the naming hierarchy that refers to groupings of networks based on organization type or geography.

domain name

A unique identifier for an Internet site or Internet Protocol (IP) network address, consisting of at least two segments separated by periods. Enterprises must register top-level domains with the Web Internet Registry and pay a yearly fee to maintain the registry.

Domain Name System (see DNS)

DOMF (Distributed Object Management Facility)

DOMF is based on the Common Object Request Broker Architecture (CORBA) standard from the Object Management Group (OMG). It provides a location-transparent object-communication mechanism across heterogeneous networks.

Domino Enterprise Connection Services (see DECS)

Domino Runtime Services (see DRS)

DOMS (Distributed Object Management System)

DOMSAT (domestic communications satellite)

DoS (denial of service)

The inability of a Web site or other server to respond to legitimate connections; blockage of access to a target Internet site.

DOS (Disk Operating System)

A command-driven PC operating system (OS) that resides in main memory and uses disks for data storage. It was once the standard OS for IBM-compatible PCs; early versions of Microsoft Windows ran on top of DOS.

dot-com

Slang term for a company that conducts all or most of its business over the Internet. The term derives from the ".com" characters that appears at the end of many commercial Web site addresses.

dot pitch

On a display screen or picture tube, the distance between two dots of the same color. Generally, the finer the dot pitch (typical values are 0.26 to 0.28 mm) the smoother the image, but smaller is not necessarily always better. Since there are a variety of different phosphor coating technologies that dictate the distance between adjacent phosphor triads or strips, there are considerable variations in how different manufacturers measure dot pitch.

dots per inch (see dpi)

double-byte character set (see DBCS)

double supertwisted nematic (see DSTN)

DOV (data over voice)

A technology that transmits data and voice simultaneously over twisted-pair copper wiring and is used primarily with local Centrex services or a special customer premises private branch exchange (PBX). Data rates for Centrex operation are 9.6 kilobits per second (Kbps) and 19.2 Kbps.

downlink

The portion of a satellite circuit extending from the satellite to the earth. Compare with uplink.

download

The process of bringing a file down to a computer through a network and typically from a server, or some other computing device. Download times can be greatly effected by the method of connection to the network.

Download Fun

A specification that supports secure downloading of consumer-oriented content using Internet-enabled mobile devices. Developed by Openwave, Download Fun is part of the GSM Association's Mobile Services (M-Services) initiative. See M-Services.

downtime

The total time a system is out of service.

DP (data processing)

DP (distribution point)

DPA (Distributed Print Architecture)

International Organization for Standardization (ISO) standard 10175, which defines print objects, attributes and protocols for submitting print requests across heterogeneous systems. Designed to provide a distributed printing system with high levels of interoperability, robustness, reliability and stability.

DPCM (Differential Pulse Code Modulation)

DPC/OPC (destination point code/origination point code)

An Integrated Services Digital Network (ISDN) term.

DPE (Disk Processor Enclosure)

dpi (dots per inch)

A measurement of resolution; e.g., the number of pixels per inch on a cathode ray tube display.

DPL (Distributed Program Link)

DPN (Data Packet Network)

A public packet networking service from Nortel Networks (formerly Northern Telecom).

DPNSS (Digital Private Network Signaling System)

A standard developed by British Telecommunications and other U.K. private branch exchange (PBX) suppliers. It enables PBXs from different vendors to be connected with E1 lines to pass calls transparently between each PBX.

DPP (distributed parallel processing)

Distribution of a job to whichever processor or network is best equipped to handle the job at that time.

DPSK (Differential Phase-Shift Keying) See phase modulation.

DQDB (Distributed Queue Dual Bus)

The Media Access Control technique used by the IEEE 802.6 Metropolitan Area Networking (MAN) standard. It is based on the Queued Packet Synchronous Exchange (QPSX) technology conceived at the University of Western Australia in 1985, and further developed by QPSX Communications. DQDB is a very high-speed (150 Mbps), distance-insensitive network that transports both voice and data. It uses a form of asynchronous time division multiplexing. By watching for empty cells on the buses, it avoids the need to pass tokens, as in Fiber Distributed Data Interface (FDDI). DQDB supports data rates of 45 Mbps to 155 megabits per second (Mbps) and isochronous circuits up to 1.44 Mbps. DQDB, which was designed for MAN usage, complements emerging asynchronous transfer mode (ATM) networks on the wide-area network (WAN) side, as it uses the same format and cell size as ATM.

DR (disaster recovery)

1. The use of alternative network circuits to re-establish communications channels in the event that the primary channels are disconnected or malfunctioning.
2. Methods and procedures for returning a data center to full operation after a catastrophic interruption (e.g., including recovery of lost data).

DR (distributed request)

A single read-only request to multiple data sources.

DRAM (dynamic random-access memory)

A computer memory chip that requires electronic refresh cycles to preserve data stored for manipulation by logic chips.

DRDA (Distributed Relational Data Architecture)

IBM's architecture for enterprisewide data access, announced in July 1990. The architecture was designed to homogenize data definition and access across IBM's different hardware/software platforms.

DRM (digital rights management)

Trusted exchange of digital information over the Internet whereby the user is granted only the privileges that the document sender allows.

DRM (distributed resource management)

An evolving discipline consisting of a set of software, hardware, network tools, procedures and policies for enabling distributed enterprise systems to operate effectively in production. DRM embraces solutions for the daily monitoring, resource planning, system administration, change management, operations, performance and other initiatives that are needed to maintain effective productivity in a distributed networked computing environment.

drop

A connection point between a communicating device and a communications network.

DRP (disaster recovery planning)

Planning to ensure the timely recovery of information technology assets and services following a catastrophe, such as fire, flood or hardware failure.

DRP (distribution requirements planning)

The process of assessing from which location products and services should be deployed, and determining the stock-keeping unit (SKU) and location-level replenishment plan.

DRP (Distribution Resource Planning)

DRS (Domino Runtime Services)

Web-based applications that work offline or online in Netscape Navigator or Internet Explorer. These IBM/Lotus tools provide automatic code and data synchronization with the server, preserve the database's security model and do not interfere with other browser-based applications.

dry line

A T1 line not providing span power.

DS (decision support)

DS (Netscape Directory Server)

DSA (Digital Signature Algorithm)

Developed by the National Institute of Standards and Technology (NIST) for use with the Digital Signature Standard.

DSA (directory system agent)

The electronic-messaging component that receives requests for addresses, looks them up and returns the address.

DSA (Dynamic Server Architecture)

From Informix Software, DSA is a parallel, scalable architecture that supports large databases running on massively parallel, loosely coupled, symmetric multiprocessing (SMP) and uniprocessor computers. DSA is designed to be independent of platforms and transparent to applications.

DSC (digital selective calling)

DSC2 (Deschutes Slot 2)

DSI (digital speech interpolation)

When speech is digitized, it can be cut into slices such that no bits are transmitted when a person is silent. As soon as speech begins, bits flow again.

DSL (digital subscriber line)

A technology for high-speed network or Internet access over voice lines. There are various types, including asymmetric DSL (ADSL), high-bit-rate DSL (HDSL), symmetric DSL (SDSL) and very-high-bit-rate DSL (VDSL). The whole group is sometimes referred to as "xDSL."

DSLAM (digital subscriber line access multiplexer)

DSLO (Distributed Systems License Option)

An IBM software license option that allows customers to copy certain IBM-licensed material for the purpose of installing multiple systems.

DSM (distributed systems management)

A technology for managing the interconnected parts of a system. As managed items — i.e., components of applications, nodes, links or subsystems — become active, they must notify their manager of their status. DSM tools are capable of dealing with a limited number of distinct elements and require a strong directory.

DSML (Directory Services Markup Language)

DSMPU (digital signal-processing microprocessor)

DSN (Distributed Systems Network)

A network architecture developed by Hewlett-Packard.

DSOM (Distributed System Object Model)

An IBM distributed-object technology that originated on OS/2 and AIX. DSOM is compliant with the Object Management Group's Common Object Request Broker Architecture (CORBA).

DSP (digital signal processor)

A specialized computer chip optimized for high data rates needed to process digitized wave forms, particularly those derived from analog sources. Often used in equipment requiring voice digitization.

DSP (Directory Synchronization Protocol)

DSRI (Digital Standard Relational Interface)

Proprietary, low-level application programming interface to Rdb, originally from Digital Equipment (and now part of the Compaq operation acquired by Hewlett-Packard).

DSS (decision support system)

A system designed to support strategic (vs. operating) decisions. Decision support systems enable the computer rather than the user to make decisions. The system

tends to be user-friendly and emphasizes ad hoc query, reporting and analysis capabilities. This is in contrast to online transaction processing, which focuses on low-cost, fast-response and predictably structured applications.

DSSI (Digital Storage Systems Interconnect)

DSSS (direct sequence spread spectrum)

A technology for third-generation wireless LAN (WLAN) systems. It has the ability to deliver twice the bandwidth per node, per access point than frequency hop (FH) and is more resilient to certain types of multipath interference.

DSSSL (Document Style Semantics and Specification Language)

DSTN (double supertwisted nematic)

A type of flat-panel display screen.

DSU (data service unit)

A simplified modem for the transmission of digital data over a private line or for limited distance communications over the public switched telephone network (PSTN), where it is not necessary to comply with all the requirements for a high-speed modem.

DSU (disk storage unit)

DSVD (digital simultaneous voice and data)

Technology used in some modems to integrate voice with data for simultaneous use.

DSx

In telephony, one of many digital signal levels, each providing a specific bandwidth, including:

- DS-0: Digital Signal Level 0, a standard 64 kilobit per second (Kbps) digital voice signal or channel.
- DS-1: Digital Signal Level 1, in which 24 voice channels are multiplexed into one 1.544 megabit per second (Mbps) (United States) or 2.048 Mbps (Europe) T1 digital channel.
- DS-1C: Digital Signal Level 1C; a 3.152 Mbps digital signal carried on a T1 C facility.
- DS-2: Digital Signal Level 2; the 6.312 Mbps digital signal carried on a T2, or 8.448 Mbps digital signal carried on an E2 (CEPT) facility.
- DS-3: Digital Signal Level 3; the 44 Mbps digital signal carried on a T3 facility, or 34.368 Mbps digital signal carried on an E3 (CEPT) facility.
- DS-4: Digital Signal Level 4; the 273 Mbps digital signal carried on a T4, or 139.264 Mbps digital signal carried on an E4 (CEPT) facility.

DSX-1 Interface

The channel service unit (CSU) interface to which a T1 line is attached.

DT (dumb terminal)

DTA (desktop administrator)

DTD (document type definition)

DTDs are Extensible Markup Language (XML)-defined document grammars. The data elements/markup tags, along with any other defined data element attributes, can be thought of as the document's metadata. DTDs can also be thought of as providing a "schema" and "data dictionary" for the document. Using a DTD, an XML parser can validate a document for conformance to a particular document type.

DTE (data terminal equipment)

End-user devices — such as terminals and computers that connect to data circuit-terminating equipment (DCE) — that either generate or receive the data carried by the network. In RS-232-C connections, designation as either DTE or DCE determines the signaling role in handshaking. In an ITU X.25 interface, DTE refers to the equipment or devices that manage the interface at the user premises.

DTIF (Digital Table Interchange Format)

The format used in Compound Document Architecture (see CDA) for the storage and interchange of documents that contain data tables, formulas and spreadsheets.

DTL (diode-transistor logic)

DTMF (dual-tone multifrequency signaling)

The basis for operation of touchtone telephones. DTMF is a method of signaling in which a matrix combination of two tones is used to transmit numerical information.

DTP (desktop publishing)

DTP (Distributed Transaction Processing)

DTV (digital television)

Dual 56

Two switched 56 kilobit per second (Kbps) calls made between videoconferencing equipment to enable data transfer at 112 Kbps. The videoconferencing equipment performs a two-channel inverse-multiplexing procedure to assure channel alignment.

dual-attached concentrator (see DAC)

dual-attached station (see DAS)

dual in-line memory module (see DIMM)

dual in-line package (see DIP)

dual-tone multifrequency signaling (see DTMF)

dumb terminal

A terminal that does not performing local processing of entered information, but serves only as an input/output device for an attached or network-linked processor.

dump

To transfer all information from a record to another storage medium, e.g., copying from memory to a printer.

DUNS (Data Universal Numbering System)

A system of classification that uses a nine-digit number assigned by Dun and Bradstreet Information Services to identify unique business entities.

DUOW (distributed unit of work)

A single transaction with updates to multiple data sources, requiring a two-phase commit (see 2PC).

DUP (data user part)

duplex circuit

A circuit used for transmission in both directions at the same time.

Duplex Ethernet

Ethernet at 20 megabit per second (Mbps) speeds using two data paths. It is only suitable for use with Ethernet switching hubs.

duplex signaling (see DX)

duplex transmission

Simultaneous, two-way, independent transmission. Often called full-duplex transmission (FDX).

DVD (digital versatile disc)

An optical disk with the ability to store large quantities of video data.

DVD-ROM (digital versatile disc read-only memory)

A read-only format of digital versatile disc (DVD) designed specifically for the PC market. Physically, DVD discs take on the same 5 1/4-inch footprint used in CD-ROMs. DVD discs, however, are loaded much more densely with data, and storage capacities start at 4.7 gigabytes.

DVI (digital video interface)

DVMRP (Dynamic Vector Multicast Routing Protocol)

An internetworking gateway protocol based on Routing Information Protocol (RIP), which uses a dense-mode Internet Protocol (IP) multicast scheme.

DW (see data warehouse)

DWDM (dense wave division multiplexing)

Also called wave division multiplexing (WDM). In fiber-optic networks such as Synchronous Optical Network (SONET), DWDM permits fiber-optic cables to carry up to eight wavelengths of light, thus multiplying their capacity by the number of wavelengths carried. DWDM has been applied to a number of international submarine optical cables to increase capacity.

DX (duplex signaling)

A signaling system that occupies the same cable pair as the voice path yet does not require filters.

DXI (Data Exchange Interface)

A frame-based protocol for customer premises equipment to access Switched Multimegabit Data Service (SMDS). It is based on High-Level Data Link Control (HDLC) format and designed to operate over serial lines.

DXT (Data Extract)

An IBM-licensed program for data extraction from a variety of Multiple Virtual Storage (MVS) files.

dyadic

A multiprocessor design term that IBM introduced with the 3081. Multiple Virtual Storage (MVS) sees a dyadic processor as two processors running as one.

dynamic adaptive routing

Automatic selection and use of alternative communications paths among two or more midrange systems of the same supplier in the event of a congested, faulty or downed circuit within the preferred data path.

dynamic address translation (see DAT)

dynamic bandwidth allocation

The process of determining current traffic loads over a channel, and automatically increasing or decreasing the bandwidth of the channel to optimize overall utilization efficiency.

dynamic content

Web site content that is continually refreshed to provide new or updated information to attract new viewers and to keep prior viewers returning to the site.

dynamic database restructuring

The ability to change the relational-database structure, table capacities and security without unloading and reloading the database.

Dynamic Data Exchange (see DDE)

Dynamic Domain Name System (see DDNS)

dynamic hardware reconfiguration

The ability to add or delete midrange system hardware components and modify operating-system hardware configuration tables by operator control without turning power off and reinitializing the operating system.

Dynamic Host Configuration Protocol (see DHCP)

Dynamic HTML (see DHTML)

Dynamic Link Library (see DLL)

dynamic RAM (see DRAM)

dynamic routing

A method of wide-area network transmission that uses a router to select the most appropriate path for each section of data packet transmission along a network.

Dynamic Server Architecture (see DSA)

Dynamic Vector Multicast Routing Protocol (see DVMRP)

E

E1

A European standard digital facility with a transmission speed of 2.048 megabits per second (Mbps). For digital signal level 1 (DS1), 24 voice channels are multiplexed onto one E1 channel. In the United States, a 1.544 Mbps channel is used (i.e., T1).

E.164

The International Telecommunications Union (ITU) recommendation for international telecommunications numbering, especially for Integrated Services Digital Network (ISDN) and Switched Multimegabit Data Service (SMDS).

E3

A European standard digital facility with a transmission speed of 34.368 megabits per second (Mbps). For digital signal level 3 (DS3), 32 voice channels are multiplexed onto one E3 channel. In the United States, a 44 Mbps channel is used (i.e., T3).

E-4GL (enterprise fourth-generation language)

EA (Enterprise Agreement)

A Microsoft volume software agreement.

E-AD (enterprise application development)

A category of tools designed for use in developing enterprise (rather than departmental or workgroup) applications. Platform-specific E-AD tools are optimized for specific environments (e.g., the AS/400 or Unix), generally at the expense of supporting more heterogeneous environments.

EAI (enterprise application integration)

An emerging category of products that provide messaging, data transformation, process flow and other capabilities to simplify the integration of enterprise resource planning, legacy and other applications.

EAM (enterprise asset management)

EAM systems, previously inclusive of Computerized Maintenance Management Systems (CMMS) functionality, have traditionally been a key tool in maintenance, repair and operations (MRO) procurement. EAM is a key strategy to increase plant capacity, using IT in lieu of new construction in large, asset-intensive enterprises. It integrates key open control systems (OCSs), enterprise resource planning (ERP) and maintenance activity and functions to reduce downtime without significantly increasing maintenance spending.

EAM (equipment asset management)

The planning and control of equipment purchase and use so as to maximize return on investment (ROI) and minimize total cost of ownership (TCO).

EAN (European Article Numbering)

A retail bar code system (the European equivalent of the UCC system in the United States) managed by EAN International, a group founded in 1974 by 12 European countries and set up as a not-for profit international association under Belgian law in 1977. Today, EAN International has 97 member organizations representing 141 countries.

EAO (enterprise applications outsourcing)

EAO divides into four major domains: IT outsourcing (ITO), business process outsourcing (BPO), application service provider (ASP) and business service provider (BSP). Each domain reflects a different approach to outsourcing with respect to the degree of business responsibility transferred to the external service provider (ESP) and the autonomy of the enterprise in adjusting the application and business processes to its needs.

EAROM (electrically alterable read-only memory)

earth station

Ground-based equipment used to communicate via satellites. See ground station.

EAS (Extended Area Service)

An option whereby the telephone subscriber can pay a higher flat rate to obtain wider geographical coverage without additional per-call charges.

EAS (Enterprise Agreement for Software)

EASDAQ (European Association of Securities Dealers Automated Quotations)

Authorized by the Belgian Banking and Finance Commission, the EASDAQ stock exchange has a small market but high standards of regulation. Recent European Union directives dictate that authorization in Belgium permits EASDAQ to operate its trading system in all other member states. The EASDAQ Rule Book has been closely modeled on National Association of Securities Dealers Automated Quotations (NASDAQ's) rules, allowing companies to comply with the information requirements of the Securities and Exchange Commission (SEC) in the United States, thus easily allowing companies to dual-list on NASDAQ and EASDAQ.

EATE (electronic automatic test equipment)

EBCDIC (Extended Binary-Coded Decimal Interchange Code)

An 8-bit alphanumeric code developed by IBM. Similar to (and now largely eclipsed by) American Standard Code for Information Interchange (ASCII).

e-bill (electronic bill)

A bill presented on the Internet.

EBIS (enterprise business intelligence suite)

A suite that offers multiple styles of common business intelligence functionality, including ad hoc query, reporting, charting, multidimensional viewing (OLAP) and light analysis (e.g., trending).

e-books (electronic books)

e-books are approximately the size of a large paperback book and contain a high-resolution screen with local storage and limited processing capability. Users buy the device and subscribe to a service that allows them to download books to the device, directly from a Web site.

EBP (electronic bill presentment)

The ability for consumers to view their bills electronically (for example, on the Web). Unlike electronic bill presentment and payment (EBPP), EBP does not include online payment capability. See EBPP.

EBPP (electronic bill presentment and payment)

The ability for consumers to view and pay their bills online (for example, via the Web or e-mail).

EBR (electronic batch record)

A transaction that captures all of the relevant steps and measurements associated with the production of a product that is produced using batch-manufacturing methods.

EBT (electronic benefits transfer)

EBTF (e-business task force)

A senior management-led team responsible for developing a coherent e-business strategy. It unites elements from marketing, advertising, circulation, design, production, editorial, research and administration management.

e-business (electronic business)

Any Internet-enabled business activity that transforms internal and external relationships to create value and exploit market opportunities driven by the new rules of the "connected economy."

e-business platform

The tools, methodology and engine necessary to provide the basic building blocks for e-business applications.

ebXML (Electronic Business XML)

A joint project of the United Nations Center for Trade Facilitation and Electronic Business (UN/CEFACT) and the Organization for the Advancement of Structured Information Standards (OASIS), whose goal is to develop a framework for consistent use of XML in B2B application standards. The ebXML framework provides a dictionary of common business objects (e.g., names and postal addresses) and an overarching protocol for communicating transactions. Industry consortia and other standards development groups are invited to use this framework to define application standards. Among the aims of the initiative is to lower the entry cost for e-commerce, particularly for small and midsize enterprises and developing nations.

EC (engineering change)

A revision to a parts list, bill of material, engineering drawing or engineering part model authorized by the engineering department. Changes are usually identified by a control number and are made for safety, cost reduction or functionality reasons. Also called ECO (engineering change order) or ECN (engineering change notice).

EC (European Commission)

EC (see e-commerce)

ECAD (electronic computer-aided design)

ECAE (electronic computer-aided engineering)

E carrier services

High-speed, private-line digital transmission services, defined by the European Conference of Postal and Telecommunications Administrations (CEPT) for Europe, that provide large-volume transmission between two stations. E1, the most common e-carrier service, operates at an aggregate data rate of 2.048 megabits per second (Mbps) and consists of 24 separate channels. (E2 and E3 are rated at 8.448 Mbps and 34.368 Mbps, respectively.) Only E1 and E3 are widely supported. In the United States, T-carrier services are provided, although the corresponding data rates are not identical.

E-CASE (enterprise computer-aided software engineering)

A category of enterprise application development (E-AD) tools that are generally noninterpretive, generate COBOL or C, and may have integrated modeling capabilities in their integrated development environments.

e-cash (electronic cash)

Currency that can be loaded onto smart cards, PCs, remote servers or handheld devices and used to purchase goods and services. It is typically used for low-value purchases and allows anonymous purchasing.

e-catalog (electronic catalog)

An aggregation mechanism that presents goods or services for sale and enables users to buy them in electronic marketplaces.

ECC (Elliptic Curve Cryptosystem)

An cryptographic algorithm forwarded for commercial use by security vendor Certicom. It is implemented using public-key algorithms combined with elliptic curves. The curves construct "elements" and "rules of combining" to produce groups which, in turn, create cryptographic algorithms.

ECC (enterprise cooperation community)

Enterprise cooperation roles represent the knowledge management (KM) emphasis on sharing. They form a community with other enterprise cooperation roles in other enterprises sharing best practices in the co-evolution of processes and technology. The ECC is linked together through the technology and practices it is championing. This approach requires that all roles use the technology. Enterprise cooperation roles are leveraged when a few users learn new practices and technology, and they pass on this knowledge to more performance-focused roles, and ultimately to the entire white-collar staff.

ECC (error-correcting code)

The diagnostic code used to correct data-storage errors and isolate hardware failures. Based on a concept of simultaneous polynomial equations, the read-back process generates a correction profile over the incorrect data. All ECCs have a very small but finite failure rate (i.e., some uncorrectable errors will either appear as

correctable or appear as having no error at all). In either situation, bad data is passed as verified and valid.

ECD (electrochromeric display)

ECD (energy conversion device)

e-channel customer relationship management (see e-CRM)

echo

A wave that has been reflected or otherwise returned with sufficient magnitude and delay for it to be perceived as a wave distinct from the one directly transmitted.

echo cancellation

A technique used in higher-speed modems that allows for the isolation and filtering of unwanted signal energy resulting from echoes from the main transmitted signal.

echo check

A method of checking data transmission accuracy, whereby the data received is returned to the sending end for comparison with the original data.

echoplex

Pertaining or referring to the printing of keyboarded characters on return of the signal from the other end of the line, using full-duplex transmission, to ensure that the data was received correctly at the other end.

echo return loss (see ERL)

echo suppressor

A mechanism used to suppress interference on long-distance analog connections; it suppresses the transmission path opposite in direction to the one in use. This feature is necessary for voice transmission but often interferes with data transmission.

ECKD (extended count key data)

ECL (emitter-coupled logic)

In chip design, ECL is a type of bipolar transistor characterized by extremely fast switching speeds.

ECM (enterprise client migration)

ECMA (European Computer Manufacturers Association)

An industry trade group headquartered in Geneva, Switzerland. The ECMA has developed more than 200 standards, including the Portable Common Tools Environment (PCTE).

ECMAScript

A standardized version of JavaScript.

ECML (Electronic Commerce Modeling Language)

ECN (Electronic Commerce Network)

A catalog-hosting environment managed by MCI that enables suppliers to shift catalog and transaction management off-site. Suppliers need only to have Internet access and provide updates in either batch or real-time to their catalogs. Buyers access ECN to view and interact with supplier catalogs.

ECN (electronics communications network)

An anonymous, electronic market maker that automatically pairs up and executes matching buy and sell orders.

ECO (electronic-commerce outsourcing)

The partial or complete outsourcing of the technologies and business processes a company requires to engage in electronic commerce with its trading partners.

ECO (engineering change order)

e-COMM

A French consortium and the “smart card” electronic-payment implementation it supports, which is compliant with the international Secure Electronic Transaction (SET) standard.

e-commerce (electronic commerce)

The use of communication technologies to transmit business information and transact business. Taking an order over the telephone is a simple form of e-commerce. Internet commerce is also e-commerce, but is only one of several advanced forms of e-commerce that use technology, integrated applications and business processes to link enterprises.

e-community (electronic community)

Groups of people collaborating and sharing ideas over an electronic network. Communities optimize their collective power by affiliation around a common interest, by the compression of the time between member interactions (i.e., communicating in real time), and by asynchronous “postings” that potentially reach more participants and allow for more reflection time than real-time interactions.

economic and monetary union (see EMU)

economic life

The remaining period over which the property is expected to be economically usable with normal repairs and maintenance.

economic value added (see EVA)

ECP (electronic check presentment)

The electronic presentation of check data to processing banks. In the first ECP phase, one bank presents a second bank with electronic check information based on the check’s magnetic-ink character recognition (MICR) line, which states the correct customer account and the amount of the payment. Electronic presentment is then followed by the paper check, speeding the clearing process by one day and enabling banks to use credited account balances a day earlier — a significant benefit, especially to large banks. In the second ECP phase, the check is imaged at the first bank of collection; the paper or the image is not passed to the second bank unless the second bank requests it.

ECR (efficient consumer response)

A consumer goods (primarily grocery) initiative aimed at reducing inefficient practices and waste in the supply chain.

e-CRM (e-channel customer relationship management)

The integration of electronic channels (e-channels) — notably the Web — into an enterprise's overall customer relationship management (CRM) strategy. A subset of CRM, e-CRM comprises the business strategies and technologies that leverage customer-facing, e-channel applications to develop more profitable customer relationships. E-CRM involves using the Web to support CRM with the goal of driving consistency within all channels relative to sales, customer service and marketing initiatives. It can support a seamless customer experience and maximize customer satisfaction, customer loyalty and revenue.

ECTF (Enterprise Computer Telephony Forum)

ECTRA (European Committee for Telecommunications Regulatory Affairs)

The European Conference of Postal and Telecommunications Administrations (CEPT) committee responsible for dealing with regulatory issues predominantly relating to nonradio telecommunications.

EDA (electronic design automation)

The use of a computer to design and simulate performance of electronic circuits on a chip. Similar to automatic test equipment, which tests primarily chips in electronic systems.

EDA (Embedded Document Architecture)

EDA/SQL (Enterprise Data Access/Structured Query Language)

A product from Information Builders, EDA provides a common interface between a wide range of Structured Query Language (SQL) programs and SQL databases. Queries on data from different types of databases can be queried simultaneously using EDA.

EDD (electronic document distribution)

Distribution of information without the use of paper. The goal is to enable viewers to actively interact with a document, directing the document to display information on request. The complexity of this task is compounded by the addition of media unsupported by print (e.g., sound and animation). EDD market problems are twofold: limiting technology and lack of acceptance by end users.

EDGE (Enhanced Data Rates for GSM Evolution)

A radio interface technology with enhanced modulation designed to provide Global System for Mobile Communications (GSM) and time division multiple access (TDMA) networks with the capacity to rival third-generation (3G) cellular networks. It is an integral part of the migration of GSM and TDMA networks to 3G. (Many vendors and industry groups substitute "Global" for "GSM" in the expanded term, to avoid the impression that the technology is applicable only to GSM networks.) EDGE is being developed to support mobile services with maximum data rates of around 384 Kbps. In real-life conditions, throughputs will be considerably lower (e.g., 64 Kbps) but much faster than today's GSM and TDMA networks. EDGE can allow operators without a Universal Mobile Telecommunications System (UMTS) license to stay competitive in wireless data markets. GSM network operators also may deploy it,

even if they win a 3G license, to provide low-cost services to automotive and machine-to-machine applications.

edge gateway (see EGW)

EDI (electronic data interchange)

The electronic exchange of trading documents (e.g., invoices and orders) to facilitate e-commerce. Originally conducted only through value-added networks, EDI is gradually moving to the Internet.

EDIF (Electronic Data Interchange Format)

A standard for defining the exchange of electronic data.

EDIFACT (Electronic Data Interchange for Administration, Commerce and Transportation)

A United Nations initiative to produce a universal standard for a total approach to electronic data interchange including syntax, data elements and messages.

EDL (Enterprise Definition Language)

EDL defines classes, which are user-specified groupings of storage objects based on a combination of node, path, file name, ownership or other filters. Classes are used to collect, analyze and manage storage resources by application, owner, location, node or some other installation-defined criteria.

EDM (electronic document management)

EDM (enterprise data model)

EDM (EMC Data Manager)

Backup and recovery software for the relational database management system (RDBMS) environment developed by EMC. It is considered to be a high-end replacement for EMC's Epoch Systems.

EDM (enterprise desktop manager)

A software application used to manage the total desktop environment. It includes functions such as software distribution and installation, configuration management, change management, and hardware and software inventory.

EDO (extended data output)

A fairly simple, pin-compatible derivative of standard fast-page mode asynchronous memory. EDO can improve memory performance by as much as 20 percent or 30 percent by shortening the page mode cycle of traditional memory. It can be used for dynamic random-access memory (DRAM) or video random-access memory (VRAM). It does not command a significant premium over standard page-mode DRAM.

EDP (electronic data processing)

EDP (enterprise demand planning)

Synchronization and consolidation of the demand-planning process across functions, using a single integrated repository for demand information and a common analytical architecture for the core forecasting and planning activities.

EDS (Electronic Data Systems)

EDUCOM

A consortium of leading colleges and universities seeking to transform education through the use of information technology.

EEMA (European Electronic Messaging Association)

EEPROM (electrically erasable programmable ROM)

A type of programmable read-only memory (ROM) that can be erased or updated by electrical signal, often done remotely.

e-ERP (electronic ERP)

The Web-enabled elements of an enterprise resource planning (ERP) business strategy that improve shareholder and customer value by integrating interenterprise manufacturing, financial and distribution functions to dynamically balance and optimize the resources of the enterprise together with its trading partners.

e-ESP (e-business external service provider)

A Web-focused external service provider (ESP). An e-ESP typically delivers e-business expertise focused on strategy or implementations, but rarely on both.

EFCI (Explicit Forward Congestion Indicator)

In an asynchronous transfer mode (ATM) network, one of two feedback mechanisms used by the available bit rate (ABR) service category to inform the source of the network resources available to it.

EFF (Electronic Frontier Foundation)

effective address

A logical or actual location in memory.

effective isotropic radiated power (see EIRP)

e-fingerprint (electronic fingerprint)

A biometric security solution used for identification and verification of a user's identity.

e-finance (electronic finance)

Also known as "electronic financial services" (EFS), e-finance is electronically enabled access to financial services. It is not a delivery (i.e., institution-centric) vehicle, but a vehicle of access (i.e., customer-centric).

e-form (electronic form)

An automated and interactive template for capturing, processing, displaying and creating output from defined sets of business data.

EFQM (European Foundation for Quality Management)

EFS (electronic financial services)

See e-finance.

EFT (electronic funds transfer)

The electronic exchange of information between financial institutions, which results in debits and credits.

EFT/POS (electronic funds transfer/point of sale)

Allows direct transfer of funds during a retail transaction.

EGA (enhanced graphics adapter)

EGP (Exterior Gateway Protocol)

The Transmission Control Protocol/Internet Protocol (TCP/IP) used by exterior routers to move data from one autonomous system to another.

EGW (edge gateway)

The perimeter switching technology that offers users the interface to a variety of virtual public data services.

EHLLAPI (Emulator High-Level Application Program Interface)

An IBM communications feature.

EI (enterprise integration or integrator)

EIA (enterprise information architecture)

Enterprise-specific definition of rules defining the use of middleware in third-generation office systems. The EIA rules include application programming interfaces (APIs) and protocols addressing interoperability and portability of applications, whether acquired or developed internally. The EIA is a living specification requiring constant management and regular review for enhancement.

EIA (Electronic Industries Alliance)

A U.S. electronics standards group; its telecommunications sector is represented by the Telecommunications Industry Association (TIA). Standards defined by EIA and EIA/TIA include:

1. EIA/TIA-232: Standard for unbalanced circuits up to 64 kilobits per second (Kbps), formerly RS-232.
2. EIA/TIA-449: Standard similar to EIA/TIA 232, but which can run at speeds of up to 2 megabits per second (Mbps). Formerly RS-449.
3. EIA-530: Standard for balanced (formerly RS-422) and unbalanced (formerly RS-423) transmission.
4. EIA/TIA-586: Standard describing the various grades of unshielded twisted-pair (UTP) cabling.

EIAJ (Electronic Industries Association of Japan)

EIA/TIA (Electronic Industries Alliance/Telecommunications Industry Association)

See EIA.

EIDE (Enhanced Integrated Drive Electronics)

An interface that supports larger capacities than standard Integrated Drive Electronics (IDE) and allows for more peripherals to attach to the interface, providing some incremental flexibility and expandability.

EIGRP (Enhanced Interior Gateway Routing Protocol)

An Interior Gateway Protocol (IGP) that transfers packets from one network to another adjacent one.

EIM (enterprise information management)

EIM is an architecture-management approach for resolving architectural differences within office information system (OIS) implementations. EIM is a tool for strategic planning comprising the enterprise information architecture (EIA) and a set of strategic IS management practices. By applying EIM to their enterprises, users have been able to migrate from the incompatible systems of the 1980s to the interoperable systems of the 1990s.

EIRP (effective isotropic radiated power)

The combination of transmitted power and antenna gain.

EIS (executive information system)

An application program specifically designed for use by the corporate executive. Presentation of material is often structured after the "board briefing book" concept. Detailed information on the summarized charts is often made available by using a concept known as "drilling." The EIS acts as a usable interface to a database of company information. It automates high-level analysis and reporting, and typically has a user-friendly graphical interface.

EISA (Extended Industry Standard Architecture)

Originally developed as an alternative 32-bit master bus to IBM's Micro Channel master bus for its PS/2 family of microcomputers. Unlike the Micro Channel, one of the design goals of the EISA bus was to enable the use of add-on cards developed for IBM's PC and PC/AT computers.

EJB (Enterprise JavaBeans)

The EJB specification defines a way to build server-side, transactional Java-based components. It was developed by Sun Microsystems in collaboration with IBM, Netscape, Oracle and other vendors.

EKA (enterprise knowledge architecture)

The table of contents, or "yellow pages," for information bases accessible through the information services of the intranet, including e-mail and Lotus notes. It enables an end user to navigate to enterprise resources using a hyperlink to the specific Web page, narrowing the search before launching a search inquiry.

EL (electroluminescent)

ELA (enterprise license agreement)

ELAN (emulated LAN)

The emulated local-area networks (LANs) that occur in a campus environment where asynchronous transfer mode (ATM) and a packet-based LAN technology are installed in the same network. ELANs are the ATM logical subnets that emulate LANs, which have been standardized and operational since the LAN emulation (LANE) 1.0 specification, and later were advanced in LANE 2.0 and Multiprotocol Over ATM (MPOA). Multiple ELANs may comprise many Ethernet or token ring networks.

e-learning

Network-enabled learning that relies on digital content, experienced through a technology interface. Collaboration is a desirable feature but not a requirement. E-learning can be a subset of distributed learning (see DL).

electrically erasable programmable ROM (see EEPROM)

electromagnetic interference (see EMI)

electromagnetic spectrum

The entire range of wavelengths or frequencies of electromagnetic radiation extending from gamma rays to the longest radio wave and including visible light. See radio frequency.

electronic batch record (see EBR)

electronic bill presentment (see EBP)

electronic bill presentment and payment (see EBPP)

electronic business (see e-business)

Electronic Business XML (see ebXML)

electronic cash (see e-cash)

electronic catalog (see e-catalog)

electronic check presentment (see ECP)

electronic commerce (see e-commerce)

Electronic Commerce Network (see ECN)

electronic-commerce outsourcing (see ECO)

electronic-channel CRM (see e-CRM)

electronic data interchange (see EDI)

Electronic Data Interchange for Administration, Commerce and Transportation (see EDIFACT)

Electronic Data Interchange Format (see EDIF)

electronic design automation (see EDA)

electronic document distribution (see EDD)

electronic financial services (see e-finance)

electronic form (see e-form)

electronic funds transfer (see EFT)

electronic funds transfer/point of sale (see EFT/POS)

Electronic Industries Alliance (see EIA)

electronic mail (see e-mail)

electronic market (see e-market)

electronic marketplace (see e-marketplace)

electronic marketplace manager (e-marketplace manager)

electronic medical record (see EMR)

electronic meeting system

A software product that can make midsize and large meetings more productive by supporting and automating all or some of these meeting-management activities: agenda planning and management, brainstorming, voting, commenting, evaluating alternatives and some task-specific support (e.g., project planning).

electronic messaging

The sending and receiving of messages and, increasingly, data through a network. Like paper messaging, it is accomplished through a number of tasks, including composition, assembly, addressing, posting, sorting, routing and delivery.

Electronic Messaging Association (see EMA)

electronic messaging utility (see EMU)

electronic output strategy (see EOS)

electronic signature

A traceable e-mail or a biometric applied to a message. The biometric may be based on digitized handwriting (handwriting that is converted by cryptography into a digital signature) or a biometric (e.g., a fingerprint that can be combined with a hash or digest of the message to show the signer's intent). The electronic signature cannot be removed and applied to other documents to forge a signature.

electronic software distribution (see ESD)

electronic switching system (see ESS)

electronic tag (see e-tag)

electronic tandem network (see ETN)

electronic whiteboard

A large, touch-sensitive screen attached to a personal computer (PC). Users view the PC's output on the screen and control it by touching a finger (or a specially designed inkless pen) to the menu options and icons.

elevation

The angle between the horizon and an antenna's beam.

ELF (extremely low frequency)

ELFExT (equal-level far-end crosstalk)

Elgamal

A public-key system that gets its security from the difficulty of calculating discrete logarithms in a finite field.

Elliptic Curve Cryptosystem (see ECC)

ELS (enterprise library service)

EMA (Electronic Messaging Association)

An electronic-mail industry organization and forum for development standards and policies for such systems as e-mail, voice mail and facsimile. EMA is now a forum of the Open Group.

EMA (Enterprise Management Architecture)

e-mail (electronic mail)

Any communications service that permits the electronic transmission and storage of text messages and attached or enclosed files. Some e-mail systems are limited to communication between end users on the same network; others have gateways that allow end users to send messages to other designated computer systems or worldwide over the Internet. Once sent, e-mail messages are stored in electronic mailboxes until the recipient retrieves them. Most Internet service providers also provide e-mail services.

e-mail response management system (see ERMS)

e-mall

A Web site that maintains catalogs from multiple suppliers. Buyers enter the site, view many suppliers at once and purchase products or services. E-malls often charge a fee for tenancy or membership, may take title to the goods themselves, and are selling environments based on traditional concepts of print and broadcast advertising that entice visitors to buy.

e-market

A category or segment of business conducted over the Internet. Online book selling, for example, might be described as an e-market (in which the competitors include Amazon.com and barnesandnoble.com). This term should not be confused with "e-marketplace" (see separate entry), which refers to online buying environments that aggregate products or services from multiple suppliers.

e-market maker (see EMM)

e-marketplace

A Web site that provides an online buying environment where customers can select from multiple suppliers' products or services. E-marketplaces aggregate supplier content and often provide decision support tools to enable buyers to make informed

decisions. Examples of e-marketplaces include those focused on consumer buyers (e.g., auction sites such as eBay) and business-to-business sites focused on specific industry segments (e.g., PlasticsNet).

e-marketplace manager (see EMM)

Embedded Structured Query Language (see ESQL)

EMC (event management center)

Command centers used by healthcare organizations (HCOs) to gather information related to failures and interruptions occurring inside and outside the enterprise, and to launch countermeasures if necessary.

EMC Data Manager (see EDM)

EMEA (Europe, Middle East and Africa)

EMEA (European Agency for the Evaluation of Medicinal Products)

The European system, created in 1995, for the authorization of medicinal products.

e-merchandising (electronic merchandising)

emergency access

An emergency alarm system integrated into the private branch exchange (PBX) switching system that can be implemented by uniquely ringing all station instruments to indicate an emergency condition.

emergency dialing

Allows all system users the ability to use preassigned two-or three-digit numbers that translate to outside exchange numbers (e.g., local police, fire department or ambulance service).

EMF (enterprise middleware framework)

EMI (electromagnetic interference)

The energy given off by electronic circuits and picked up by other circuits. Based on the type of device and operating frequency, EMI can be reduced by shielding. Minimum acceptable levels are detailed by the Federal Communications Commission (FCC).

emitter-coupled logic (see ECL)

EMM (e-market maker)

An intermediary (also known as an "e-marketplace manager") that develops a business-to-business (B2B) e-marketplace of buyers and sellers within an industry, geographic region or affinity group. EMMs enter supply chains introducing new efficiencies and new ways of selling and purchasing products and services by providing content, value-added services and often e-commerce capabilities.

emoticon

A text rendition of a face viewed sideways, created from punctuation marks. Used in e-mail and Usenet newsgroups to denote an emotion. (From "emotive icon.")

EMPAC (Enterprise Maintenance Planning and Control)

An enterprise asset management (EAM) product from Indus International.

EMPI (enterprise master person index)

The critical first step to safely sharing information on patients and members within the integrated delivery system (IDS). At a minimum, EMPIs uniquely identify patients, persons or members, and cross reference patient, person or member identification numbers to link patient information stored in disparate systems.

employee relationship management (see ERM)

employee self-service (see ESS)

EMR (electronic medical record)

Used to compile computerized patient healthcare information, EMR generally deals with information only in the form of document images or text formatted for output to a printer or for video display. It does not ordinarily handle discrete data such as is usually stored in a database management system.

EMS (electronics manufacturing service)

EMS (Enhanced Messaging Service)

A messaging standard developed by the 3G Partnership Project (see 3GPP). EMS uses features defined in the Short Message Service (SMS) specification to enhance the user experience when sending messages. This cross-vendor initiative has support from Alcatel, Ericsson, Motorola and Siemens, which have been the main proponents of EMS in the standards bodies, with additional support from mobile-phone vendors such as NEC, Panasonic (Matsushita), Sagem and Trium (Mitsubishi).

With EMS, software added to the mobile phone enables standard SMS parameter fields (such as the "user data header") to be used to send binary-encoded and concatenated messages. Thus, a suitably capable device can display enriched content such as text enhancements (including italics and underlining), sounds, and static or animated images.

EMS (Enterprise Management System)

EMS (Enterprise Messaging Server)

Former name for Microsoft Exchange, an enterprisewide messaging and mail system. Rolled into Exchange, it and other planned products will ship as a single system: a workgroup-focused, enterprise-scoped platform for standards-based messaging, groupware, work flow, document management and "information exchange" through a single, multimedia mailbox.

EMS (equipment management system)

The subsystem of a warehouse management system (WMS). It controls automated material-handling equipment such as carousels, pick to light, in-line scales, and conveyors. The EMS is designed to provide a standardized interface between the warehouse management system and the automated material-handling equipment.

EMS (Expanded Memory Specification)

EMU (economic and monetary union)

The European process of standardizing on a single currency (i.e., the euro).

EMU (electronic messaging utility)

A Gartner concept describing a solid base of interworking electronic messaging provided as a utility service like electricity. The EMU is sturdy enough to bear the anticipated increase in number and size of messages, sufficiently generalized to ensure readability of attachments and well-connected to the world beyond the boundary of the enterprise. The EMU provides open interfaces to make its services available to end users and applications.

emulate

To imitate one system with another, so that the imitating system accepts the same data, executes the same computer programs and achieves the same results as the imitated system.

emulated LAN (see ELAN)

Emulator High-Level Application Program Interface (see EHLLAPI)

EMV (Europay-MasterCard-Visa)

An emerging general payment standard for smart cards.

EN (end node)

A node that only sends and receives information and cannot route and forward information to another node.

encapsulating bridge

A bridge used to connect two segments (built on one type of medium) with a different type of medium. An example of a bridge of this sort is where an Fiber Distributed Data Interface (FDDI) ring acts as a backbone for the interconnection of several bridged local-area networks (LANs).

encapsulation

The binding and the hiding of the underlying implementation of an object's data and operations. The data is wrapped in a particular type of protocol header. The set of operations that is accessible is the object's interface. See tunneling.

encryption

The process of systematically encoding a bit stream before transmission so that an unauthorized party cannot decipher it.

end node (see EN)

End System to Intermediate System (see ES-IS)

end-to-end services

Services offered by the lower three layers of the Open Systems Interconnection (OSI) Reference Model.

end-to-end storage management

A component of an overall systems-management infrastructure that improves application availability by reducing the number of storage-related outages and minimizing the length of the outages when they occur.

end user

An individual who uses a computer to perform a business or personal activity. Technical personnel are generally not considered end users when they are programming or operating the computer for technical purposes, though they are when they perform other tasks.

end-user computing (see EUC)

end-user training (see EUT)

engineering change (see EC)

engineer-to-order (see ETO)

Enhanced Data Rates for GSM (or Global) Evolution (see EDGE)

Enhanced Integrated Drive Electronics (see EIDE)

Enhanced Interior Gateway Routing Protocol (see EIGRP)

Enhanced Messaging Service (see EMS)

Enhanced Parallel Sysplex Offering (see EPSO)

enhanced phone

A voice-centric device designed to deliver data content via network-based — e.g., Wireless Application Protocol (WAP) — Web delivery mechanisms that offers only minimal offline capability, such as contact management.

enhanced private switched communications service (see EPSCS)

ENQ (enquiry)

In data communications, a request for a response from another terminal. It is used to obtain identification or an indication of the other station's status.

ENS (enterprise nervous system)

Gartner's term for the intelligent network that provides unifying connectivity among people, application systems and devices in different locations and business units across a virtual enterprise. The emerging ENS is based on the traditional enterprise network, but it is an evolution of that network, providing value-added functions that elevate the role of the network well beyond that of plain communication. Whereas a conventional network simply aims to transfer data between sending application systems and explicitly defined destinations, an ENS offloads work from the application systems because it:

- Offers enhanced quality-of-service for communication

- Transforms messages
- Redirects messages as appropriate, using logical business rules
- May track and control business processes

enterprise application outsourcing (see EAO)

enterprise application development (see E-AD)

enterprise application integration (see EAI)

enterprise applications

Software products designed to integrate computer systems that run all phases of an enterprise's operations to facilitate cooperation and coordination of work across the enterprise. The intent is to integrate core business processes (e.g., sales, accounting, finance, human resources, inventory and manufacturing). The ideal enterprise system could control all major business processes in real time via a single software architecture on a client/server platform. Enterprise software is expanding its scope to link the enterprise with suppliers, business partners and customers.

enterprise asset management (see EAM)

enterprise business intelligence suite (see EBIS)

enterprise-class

A term referring to the ability of a given tool or product to handle complex processes or services.

enterprise computer-aided software engineering (see E-CASE)

enterprise console

A key component of event-management systems, comprised of applications' component maps, autodiscovery mechanisms, agents that send information to a manager and a correlation engine (at the agent level, the mid-level manager and the console level).

enterprise content

An all-encompassing term for print and digital (i.e., text, video and audio) information resources, internally produced (i.e., internal documents and discussion groups) and externally available (i.e., news, research and reference).

enterprise cooperation community (see ECC)

Enterprise Data Access/Structured Query Language (see EDA/SQL)

Enterprise Definition Language (see EDL)

enterprise demand planning (see EDP)

enterprise desktop manager (see EDM)

enterprise grade

Products that integrate into an infrastructure with a minimum of complexity and offer transparent proxy support.

enterprise information architecture (see EIA)

enterprise information management (see EIM)

Enterprise JavaBeans (see EJB)

enterprise knowledge architecture (see EKA)

Enterprise Maintenance Planning and Control (see EMPAC)

enterprise master person index (see EMPI)

Enterprise Messaging Server (see EMS)

enterprise messaging utility (see EMU)

enterprise nervous system (see ENS)

enterprise operations center (see EOC)

enterprise performance management (see EPM)

enterprise portal (see EP)

Enterprise Ready Applications (see ERA)

enterprise resource planning (see ERP)

enterprise resource planning II (see ERP II)

enterprise scheduling system (see ESS)

Enterprise Server (see ES)

Enterprise Server Offering (see ESO)

enterprise server platform (see ESP)

Enterprise Service Agreement (see ESA)

enterprise solutions

Solutions designed to integrate multiple facets of a company's business through the interchange of information from various business process areas and related databases. These solutions enable companies to retrieve and disseminate mission-critical data throughout the organization, providing managers with real-time operating information.

Enterprise Storage Array (see ESA)

Enterprise Storage Server (see ESS)

enterprise support management (see ESM)

Enterprise Systems Architecture (see ESA)

Enterprise Systems Connection (see ESCON)

Enterprise TCO (see ETCO)

enterprise user administration (see EUA)

Enterprise Volume Manager (see EVM)

enterprisewide web (see EWW)

entropy encoding

A lossless compression method that treats the entire data stream as a series of bits and takes no account of the information being compressed.

Entry Server Offering (see ESO)

envelope delay

An analog line impairment that conditioning is meant to overcome. It is a variation of signal delay with frequency across the data channel bandwidth.

environmental, safety & health (see ES&H)

EOC (enterprise operations center)

The concept of managing multitechnology computing platforms. Not necessarily a centralized facility, but rather the organizational and process management of enterprise IT assets and operations.

EOD (end of data)

EOF (end of file)

EOM (end of message)

EOM (Enterprise Output Management)

E-OO4GL (enterprise object-oriented fourth-generation language)

E-OOA&D (enterprise object-oriented analysis and design)

EOP (end of partition)

EOQ (economic order quantity)

EOS (electronic output strategy)

An enterprise's integrated policies, procedures and infrastructure for printing and publishing electronically generated documents.

EOT (end of tape)

EOT (end of transmission)

EOTD (Enhanced Observed Time Difference)

The European Telecommunications Standards Institute's (ETSI's) standard for location technology. Location is calculated at a central computer using the time difference between the arrival of signals from the handset and from the location measurement unit.

EP (enterprise portal)

Internet technologies that provide windows into enterprise information, applications and processes. EPs go by many names, including corporate portals, business portals and enterprise information portals. There are two types: horizontal enterprise portals (HEPs) and vertical enterprise portals (VEPs).

EPABX (electronic private automatic branch exchange)

e-partner selling (see ESE)

EPC (event-driven process chain)

EPD (electronic product definition)

EPF (enhanced platform feature)

EPHOS (European Procurement Handbook for Open Systems)

A set of profiles published by the European Commission to aid the buying process of Open Systems Interconnection (OSI) systems.

EPIC (Explicitly Parallel Instruction Computing)

An effort by Intel and Hewlett-Packard to create an open terminology that the industry would use to represent the latest iteration of wide-word technology. It is the core technology used for IA-64.

EPM (enterprise performance management)

The process of monitoring performance across the enterprise with the goal of improving business performance. An EPM system integrates and analyzes data from many sources, including, but not limited to, e-commerce systems, front-office and back-office applications, data warehouses and external data sources. Advanced EPM systems can support many performance methodologies such as the balanced scorecard.

EPM (Enterprise Production Management)

EPMS (Electronic Publishing Management System)

EPOS (electronic point of sale)

EPP (Enhanced Parallel Port)

e-print procurement

A Web-enabled ordering mechanism that automates many of the steps involved in ordering and managing printed materials.

e-procurement

A set of applications and business-to-business information management processes that support the purchase of goods and services over the Internet.

EPROM (erasable programmable read-only memory)

Memory chips programmed after manufacture which may be reprogrammed at a later date.

EPS (Encapsulated PostScript)

EPSCS (enhanced private switched communications service)

EPSCS is a service designed to provide customized private communications to customers. Switching centers are located on telephone company premises and may be shared with other EPSCS customers or other telephone company services. Private-line channels between switching centers and a customer's premises are dedicated to specific customers.

EPSO (Enhanced Parallel Sysplex Offering)

An IBM package designed to help users migrate to a Parallel Sysplex environment through the packaging of hardware, maintenance and services. Unlike IBM's Enterprise Server Offering (ESO), it does not include software.

e-purse (electronic purse)

An application that allows value (e.g., e-cash) to be loaded into a smart card or handheld device that then can be used to make purchases. A device or card can hold multiple e-purse applications designed for specific uses (e.g., one e-purse on a student card or device can be restricted to purchases at a bookstore).

equal access

The U.S. Department of Justice ruling, effective September 1984, that requires most local telephone companies to offer the same quality of connection, at the same rates, to all common carriers.

equalization

Compensation for the attenuation (signal loss) variation with frequency (attenuation equalization) and propagation time variations with frequency (delay equalization).

equipment asset management (see EAM)

equipment management system (see EMS)

ER (entity relationship)

ER (explicit rate)

In an asynchronous transfer mode (ATM) network, one of two feedback mechanisms used by the available bit rate (ABR) service category to inform the source of the network resources available to it.

ERA (Enterprise Ready Applications)

The industry standard unit used for calculating line voice capacity and traffic; it is used in making predictions about randomly arriving workload and the resources needed to handle it.

erasable optical storage device

An optical storage device on which data can be rewritten, either after bulk erasure or spot erasure.

erasable programmable read-only memory (see EPROM)

ERC (European Radiocommunications Committee)

Established in 1990, in response to concern expressed by the Commission of the European Community (CEC) at the slow progress being made in European frequency harmonization. The ERC brings together the radio regulatory authorities of the European Conference of Postal and Telecommunications Administrations (CEPT) countries.

ERD (entity-relationship diagram)

e-retailing

Online retailing; also known as "e-tailing."

ERL (echo return loss)

Attenuation of echo currents in one direction caused by telephone circuits operating in the other direction.

Erlang

In telecommunications, a unit of traffic intensity. One Erlang is the intensity at which one traffic path would be continuously occupied (e.g., one call-hour per hour or one call-minute per minute). Generally referred to as 36 CCS. Related terms include:

- Erlang B: A traffic-engineering formula used when traffic is random and there is no queuing. It assumes that blocked callers either automatically use another route or blocked calls disappear entirely.
- Erlang C: A traffic engineering formula used when traffic is random and queuing is provided. It assumes that all callers will wait indefinitely until a line becomes available.

ERM (employee relationship management)

ERM focuses on optimizing the employee's "total employment experience" — including both the human and technology aspects of that experience. ERM includes manager and employee interactions, the formal business tasks required manage employee relationships, and the technology used to manage the employee experience. Thus, ERM is most closely aligned with the human capital management (HCM) focus area of workforce management. See HCM.

ERMES (European Radio Message System)

A program sponsored by the Commission of the European Community (CEC) for a Pan-European radio (i.e., paging) message system.

ERMS (e-mail response management system)

A customer service and support (CSS) software feature for handling and managing e-mail from customers. It integrates e-mail and Web forms into the incoming call flow to ensure that these inquiries and transactions are handled on a timely basis.

ERO (enterprise resource optimization)

ERO (European Radiocommunications Office)

Established in 1991, the ERO is responsible for administrative regulatory radio and spectrum matters. The ERO is a permanent organ of the European Conference of Postal and Telecommunications Administrations (CEPT) and is managed by the European Radiocommunications Committee.

ERP (enterprise resource planning)

1. A business strategy that integrates manufacturing, financial and distribution functions to dynamically balance and optimize the enterprise's resources.
2. An integrated application software suite that balances manufacturing, distribution and financial business functions. ERP is the technological evolution of manufacturing requirements planning (MRP) II through the introduction of relational database management systems (RDBMSs), computer-aided software engineering (CASE), fourth-generation language (4GL) development tools and client/server architecture. When fully implemented, ERP can enable enterprises to optimize their business processes and allows for necessary management analysis and appropriate decision making in a quick and efficient manner. As more-robust technology is implemented, ERP improves an enterprise's ability to react to market changes.

ERP II (enterprise resource planning II)

An application and deployment strategy that expands from traditional enterprise resource planning (ERP) functionality to achieve integration of an enterprise's key internal and external collaborative, operational and financial processes. ERP II starts as an application strategy, setting a vision for the integration of all enterprise-centric, commerce-oriented business processes, without requiring a single-vendor strategy. As a deployment strategy, ERP II enables enterprises to determine the degree of single-vendor-centricity needed to fulfill enterprise process requirements and — through native integration capabilities — to include best-of-breed components from multiple vendors. ERP II includes capabilities specific to the enterprise, as well as the ability to connect the enterprise to key business partners directly or through a private e-marketplace. The process domain of ERP II includes all collaborative, operational and financial processes that have the enterprise at the center. See ERP and e-marketplace.

error

In data communications, any unwanted change in the original contents of a transmission.

error burst

A concentration of errors within a short period of time as compared to the average incidence of errors. Retransmission is the normal correction procedure in the event of an error burst.

error control

An arrangement that detects the presence of errors. In some systems, refinements are added that correct the detected errors, either by operations on the received data or by retransmission from the source.

error-correcting code (see ECC)

error rate

The ratio of the number of bits, elements, characters or blocks incorrectly received to the total number of bits, elements, characters or blocks transmitted.

ERS (evaluated receipt settlement)

The process for paying for goods on receipt, rather than upon invoicing (sometimes referred to as “two-way matching”).

ERSO (Electronics Research and Service Organization)

ES (Enterprise Server)

Netscape's Hypertext Transport Protocol (HTTP) server.

ES&H (environmental, safety and health)

A collection of software applications dealing with regulatory compliance for Environmental Protection Agency (EPA), Occupational Health and Safety Administration (OSHA), state and local reporting.

ESA (Enterprise Service Agreement)

IBM umbrella contracts that bundle different services to different customer needs.

ESA (Enterprise Storage Array)

A Small Computer Systems Interface (SCSI) disk array introduced by Digital Equipment (now part of Hewlett-Packard).

ESA (Enterprise System Adapter)

An interface from Ariba that integrates with major enterprise resource planning (ERP) vendors' offerings allowing for the exchange of data between Operating Resource Management System (ORMS) and back-end systems.

ESA (Enterprise Systems Architecture)

ESA/370 is IBM's System/370 extended architecture for the 3090 and 4381 processors, and is supported by MVS/ESA (Multiple Virtual Storage/ESA). The major ESA feature is the support of data spaces in addition to the program-and-data address spaces of System/370.

Esc (escape) key

A control character whose function varies with the software or type of terminal. It generally serves to transfer control to another part of the program or to exit the program.

escape character

A symbol that indicates that the next character is a member of a different character set or a control character.

ESCON (Enterprise Systems Connection)

A high-speed fiber-optic serial channel for IBM's ES/9000 processors announced in September 1990. ESCON was initially based in part on a fiber-optic link operating at a speed of 200 megabits per second (Mbps) regardless of the driver light source, but has been driven much faster.

ESD (electronic software distribution)

ESD enables software to be installed by transmitting it over a network. It is designed to help users distribute programs and files in their environments. The development of client/server and mobile client/server applications has made ESD a critical requirement. Without an effective means of automating the distribution and installation of software, most applications of client/server and mobile client/server technology will not be viable.

ESD (electrostatic discharge)

ESDI (Enhanced Small Device Interface)

ESDS (electronic software delivery system)

ESE (extended selling enterprise)

Also known as e-partner selling, a group of applications and technologies provided by the enterprise to assist third-party selling-channel partners — e.g., brokers, agents, distributors and value-added resellers (VARs) — in achieving selling objectives. E-partner selling is a component of partner relationship management (PRM), which is itself a component of customer relationship management (CRM). Specifically, it is an enterprisewide business strategy designed to optimize profitability, revenue and partner satisfaction by organizing the enterprise around partner segments, fostering partner-satisfying behaviors and linking processes from customers to partners through suppliers.

e-service

Applications and tools that empower customers, partners and prospects for self-service via the Web, Internet, intranet or extranet. Interactive customer service Web sites are integrated with the front-end servicing applications (customer service, sales and marketing), back-end databases and the contact center, and facilitate interaction between users and the enterprise.

e-services

Another name for Web services (see separate entry).

ESF (Extended Super Frame)

AT&T's and Bell Communications Research's framing standards that improve networking performance monitoring. ESF is a T1 format that is comprised of 24 frames of 192 bits each. A 193rd bit is used for link control and error checking.

e-signature (see electronic signature)

ESIOP (Environment-Specific Inter-ORB Protocol)

ES-IS (End System to Intermediate System)

Open Systems Interconnection (OSI) routing protocol defining the contact between systems at the center (i.e., hosts) and in the middle of the network.

ESM (enterprise support management)

A software tool that can assist in automating basic and advanced e-mail-handling functions.

ESM (European Satellite Multimedia Services)

ESMR (Enhanced Special Mobile Radio)

ESMS (enterprise system managed storage)

ESMTP (Extended Simple Mail Transfer Protocol)

Extensions to the Simple Mail Transfer Protocol (SMTP) that provide faster delivery and message status.

ESO (Enterprise Server Offering)

A marketing package for IBM's S/390 complementary metal-oxide semiconductor (CMOS) offering, which combines hardware, software, maintenance and services (formerly called "Entry Server Offering").

ESP (enterprise server platform)

An inherently distributable, multiply interoperable server engineered for use within the context of a larger information technology architecture. It transparently provides services to any authorized client. ESP is a collection of services that enable enterprises to manage and leverage the benefits of personal, workgroup and enterprise information processing resources.

ESP (Enterprise Storage Platform)

ESP (external service provider)

A company that provides services such as consulting, outsourcing and software services (including system integration and application service provision). An ESP, which is a separate legal entity from the contracting company, supplements the skills and resources of that company's in-house IS department.

ESPRIT (European Strategic Program for Research and Development of Information Technology)

A research and development (R&D) program, funded by the European Community, that was chartered to create a technology base for the development of products, processes and services for European companies in an effort to gain world information market share.

ESQL (Embedded Structured Query Language)

SQL statements inserted within a program during its preparation that will not change when the program is run.

ESS (electronic switching system)

A circuit switch environment that has supported delivery of services through telecommunications networks for most of this century. The architecture is now changing over to asynchronous transfer mode (ATM).

ESS (employee self-service)

Once simply defined as employee access to human resources (HR) for review or update of benefits or address information that had previously been in an HR staff-

controlled domain. The definition has now expanded to include “look up” or edit access to many other areas such as employee time tracking, direct deposit account changes for payroll, and skills and training updates. The boundaries of this new “open” access will be stretched even further as companies demand increased productivity and real-time information exchange.

ESS (Enterprise Storage Server)

A storage disk subsystem from IBM.

ESS (enterprise scheduling system)

A complex system that interacts with enterprise applications such as enterprise master person indices (EMPIs), computer-based patient record systems (CPRs), clinical decision support systems (CDSSs) and departmental scheduling systems. To achieve a successful ESS implementation, a care delivery system (CDO) must consider not only the features that the ESS supports, but also how it interoperates with the CDO’s environment.

ESS (enterprise storage service)

Technology that provides data storage for application service providers (ASPs). It is geared to online or Web-facing data needs.

ESW (externally sourced worker)

ETACS (Extended TACS)

See TACS.

e-tag (electronic tag)

A machine-readable tag with an embedded chip that provides information via radio signals. E-tags, also known as radio frequency identification (RFID) tags, are a form of auto-identification technology. See RFID and auto-identification technology.

e-tailing

Online retailing; short for “e-retailing.”

ETCO (Enterprise TCO)

A new Institute of Management Accountants (IMA) work point accounting standard that adds direct and indirect costs to the total cost of ownership (TCO) of IT infrastructure framework, including direct and indirect occupancy costs, home office setup, direct and indirect non-IT equipment, and direct and indirect non-IT support costs.

ETG (European Telecommunications Group)

Ethernet

A baseband local-area network (LAN) originally developed by Xerox and supported by Intel, Digital Equipment (later acquired by Compaq Computer) and Hewlett-Packard. It has a bus topology with carrier sense multiple access with collision detection (CSMA/CD) access control. Ethernet is not identical to Institute of Electrical and Electronics Engineers (IEEE) 802.3. Related terms include:

- Ethernet address: 48-bit code for Layer 2 networking maintained by the IEEE and hardwired into network adapters; also called MAC address.

- Ethernet, thick: Ethernet coaxial cable suitable for networks that are medium/large or with widely spaced nodes.
- Ethernet, thin: Ethernet coaxial cable suitable only for small networks with closely connected nodes.

ETL (extraction, transformation and loading)

Tools for extracting data and its metadata from one data store, transforming the record structure and content of this data, and loading the transformed data to another data store. These tools are sometimes referred to as extraction/transformation/transport (ETT) technology.

ETN (electronic tandem network)

A private telecommunications network in which calls are automatically switched over specific trunks.

ETNO (European Public Telecommunications Network Operators' Association)

Established in 1992, by European fixed public telecommunications network operators, ETNO has taken over the operational interests relating to telecommunications, which were formerly the responsibility of the European Conference of Postal and Telecommunications Administrations (CEPT).

ETO (engineer-to-order)

Configurable product offerings that consist of standard and custom-engineered components.

ETS (electronic trading system)

ETS (Engineering and Technical Systems)

ETSI (European Telecommunications Standards Institute)

A not-for-profit enterprise whose mission is to produce the telecommunications standards that will be used throughout Europe. Some of the standards developed by the ETSI may be adopted by the European Commission as the technical base for directives or regulations. The ETSI's main task is to remove any possible variation from a global standard and to focus on a defined European-specific set of requirements. The ETSI also ensures that there is interoperability between standards, such as Integrated Services Digital Network (ISDN), Global Systems for Mobile Communications (GSM) and Universal Mobile Telecommunications System (UMTS).

ETT (extraction/transformation/transport)

See ETL.

EU (European Union)

EUA (enterprise user administration)

A category of security tools (also known as consolidated user administration or consolidated security administration tools) that enable security administrators to more easily manage a large number of permissions on behalf of users. Ideally, adding a new user might be achieved with a single command. EUA tools reduce the

workload of the security administrator, and reduce turnaround time between request and implementation — in some cases to less than 24 hours.

EUC (end-user computing)

Computer operations or activities carried out by the end user, often incurring hidden costs.

EULA (End-User License Agreement)

EUNIS (European University Information Systems Organisation)

euro

The pan-European currency of European economic and monetary union (EMU).

European Article Numbering (see EAN)

European Association of Securities Dealers Automated Quotations (see EASDAQ)

European Committee for Telecommunications Regulatory Affairs (see ECTRA)

European Computer Manufacturers Association (see ECMA)

European Conference of Postal and Telecommunications Administrations (see CEPT)

European Procurement Handbook for Open Systems (see EPHOS)

European Public Telecommunications Network Operators' Association (see ETNO)

European Radiocommunications Committee (see ERC)

European Radiocommunications Office (see ERO)

European Radio Message System (seeERMES)

European Strategic Program for Research and Development of Information Technology (see ESPRIT)

European Telecommunications Standards Institute (see ETSI)

European Workshop on Open Systems (see EWOS)

EuroSmart

A method developed by Gartner that provides users with a way of estimating the size of economic and monetary union (EMU) information technology (IT) needs.

EUT (end-user training)

Classroom, desktop-based or printed manual instruction in the operation of computer technology.

EUTELSAT (European Telecommunications Satellite Organization)

EVA (economic value added)

A methodology developed by Stern Stewart & Co. — a New York management consulting firm — that measures corporate economic performance. It views investments (including IT investments) in terms of their impact on shareholder wealth — that is, the increase (or decrease) in enterprise wealth that results from an investment.

even parity check/odd parity check

A test determining whether the number of digits in a group of binary digits is even (i.e., even parity check) or odd (i.e., odd parity check).

EVM (Enterprise Volume Manager)

A product that offers point-in-time copy and other advanced functions for Hewlett-Packard's (formerly Compaq's) RA8000 and ESA12000 RAID storage systems.

e-voucher (electronic voucher)

An application that allows the sending of an electronic version of a voucher to a consumer's handheld device or mobile phone. Users can then "carry" the e-voucher in their devices for use at an online store or a traditional business.

EW (early warning)

EW (electronic warfare)

EW3 (Easy World Wide Web)

e-wallet (electronic wallet)

Residing either on a device or on a server, this software application stores personal information (e.g., passwords and shipping addresses), digital certificates and information for a variety of payment instruments (e.g., credit card or e-cash) used for e-commerce transactions. The information can be automatically applied to payment and other Web transactions.

EWDP (Enhanced Web Development Package)

EWOS (European Workshop on Open Systems)

When fully operational, this will provide a flexible, responsive and open framework for the rapid development of information technology (IT) standards.

EWV (enterprisewide web)

Internal implementations of Web-based system components and standards that support basic intraenterprise publishing, broadcasting and information-sharing capabilities.

exchange

The assembly of equipment in a communications system that controls the connection of incoming and outgoing lines and includes the necessary signaling and supervisory functions. Different exchanges, or switches, can be colocated to perform different functions (e.g., local exchange and trunk exchange). Special types of exchanges that may be installed within an enterprise include:

- Private exchange (PX): An exchange serving a particular organization and having no means of connection with a public exchange.
- Private automatic exchange (PAX): A dial telephone exchange that provides private telephone service to an organization and that does not allow calls to be transmitted to or from the public telephone network.
- Private branch exchange (PBX): private automatic telephone exchange that provides for the transmission of calls internally and to and from the public telephone network. Also called private automatic branch exchange (PABX).
- Private automatic telex exchange (PATX): Provides telex service within and without an enterprise (largely obsolete).

exchange area

A calling area served by a local telecommunications company central office switch; may be comprised of one or more area code and exchange (NXX) codes.

exchange line socket

A two- or four-wire socket providing physical access to the public switched telephone network (PSTN).

exciter

The transmitter component that modulates and drive a signal. It is used in uplinks.

executive information system (see EIS)

ExHPDM (Extended High Performance Data Mover)

expandor

A transducer that, for a given amplitude range of input voltages, produces a larger range of output voltages.

expansion card

Circuit boards that can be inserted into the computer to increase its capabilities. Also called expansion board.

expert system

A software system that can learn new procedures by analyzing the outcome of past events, or that contains a knowledge base of rules that can be applied to new data or circumstances not explicitly anticipated by the developer. Applications include network management, database management and data mining, computer vision and image processing, speech recognition, biometrics and software for complex evaluation in such fields as petroleum geology.

Explicit Forward Congestion Indicator (see EFCI)

explicit input

The definition of model dimensions using explicit numbers as opposed to variables.

Explicitly Parallel Instruction Computing (see EPIC)

explicit rate (see ER)

explicit tagging

A frame is classified as belonging to a particular virtual local-area network (VLAN), based on a VLAN tag value that is included in the frame.

explosion

An extension of a bill of material into the total of each of the components required to manufacture a given quantity of upper-level assembly or subassembly.

Extended Architecture (see XA)

Extended Area Service (see EAS)

Extended Binary-Coded Decimal Interchange Code (see EBCDIC)

extended data output (see EDO)

Extended Graphics Array (see XGA)

Extended Industry Standard Architecture (see EISA)

Extended Java VM

A Java virtual machine (VM) certified by Sun Microsystems as conforming to the VM specification but adding functionality or application programming interfaces (APIs) that may be proprietary.

Extended Recovery Facility (see XRF)

extended remote copy (see XRC)

Extended Super Frame (see ESF)

Extensible Form Definition Language (see XFDL)

Extensible Markup Language (see XML)

Extensible Markup Language/Electronic Data Interchange (see XML/EDI)

Extensible Stylesheet Language (see XSL)

Extensible Stylesheet Language Transformations (see XSLT)

Exterior Gateway Protocol (see EGP)

external function testing

The verification of the specifications and performance of devices located outside of the computer.

external service provider (see ESP)

extraction, transformation and loading (see ETL)

extraction/transformation/transport (ETT) — see ETL.

extranet

A collaborative, Internet-based network that facilitates intercompany relationships by linking an enterprise with its suppliers, customers or other external business partners. Extranets use Internet-derived applications and technology to provide secured extensions of internal business processes to external business partners. See intranet.

extranet VPN (extranet virtual private network)

Extranet VPNs connect business partners, enterprises in the distribution chain or even customers to the enterprise network.

F

FA (financial administrator)

FA (fixed assets)

A system that records the financial value of enterprise assets and calculates depreciation expenses.

fabric-attached storage (see FAS)

face recognition

A biometric technique that uses the geometry of a person's face to authenticate identity. Limited deployments, such as those for check cashing, are in place.

Face Recognition Technology (see FERET)

facilities administration and control

Enables PBX users to administer the assignment of parameters that determine user calling privileges, such as restriction levels and authorization codes.

facilities assurance reports

Enables PBX users to obtain an audit trail of the data generated by the automatic circuit assurance feature.

facilities management (see FM)

facilities restriction level

The parameter associated with each authorization code, each station at a private branch exchange (PBX), each incoming tie trunk group from subtending PBXs, and attendants as a whole, which determines the types of calls and types of facilities within the privileges of the associated user.

facility

A transmission path between two or more points, provided by a common carrier.

facsimile (see fax)

fading

1. A phenomenon, generally of microwave or radio transmission, where atmospheric, electromagnetic, or gravitational influences cause a signal to be deflected or diverted away from the target receiver.
2. The reduction in intensity of the power of a received signal.

failsafe

Internal circuitry that monitors a system and shuts it down or alerts the operator in the event of a problem.

fail softly

When a piece of equipment fails and the system falls back to a degraded mode of operation rather than shutting down.

failure

Any anomaly of a system that results in an impairment of a mission-critical business process and:

1. A hard stop (abend) of any duration.
2. Incorrect calculations causing incorrect output.

Fairchild Advanced Schottky TTL (see FAST)

fair market value (see FMV)

FAQ (frequently asked questions)

A list of questions with their answers posted regularly over the Internet. Usually applied to Usenet.

far-end crosstalk

Crosstalk that travels along a circuit in the same direction as the signals in that circuit. See near-end crosstalk.

FAS (fabric-attached storage)

An umbrella term used by Gartner to encompass emerging storage area network (SAN) and network-attached storage (NAS) architectures. Gartner believes that FAS will continue to gain share because, unlike direct-attached storage (DAS), its architectures permit many servers and clients to share common storage assets, improve storage utilization efficiency, and expand client access to mission-critical data, while reducing total cost of ownership. See DAS, NAS and SAN.

FAST (Fairchild Advanced Schottky TTL)

A type of integrated circuit.

Fast Ethernet

The Institute of Electrical and Electronics Engineers (IEEE) standard based on 802.3 but running at 100 megabits per second (Mbps) instead of 10 Mbps and using 100Base-T or 100Base-F families of cabling standards.

fast-packet switching

A generic term for improved packet-switching technologies such as frame relay and cell relay. Fast-packet techniques feature less functionality than traditional X.25 packet-switching for higher packet-switching speeds and lower processing costs.

FAT (file allocation table)

An index for mapping logical files to physical locations on a disk storage device.

fat client

In client/server computing, a client is said to be "fat" (or "thick") when it handles a high amount of the application logic and data management tasks, rather than having these tasks executed remotely on the server. See client/server and thin client.

fault detection and isolation

Online diagnostics that detect and isolate faults in real time, prevent contamination into other areas, and attempt to retry operations.

fault tolerant or fault tolerance (see FT)

fax (facsimile)

The system or equipment used for the transmission of images, usually over the public telephone network. The image is scanned at the transmitter, reconstructed at the receiving station, and duplicated on some form of paper.

FBIA (financial business intelligence applications)

FBU (functional business unit)

FC (feature code)

FC (Fibre Channel)

A high-speed serial communication protocol developed by IBM and others; usually used for storage subsystems. The transmission medium may be fiber optics or copper.

FC25 (Fibre Channel 25 MB/sec.)

FC-AL (Fibre Channel-Arbitrated Loop)

A subset of the Fibre Channel Standard that was defined to support network storage. It is an inexpensive technology that supports up to 126 devices that can be configured in a fibre channel ring network and arbitrate for network access. Only one pair of devices can talk at one time on the loop. Each port on the loop sees all messages and ignores those not intended for it.

FCC (Federal Communications Commission)

The U.S. federal agency responsible for regulating interstate telecommunications, as well as international telecommunications, aspects of cellular communications and broadcasting. The FCC was established by the Communications Act of 1934.

FC-EL (Fibre Channel-Enhanced Loop)

An expanded version of FC-AL aimed at extending loop topologies to support additional applications.

FC-IC (Fibre Channel Interconnect)

FCIF (Full Common Intermediate Format)

Part of the H.261 worldwide video code standard, which is designed to accommodate both the North American National TV System Committee (NTSC) and European PAL protocols, that is, the accommodation is accomplished by using the NTSC frame rate and the PAL resolution in a compromise called the Common Intermediate Format (CIF), of which there are two versions: Full CIF (FCIF) and Quarter CIF (QCIF). Full

CIF pictures will be sharper in the future, but at the cost of additional processing power requirements.

FCLC (Fibre Channel Loop Community)

FCO (field change order)

FCP (Foundation for Cooperative Processing)

FCS (Fibre Channel Standard)

A definition of Fibre Channel specifications undertaken by the American National Standards Institute (ANSI) Technical Committee, now known as X3T11. The developers wanted to provide an extremely flexible method of information transfer that can support both channel-style and network-style peripheral interconnections.

FCS (first customer shipment)

The stage at which a product is released from development, and ready for beta testing.

FCS (frame check sequence)

A checking code used to detect data corruption.

FCSS (Fibre Channel RAID Storage Server)

A storage solution for environments that require Fibre Channel attachment. This storage system can be attached to servers running AIX, Sun Solaris, HP/UX, Windows NT/2000, or Novell NetWare — and it can be shared simultaneously by two operating systems, such as Unix and Windows NT/2000. Multiple server attachments can be implemented via the attachment of the IBM Storage Area Network (SAN) Fibre Channel Switch or the IBM Fibre Channel Storage Hub.

FD or FDX (full duplex)

Refers to a communications system or equipment capable of transmitting simultaneously in both directions. See half duplex.

FDC (financial data control)

FDC (First Data Corp.)

FDD (flexible disk drive)

A low-cost, low-capacity (usually 1.4 megabytes or less) removable disk storage device. See floppy disk.

FDDI (Fiber Distributed Data Interface)

An American National Standards Institute (ANSI) standard for 100 megabits per second (Mbps) fiber-optic local-area networks (LANs). Incorporates token processing and supports circuit-switched voice and packetized data.

FDDI XTP (Fiber Distributed Data Interface Express Transfer Protocol)

Express Transfer Protocol for use with FDDI.

FDM (frequency division multiplexing)

The division of a transmission frequency range into narrower bands to create two or more channels, enabling each data source to have its own channel.

FDMA (frequency division multiple access)

Communicating devices at different locations sharing a multipoint or broadcast channel by means of a technique that allocates different frequencies to different users.

FDR (Fast Dump Restore)

FDRSOS (Fast Dump Restore Safeguard Open Storage)

FE (field engineer)

An engineer who can work on site.

FERC (Federal Energy Regulatory Commission)

feature-based modeling

A modeling technique that allows the creation of manufacturing form features, e.g., through holes, ribs or slots, when defining mechanical component models.

feature-driven manufacturing

The ability to automatically create optimized process instructions and machine tool paths by utilizing mechanical component form features.

FEC (forward error correction)

FECN (forward explicit congestion notification)

Federal Communications Commission (see FCC)

Federal Information Processing Standard (see FIPS)

FEDI (financial electronic data interchange)

Information that passes between banks and companies that relates directly to payments and bank balances.

feedback

A method of signal regeneration involving a coupling from a high-level point in an amplifier to a lower-level point in the same or previous stage, in such a manner as either to increase or decrease apparent gain of the amplifier.

feed horn

A small antenna, located at the focal point of a reflector antenna, which gathers the signal.

FEMA (Federal Emergency Management Agency)

FEP (front-end processor)

A dedicated communications system that intercepts and handles activity for the host. It may perform line control, message handling, code conversion, error control, and such application functions as control and operation of special-purpose terminals. See communication controller.

FEPI (front-end programming interface)

FERET (Face Recognition Technology)

A comprehensive U.S. government face recognition test, used for personnel security.

FERRAM (ferroelectric random-access memory)

An early form of random access memory used in early IBM mainframes. It consisted of tiny ferrite rings that could be magnetized by electric pulses to indicated a binary "1" bit. Also known as core memory.

ferrule

The tip of a fiber-optic connector.

FET (field-effect transistor)

FFA (field force automation)

FFAPI (File Format Application Programming Interface)

The file format used only by Microsoft Mail is held as proprietary to Microsoft. In order to read or write such a file, the FFAPI utilities provided by Microsoft must be used. The utility creates the message header, or envelope, by writing field tags and data to the file in the Microsoft Mail structured data format. Another utility extracts the data by parsing the data structure.

FFS (fee for service)

FFT (fast Fourier transform)

FH (frequency hop)

FHSS (frequency-hopped spread spectrum)

FI (financial institution)

FIB (forward indication bit)

An Integrated Services Digital Network (ISDN) term.

Fiber Distributed Data Interface (see FDDI)

Fiber Distributed Data Interface Express Transfer Protocol (see FDDI XTP)

Fiber Link Around the Globe (see FLAG)

Fiber Optic 10BASE-F

10 megabits per second (Mbps) over fiber; an Institute of Electrical and Electronics Engineers (IEEE) standard for 802.3.

Fiber Optic Inter-Repeater Link (see FOIRL)

fiber optics

A high-bandwidth transmission technology that uses light to carry digital information. One fiber telephone cable carries hundreds of thousands of voice circuits. These cables, or light guides, replace conventional coaxial cables and wire pairs. Fiber transmission facilities occupy far less physical volume for an equivalent transmission

capacity, which is a major advantage in crowded ducts. Optical fiber is also immune to electrical interference.

fiber-optic waveguides

Thin filaments of glass through which a light beam can be transmitted for long distances by means of multiple internal reflections. Occasionally, other transparent materials, such as plastic, are used.

fiber to the building (see FTTB)

fiber to the curb (see FTTC)

fiber to the home (see FTTH)

Fibre Channel (see FC)

Fibre Channel-Arbitrated Loop (see FC-AL)

Fibre Channel Connectivity (see FICON)

Fibre Channel-Enhanced Loop (see FC-EL)

Fibre Channel Standard (see FCS)

FICON (Fibre Channel Connectivity)

An IBM channel architecture introduced in 1998.

field engineer (see FE)

FIFO (first in, first out)

A method of transmitting data through a switch to minimize delay.

file

A collection of bytes containing data, text or other information, or software that is stored and accessed as a coherent unit. Examples include: data file or table, spreadsheet, document, image, program, audio and video.

file allocation table (see FAT)

file server

A computer containing files available to all users connected to a local-area network (LAN). In some LANs, a microcomputer is designated as the file server, while in others it is a computer with a large disk drive and specialized software. Some file servers also offer other resources such as gateways and protocol conversion.

File Transfer, Access and Management (see FTAM)

File Transfer Protocol (see FTP)

filter

A circuit designed to transmit signals of frequencies within one or more frequency bands, and to attenuate signals of other frequencies.

filtering

1. Restricting the results returned to a user seeking information through a Web search engine or other information retrieval system. It is often referred to more precisely as "content filtering" or "information filtering." Information may be filtered for relevance, using algorithms that analyze factors such as a term's context and its proximity to other terms. Content may also be filtered for appropriateness — for example, excluding material inappropriate for children.
2. A communications term, which refers to allowing only a single band of frequencies to pass — or to isolating and blocking unwanted signal energy (for example, echoes and other signal interference) — to improve transmission quality. Also known as "signal filtering."

financial analytical applications

A subset of business intelligence with a specific emphasis on financial process, including, but not limited to, budgeting, forecasting, expense allocation and cost/revenue analysis.

financial footprint

The amount of money paid, monthly or yearly, to a vendor to support a particular system or application. Because most hardware and software is easily upgradable, financial-footprint management involves managing a stream of recurring payments instead of physical assets.

Financial Products Markup Language (see FpML)

finger

A Unix command that traces any user logged onto the system. Widely used on the Internet.

fingerprint recognition technology

A technology by which a user places his or her finger on a small glass plate at which point the system captures a high-resolution optical image of the fingerprint, typically using a charge-coupled-device (CCD) camera. No ink is involved. The system then converts the image into a template containing a mathematical representation based on features of the image such as the minutiae (i.e., the points at which the ridges branch or end). The technology typically takes three to four samples of a fingerprint to make a template for enrollment into the system. For user verification, the system takes a live scan of the fingerprint, which it then compares to the stored template for the user. If the match is close enough (determined by an application-defined threshold), verification is successful.

finite loading

Conceptually, the term means putting no more work into a factory than the factory can be expected to execute. The term usually refers to a computer technique that involves automatic shop priority revision to level load operation by operation.

FIPS (Federal Information Processing Standard)

A set of standards produced by the National Institute of Standards and Technology setting forth protocols, encryption, interoperability, hardware, and other specifications for the federal government.

FIRE (Flexible, Intelligent Routing Engine)

A set of application-specific integrated circuits (ASICs) developed by 3Com to add wire speed routing and policy capabilities to its CoreBuilder 3500 and 9000 series and products.

firewall

An application or an entire computer (e.g., an Internet gateway server) that controls access to the network and monitors the flow of network traffic. A firewall can screen and keep out unwanted network traffic and ward off outside intrusion into a private network. This is particularly important when a local network connects to the Internet. Firewalls have become critical applications as use of the Internet has increased.

FireWire

Also known as Institute of Electrical and Electronics Engineers (IEEE) 1394, this is a high-speed serial bus similar to the Universal Serial Bus (USB), but much faster and more expensive to implement. Its primary supporters are Apple Computer and consumer electronics firms such as Fuji, Sony, Hitachi and Nintendo. These firms will use FireWire as the input/output (I/O) bus for next-generation digital products: camcorders, VCRs, TVs, digital cameras and game players.

firm planned order

A planned order that can be frozen in quantity and time. The computer is not allowed to automatically change it; this is the responsibility of the planner in charge of the item that is being planned. This technique can aid planners working with material requirements planning (MRP) systems to respond to material and capacity problems by firming up selected planned orders. Additionally, firm planned orders are the normal method of stating the master production schedule.

firmware

A category of memory chips that hold their content without electrical power and include ROM, PROM, EPROM and EEPROM technologies. Firmware becomes "hard software" when holding program code. The contents are generally permanent or semipermanent control coding implemented at a microinstruction level for an application program, instruction set, operating routine, or similar user-oriented function.

FIRST (Forum of Incident Response and Security Teams)

A nonprofit volunteer group whose goal is to foster cooperation and coordination in incident prevention across diverse sectors, and prompt rapid reaction to incidents. Through an annual conference and its Web site at www.first.org, the group promotes information sharing among its members and the incident response community at large.

first customer shipment (see FCS)

first-generation office systems

The class of office information systems (OIS) products dominated by dedicated word processors, such as Digital Equipment's DECmate, IBM's DisplayWriter and Wang's OIS. The period began with the first word processors (late 1970s) and lasted until the introduction of the first integrated office systems (1983).

first in, first out (see FIFO)

FISU (fill in signal unit)

An Integrated Services Digital Network (ISDN) term.

FIX (Financial Information Exchange)

fixed-measure trade item

An item always produced in the same predefined version (e.g., type, size, weight, content and design) that may be sold at any point in the supply chain.

FIXML (Financial Information Exchange Markup Language)

flag

A data bit or group of bits that can be set on or off to signal the status of a system or process.

FLAG (Fiber Link Around the Globe)

An undersea fiber-optic system using two fiber pairs, each operating at 5 gigabits per second (Gbps), with capacity for 120,000 64-kilobit per second (Kbps) digital circuits.

flame

An insult or angry remark in Internet e-mail or Usenet messages. A flame may contain obscene language, sarcasm or derogatory comments. It may be "shouted" by placing the words in capital letters. Flaming is considered rude and inconsiderate, but it frequently occurs in newsgroups and e-mail.

flash memory

A nonvolatile storage chip developed by Intel that enables easy electrical erasability and reprogramming, often used to remotely update routers or modems with new versions of software.

flat-file database

A database that is comprised of a file or several files without any attendant, application-independent server logic. Flat-file databases can be used by multiple concurrent users if hosted on a shared file-server.

flat-panel display (see FPD)

flat rate

A fixed payment for service within a defined area, independent of use, with an additional charge for each call outside the area.

Flexible, Intelligent Routing Engine (see FIRE)

flexible route selection (see FRS)

floating-point unit (see FPU)

floppy disk

A reusable, portable storage device that generally holds 1.4 megabytes of data. Floppy disks, sometimes called flexible disks or diskettes, were the primary method of software and data distribution beginning in the 1970s. They were originally eight inches wide and enclosed in a flexible envelope; a 5.25-inch version was later

introduced, and finally a 3.5-inch floppy disk encased in rigid plastic became the standard. CD-ROMs have become a popular alternative to floppy disks, especially for larger software and multimedia files, given their higher storage capacity. As network speeds have increased and Internet access has expanded, more software and data are now transmitted directly rather than stored.

FLOPS (floating-point operations per second)

flow control

Management of data transmission; an important part of asynchronous transfer mode (ATM).

Flow Control Layer

Layer 5 of the Systems Network Architecture (SNA) architectural model, which determines and manages interactions between session partners. Corresponds to the Session Layer of the Open Systems Interconnection (OSI) model.

flow manufacturing (see FM)

flow-through distribution

A process that allows the seamless flow of product and information through the supply chain.

FM (facilities management)

1. Entering into an agreement with a service supplier to manage internal company facilities such as telecommunications or data-processing services. Facilities management does not involve the transfer of ownership of facilities to the service provider.
2. For government enterprises, FM is the most prevalent type of contractual relationship where the vendor assumes responsibility for one or more information technology (IT) functions. Unlike outsourcing, it refers to the vendor operating and managing the assets (usually hardware and software), but not taking ownership. In a government-owned, contractor-operated deal, the government owns the equipment and operating systems, and the contractor operates the equipment with its personnel. Usually, the equipment is maintained in a government facility, but it can reside in a vendor-owned facility. In a government-owned, government-operated arrangement, the government owns the equipment and manages the operation using contractor personnel.

FM (financial management)

FM (flow manufacturing)

Flow manufacturing is a form of techniques and processes that make possible continuous throughput within selected manufacturing operations. FM requires a reallocation of work steps, direct management of the time required for these work steps, and a redesign of typical material requirements planning (MRP) floor operations to achieve a reduction in productions costs, work in progress (WIP) and time to market. FM is synonymous with flexible, lean and synchronous manufacturing and is very similar to continuous flow operations.

FM (frequency modulation)

One of three ways of modifying a sine wave signal to make it carry information. The sine wave or "carrier" has its frequency modified in accordance with the information to be transmitted. The frequency function of the modulated wave can be continuous or discontinuous.

FMA99 (Gramm-Leach-Bliley Financial Services Modernization Act of 1999)

FMS (financial management system)

A collection of integrated applications and technologies designed to provide a financial solution specific to organizational requirements.

FMV (fair market value)

The price for which equipment could be sold in an arm's length transaction between unrelated parties.

FNC (Federal Networking Council)

Focus

A fourth-generation language (4GL) from Information Builders. It belongs to a class of software called information generators. Focus boasts easy-to-use, English-language commands with online help and error correction. Focus features "talk" windows, which improve end-user productivity. It also provides integrated reporting, ad hoc queries, graphs, statistics, financial modeling and a spreadsheet for building decision support systems.

FODS (front-office decision support)

FOIRL (Fiber Optic Inter-Repeater Link)

Often known as Ethernet over fiber, FOIRL is made up of dual fiber links (transmit and receive), as defined within Institute of Electrical and Electronics Engineers (IEEE) 802.3.

foreign exchange (see FX)

formal standards

Specifications or styles that are approved by vendor-independent standards bodies, such as the American National Standards Institute (ANSI), the International Organization for Standardization (ISO), the Institute of Electrical and Electronics Engineers (IEEE) and the National Institute for Standards and Technology (NIST).

Format Specification Document Participant

A format-specific Linear Tape Open (LTO) license package for vendors that want competitive intelligence on the LTO technology.

formatting

The preparation of a storage medium with guidance information, synchronization information, and a structure for keeping or collecting information for a directory; this collection of material placed on the disk before user data is written is called a "format," and frequently also includes room for error correction check sums and rewriting of bad or updated sectors.

formatting code

Formatting code (e.g., bold or italic) is rudimentary markup that applications convert to presentation characteristics (e.g., typeface and colors).

form factor

The size and shape of a product. For example, most 5.25-inch Winchester drives have the same dimension so that they can fit interchangeably into computer cabinets. For the same reason, 5.25-inch optical drives are expected to have the same form factor as their predecessor Winchesters.

FORTRAN (Formula Translator)

A programming language developed primarily for numeric computations and chiefly used in mathematics, science and engineering. It was introduced in the 1950s as the first high-level language (i.e., closer to natural language than to machine language).

Forum of Incident Response and Security Teams (see FIRST)

forward channel

The communications path carrying voice or data from the call initiator to the called party.

forward error correction

A technique for correcting errors incurred in transmission over a communications channel by the receiver, without requiring the retransmission of any information by the transmitter. It typically involves a convolution of the transmitted bits and the appending of extra bits, using a common algorithm by both the receiver and transmitter.

forward indication bit (see FIB)

FOTS (fiber-optic transmission systems)

See fiber-optic waveguides.

four-wire circuit

A circuit using two pairs of conductors, one pair for the outbound channel and the other pair for the return channel.

FP (FrontPage)

A Hypertext Markup Language editor.

FP (function point)

Function points measure the size of an application system based on the functional view of the system. The size is determined by counting the number of inputs, outputs, queries, internal files and external files in the system and adjusting that total for the functional complexity of the system. Function point analysis, originally developed at IBM, has as an advantage its focus on measuring software produced in terms of functionality delivered to the end user, rather than in terms of development deliverables, which have no direct bearing on the end user.

FPD (flat panel display)

A core component for notebooks, liquid crystal display (LCD) flat panel displays are broadening their applications on the commercial desktops. There are five key FPD technologies available: LCD, gas plasma, electroluminescent (EL) display, field

emission display and digital micromirror devices. The most mature is LCD, which includes active-matrix liquid crystal display (AMLCD) and passive-matrix liquid crystal display (PMLCD).

FPDA (focus, plan, define, act)

FPE (FrontPage Extension)

Web server support for certain specific features of Microsoft's FrontPage (a Web site editor).

FPGA (field-programmable gate array)

FPLA (field-programmable logic array)

FpML (Financial Products Markup Language)

An Extensible Markup Language (XML)-based computer language introduced by J.P. Morgan and PricewaterhouseCoopers in June 1999.

FPS (frames per second)

FPU (floating-point unit)

A processor specially designed to manipulate floating-point values rather than integer values.

FR (frame relay)

An American National Standards Institute (ANSI) standard (T1S1) for an Integrated Services Digital Network (ISDN) packet-mode bearer service that defines a user-to-network interface. The two main benefits are bandwidth on demand and integrated access. The standard currently addresses data communications speeds up to 2 megabits per second (Mbps) over permanent virtual circuits. Frame relay transmits data packets at high speeds across a digital network encapsulated in a transmission unit called a frame. It requires a dedicated connection during the transmission period. It is used on wide area networks (WANS) and also in private network environments with leased lines over T-1 lines. Frame relay is faster than traditional networks, because it was designed for today's reliable circuits and performs less rigorous error detection. When circuits are less reliable, a great deal of network traffic is dedicated solely to correcting errors.

Fractional T1

Telecommunication service providing permanent leased lines operating at a bit rate between 64 kilobits per second (Kbps) and 1.544 megabits per second (Mbps). See T1.

FRAD (frame relay assembler/disassembler)

A communications device that formats outgoing data into the format required by a frame relay network. It performs a similar function in a frame relay network to a packet assembler/disassembler (PAD) in a packet switched network, i.e., it converts an asynchronous data stream into data units that can be transmitted over a frame relay network and vice versa. This enables asynchronous devices such as personal computers (PCs) to communicate over a frame relay network.

FRAM (ferroelectric RAM)

frame

1. In High-Level Data-Link Control (HDLC), the sequence of contiguous bits bracketed by and including opening and closing flag (01111110) sequences.
2. In data transmission, the sequence of contiguous bits bracketed by and including beginning and ending flag sequences.
3. A set of consecutive digit time slots in which the position of each time slot can be identified by reference to a frame alignment signal.
4. In a time division multiplexing (TDM) system, a repetitive group of signals resulting from a signal sampling of all channels, including any additional signals for synchronizing and other required system information.

frame check sequence (see FCS)

frame relay (see FR)

frame relay assembler/disassembler (see FRAD)

frame-relay-to-ATM service interworking (see FRASI)

Frame Relay User Network Interface (see FUNI)

framework

A style guide that defines the look, feel and interoperability of software applications.

framing

A control procedure used with multiplexed digital channels, whereby bits are inserted so the receiver can identify the time slots allocated to each channel. Framing bits can also carry alarm signals indicating specific alarm conditions.

FRASI (frame-relay-to-ATM service interworking)

A network-based function that enables frame relay traffic to be mapped to asynchronous transfer mode (ATM). This enables customers to have frame relay access services in some locations and ATM in others, while still permitting intercommunication between them.

free MIPS

Excess mainframe capacity.

Free Software Foundation

A license and development project on which many of today's open-source products, including Linux, is based. Free software licensing includes beta (recently developed software that is tested under customers' actual usage conditions but has not been made generally available), trial (generally available software that is provided to a customer for a distinct period of time, which, if the software performs to its specifications, customers are typically obligated to pay for after the trial), and free (software that is provided to a customer with no initial fees and which is usually provided in a bundled offering with other products and services; ongoing support fees for free software are usually chargeable).

frequency

An expression of how frequently a periodic wave form or signal repeats itself at a given amplitude. It can be expressed in hertz (Hz), kilohertz (kHz), megahertz (MHz), etc.

frequency coordination

International procedure to prevent interference between new and existing radio communications services. Under International Telecommunication Union (ITU) Radio Regulations, potential operators must consult countries and administrations whose services might be affected.

frequency division multiple access (see FDMA)

frequency division multiplexing (see FDM)

frequency modulation (see FM)

Frequency Shift Keying (see FSK)

frequency-to-voltage (see F/V)

frequently asked questions (see FAQ)

Fresnel zone

Zone of clearance used in calculations during the setting up of a microwave link. Although not directly between the transmitter and receiver, an object near the line-of-sight path can interfere with the propagated signal.

front-end processor (see FEP)

front-office suite

A set of integrated applications employing a common architecture and referencing a common database that enable customer service and support (CSS), sales and marketing.

FrontPage (see FP)

FrontPage Extension (see FPE)

frontware

Software that runs on a client workstation to provide better end-user interfaces for application programs running elsewhere.

FRS (flexible route selection)

Used with Centrex service to achieve least-cost routing.

FRU (field-replaceable unit)

FS (Future System)

FS was thought by some factions within IBM to be the successor architecture to S/370. It featured one-level addressing, an object orientation, interactive interfaces for operations and programmers, and high-level input/output (I/O).

FSCC (financial services call center)

FS/D (field service/dispatch)

FSDM (financial services data model)

FSK (Frequency Shift Keying)

A method of modulation that uses two different frequencies to distinguish between a mark (digital 1) and a space (digital 0) when transmitting on an analog line. FSK is used in modems operating at 1200 bits per second (bps) or slower.

FSN (forward sequence number)

FSP (financial-services provider)

FT (fault tolerant or fault tolerance)

Fault tolerance enables users to maintain continuity if the process they are attached to fails. Once the process fails on the server, another process is launched to resume the user session from where it left off, with little to no loss in productivity.

FTAM (File Transfer, Access and Management)

The Open Systems Interconnection (OSI) protocol for file transfer (i.e., the communication of an entire file between systems), file access (i.e., the ability to remotely access one or more records in a file) and management (e.g., the ability to create/delete or name/rename a file).

FTE (full-time equivalent)

FTP (File Transfer Protocol)

A Transmission Control Protocol/Internet Protocol (TCP/IP) standard used to log onto a network, list directories and copy files. That is, it provides authentication of the user and lets users transfer files, list directories, delete and rename files on the foreign host, and perform wild-card transfers.

FTPI (flux transitions per inch)

FTTB (fiber to the building)

Fiber-optic access to an individual building for telephony, video, and other multimedia services where several users share the same fiber termination. FTTB access terminates typically in a building basement from which access to multiple end users is implemented through an in-building digital home network (DHN).

FTTC (fiber to the curb)

Fiber-optic distribution of telephony, telecommunications, and multimedia and media services, such as cable TV, to a point outside the customer's premises.

FTTH (fiber to the home)

Fiber-optic access to users' homes for telephony, video, and other multimedia services where each home has a direct fiber connection.

Full Common Intermediate Format (see FCIF)

full duplex (see FD)

full-text index

An index containing every word (with the possible exception of stop words) in a collection of documents, for use by a full-text search engine. See "search engine," "search index" and "stop word."

functional profile

A defined protocol, together with options applicable to specific functions, which can be implemented as a product. Also known as a functional standard or protocol stack.

functional split

A division within an automatic call distributor (ACD) that allows incoming calls to be directed from a specific group of trunks to a specific group of agents.

functional standard (see functional profile)

functional test

A test carried out under normal working conditions to verify that a circuit or a particular part of the equipment functions correctly.

function point (see FP)

FUNI (Frame Relay User Network Interface)

A variable frame length version of asynchronous transfer mode (ATM).

FURPS (functionality, usability, reliability, performance and support)

Future System (see FS)

fuzzy logic

This reasoning paradigm deals with approximate or imprecise information by enabling variables to be described (often linguistically) and acted upon in terms of their degree of membership in predetermined sets. Control systems in electronic equipment and consumer products and other embedded control systems are among the most popular applications.

F/V (frequency-to-voltage)

A signal-conversion process wherein frequency variations are converted to amplitude variations.

FX (foreign exchange)

U.S. term that connects a customer's location to a remote exchange. This service provides the equivalent of local service from the distant exchange.

G

G2C (government-to-business)

Connecting or flowing between government and private-sector enterprises, typically through online or network-enabled channels.

G2C (government-to-constituent)

Connecting or flowing between governments and their constituents, typically through Internet or other network-enabled channels.

G2G (government-to-government)

Intergovernmental, typically with reference to Internet or other network-enabled interactions, communications or information sharing.

G2 (Generation 2)

IBM's second generation of 9672 mainframe models.

G3 (Generation 3)

IBM's third generation of 9672 mainframe models.

G4 (Generation 4)

IBM's fourth generation of 9672 mainframe models.

G5 (Generation 5)

IBM's fifth generation of 9672 mainframe models.

G6 (Generation 6)

IBM's sixth generation of 9672 mainframe models, introduced in May 1999.

G.7x series

A series of International Telecommunication Union (ITU) standards for voice and data communications (see **ITU**). The series includes:

- G.703: Specifications for plesiochronous digital hierarchy (see **PDH**).
- G.709: Specifications for synchronous digital hierarchy (see **SDH**).
- G.711: Specifications for digital speech encoding based on a pulse code modulation (see **PCM**) digitizing algorithm using logarithmic encoding.
- G.721: Specifications for digital speech encoding based on an adaptive differential pulse code modulation (see **ADPCM**) technique.
- G.723: Specifications for the voice component of the H.323 videoconferencing suite (see **H.x series**).
- G.728: Specifications for digital speech encoding using low bit rates based on a sector generalization scheme (which yields lower sound quality than G.711).

GA (general availability)

The point at which a new product is available on the open market.

GaAs (gallium arsenide)

A chemical used in semiconductor manufacturing.

gain

Denotes an increase in signal power in transmission from one point to another, usually expressed in decibels.

gain sharing

Describes a contract that defines the vendor's contribution to the customer in terms of specific benefits to the customer's business. Such a contract also defines the payment the customer will make according to the vendor's performance in delivering these benefits. Gain-sharing contracts require the development of a delivery paradigm that links a customer's business metrics to a vendor's IT solution. Key elements of this paradigm include:

- Business metric definition and selection
- Client metric benchmarking
- Development of key performance indicators
- Investment options evaluation
- Gain-sharing contract development
- Financial engineering
- Delivery of services
- Re-evaluation and adjustment of metrics

gallium arsenide (see GaAs)

GAP (Groupe d'Analyse et Prevision)

An ad hoc working group, made up of industrial, member state and commission representatives in the European Union for the study and proposal of common approaches to various communication issues. Reports by the group have covered Integrated Services Digital Network (ISDN), broadband technologies and networks, mobile communications and Open Network Provision (ONP).

GARP (Generic Attribute Registration Protocol)

Part of the Institute of Electrical and Electronics Engineers (IEEE) 802.1p extension to its 802.1d (spanning tree) specification. it includes:

- GARP Information Declaration (GID): The part of GARP that generates data.
- GARP Information Propagation (GIP): The part of GARP that distributes data.
- GARP Multicast Registration Protocol (GMRP): Provides a mechanism that allows participants to dynamically register and de-register information with the Media Access Control (MAC) bridges attached to the same local-area network (LAN) segment.

GARP Information Declaration (GID)

See GARP.

GARP Information Propagation (GIP)

See GARP.

GARP Multicast Registration Protocol (GMRP)

See GARP.

gate assignments

Used in the context of automated call distribution (ACD) equipment. Gates are made up of trunks that require similar agent processing. Individual agents can be reassigned from one gate to another gate by the customer via the supervisory control and display station. Also called splits.

gateway

1. A computer that sits between different networks or applications. The gateway converts information, data or other communications from one protocol or format to another. A router may perform some of the functions of a gateway. An Internet gateway can transfer communications between an enterprise network and the Internet. Because enterprises often use protocols on their local-area networks (LANs) that differ from those of the Internet, a gateway will often act as a protocol converter so that users can send and receive communications over the Internet.
2. A product or feature that uses proprietary techniques to link heterogeneous systems.

gateway server

server designed to transform data streams to better match device capabilities. For example, Wireless Application Protocol (WAP) gateway servers convert HTML to Wireless Markup Language (WML) for wireless devices, and a number of products can reformat HTML for devices such as mobile phones and personal digital assistants (PDAs). Today, HTML-based servers predominate. While HTML can be made aware of a unique device requesting content, more often "shadow" Web server applications are created to draw off and reformat the native content.

Gb (gigabit)

Approximately one billion bits of data (1,073,741,824, to be exact).

GB (gigabyte)

Approximately 1 billion bytes of data (actually 1,073,741,824 bytes).

Gbps (gigabits per second)

A measure of data transmission speed (see Gb).

GCOS (General Comprehensive Operating System)

A Groupe Bull mainframe operating system.

GCR (group code recording)

GDI (Graphics Device Interface)

A Microsoft Windows interface.

GDMO (Guidelines for the Definition of Managed Objects)

An International Organization for Standardization (ISO) common object modeling technology standard, part of the Open Systems Interconnection (OSI) Common Management Information Protocol (CMIP). A quasi-object database that “abstracts” objects but does not embed procedures (i.e., the objects have persistence and class, but not inheritance).

GDPS (Geographically Dispersed Parallel Sysplex)

An IBM mainframe feature for continuous availability and disaster recovery across multiple, geographically dispersed sites.

GDS (Global Directory Service)

A Distributed Computing Environment (DCE) directory service that provides a pointer to resources between cells.

GEM (global electronic marketplace)

GEM (Global Enterprise Manager)

GEMMS (Global Enterprise Manufacturing Management System)

A process manufacturing product once offered by Datalogix (which was acquired by Oracle in 1997).

GEMS (Global Enterprise Management of Storage)

A family of storage management products from Legato Systems.

general availability (see GA)

General Inter-ORB Protocol (see GIOP)

General Security Services Application Programming Interface (see GSS-API)

Generic Attribute Registration Protocol (see GARP)

genetic algorithm

An optimization technique that applies the principles of natural selection and genetics. A genetic algorithm generates option variations through random “mutation.” Successful variations survive iterations of the algorithm. The technique is used to optimize plans, neural networks and other algorithms.

geo-coding

Identifying narrow locations using ZIP codes or other identifiers (depending on country) to allow mapping, planning and execution of deliveries.

Geographically Dispersed Parallel Sysplex (see GDPS)

GFLOPS (gigaFLOPS)

A measure of computing performance, equal to 1 billion floating-point operations per second (FLOPS); e.g., 50 GFLOPS equals 50 billion FLOPS.

GHz (gigahertz)

A frequency unit equal to one billion cycles per second.

GID (GARP Information Declaration)

See GARP.

GIF (Graphics Interchange Format)

An integrated Web component of Hypertext Markup Language (HTML) into which graphics files can be converted so that programs can recognize them.

gigabit (see Gb)

Gigabit Ethernet

The Institute of Electrical and Electronics Engineers (IEEE) standard (802.3z) for transmission of standard Ethernet traffic at 1 gigabit per second (Gbps) speeds.

gigabyte (see GB)

gigaFLOPS (see GFLOPS)

gigahertz (see GHz)

giga-point-of-presence (see gigaPOP)

gigaPOP (giga-point-of-presence)

A concept developed by the Internet 2 Architecture Planning Group. A gigaPOP has lower-speed Internet Protocol (IP) routing capabilities similar to today's Internet service provider (ISP) environments at the fringes of the backbone networks, and an asynchronous transfer mode (ATM) switching core that can carry IP, X.25, frame relay and ATM traffic, eliminating the need for separate and costly networks to support a particular traffic type.

GIOP (General Inter-ORB Protocol)

A protocol that defines a small set of messages and data formats. It enables a client-side object request broker (ORB) to forward object method invocations to another, server-side ORB and receive back replies and error messages.

GIP (GARP Information Propagation))

See GARP.

GIPS (billion instructions per second)

A metric used to quantify system processing power.

GIS (geographic information system)

GL (general ledger)

A permanent record of accounting transactions summarized according to an enterprise's accounting and organizational structure. A GL system is designed to summarize entries from subledger systems and produce financial statements and reports.

GL (graphics library)

A component of a graphical user interface (GUI) that provides a high-level graphics programming interface.

G Lite

An International Telecommunication Union (ITU) specification (G.992.2) for a reduced-speed, splitterless version of asymmetric digital subscriber line (ADSL).

Global Directory Service (see GDS)

Global Enterprise Management of Storage (see GEMS)

Global Enterprise Manufacturing Management System (see GEMMS)

GlobalOne

A company that provides global voice and data communications services. Originally launched as a joint venture by Deutsche Telecom, France Telecom and Sprint in January 1996, it was wholly taken over by France Telecom in January 2000.

Global Navigational Satellite System (GLONASS)

See GPS.

Global Positioning Satellite System (GPSS)

See GPS.

Global Positioning System (see GPS)

Global System for Mobile Communications (see GSM)

GloBand

The name given to a set of European network services that offer digital dial-up bandwidth on demand in 64 kilobit per second (Kbps) increments, accessed from customer premises over Primary Rate Interface (PRI) lines. These services are offered under different names in each participating country.

GLONASS (Global Navigational Satellite System)

See GPS.

GMLC (Graduated Monthly License Charge)

The traditional S/390 software pricing structure offered by IBM.

GMR (giant magnetoresistor)

GMRP (GARP Multicast Registration Protocol)

See GARP.

GN (Generation N)

GNS (Global Network Services)

GNU (GNU's Not Unix)

A project launched in 1984 to develop a free, complete Unix-like operating system. Variants of the GNU operating system (which use the kernel Linux) are now widely in use. Although these systems are often referred to as "Linux systems," they are more accurately termed "GNU/Linux systems."

GNU ZIP

GNU's compression utility for Unix-like operating systems, available as freeware via the Internet.

GOLC (Growth Opportunity License Charge)

An IBM software pricing program.

GOLD

Integrion's GOLD specification encompasses a range of messages and transactions used by financial institutions to communicate with end users. It supports online services and batch protocol.

gopher

A text-based distributed document delivery and retrieval system once used over the Internet. It predates, and has been superseded by, hypertext documents on the Web.

GOSIP (Government Open Systems Interconnection Profile)

A standard issued by the National Institute of Standards and Technology (NIST), specifying the details of an interoperable Open Systems Interconnection (OSI) configuration for networking products procured by the U.S. government.

Government Open Systems Interconnection Profile (see GOSIP)

government-to-business (see G2B)

government-to-constituent (see G2C)

government-to-government (see G2G)

GPO (group purchasing organization)

An organization that provides collective buying service negotiations for its aggregate members with distributors and manufacturers of medical supplies.

GPRS (general packet radio service)

GPRS is a packet-oriented overlay to Global System for Mobile Communications (GSM) networks supporting connection- and connectionless-oriented services and diverse quality-of-service mechanisms. The theoretical maximum speed can be as high as 171.2 Kbps using all eight time slots and CS 4 channel speeds. Real-life user throughput is expected to be much lower — less than or equal to 56 Kbps.

GPRS Roaming Exchange (see GRX)

GPS (Global Positioning System)

A technology for assessing the precise location of any compatible receiver unit, using satellites to provide 24-hour positioning information regardless of the weather. GPS works on the principle of triangulation: By knowing its distance from three or more satellites, the receiver can calculate its position by solving a set of equations. While the technology is most commonly known as GPS, the satellite constellation used by the U.S. government (and most commercial GPS equipment) is known as Global Positioning Satellite System (GPSS). Until May 2000, this system suffered from the U.S. government's controlled dilution of precision (DOP), which reduced location accuracy to 100 meters. With DOP now turned off, accuracy has improved to 10

meters. The Russian government also runs a global positioning system called Global Navigational Satellite System (GLONASS), while Galileo is the name planned for the yet-to-be-created European global navigation satellite system.

GPSS (Global Positioning Satellite System)

See GPS.

GPWW (group practice without walls)

An umbrella organization for a group of independent physician practices that performs certain business operations (e.g., technology procurement).

grade of service

Quality of telephone service provided by a system. It is described in terms of the probability that a call will encounter a busy signal during the busiest hour of the day.

Graduated Monthly License Charge (see GMLC)

granularity

The ability to increase a system's capacity and performance through incremental processor expansion.

graphical shell

A simple graphical user interface to a character-oriented command language, such as DOS.

graphical user interface (see GUI)

Graphics Interchange Format (see GIF)

graphics library (see GL)

gray scale

A range of gray tones used to create a monochrome image.

ground circuit

A circuit in which energy is carried one way over a metallic path and returned through the earth.

ground start

A telephony term describing a signaling method whereby one station detects that a circuit is grounded at the other end.

ground station

An assemblage of communications equipment, including a signal generator, transmitter, receiver and antenna, that receives (and usually transmits) signals to and from a communications satellite. Also called earth station.

group band modem

A modem designed for ITU V.35, V.36, or V.37 standard for transmission over group band (60 to 108 kHz) circuits.

Groupe d'Analyse et Prevision (see GAP)

Groupe Speciale Mobile (see GSM)

groupware

Software that supports interpersonal processes and the objects with which people commonly work. Groupware was originally coined to describe a new class of applications designed to provide electronic support for groups of individuals working together toward a common goal. The term has been applied to applications ranging from unstructured electronic mail to rigorously structured workflow systems. Groupware is more useful as a concept when it is broken down into three major stages: communication, coordination and cooperation. In this light, groupware applications can be viewed in terms of the degree of structure in the group interaction and in the complexity of that structure, and the rigor with which the activity itself is monitored.

GRS (Global Resource Serialization)

GRX (GPRS Roaming Exchange)

A standard defined by the GSM Association for global packet radio service (GPRS) roaming. The standard enables mobile Internet services to be accessed through an Internet Protocol network managed and operated by a third party.

GSCM (global supply chain model)

GSIC (graphics-specific integrated circuit)

GSM (Global System for Mobile Communications)

GSM (formerly Groupe Speciale Mobile) indicates digital mobile telephone networks, based on digital technology, which operate at 900MHz, 1,800MHz and 1,900MHz frequencies. In Europe, GSM utilizes the 905-915MHz and 950-960MHz reserved spectrum to provide the capability of roaming over 18 countries with competition both within and between countries. GSM 1900 is the North American version of GSM. This allows the standard to be used in the 1,900MHz frequency band, which the U.S. Federal Communications Commission (FCC) and Industry Canada have allocated for personal communication services (PCS). GSM also refers to the working party, now a technical committee responsible to the European Telecommunications Standards Institute (ETSI), which is elaborating the standard.

GSM 1900

The North American version of Global System for Mobile Communications (GSM). It is a modification of the European specification that allows use of the standard in the 1,900-megahertz (MHz) frequency band, which the Federal Communications Commission (FCC) and Industry Canada have allocated for personal communications services (PCS). (GSM in Europe operates at 900 MHz.)

GSM Anywhere

An initiative, supported by the European Telecommunications Standards Institute (ETSI), to provide GSM network functionality using frequencies outside of the existing 900MHz, 1,800MHz and 1,900MHz frequency allocations. An example would be the reuse of existing analogue cellular phone frequencies in the 450MHz band in the Nordic countries.

GSM Association

An industry group representing wireless network operators that use Global System for Mobile Communications (GSM) technology, as well as GSM technology manufacturers and suppliers. Founded in 1987, the association has played a pivotal role in the development of the GSM platform and the global wireless industry. See GSM.

GSP (graphics system processor)

GSR (Gigabit Speed Router)

GSS-API (General Security Services Application Programming Interface)

An information security standard described in Internet Engineering Task Force (IETF) Request for Comment (RFC) 1508 that defines the formats for access control, user identification, etc.

GUI (graphical user interface)

A graphics-based operating system interface that uses icons, menus and a mouse (to click on the icon or pull down the menus) to manage interaction with the system. Developed by Xerox, the GUI was popularized by the Apple Macintosh in the 1980s. At the time, Microsoft's operating system, MS-DOS, required the user to type specific commands, but the company's GUI, Microsoft Windows, is now the dominant user interface for personal computers (PCs). A comprehensive GUI environment includes four components: a graphics library, a user interface toolkit, a user interface style guide and consistent applications. The graphics library provides a high-level graphics programming interface. The user interface toolkit, built on top of the graphics library, provides application programs with mechanisms for creating and managing the dialog elements of the windows, icons, menus, pointers and scroll bars (WIMPS) interface. The user interface style guide specifies how applications should employ the dialog elements to present a consistent, easy-to-use environment (i.e., "look and feel") to the user. Application program conformance with a single user interface style is the primary determinant of ease of learning and use, and thus, of application effectiveness and user productivity.

Guidelines for the Definition of Managed Objects (see GDMO)

Gunn diode

Named for IBM scientist John Gunn. Until recently, it was the standard form for generating microwave energy. It produces only a single fixed frequency.

H

H.100, H.221, etc. — see H.x series

H2GF (HiperLAN2 Global Forum)

A consortium created to promote the European Telecommunications Standards Institute's (ETSI's) High-Performance Radio LAN Type 2 (HIPERLAN/2) standard (which H2GF styles as "HiperLAN2"). See HIPERLAN/2.

HA (high availability)

A high probability that a system at any given time will be operational for useful work and has the ability to recover quickly and with minimum disruption to the user in case of failure. Those characteristics of a data processing system that reduce or eliminate its vulnerability to unscheduled outages, such as power failures, code defects or hardware failures.

HA-API (Human Authentication Application Programming Interface)

A biometric specification developed by SAFLINK — a private company that provides biometric identification and authentication software products and services for enterprisewide network security applications — under contract to the U.S. Department of Defense. In 1998, it transitioned to an industry-working group. HA-API is the first universal standard in force and is the current, de facto, standard high-level biometric API.

HACMP (High Availability Clustered Multiprocessing)

An IBM RS/6000 product originally designed to provide a high-availability configuration (by enabling automated failover from one RS/6000 to another). HACMP comprises:

1. Cluster management software to restart an application or subsystem on an alternate RS/6000 processor after a failure of the primary processor.
2. Distributed lock manager software to coordinate and synchronize process execution on multiple processors in a cluster.
3. Configuration and topology software management tools to enable developers to build cluster-aware applications.

HADA (high-availability disk array)

HAL (hardware abstraction layer)

Operating-system protection that prevents unauthorized access to the operating-system kernel.

half duplex (see HD)

hand geometry recognition

A biometric that uses the shape of a person's hand to authenticate identity.

Handheld Device Markup Language (see HDML)

handler

A program with the sole function of controlling a particular input, output, storage device, file or interrupt facility.

handling unit

Goods or aggregation of goods bundled together for distribution and logistics purposes. May include an individual item in a carton, combined items on pallets and skids, or items transferred in independently identified containers, such as ocean containers, rail cars or trucking trailers.

handoff

The transfer of duplex signaling as a mobile terminal passes to an adjacent cell in a cellular radio network.

handset

The portion of the telephone containing the transmitter and receiver.

handshaking

The exchange of predetermined signals for control when a connection is established between two modems or two other devices.

handwriting recognition

Systems using pattern matching to convert handwritten letters into corresponding American Standard Code for Information Interchange (ASCII) values. The two subfields are static recognition of handwritten documents such as forms, and dynamic recognition of real-time handwriting (e.g., for personal digital assistants).

HARC (Hitachi Asynchronous Remote Copy)

A remote copy function for Hitachi Data Systems storage products. HARC allows movement of large amounts of data over any distance with full data integrity and minimal impact on performance. It duplicates data files directly from a primary subsystem to a secondary subsystem without using valuable server processor cycles.

hard disk

The main data storage area of computing devices. The hard disk is where the operating system and other software are permanently stored on a computer. Hard disks typically have a larger storage capacity and faster data retrieval capabilities than floppy disks.

hardware

Machinery and equipment associated with computing devices. A computer is composed of both hardware and software. The software provides the instructions, and the hardware performs the processing.

hardware abstraction layer (see HAL)

hardware device module (see HDM)

hardware RAID

All hardware redundant array of independent disks (RAID) products employ software instructions that specify the required data movements and parity computations. The term "hardware RAID" generally implies a dedicated disk array controller board — with its own microprocessor, memory and microcode — that implements the RAID functions and computations. However, the concept can be extended to storage servers that perform the RAID operations using standard computer hardware and operating system software. The crucial point is that a hardware RAID solution performs its RAID operations in a processor that is separate from the application processor, and that provides nonvolatile memory for the data and instructions it receives from the host system.

hardware token

A small device used to authenticate user access to an information system. Hardware tokens are distinguished from smart cards in that they do not require any type of reader or sensor. They are synchronized with an authentication server at the time they are issued, and they remain synchronized with the server for the life of the token. The token generates and displays a one-time password, which the user types into the device he or she uses to access the system, along with his or her personal identification number.

The principal advantage of hardware tokens is their portability. They enable authentication from any device without imposing any special software or hardware requirements on the device. The principal disadvantage is the requirement that users carry the tokens with them, although credit-card-size tokens and key-fob form factors reduce the nuisance factor of this burden.

hard-wired

1. A communications link, whether remote phone line or local cable, that permanently connects two nodes, stations or devices.
2. Electronic circuitry that performs fixed logical operations by virtue of unalterable circuit layout, rather than under computer or stored-program control.

harmonic distortion

The presence of harmonic frequencies in a received signal due to nonlinear characteristics of a transmission line.

Harmonized Tariff Code

A code to numerically describe all articles in international trade managed by the World Customs Organization.

hashing

An algorithm that generates a unique "checksum" based on the content of transaction data so that it can be verified by recipients using the same algorithm in reverse to ensure the content was not modified.

HASP (Houston Automatic Spooling Program)

Batch-oriented IBM software that provides supplementary job management, such as job spooling, and controls communications between local and remote processors and Remote Job Entry (RJE) stations.

HBA (host bus adapter)

HBT (heterojunction bipolar transistor)

HCF (high-cost fund)

An access charge paid by interexchange carriers (IXCs), HCF was developed by the Federal Communications Commission (FCC) to help local exchange carriers (LECs) pay for their installed network infrastructures and for new network developments after the splitting up of AT&T in 1984.

HCFA1500 (Healthcare Financing Administration Codes, labeled 1500)

This Uniform Billing scheme for physician practices consists of information with uniform, prescribed content and formats used by virtually every payer for purposes of reimbursement and historical documentation. It includes details regarding procedures, tests and treatments from every point of care within the scope of the organization that owns the data.

H channel

A transmission channel, defined in the Integrated Services Digital Network (ISDN) standards, made up of multiple B channels. Currently defined H channels include:

1. H0: Comprised of 6 B channels to create a 384 kilobit per second (Kbps) ISDN channel.
2. H10: Comprised of 23 B channels to create a 1472 Kbps ISDN channel.
3. H11: Comprised of 24 B channels to create a 1536 Kbps ISDN channel.
4. H12: Comprised of 30 B channels to create a 1920 Kbps ISDN channel.

HCIS (healthcare information system)

HCM (human capital management)

A set of human resources practices that focus on providing specific competencies in three areas: workforce acquisition, workforce management and workforce optimization.

HCO (healthcare organization)

A general term covering a variety of enterprises that use healthcare information systems. HCOs include care delivery organizations (CDOs) such as hospitals and physician practices, as well as healthcare payer organizations such as health plans. See CDO.

HCPCS (Healthcare Procedural Classification System)

One of the standard code formats and definitions included in the Uniform Billing Codes.

HD or HDX (half duplex)

A circuit designed for transmission alternately in either direction but not in both directions simultaneously. See full duplex.

HDA (head disk assembly)

A sealed assembly containing magnetic disks, head and access arms. By reducing contamination, a sealed HDA enables the head to fly closer to the disk surface, increasing the areal density that can be obtained.

HDCCD (high-density CD)

HDD (hard disk drive)

HDL (hardware description language)

HDLC (High-Level Data Link Control)

A bit-oriented data link protocol developed by the International Organization for Standardization (ISO) and used in Open Systems Interconnection (OSI).

HDM (hardware device module)

Processes most of the input/output (I/O) request locally with direct memory access, and notifies the operating-system module when the operation is complete.

HDM (high-density module)

HDML (Handheld Device Markup Language)

A device- and network-independent application language developed by Unwired Planet for Web programming on a handheld device with limited memory and display such as a cellular phone or an organizer.

HDO (help desk outsourcing)

HD-ROM (high-density read-only memory)

HDS (Hitachi Data Systems)

HDSL (high-bit-rate digital subscriber line)

One of several digital subscriber line (DSL) technologies. HDSL (also known as as "high-speed digital subscriber line") provides network or Internet access over voice lines at 1.5 megabits per second.

HDTV (high-definition television)

HDX (half duplex) — see HD

head disk assembly (see HDA)

head end

The originating point in a one-to-many system such as a TV cable network or a central information-gathering point of an Internet service provider (ISP).

header

The initial portion of a message, which contains any information and control codes that are not part of the text (e.g., routine, priority, message type, destination addressee and time of origination).

header record

Data containing common, constant or identifying information for a group of records that follow.

head-mounted display (see HMD)

Healthcare Financing Administration Codes, labeled 1500 (see HCFA1500)

healthcare organization (see HCO)

Healthcare Procedural Classification System (see HCPCS)

Health Information and Management Systems Society (see HIMSS)

Health Insurance Portability and Accountability Act (see HIPAA)

Health Level Seven (see HL7)

Health Plan Employer Data and Information Set (see HEDIS)

HEDIS (Health Plan Employer Data and Information Set)

A set of standardized performance measures designed to provide purchasers and consumers with the information they need to reliably compare healthcare organizations' performance. HEDIS 99 contains measures that cover prevention, acute and chronic care — including mental health and chemical dependency — across a full range of care settings (physician office, clinics and hospital outpatient care, inpatient acute and nonacute care, and behavioral health).

helical-scan recording

In tape recording, when the tape is wrapped around a transport drum at an angle, track length is created that is nearly 10 times longer than the tape width. These diagonal tracks can increase track density and data transfer rates.

help desk

The first point of contact for all technical and end-user support issues, it includes Tier 1 and Tier 2 support levels. Tier 1 is the first point of contact. Tier 2 help desk analysts have more in-depth technical knowledge or specialized expertise.

help system

A consistent, system-supplied mechanism for system and application programs to offer online, context-sensitive help to end users.

Henderson Strategic Alignment Model

Developed by John Henderson of Boston University, this model provides a helpful framework for the CIO to understand the transition to a new management approach appropriate for the times. In this model, business strategy serves as a demarcation between current and future business platforms. The process of developing business strategy, including the requisite dialog between the IS organization and business units, leads to derivation of future business and IT architectures that will serve the enterprise.

HEP (horizontal enterprise portal)

A portal that enables a broad population of users across an enterprise.

hertz (see Hz)

heterogeneous

A system composed of hardware, operating systems, middleware or applications from several different vendors.

heterogeneous (computer) network

A system of different host computers, such as those of different manufacturers. See homogeneous (computer) network.

Heterogeneous LAN Management (see HLM)

heterogeneous multiprocessing (see HMP)

Hewlett-Packard (see HP)

hexadecimal

A number system with 16 members represented by 0 through 9 followed by A through F. Used to identify the 16 possible bit patterns of a half byte; two hex digits represent one byte. Synonymous with hex.

HF (high frequency)

The portion of the electromagnetic spectrum typically used in shortwave radio applications — frequencies approximately in the 3 to 30 megahertz (MHz) range.

HFC (hybrid fiber coaxial)

Network architecture that consists of fiber in the backbone and coax as the access. This type of architecture permits very-high-speed services into the home for applications such as video.

HHIS (home-health information system)

HI (healthcare infomediary)

An entity that captures healthcare usage data, which then can be used to profile consumer purchasing and usage patterns. Once it creates a consumer profile, a HI can act as an agent that connects the consumer to healthcare entities that offer services aligned with the consumer's lifestyle.

HIAA (Health Insurance Association of America)

hierarchical database

A database that is organized in a tree structure in which each record has one owner. Navigation to individual records takes place through predetermined access paths.

hierarchical storage management (see HSM)

high availability (see HA)

High Availability Cluster Multiprocessing (see HACMP)

high-bit-rate digital subscriber line (see HDSL)

high-cost fund (see HCF)

high frequency (see HF)

High-Level Data Link Control (see HDLC)

high-level language (see HLL)

High-Level Language Application Program Interface (see HLLAPI)

highly parallel systems (see HPS)

High-Performance File System (see HPFS)

High-Performance Parallel Interface (see HIPPI)

High-Performance Radio LAN Type 1 (see HIPERLAN/1)

High-Performance Radio LAN Type 2 (see HIPERLAN/2)

High-Performance Routing (see HPR)

High-Performance Space (see Hiperspace)

High Performance Transaction System (see HPTS)

high-speed circuit switched data (see HSCSD)

high-speed digital subscriber line (see HDSL)

high-speed local-area network (see HS LAN)

High-Speed Serial Interface (see HSSI)

high-speed subscriber data line (see HSDL)

High-Speed Token-Ring Alliance (see HSTRA)

HIM (health information management)

HIMSS (Health Information and Management Systems Society)

An industry association that offers a variety of publications, educational programs and services related to healthcare information systems. Its members contribute to the development of such technologies as telemedicine, computer-based patient records, community health information networks and portable/wireless healthcare computing.

HIPAA (Health Insurance Portability and Accountability Act)

HIPAA, passed by the U.S. congress in August 1996, calls for the use of electronic data interchange (EDI) in healthcare transactions, and for protecting the privacy of patient healthcare information. Once the April 2003 privacy compliance deadline has passed, healthcare organizations (HCOs) face fines of up to \$250,000 and 10 years imprisonment for wrongfully disclosing patient information. They also risk having claims rejected if they do not conform to the EDI requirements by October 2003. The HIPAA directives — for EDI in particular — have major IT implications for HCOs, many of which have devoted considerable time and resources to system compliance efforts.

HIPERLAN/1 (High-Performance Radio LAN Type 1)

A wireless-LAN standard from the European Telecommunications Standards Institute (ETSI). HIPERLAN/1 is designed to provide high-speed communications (20 Mbps)

between portable devices broadcasting in the 5-gigahertz radio band. It is intended to allow flexible data networks to be created without the need for a wired infrastructure. It can also be used as an extension of a wired LAN.

HIPERLAN/2 (High-Performance Radio LAN Type 2)

A 5-gigahertz wireless-LAN standard, developed by European Telecommunications Standards Institute (ETSI) as follow-up to the HIPERLAN/1 standard it introduced in the 1990s. HIPERLAN/2, which provides access at speeds up to 54 Mbps at physical layer, competes directly with IEEE's 802.11a standard, which also operates at 5 gigahertz. Both standards are almost identical at the physical layer. The main difference lies in their media access control (MAC) layers. For 802.11a the MAC layer is like Ethernet, while for HIPERLAN/2 it is more like an asynchronous transfer mode (ATM) interface.

HiperLAN2 Global Forum (see H2GF)

HiPerLinks (high-performance links)

First introduced with the IBM G3, HiPerLinks use the same connection technology as the coupling links, but with improved efficiency due to a reduced latency time. The effect of the HiPerLinks on small data transfers is negligible, but on larger data transfers, the link capacity can be improved by up to 20 percent.

Hiperspace (High-Performance Space)

Introduced with ESA/370, Hiperspace refers to an address space occupied only by data and to which access is controlled by hardware via special instructions.

HIPPI (High-Performance Parallel Interface)

An American National Standards Institute (ANSI) standard for high-speed channels used in I/O-intensive systems, including supercomputers.

HIS (hospital information system or healthcare information system)

The IT applications used to manage hospital operations (e.g., patient financials, registration, scheduling, general financials, back-office systems and order communications).

hit

An individual visit to a Web site or Web page, expressed as a measure of its popularity (i.e., Web traffic volume). For example, a site that had 48,000 visitors during a 24-hour period might be said to have averaged 2,000 "hits" per hour during that period.

Hitachi Asynchronous Remote Copy (see HARC)

Hitachi Extended Remote Copy (see HXRC)

Hitachi Online Data Migration (see HODM)

Hitachi Remote Copy (see HRC)

HL7 (Health Level Seven)

A set of application-level standard for community health information network (CHIN) initiatives, widely used in hospitals.

HLL (high-level language)

A procedural programming language that offers a higher level of abstraction than assembly language. Examples of HLLs include COBOL, FORTRAN, PL/I, C, Pascal and Ada. The term is interchangeable with “third-generation language” (see 3GL).

HLLAPI (High-Level Language Application Programming Interface)

IBM’s high-level language multiprotocol applications interface.

HLM (Heterogeneous LAN Management)

A proprietary network management draft specification developed jointly by 3Com and IBM that specifies using Common Management Information Protocol (CMIP) over Logical Link Control (LLC) to provide network management of devices on mixed-media local-area networks (LANs). Also known as CMOL.

HMBR (Hitachi Multihost Backup/Restore)

HMD (head-mounted display)

A display mounted close to the eye on a headset or goggles typically uses optics to create the illusion of a larger screen further away from the eye. Most techniques involve a small liquid crystal display (LCD), usually monochrome and low resolution (other approaches include light-emitting diode — LED — or electroluminescent technologies). Applications include industrial inspection and maintenance. The 3-D head-mounted displays used in immersive virtual reality systems are typically based on dual LCDs with a distinct image for each eye (offset to provide depth perception), updated at 30 frames per second. This is coupled with head tracking so that the image changes appropriately in response to head movement.

HMDE (Hitachi Multiplatform Data Exchange)

HMI (human machine interface)

HMMP (Hypermedia Management Protocol)

This protocol is a part of the Web-Based Enterprise Management (WBEM) proposal. It operates over the Hypertext Transport Protocol (HTTP) standard, collecting and transporting device management data to populate a management database. Initially, HMMP will complement Simple Network Management Protocol (SNMP), not replace it, although that is the long-term objective.

HMMS (Hypermedia Management Schema)

A schema that was a part of the Web-Based Enterprise Management (WBEM) proposal. It describes what is to be contained in an object model, outlining a data model for representing the relationship between network elements, not just the elements themselves. The schema relies on basic object-oriented constructs, establishing classifications of managed elements in the network environment. The four classes are: system elements, application components, resource components and network components. HMMS has since been renamed Common Information Mode (CIM) by the Distributed Management Task Force (DMTF).

HMO (health maintenance organization)

HMOM (Hypermedia Object Manager)

Included within the Web-Based Enterprise Management (WBEM) proposal, HMOM is an example of a specification designed to use the data model — as described by

Hypermedia Management Schema (HMMS) — in developing a repository of object instances, each representing network elements. HMOM serves as a controller to management data (allowing network managers and management applications to observe and manipulate this data) through a series of management protocols and interfaces.

HMP (heterogeneous multiprocessing)

An operating-system (OS) shell that runs the mainframe OS as a mandatory partition with the optional ability to run Unix or NT.

HMP (Hierarchical Multiprocessing)

HMRS (Hitachi Multihost Resource Sharing)

HODM (Hitachi Online Data Migration)

A controller-based feature that requires the target volume(s) to be contained within a Symmetrix or 7700 subsystem. Depending on customer configuration, the recabling required to support data migration may or may not be disruptive.

HOLAP (hybrid online analytical processing)

A means of combining data from relational online analytical processing (ROLAP) and multidimensional database management system (MDDDBMS) sources through the same application.

holding time

The length of time a communications channel is in use for each transmission. It includes message time and operating time. Also called connect time.

home page

The top-level page or entry point into a Web site.

HomeRF

A 2.4GHz wireless LAN technology based on a frequency hopping modulation scheme. HomeRF is designed to transport voice, video and data at an attractive price for consumer purchases.

homogeneous network

A system of similar computers, such as those of one model by the same manufacturer.

homologation

Conformity of a product to international standards.

hop count

Measures the number of stops a packet makes from source to destination.

HOPS (HTTP operations per second)

host

Any computer connected to the Internet that has a domain name associated with it. It provides the services and facilities used by the other computers and terminals.

HOST (Healthcare Open Systems and Trials)

host computer

The primary or controlling system in a multiple computer network operation.

hosting

Storage of data on a server for later access. Typically, the hosting server provides services and tools that expedite access by individuals or other servers. The popularity of the World Wide Web has given rise to an industry of hosting services that store large amounts of data that can be accessed from the Internet 24 hours a day, seven days a week.

host interface

The link between a communications processor or network and a host computer.

host processor

A mainframe attached to a network for network users' access.

hot standby

Alternate equipment ready to take over an operation quickly if the equipment on which it is being performed fails.

hot swap

To exchange a component or a module in a networking device without powering down.

HOTT (Hands On Technology Transfer)

Houston Automatic Spooling Program (see HASP)

HP (Hewlett-Packard)

A company founded in Palo Alto, California, by Stanford University classmates William Hewlett and David Packard in 1939. It originally made audio oscillators and grew to become a multibillion-dollar computer vendor. HP acquired Compaq Computer in 2002.

HPC (handheld PC)

HPC (high-performance computing)

HPCC (High-Performance Computing and Communications)

HPFS (High-Performance File System)

The data-mapping protocol used for disk storage in IBM's OS/2 operating system. It replaced the file allocation table (FAT) system used in PC-DOS.

HPNA (Home Phoneline Networking Alliance)

An alliance to advance technologies for high-speed local-area networking over conventional phone wires. It was founded in 1998 by 3Com, Advanced Micro Devices, AT&T Wireless Services, Compaq Computer, Conexant Systems, Hewlett-Packard, IBM, Intel, Lucent Technologies and Tut Systems.

HPPD (hours per patient day)

HPR (High-Performance Routing)

IBM's technology to enable users to send Systems Network Architecture (SNA) traffic over frame and cell-based networks, making for more efficient communications. For example, the HPR version of Advanced Peer-to-Peer Networking (APPN) is generally superior to Transmission Control Protocol/Internet Protocol (TCP/IP), which uses reactive congestion control, while APPN is predictive, resulting in higher network utilization and throughput for APPN. Because it is newer, APPN can better take advantage of increasingly error-free, high-speed digital transmission facilities. TCP/IP's domain name service and limited (now insufficient) addressing capacity are inferior in performance to APPN's topology database and use of logical unit names. APPN supports a prioritization/class-of-service mechanism for different types of network applications, while individual router vendors have been forced to add their own nonstandard prioritization capabilities to TCP/IP.

HPS (highly parallel systems)

Hardware configurations having from 10 to 99 central processing units (CPUs).

HPTS (High Performance Transaction System)

A check-processing system from Check Solutions (Memphis, Tennessee).

HP-UX

Hewlett-Packard's System V-based Unix implementation, with real-time extensions added.

HR (human resources)

HRC (Hitachi Remote Copy)

A controller-based remote-copy capability that Hitachi Data Systems (HDS) is evolving into a competitive alternative to Symmetrix Remote Data Facility (SRDF).

HR disintermediation

The practice of bypassing the HR department to define and implement technology practices to support human capital management (HCM) issues. When no clear HCM strategy is in place, business leaders formulate departmental plans and adopt the technologies they feel they need to support individual and group responsibilities for enterprise performance.

HRE (Hardware Routing Engine)

HREF (Hypertext Reference)

In Hypertext Markup Language (HTML), HREF is used to assign a hypertext address to a link.

HRIS (human resource information system)

HRMS (human resource management system)

Business applications for the management of human resources (HR) transactions, best practices and enterprise reporting. Functions typically include core HR tracking, payroll and benefits. The scope is often extended to include recruiting, competency management, training, time management, performance management and self-service offerings.

HRMS (Human Resource Management System)

A PeopleSoft product.

HSC (High-Speed Connect)

HSCSD (high-speed circuit switched data)

HSCSD is a Global System for Mobile Communications (GSM) bearer service, already deployed in some networks (notably in Scandinavia, the United Kingdom and Hong Kong). It permits circuit switched connections with speeds of up to 56 Kbps. This is achieved by channel-bonding time slots to obtain the requested speed. Typically, users will be allowed between two and four time slots in a real-life network.

HSDL (high-speed subscriber data line)

A local phone line with full-duplex TI capabilities. As corporate intranets grow, bigger "pipes" are needed to maintain network performance. The use of inverse multiplexers that combine multiple E1 circuits will be the best way to deliver that bandwidth between the user site and the carrier's central office. But the use of HSDL technology, as an alternative to inverse multiplexing of E1 circuits, will be a "dark horse" alternative to inverse multiplexing. Unlike inverse multiplexing, HSDL would require new support systems and maintenance procedures.

HS LAN (high-speed local-area network)

A LAN operating at four megabits (Mbps) per second or more to facilitate rapid exchange of files and mail within a broader departmental or divisional environment.

HSM (hierarchical storage management)

A storage management technology that can be used to identify inactive data and move it to near-line storage, automate the retrieval process, and migrate the data back to the primary storage medium and provide access for the user.

HSSI (High-Speed Serial Interface)

Standard for a serial interface used to connect to T3 lines. HSSI operates over a shielded cable at speeds of up to 52 megabits per second (Mbps) and distances of up to 50 feet. Functionally, it serves the same purpose as lower-speed serial interfaces such as V.35 and RS232, in that it provides the interface to DCE for wide-area-network (WAN) communications.

HSTR (High-Speed Token Ring)

HSTRA (High-Speed Token Ring Alliance)

A consortium of vendors formed in 1997 as a re-creation of the Alliance for Strategic Token Ring Advancement and Leadership (ASTRAL). HSTRA was created to draft standards for 100 megabits per second (Mbps) token ring and Gigabit token ring and present them to the Institute of Electrical and Electronics Engineers (IEEE) 802.5 committee.

HTLOC (hundred thousand lines of executable code)

HTML (Hypertext Markup Language)

A document-formatting language derived from the Standard Generalized Markup Language (SGML), predominately used to create Web pages. The user's browser interprets HTML commands and formats the page layout, fonts and graphics on the screen. One of the more powerful features of HTML is its ability to create hyperlinks

that enable the user to navigate between documents and files with a single click. HTTP is also sometimes used for messaging attachments as a way of supporting rich text formatting across product boundaries.

HTML that is generated by a program or service is considered “dynamically generated HTML,” which is often confused with a Netscape and Microsoft technology called Dynamic HTML (DHTML). DHTML offers client-side mechanisms for enhancing the capabilities of the Web browser and HTML documents. Dynamically generated HTML might contain DHTML, but they are not synonymous terms.

HTTP (Hypertext Transport Protocol)

The Internet standard for accessing and exchanging documents on the Web. It is a client/server protocol used to connect to Web servers.

HTTPS (Secure Hypertext Transport Protocol)

See S-HTTP.

hub

A central device, usually in a star topology local-area network (LAN), to which each station's wiring is attached. Also called wiring concentrator. See intelligent hub.

Huffman Code

A code used for one-dimensional data compression in the International Telecommunication Union Telecommunications Standards Sector (ITU) Group III digital facsimile standard.

hum

Spurious electrical interference, usually picked up from a conventional alternating current power supply.

HVP (healthcare vertical portal)

An Internet-based vendor that provides a foundation for creating e-business and e-health functions in conjunction with a care delivery organization (CDO) by providing one or more of the following: organized access to focused aggregate healthcare content, connectivity solutions for linking business partners and healthcare stakeholders, e-commerce functions for managing business transactions, and linkages to improve supply chain management (SCM) operations using the Internet as the communication and transaction medium.

HW (see hardware)

HXRC (Hitachi Extended Remote Copy)

A remote copy function for Hitachi Data Systems storage products. HXRC uses an asynchronous copy approach to deliver high data integrity with minimal disruption and performance impact to the primary system. It is compatible with industry-standard Extended Remote Copy (XRC).

H.x series

A series of computing standards from the International Telecommunication Union. It includes:

- H.100: A hardware standard merging Signal Computing System Architecture (SCSA) and Multivendor Integration Protocol.

- H.221: A standard describing a method of inverse multiplexing for videoconferencing terminals, to be used with Px64 videoconferencing.
- H.261: A standard for video compression (also known as px64). H.261 will be the worldwide standard and therefore must accommodate both the North American National Television System Committee (NTSC) and European Phase Alternate Line (PAL) protocols.
- H.263: The video component of the H.323 videoconferencing suite.
- H.263+: A standard for higher compression of video at 128 Kbps.
- H.320: An international standard specification for audioconferencing and videoconferencing that comprises many standards, including H.261 standard, the compression schema for video transmissions.
- H.323: A standard for audio- and videoconferencing. It is a videoconferencing suite that has G.723 voice, T.120 collaborative and H.263 video components in a single session.
- H.324: An international standard protocol for "plain old telephone service" (POTS).

hybrid business

A brick-and-mortar business that has responded to Internet threats by creating a Web front end with links to back-end systems, or a dot-com (i.e., a virtual company) that is creating traditional infrastructure (e.g., a warehouse and logistics system) to meet customer expectations.

hybrid card (see combi card)

hybrid circuit

A circuit having four sets of terminals arranged in two pairs. It is designed so that there is high loss between the two sets of terminals of a pair when the terminals of the other pair are suitably terminated. Hybrid circuits are commonly used to couple four-wire circuits to two-wire circuits.

hybrid fiber coaxial (see HFC)

hybrid modeling

A term, coined by Gartner, that is used to signify second-generation, dimension-driven, constraint-based solids modeling technology beyond first-generation parametric modeling. Hybrid modelers offer multiple design input mechanisms, flexible constraint management, and robust interoperability with legacy computer-aided design (CAD) data.

hybrid private/public networking

The creation of a network using both private leased lines and public switched facilities (digital dial-up bandwidth).

Hype Cycle

A Gartner model designed to help clients make intelligent decisions about when to implement emerging technologies. The Gartner Hype Cycle provides not only a scorecard to separate hype from reality, but also a model that can be used to decide when it makes sense for an enterprise to move forward with a new technology. The five phases in the Hype Cycle are:

- Technology trigger
- Peak of inflated expectations
- Trough of disillusionment
- Slope of enlightenment
- Plateau of productivity

hyperlink

An area on a Web page that, when clicked on with a mouse, will transport the user to another Web page. Also called "links" or "hot links," hyperlinks are analogous to hypertext. Hyperlinks are commonly used on the Web to provide navigation, reference and depth where published text cannot. A hyperlink can be created from text or from a graphic.

hypermedia

Hypertext for multimedia documents. Can include audio and video as well as text and graphics.

Hypermedia Management Protocol (see HMMP)

Hypermedia Management Schema (see HMMS)

Hypermedia Object Manager (see HMOM)

Hypermedia/Time-Based Structuring Language (see HyTime)

hyperprotocol

A file transfer protocol that sends data in a steady stream, rather than in packets, with built-in error correction and data compression.

hypertext

Software technology used to create and store simple and complex navigational paths across computerized data. When the text is "clicked on," it can enable a user to navigate within or between Web pages.

Hypertext Markup Language (see HTML)

Hypertext Reference (see HREF)

Hypertext Transport Protocol (see HTTP)

HyTime (Hypermedia/Time-Based Structuring Language)

An extension to Standard Generalized Markup Language (SGML) providing structure for multimedia documents. It facilitates the creation of pointers and the description of spatial and temporal relationships between parts of the document. Developed by

the American National Standards Institute (ANSI), now maintained by the International Organization for Standardization (ISO). See SGML.

Hz (hertz)

A unit of frequency equal to one cycle per second.

I

IA (Intel Architecture)

The acronym used in the names of Intel processor architectures, such as IA-32 and IA-64.

IA (intelligent agent)

A software application that assists users with information mining and retrieval to delegate work the user would otherwise have to do. Other names for IAs include "autonomous agents," "intelligent assistants," "bots" and "virtual assistants."

IA-32

Intel proprietary complex instruction set computer (CISC) architecture.

IA-64

An Explicitly Parallel Instruction Computing (EPIC)-based architecture from Intel, code-named Merced.

IAA (insurance application architecture)**IAAI (Innovative Applications of Artificial Intelligence)**

A yearly conference focused on deployed applications of artificial intelligence (AI) technology. It runs in conjunction with the American Association of Artificial Intelligence conference.

IAB (Internet Activities Board)

The technical policy- and standards-setting body for the Internet suite of protocols. It consists of two task forces: Internet Engineering Task Force (IETF) and Internet Research Task Force (IRTF).

IAC (interapplication communication)**IAC (Internet application component)****IAD (integrated access device)****IAE (integrated application environment)**

An environment with a strong integrated development environment (IDE), application server, middleware, and the "glue" (or framework) to integrate all application development (AD) facilities. See IDE.

IANA (Internet Assigned Numbers Authority)

The part of the Internet Activities Board (IAB) that allocates Internet Protocol (IP) address identifiers, and maintains a database of assigned IP identifiers.

i-appli

Service applications launched in January 2001 by NTT DoCoMo, for use with its popular i-mode wireless service. Use of the applications, which include video games and online financial services, requires Java-based handsets. See i-mode.

IAS (Internet Authentication Service)

A certification authority and online authentication service for closed communities of Internet users, announced by Intel on October 12, 1999. The credentials are designed to be branded by healthcare organizations that have the databases and real-world identity verification procedures to act as the identity registration authorities for their well-defined communities.

IB (integration broker)

Also called an interface engine or a message broker, an IB is a third-party intermediary that facilitates interactions between applications. IBs minimally provide message transformation and routing services. They mostly communicate program to program; they integrate previously independent applications at the application-logic level of the software design.

IBG (interblock gap)

Space left unused between consecutive data blocks.

IBM Advanced Identification Services (see IBM AIS)

IBM AIS (IBM Advanced Identification Services)

An application programming interface (API) drafted by IBM for lower-level application interfaces and presented to the International Computer Security Association (ICSA) for large-scale, generic biometric applications.

IBSTPI (International Board of Standards for Training, Performance and Instruction)

IBX (Internet Business Exchange)

IC (integrated circuit)

An assembly of electronic circuits contained on a single piece of semiconductor material.

IC (internal coupling)

IC (Internet Connector)

ICA (Independent Computing Architecture)

A protocol that provides users with integrated access to the local hard drive, printer and serial port resources.

ICA (Intelligent Console Architecture)

ICA (International Communications Association)

ICANN (Internet Corporation for Assigned Names and Numbers)

I-CASE (integrated-computer-aided software engineering)

Refers to the integration of lower-CASE (computer-aided software engineering) tools with upper-CASE tools. (Upper-CASE tools are used in the analysis and design phases of a system's life cycle. Lower-CASE tools take the information contained in the upper-CASE tools and rapidly generate whole applications, or parts of an application.) I-CASE creates a bidirectional link that transfers information back and forth.

ICB (Integrated Cluster Bus)

The IBM proprietary high-speed coupling bus link between 9672 G5 servers and coupling facilities (CFs).

ICC (Internal Coupling Channel)

ICCP (Institute for Certification of Computer Professionals)

A certification body for information technology professionals. Based in Des Plaines, Illinois, ICCP offers the Certified Computing Professional (CCP) credential. See CCP.

ICD (International Classification of Diseases)

ICD (International Code Designator)

One of two formats from the Asynchronous Transfer Mode (ATM) Forum for address usage. See DCC.

ICD-9 (International Classification of Diseases, Ninth Revision)

ICD-9-CM (International Classification of Diseases, Ninth Revision, Clinical Modification)

ICDA (Integrated Cached Disk Array)

A family of storage products for the IBM AS/400 from EMC.

ICDS (Intelligent Content Distribution Service)

Technology developed by AT&T and partners InfoLibria, Inktomi and Novell, geared to the needs of application service provisioning (ASP) content distributors. It enables distribution and replication of content in AT&T's network by archiving, routing and caching to improve performance (i.e., by moving content closer to users) and to support Web events (e.g., multicasting, event broadcasting and distance learning).

ICE (Information and Context Exchange)

A proposed Extensible Markup Language (XML) standard from the World Wide Web Consortium (W3C) that provides a common protocol, vocabulary and management model for information providers and recipients to use when sharing and reusing information. ICE components include:

- Subscription management (including start and stop dates)
- Time and frequency of delivery
- Delivery method (push or pull)
- Data delivery, detailing what items must be sent to fulfill a subscription
- Event logs diagnosing problems

- Miscellaneous functions, including sending messages destined for system administrators and troubleshooting

ICE (integrated computing environment)

ICF (internal coupling facility)

ICMF (Integrated Coupling Migration Facility)

IBM's internal Coupling Facility product for Parallel Sysplex support.

ICMP (Internet Control Message Protocol)

A route management protocol that is part of the Internet Protocol (IP), handling error and control messages.

ICMP Router Discovery Protocol (see IRDP)

iCOMP (Intel Comparative Microprocessor Performance)

A proprietary Intel test suite and performance measurement that is applicable only to Intel processors.

icon

A symbol or pictorial representation of an object or idea in graphic interfaces, used to represent the different functions or applications available to the user. A mouse is typically used to select the desired operation by pointing to one of the icons on the screen.

ICP (Internet Cache Protocol)

ICP (Internet content provider)

ICPR (Internet commercial publishing revenue)

The sum of advertising revenue generated from classified ads, display ads, banner ads and cooperative marketing sources, and subscription revenue generated for a multiple-issue schedule or on a per content use basis from Internet publications and directory sources.

ICQ

An online instant-messaging program introduced in November 1996 by Israeli firm Mirabilis, which was acquired by America Online (AOL) in June 1998. ICQ, which surpassed 100 million registered users in May 2001, is similar to AOL's popular Buddy List and Instant Messenger programs. Individuals use it to chat, send e-mail, perform file transfers and play computer games. The name ICQ (pronounced "I seek you") is derived from the amateur-radio term "CQ," which means "seeking conversation."

ICR (intelligent character recognition)

A technology that employs either software alone or software and hardware to automatically recognize and translate raster images into structured data.

ICSA (International Computer Security Association)

ICT (information and communication technology)

ICX (Intercartridge Exchange)

IDA (integrated digital access)

A British Telecom ISDN precursor.

IDA (intelligent drive array)

IDAPI (Integrated Database Application Programming Interface)

An application programming interface (API) for data integration first promoted by Borland, IBM, Novell and WordPerfect in November 1992 to counter Microsoft's Open Database Connectivity.

IDARS (integrated document archive and retrieval system)

A consolidated system for storage, access, management and viewing of data that is often print-stream-originated. Leading uses of IDARS include mission-critical customer service support, electronic bill presentment, management and distribution of report data (e.g., mainframe output, transaction logs and financial reports) and long-term archiving of historical data.

IDC (Internet data center)

Provides server outsourcing, hosting and colocation services, as well as IP and broadband connectivity, virtual private networks (VPNs), and other network and transport services.

IDD (international direct dial)

IDE (integrated development environment)

Environments for writing application logic and designing application interfaces. They differ from integrated application environments (IAEs) in their lack of solutions such as application servers (with a runtime framework or middleware component) and development frameworks (e.g., with integrated testing, project and process management, software configuration management, and component design and assembly). See IAE.

iDEN (Integrated Digital Enhanced Network)

A technology that allows users to access phone calls, two-way radio transmissions, paging and data transmissions from one wireless device. Developed by Motorola, iDEN is based on time division multiple access (see TDMA). Services based on the technology are available in North America (offered by Nextel), South America and parts of Asia.

IDEA (International Data Encryption Algorithm)

IDEF (Integrated Definition Methodology)

The most commonly used representation standard for data models.

Identrus

A global trust enterprise formed in 1998 by eight leading banks to provide certification authority services for business-to-business e-commerce. Originally called the Bank Global Trust Enterprise, the initiative was renamed Identrus in 1999.

IDH (integrated data hub)

IDIQ (indefinite delivery/indefinite quantity)

Contracts that enable government customers to buy information technology (IT) services from preselected vendors at prenegotiated rates. These contracts, developed and administered by individual government organizations to meet their own IT service requirements, are usually available to all federal agencies for a small (1 percent) administrative fee. The contracts typically cover a broad scope of IT services, and are awarded to several "prime" contractors. Each prime contractor team includes a group of subcontractors that enable it to provide the full suite of required services.

IDL (interface definition language)

A standard language for defining objects' abstract descriptions in terms of their external interfaces (i.e., methods and parameters). The IDL compiler will create relevant runtime static and dynamic interface binding information.

IDM (integrated document management)

A vital class of middleware services that integrates library services, document-manufacturing and document-interchange technologies with critical business-process applications. The term "integrated" describes the transformation of document management from an end-user application to a network-based service integrated with a full complement of end-user personal-productivity and custom-developed applications.

IDMS (Integrated Database Management System)

The database management system introduced by Cullinet in 1973, based on technology acquired from B.F. Goodrich. Computer Associates (CA) acquired Cullinet and has maintained the product line (also called "CA-IDMS") since 1989.

IDN (Integrated Digital Network)

A network employing both digital switches and digital transmission.

IDNX (Integrated Digital Network Exchange)

IDOM (integrated document and output management)

Gartner's term for an integrated collection of technologies, architectures and services aimed at applying the value of information in documents to business practices. IDOM takes a digital approach to reducing the burden and cost of paper-based documents.

IDOM suite

A Gartner concept that describes the convergence of digital document-processing technologies into integrated product portfolios. An IDOM suite is an aggregation of "point" technologies, but it is also a product architectural foundation built on a core set of backbone services providing the basics of cross repository query, common administration, unified system management and integrated application development environments.

IDRP (Interdomain Routing Protocol)

IDS (integrated delivery system)

A coordinated system for healthcare delivery that includes hospitals, clinics and physician practices.

IDS (Internet document security)

Services that provide a core set of capabilities required to enable business communications to move in a secure electronic form over the Internet.

IDS (intrusion detection system)

A software product or hardware device that monitors the events occurring in a computer system or network and analyzes them for signs of intrusion. By applying the latest security and attack expertise to separate a relatively few suspicious events from a vast amount of benign activity, an IDS enables more-effective network security administration and facilitates timely response.

ISDL (ISDN digital subscriber line)

IE (information engineering)

A methodology for developing an integrated information system based on the sharing of common data, with emphasis on decision support needs as well as transaction-processing (TP) requirements. It assumes logical data representations are relatively stable, as opposed to the frequently changing processes that use the data. Therefore, the logical data model, which reflects an organization's rules and policies, should be the basis for systems development.

IE (Internet Explorer)

Microsoft's Web browser, now partially integrated into Windows.

IEC (International Electrotechnical Commission)

A standards-setting body, often jointly responsible with the International Organization for Standardization (ISO) for Open Systems Interconnection (OSI) standards.

IEEE (Institute of Electrical and Electronics Engineers)

An organization of engineers, scientists and students involved in electrical, electronics and related fields. IEEE also functions as a publishing house and standards body.

IEEE 802

An Institute of Electrical and Electronics Engineers standard for interconnection of local-area networking equipment dealing with the physical and link layers of the International Organization for Standardization (ISO) model for Open Systems Interconnection (OSI).

IEEE 802.1p (priority queues)

A standard that provides for up to eight distinct priority queues within local-area network (LAN) switches.

IEEE 802.1Q or IEEE 802.1q (bridged virtual local-area networks)

A standard for the identification of virtual LANs, to increase the interoperability between different vendors for switch trunking and server connections.

IEEE 802.3

A standard for CSMA/CD (carrier sense multiple access with collision detection) LANs such as Ethernet.

IEEE 802.4

A standard for token bus LANs.

IEEE 802.5

A standard for token ring LANs.

IEEE 802.6

A standard for metropolitan-area networking.

IEEE 1394 (see FireWire)

IEF (Information Engineering Facility)

A life-cycle management computer-aided software engineering (CASE) product offered by Texas Instruments that focuses on Customer Information Control System/Database-2 (CICS/DB2) applications.

IEFM (integrated e-form management)

A Gartner concept that describes a system designed for creating and administering e-form applications using installed databases, messaging, document management and workflow infrastructures.

IESG (Internet Engineering Steering Group)

An organization appointed by the Internet Activities Board (IAB) to manage the operation of the Internet Engineering Task Force (IETF).

IETF (Internet Engineering Task Force)

A subgroup of the Internet Society responsible for recommending protocols and procedures used over the Internet. It is composed of representatives from vendor, government and academic communities, and is divided into six subcommittees (with further divisions into working parties). The chairperson sits on the Internet Activities Board (IAB). IETF working group meetings are open to the public.

IEW (Information Engineering Workbench)

KnowledgeWare's upper-CASE (computer-aided software engineering) product that includes planning, analysis and design functions.

IFPUG (International Function Point Users Group)

IFRB (International Frequency Registration Board)

Within the International Telecommunication Union (ITU), the IFRB is responsible for the maintenance of an international list of radio frequency usage and the allocation of new frequencies.

IFX (Interactive Financial Exchange)

A financial services specification published by the Banking Industry Technology Secretariat (BITS).

IGBT (insulated-gate bipolar transistor)

IGES (Initial Graphics Exchange Specification)

A standard for the exchange of computer-aided design (CAD) geometry. IGES has been largely superseded by STEP (Standard for the Exchange of Product Model Data) and PDES (Product Design Exchange using STEP). A vendor-neutral method of

representing parts, geometrics and product dimensions, used as an intermediate system for transfer between specific computer-aided design (CAD) products.

IGFET (insulated-gate field-effect transistor)

IGP (Interior Gateway Protocol)

A Transmission Control Protocol/Internet Protocol (TCP/IP) used by interior routers to move information within an autonomous system or group of networks under the control and authority of a single entity. IGP transfer packets from one network to an adjacent one. Examples are: Enhanced Interior Gateway Routing Protocol (EIGRP), open shortest path first (OSPF) and Routing Information Protocol (RIP).

IGRP (Interior Gateway Routing Protocol)

Cisco Systems proprietary routing algorithm.

IHV (independent hardware vendor)

IIA (Information Industry Association)

IIN (IBM Information Network)

IIOP (Internet Inter-ORB Protocol)

Part of the Object Management Group's Common Object Request Broker Architecture (CORBA), IIOP is the protocol used for communication between CORBA object request brokers (ORBs) over Internet Protocol networks. It is designed to enable systems using CORBA middleware from any vendor to communicate with objects from any other vendor's CORBA product.

IIS (Internet Information Server)

Microsoft's Hypertext Transport Protocol (HTTP) server, originally packaged with the Windows NT operating system.

IKE (Internet key exchange)

IL (Intermediate Language)

ILEC (incumbent local-exchange carrier)

A local-exchange carrier (LEC) that existed prior to entry of competitive LECs (CLECs) in the wake of telecommunications deregulation. (See CLEC.)

ILM (IBM License Manager)

ILPT (instructor-led practical training)

A training method that has some lectures and discussions, but concentrates on giving trainees many chances to practice what they are learning in a risk-free situation.

ILS (Internet locator service)

ILTC (instructor-led training in the classroom)

The most common form of training. It uses lecture sessions and discussions to convey information. It is most commonly used for briefings and high-level education, but many commercial training courses also use this method.

IM (information management)

A method of using technology to collect, process and condense information with a goal of efficient management. Most large enterprises have a central IM function to facilitate this coordination. The primary technologies needed are contained in a set of modeling tools that either have or interface to a production-worthy repository where the information is stored and managed. The repository and tools must be capable of receiving information in a "top-down," "bottom-up" or "middle-out" evolutionary manner.

IM (see instant messaging)

IM (interactive marketing)

Integrated IM systems consist of two or more of the following functions: personalization, ad management, Web measurement, content management and commerce transaction management.

IMA (Institute of Management Accountants)

IMA (inverse multiplexing over asynchronous transfer mode)

IMAC (installations, moves, adds and changes)

Image and Scanner Interface Specification (see ISIS)

ImagePlus

A document image package, introduced by IBM in 1988, that represented the commercial versions of image systems developed jointly by IBM and two customer "partners," USAA and Citibank. ImagePlus provided generic application software for using document images in common user environments.

image support

Hardware (scanner, workstation, printer) and software support for image as a system-recognized information type. Typically, although not necessarily, support for optical storage devices is included.

image understanding

Image understanding systems analyze captured still video images and extract their content (e.g., color, texture, shape) for indexing or other action. Image understanding is used for content-based retrieval of image archives.

IMAP (integrated multiservice access platform)

A scalable and flexible network access technology that can host different types of functionality, regardless of the network source. An IMAP provides access to voice, data and multimedia services through a single, integrated platform that incorporates functions such as switching, routing and multiplexing. It can be located either at the core of the network close to the central office, or at the subscriber demarcation point. IMAP solutions are becoming popular in North America, Europe and some Asia/Pacific countries. The target markets for the deployment of advanced IMAP solutions are in high-subscriber-density areas, and in the new installations of incumbent and alternative network operators.

IMAP (Internet Message Access Protocol)

A protocol used to access e-mail or bulletin board messages from a (possibly shared) mail server. IMAP allows a client e-mail program to access remote message stores as if they were local. E-mail stored on an IMAP server can be manipulated from a workstation at the office, a desktop computer at home or a notebook computer while traveling, without requiring the transfer of messages or files back and forth between these computers. Details of the IMAP specification can be found at www.imap.org.

IMDBS (in-memory database system)

A database design that obviates navigating the complexities of multilevel stores compared with traditional random-access memory (RAM)-and-disk-aware relational database management systems (RDBMSs). IMDBS begins with the assumption that all data is resident in memory. In an IMDBS, pointers point directly to the data and do not need to be translated to disk addresses and blocks. This technology bolsters database application performance for embedded applications and tactical requirements where performance gains can be traded off against traditional RDBMS strengths.

IML (initial microprogram load)

IMLC (Indexed Monthly License Charge)

An S/390 software licensing arrangement offered by IBM. It yields charges based on the machine capacity in millions of service units (MSUs), if the machine was not previously available, and has capacity greater than the newly defined upper limit to the Graduated Monthly License Charge (GMLC) Group 80 — approximately 480 million instructions per second (MIPS).

i-mode

NTT DoCoMo's mobile system that allows users to view specially formatted Web sites, receive e-mail, and access financial, travel and news information via their mobile phones. A key feature of i-mode is that it offers constant connection to the Internet.

IModem

A data transmission protocol with no error correction; intended for high-speed modems with built-in error correction.

impedence

A measure of the electrical property of resistance, expressed in ohms.

IMPL (initial microcode program load)

implementation inheritance

A relationship among classes in which a subclass shares, overrides or supplements operations or data values from one or more superclasses. A subclass is a specialization of one or more superclasses.

implicit tagging

A frame is classified as belonging to a particular virtual local-area network (VLAN) based on the data content of the frame and/or the receiving port.

implode

Compression of detailed data into a summary-level record or report.

IMS (Information Management System)

The name for two IBM Multiple Virtual Storage (MVS) products, a database management system (IMS/DB) and a transaction-processing monitor (IMS/TM), which are capable of accessing either Database-2 (DB2) or IMS/DB or both. IMS applications run in what are called message-processing regions, each in its own address space. They communicate as needed with the IMS control region, which can be located in a separate MVS address space.

IMS/DB (IMS/Database Manager)

A Multiple Virtual Storage (MVS) database management system (DBMS) that supports the hierarchical data model (with optional extensions for some network data model features). IMS/DB may be used with either Information Management System/Transaction Monitor (IMS/TM) or Customer Information Control System (CICS) to provide online access to the database, or it may be used without a monitor for batch processing.

IMS/DC (IMS/Data Communication)

IMS/DC is the part of Information Management System (IMS) that is used to create online transaction processing (OLTP) applications. Roughly comparable to Customer Information Control System/Virtual Storage (CICS/VS), IMS/DC became known as IMS/ESA TM (Information Management System/Enterprise Systems Architecture Transaction Manager) when ESA was introduced (see IMS/ESA TM). IMS/DC has a multiaddress space architecture and is quite fast-which explains why IBM tends to run its DB2 benchmarks with IMS/DC rather than CICS.

IMS/ESA (IMS/Enterprise Systems Architecture)

IMS/ESA may be used with either IMS/DC (Information Management System/Data Communication) or CICS/VS (Customer Information and Control System/Virtual Storage) to provide online access to the database.

IMS/ESA TM (IMS/ESA Transaction Manager)

The part of Information Management System (IMS) needed to create online transaction processing (OLTP) applications. Until the announcement of Information Management System/Enterprise Systems Architecture (IMS/ESA), IMS/TM was a feature of IMS, not a product separate from Information Management System/Database Manager (IMS/DB). IMS/TM, once called Information Management System/Data Communication (IMS/DC), has a multiaddress-space architecture. It is quite fast, which is why IBM runs its Database-2 (DB2) benchmarks with IMS/DC rather than Customer Information Control System (CICS).

IMS/FP (IMS/FastPath)

A functional extension to IBM's Information Management System (IMS) that improves performance for specific transactions, and features high data availability and specialized facilities for very large databases. It was introduced as a separately priced feature of IMS v.1.1.4. With IMS v.1.1.3, FastPath became part of Information Management System/Data Communication (IMS/DC).

IMS/VS (IMS/Virtual Storage)

A common IBM host operating environment, usually under the IBM Multiple Virtual Storage (MVS) operating system, oriented toward batch processing and telecommunications-based transaction processing.

IMT-2000 (International Mobile Telecommunications-2000)

The ITU's name for a family of third generation (3G) cellular standards. IMT-2000 is aimed at providing a standard framework for worldwide wireless access that links the diverse system of terrestrial- and satellite-based networks.

IMTS (international message telephone service)

IN (intelligent network)

An architecture being standardized by the European Telecommunications Standards Institute (ETSI), the International Telecommunication Union (ITU) and the American National Standards Institute (ANSI). INs originated from Ameritech's Feature Node/Service Interface concept of the late 1980s. The premise of an IN is to have a "dumb" switch controlled by a computer for which new applications (services) can be created.

incumbent local-exchange carrier (see ILEC)

INDA (investigational new drug application)

indefinite delivery/indefinite quantity (see IDIQ)

Independent Computing Architecture (see ICA)

independent software vendor (see ISV)

Indexed Monthly License Charge (see IMLC)

Indexed Sequential Access Method (see ISAM)

industrial computer

A personal or process control computer that is designed to withstand the rigors of the factory floor. Some industrial computers are configured so that maintenance and cold start-up are relatively simple. These devices are used for applications such as data collection, monitoring and programming.

Industry Standard Architecture (see ISA)

INEWS (integrated electronic warfare system)

information architecture

All the sources of information — including paper, graphics, video, speech and thought — that define the enterprise are represented by this layer of applications architecture. It also defines the sources and destinations of information, its flow through the enterprise, as well as the rules for persistence, security and ownership.

Information and Content Exchange (see ICE)

information engineering (see IE)

Information Engineering Facility (see IEF)

Information Engineering Workbench (see IEW)

information filtering

See "filtering."

information (knowledge) assets

Information relevant to an enterprise's business function, including captured and tacit knowledge of employees, customers or business partners; data and information stored in highly-structured databases; data and information stored in textual form and in less-structured databases such as messages, e-mail, workflow content and spreadsheets; information stored in digital and paper documents; purchased content; and public content from the Internet or other sources.

information management (see IM)

Information Management System (see IMS)

information oscillation

A concept that identifies the need to manage information in the supply chain, taking into account such factors as time zones, shipping and receiving schedules, and open and closed days. As near-real-time event information is provided in collaborative environments, work process rules must account for escalation processes to allow for human intervention in continuous operations.

information path

The functional route by which information is transferred.

information portal (see enterprise portal)

information repository

Holds all the information needed to create, modify and evolve a software system, including information on the problem to be solved, problem domain, emerging solution, software process being used, project resources and history, and organizational context.

information resource center (see IRC)

Information Resource Dictionary System (see IRDS)

information retrieval (see IR)

information superhighway

A proposed high-speed computing and communications network to deliver ubiquitous voice, telephony data, video graphics and more. The Internet itself was originally cited only as a model for the information superhighway, though the popularity of the Web made it the default successor to the concept. See Web and Supranet.

information systems (see IS)

Information Systems Management Architecture (see ISMA)

information technology (see IT)

Information Technology Infrastructure Library (see ITIL)

Information Technology Steering Committee (see ITSTC)

infrared

The portion of the electromagnetic spectrum above radio frequencies but below visible light, with wavelengths between 10 nanometers and 780 nanometers.

infrared communications (see IR communications)

Infrared Data Association (see IrDA)

infrastructure resource management (see IRM)

Initial Graphics Exchange Specification (see IGES)

INMARSAT (International Maritime Satellite Organization)

INMARSAT's work now increasingly covers aeronautical and mobile as well as maritime satellite communications.

in-memory database system (see IMDBS)

INMS (integrated network management system)

Innovative Applications of Artificial Intelligence (see IAAI)

in-place conversion

A data conversion technique used in countries where the magnitude and format of the euro and the local currency are similar. The conversion consists of replacing local currency values in the database with their euro equivalents.

input/output (see I/O)

input/output control system (see IOCS)

Inquire/Text

Content-based retrieval and document management product for IBM Multiple Virtual Storage (MVS) and Virtual Machine (VM) environments from Infodata of Falls Church, Va.

insertion loss

The difference in the amount of power received before and after a device is inserted into a circuit or a call is connected.

INSM (integrated network and systems management)

in-sourcing

The process of separating the IS organization from the enterprise, usually as a business unit measured by its own profit and loss. The in-sourced IS organization provides the enterprise with IT services on a business-rules basis, e.g., semiformal contracts, service-level agreements (SLAs) and definition of tariffs for services.

INSPECT

Gartner's iterative framework for planning and executing application change. The letters stand for: Inventory, Scope, Parse, Examine, Consider options, Tactical solutions.

instant messaging

A communication service in which short messages appear in pop-up screens the instant they are received, thereby commanding the recipient's immediate attention. Examples include ICQ and AOL Instant Messenger.

Institute for Certification of Computer Professionals (see ICCP)

Institute of Electrical and Electronics Engineers (see IEEE)

instructor-led practical training (see ILPT)

instructor-led training in the classroom (see ILTC)

insured ate

In an asynchronous transfer mode (ATM) network, the long-term agreed rate of data throughput.

INTAP (Interoperability Technology Association for Information Processing, Japan)

The Japanese organization concerned with conformance testing for Open Systems Interconnectivity (OSI) products.

integrated application environment (see IAE)

Integrated Cached Disk Array (see ICDA)

integrated circuit (see IC)

Integrated Cluster Bus (see ICB)

integrated computer-aided software engineering (see I-CASE)

Integrated Coupling Migration Facility (see ICMF)

Integrated Database Application Programming Interface (see IDAPI)

Integrated Database Management System (see IDMS)

Integrated Definition Methodology (see IDEF)

integrated delivery system (see IDS)

integrated development environment (see IDE)

integrated digital access (see IDA)

Integrated Digital Enhanced Network (see iDEN)

Integrated Digital Network (see IDN)

integrated document and output management (see IDOM)

integrated document archive and retrieval system (see IDARS)

integrated document management (see IDM)

integrated e-form management (see IEFM)

integrated e-mail

A facility that enables PC users to create, send, receive, read, forward and file mail through the midrange system mail system directly from an application, without explicitly invoking the mail system.

Integrated IS-IS

A routing protocol based on Intermediate System to Intermediate System (IS-IS) but with support for Internet Protocol (IP) and other protocols. Formerly known as Dual IS-IS.

integrated multiservice access platform (see IMAP)

integrated office system (see IOS)

Integrated On-Demand Network (see ION)

integrated plant system (see IPS)

integrated publishing architecture (see IPA)

Integrated Services Digital Network (see ISDN)

Integrated Services over Specific Link Layers (see ISSLL)

integrated service unit (see ISU)

integrated software

A business software productivity program that incorporates a number of applications (typically word processing, database management, spreadsheet, graphics and communications) into one product, allowing data sharing between all or most modules.

Integrated Voice/Data Terminal (see IVDT)

integration broker (see IB)

integration middleware

Middleware for business-to-business (B2B) application integration that includes integration brokers, Web integration servers and portal servers used to transform, route and process Extensible Markup Language (XML) documents.

integration service marketplace

One of three types of emerging e-marketplaces identified by Gartner (along with commodity and business service marketplaces). An integration service marketplace provides a hosted translation, transformation, routing and business process workflow solution to enable enterprise interaction with multiple e-marketplaces and allow marketplace-to-marketplace interaction (see business service marketplace and commodity marketplace).

integration technologies

Required to facilitate data sharing and integration throughout all customer-facing applications.

integrative policy group (see IPG)

Integriion

A private company owned by 18 banks and IBM. Its purpose is to develop and maintain highly scalable and secure online banking solutions for the large-bank marketplace.

integrity

An operating system's ability to ensure that only authorized users can enter a privileged state (supervisor mode, protect key zero, superuser or root). Integrity is a prerequisite for security in an operating system. The term is not synonymous with security, but security requires it. In database technology, it is a broad term denoting the correct state of the database.

Intel Architecture (see IA)

intellectual assets

Intangible assets including employees' knowledge; data and information about processes, experts, products, customers and competitors; brand names and image; and intellectual property, such as patented, trademarked or copyrighted materials, as well as regulatory licenses.

intellectual (knowledge) capital

Intangible assets of an enterprise that are required to achieve business goals, including knowledge of employees; data and information about processes, experts, products, customers and competitors; and intellectual property such as patents or regulatory licenses.

intellectual property

Intellectual property traditionally includes assets that are protected through regulatory methods such as patents, copyrights and regulatory licenses; however, this protection is being expanded to include software and business processes when these can be demonstrated to be original, novel and non-obvious. Customer intelligence and business intelligence may be considered intellectual "property" by its owner, depending on its value to enterprise competitiveness and its integration into business processes.

intelligent agent (see IA)

intelligent character recognition (see ICR)

Intelligent Content Distribution Service (see ICDS)

intelligent hub

An evolution of the wiring concentrator, an intelligent hub can also act as a platform for chassis-mounted bridges, routers, terminal servers, gateways, and print servers, and can be fully managed.

Intelligent Input/Output (see I₂O)

intelligent matching

Refers to the process of identifying a set of likely matches in a typical registration system rather than attempting to determine an exact match. Intelligent matching seeks closeness rather than exactness of matches. Typical registration systems in integrated delivery systems (IDSs) use name or social security number (SSN) to obtain an exact match. By using phonetic names and other variables, intelligent matching allows for variability in spelling and thus, an increased number of matches. The result is a list of most likely matches.

intelligent network (see IN)

Intelligent Printer Data Stream (see IPDS)

intelligent terminal

A terminal with some logical capability; a remote device that is capable of performing some editing or other function upon input or output data.

intelligent workstation (see IWS)

interactive

In communications, a term describing time-dependent data communications, typically where a user enters data and then awaits a response message from the destination before continuing.

Interactive Financial Exchange (see IFX)

interactive marketing (see IM)

interactive selling system (see ISS)

Interactive System Productivity Facility (see ISPF)

interactive television (see interactive TV)

Interactive Terminal Interface (see ITI)

interactive TV

Provides two-way access to entertainment, communication, information and transaction services through cable, telephone or wireless networks using a low-cost addition to a television set. See Web TV.

interactive voice response (see IVR)

interblock gap (see IBG)

interchange point

A location where interface signals are transmitted.

intercom

An internal communications system that enables calling within the same building, but not outside the system.

interexchange carrier (see IXC)

interexchange channel (see IXT)

interface

The boundary between two pieces of equipment across which all the signals that pass are carefully defined. The definition includes the connector signal levels, impedance, timing, sequence of operation and the meaning of signals.

interface definition language (see IDL)

interface inheritance

An object request broker (ORB) technique for reusing the interface definition language (IDL) from another object implementation (program) in a new one. Unlike implementation inheritance, this does not make data and code visible and usable by another object.

Interior Gateway Protocol (see IGP)

Interior Gateway Routing Protocol (see IGRP)

InterLATA

A United States term meaning between different local access and transport areas.

interleaving

A multiprogramming technique in which segments of one program are inserted into other programming so that during delays in the processing of one of the programs, the other program can be processed.

Intermediate-System-to-Intermediate-System (IS-IS)

A routing method among intermediate systems that requires the end systems, rather than the intermediate systems, to be responsible for providing error correction. It reduces complexity and overhead of routing protocols.

internal rate of return (see IRR)

International Code Designator (see ICD)

International Electrotechnical Commission (see IEC)

International Frequency Registration Board (see IFRB)

International Information Systems Security Certification Consortium (see ISC2)

International Maritime Satellite Organization (see INMARSAT)

International Mobile Telecommunications-2000 (see IMT-2000)

International Organization for Standardization (see ISO)

International Organization for Standardization/International Electrotechnical Commission (see ISO/IEC)

International Telecommunication Union (see ITU)

international trade systems (see ITS)

Internet

A loose confederation of independent yet interconnected networks that share information using a standard set of protocols. The founding principles of the Internet can be traced back to the early 1960s, when the U.S. Department of Defense Advanced Research Projects Agency began to conduct research into packet-switching technology. Since then, it has grown to become a global "network of networks" connecting millions of users worldwide. These networks are connected through "gateways," which enable the transfer of electronic data worldwide using a common address system and a common communications protocol called Internet Protocol (IP). This, together with the Internet's ubiquity, makes it an excellent tool for distributing and sharing information.

The Internet is best known for (and also sometimes mistaken as being synonymous with) the World Wide Web (known as "the Web" for short), but the Web is just one of many applications on the Internet. In addition to Web access, Internet services include e-mail, file transfer (using File Transfer Protocol) and newsgroups. Many organizations contribute to the ongoing development and standards efforts of the Internet, including Internet Society, The Internet Engineering Task Force, the Internet Architecture Board and the Internet Assigned Numbers Authority (which assigns IP addresses).

Internet 2

An initiative by the research and academic community to build a 2.4-gigabit-per-second (Gbps) network. It outlines a plan for network service providers to converge their data networks and then, much later, their voice and data networks. It has specific plans for creating a closed user group of research and academic institutions that is connected via a high-speed network offering varying classes of services as well as guaranteed quality of service.

Internet Activities Board (see IAB)

Internet address (see IP address)

Internet Assigned Numbers Authority (see IANA)

Internet Authentication Service (see IAS)

Internet-based digital stamping

A technology that will let users download postage from the Internet, then print it on envelopes. It requires an Internet connection and the stamping application, and may

require an optional hardware component that resides between the PC and printer and contains the downloaded postage. Enterprises can also access postage directly from vendors' Web sites during the stamping process, so the postage is never stored locally.

Internet commercial publishing revenue (see ICPR)

Internet Control Message Protocol (see ICMP)

Internet data center (see IDC)

Internet document security (see IDS)

Internet-enabled

Linked to the Internet through Internet protocols.

Internet Engineering Steering Group (see IESG)

Internet Engineering Task Force (see IETF)

Internet Explorer (see IE)

Internet Information Server (see IIS)

Internet Inter-ORB Protocol (see IIOP)

Internet kiosks

Stand-alone terminals designed for accessing information via the Internet typically comprised of hardware, software, touch screens, printers, and other peripheral devices.

Internet Message Access Protocol (see IMAP)

Internet Protocol (see IP)

Internet Protocol virtual private network (see IP VPN)

Internet Relay Chat (see IRC)

Internet Research Task Force (see IRTF)

Internet sales outlet (see ISO)

Internet Server API (see ISAPI)

Internet service provider (see ISP)

Internet Society (see ISOC)

Internet telephony

Use of Internet protocols for transmitting two-way audio signals in real time, as an alternative to traditional telephone carriers.

Internetworking Operating System (see IOS)

Internetwork Packet Exchange (see IPX)

Internet Protocol address (see IP address)

interoperability

The ability for a device from one manufacturer to work with one from another.

Interoperability Technology Association for Information Processing, Japan (see INTAP)

interpreter

A computer program that translates and executes each source language statement before translating and executing the next one. Also known as incremental coupler.

interprocess communications (see IPC)

interrupt

A jump out of one software program into another due to an external event. A mechanism is usually provided to store the information needed for a return to the interrupted program.

interrupt request (see IRQ)

Interswitching System Interface (see ISSI)

IntraLATA

United States term meaning within a single local access and transport area (LATA).

intranet

A network internal to an enterprise that uses Internet technology and protocols. It is not necessarily connected to the Internet and is commonly secured from it using firewalls. Intranets are often used by companies for informational purposes, such as distributing internal announcements or displaying job postings, internal directories and organizational charts.

intranet semisuites

Web and Internet standards-based products that provide integrated capabilities to end users including e-mail-based notification (to notify team members of changes or updates in objects of interest — a publish-and-subscribe mechanism), work and project-tracking tools, chat, instant messaging, real-time broadcasting ("push") and T.120-based screen and application sharing.

intranet VPN (intranet virtual private network)

A virtual private network (VPN) that secures portions of an internal network from insiders who do not have the required access privileges.

intrusion detection system (see IDS)

INTUG (International Telecommunications Users Group)

inverse multiplexing

Several combined, lower-speed circuits that make up one circuit for greater bandwidth. Inverse multiplexing also pulls together and synchronizes multiple channels at the receiving end of data, voice, or video transmission.

investigational new drug application (see IND)

I/O (input/output)

The activity of sending information to or from peripheral devices, terminals, direct-access storage devices (DASDs), tape drives and printers. Physical I/O performance lags behind that of memory and logical technologies.

I/O bound

Refers to programs with a large number of I/O (input/output) operations, which slow the central processing unit (CPU).

I/O channel

Equipment forming part of the input/output system of a computer. Under the control of input/output (I/O) commands, the channel transfers blocks of data between main storage and peripherals.

I₂O (Intelligent Input/Output)

The I₂O initiative seeks to change server input/output (I/O) from a tightly related driver-device structure to a more abstract, layered, message-oriented structure. It is being fostered by the I₂O Special Interest Group (I₂O SIG). The I₂O architecture is designed to facilitate the introduction of new I/O devices while offloading most I/O processing to intelligent I/O processors.

IOCS (input/output control system)

A set of programs used in an operating system that handles all data transfers including opening and closing files.

ION (Integrated On-Demand Network)

The enterprise-oriented part of Sprint's overall strategy to migrate from various transmission, access and switching systems to a Synchronous Optical Network (SONET)/dense wave-division multiplexing (DWDM) enhanced-transmission platform and an asynchronous transfer mode (ATM) core-switching fabric. It is intended to ultimately deliver all local service, local access and long-distance voice and data network services.

IOP (input/output processor)

IOS (integrated office system)

The label applied to second-generation office information systems (OISs) originating from the tight integration between the bundled applications, usually word processing, filing, e-mail, calendars, scheduling and basic decision support.

IOS (Internetworking Operating System)

The proprietary software run by Cisco routers and access servers. It supports both local-area network (LAN) and wide-area network (WAN) protocols. It originated from code developed at Stanford University in the mid-1980s.

IP (Internet Protocol)

Transmission Control Protocol/Internet Protocol (TCP/IP) tracks the address of nodes, routes outgoing messages, and recognizes incoming messages. Current networks consist of several protocols, including IP, Internetwork Packet Exchange (IPX), DECnet, AppleTalk, Open Systems Interconnection (OSI) and LLC2. This wide diversity of protocols results from application suites that assume their own particular protocols. Collapse from this wide variety is inevitable, but users will only be able to reduce this diversity, not eliminate it. Most users will collapse networks into two main protocols: IP and IPX. Installed-base applications and the pain of change will prevent a total reduction to a single backbone protocol.

IPA (integrated publishing architecture)

A Gartner concept that defines the creation, assembly and production functions to support the complete document life cycle. Creation includes document component modeling, authoring, content identification, revision, review and approval; assembly includes document aggregation modeling, on-demand document construction and database interchange; and production includes composition, layout and file formatting for multiple communication channels.

IP address (Internet Protocol address)

A unique number assigned by an Internet authority that identifies a computer on the Internet. The number consists of four groups of numbers between 0 and 255, separated by periods (dots). For example, 195.112.56.75 is an IP address.

IPAM (intellectual property asset management)

IPC (interprocess communications)

A system that lets threads and processes transfer data and messages among themselves; used to offer services to, and receive them from, other programs.

IPDS (Intelligent Printer Data Stream)

IBM format for sending files to a laser printer. It provides an interface to all-points-addressable printers that make possible the presentation of pages containing a mix of different types of data, such as high-quality text, raster image, vector graphics and bar codes. In addition, IPDS provides commands for the management of printing resources such as fonts and overlays; for the control of device functions such as paper sourcing and stacking; for the comprehensive handling of exception functions; and for a complete acknowledgment protocol at the data stream level. IBM has positioned IPDS as the ubiquitous enterprise data stream.

IPF (Itanium Processor Family)

The family of 64-bit, Itanium-branded processors from Intel, including Itanium and Itanium 2. (See Itanium.)

IPG (integrative policy group)

IPGs are responsible for the formulation and maintenance of all policy and procedures regarding an enterprise's intranet. Cross-functional representation on this team is critical and should include the IS department, business units and corporate communications. While the information systems (IS) department should play a major role, facilitation of the team should be provided by someone from a key business unit, if possible.

IPI (Intelligent Peripheral Interface)

IPL (initial program load)

IPM (impulses per minute)

The interruption rate for call progress tones.

IP multicast

A routing technique that enables propagation of Internet Protocol (IP) traffic from one source to many destinations.

IP next generation (see IPng)

IPng (IP next generation)

A new standard intended to replace Internet Protocol (IP) under the auspices of the Internet Engineering Task Force (IETF) under RFC 1752. It offers 128-bit addressing to overcome the limitations of the numbers of IP addresses. It also has facilities for mobile logins and includes authentication.

IPNS (ISDN PBX Networking Specification Forum)

A forum established by major European vendors with the aim of contributing to European and world standards organizations in the area of Integrated Services Digital Network (ISDN) private branch exchange (PBX) networking.

IP/PBX (Internet Protocol Private Branch Exchange)

IPS (integrated plant systems)

A set of integrated computerized systems for open control, production management, production information management and analysis, model-based control and optimization, plant resource planning and plant resource reconciliation functions. IPSs combine open control system (OCS) and manufacturing execution system (MES) functionality in the broader context of enterprise and supply chain management (SCM) applications.

IPsec (IP Security)

A working group of the Internet Engineering Task Force (IETF) that is developing a security standard for Internet Protocol (IP). IPsec is also the name of the standard itself. It defines protocols for authentication, privacy and data integrity based on encryption and X.509 digital certificates.

IP Security Option (see IPSO)

IPSO (IP Security Option)

A U.S. Government specification defining an optional field in the Internet Protocol (IP) packet header for hierarchical packet security levels on a per-interface basis.

IP subnet

An Internet Protocol (IP) subnet or subnet mask is a way to subdivide a network into smaller networks to allow a large number of computers on a network with a single IP address.

IP Telephony

Voice traffic or data traffic that has traditionally been treated like voice traffic — it is transmitted over an Internet Protocol (IP) network. Three distinct types of IP telephony traffic exist: real-time voice, non-real-time voice and fax.

IPV (in-place value)

IPv6 (Internet Protocol Version 6)

A set of specifications from the Internet Engineering Task Force. IPv6 offers increased network-addressing capacity compared with earlier versions of Internet Protocol, and supports mechanisms for quality of service.

IP VPN (Internet Protocol virtual private network)

An Internet Protocol network that delivers private, enterprise-focused data communications services using a public-network infrastructure. See VPN.

IPX (Internetwork Packet Exchange)

A widely used routing protocol, based on Xerox's XNS, developed by Novell and implemented in Novell's NetWare. From a user perspective, an IPX implementation allows concurrent dual access to both an application server and a native NetWare file server without imposing memory overhead on client workstations.

IQS (Internet quality of service)

IR communications (infrared communications)

Cableless communication connections similar to household remote control devices.

IR (information retrieval)

Structured or unstructured data searching to retrieve information specific to a computer user's needs. IR tools index documents based on textual content. Web search engines are an example of IR technology. See "search engine."

IR (infrared)

IRC (information resource center)

A service organization whose primary mission has been to respond to requests for specific information and to organize and maintain content collections. More recently, its role has expanded to the delivery of desktop content resources.

IRC (Internet Relay Chat)

A real-time conferencing service, organized in channels, on the Internet.

IrDA (Infrared Data Association)

An international organization that produces a standard for infrared data transmission (up to 4 Mbps). Because of the low cost of this technology, it is now embedded in a number of devices (e.g., in the head of handheld devices, such as cellular phones).

IRDP (ICMP Router Discovery Protocol)

Similar to End System to Intermediate System (ES-IS) but used with Internet Protocol (IP). IRDP enables a host to determine the address of a router it can use as a gateway.

IRDS (Information Resource Dictionary System)

A standard data dictionary specification from the American National Standards Institute (ANSI) X3H4 Committee.

Iris recognition

A biometric technique that uses the colored ring in a person's eye (i.e., the iris) to authenticate identity. Iris recognition is a relatively new biometric technique that has been used in trials in automated teller machines.

IRLM (IMS Resource Lock Manager)

IRM (infrastructure resource management)

A methodology to manage and optimize IT infrastructure as one complex system.

IRQ (interrupt request)

A hardware device inside of the computer that stops the microprocessor and orders it to wait for further instructions. PCs have a maximum of 15 IRQs, prioritized by importance.

IRR (internal rate of return)

IRR analysis determines the interest rate and then compares this rate to the "risk-adjusted rate of return."

IRSG (Internet Research Steering Group)

IRTF (Internet Research Task Force)

A committee of network experts concerned with long-term research into Transmission Control Protocol/Internet Protocol (TCP/IP) and the Internet. It is further divided into research groups. The chairperson sits on the Internet Activities Board (IAB).

IS (information systems)

The use of and investment in computer technology by the principal or centralized organization formally charged with the responsibility for computer technology. Often the IS organization is led by a chief information officer (CIO), vice president of the IS organization or director of data processing. IS is also often the formal name of the department within an enterprise that is responsible for information technology. Other common names for the IS department are:

- Management information systems (MIS)
- Data processing (DP)
- Information processing (IP)

ISA (Industry Standard Architecture)

Bus architecture originally developed and evolved by IBM for personal computers, now in the public domain. Now largely superseded by the Peripheral Connect Interface (PCI).

ISA (Instrument Society of America)

ISAM (Indexed Sequential Access Method)

Disk access method that stores data sequentially, maintaining an index of key fields to all the records in the file.

ISAPI (Internet Server API)

A proprietary Web server application programming interface (API) from Microsoft. It is geared to provide direct function-level access to its Internet Information Server.

ISB (Integrated System Bus)

ISC2 (International Information Systems Security Certification Consortium)

The non-profit organization that administers the Certified Information Systems Security Professional (CISSP) certification program.

iSCSI (SCSI Over IP)

A network transport standard — supported by Cisco Systems, Adaptec and IBM — that enables storage area networks to be built using Small Computer Systems Interface (SCSI) protocols. iSCSI works by encapsulating SCSI data in Internet Protocol (IP) packets for transport over IP-based networks.

ISDN (Integrated Services Digital Network)

A technical standard and design philosophy for digital networks. ISDN provides high-speed, high-bandwidth channels to every subscribers on the public switched telephone network, achieving end-to-end digital functions with standard equipment interface devices. ISDN networks enable a variety of mixed digital transmission services to be accommodated at a single interface.

ISDN PBX Networking Specification Forum (see IPNS)

ISDN user part (see ISUP)

I series

International Telecommunication Union (ITU) recommendations for Integrated Services Digital Network (ISDN).

ISIS (Image and Scanner Interface Specification)

A common interface standard that allows consistent scanner control dialogue.

ISL (Integrated Services Layer)

ISM (industrial, scientific and medical)

ISMA (Information Systems Management Architecture)

An IBM process model introduced in 1979 that identified 11 key process groups and 42 subprocesses for information technology (IT) management.

ISO (independent service organization)

ISO (International Organization for Standardization)

A voluntary, nontreaty organization established in 1949, as a technical agency of the United Nations, to promote international standardization in a broad range of industries. ISO's Open Systems Interconnection (OSI) Reference Model establishes guidelines for network architectures.

ISO (Internet sales outlet)

A third-party Web site that attracts visitors looking to buy goods or services. ISOs make money by selling links or ads that lead directly to Web merchant sites or by selling products or services on behalf of Web merchants.

ISO 9000

An international standard for quality control, also known as BS5750 and EN29001 (the three standards are identical, but are numbered and published differently by different standards bodies). The worldwide standard is published by the International Organization for Standardization (ISO), and is the one generally referenced. ISO 9000 is a generic standard that any enterprise or individual department can use. It is not specific to the IT industry. The standard is subdivided into three parts — ISO 9001, ISO 9002 and ISO 9003. Enterprises may use some or all of the parts, depending on the nature of their business. Any enterprise can claim that it complies with ISO 9000. To make the claim credible, an external assessor from an accreditation body must evaluate the enterprise's quality system. ISO 9000 certification does not guarantee quality; it guarantees consistency of approach.

ISOC (Internet Society)

Originating from the U.S. academic community, that ISOC is now a body dedicated to the development of Internet standards via its subgroup, the Internet Engineering Task Force (IETF).

isochronous

A transmission technique characterized by synchronization of the transmitter and receiver, so that packets are delivered to the destination at regular intervals, such as that required by voice or video.

isochronous Ethernet

An extension to the Ethernet standard combining a standard Ethernet channel with 96 64 kilobit per second (Kbps) voice grade channels. It is intended for multimedia applications.

isochronous transmission

A transmission where the signal recurs at known, equally spaced time intervals relative to a single time reference. It is a form of periodic transmission.

ISODE (ISO Development Environment)

A software package that allows the use of TP4 (Transaction Processing Facility 4) and high-layer International Organization for Standardization/Open Systems Interconnection (ISO/OSI) protocols over Transmission Control Protocol/Internet Protocol (TCP/IP).

ISO/IEC (International Organization for Standardization/International Electrotechnical Commission)

An international body that controls cabling and wiring standards.

ISP (Internet service provider)

A company that provides Internet access to its customers. The majority of ISPs are too small to purchase access directly from the network access point (NAP), and instead buy pieces of bandwidth that are available from larger ISPs. ISPs are different from online services, although these services sometimes also provide

access to the Internet. Online services provide access to exclusive content, databases and online discussion forums that are not available outside the service.

ISPBX (integrated services private branch exchange)

ISPF (Interactive System Productivity Facility)

ISPF is an IBM-licensed program that is used for writing application programs. It serves as a full-screen editor and dialogue manager and provides a means of generating standard screen panels and interactive dialogues between the application programmer and the terminal user.

ISS (interactive selling system)

An ISS is the integration of a marketing encyclopedia system (MES), a sales configuration system, a proposal generation system and an order management system under a common user interface. It is used by organizations to streamline and enhance the selling process. The ISS effectively combines the application functionality that helps enterprises to configure solutions that meet customer needs, while avoiding data entry and business process errors. With an ISS, the salesperson can sit in front of the customer and pull up data to prove value, configure an order, create a proposal, and then turn the proposal into an order and book the order.

ISS (Internet Security Systems)

ISSI (Interswitching System Interface)

The standard interface between Switched Multimegabit Data Service (SMDS) switches.

ISSLL (Integrated Services over Specific Link Layers)

A working group given the task of providing mappings and interfaces between the advanced services that are available with a number of technologies. Two specific tasks are: looking to negotiate between Resource Reservation Protocol (RSVP) and asynchronous transfer mode (ATM) class-of-service queues, and between Institute of Electrical and Electronics Engineers (IEEE) 802.1p priority queues and asynchronous transfer mode (ATM).

ISSS (Information Society Standardization System)

ISU (integrated service unit)

A single device combining the functions of a channel service unit (CSU) and a data service unit (DSU).

ISUP (ISDN user part)

Provides Integrated Services Digital Network (ISDN) call setup and control and is harmonized to work with ISDN access protocol (ISDN term).

ISV (independent software vendor)

A software producer that is not owned or controlled by a major IT vendor. An ISV is a company whose primary function is to distribute software. Major hardware, operating-system and database vendors that also sell software (such as IBM, Microsoft and Oracle) are not ISVs, nor are companies in nontechnology industries (such as financial services) that may also sell software products.

ISVs typically offer products that the primary vendor (i.e., IBM) does not offer, allowing clients of that vendor to round out their software needs. ISVs create price competition and also increase the pace of technology innovation in their markets.

IT (information technology)

This is the common term for the entire spectrum of technologies for information processing, including software, hardware, communications technologies and related services. In general, IT does not include embedded technologies that do not generate data for enterprise use.

ITAA (Information Technology Association of America)

Itanium

A new processor generation introduced by Intel in 2001, based on IA-64 architecture. Designed for high-end, enterprise-class servers and workstations, it offers increased performance over previous Intel processor generations through expansion from 32- to 64-bit technology, and the introduction of Explicitly Parallel Instruction Computing (EPIC) design. See EPIC and IA-64.

Itanium Processor Family (see IPF)

IT asset management

A systematic approach to managing IT assets, including information systems (IS) department staff, end users performing IT support, technology procurement teams, suppliers, facilities, hardware and software.

IT consulting

An area of consulting that includes systems architecture design/development and IS organizational planning, as well as technical consulting — including technology assessment, and hardware and software tuning.

IT decision support

The creation and management of information sources, reporting processes and automated tools that enable IT executives to implement profitable IT strategies.

item master file

Typically, this computer file contains identifying and descriptive data, and control values (e.g., dead time and lot sizes). It may also contain data on inventory status, requirements and planned orders. There is normally one record in this file for each stock-keeping unit. Item master records are linked together by product structure records, thus defining the bill of material.

IT governance

The set of rules and agreements that permit the enterprise to make decisions, resolve disputes and allow people to work together effectively across departments and other business boundaries. This may include informal relationships as well as service-level agreements and service contracts and may be resolved within steering committees, user councils and other formal or virtual organizational structures.

ITI (Interactive Terminal Interface)

A packet assembler/disassembler (PAD) supporting network access by means of asynchronous terminals.

ITIL (Information Technology Infrastructure Library)

An initiative developed by the Central Computing and Telecommunications Agency consultancy for the government of the United Kingdom. It offers a set of best practices in 24 service delivery and IT service support areas, including help desk, problem management, change management, software distribution and cost control.

ITIL security management

The part of the Information Technology Infrastructure Library (ITIL) designed to guarantee continuity of the IT organization (internal scope) and the quality of IT services (external scope) through service-level agreements (SLAs).

IT infrastructure

The underlying technological components that constitute an organization's systems architecture. The seven components of IT infrastructure are hardware, operating system, network, database, development environment, user interface and application.

ITIS (information technology integration services)

ITOA (IT Overview Analysis)

Using multiple key metrics, Gartner's ITOA integrates data from across several IT functional areas to provide a management-level, integrated overview of IT performance and cost.

ITO (IT outsourcing)

The contractual vehicle through which end users externalize life cycle service and support operations for their IT infrastructure. Outsourcing can be partial (i.e., modular or selective) or total, and can involve not only operations but also the acquisition of customer assets and personnel. IT outsourcing is divided into five primary market segments: data center operations, network operations, client/server operations, application management and desktop management.

IT outsourcing (see ITO)

IT Overview Analysis (see ITOA)

ITS (international trade systems)

Execution systems designed to automate the import/export business process. The basic functional components are trade documentation generation and transmission, and regulatory compliance validation.

ITSEC (Information Technology Security Evaluation and Certification)

ITSTC (Information Technology Steering Committee)

The committee that administers the European Standards Institute/European Electrical Standards Institute's (CEN/CENELEC's) information technology (IT) program. It uses ad hoc expert groups.

ITT (invitation to tender)

ITU (International Telecommunication Union)

An international organization, headquartered in Geneva, Switzerland, within which governments and the private sector coordinate global telecommunications networks and services.

ITV (see interactive TV)

IUnknown

The base interface in Microsoft's Component Object Model (COM) interface hierarchy. All COM interfaces are derived from IUnknown, which provides the mechanism for specifying an object's life span and navigating multiple interfaces.

IVDT (Integrated Voice/Data Terminal)

A device that features a keyboard/display and a voice telephone instrument. Many contain varying degrees of local processing power, ranging from full personal computer (PC) capacity to directory storage for automatic telephone dialing. They may be designed to work with a specific customer-premises private branch exchange (PBX) or to be PBX independent.

IVDW (intelligent voice/data workstation)

IVR (interactive voice response)

A voice/call-processing option for improving call center functionality and integration. It enables callers to have more flexibility to access information or leave messages. Use of this option can "offload" call volume from agents to the IVR or improve load balancing by having agents handle recorded messages during slow periods. A slowly growing number of IVR developers are now using speech recognition in their applications.

IVRS (interactive voice response system)

IV&V (independent verification and validation)

IWS (intelligent workstation)

A term synonymous with programmable workstation (e.g., a personal computer).

IXC (interexchange carrier)

A long-distance telephone company in the United States that provides service between local access and transport areas (LATAs).

IXT (interexchange channel)

A direct circuit between telephone exchanges.

J

J2EE (Java 2 Platform, Enterprise Edition)

A Sun Microsystems platform specification and branding initiative that provides a unifying umbrella for enterprise-oriented Java technologies. J2EE focuses on server-side, multitier services. J2EE includes the Java Server Pages, Java Servlets and Enterprise JavaBeans programming models, a number of protocols and application programming interfaces, a reference implementation, a test suite, and an application model.

J2ME (Java 2 Platform, Micro Edition)

An edition of Sun's Java platform that focuses on small-form-factor devices, such as personal digital assistants, pagers and cell phones.

J2SE (Java 2 Platform, Standard Edition)

The basic Java platform, directed at desktop users.

jabber

An Ethernet data packet with more than 1,518 bytes and a faulty frame check sequence (see FCS).

jack

A device used generally for terminating the permanent wiring of a circuit, access to which is obtained by the insertion of a plug.

JAD (joint application development)

An application development process aimed at achieving high levels of functional quality through the participation of prospective end users. It is especially effective in developing user interface requirements.

JAR (Java Archive)**Java**

The term "Java" can be applied to Sun Microsystems' Java platform or to its Java programming language. The Java platform is made up of a set of technologies that provide cross-platform, network-centric computing solutions. The programming language is simply one aspect of the Java platform. The elements of the Java platform include the Java Virtual Machine (JVM), which provides a uniform Java byte code emulator for Java's cross-platform runtime environment; the Java programming language, which provides a robust, object-oriented language for constructing Java components and applications; and the standard Java class library packages, which provide sets of reusable services that promote consistency among components and applications.

The Java programming language is based on C and extends and complements the basic capabilities of Hypertext Markup Language (HTML). Originally developed as a

language for consumer-oriented devices such as TV set-top boxes, Java became a viable alternative to other programming languages with the rapid growth of the Internet. Java permits the creation of applications and application modules (called "applets") that run in the JVM on the browser. Browsers from both Netscape and Microsoft have a JVM. Java's platform independence and security are designed in, rather than added on, so applications can run on a wide variety of desktop platforms as long as they can run a Java-enabled browser.

Java 2 Platform, Enterprise Edition (see J2EE)

Java 2 Platform, Micro Edition (see J2ME)

Java 2 Platform, Standard Edition (see J2SE)

Java API for XML Messaging (see JAXM)

Java API for XML Processing (see JAXP)

Java applet

A small piece of Java code that implements a specific function. Applets may run on a server or be downloaded and run on the client's machine.

JavaBeans

Similar to Microsoft's ActiveX, this Sun Microsystems' technology that enables external applications access each other's services.

JavaCard

A set of Java application programming interfaces for smart cards.

JavaCard Open Platform

Visa's version of the JavaCard application programming interface set, with added code to make it more secure for financial transactions.

Java Database Connectivity (see JDBC)

Java Interface for OLAP (see JOLAP)

Java Messaging Service (see JMS)

Java Native Interface (see JNI)

JavaScript

A scripting language targeted specifically to the Internet. It is the first scripting language to fully conform to ECMAScript, the Web's only standard scripting language. Despite its name, JavaScript is not a derivative of Java; its origin is Netscape's Livescript language. JavaScript is, in fact, closer to C/C++ in syntax than it is to Java.

Java Server Pages (see JSP)

Java servlet

A Java program that operates in conjunction with a Web server, and can output Hypertext Markup Language (HTML) to a browser or even communicate with Java

applets. Servlets offer an alternative to using Common Gateway Interface (CGI) and server application programming interfaces (SAPIs) to communicate with Web server processes.

Java Telephony Application Programming Interface (see JTAPI)

Java Virtual Machine (see JVM)

JAXM (Java API for XML Messaging)

An application programming interface (API) from Sun Microsystems. JAXM is used to build Java-based e-commerce applications that can generate and exchange Extensible Markup Language (XML) messages.

JAXP (Java API for XML Processing)

A Sun Microsystems application programming interface (API) that enables Extensible Markup Language (XML) documents to be read, manipulated and generated in Java-based applications.

JBIG (Joint Bitonal Image Group)

An standard for black-and-white and gray-scale compression.

JCAHO (Joint Commission on Accreditation of Healthcare Organizations)

JCALs (Joint Continuous Acquisition and Life-Cycle Support)

JCL (Job Control Language)

A language used to communicate with IBM's MVS operating system. JCL is used to tell the system who a user is, what programs are being invoked and what resources will be needed.

JCP (Java Community Process)

JDBC (Java Database Connectivity)

A standard that provides connectivity between Sun Microsystems' Java platform and a range of database management systems. JDBC provides a call-level application programming interface for database access based on Structured Query Language.

Jdirect

Microsoft's application programming interface (API) for directly accessing Win32 API functions.

JDK (Java Development Kit)

JEIDA (Japan Electronic Industry Development Association)

JES (Job Entry Subsystem)

The component of IBM's MVS operating system that reads in jobs, interprets their Job Control Language, schedules their execution, spools the print, and produces the output at the appropriate destination.

JFC (Java Foundation Classes)

JFlow

Officially called Workflow Management Facility (WMF), JFlow was designed to provide a workflow framework and interfaces for the interaction with the workflow process execution environment. In the JFlow model, a workflow system appears externally as a number of different "objects" that must be implemented or simulated. Each invocation of a service creates a "performer object" that knows a reference to a "request object" it must communicate with when it is finished.

Jini

The first distributed-computing product for the consumer market, introduced by Sun Microsystems in July 1998. The system enables any devices that use Sun's Java programming language to communicate and work with one another.

Jiro

A Sun Microsystems' platform that implements Java and Jini technologies to provide an "open storage management platform." The platform is intended to allow multiple storage systems from multiple vendors to work together more cohesively.

JIT (just in time)

The concept of reducing inventories by working closely with suppliers to coordinate delivery of materials just before their use in the manufacturing process.

JITT (just-in-time training)

A training methodology in which users are trained in a technology just when they begin using it.

jitter

A slight movement of a transmission signal in time or phase, which can introduce errors and loss of synchronization for high-speed synchronous communications. See phase jitter.

JMAPI (Java Management Application Programming Interface)**JMS (Java Messaging Service)**

Part of Sun Microsystems' Java 2 Platform, Enterprise Edition (J2EE), JMS is an application programming interface that connects Java programs to messaging middleware, such as IBM's MQSeries and Tibco Software's Rendezvous.

JNDI (Java Naming and Directory Interface)**JNI (Java Native Interface)**

An application programming interface developed by Sun Microsystems that defines a standardized way of naming and calling native functions created in a non-Java language. This is necessary for Java to access pre-existing functionality created with more-historical language environments and languages tuned to specialized purposes. With the JNI, the Java Virtual Machine can locate and invoke native methods, or those native methods can perform Java-specific functions such as creating Java objects, calling Java methods, and even processing Java objects.

Job Control Language (see JCL)**Job Entry Subsystem (see JES)**

joint application development (see JAD)

Joint Bitonal Image Group (see JBIG)

Joint European Standards Institute

The European Standards Institute (CEN) and the European Electrical Standards Institute (CENELEC) combined.

Joint Photographic Experts Group (see JPEG)

JOLAP (Java Interface for OLAP)

A Java application programming interface (API) for the Java 2 Platform, Enterprise Edition (J2EE) environment. JOLAP supports the creation and maintenance of online analytical processing (OLAP) data and metadata, independent of vendor.

Josephson junction

A type of circuit capable of switching at very high speeds when operated at a temperature approaching absolute zero.

JPEG (Joint Photographic Experts Group)

A joint International Organization for Standardization/International Telecommunication Union (ISO/ITU) standard for still-image compression.

JRMP (Java Remote Method Protocol)

A wire protocol for Remote Method Invocation (RMI).

JScript (see JavaScript)

JSP (Java Server Pages)

A mechanism that allows Java source code to be embedded into Hypertext Markup Language (HTML) files. This source code is executed on the Web server (by servlets), and the resulting HTML is output to a Web browser. JSP allows many types of developers to interact with both HTML and Java in designing the interface to Java applications.

JSR (Java Specification Request)

JTAPI (Java Telephony Application Programming Interface)

An application programming interface for developing Java-based telephony applications.

JTS (Java Transaction Service)

jukebox

An automatic media handler for an optical disk drive, also called a library. An optical jukebox is designed to move optical platters in and out of optical drives. The intent is to provide a large amount of easily accessed storage in a "near-line" fashion. Jukeboxes use robotics. Similar in concept to music jukeboxes, the robot arm locates the appropriate disk and, if it is not already mounted in a drive, clears the drive and loads the selected disk.

jumper

A patch cable or wire used to establish a circuit, often temporarily, for testing or diagnostics.

just in time (see JIT)

just-in-time training (see JITT)

JVM (Java Virtual Machine)

The virtual machine (VM) that runs in the Java environment. See VM and Java.

K

Ka-band

The portion of the electromagnetic spectrum allotted for satellite transmission, with frequencies in the 20 to 30 gigahertz (GHz) range.

Kb (kilobit)

Approximately one thousand bits of data (1,024, to be precise).

KB (kilobyte)

Approximately one thousand bytes of data (1,024, to be precise). Each byte contains eight bits.

KBE (knowledge-based engineering)

A system that encapsulates design knowledge that can be recalled, reused and extended to create product designs.

Kbps (kilobits per second)

A measure of data transmission speed — 1,024 bits of information per second.

KBS (knowledge-based system)

See expert system.

KBTU (one thousand British thermal units)

A measure of radiated heat used in determining cooling-system requirements.

KCR (knowledge-centric re-engineering)

The application of business process re-engineering (BPR) and change enablement methodologies in support of enterprisewide knowledge management (KM), thereby effecting major cultural and process changes that are fundamental to the management of the enterprise's competitive position.

Kerberos

An authentication system used for dial-up, remote or Internet connections. An Internet Engineering Task Force (IETF) standard, Kerberos works by having a central server grant a "ticket" honored by all networked nodes running Kerberos.

Kermit

A packet-oriented public-domain file transfer protocol designed at Columbia University.

kernel

The heart of an operating system, a kernel is the part of the operating system that interconnects with the hardware. With Open Systems Interconnection (OSI) software intended for use in Unix environments, the kernel's functional units are often included as a function library.

key generation unit

A portable, plug-in unit that generates a key for insertion into an encryption device.

key performance indicator (see KPI)

keyword

A word specified in a query placed to an information retrieval system, such as a Web search engine.

keyword search

Searching for documents based on one or more words or phrases specified by a computer user. In many search engines, a string of keywords entered without quotation marks is treated as a Boolean "and" search (see "Boolean"), while enclosing the string in quotation marks restricts the search to the complete phrase. See "keyword" and "search engine."

KHz (kilohertz)

A unit equal to 1,000 hertz (i.e., 1,000 cycles per second).

KIF (Knowledge Interchange Format)

kilobit (see Kb)

kilobits per second (see Kbps)

kilobyte (see KB)

kilohertz (see KHz)

kilovolt ampere (see KVA)

Kinetics Internet Protocol (see KIP)

kiosk

A self-service terminal typically used for one of three functions: to support noncash transactions (such as ordering tickets or making reservations), to dispense noncash items (e.g., documents, tickets or coupons), or to provide access to information (e.g., rate quotations, product information or interactive sessions with product specialists). Kiosks typically do not dispense cash.

KIP (Kinetics Internet Protocol)

Allocates Internet Protocol (IP) addresses to AppleTalk computers.

KLOC (thousand lines of code)

A metric that serves as an indicator of the scale of programming or system compliance efforts (e.g., 5 KLOCs equals 5,000 lines of code).

KM (knowledge management)

A business process that formalizes the management and use of an enterprise's intellectual assets. KM promotes a collaborative and integrative approach to the creation, capture, organization, access and use of information assets, including the tacit, uncaptured knowledge of people.

K-map (knowledge map)

A representation of concepts and their relationships (e.g., hierarchy, taxonomy and network). A K-Map is a navigational aid that enables a user to hone in rapidly on the desired concept, and then follow links to relevant knowledge sources (information or people).

KM architecture

The technology infrastructure that supports an enterprise's knowledge management (KM) initiatives, including physical (e.g., intranet, knowledge retrieval tools) and logical (e.g., knowledge maps) dimensions.

KM infrastructure

The people, processes and automated systems required to support knowledge management (KM).

KM IT manager

Implements and oversees the technology that supports the enterprise knowledge architecture, and ensures the technical integrity of knowledge content and automated processes.

KM strategy

A declaration of how the enterprise will use knowledge to compete, and how knowledge management (KM) will support the enterprise's business strategies. A KM strategy may be knowledge-focused if the enterprise's market value is predominantly composed of intellectual capital, or knowledge-enabled if the enterprise has other valuable assets in addition to intellectual capital.

knowledge access

One of the five activities of the knowledge management (KM) process framework. Knowledge access is the retrieval or dissemination of knowledge to users.

knowledge architect

the individual who oversees implementation of the enterprise's knowledge architecture, who leads the "knowledge architecture team" in identifying, organizing and providing access to scattered, heterogeneous information in digital and paper form, and who leads the knowledge audit to determine and continually re-evaluate the specific knowledge needs of users and their business processes. The knowledge architect defines knowledge processes and identifies the technology requirements for creating, capturing, organizing, accessing and using knowledge assets.

knowledge assets

Information relevant to an enterprise's business function, including the captured and tacit knowledge of employees, customers or business partners; data and information stored in structured databases; data and information stored in textual form and unstructured databases (e.g., e-mail and workflow systems); information stored in digital and paper documents; purchased content; and public content from the Internet or other sources.

knowledge audit

A formal determination and evaluation of how and where knowledge is used in business processes. The knowledge audit identifies implicit user needs, as well as explicit information stores. With the audit, enterprises can identify and evaluate all information resources and workflows, and determine enterprise user access

requirements. Access requirements will vary widely, from wide access (e.g., policies and procedures) to extremely limited access (e.g., payroll processing). The knowledge audit is a rigorous process using questionnaires, interviews and resource descriptions.

knowledge base

The knowledge, which may include assertion, rules, objects and constraints, used by a knowledge-based or expert system. Its organization is based on knowledge representations. The developer or user of the system may be unaware of the underlying knowledge representations, seeing only the domain knowledge representations.

knowledge-based engineering (see KBE)

knowledge-based system (KBS)

See expert system.

knowledge capital

Intangible assets of an enterprise that are required to achieve business goals, including employee's knowledge; data and information about processes, products, customers and competitors; and intellectual property such as patents or regulatory licenses.

knowledge capture

One of the five activities of the knowledge management process framework. Knowledge capture makes tacit knowledge explicit, i.e., it turns knowledge that is resident in the mind of the individual into an explicit representation available to the enterprise.

knowledge-centric re-engineering (see KCR)

knowledge community

A group of people within an enterprise who engage in knowledge-sharing activities in support of a common work interest (shared responsibility for a business process, a product or service, or a project). The KC may include people from multiple disciplines within the enterprise, as well as extended-enterprise participants (service providers, supply-chain partners or customers).

knowledge content owners

Individuals who oversee the definition and delivery of knowledge content for their business processes to the knowledge management environment, and define access privileges to their knowledge resources. Human-resources executives are an example of knowledge content owners for benefits information resources; as such, they are responsible for ensuring that appropriate benefits content is available to support knowledge user and resource requirements, and that appropriate users are granted or denied access to it.

knowledge content specialists

Individuals who refine knowledge content from the originating owner into the specific product that the knowledge users require. These specialists have strong skills in the business process and its knowledge resources, as well as skill in organizing and filtering knowledge into a highly accessible and usable form. An extension to this

definition is the external knowledge content specialist, who manages access to external content that the enterprise has purchased or subscribed to.

knowledge creation

One of the five activities of the knowledge management process framework. Knowledge creation results in new knowledge, or new assemblies of existing knowledge.

knowledge management (see KM)

knowledge management infrastructure (see KM infrastructure)

knowledge management IT manager (see KM IT manager)

knowledge map (see K-map)

knowledge organization

One of the five activities of the knowledge management process framework. Knowledge organization is the classification and categorization of knowledge for navigation, storage and retrieval. This includes knowledge maintenance.

knowledge representation

Structures used to store knowledge in a manner that relates items of knowledge to one another, and that permits an inference engine to manipulate the knowledge and its relationships.

knowledge sharing

An activity that spans three components of the knowledge management (KM) process — knowledge capture, knowledge organization and knowledge access — to enable people to share knowledge across the boundaries of geography and time.

Knowledge Sharing Effort (see KSE)

knowledge use

One of the five activities of the knowledge management process framework. Knowledge use is the application of knowledge to business decisions or opportunities. Use is also recursive, and continually generates feedback that affects and is integrated into the other knowledge activities.

knowledge users

Participants in knowledge management programs. They fill the dual roles of applying knowledge in their work tasks and contributing their own knowledge and insight to the enterprise's knowledge content.

knowledge work management

A discipline that promotes an integrated approach to identifying, managing and sharing all of an enterprise's information assets. Knowledge work management focuses on extending knowledge management to business processes (i.e., policies and procedures as well as unwritten rules).

knowledge workplace

This represents the intersection of three key trends: the leverage of intellectual capital, the virtualization of the workplace and the shift from hierarchical to organic

models of management. The focus is on knowledge as the primary source of competitive advantage.

KPI (key performance indicator)

A high-level measure of system output, traffic or other usage, simplified for gathering and review on a weekly, monthly or quarterly basis. Typical examples are bandwidth availability, transactions per second and calls per user. KPIs are often combined with cost measures (e.g., cost per transaction or cost per user) to build key system operating metrics.

KQML (Knowledge and Query Management Language)

KSE (Knowledge Sharing Effort)

An initiative for building standard knowledge representation frameworks, led by a consortium of 18 university and industry research centers. It addresses the entire issues of standard knowledge representation schemes, standard interfaces between objects, the definition of a language that will map between various knowledge representation schemes, and work on standard application-specific ontologies.

KTS (key telephone system)

Ku-band

A portion of the electromagnetic spectrum used increasingly for satellite communications, with frequencies in the 12 to 14 gigahertz (GHz) range.

KVA (kilovolt ampere)

KW (kilowatt)

KWH (kilowatt hour)

L

L2F (Layer 2 Forwarding)

L2F, developed by Cisco, is a tunneling protocol that encapsulates Point-to-Point Protocol (PPP) packets in IP packets. This provides the significant benefit of being able to use private IP addresses across a virtual private network (VPN) by hiding them (via encapsulation) from the public network. This concept is important because many enterprises make use of private addressing schemes that conflict with public Internet addressing.

L2TP (Layer 2 Tunneling Protocol)

A protocol that combines the features of Microsoft's Point-to-Point Tunneling Protocol (PPTP) and Cisco Systems' Layer 2 Forwarding (L2F) to support multiprotocol or private-address Internet Protocol (IP) traffic across a public IP network. L2TP is a mature Internet Engineering Task Force standard that has been widely implemented. (See L2F and PPTP.)

laboratory information management system (see LIMS)**LAN (local-area network)**

A geographically limited communication network that connects users within a defined area. A LAN is generally contained within a building or small group of buildings and is managed and owned by a single enterprise. The shorter distances within a building or campus enable faster communications at a lower cost than wide-area networks (WANs). Although an increasing number of LANs use Internet standards and protocols, they are normally protected from the public Internet by firewalls.

LANs are generally used to perform the following functions:

- Send output to printers attached to the network.
- Transfer data or software to or from other systems attached to the network.
- Send e-mail to other users on the network.
- Access wider-area networks, including the Internet, via a direct connection from the network, for external file transfer, e-mail, facsimile, group collaboration and videoconferencing.

LAN bridging

The connection of multiple physical local-area networks (LANs) to support a single logical LAN environment.

LANClient Control Manager (see LCCM)**LANDesk Configuration Manager (see LCM)****LANDesk Management Suite (see LDMS)**

LANE (Local-Area Network Emulation)

An asynchronous transfer mode (ATM) standard proposed by the ATM Forum. It makes ATM networks resemble Ethernet or token ring local-area networks. LANE resides at the Media Access Control (MAC) layer of the OSI stack, and does not recognizing the Network Layer at all.

LANE 2.0 (Local-Area Network Emulation 2.0)

In July 1997, the Asynchronous Transfer Mode (ATM) Forum ratified a standard that split the Local-Area Network Emulation (LANE) standard into two sections — LANE User-to-Network Interface (LUNI) and LANE Network-to-Network Interface (LNNI). This added considerable functionality to campus networks while maintaining interoperability with traditional routed networks and LANE 1.0. LANE 2.0 must be implemented to run Multiprotocol over ATM (MPOA). While most of the market's attention has been placed on the LNNI features of redundancy and scalability, only the LUNI portion of LANE 2.0 was ratified at this time. Most vendors already provide LNNI features in their campus products.

LAN Emulation (see LANE)

LAN Emulation 2.0 (see LANE 2.0)

LAN performance monitor

A tool, or set of tools, for local or remote monitoring of local-area network (LAN) traffic and problems.

LAN switching (see Layer 2 switching)

LANE User-to-Network Interface (see LUNI)

LANE Network-to-Network Interface (see LNNI)

LAP-D or LAPD (Link Access Procedure — D Channel)

An Integrated Services Digital Network (ISDN) data link layer procedure, used for signaling and control on the D channel of an ISDN network.

laptop computer

A portable personal computer (PC) that can be easily used in multiple environments. The portability of the laptop enables individuals to take all of their software, files and applications with them, whether they are in the office, at home or on an airplane. At minimum, laptops include a keyboard, a flat display screen, a central processing unit (CPU), a battery and AC adapter, and a hard disk drive. They are frequently also equipped with compact disc read-only memory (CD-ROM) or floppy disk drives, modems and a variety of communication ports.

Given their small size, laptop computers generally do not have the same performance levels as desktop PCs. For example, laptops usually have smaller keyboards and display screens than those used on larger desktop computers. To compensate for this sacrifice, laptop computers usually have ports to connect external devices such as monitors or keyboards.

LAR (large-account reseller)

large-scale integration (see LSI)

laser

A device that emits a highly coherent beam of light. The term stands for "light amplification by stimulated emission of radiation." A typical laser has an active medium, which emits light, and a cavity structure, which selects certain wavelengths and directions for the emitted light. Lasers convert electrical energy into radiant energy in the visible or infrared parts of the spectrum, emitting light with a small spectral bandwidth. For this reason, they are widely used in fiber-optic communications, particularly as sources for long-haul links.

laser disk

An storage medium that uses laser technology to record and retrieve data.

LATA (local access and transport area)

A local telephone network area controlled and operated by a local-exchange carrier. A LATA is one of approximately 200 local telephone companies serving areas generally encompassing the largest standard metropolitan statistical areas. Circuits and calls within a LATA (intra-LATA) are generally the sole responsibility of the LEC, while interexchange carriers (IXCs), such as AT&T or MCI, handle circuits and calls that cross LATA boundaries (inter-LATA).

latency

The time interval during which a network station seeks access to a transmission channel and access is granted or received.

layer

In the Open Systems Interconnection (OSI) reference model, a layer is a collection of related network-processing functions that constitute one level of a hierarchy of functions. See OSI.

Layer 2

The term used to describe the Media Access Control layer Open Systems Interconnection (OSI) reference model. See OSI.

Layer 2 Forwarding (see L2F)**Layer 2 switching**

Layer 2 switching, commonly called LAN (local-area network) switching, is the transparent connection of at least two LAN segments to form a single, logical LAN. In this type of switching, local traffic stays within its respective segment, and only traffic destined for another segment will be transmitted by the switch. Common switching protocols include 802.1D Spanning Tree 802.1Q and IEEE 802.1P (see 802.x series).

Layer 2 Tunneling Protocol (see L2TP)**Layer 3**

The term used to describe the network layer of the Open Systems Interconnection (OSI) reference model, where protocols such as Internet Protocol (IP) reside. See OSI.

Layer 3 switching

1. Circuit- or hardware-based routing: This provides network-layer routing of such protocols as Internet Protocol (IP) and Internetwork Packet Exchange (IPX) at wire speed performance of more than one gigabit per second full duplex per port.
2. Route once, switch many: This is a tactical technology implemented by local-area network (LAN) vendors to use switches to offset performance and latency problems associated with older routers. With the switch "front ending" a router, the packets go through the switch and the switch allows the first packet to go to the router, which performs address resolution and frame regeneration, and then forwards the packet back to the switch toward the destination. Subsequent packets in a flow are handled by the switch, with no help from the router, using the information that was learned by the switch from the first routed packet.
3. Fully proprietary implementations: These implementations are numerous, but are falling by the wayside in campus LANs due to their complexity, their need for constant management and their divergence from industry direction. Examples of such implementations include switching between virtual local-area networks (VLANs) at Layer 2 and desktop software for shortcuts between VLANs or subnetworks.

Layer 4

The transport layer of the Open Systems Interconnection (OSI) reference model, where such protocols as Transmission Control Protocol (TCP) and User Datagram Protocol (UDP) reside. See ISO.

Layer 4 switching

An industry term that is misleading and incorrect. While it is physically impossible to switch in Layer 4, Layer 4 "switching" is, by definition, the control of traffic through the use of policies or filters using Layer 4 or transport layer protocol port IDs to differentiate and act upon application-specific traffic.

LBO (line build out)

A feature of some channel service units (CSUs). Switchable attenuators adjust signal strength as needed when the CSU is too close to a T1 repeater.

LCC (leadless chip carrier)

LCCM (LANClient Control Manager)

An IBM product designed to make systems management simpler and less time-consuming, thereby lowering the total cost of ownership of IBM systems.

LCD (liquid crystal display)

A low-powered, flat-panel display technology. LCD displays create images using liquid crystal molecules controlled by an electrical field.

LCD (lowest common denominator)

One way to build portable applications is to support only those functions that are provided on all of the target platforms. This "LCD" approach enables programs to run on numerous platforms, but does so at the expense of forgoing the use of unique, added-value features that may be offered on individual platforms.

LCM (LANDesk Configuration Manager)

A desktop management tool from Intel intended to facilitate operating-system and application upgrades.

LCMS (learning content management system)

An integrated set of technologies used to manage all aspects of learning content, including authoring or acquisition, content history, auditing, replacement and deletion. An LCMS requires that content be broken into "learning objects," which represent learning events such as classes. These objects can be organized into courses, courses into curricula, curricula into student learning agendas, and so on. Traditional content management software can perform these tasks, but requires customization for the e-learning focus; hence, a new LCMS software category has emerged. See e-learning.

LCR (least-cost routing) See ARS (automatic route selection).

LCR (Lifetime Clinical Record)

A computer-based patient record system from Siemens Medical Solutions Health Services. LCR provides a longitudinal view of patients' lifetime clinical histories.

LDAP (Lightweight Directory Access Protocol)

A server-to-server interface for directory information exchange among directories, devised as a low-cost, simpler implementation of the X.500 Directory Access Protocol. It facilitates the implementation of replication and chaining among dissimilar directories. Proposed by the University of Michigan, it was adopted by Netscape in 1996 for directory lookup, and has become the preferred access path for looking up directory information not only in X.500 directories, but also in many other directory structures on the Internet.

LDMS (LANDesk Management Suite)

A desktop management tool for software distribution, inventory and remote control from Intel.

LDO (low dropout)

leaky PBX

A private branch exchange (PBX) that allows incoming private-line traffic to access the local exchange.

learning content management system (see LCMS)

learning management system (see LMS)

leased line

A dedicated circuit, typically supplied by the telephone company or transmission authority, that permanently connects two or more user locations and is for the sole use of the subscriber. Such circuits are generally voice grade in capacity and in the range of frequencies they support. Also called a private line, tie line or dedicated facility.

LEC (local-exchange carrier)

The carrier or local telephone company that handles local access and transport area (LATA) traffic (e.g., a Bell operating company or independent local telephone company). See LATA.

LED (light-emitting diode)

A semiconductor that produces light when activated.

legacy application or system

An information system that may be based on outdated technologies, but is critical to day-to-day operations. Replacing legacy applications and systems with systems based on new and different technologies is one of the information systems (IS) professional's most significant challenges. As enterprises upgrade or change their technologies, they must ensure compatibility with old systems and data formats that are still in use.

LEI (Lotus Enterprise Integrator)

Lempel-Ziv

A compression algorithm developed in the 1970s and selected as the basis for the International Telecommunication Union's V.42 bis compression standard.

Lempel-Ziv-Stac (see LZS)

LEN (Low-Entry Networking)

An IBM networking architecture introduced in 1986 as an Advanced Program-to-Program Communications (APPC) application and a potential low-end alternative to Systems Network Architecture (SNA). See APPN and SNA.

LEO (low earth orbit)

LEP (light-emitting polymer)

A display technology based on the ability of certain plastics to glow when charged with an electric current. Still in the early stages of development, the technology has the long-term potential to enable the creation of flexible displays that could be rolled up and placed in a jacket pocket.

LES (Logistics Execution System)

An SAP product.

lessee

An entity that contractually obtains use of equipment in return for payments over a specified time.

lessor

An entity that conveys possession of specified property to a lessee for a period of time.

level

1. In data management structures or communication protocols, the degree of subordination in a hierarchy.
2. Measurement of signal power at a specific point in a circuit.

LF (low frequency)

Generally indicates frequencies between 3 and 300 kilohertz (kHz).

library

A data management system for documents frequently, though not necessarily, organized in a hierarchy of "folders" and "drawers." Also called a "file cabinet."

LIDM (line impact dot matrix)

A printing technology using a ribbon and an array of impact elements; suitable for making carbon copies.

Lifetime Clinical Record (see LCR)

LIFF (line impact fully formed)

A printing technology using a ribbon and an array of character wheels; suitable for making carbon copies.

LIFO (last in, first out)

light-emitting diode (see LED)

lightwave communications

A term sometimes used in place of "optical communications" to avoid confusion with visual information and image transmission such as facsimile or television. See fiber optics.

Lightweight Directory Access Protocol (see LDAP)

LIMS (laboratory information management system)

Applications used to manage the collection of samples, collection and formatting of test results, and the reporting of results by sample or product category. LIMS applications may be generic, product/manufacturing-specific, or environmental-, medical- or R&D-focused.

line

1. A communications path between two or more points, including a satellite or microwave channel.
2. In data communications, a circuit connecting two or more devices.
3. A transmission path from a nonswitching subscriber terminal to a switching system.

line-conditioned data stream (see LCDS)

Linear Tape Open (see LTO)

line balancing

The optimization of the assignment of operations to workstations in an assembly line to minimize idle time and the number of workstations required.

line build out (see LBO)

line conditioning (see conditioning)

line control unit (see LCU)

line driver

A communications transmitter/receiver used to extend the transmission distance between terminals and computers that are directly connected. It acts as an interface between logic circuits and a two-wire transmission line.

line hit

Electrical interference causing the introduction of undesirable signals on a circuit.

line impact dot matrix (see LIDM)

line impact fully formed (see LIFF)

line level

Signal strength on a transmission channel.

line load control

Equipment in a telephone system that provides a means by which essential paths may be ensured continuity of service under overloaded conditions. This is generally accomplished by temporarily denying originating service to some or all of the nonessential lines.

line loading

The process of installing loading coils in series with each conductor on a transmission line, usually 88 millihenry coils installed at 6,000-foot intervals.

line of business (see LOB)

line of sight (see LOS)

Line Printer Daemon (see LPD)

line processing

The processing of transactions as they occur, with no preliminary editing or sorting before they enter the system. See batch processing.

line speed

The maximum data rate that can be reliably transmitted over a line.

line switching

See circuit switching.

line terminating equipment (see LTE)

linguistic analysis (see natural-language processing)

link

1. A physical circuit between two points.
2. A conceptual (or virtual) circuit between two users of a packet switched (or other) network that enables them to communicate, even when different physical paths are used.
3. See hyperlink.

Link Access Procedure — D Channel (see LAP-D)

link layer

Layer 2 of the Open Systems Interconnection (OSI) reference model. This layer is concerned with the transmission of data between adjacent network nodes. See OSI.

link redundancy level

The ratio of the actual number of paths to the minimum number of paths required to connect all nodes of a network.

link state protocol

Any routing protocol derived from the Dijkstra algorithm such as Shortest Path First (SPF), Open Shortest Path First (OSPF) and Intermediate System to Intermediate System (IS-IS). Routers that use link state protocols update each other and learn network topology by periodically broadcasting link state data across the network. The data from each router includes only the cost and identification of those networks directly connected to that router. All routers on a network compile tables of routers and connections and can then calculate optimal paths from themselves to each link.

Linpack

An early benchmark for scientific applications, now replaced by the SPECmark.

Linux

A version of Unix developed by Linus Torvalds, who sought to champion openness and counter the closed nature of vendor-proprietary operating systems (OSs). The Linux credo says that the software must be made available as open-source code, enabling anyone to read the coding rules and submit improvements. The source code is accessible free on the Web or is otherwise virtually given away, and vendors derive their revenue from utilities and support.

liquid crystal display (see LCD)

LIS (laboratory information system)

LISP (List Processing)

An object-oriented programming language.

little-endian

A method of storing or transmitting data where the most significant bit or byte is presented last. See big-endian.

LLC (Logical Link Control)

A protocol defined in the Institute of Electrical and Electronics Engineers (IEEE) 802 standard, common to all of IEEE's local-network standards, for data link-level

transmission control. It is the upper sublayer of the IEEE Layer 2 protocol that complements the Media Access Control (MAC) protocol (IEEE 802.2).

LMDS (local multipoint distribution services)

A microwave-based wireless technology that operates at around 28GHz. In the United States and other countries, it is used for fixed high-speed data, Internet access, and advanced telephone and entertainment services in metropolitan areas.

LMM (local modifications memory)

A technology that remembers manual changes to the graphical user interface (GUI) design and reapplies them should the screen need to be reconverted and generated.

LMR (longitudinal medical record)

A record of a person's inpatient and outpatient healthcare, from birth to death, regardless of when and where such care was obtained.

LMS (learning management system)

A full infrastructure on which e-learning can be built and delivered. An LMS has six main components:

- Registration capabilities (including curriculum, courses and instructional responsibilities)
- Administration of curriculum and courses
- Student skills and records management
- Student interfaces to courseware, including the ability to launch a course or interact with a learning content management system (LCMS)
- Learning program administration (including test and assessment capabilities, certification, instructor assignments and regulatory requirements)
- Application programming interfaces to external systems (such as human-resources systems)
- Learning portals (the use of portal technologies to provide the user interface for e-learning)
 - See courseware, e-learning and LCMS.

LNA (low-noise amplifier)

LNNI (LANE Network-to-Network Interface)

The interface between two Local-Area Network Emulation (LANE) servers. It is part of the LANE 2.0 specification.

LNP (local number portability)

The ability to change to a different local phone service provider while retaining the same phone number.

LNP (logistics network planning)

The class of tools required to analyze the trade-offs among inventory quantities, number and location of warehouses, and transportation costs to most profitably support a desired level of customer service. LNP is a proven scientific method for

analyzing the required cost and service levels that warehouses need to meet specified customer service objectives.

LNTT (line nonimpact thermal transfer)

A category of printer technology.

load

1. To copy a program into the memory of a computing device so that it can later be used for processing.
2. To add inductance to a transmission line to minimize amplitude distortion (see loading coil).

load balancing

The ability of processors to schedule themselves to ensure that all are kept busy while instruction streams are available.

load balancing for station/trunk lines

Traffic balancing may be required when system traffic limits are approached. This feature provides the capability, during installation or on an in-service system, to change specific station and trunk terminations on the private branch exchange (PBX) system switching network with minimum installer effort and without requiring number changes for the purpose of balancing the traffic load on the switch network.

loading coil

An induction device employed in local loops, generally those exceeding 5,500 meters in length, that compensates for the wire capacitance and serves to boost voice grade frequencies. They are often removed for new-generation, high-speed, local-loop data services as they can distort data signals at higher frequencies than those used for voice.

LOB (line of business)

A corporate subdivision focused on a single product or family of products.

LOC (line of code)

A unit used in measuring or estimating the scale of programming or code conversion efforts.

local access and transport area (see LATA)

local-area network (see LAN)

local central office (see local exchange)

local-exchange carrier (see LEC)

local line (see loop)

local modifications memory (see LMM)

local multipoint distribution services (see LMDS)

local number portability (see LNP)

local positioning system (see LPS)

local telephone exchange (see LTE)

local trunk

A trunk between local exchanges.

location broker

Middleware that automatically matches a client query with whichever server has the most suitable services, data and computing resources to accomplish the task.

location services

Services that enhance applications for mobile telephones and devices by exploiting information about where the user is located. This information is detected through technologies deployed in the carrier's infrastructure and in the handset. Location methods used by these technologies include cell of origin, time of arrival (TOA) angle of arrival (AOA), Enhanced Observed Time Difference (EOTD) and Global Positioning System (GPS). (See cell of origin, TOA, AOA, EOTD and GPS.)

lock

A means of preventing two users from trying to update the same data simultaneously.

LOCT (Layered Open Crypto Toolkit)

A cryptographic framework introduced by RSA Security in 1996.

logical channel number

The number assigned to a virtual call on a packet-switched network.

Logical Link Control (see LLC)

Logical Observation Identifier Names and Codes (see LOINC)

logical terminal

A device addressable by its logical function rather than its physical address. Translation from logical to physical addresses is achieved by a common routine using a table.

Logical Unit (see LU)

Logical Unit 6.2 (see LU 6.2)

Logical Volume Series (see LVS)

login or logon

The procedure by which a user begins a terminal session.

logistics execution system (see LES)

logistics network planning (see LNP)

logistics networks

Logistics networks are like computer networks that can be configured by different hubs, routers and transmission lines. (The logistics network corollaries are warehouses, replenishment rules and trucking companies.) Tools exist to model and optimize the performance of the network. These systems rationalize and quantify the role of each warehouse in satisfying corporate objectives. These systems run on desktops or workstations and should be considered end-user power tools.

logoff or logout

The procedure by which a user ends a terminal session.

LOINC (Logical Observation Identifier Names and Codes)

A system of names and codes for identifying laboratory and clinical observations.

long-haul

Long distance — telephone circuits that cross out of the local exchange.

longitudinal balance

A measure of the electrical balance between the two conductors (tip and ring) of a telephone circuit. Specifically, it is the difference between the tip-to-ground and ring-to-ground AC signal voltages, expressed in decibels.

longitudinal medical record (see LMR)

longitudinal redundancy check (see LRC)

look and feel

The appearance and behavior of a graphical user interface to the end user (who sees it as part of an application), determined by the tools and style guide provided by the vendor (and by whether the software developer obeys these guidelines).

loop

A local circuit between an exchange and a subscriber telephone station. Also called subscriber loop or local line.

loopback

A test of the performance and quality of a line or terminating equipment. See analog loopback and digital loopback.

loosely coupled multiprocessing

A configuration of several processors, each with its own memory, that execute user and operating-system code independently.

LOS (line of sight)

A characteristic of some open-air transmission technologies where the area between a transmitter and a receiver must be clear and unobstructed. This applies to microwave, infrared and open-air laser-type transmissions that operate at frequencies which transmit through the air in a perfectly straight line. A clear, open-air, direct transmission path is one that is free of obstructions such as buildings, but is in some cases impeded by adverse weather or environmental conditions.

loss (transmission)

The decrease in the energy of signal power in transmission along a circuit due to the resistance or impedance of the circuit or equipment.

lossless compression

A data compression technique where the recovered data after decompression is precisely the same as before compression.

lossy compression

A data compression technique where the recovered data after decompression is different from the data that was compressed, though it will not be perceived as such by an observer.

Low-Entry Networking (see LEN)

lower-CASE

A category of computer-aided software engineering (CASE) tools used for application generation rather than application modeling.

lowest common denominator (see LCD)

low frequency (see LF)

LP (load point)

LPAR (logical partition)

LPD (Line Printer Daemon)

A Unix print server protocol.

LPS (local positioning system)

Technology used to pinpoint indoor position, similar to the global positioning system (GPS) used to locate outdoor position. LPS triangulates signals from cell controllers to proprietary long-range, long-life, low-cost radio frequency electronic tags. By calculating the distance between cell controllers, electronic tags and several different antennae, equipment or personnel inside an enterprise can be instantly located.

LRC (longitudinal redundancy check)

A data communications error-trapping technique in which a character is accumulated at both the sending and receiving stations during the transmission and is compared for an equal condition, which indicates a good transmission of the previous block.

LRU (least recently used)

LSC (low-speed connection)

LSI (large-scale integration)

Between 3,000 and 100,000 transistors on a chip.

LSPR (Large System Performance Reference)

LSSU (link status signal unit)

LTE (line terminating equipment)

Devices that originate or terminate signals.

LTE (local telephone exchange)

The exchange in which subscribers' lines terminate. It has access to other exchanges and to national trunk networks. Also called local central office or end office.

LTl (less than truckload)

A logistics term denoting a segment of the shipping market.

LTO (Linear Tape Open)

Technology from Hewlett-Packard, IBM and Seagate Technology, designed to establish an open tape architecture with road maps extending out four generations of drives. The consortium developed two formats to address the needs of read-intensive applications, which require fast data access, as well as write-intensive applications, which require high capacity.

LU (Logical Unit)

IBM's term for a logical node that is the user's "port" into a Systems Network Architecture network. LU 1 is a high-performance print data stream and a 3270 data stream. LU 6 enables host-to-host data exchange, and provides a peer-to-peer data stream.

LU 6.2 (Logical Unit 6.2)

An IBM Systems Network Architecture protocol. LU 6.2 is a device-independent process-to-process protocol; it also supports asynchronous (store-and-forward) networking. It provides the facilities for peer-to-peer communication between two programs. Among other things, this means that programs and small-to-midrange computer systems can use LU 6.2 to establish and control sessions with each other, without host mainframe participation. When LU 6.2 is installed in devices such as PCs and cluster controllers, required functionality, such as multiple sessions among those devices and S/370 hosts, becomes available.

LUN (logical unit number)

LUNI (LAN User-to-Network Interface)

The section of Local-Area Network Emulation (LANE) 2.0 that provides support for a multicast function and the multiplexing of extended LANs within a single virtual circuit.

lurker

A user who does not participate in an online discussion (e.g., in a chat room), but merely observes the activity.

LVS (Logical Volume Series)

A line of Amdahl storage systems.

LZS (Lempel-Ziv-Stac)

A data compression technique developed by Stac Software (now Previo), based on the Lempel-Ziv compression algorithm.

M

M&A (mergers and acquisitions)

MAC (Media Access Control)

An IEEE protocol defining the methods used to gain access to the physical layer of a LAN (i.e., Layer 1 of the OSI model — see OSI).

MAC (message authentication code)

A way of confirming that a message has not been tampered with.

MAC (moves, adds and changes)

General term for the routine work performed on computer equipment in an enterprise, including installations, relocations and upgrades.

MACRS (Modified Accelerated Cost Recovery System)

The tax depreciation system introduced by the Tax Reform Act of 1986. It became effective for all servers placed in service after year-end 1986.

MAD (mass access device)

A non-PC device that provides Internet access (e.g., a set-top box, Web phone or personal digital assistant).

MAD (mergers, acquisitions and divestitures)

Magic Quadrant

A two-dimensional graphical framework developed by Gartner that places a particular set of vendors from a specific technology industry sector into a strategic matrix. Gartner analysts use multiple objective and subjective criteria to evaluate individual vendors, presented on two axes:

Completeness of vision (x-axis). Measures the depth and breadth of a vendor's goals, its knowledge of the markets and customers it serves, and how it is positioned to address future industry scenarios.

- Ability to execute (y-axis). Measures the ability of the vendor to execute its vision. Focuses on the firm's management team and financial stability; sales channels; quality of research and development; installed product base in the market; service and support reputation; and track record in delivering products on time.

magnetic-ink character recognition (see MICR)

magneto-optic (see MO)

main distribution frame (see MDF)

mainframe

A large-capacity computer system with processing power that is significantly superior to PCs or midrange computers. Traditionally, mainframes have been associated with centralized, rather than distributed, computing environments. Skilled technicians are required to program and maintain mainframes, although client/server technology has made mainframes easier to operate from the user's and programmer's perspectives. They are generally used by large organizations to handle data processing for enterprisewide administrative tasks like payroll or accounts payable.

mainframe data center (see MDC)

mainstream notebook

A computer system that meets all the criteria for a notebook PC but is designed to be the best compromise between all-inclusive functionality and light weight. Mainstream notebooks weigh between 4.5 and 6 pounds with the weight-saver and battery modules. Mainstream notebooks often have a single bay for the inclusion of a peripheral, such as a CD-ROM.

maintenance, repair and operations (see MRO)

MAN (metropolitan-area network)

A type of network that evolved from local-area network (LAN) designs, but is optimized for longer distances (more than 50 kilometers), greater speeds (more than 100 megabits per second) and diverse forms of information (e.g., voice, data, image and video).

managed network services (see MNS)

managed object

A data processing or data communications resource that may be managed through the use of an Open Systems Interconnection (OSI) management protocol. The resource itself need not be an OSI resource. A managed object may be a piece of equipment, a software component, an abstract collection of information or a combination of all three.

managed security service provider (see MSSP)

managed-service provider (see MSP)

management services organization (see MSO)

management consulting

Strategic consulting focused on high-level corporate or business unit strategy (e.g., deciding what businesses to participate in or whether to make an acquisition), or on operational improvement (e.g., improving customer service or determining the most effective type of retail delivery system).

Management Information Base (see MIB)

management information systems (see MIS)

management service provider (see MSP)

Manchester encoding

A method of coding that alters the binary signal by dividing the bit period into halves and using a change in voltage to differentiate the two halves. A binary "1" is indicated by having a high voltage during the first interval and low voltage during the second. A binary "0" is the reverse.

mandatory security controls

An operating-system security rating of B1 or higher based on U.S. Department of Defense trusted-computer-system evaluation criteria.

man-machine interface (see MMI)

Manufacturing Accounting and Production Information Control System (see MAPICS)

Manufacturing Automation Protocol (see MAP)

manufacturing execution systems (see MES)

Manufacturing Execution Systems Association (see MESA)

manufacturing lead time

The total time required to manufacture an item. Included are order preparation time, queue time, setup time, runtime, move time, inspection time and put-away time.

Manufacturing Message Service (see MMS)

manufacturing planning

Definition of the weekly or daily production and machine schedules across multiple plants or lines to meet orders and forecast demand. Some manufacturing-planning modules also incorporate materials planning.

manufacturing resource planning (see MRP II)

manufacturing scheduling

Generation of plant-level execution schedules by product and resource (e.g., line and machine) and resolution of day-to-day capacity bottlenecks. Scheduling applications normally include a more granular level of resource information, and will provide such functionality as sequence dependent setup, tank scheduling and point-of-use material availability.

MAOSCO (Multi-Application Operating System Consortium)

A vendor consortium formed in 1997 to oversee the Multi-Application Operating System (MULTOS) for smart cards. See MULTOS.

MAP (Manufacturing Automation Protocol)

An application-specific protocol based on Open Systems Interconnection (OSI) standards. It is designed to allow communication among competing vendors' digital products in the factory.

MAPI (Messaging Application Programming Interface)

The programming interface specification that enables an application to send and receive mail over a Microsoft Mail messaging system. It was designed to separate the mail engine from the mail client.

MAPICS (Manufacturing Accounting and Production Information Control System)

A manufacturing enterprise resource planning (ERP) product, originally introduced by IBM as software for the AS/400 and earlier System/36 and System/38 systems. MAPICS was later taken over by Marcam, and separated into an independent company in 1997.

mapping

The transcription of functions into terms that make them equivalent on two different systems. In network operations, it is the logical association of one set of values, such as addresses, on one network with quantities or values of another set, such as devices, on a second network (e.g., name-address mapping).

MAP/TOP (Manufacturing Automation Protocol/Technical Office Protocol)

MAPS (Microsoft Authorized Premier Support)

mark

A signal condition corresponding to the "on" state (binary 1), which usually occurs when current flows or voltage exceeds a specified value.

market killers (see market spoilers)

market of one

Continual customization of content, services and interactions with customers to deliver exactly what they need, (creating the sense that each customer is a "market of one").

market spoilers (also market killers)

Web-based businesses that aggregate information about a market and its suppliers, present the aggregated information to consumers via a Web site, and offer decision support to allow customers to differentiate offerings based on independent validation of competitors' services and features. These businesses diminish the advantage of suppliers that compete through brand identity or reputation.

marketing

The domain of customer relationship management (CRM) comprising the selection, acquisition, retention and extension of customer relationships. As marketing models change and new strategies are created that add value to customer relationships, use of technology-enabled marketing will emerge through the Web.

marketing automation system (see MAS)

marketing content management (see MCM)

marketing customer information file (see MCIF)

marketing database system (see MDS)

marketing encyclopedia system (see MES)

MAS (marketing automation system)

Systems that help marketers execute multichannel marketing campaigns by providing a scripting environment for authoring business rules and interfaces to a variety of third-party applications.

MAS (Microsoft Authorized Support)

mass-access device (see MAD)

massively parallel processing (see MPP)

master enterprise identifier (see MEI)

Master License Agreement (see MLA)

master station

The unit having control of all other terminals on a multipoint circuit for purposes of polling or selection.

master terminal

A reserved workstation used by the system manager that can initiate conversations for network management tasks.

matchmakers

Web sites that refer buyers to Web merchants willing to sell products or services at prices specified by the buyers.

material requirements planning (see MRP)

materials management

A term to describe the grouping of management functions related to the complete cycle of material flow, from the purchase and internal control of production materials, to the planning and control of work in process, to the warehousing, shipping and distribution of the finished product. It differs from materials control in that the latter term, traditionally, is limited to the internal control of production materials.

materials management information system (see MMIS)

Mathematical Markup Language (see MathML)

MathML (Mathematical Markup Language)

One of the first four Extensible Markup Language (XML) specifications approved by the Worldwide Web Coalition in 1998, along with XML, Document Object Model (DOM) and Synchronized Multimedia Integration Language (SMIL).

matrix

The portion of a switch architecture where input and output leads meet, any pair of which may be connected to establish a through circuit. Also called a switching matrix.

MAU (media access unit)

A network device that, along with the cable, forms the physical level (Layer 1) of the Open Systems Interconnection (OSI) seven-layer reference model. Its functions are signaling, encoding and medium-handling. See OSI.

MAU (multistation access unit)

A wiring concentrator in a token ring network that can connect up to eight lobes to a ring.

MAW (Microsoft at Work)

maximum burst size (see MBS)

MB (megabyte)

Approximately million bytes (1,048,576, to be precise). This is the common unit used to specify the size of computer files, storage capacity or random-access memory (RAM). The relationship of the size of a file or accumulation of files to the size of the RAM and storage space is critical, because it determines the ability of a computing device to process a file.

Mb (megabit)

approximately one million bits of computer data (1,048,576, to be precise).

MBE (molecular beam epitaxy)

A semiconductor manufacturing technology.

Mbps (megabits per second)

A measure of data transmission speed.

MBR (mean bit rate)

The number of bits transmitted per time unit averaged over the period of the session.

MBS (maximum burst size)

In asynchronous transfer mode (ATM), the number of cells a source is allowed to send at the peak cell rate.

m-business (mobile business)

Gartner uses this term to describe the new business models and opportunities enabled by the extensive deployment of mobile and wireless technologies and devices (e.g., Bluetooth, WAP, UTMS, GPRS, EDGE, smart phones and e-purses) in support of the inherent mobility of most people's work styles and lifestyles. M-business is a key factor driving the emergence of the Supranet (see separate entry). The value proposition of m-business is that the user can benefit from information or services at any time or in any place. To successfully prepare for m-business, business and IT managers must address three key challenges:

- Understanding the business opportunities of mobility in terms of services and information in various contexts (e.g., business-to-business, business-to-consumer and consumer-to-consumer)
- Preparing enterprise infrastructures for applications that involve customers and employees using the Supranet to conduct transactions

- Understanding the security and management implications of using mobile devices within the enterprise

MCA (Micro Channel Architecture)

IBM's bus architecture developed for the Personal System/2 computer. It is no longer used in IBM's PC products.

MCAD (mechanical computer-aided design)

MCAE (mechanical computer-aided engineering)

MCAP (Managed Care Appropriateness Protocol)

MCIF (marketing customer information file)

A system has some or all of the characteristics of a data warehouse, but is designed specifically to support marketing activities.

MCIS (Microsoft Commercial Internet System)

A Microsoft system that provides user management services for large hosted Web services.

MCM (marketing content management)

A category of applications that help enterprises respond rapidly to unfolding business circumstances by applying the optimal combination of marketing content across multiple channels. MCM databases provide enterprises with an overview of all available marketing content.

MCM (multichip module)

MCNS (Multimedia Cable Network System)

Formed by Comcast Cable Communications, Cox Communications, Time Warner Cable and Tele-Communications (which was later acquired by AT&T).

MCO (managed care organization)

m-commerce (mobile commerce)

A category of commerce that includes any purchase transaction completed using a wireless device, such as a cellular phone, PC or personal digital assistant. M-commerce includes paying for a subscription to get content "pushed" to a mobile device, purchasing a product via a mobile device or using such a device to obtain a service for which a fee is charged. Purchases that are researched or arranged via a wireless device, but completed and settled by other means, are classified as mobile-enabled transactions.

MCP (Microsoft Certified Professional)

A Microsoft certification that requires passing a single operating-system test. MCPs can further specialize in Internet technologies to attain an MCP with Internet or site-building specialization, or can continue taking the exams necessary to become a Microsoft Certified Systems Engineer (MCSE).

MCR (minimum cell rate)

An asynchronous transfer mode (ATM) parameter used to determine the minimum number of cells permitted in a specific time period.

MC/S (mobile client/server)

An approach used to transport information between mobile computers and host systems. At a minimum, this approach provides communications and database replication and synchronization.

MCSC (Microsoft Certified Support Center)

MCSD (Microsoft Certified Software Developer)

MCSE (Microsoft Certified Systems Engineer)

MCSS (Mission Critical Server Suites)

A service program from Hewlett-Packard.

MCU (microcontroller unit)

MCU (multipoint control unit)

MD5

A message digest hash function.

MDC (mainframe data center)

The centralized, large-volume processing and storage facilities built around standard mainframe computing platforms.

MDDBMS (multidimensional database management system)

A database management system that stores and manages data in dimensional arrays, indexed by dimensions and measured over time.

MDF (main distribution frame)

A wiring arrangement that connects outside lines on one side and internal lines from exchange equipment on the other.

MDF (Message Development Framework)

A detailed formal methodology for Health Level Seven (HL7) message development, adopted by the HL7 organization in 1997.

MDF (Multiple Domain Feature)

A logical-partitioning feature in Amdahl mainframes.

MDNS (managed data network services)

MDS (marketing database system)

A database system designed to meet the specialized analytical and application needs of marketers.

MDS (multipoint distribution system)

MDT (mobile data terminal)

mean bit rate (see MBR)

mean time between failures (see MTBF)

mean time to repair (see MTTR)

Measured Usage License Charge (see MULC)

Media Access Control (see MAC)

media access unit (see MAU)

Media Gateway Control Protocol (see MGCP)

media interface connector (see MIC)

media objects

Non-Hypertext Markup Language (HTML) files or applications that can be displayed or executed as part of an HTML document. Examples include graphic, audio and video files, and Java applets.

Medical Logic Module (see MLM)

medical management

An umbrella term that encompasses the use of IT for health, disease, care and case management functions. Medical management strategies are designed to modify consumer and provider behavior to improve the quality and outcome of healthcare delivery.

medium

1. The material on which data is recorded (e.g., magnetic tape or diskette).
2. Any material substance that is, or can be, used for the propagation of signals, usually in the form of modulated radio, light or acoustic waves. Examples include optical fiber, cable, wire, water or air.

medium frequency (see MF)

medium-scale integration (see MSI)

megabit (see Mb)

megabits per second (see Mbps)

megabyte (see MB)

megaFLOPS (see MFLOPS)

megahertz (see MHz)

megaportal

A portal that attempts to serve the entire Internet community (in contrast to a vertical Internet portal, which targets a niche audience).

MEI (master enterprise identifier)

An identifier that is unique for each person.

MEL (MULTOS Executable Language)

The proprietary programming language for the MULTOS smart-card operating system.

MELP (Microsoft Enterprise License Pak)

memory in cassette (see MIC)

Merced

Intel's code name for its first 64-bit (IA-64) processor.

MES (manufacturing execution system)

1. Computerized systems that formalize production methods and procedures within the manufacturing environment, providing online tools to execute work orders.
2. A system that electronically distributes and consolidates up-to-date marketing information into a highly cross-referenced, single-source repository. An MES enables users to quickly locate and display information, thus shortening sales cycles and decreasing printing and distribution costs. An MES contains a database capable of storing all types of information that will be incorporated into the system (e.g., audio, video, sound, text and graphics). The system should have the capability to be used unconnected to a server and should include a remote communications mechanism for electronically downloading and uploading information.

MESA (Manufacturing Execution Systems Association)

A trade association that promotes the adoption of manufacturing execution system (MES) application software.

mesh network

A net-like structure where at least two paths exist to all stations.

message

A sequence of characters used to convey information or data. In data communications, messages are usually in an agreed format with a heading, which establishes the destiny of the message, and text, which consists of the data to be sent.

message authentication

A function in which the device determines if the received message arrived from its stated source and in unaltered form. The actual message need not be encrypted, but its authentication code must be.

message authentication code (see MAC)

message broker

A logical hub that copies and resends messages to one or more destinations. As a value-adding third party between information sources and information consumers, it can complement a service-oriented architecture (SOA). Like an SOA, a broker is a design abstraction that may be implemented using component software for some or all of the connections. The interface from a message broker to the application may use an object request broker (ORB) or object transaction monitor (OTM); a request

to the message broker may be implemented as a series of method calls to participating components.

Message Development Framework (see MDF)

message feedback

A method of checking the accuracy of data transmission. The received data is returned to the sending end for comparison with the original data, which is stored there for this purpose. Also called information feedback and loop checking.

message format

Rules for the placement of portions of a message, such as the heading, address, text, end-of-message indication and error-detecting bits.

Message Handling Service — see “MHS (Message Handling Service)”

Message Handling System — see “MHS (Message Handling System)”

message numbering

The identification of each message within a communications system through the assignment of a sequential number.

message-oriented middleware (see MOM)

message passing

Services performing a simple, one-way transfer operation between two programs. Like the other one-way messaging models, message passing generally leaves the sending program unblocked. Also, as with all forms of messaging, message passing is usually connectionless, which means that the sending application does not have to explicitly establish and manage a connection with the message's intended recipient. Although message passing is not inherently a two-way communication model, two-way communication can be accomplished by sending the response in a separate message.

message queuing

The message-queuing or store-and-forward model is basically the message-passing model with one additional feature. Message queuing is asynchronous in a manner similar to a traditional postal system — i.e., the recipient need not be available when the message is sent. Message queuing stores messages at an intermediate node on the network in a queue and then forwards the messages to the intended targets.

message-queuing middleware (see MQM)

Message Security Protocol (see MSP)

message switching

The technique of receiving a message, storing it until the proper outgoing line is available, and then retransmitting it. Unlike in-line switching, no direct connection between the incoming and outgoing lines is established.

message transfer agent (see MTA)

Message Transfer Part (see MTP)

message unit (see MU)

message warehouse

A message broker service that temporarily stores messages to be analyzed or retransmitted at a later time.

messaging

In e-mail, messaging consists of moving messages from user to user, application to application or place to place, providing a service analogous to that of the paper postal service and providing an infrastructure along which many other objects may be moved. The term is sometimes applied, with an entirely different meaning, to real-time interprocess communications in a transaction-processing environment.

Messaging Application Programming Interface (see MAPI)

messaging device

A data-centric device designed primarily to process electronic messages. A basic messaging device weighs less than 1 pound and is designed for one-way outbound messaging. Limited responses, such as acknowledgments, are also possible but are not common. A two-way messaging device weighs less than 1 pound and is designed to send and receive messages.

Messaging Hypertext Markup Language (see MHTML)

meta content framework (see MCF)

metadata

Information about data, including attributes such as description, length and location. For example, the fact that a field in a given database is called "client number" and is five characters long is metadata. By comparison, a given instance of the client number "123456" in that database field is just "data."

Metadata Interchange Format (see XMI)

metadirectory

A directory that acts as a superset of all other directories. Metadirectories have evolved from stand-alone products to services that enable a given directory to synchronize and exchange information with other data repositories.

metatag

As a noun or verb, this term refers to the tagging of documents with metadata (such as keywords, dates, categories or creators) to enhance search efficiency. See metadata.

metalanguage

A language used to describe a language. A metalanguage defines a language's constructs, such as character sets, syntax and valid sequences.

metallic circuit (see loop circuit)

metric

A standard of measurement.

metropolitan-area network (see MAN)

MExE (Mobile Station Application Execution Environment)

A wireless protocol, currently under development by the European Telecommunications Standards Institute (ETSI), which is designed to be incorporated into smart phones for executing user- or service provider-specific applications. MExE is designed as an application execution environment on the mobile device.

MF (medium frequency)

Frequencies in the range between 300 kilohertz (kHz) and 3 megahertz (MHz).

MFC (Microsoft Foundation Class)

MFI (Multifunction Interpreter)

A gateway that handles connectivity between the Defense Message System (DMS) and other systems.

MFLOPS (megaFLOPS)

A performance measurement unit for floating-point-intensive applications, equal to one million floating-point operations per second (FLOPS)

MFP (multifunction product)

A network-attached document production device that combines two or more of the copy, print, scan and fax functions.

MFS (metropolitan fiber system)

MGCP (Media Gateway Control Protocol)

An Internet Engineering Task Force (IETF) signaling protocol proposal allowing a bridge between classic telephone networks and Internet (i.e., IP-based) infrastructures.

MHEG (Multimedia and Hypermedia Information Coding Expert Group)

An International Organization for Standardization (ISO) and International Telecommunication Union (ITU) standard suitable for the interchange of multimedia documents using HyTime.

MHS (Message Handling Service)

A Novell NetWare utility that provides a common format for exchanging information among applications. It is typically used for e-mail.

MHS (Message Handling System)

Electronic mail implemented according to Open Systems Interconnection (OSI) specifications, based on International Telecommunication Union (ITU) X.400 recommendations.

MHTML (Messaging Hypertext Markup Language)

A language that is capable of packaging externally referenced image files within an HTML page, potentially increasing the bulk of the page significantly.

MHz (Megahertz)

A unit equal to 1 million hertz (1 million cycles per second).

MIB (Management Information Base)

A Simple Network Management Protocol (SNMP) flat-file, nonrelational database that describes devices being monitored. Network management platforms monitor nodes by reading the value of the managed resources in the MIB. Management platforms can effect changes in managed resources by altering MIB values (e.g., by establishing thresholds beyond which alerts are created). The MIB has been published as IETF RFC 1604, "Definitions of Managed Objects for Frame Relay Service."

MIC (media interface connector)

A fiber distributed data interface (FDDI) connector.

MIC (memory in cassette)

A media feature with which Sony has differentiated its Advanced Intelligent Tape (AIT) drive. It is a 16-kilobit E2PROM chip that indexes file locations, thus enabling faster access to data, and stores metadata about the media, allowing enterprises to track media life cycles in the physical cartridge.

MIC (microwave integrated circuit)

MICR (magnetic-ink character recognition)

Machine recognition and digitization of magnetically charged characters printed on paper (typically bank drafts and deposit slips).

microbrowser

A browser that differs from a typical browser in its storage size and scaled-down functionality. Microbrowsers perform client-side functions that render Web content to a particular device. They share responsibility for this task with back-end servers by controlling the presentation layer at the server level, the client level, or as a hybrid of the two.

Micro Channel Architecture (see MCA)

microcode

The microinstructions, especially of a microprocessor, that govern the details of operation. Microcoded functions can improve performance but add a layer of complexity. For example, microcode errors appear to software as being hardware failures.

microfilm

A high-resolution film used to record images reduced in size from the original.

micro-marketplace (see MM)

microprocessor

A central processing unit (CPU) on a single chip, also known as a microprocessing unit (MPU). Desktop and portable computers typically contain one microprocessor, while more powerful computers often make use of multiple microprocessors.

Microsoft Certified Professional (see MCP)

Microsoft Cluster Server (see MSCS)

Microsoft Commercial Internet System (see MCIS)

Microsoft Data Engine (see MSDE)

Microsoft Management Console (see MMC)

Microsoft Message Queue Server (see MSMQ)

Microsoft Transaction Server (see MTS)

microwave

1. The portion of the electromagnetic spectrum above about 760 megahertz (MHz).
2. A term describing high-frequency transmission signals and equipment that employ microwave frequencies, including line-of-sight open-air microwave transmission and, increasingly, satellite communications.

middleware

The software “glue” that helps programs and databases (which may be on different computers) work together. Its most basic function is to enable communication between different pieces of software.

midrange computer

A computer with an architecture similar to that of the minicomputer, which is used for multiple users.

MIF (management information file)

million instructions per second (see MIPS)

million floating-point operations per second (MegaFLOPS) — see MFLOPS

MILNET (military network)

MIL-SPEC (military specification)

MIL-STD (military standard)

MIMD (multiple instruction, multiple data)

A design for parallel computers characterized by the simultaneous execution of many different instruction streams (programs), each of which handles different data.

MIME (Multipurpose Internet Messaging Extensions)

MIME functions enable the transport of attachments and non-ASCII text via the Simple Message Transport Protocol (SMTP). They cover most (but not all) of the features of X.400, and interface with X.400 via the MIME-X.400 Enhanced Relay (MIXER) specification.

MIME Object Security Services (see MOSS)

MIMIC (millimeter-wave monolithic integrated circuit)

minicomputer

A computing device that is typically more powerful than a PC, but less powerful than a mainframe, and is therefore often referred to as "midrange." A minicomputer can support or more users.

MiniMAP

A simplified subset of the Manufacturing Automation Protocol (MAP) covering Open Systems Interconnection (OSI) Layers 1, 2, and 7 only.

minimum cell rate (see MCR)

Minimum Interoperability Specification for PKI Components (see MISPC)

Minitel

Minitel, which appeared in France in the 1970s, was the world's first electronic information system for public use — a forerunner of today's Internet.

MIPS (million instructions per second)

An approximate measure of a computer's raw processing power. MIPS figures can be misleading because measurement techniques often differ, and different computers may require different sets of instructions to perform the same activity.

MIPS RISC

A reduced instruction set computer (RISC) processor architecture designed and licensed by MIPS Computer Systems (now part of Silicon Graphics). Used by a number of midrange suppliers, it was the basis for the Advanced RISC Computing (ARC) definitions endorsed by the now-defunct Advanced Computing Environment (ACE) consortium.

mirroring

Replicating data on different computers or devices to provide backup or improved performance. In a storage sense, "mirroring" refers to duplicating data on multiple disk storage devices to provide backup in the event of a system failure. In an Internet sense, "mirroring" refers to replicating the contents of a Web site at another site to minimize load on the main server, or to speed up connectivity in a specific geographic region.

MIS (management information systems)

This was once the predominantly used name for the central data-processing organization in an enterprise. Today, information systems (IS) is the more commonly used term.

MISPC (Minimum Interoperability Specification for PKI Components)

An initiative, led by the National Institute of Standards and Technology (NIST), to address interoperability and cross-certification among certificate authorities. It seeks to encourage the development of commercial, off-the-shelf software for public-key infrastructures (PKIs). The intent is to leverage standards and the activities of the public-key community of interests. The effort also recommends support for multiple digital signature and nonrepudiation methods as a way to ensure interoperability among products. Interoperable testing is to be done by members of the CommerceNet consortium. NIST also addresses the issues of certification authority policies, practice statements and accreditation of certificate authorities as they relate to interoperability.

MISSI (Multilevel Information System Security Initiative)

mission-critical

A term used to describe applications that are critical to the survival of an enterprise. Mission-critical services require a combination of several factors, such as availability, reliability, serviceability and performance. Each of these must be weighted in importance to fit the particular mission to be supported.

Mission Critical Server Suites (see MCSS)

MITI (Ministry of International Trade and Industry)

M-JPEG (Motion JPEG)

A version of Joint Photographic Experts Group (JPEG) for compression of motion video, as opposed to still images. Frame-to-frame redundancy is ignored, resulting in less compression than Motion Picture Experts Group (MPEG). The advantages are simplicity and fast access to individual images.

MLA (Master License Agreement)

A Novell licensing option.

MLC (monthly license charge)

A pricing structure for software licensing. Users can usually terminate their software licenses with a 30-day notice.

MLM (Medical Logic Module)

The Arden Syntax for MLMs is a de facto standard in healthcare for coding clinical information (e.g., to generate clinical alerts, suggest interpretations or diagnoses, guide compliance with protocols). Each MLM encodes only one decision and evaluates as true or false. Although an MLM can call another MLM, in practice, this rarely occurs.

MLR (Multichannel Linear Recording)

A technology developed jointly by Tandberg, IBM and Imation based on quarter-inch tape cartridge recording.

MM (micro-marketplace)

A narrowly focused market that aggregates multiple vendor offerings, content and value-added services (such as comparison of features) to enable buyers within a particular industry, geographic region or affinity group to make informed purchasing decisions.

MM (multimedia)

Applications and technologies that manipulate text, data, images, sound and full-motion-video objects. Given the usage of multiple formats, multimedia is capable of delivering a stronger and more engaging message than standard text. Multimedia files are typically larger than text-based information and are therefore usually stored on CD-ROMs. Games and educational software commonly use multimedia.

MMC (Microsoft Management Console)

A Windows-based multiple document interface application that uses Internet technologies. MMC does not provide any management functionality, but rather a common management environment. It relies on independent software vendors

(ISVs) developing snap-ins to provide management functionality. Snap-ins are Microsoft- or ISV-developed software modules that integrate with MMC to enhance its functionality.

MMDS (multichannel multipoint distribution service)

A fixed wireless technology, commonly referred to as “wireless cable TV” or “wireless xDSL,” that operates between 2.5 gigahertz (GHz) and 2.7GHz. MMDS is used for broadcasting, personal communications and interactive media services in metropolitan areas.

MMI (man-machine interface)

another term for an end-user interface.

MMIS (materials management information system)

A software suite packaged as an integrated offering to meet materials management, human-resources and back-office needs. At a minimum, MMISs should be designed to interface readily with other mission-critical information systems in the enterprise.

MML (Mathematical Markup Language) — see MathML

MMMC (minimum monthly maintenance charge)

MMS (Manufacturing Message Service)

Part of Layer 7 in the Open System Interconnection (OSI) seven-layer network protocol stack. MMS creates a common communications interface between application software and factory devices such as programmable logic controllers, numerical controllers and robots.

MMS (Multimedia Messaging Service)

A mobile-messaging standard jointly defined by the 3G Partnership Project (see 3GPP) and the Wireless Application Protocol Forum (see WAP Forum). MMS provides a broad set of new features that cannot be delivered through traditional wireless-messaging standards, such as the ability to send and receive messages containing rich multimedia content, and to send messages to both mobile phone numbers and e-mail addresses. Unlike Enhanced Messaging Service (see EMS), MMS does not draw on established messaging technology such as Short Message Service (see SMS). Instead, it requires new infrastructure to be deployed by the network operators, and new functionality in mobile terminals.

MMU (memory management unit)

MNCRS (Mobile Network Computing Reference Specification)

A standard proposed in 1997 to enable network computers to dynamically adjust to a broad range of user devices under varying connectivity conditions (i.e., low bandwidth, offline and online), all without intervention on the part of the application programmer.

M/NET

An electronic-commerce (e-commerce) initiative acquired by PSI through the purchase of the ARM Group. It helps track purchases and inventory.

MNS (managed network services)

The vendor delivery of primarily operational support for a new environment in which the hardware assets, financial obligations and personnel still remain on the books of the customer.

MO (magneto-optic)

A type of storage technology that uses magnetization produced by a focused light beam.

mobile business (see m-business)

mobile client/server (see MC/S)

mobile commerce (see m-commerce)

mobile earth station

A radio transmitter or receiver situated on a ship, aircraft or other vehicle and used for satellite communications.

Mobile IP

Defines a packet-forwarding mechanism for mobile and remote hosts so that remote users can connect to their networks over the Internet. It can work with Dynamic Host Configuration Protocol (DHCP). See ATMP.

mobile middleware

Middleware is the software layer that helps programs and databases on different computers work together. Mobile middleware enables the implementation of distributed applications connecting mobile and enterprise applications over wireless networks. Many 100-percent wireless products created to fill in the missing functionality are now merging into prevalent application integration and middleware products, such as application servers. As vendors of traditional middleware try to offer comprehensive wireless solutions, they will either build them internally, or acquire or partner with specialized wireless vendors.

Mobile Network Computing Reference Specification (see MNCRS)

mobile payment (see m-payment)

mobile satellite communications

Satellite-based services to ships, motor vehicles and aircraft.

mobile satellite service (see MSS)

Mobile Services (see M-Services)

Mobile Station Application Execution Environment (see MExE)

mobile telephone service (see MTS)

mobile telephone switching office (see MTSO)

mobile terminal

Any mobile device that can connect to a wireless network. Data-centric examples include wireless personal digital assistants (PDAs). Voice-centric examples include basic and enhanced mobile phones. (See PDA, basic phone, enhanced phone, smart phone and Web phone.)

mobile virtual network operator (see MVNO)

mobile/wireless portal

A Web site with a wide range of content, services and links designed for mobile devices. It acts as a value-added middleman by selecting the content sources and assembling them in a simple-to-navigate (and customizable) interface for presentation to the end-user's mobile device.

MOCVD (metal organic chemical vapor deposition)

modal bandwidth

Optical fiber capacity measured in megahertz per kilometer.

modal dispersion

Signal distortion in an optical-fiber cable that causes the light to spread so much that the signal cannot be received.

mode

1. The most common or frequent value in a group of values.
2. One of many energy propagation patterns through a waveguide.

modem (modulator-demodulator)

A conversion device that allows a computing device to transmit information over analog communication lines (for example, traditional phone lines) by converting information that is digitally stored on the computer into transmission signals. The transmitting-end modem modulates digital signals received locally from a computer or terminal and sends analog signals over the line. The receiving-end modem demodulates the incoming signal, converting it back to its original (i.e., digital) format and passes it to the destination business machine.

modem-sharing unit (see MSU)

Modified Accelerated Cost Recovery System (see MACRS)

modified final judgment (see MFJ)

modular

1. Made up of replaceable units that can be added together to make a system larger or modify its capabilities.
2. A type of cable connector used in telephone and data communication systems.

modulation

The application of information onto a carrier signal by varying one or more of the signal's basic characteristics (frequency, amplitude or phase); the conversion of a signal from its original (e.g., digital) format to analog format. Specific types include:

- AM (amplitude modulation)
- ADPCM (adaptive differential pulse code modulation)
- FM (frequency modulation)
- PAM (pulse amplitude modulation)
- PCM (pulse code modulation)
- PWM (pulse width modulation; also called PDM, or pulse duration modulation)

modulator

A device that converts a voice or data signal into a form that can be transmitted.

modulator-demodulator (see modem)**MOF (Meta Object Facility)****MOLP (Microsoft Open License Pak)****MOM (manufacturing operations management)****MOM (message-oriented middleware)**

A model that programs the delivery of a message or a reply that must be deferred. MOM differs from other forms of program-to-program middleware, such as remote procedure calls (RPCs) and conversational services, in that MOM communication is connectionless — i.e., the sending and receiving programs do not interact directly. A program sends the message to the MOM, which then takes responsibility for delivering it to the proper receivers.

MOMA (Message-Oriented Middleware Association)**Mondex**

A maker of electronic cash systems for smart cards. The main difference between Mondex and its major stored-value competitors is that Mondex is a peer-to-peer system that operates just like cash — without a back-end settlement system. Value is exchanged from one card to another using readers that can be as small as a key ring device.

monitor

Computer hardware used for displaying digital output.

monomode

In fiber-optic cable, this enables light to enter at only one angle. This provides high bandwidth at the expense of requiring a light source with a narrow spectral width. Also called single-mode fiber. See multimode.

monthly license charge (see MLC)

Moore's Law

The idea that computing power doubles every 18 months, originally conceived by Intel co-founder Gordon Moore in 1965.

MOS (metal-oxide semiconductor)

Mosaic

A user-friendly Internet front end developed at the University of Illinois. The prototype for most of today's Web browsers.

MOSFET (metal-oxide semiconductor field-effect transistor)

MOSP (Multiple Operating System PR/SM Charge Option)

An IBM licensing option.

MOSPF (Multicast OSPF)

Extensions to open shortest path first (OSPF) routing that allow Internet Protocol (IP) multicasts.

MOSS (MIME Object Security Services)

An Internet secure messaging method.

Motif

A graphical user interface (GUI) specified by the Open Software Foundation (OSF) and built on the Massachusetts Institute of Technology's (MIT's) X Windows.

Motion JPEG (see M-JPEG)

Motion Picture Experts Group (see MPEG)

moves, adds and changes (see MAC)

MP (multiprocessor)

A computer that incorporates multiple processors with access to common storage.

MP3 (MPEG Layer 3)

An open Motion Picture Experts Group (MPEG) standard for digital audio compression. It offers significant compression while retaining good to excellent audio quality. Files compressed with MP3 can be transmitted over the Internet using a low-bandwidth connection. Standard, uncompressed audio — as recorded on a compact disc (CD) — requires about 740 megabytes for a 74-minute CD. MP3 can compress this data size by approximately 12 to 1. The size and quality of MP3 files varies depending on the sampling rate used in conversion. A rate of 128 kilobits per second (Kbps) results in an MP3 file that is approximately one megabyte per minute of music, with near-CD-quality audio. Dropping the sampling rate to 80 Kbps or 64 Kbps results in smaller files with audio quality that is reasonable, but inferior to that of a CD.

m-payment (mobile payment)

A payment initiated or completed through a wireless device. As a point of entry into retail payments, many carriers are targeting the underdeveloped market for micropayments (payments of less than \$10) for digital content and physical goods.

MPC (massively parallel computer)

MPC (Multimedia PC)

Specifications for a multimedia PC published by the Multimedia PC Marketing Council in 1990.

MPDSP (microprogrammable digital signal processing)

MPE (Multiprogramming Executive)

A multitasking operating system used on the Hewlett-Packard 3000 series.

MPEG (Motion Picture Experts Group)

A digital video standard for compression of full-motion images. The compression ratios achieved with MPEG encoding make it an ideal standard for delivery of digital video data.

MPEG-1 deals with mono and stereo sound coding, at sampling frequencies commonly used for high-quality audio. MPEG-2 contains an extension to lower sampling frequencies, providing better sound quality at the low bit rates, and an extension for multichannel sound. MPEG-3 and MPEG-4, with further improvements, are under development. Both MPEG-1 and MPEG-2 have a three-layer structure. Each layer represents a family of coding algorithms. Layer 3 deals with sound encoding, and can't be used by itself to encode audio files. In this form, it is known as MP3.

MPEG Layer 3 (see MP3)

MPI (master person index)

MPLA (Multiport Link Aggregation)

A 3Com switch technology.

MPLS (Multiprotocol Label Switching)

A protocol that helps support quality of service in IP networks. A router labels packets to assign different levels of service based on different priority levels. This helps ease congestion for high-priority network traffic, such as that needed for mission-critical applications.

MPN (multiprotocol network)

MPOA (multiprotocol over ATM)

A routing option that combines the second phase of local-area network (LAN) emulation with a route server, a route set cache of protocols, and route cache devices or edge devices that maintain route tables updated by the server.

MPP (massively parallel processing)

An architecture that uses hundreds or thousands of parallel processors.

MPP (Multichannel Point-to-Point Protocol)

A protocol developed by Ascend Communications (later acquired by Lucent Technologies). MPP is similar to PPP (Point-to-Point Protocol) but operable over multiple network channels in an inverse multiplexed scenario.

MPPC (multipurpose parallel computing)

MPPL (Multipurpose Processing Language)

MPR (microperipheral)

MPS (master production scheduling)

MPU (microprocessing unit)

See microprocessor.

MQI (Message Queue Interface)

MQM (message-queuing middleware)

A model to store and forward messages, thereby enabling asynchronous operations.

MQS (Message Queuing Series)

MRI (magnetic resonance imaging)

MRO (maintenance, repair and operations)

The activities associated with the operation and repair of any facility, equipment or asset, and the material purchased to support these activities.

MRP (material requirements planning)

An early category of manufacturing business software, which focused only on planning manufacturing materials and inventories, and did not integrate planning for other resources, such as people and machine capacity.

MRP II (manufacturing resource planning)

A method for effective planning of all the resources of a manufacturing company. Ideally, it addresses operational planning in units and financial planning in dollars, and has a simulation capability to answer "what if" questions. It is made up of a variety of functions, each linked together: business planning, production planning, master production scheduling (MPS), material requirements planning (MRP), capacity requirements planning (CRP) and the execution systems for capacity and priority. Outputs from these systems can be integrated with business reports (e.g., business plans, purchase commitment reports, shipping budgets or inventory projections). MRP II is a direct outgrowth and extension of MRP. MRP II is migrating toward enterprise resource planning (ERP).

ms (millisecond)

MSB (most significant bit)

MSCS (Microsoft Cluster Server)

Previously known as Wolfpack Phase 1, this provides two-node server failover for enterprises seeking high-availability NT Server (NTS) solutions. Specific certified hardware configurations are required to support MSCS.

MSDE (Microsoft Data Engine)

A desktop database engine that works as an extension of SQL Server 7.0 to store data for offline application components.

MSDN (Microsoft Developer Network)

MS-DOS (Microsoft Disk Operating System)

An operating system written by Microsoft for personal computers that was the basis for IBM's PC-DOS. MS-DOS was used by all IBM-PC-compatible machines. It was originally designed to organize how a computer reads, writes and interacts with its disks — floppy or hard — and how it talks to its various input/output (I/O) devices, such as keyboards, screens, printers and modems. Until the introduction of Windows, the most popular operating system for PCs was MS-DOS. Unlike its successors, MS-DOS was character-based and did not use a graphical user interface.

MSDSS (Microsoft Directory Synchronization Services)

MSE (midsize enterprise)

M-Services (Mobile Services)

The Global System for Mobile Communications (GSM) M-Services initiative, announced by the GSM Association in June 2001, is designed to provide a framework outlining the minimum specification for new GSM mobile handsets. The GSM Association believes one barrier to mass-market adoption of Internet-enabled mobile services is the lack of a single set of comprehensive specifications for mobile-terminal manufacturers to adhere to.

MSF (Multiple Server Feature)

An Amdahl S/390 complementary metal-oxide semiconductor (CMOS) feature that allows a single system to be divided into smaller systems, each with a unique serial number.

MSI (medium-scale integration)

Fewer than 100 circuits built onto a single chip. MSI is used frequently in third-generation systems.

MSMQ (Microsoft Message Queue Server)

Originally called Falcon, MSMQ is a message-queuing middleware (MQM) product, or, more accurately, an operating-system service that provides store-and-forward connectivity between two application programs.

MSN (Microsoft Network)

MSN (multiservice network)

MSNF (Multisystem Networking Facility)

A facility of IBM's Network Control Program (NCP) used to control a multiple-domain network. It was an optional feature of Telecommunications Access Method (TCAM) and Virtual Telecommunications Access Method (VTAM) version I.

MSO (management services organization)

A service organization of an integrated delivery system or hospital that provides management services for multiple affiliated physician practices and clinics.

MSP (management service provider)

Also known as "managed-service providers," MSPs deliver network, application, system and e-management services across a network to multiple enterprises, using a

“pay as you go” pricing model. A “pure play” MSP focuses on management services as its core offering. In addition, the MSP market includes offerings from other providers — including application service providers (ASPs), Web hosting companies and network service providers (NSPs) — that supplement their traditional offerings with management services. (See ASP and NSP.)

MSP (Message Security Protocol)

An X.400-like protocol that can be used over Transmission Control Protocol/Internet Protocol (TCP/IP) networks.

MSS (mobile satellite service)

Mobile earth stations (e.g., on ships, survival stations and positioning radio beacons) that provide location identification using satellite service.

MSSP (managed security service provider)

An MSSP provides outsourced monitoring and management of security devices and systems. Common services include managed firewall, intrusion detection, virtual private network, vulnerability scanning and anti-viral services. MSSPs use high-availability security operation centers (either from their own facilities or from other data center providers) to provide 24x7 services designed to reduce the number of operational security personnel an enterprise needs to hire, train and retain to maintain an acceptable security posture.

MSU (message signal unit)

MSU (million service units)

MSU (modem-sharing unit)

A device that permits two or more terminals to share a single modem.

MSW (most significant word)

MTA (message transfer agent)

The storing and forwarding portion of an electronic-mail system.

MTBF (mean time between failures)

The cumulative average time that a manufacturer estimates will elapse between failures in a computer component or system.

MTF (mean time to failure)

MTO (master terminal operator)

MTP (Message Transfer Part)

An ISDN term for the mechanism that ensures reliable functional signal message transmission. It corresponds to the first three layers of the Open Systems Interconnection (OSI) reference model (see OSI).

MTS (Microsoft Transaction Server)

Microsoft's NT-based transactional middleware.

MTS (mobile telephone service)

Telephone service provided between mobile stations and the public switched telephone network; radio transmission provides the equivalent of a local loop. See cellular radio.

MTSO (mobile telephone switching office)

A U.S. term for the central switch that controls the entire operation of a cellular system. It is a sophisticated computer that monitors all cellular calls, keeps track of the locations of all cellular-equipped vehicles traveling in the system area, arranges handoffs and keeps track of billing information.

MTTR (mean time to repair)

An estimated average time required to do repairs on equipment.

MTX (mobile telephone exchange)

MU (message unit)

The U.S. term for the unit of measure for charging local calls. It details the length of call, the distance called and the time of day.

mu-chip

A tiny radio frequency identification (RFID) tag, or "e-tag," introduced in 2001 by Hitachi through its in-house venture company, Mu-Solutions. It measures 0.4 square millimeters and is thin enough (60 microns) to be embedded in paper. It can store 128 bytes, which must be written at the time of manufacturing and cannot be changed. See RFID and e-tag.

MUI (multimedia user interface)

mu-law encoding

Encoding according to International Telecommunication Union (ITU) recommendation G.711, used with 24-channel (PCM) systems in the United States.

MULC (Measured Usage License Charge)

An S/390 software licensing arrangement offered by IBM. It yields charges based on the amount of processing capacity being used by selected software products.

Multi-Application Operating System (see MULTOS)

Multi-Application Operating System Consortium (see MAOSCO)

Multibus

Intel's central path (channel) for transmitting electrical signals and data in 8- and 16-bit computer systems.

multicast

Multicast, as opposed to broadcast, is a signal transmitted to only a subset of potential destinations.

Multicast OSPF (see MOSPF)

Multichannel Linear Recording (see MLR)

multichannel multipoint distribution service (see MMDS)

Multichannel Point-to-Point Protocol (see MPP)

multidimensional database management system (see MDDBMS)

multidrop

A communications arrangement where multiple devices share a common transmission channel, though only one may transmit at a time. See multipoint.

Multifunction Interpreter (see MFI)

multifunction product (see MFP)

multileaving

The transmission of a variable number of data streams between user devices and a computer, usually via bisynchronous facilities and using bisynchronous protocols.

multimedia (see MM)

Multimedia and Hypermedia Information Coding Expert Group (see MHEG)

Multimedia Cable Network System (see MCNS)

Multimedia Messaging Service (see MMS)

Multimedia PC (see MPC)

multimedia workstation

A workstation that can process multimedia applications.

multimode

In fiber-optic cable, the ability to transmit multiple frequencies of light. See monomode.

multimode fiber

A fiber supporting propagation of multiple modes. The cable has a core diameter of 50-100 microns. It causes more distortion and gives less bandwidth than single-mode (monomode) fiber.

multiple console operation

A private branch exchange (PBX) supporting more than one attendant's position to handle heavy traffic. Call traffic is distributed evenly among the consoles in use.

multiple customer group operation

A public branch exchange (PBX) that can be shared by several different companies, each having separate consoles and trunks. Stations are assigned to one company or the other and are then capable of reaching only that company's trunks and attendants.

multiple instruction, multiple data (see MIMD)

Multiple Domain Feature (see MDF)

Multiple Server Feature (see MSF)

Multiple Virtual Storage (see MVS)

Multiple Virtual Storage/Enterprise Systems Architecture (see MVS/ESA)

multiplexer (see MUX)

multiplexing

Division of a transmission facility into two or more channels either by splitting the frequency band transmitted by the channel into narrower bands, each of which is used to constitute a distinct channel (frequency division multiplexing), or by allotting this common channel to several different information channels, one at a time (time division multiplexing).

multipoint

Pertaining to or referring to a communications line to which three or more stations are connected. It implies that the line physically extends from one station to another until all are connected. See point to point.

multipoint line

A single communications channel (typically a leased telephone circuit) to which more than one station or logical unit is attached, though only one can transmit at a time. Such arrangements usually require a polling mechanism under the control of a master station to ensure that only one device transmits data at a time. Also called a multidrop line.

Multiport Link Aggregation (see MPLA)

multiprocessor (see MP)

multiprogramming

A computer system operation whereby a number of independent jobs are processed together. Rather than allow each job to run to completion in turn, the computer switches between them so as to improve the use of the system hardware components.

Multiprogramming Executive (see MPE)

Multiprotocol Label Switching (see MPLS)

multiprotocol over ATM (see MPOA)

Multipurpose Internet Messaging Extensions (see MIME)

multistation access unit (see MAU)

Multisystem Networking Facility (see MSNF)

multitasking

Two or more program segments running in a computer at the same time.

multithreading

Concurrent processing of more than one message (or similar service request) by an application program.

multiuser

A term that describes a computer that can support several workstations operating simultaneously.

multivendor network management

Support for integration of network management on a common workstation. The network management facility should enable user organizations or third-party vendors to integrate additional network management tools.

MULTOS (Multi-Application Operating System)

A smart-card operating system originally created by Mondex, and now overseen by the MAOSCO consortium. See MEL.

MULTOS Executable Language (see MEL)

MUX (multiplexer)

A device that combines inputs from two or more terminals, computer ports or other multiplexers, and transmits the combined data stream over a single high-speed channel. At the receiving end, the high-speed channel is "demultiplexed," either by another multiplexer or by software.

MVC (multiport video controller)

MVIP (Multivendor Integration Protocol)

MVLP (Microsoft Variable License Pak)

MVNO (mobile virtual network operator)

A company that provides mobile network services through a contract with a mobile network operator. At the low end of the MVNO range is the company that offers mobile-telephony services through an agreement with a mobile-telephony spectrum owner, with the new entrant controlling the branding, marketing and tariff structure of the offering to customers. At the higher end is the most widely accepted definition of an MVNO: a company with its own network code, issuing its own subscriber identity module cards and controlling elements of network infrastructure, such as the home location register and mobile switching center.

MVS (Multiple Virtual Storage)

IBM's flagship operating system. Essentially all the device support, software functions, time-sharing aids and reliability improvements that IBM produces are positioned most firmly within MVS. Many Virtual Machine (VM) microcode "assists" were designed so that MVS guests can run more effectively in that environment. Of all of IBM's software products, MVS has by far received the greatest investment for development, documentation and support.

MVS/370

A version of MVS limited to 24-bit addressing and lacking the dynamic channel subsystem. Dropped from support in the early 1990s.

MVS/ESA (Multiple Virtual Storage/Enterprise Systems Architecture)

A version of MVS introduced by IBM in 1988, capable of addressing up to 16 terabytes of data.

N

N+1 disk shadowing/mirroring

A system- or storage-subsystem-level storage facility that provides improved disk reliability without the cost of two-times disk shadowing.

NACHA (National Automated Clearing House Association)**Nagle's algorithm**

Two control algorithms that may be used in Transmission Control Protocol (TCP)-based networks; one reduces the sending window, the other limits small datagrams.

NAI (Network Associates Inc.)**nailed-up connection**

A slang term for a permanent, dedicated path through a switch. This often refers to lengthy, regular data transmissions going through a private branch exchange (PBX).

NAK (negative acknowledgment)

A return signal that reports an error in the method of error control that relies on repeating any message received with detectable errors.

NAM (number assignment module)

The electronic memory in the cellular phone that stores the telephone number and electronic serial number. Phones with dual- or multi-NAM features offer the user the option of registering the phone so that it will have two or more phone numbers.

named pipes

One of several operating-system facilities for interprocess communications.

naming service

A program that provides a mapping between a logical name and a physical address in a network. It can be extended by a directory service.

NAP (network access point)

The point from which an Internet service provider (ISP) drops down its lines and establishes a peering arrangement to provide Internet connectivity to customers.

narrowband channels

Sub-voice-grade pathways characterized by a speed range of 100 to 200 bits per second.

Narrowband ISDN (see N-ISDN)**narrowband PCS (see N-PCS)**

NAS (Netscape Application Server)

NAS (NetWare Access Server)

NAS (network-attached storage)

A category of storage products in which the requisite hardware and software comes bundled into an integrated product, which is optimized for use as a dedicated file or storage management server attached to the enterprise's network. Ideally, NAS is platform- and OS-independent, appears to any application as another server, can be brought online without shutting down the network and requires no changes to other enterprise servers.

NASD (network-attached storage device)

See NAS.

NASI (NetWare Asynchronous Service Interface)

A legacy terminate and stay resident (TSR) program, developed by Novell to allow an application to access a modem in a network-based modem pool via a communications port (COM) redirector on a workstation.

NAT (network address translation)

A function that translates the Internet Protocol (IP) addresses used in an internal IP network's addressing structure to fewer, external Internet IP addresses.

NATA (North American Telecommunications Association)

National Computer Security Center (see NCSC)

national currency unit (see NCU)

national healthcare identifier (see NHID)

National Industrial Information Infrastructure Protocols (see NIIIP)

National Institute of Standards and Technology (see NIST)

national language support (see NLS)

National Security Agency (see NSA)

native code compiler for Java (see NCCJ)

natural-language processing (see NLP)

natural-language search (see NLS)

NAU (Network Addressable Unit)

Any presentation-level service in IBM's Systems Network Architecture (SNA).

NAV (Norton Antivirus)

NBS (Narrowband Sockets Specification)

NC (network computer)

Often called a "thin client," a network computer is a limited-function desktop computer that is designed to easily connect to networks. NCs include a keyboard, mouse, monitor, memory and a network connection but only limited, if any, local disk storage. When end users want to access software or databases using an NC, they would use a graphical user interface (GUI), much as they do now with a PC. However, the software would be downloaded from a central server instead of being resident on the desktop PC; large databases would also be maintained on the central server. All software backups, upgrades and maintenance would therefore be performed centrally on the network server.

NC (Numerical Control)

NCA (Network Computing Architecture)

Oracle's implementation of an object transaction manager (OTM).

NCCJ (native code compiler for Java)

Translates byte code into a file format and binary-code representation that can be linked (using a static linker) with pre-compiled libraries and resources to form the executable program. An NCCJ provides the opportunity to optimize the structure of the program code to eliminate redundant code as well as to fine tune code sequences (e.g., loops, jumps and substitutions).

NCCS (network computing client/server)

An emerging set of technologies and architectures that will enable dynamic (and, to some extent, adaptive) application deployment, execution and management. NCCS is largely based on existing elements including Internet-derived technologies — e.g., Hypertext Markup Language (HTML), Hypertext Transport Protocol (HTTP), Java, ActiveX and Internet Inter-ORB Protocol (IIOP); various forms of middleware — e.g., message-oriented middleware (MOM), Common Object Request Broker Architecture (CORBA) and Component Object Model + (COM+); and architectures (e.g., multitier client/server). However, it will foster a whole set of new technologies and architectural patterns.

NCD (networked computing device)

NCDM (National Center for Database Marketing)

NCF (network-computing framework)

NCGA (National Computer Graphics Association)

N-channel metal-oxide semiconductor (see NMOS)

NCL (Network Control Language)

NCOP (network code of practice)

A code of standard practices for network design, intended to maintain technical standards.

NCP (Network Control Program)

The IBM-licensed program that controls the operation of an IBM 37xx front-end processor (FEP). Communication with the host is through Virtual Telecommunications

Access Method (VTAM) via a channel interface, and communications with terminals or another FEP is via telecommunications lines. NCP offloads certain line protocol and routing functions from the host central processing unit (CPU).

NCPDP (National Council for Prescription Drug Programs)

NCQA (National Committee for Quality Assurance)

An accreditation body for healthcare plans.

NCRP (Network Computer Reference Platform)

A specification defining the minimum standards for a network of computers.

NCS (National Communications System)

NCSA (National Center for Supercomputing Applications)

NCSC (National Computer Security Center)

A branch of the U.S. National Security Agency

NCTP (Next Compatible Tape Product)

NCU (national currency unit)

A generic term for a unit of local currency, usually referenced in the context of euro conversion calculations or compliance efforts.

NCVHS (National Committee on Vital and Health Statistics)

NDA (new drug application)

NDA (nondisclosure agreement)

NDC (networked data center)

A service that goes beyond making data center functionality available on a network. An NCC employs networking technology to treat multiple data centers and the network as a single system to efficiently access and process applications.

NDF (network dynamic functionality)

A software development and execution technology that allows applications to be written to a model designed from the outset to achieve platform independence. It affords dynamic movement and invocation of code resources over a network at runtime.

NDIS (Network Device Interface Specification)

Specification for the Ethernet adapter/NetBIOS liaison.

NDL (Network Data Language)

A skeletal standard developed by the Conference on Data Systems Languages (CODASYL) for an application programming interface (API) to network database management systems (DBMSs). It has since been abandoned.

NDMP (Network Data Management Protocol)

NDS (Novell Directory Services)

A globally oriented service that maintains information about network resources (including users, groups, servers, volumes, printers and multiple servers) and provides a naming service. NDS replaces the binary file used in previous versions of Novell's NetWare network operating system. (Previously called NetWare Directory Service.)

near-end crosstalk (see NEXT)

near-letter quality (NLQ)

NEDA (National Electronic Distribution Association)

negative acknowledgment (see NAK)

Neper

The basic unit of a logarithmic scale used for the expression of ratios of voltages, currents and similar quantities.

NEPP (network-enabled publishing processes)

Network-based, intranet-based or groupware applications that are collaborative and dynamic in nature; they are replacing closed, static and segmented print-based processes. NEPPs enable publishers to issue content in multiple formats (i.e., multichannel publishing), and allow an effective way to create, archive, access and market content through electronic sources. NEPPs form the basic building blocks of a new comprehensive production architecture (CPA) emerging for commercial publishers.

NES (Netscape Enterprise Server)

.NET

A major software initiative announced by Microsoft in July 2000, and expected to be implemented over the course of several years. At its core, .NET represents Microsoft's implementation of the Web services concept, which treats software as a set of services accessible over ubiquitous networks using Web-based standards and protocols — although Microsoft has broadly applied the ".NET" moniker to several independent technologies and initiatives that have little to do with Web services (e.g., .NET Enterprise Servers).

As a software infrastructure, .NET consists of two new programming models:

- *A Web services programming model*, which exposes programming interfaces through Internet standards. This loosely coupled model uses Hypertext Transport Protocol (HTTP) and other Internet protocols as the main underlying transport mechanisms, and also uses Extensible Markup Language (XML), Simple Object Access Protocol (SOAP), Web Services Description Language (WSDL) and Universal Description, Discovery and Integration (UDDI). Initially, most of this model will use a request/reply communications style.
- *A system programming model*, designed to supercede Microsoft's Component Object Model (COM) and the Windows application programming interfaces over the long term. This model introduces a new set of fundamental classes and a new runtime environment — the Common Language Runtime (CLR) — providing an object-oriented class hierarchy structure as part of the application runtime

environment. It also provides classes and mechanisms that enable programs to be “wrapped” as Web services, so it can ease — but is not required for — Web services development.

See Web services, SOAP, WSDL, UDDI and CLR.

NetBEUI (NetBIOS Extended User Interface)

A device driver and transport protocol, originally developed by Microsoft, that can bind with up to eight Media Access Control (MAC) drivers.

NetBIOS (Network Basic Input/Output System)

An extension of IBM’s PC BIOS. It was developed as the interface for the PC Network program, but is supported on token ring networks and a number of non-IBM systems.

NetBIOS Extended User Interface (see NETBEUI)

netiquette

The term used to describe the proper way to behave on the Internet. Users must follow netiquette to avoid being flamed. See flame.

netizen

A network citizen — someone who uses the Internet frequently and essentially “lives” part of his or her life on the Web. Netizens are different from Web surfers in that they use specific Internet services for longer periods rather than visit many sites in short increments.

net-liberated organization (see NLO)

NetMeeting

A data conferencing and groupware facility for Microsoft Windows.

Netnews (see Usenet)

net present value (see NPV)

NetView

An IBM Systems Network Architecture (SNA) network management product encompassing some of the functions of NCCF, NLDM, NPDA, NMPF and VTAM node control applications. It was a key enabler to IBM’s overall systems management strategy.

NetWare

A network operating system (NOS) from Novell. It is the dominant local-area networking system for peripheral sharing and application communications.

NetWare Asynchronous Service Interface (see NASI)

NetWare Loadable Module (see NLM)

network

Any number of computers (e.g., PCs and servers) and devices (e.g., printers and modems) joined together by a physical communications link. In the enterprise

context, networks allow information to be passed between computers, irrespective of where those computers are located. Networks provide the roads for information traffic (e.g., sending files and e-mail) within a corporate environment, and allow users to access databases and share applications residing on servers. If a network does not go outside of a company building, or campus, then it is known as a local-area network (LAN). If it has a bridge to other outside networks, usually via lines owned by public telecommunications carriers like AT&T, then it is known as a wide-area network (WAN).

network access point (see NAP)

network address translation (see NAT)

Network Addressable Unit (see NAU)

network and systems management (see NSM)

network appliance

A type of computing appliance that aids in the flow of information to other network-connected computing devices. Services that may be provided by a network appliance include firewall functions, caching, authentication, network address translation and IP address management. Examples include CacheFlow's caching appliance and WatchGuard Technologies' Firebox firewall appliance. See computing appliance.

network-attached storage (see NAS)

Network Basic Input/Output System (see NetBIOS)

Network Caching Servers

Unlike proxy servers, a network caching server can "listen" on the network and intercept protocol requests on its associated Transmission Control Protocol (TCP) ports. This alleviates the need to have the enterprise's browser configured to "know" where the server sits on the network.

network code of practice (see NCOP)

network computer (see NC)

Network Computer Reference Platform (see NCRP)

network computing

A client/server application architecture with dynamic application deployment, execution and management. Network computing is characterized by four properties: dynamic cached propagation; write once, run anywhere; automatic platform adjustment; and network context storage.

Network Computing Architecture (see NCA)

network computing client/server (see NCCS)

Network Control Program (see NCP)

network database

A database organized according to ownership of records, allowing records to have multiple owners and thus providing multiple access paths to the data. Database management systems (DBMSs) providing such capabilities are also known as CODASYL (Conference on Data Systems Languages) DBMSs.

Network Data Language (see NDL)

Network Device Interface Specification (see NDIS)

network dynamic functionality (see NDF)

networked systems management (see NSM)

network-enabled publishing processes (see NEPP)

Network File System (see NFS)

Network Independent File Transfer Protocol (see NIFTP)

Network Information Service (see NIS)

networking

The linking of a number of devices, such as computers, workstations and printers, into a network for the purpose of sharing resources. See network.

network interface card (see NIC)

network layer

In the Open Systems Interconnection (OSI) model, the logical network entity that services the transport layer. It is responsible for ensuring that data passed to it from the transport layer is routed and delivered through the network.

network management

Administrative services performed in managing a network — e.g., network topology and software configuration, downloading of software, monitoring network performance, maintaining network operations, and diagnosing and troubleshooting problems.

Network management center (see NMC)

Network Management Forum (NMF)

Now known as the TeleManagement Forum (see TMF).

network management system (see NMS)

Network Management Vector Tables (see NMVT)

Network News Transport Protocol (see NNTP)

Network OLE (see NOLE)

network operating system (see NOS)

network operations center (see NOC)

network performance tuning/configuring facilities

The ability to configure combinations of local-area networks (LANs) and wide-area networks (WANs) centrally and dynamically based on anticipation and prioritization of data traffic volumes. Physical data paths could be configured based on data traffic content (i.e., transaction type). This is particularly important for online transaction processing (OLTP) applications.

network redundancy

A communications pathway that has additional links to connect all nodes in case one link goes down.

network security

Measures taken to protect a communications pathway from unauthorized access to, and accidental or willful interference of, regular operations.

Network Series (see NS)

Network Series 8 (see NS8)

network service access point (see NSAP)

network service provider (see NSP)

network terminating unit (see NTU)

network topology

Describes the physical and logical relationship of nodes in a network, the schematic arrangement of the links and nodes, or some hybrid combination thereof.

network virtual terminal

A communications concept describing a variety of data terminal equipment (DTE), with different data rates, protocols, codes and formats, accommodated in the same network. This is done as a result of network processing where each device's data is converted into a network standard format, then converted into the format of the receiving device at the destination end.

networked data center (see NDC)

neural net or neural network

An artificial-intelligence processing method within a computer that allows self-learning from experience. Neural nets can develop conclusions from a complex and seemingly unrelated set of information.

NeWS

Sun Microsystems' initial graphical user interface (GUI) product.

newsgroup

On the Internet, one of over 16,000 discussion forums collectively known as Usenet.

NEXT (near-end crosstalk)

Unwanted energy transferred from one circuit, usually to an adjoining circuit. It occurs at the end of the transmission link where the signal source is located, with the absorbed energy usually propagated in the opposite direction of the absorbing channel's normal current flow. It is usually caused by high-frequency or unbalanced signals and insufficient shielding.

Next-Generation Internet (see NGI)

Next-Hop Resolution Protocol (see NHRP)

NFS (Network File System)

A method of sharing files across a computer network. Pioneered by Sun Microsystems, it is now a de facto standard in the Unix environment. NFS is built on Transmission Control Protocol/Internet Protocol (TCP/IP) and Ethernet.

NGI (Next-Generation Internet)

The federal government project to fund research and to build a high-speed network designed to provide performance guarantees.

NHID (national healthcare identifier)

An identification code unique to each person, mandated by the Health Insurance Portability and Accountability Act of 1996 (HIPAA), that provides a way to seamlessly aggregate data on a given person regardless of the location of the data or the systems used to access it.

NHRP (Next-Hop Resolution Protocol)

Protocol defined for the Internet Engineering Task Force (IETF) to determine shortcuts through a routed infrastructure. The protocol has been incorporated in multiprotocol over ATM (MPOA) for asynchronous transfer mode (ATM) as well as for proprietary layer 3 implementations.

NIA (National Industrial Automation)

NIC (network interface card)

The bus-specific adapter that connects an end station or server to a local-area network (LAN). It plugs into an expansion slot on a workstation or server that is to be networked and has a connector for the network cabling.

nickel metal hydride (see NiMH)

NIFTP (Network Independent File Transfer Protocol)

Part of the Color Book protocol suite, which preceded Open Systems Interconnection (OSI) but is still widely used.

NIH (not invented here)

NIIP (National Industrial Information Infrastructure Protocols)

The NIIP-Smart activity, sponsored by the National Institute of Standards and Technology, is the technical standards aspect of the U.S. Department of Commerce's Consortium for Intelligent Integrated Manufacturing Planning and Execution project, part of the Advanced Technology Program. The thrust of this effort is an object-oriented framework with modular plug-and-play capability to facilitate the integration

of open control systems (OCS), manufacturing execution systems (MES) and enterprise resource planning (ERP) applications. This effort will go beyond traditional standards activities in not only defining but demonstrating feasibility. Applicable to a broad range of discrete manufacturing from engineer-to-order to make-to-stock, this activity has broad appeal to all discrete manufacturers.

NiMH (nickel metal hydride)

A type of rechargeable battery used in portable computers and cell phones. NiMH batteries have greater charge density than nickel-cadmium batteries.

NIPRnet (Nonclassified Internet Protocol Router Network)

A subset of the Internet used by the U.S. military.

NIS (Network Information Service)

A Sun Microsystems Unix repository for user and resource configuration information.

N-ISDN (Narrowband ISDN)

Integrated Services Digital Network (ISDN) speeds up to 1.544 megabits per second (Mbps). As speeds get faster, the definition of narrowband ISDN applies to faster speeds.

NIST (National Institute of Standards and Technology)

An agency of the U.S. Department of Commerce. Previously called the National Bureau of Standards.

NLM (NetWare Loadable Module)

Software that provides additional functions to a NetWare server — for example, an application (such as a DBMS) running directly under NetWare.

NLO (net-liberated organization)

A Gartner concept describing an organizational philosophy enabled by the advent of the Internet and related Web technologies. Rather than "bolting on" Web initiatives to mainstream operations (i.e., integrating the Internet into business processes, but leaving the structure of these processes intact) an NLO uses these technologies to liberate itself from the constraints of traditional business environments that focus on local and physical infrastructures.

The goal of an NLO is to transform the enterprise into one that is more agile, and thus better able to meet customer demands. As such, the Internet is the primary enabler of liberating organizations from constraints associated with:

- Time (when business is conducted)
- Place (where business is conducted)
- Hierarchy (how people interrelate)
- Ownership (who owns the assets deployed for business)
- Information (how information is disseminated and used in human and process activities)

NLP (natural-language processing)

The process of converting narrative text to coded facts. Typically, a body of information — such as a series of publications, reports or even a collection of

messages — is analyzed and the results displayed as dominant associations or relationships, and show which topics or subjects occurred most frequently.

NLQ (near-letter quality)

A designation for high-quality dot-matrix printers. Now largely obsolete.

NLS (natural-language search)

A search in which users enter questions in natural human language, rather than using Boolean operators or other notations. An NLS search engine uses algorithms to translate the question into search logic and attempts to return results that answer the question. The success of these attempts is typically mixed, depending on the complexity and nuances of the question and sophistication of the engine's translation capabilities.

NLS (national language support)

Support for local languages in system or application user interfaces.

NLS (network licensing server)

NLSP (Network Layer Security Protocol)

nm (nanometer)

One-billionth of a meter, a microscopic measurement commonly used in electronic circuitry.

NMC (network management center)

The center used for control of a network. It may provide traffic analysis, call detail recording, configuration control, fault detection and diagnosis and maintenance.

NMF (Network Management Forum)

The former name of the TeleManagement Forum (see TMF).

NMI (nonmaskable interrupt)

NMOS (N-channel metal-oxide semiconductor)

A microelectronic circuit used for logic and memory chips and in complementary metal-oxide semiconductor (CMOS) design. NMOS transistors are faster than the P-channel metal-oxide semiconductor (PMOS) counterpart, and more of them can be put on a single chip.

NMR (nuclear magnetic resonance)

A medical imaging technology that can "see" soft tissue.

NMS (network management system)

A central set of programs providing network-based control of disparate hardware elements; also, the software and hardware used in bridging, routing, etc.

NMVT (Network Management Vector Tables)

A Systems Network Architecture (SNA) management protocol which provides alert problem determination statistics and network management data.

NNI (network-to-network interface)

NNM (network node manager)

NNTP (Network News Transport Protocol)

The discussion database standard that governs Usenet newsgroup distribution on the Internet.

NOC (network operations center)

A Web-hosting term which refers to the "central nervous system" of the data center. Without it, there would be no infrastructure to transport data to the Web. All Tier 1 Internet service providers and many major telecommunications companies have developed, or are developing, NOCs.

node

A termination point for two or more communications links. The node can serve as the control location for forwarding data among the elements of a network or multiple networks, as well as performing other networking and, in some cases, local-processing functions. Nodes can be host processors, communications controllers, cluster controllers, workgroup computers or terminals.

noise

Unwanted electrical signals, introduced by circuit components or natural disturbances, that tend to degrade the performance of a communications channel.

NOLE (Network OLE)

An early name for Microsoft's Distributed Component Object Model (DCOM).

nonblocking

Describes the state of a telephone switch where a through traffic path always exists for each attached station. Generically, it describes a switch or switching environment designed to never experience a busy condition due to call volume.

Nonclassified Internet Protocol Router Network (see NIPRnet)

nonpublic personal information (see NPPI)

non-real-time variable bit rate (see nrtVBR)

nonrepudiation

A security feature under which the origin of data cannot be denied, and can be proven to an independent third party.

Non-return-to-zero-inverted (see NRZI)

NonStop SQL (NonStop Structured Query Language)

Relational database management system (RDBMS) software from Compaq Computer (now Hewlett-Packard). NonStop SQL supports near-linear expandability and high uptimes for applications by tolerating single points of failure. NonStop SQL was the first DBMS to support engine-based (as opposed to front-end-tool-based) distributed databases.

Nonuniform Rational B-Spline (see NURBS)

nonvolatile

Term used to describe a data store that retains data when the power is off.

nonwireline cellular carrier

A U.S. term referring to Block A carriers. Nonwireline or Block A systems operate on radio frequencies from 824 to 848 megahertz (MHz).

normalization

Translating all source messages into a standard format before making them visible to other processes or transforming them a second time into the formats required by the destinations.

NOS (network operating system)

A set of software utilities that, working in conjunction with an operating system, provides the local-area network (LAN) user interface and controls network operation. A network operating system communicates with the LAN hardware and enables users to communicate with one another and to share files and peripherals. Typically, a NOS provides file-to-print services, directory services and security.

notary

An Internet standard defining message status capabilities and delivery notification.

notebook

A computer system designed for portability. It comes with a battery and typically measures 8.5 inches by 11 inches and weighs less than 8 pounds with the battery and weight-saver modules. Notebooks use flat-panel color screens of Super Video Graphics Array (SVGA) resolution or higher. They offer expansion through PC-Card technologies and have specialized integrated pointing devices. Types of notebooks include:

- **Desktop Alternative:** This is a computer system that meets all the criteria for a notebook PC but is designed to replicate the functionality of a desktop. It weighs 6 pounds or above. The screen can be as large as 15 inches or 16 inches with SVGA resolution or higher. Target markets include engineers and end users wanting to travel carrying minimal weight.
- **Mainstream:** This is a computer system that meets all the criteria for a notebook PC but is designed to be the best compromise between all-inclusive functionality and light weight. Mainstream notebooks weigh between 4.5 and 6 pounds with the weight-saver and battery modules. Mainstream notebooks often have a single bay for the inclusion of a peripheral, such as a CD-ROM.
- **Ultraportable:** This is a computer system that meets all the criteria for a notebook PC but is lighter and may not have an internal floppy disk drive. It typically weighs 4 pounds or less with the battery and weight-saver modules. The keyboard and screen are often compromised to meet weight targets and the unit must be augmented with a standard keyboard and mouse for long-term use.

Novell Directory Services (see NDS)

Novell Single Sign-On (see NSSO)

NOW Index

A Gartner metric that measures normalized cost and work produced in an IS facility.

NPA (Network Professional Association)

N-PCS (narrowband personal communications services)

N-PCS is similar to paging since it is used for short-message applications with a small "palm-of-the-hand" form factor. The addition of a response channel enables two-way paging capability and allows N-PCS to serve users that have more robust data and reliability requirements. N-PCS has been allocated a total of three megahertz (MHz) of spectrum in the 900-MHz block by the Federal Communications Commission (FCC).

NPPI (nonpublic personal information)

Any identifiable financial information provided by a consumer resulting from a transaction made by, or a service provided to, the consumer or "otherwise obtained" by the financial services provider (FSP). Specifically includes lists, descriptions or groupings of consumers derived from NPPI held by the FSP.

NPSI (NCP Packet Switching Interface)

NPV (net present value)

Given an initial investment, a series of expected cash flow returns per period and an internal opportunity cost of capital (OCC), NPV analysis helps determine whether an investment should be made.

NRC (nonrecurring charge)

NRE (nonrecurring engineering)

nrtVBR (non-real-time variable bit rate)

An asynchronous transfer mode (ATM) service category, defined by the ATM Forum, that provides latency or bandwidth guarantees for traffic with no strict timing requirements. Good uses of this service are for legacy Systems Network Architecture (SNA) traffic or frame relay committed information rate (CIR) trunking.

NRZ (non-return-to-zero)

NRZI (non-return-to-zero inverted)

Encoding by inverting a signal on a high-voltage (on) pulse, leaving the signal unchanged for the low (off).

NS (Network Series)

A tape drive based on Travan technology. The features that differentiate it from Travan are read-after-write functionality and hardware compression.

NS8 (Network Series 8)

A tape drive based on Travan technology. The feature that differentiates it from NS is native 8 gigabyte capacity.

NSA (National Security Agency)

A U.S. government agency responsible for electronic intelligence.

NSAP (network service access point)

In the semantics of the Open Systems Interconnection (OSI) addressing scheme, the point at which the OSI network service is made available to the transport entity.

NSAPI (Netscape Server API)

NSF (National Science Foundation)

NSF (National Standard Format)

NSM (network and systems management)

The intersection of networking, network management and systems management. The vision of NSM (also known as "networked systems management") is to enable the management of a distributed set of systems in a fashion similar to that in which many centralized data centers are managed.

NSP (network service provider)

Any provider of network services, including regional Bell operating companies (RBOCs), interexchange carriers (IXCs) and Internet service providers (ISPs).

NSSO (Novell Single Sign-On)

A combination of extensions to Novell Directory Service (NDS) for storage of user sign-on information and software loaded locally on each workstation which enables "single sign-on" (SSO) to supported applications.

NSTL (National Software Testing Lab)

NT (see Windows NT)

NT1

An Integrated Services Digital Network (ISDN) Basic Rate Interface (BRI) line-terminating device at the subscriber's location that provides line maintenance access, timing and echo cancellation. NT1s may be built into other pieces of equipment or may stand alone.

NTFS (NT File System)

NTHSM (NT Hierarchical Storage Management)

NTL (narrow track longitudinal)

NTS (NT Server)

Now known as Windows 2000 Server (see Windows 2000).

NTSC (National Television System Committee)

NT Server

Now known as Windows 2000 Server (see Windows 2000).

NTU (network terminating unit)

Network equipment that connects directly to data terminal equipment.

NTW (NT Workstation)

Now known as Windows 2000 Professional (see Windows 2000).

nuclear magnetic resonance (see NMR)

null modem

A configuration that enables two pieces of data terminal equipment to intercommunicate.

NUMA (nonuniform memory access)

number assignment module (see NAM)

NURBS (Nonuniform Rational B-Spline)

An algorithm used in 3-D modeling software.

NV (see nonvolatile)

NVRAM (nonvolatile RAM)

NW/IP (NetWare Internet Protocol)

NWO (network outsourcing)

O

OAG (Open Applications Group)

An Atlanta-based, nonprofit consortium which develops standards for e-business software interoperability and integration.

OAGIS (Open Applications Group Integration Specification)

An e-business application integration standard from the Open Applications Group.

OASIS (Open Access Same-Time Information System)**OASIS (Organization for the Advancement of Structured Information Standards)**

An international nonprofit consortium that promotes open, collaborative development of e-business specifications based on public standards such as XML and SGML.

OBI (Open Buying on the Internet)

A standard for Internet-based procurement created by a U.S. consortium of vendors formed in October 1996, which released v.1.0 of OBI specification the following year. In April 1998, the consortium placed the management of the specification under CommerceNet.

object

In its simplest terms, it means nothing more than "thing" or "entity." In the world of object-oriented programming and databases, it signifies an instance of a class, such as "IBM" being an instance of the class "hardware vendor." An object is the unit of information interchange in third-generation (3GL) office systems. An object contains both content and semantics describing how the content is to be interpreted or operated on. A network object is any entity in a network (e.g., a node, printer or file server). Software objects may be files or pieces of data.

object-based middleware

Runtime software that enables objects (components) to work cooperatively with a container program or another object, even if the software is distributed across multiple computers.

object-based technologies

Technologies in which objects have encapsulation.

object class

A grouping of objects that can be described in terms of the attributes its members have in common. Generic e-mail, for example, could be an object class, because all e-mail systems have certain things in common.

object code (see object program)**Object Database Management Group (see ODMG)**

object data model

A data model based on object-oriented programming, associating methods (procedures) with objects that can benefit from class hierarchies. Thus, “objects” are levels of abstraction that include attributes and behavior. An object-oriented data model is one that extends the individual program space into the world of persistent object management and shareability.

object encapsulation

Hiding complexity. Data and procedures may be encapsulated to produce a single object.

Object File System (see OFS)

object inheritance

Inheritance defines a relationship among classes in which a subclass shares, overrides or supplements operations or data values from one or more superclasses. A subclass is a specialization of one or more superclasses.

object instance

A specific occurrence of an object. For example, a specific mail message document is an instance.

object late binding

Runtime interpretation of messages. Using late binding, objects are integrated at runtime, as opposed to compiling an integrated object. This greatly enhances flexibility.

object management

Middleware that manages the naming, location and invocation of objects in a system.

Object Management Architecture (see OMA)

Object Management Group (see OMG)

object message passing

In object-oriented systems, one object never operates on another. Instead, one object may pass a message to another object requesting invocation of a method — e.g., “print yourself,” “display yourself” or “file yourself.”

object method

Methods define what can be done with an object. Methods for an electronic-mail document may be display, send, file or print.

Object Method and Message Manager (see OM3)

object orientation, object-oriented (see OO)

object-oriented database management system (see OODBMS)

object-oriented programming (see OOP)

object-oriented system

In an object-oriented system, all data is represented as discrete objects with which the user and other objects may interact. Each object contains data as well as information about the executable file needed to interpret that data. An object-oriented system allows the user to focus completely on tasks rather than tools. Examples of object-oriented programming languages include C++ and Smalltalk.

object-oriented technology (see OOT)

object program

A fully compiled or assembled program that is ready to be loaded into the computer. It is the result of processing a source program through an assembler or compiler. Also called object code.

object-relational mapping tools (see OR mapping tools)

object request broker (see ORB)

object role modeling (see ORM)

object transaction monitor (see OTM)

OC-192 (Optical Carrier Rate 192)

A Synchronous Optical Network (SONET) specification for a 10-Gbps fiber-optic transmission (see OC-N).

OCC (opportunity cost of capital)

OCF (Open Connection Feature)

OCI (Oracle Call-Level Interface)

OC-N (Optical Carrier at Rate N)

The range of incremental rates defined for Synchronous Optical Network (SONET)-based fiber-optic transmission. A SONET/SDH term identifying the place in the hierarchy or speed of such a channel (e.g., OC-3 is 155 Mbps).

OCR (optical character recognition)

The ability of a computer to recognize written characters through some optical-sensing device and pattern recognition software.

OCS (open control systems)

1. A contemporary form of distributed control system (DCS) or programmable logic controller (PLC) based on a set of commercially available, standards-based technologies that permits the open exchange of process data with plant systems and business systems throughout the manufacturing enterprise.
2. Systems that incorporate technologies generally perceived as capable of integration among themselves and with other applications, without the use of proprietary technology. OCS refers to process control for discrete, batch and continuous-process industries as implemented on platforms that range from PLCs and DCSs to soft control implemented on general-purpose computing platforms. OCS also includes computer numerical control (CNC) and other motion controls.

OCSP (Online Certificate Status Protocol)

A method of certificate revocation.

OCT (object code translation)

OCX (OLE Custom Controls)

Visual Basic add-on components based on Microsoft's Object Linking and Embedding (OLE) set of interfaces.

ODA (Open Document Architecture)

An international standard that enables users to exchange texts and graphics generated on different types of office products.

ODBC (Open Database Connectivity)

A vendor-neutral interface based on the Structured Query Language (SQL) Access Group specifications. Announced by Microsoft in December 1991, ODBC accesses data in a heterogeneous environment of relational and nonrelational databases.

ODBO (OLE DB for OLAP)

A Microsoft application programming interface (API) specification for SAP.

ODD (optical disk drive)

ODETTE (Organization for Data Exchange Through Teletransmission in Europe)

An electronic data interchange (EDI) project for the European automotive industry.

ODIF (Open Document Interchange Format)

Defines the Open Document Architecture (ODA) format in terms of interchanged data elements.

ODM (Open Document Management)

ODMA (Open Document Management API)

Provides standard interface between document management systems and end-user applications. ODMA is becoming a desktop application integration de facto standard.

ODMG (Object Database Management Group)

A group of object-oriented database management system (OODBMS) vendors assisting the Object Management Group (OMG) in defining object standards.

ODS (Online Dynamic Server)

ODS (operational data store)

A new articulation of the perennial concept of shared production data. Different from a data warehouse, the ODS is an alternative to having operational decision support system (DSS) applications access data directly from the database that supports transaction processing (TP).

While both require a significant amount of planning, the ODS tends to focus on the operational requirements of a particular business process (for example, customer service), and on the need to allow updates and propagate those updates back to the source operational system from which the data elements were obtained. The data

warehouse, on the other hand, provides an architecture for decision makers to access data to perform strategic analysis, which often involves historical and cross-functional data and the need to support many applications.

ODSI (Open Directory Services Interface)

A standard in the Microsoft Windows operating system that provides a generic application interface to any of a number of database systems that have a service provider interface (SPI) available in the architecture. Additional SPIs may be provided by the database vendor or a third party.

OECD (Organisation for Economic Co-operation and Development)

OEM (original equipment manufacturer)

Refers to the manufacturer of a device that another vendor resells as part of a system.

OFDM (orthogonal frequency division multiplexing)

A method of modulating broadcast data signals to minimize interference among channels that are near each other in frequency. OFDM is a type of frequency division multiplexing (see FDM) often used in wireless networking.

office information system (see OIS)

OfficeVision

An IBM Systems Application Architecture (SAA) office program, introduced in 1989.

offline

1. Equipment or devices not under the direct control of the central processing unit (CPU).
2. Terminal equipment not connected to a transmission line.

offshore programming

The contracting by a company for software services to be carried out in a country other than its own — e.g., a user located in North America may choose to have applications maintenance work carried out in India.

off-the-shelf

Equipment already manufactured and available for delivery from stock.

OFS (Object File System)

A file system conceived by Microsoft as a base for the directory in Windows NT, in which the name spaces of its file system and its Exchange electronic mail (e-mail) system would be unified.

OFX (Open Financial Exchange)

A specification that creates the presentment of a bill. It defines two fundamental capabilities for billing: communication of sign-up information from consolidator to biller, and communication of bill summary information from biller to consolidator. OFX does not define standards for the look of a bill. It merely provides a control channel for information about the bills.

OIS (office information system)

The architectural reference frame for the construction of an office solution that meets the needs that originate at all levels (individual, departmental and corporate), integrating them in a functional sense.

OLAP (online analytical processing)

A definition of multidimensional business intelligence (BI) servers, OLAP originated with a Code and Date white paper that defined 12 "OLAP product evaluation rules" as the basis for selecting multidimensional products. OLAP is, in truth, only a new name for a class of BI products, some of which have existed for decades. The 12 evaluation rules for providing OLAP to user-analysts are:

1. Multidimensional conceptual view
2. Transparency
3. Accessibility
4. Consistent reporting performance
5. Client/server architecture
6. Generic dimensionality
7. Dynamic sparse matrix handling
8. Multiuser support
9. Unrestricted cross-dimensional operations
10. Intuitive data manipulation
11. Flexible reporting
12. Unlimited dimensions and levels

OLCP (online complex processing)

An extension of online transaction processing (OLTP) to include concurrent ad hoc query and batch processing.

OLE (Object Linking and Embedding)

A Microsoft protocol that enables creation of compound documents with embedded links to applications, so that a user does not have to switch from one application to another to make revisions. With OLE:

- Users can create compound documents using multiple applications.
- Compound documents may contain text and spreadsheet objects, graphic and chart objects, sound objects, and video and animated objects.
- Objects that support OLE automation may be scripted by OLE controllers, such as Visual Basic, and used in end-user-developed applications.

OLE Custom Controls (see OCX)

OLE DB for OLAP (see ODBO)

OLE DB (Object Linking and Embedding for Databases)

OLE DS (OLE Directory Services)

OLE for Process Control (see OPC)

OLEPOS

An international open standard for in-store technologies. Introduced in 1994 by a Microsoft-led group of vendors to standardize implementations on a Windows platform, OLEPOS was designed to give retailers flexibility in choosing hardware peripherals for the Windows operating system. While a step in the right direction, it was limited to a Windows environment and resulted in a competing initiative for the Java platform, JavaPOS, led by Sun Microsystems. The dueling standards resulted in a technical committee to develop a unified standard that would incorporate the efforts of JavaPOS and OLEPOS.

OLL (online learning)

OLS (online service)

OLTP (online transaction processing)

A mode of processing that is characterized by short transactions recording business events and that normally requires high availability and consistent, short response times. This category of applications requires that a request for service be answered within a predictable period that approaches "real time." Unlike traditional mainframe data processing, in which data is processed only at specific times, transaction processing puts terminals online, where they can update the database instantly to reflect changes as they occur. In other words, the data processing models the actual business in real time, and a transaction transforms this model from one business state to another. Tasks such as making reservations, scheduling and inventory control are especially complex; all the information must be current.

OLTP monitor

Midrange system software designed to enhance the efficiency of online transaction-processing applications by providing screen mapping, transaction queuing, service prioritization, communications buffering and security.

OLTP screen-formatting front end

A cooperative processing facility that offloads screen formatting from a midrange system-resident online transaction processing (OLTP) system to a PC. The application programmer should be able to define a single set of screen-formatting maps without regard to whether the formatting is performed at the midrange system or at the PC.

OM3 (Object Method and Message Manager)

HIE's integration broker (IB) with a distributed component architecture, notable within the healthcare application integration space for its bus topology, publish-and-subscribe logical architecture and support for normalization.

OMA (Object Management Architecture)

An Object Management Group architecture that defines and organizes the types of objects that provide standard services for distributed, object-oriented applications. In essence, the OMA provides a conceptual description of what distributed objects are and which infrastructure facilities are needed to support them.

OMAP (Operations and Maintenance Application Part)

An Integrated Services Digital Network (ISDN) feature that provides network management functions.

OMF (Weather Observation Markup Format)

OMG (Object Management Group)

A group, primarily of vendors, set up in 1990 to foster common definitions, understanding and standards for object-oriented computing. It has more than 100 members.

OMNIPoint (Open Management Interoperability Point)

A set of specifications, processes, and technology defined by the Network Management Forum (now the TeleManagement Forum) within its Open Management Roadmap.

OMR (optical mark recognition)

OMS (opportunity management system)

A system tied closely to the sales process; it is the framework for any sales force automation (SFA) design. All other applications are subordinate to the OMS. Transactions flow from the OMS to other applications on the users' portable computers. Applications can be integrated among vendors.

OMS (order management system)

Provides the order-taking capability within the supply chain environment. The OMS typically provides the orders to a warehouse management system for execution — i.e., picking, packing and shipping.

OMT (Object Modeling Technique)

ONA (Open Network Architecture)

ONC (Open Network Computing)

ONE (Open Network Environment)

One-Time Charge (see OTC)

online analytical processing (see OLAP)

Online Certificate Status Protocol (see OCSP)

online complex processing (see OLCP)

online documentation

A consistent, system-supplied mechanism for online documentation of system and application programs, including provision of such documentation of vendor-provided software.

online privacy-seal program (see OPSP)

online services

These provide access to exclusive data, information and communications resources. Customers of these services often must use proprietary software to access these resources. Most online services provide access to the Internet in addition to their exclusive resources. America Online, Microsoft Network and Lexis-Nexis are examples of online services.

Online Service System (see OSS)

online transaction processing (see OLTP)

online transaction-processing monitor (see OLTP monitor)

OO (object-oriented or object orientation)

An umbrella concept used to describe a suite of technologies that enable software to be highly modular and reusable; applications, data, networks and computing systems are treated as objects that can be mixed and matched flexibly rather than as components of a system with built-in relationships. As a result, an application need not be tied to a specific system or data to a specific application. The four central object-oriented concepts are encapsulation, message passing, inheritance and late binding.

OO4GL (object-oriented fourth-generation language)

OOA (object-oriented analysis)

OOA&D (OO analysis and design)

OOD (object-oriented design)

OODBMS (object-oriented database management system)

A database management system (DBMS) that applies concepts of object-oriented programming, and applies them to the management of persistent objects on behalf of multiple users, with capabilities for security, integrity, recovery and contention management. An OODBMS is based on the principles of "objects," namely abstract data types, classes, inheritance mechanisms, polymorphism, dynamic binding and message passing.

OOE (out-of-order execution)

This design preserves the order of instructions going in and out of the execution unit but allows up to three instructions to be processed at the same time (sometimes out of order if there are no dependencies), speeding up total execution. This works best for single-cycle instructions rather than multicycle instructions.

OOP (object-oriented programming)

A style of programming characterized by the identification of classes of objects closely linked with the methods (functions) with which they are associated. It also includes ideas of inheritance of attributes and methods. It is a technique based on a mathematical discipline, called "abstract data types," for storing data with the procedures needed to process that data. OOP offers the potential to evolve programming to a higher level of abstraction.

OOT (object-oriented technology)

A software design model in which objects contain both data and the instructions that work on the data. It is increasingly deployed in distributed computing.

OPAC (online public access catalog)

OPC (OLE for Process Control)

A specialized implementation of Microsoft OLE objects related to manufacturing, the specification of which is a function of the OPC Foundation.

Open Applications Group (see OAG)

Open Applications Group Integration Specification (see OAGIS)

open architecture

A technology infrastructure with specifications that are public as opposed to proprietary. This includes officially approved standards as well as privately designed architectures, the specifications of which are made public by their designers. The IBM PC, for example, was based on an open architecture, and spawned an entire industry of IBM clones.

Open Buying on the Internet (see OBI)

Open Client/Open Server

A product (and set of interfaces) created by Sybase to provide gateways to foreign information environments.

open control systems (see OCS)

Open Database Connectivity (see ODBC)

Open Directory Services Interface (see ODSI)

Open Document Architecture (see ODA)

Open Document Interchange Format (see ODIF)

Open Document Management API (see ODMA)

Open Financial Exchange (see OFX)

Open Group

Formed by the merger of the Open Software Foundation and X/Open in 1996, the Open Group's mission is to deliver assurance of conformance to open systems standards through the testing and certification of suppliers' products.

Open Management Interoperability Point (see OMNIPoint)

Open Server Offering (see OSO)

Open Shortest Path First (see OSPF)

Open Software Description (see OSD)

Open Software Fabric Initiative (see OSFI)

Open Software Foundation (see OSF)

open source

Describes software that comes with permission to use, copy and distribute, either as is or with modifications, and that may be offered either free or with a charge. The source code must be made available.

open-source software (see OSS)

open system

A system whose interfaces — for example, application programming interfaces (APIs) or protocols — conform to formal, multilateral, generally available industry standards. “Formal” implies that the standard is selected and maintained using a structured, public process (i.e., de facto standards, such as those developed by the Open Software Foundation). “Multilateral” implies that, while nothing is ever completely vendor-neutral, the standard is not controlled by a single vendor. “Generally available” implies that the specifications are fully published (preferably with source code of a reference implementation), and that anyone can readily obtain license rights for free or at low cost.

Open Systems Interconnection (see OSI)

OpenView

A network management system from Hewlett-Packard.

operating lease

A lease agreement that does not meet any of the criteria of a capital lease as defined by FASB 13 (Financial Accounting Standards Board Statement 13).

Operating Resource Management System (see ORMS)

Operating Resource Management Exchange (see ORMX)

operating system (see OS)

Operating System/2 (see OS/2)

Operating System/400 (see OS/400)

operational data store (see ODS)

operational resource management

A method for acquiring a better view into the cost of goods and services to yield enterprisewide financial control that streamlines the maintenance, repair and operations (MRO) procurement process (indirect goods) and supply chain control.

Operational Support Agreement (see OSA)

Operations and Maintenance Application Part — ISDN (see OMAP)

operations support system (see OSS)

opportunity management system (see OMS)

OPS (Oracle Parallel Server)

OPSP (online privacy seal program)

A not-for-profit entity that identify Internet companies utilizing appropriate processes and operations to protect the confidentiality of consumer information. By displaying an OPSP emblem on their Web sites, Internet companies indicate to consumers that they will follow the tenets of the OPSP by protecting any consumer information they capture. OPSP-compliant companies must also display prominent and easily understood privacy statements and operate in a manner that ensures fair and secure information practices. OPSPs generally mandate the following basic, though evolving, policies, which represent fair information practices as defined by the U.S. Department of Commerce, the Federal Trade Commission and the Online Privacy Alliance:

- Tell consumers what personal information will be collected.
- Inform consumers about how the collected information will be used.
- Tell consumers who the information will be shared with.
- Provide consumer-friendly dispute settlement.
- Communicate the process for correcting and updating personal information.
- Offer specific consequences for noncompliance, such as seal withdrawal, publicity of noncompliance and referral to government enforcement agencies. Existing OPSPs include TRUSTe (Trusted Universal Standards in Electronic Transactions), the Better Business Bureau's BBBOnline program and the American Institute of Certified Public Accountants' WebTrust.

Optical Carrier Rate 192 (see OC-192)

Optical Carrier at Rate N (see OC-N)

optical character recognition (see OCR)

optical disc

A disk read or written by light, generally laser light; such a disk may store video, audio or digital data.

optimization routines

Routines used to determine the optimal solution for a particular problem. These routines are included in supply chain execution and supply chain planning applications to reduce costs or time in the supply chain, and usually are tactically focused for use in current operations.

option analysis

A statistical technique based on the idea that changes over time will affect the cost and benefit of a future IT investment. If it has no option to change the decision, the IT organization must abide by the original decision. However, by creating an option for a decision in the future, the IT organization can choose to pursue IT investments that are then deemed profitable.

OR mapping tools (object-relational mapping tools)

Tools that attempt to automate the mapping from object-oriented (OO) programming structures to relational storage structures. They are similar in intent to application integration middleware, which is designed to enable the interoperation of heterogeneous systems. However, they are different in the following important ways:

- The mapping layer is not runtime software that can be adjusted after deployment. OR mapping is done at development time, producing a compiled translation between objects and relational database management systems (RDBMSs).
- OR mapping tools map between data models, application integration middleware maps, message protocols and occasionally message contents. In this way, OR mapping tools seek to integrate at a deeper level than most other forms of middleware.

ORB (object request broker)

Middleware that allows objects to communicate with other software.

ORDBMS (object-relational database management system)

order management system (see OMS)

Organization for Data Exchange Through Teletransmission in Europe (see ODETTE)

Organization for the Advancement of Structured Information Systems (see OASIS)

original equipment manufacturer (see OEM)

ORM (object role modeling)

A proprietary method that uses English to convey data design elements. The object is to enable developers to easily present the elements to end users who are unfamiliar with modeling notation.

ORM (operational resource management)

A method of managing of an organization's operating resources.

ORMS (Operating Resource Management System)

An Ariba e-procurement application system that provides functionality for centralizing procurement of operating resources — products and services that keep enterprises running on a day-to-day basis.

ORMX (Operating Resource Management Exchange)

A version of Ariba's ORMS offered to application service providers (ASPs), enabling them to host automated procurement systems for customers that wish to establish an ASP-based e-marketplace.

orthogonal frequency division multiplexing (see OFDM)

OS (operating system)

The operating system is the main control program that runs a computer and sets the standard for running application programs. It is the first program loaded when the computer is turned on, and it resides in memory at all times. An operating system is responsible for functions such as memory allocation, managing programs and errors, and directing input and output. It may be developed by the computer vendor or by a third-party independent software vendor.

OS/2 (Operating System/2)

An IBM operating system for personal computers that featured capabilities for large memory, multitasking and virtual machines.

OS/400 (Operating System/400)

An IBM operating system for midrange computers.

OSA (Open System Adapter)

OSA (Operational Support Agreement)

IBM's Extended Service Agreement (ESA) offering in the United Kingdom.

OSA2 (Open System Adapter 2)

OSD (Open Software Description)

A specification that enables developers to specify attributes of their products, including how various pieces work together and when they were last updated, so that desktops are using current versions of software.

OSF (Open Software Foundation)

A vendor consortium originally formed to develop multivendor standards for open systems. OSF merged with X/Open in 1996 to form the Open Group.

OSF/1

The first Open Software Foundation operating system, based on Carnegie-Mellon's Mach kernel and shipped in 1992.

OSFI (Open Software Fabric Initiative)

An initiative launched in 1999 by five storage switch vendors (Ancor Communications, Brocade Communications Systems, Gadzoox Networks, McData and Vixel) to propose standards for interoperability among competing Fibre Channel switch products, which are used in storage area networks.

OSI (Open Systems Interconnection)

A model developed by the International Organization for Standardization (ISO) for communications. It is similar in structure to Systems Network Architecture (SNA), but more open. A standard, modular approach to network design that divides the required set of complex functions into manageable, self-contained, functional layers. These layers, starting from the innermost, are:

1. Physical layer — concerned with the mechanical and electrical means by which devices are physically connected and data is transmitted.
2. Link layer — concerned with how to move data reliably across the physical data link.

3. Network layer — provides the means to establish, maintain and terminate connections between systems. Concerned with switching and routing information.
4. Transport layer — concerned with end-to-end data integrity and quality of service.
5. Session layer — standardizes the task of setting up a session and terminating it. Coordinates the interaction between end application processes.
6. Presentation layer — relates to the character set and data code that is used, and to the way data is displayed on a screen or printer.
7. Application layer — concerned with the higher-level functions that support application or system activities.

OSI management

The facilities to control, coordinate and monitor the resources that enable communications in an Open Systems Interconnection (OSI) environment.

OSIS (operating-system-independent server)

OSI TP (OSI Transaction Processing)

OSM (operating-system service module)

OSO (Open Server Offering)

A process and marketing program from IBM, similar to its long-established Entry Server Offering (ESO), but with a focus on the network-computing and operating-systems environment.

OSPF (Open Shortest Path First)

An Interior Gateway Protocol (IGP) that transfers packets from one network to an adjacent one.

OSS (Online Service System)

An online system that offers service information, error notes and downloadable patches for R/3 and other SAP products.

OSS (open-source software)

"OSS" and "free software" are terms applied to software that can be run, distributed and modified without a fee. Restrictions are often applied — in the form of an "open-source license" — to profits or commercial fees for any product built on a base of free software (e.g., the GNU General Public License).

OSS (operations support system)

A system that supports public-network-infrastructure operations, administration and maintenance, as well as network management functions (such as fault, configuration and security management). OSSs usually comprise software systems — residing on general-purpose or specialized computer systems — that support one or more business processes associated with network operations. Examples of the business processes addressed include planning and engineering, provisioning, performance management, and message collection and mediation.

OST (offline store)

OT (object technology)

OTC (One-Time Charge)

An IBM single charge for purchasing or upgrading a product, as opposed to ongoing fees.

OTM (object transaction monitor)

An application program — similar to the mainframe Customer Information Control System (CICS) in function, but not in spirit — that represents a consistent model of an application for a modular and potentially highly distributed environment.

OTP (Open Trading Protocol)

OTS (Object Transaction Service)

out-of-order execution (see OOE)

outsourcing

A contractual relationship with an outside vendor that is usually characterized by the transfer of assets, e.g., facilities, staff or hardware. It can include facilities management (for data centers or networks), application development and maintenance functions, and end-user computing services.

End-user-computing outsourcing includes all services to support end users from the workstation up to the front-end router. The scope ranges from the end user's workstation to the shared LAN and local network infrastructure of an enterprise's distributed computing architecture. It includes desktop services (implementation, operations and management of desktop hardware, software and peripherals), network systems management (implementation and management of LANs, network-enabling hardware and software) and help desks.

overflow

1. In telephony, excess traffic, on a particular route, that is offered to another (alternate) route.
2. In computing, a condition where data is presented to a buffer or port faster than it can be processed, resulting in lost data.
3. Also, in computing, a condition where a computational result is greater than the capacity of its location, resulting in an error condition.

overlapping VLAN

A virtual local-area network (VLAN) type where one user may belong to more than one VLAN at a time.

override

Seizure of a circuit that is already occupied.

overspeed

The condition in which the transmitting device runs faster than the data can be presented for transmission.

P

P2P (peer-to-peer)

A style of networking in which computers communicate directly with each other rather than depending on interactions managed via central servers and networks. A new generation of highly scalable P2P applications has emerged to handle a variety of spontaneous communications, including:

- Short real-time messages (for example, AOL Instant Messenger and SMS messages on cell phones)
- Collaborative computing
- File-sharing programs, which enable Internet users to share (such as music files) via point-to-point file transfers

P3P (Platform for Privacy Preferences)

A standard for privacy and content personalization proposed by the World Wide Web Consortium (W3C). P3P creates a language in which Web sites and Web users can communicate, and which shields reader preferences and interests.

P7

X.400 protocol.

P42

The protocol created by the National Security Agency (NSA) by which the Defense Message System (DMS) Message Security Protocol (MSP) is described under the X.400 standards.

PA (Precision Architecture)

PA (project accounting)

A system that consolidates information from other systems to account for overall costs and expenditures involved with a specific project.

P/A (physical asymmetric)

A method of system partitioning that allows different-size increments of system resources to be divided physically into two or more partitions of different sizes. Some shared resources may be available.

PABX (private automatic branch exchange)

A telephone switch located at the site of the enterprise.

packet

An information block identified by a label at Layer 3 of the International Organization for Standardization (ISO) reference model for Open Systems Interconnection (OSI). It is a collection of bits that contains both control information and data, and is the basic unit of transmission in a packet-switched network. Control information is

carried in the packet, along with the data, to provide for such functions as addressing sequence, flow control and error control at each of several protocol levels. A packet can be of fixed or variable length, but generally has a specified maximum length.

packet assembler/disassembler (see PAD)

packet format

The exact order and size of a packet's various control and information fields, including header, address and data fields.

Packet Level Protocol (see PLP)

packet overhead

A measure of the ratio of the total packet bits occupied by control information to the number of bits of data, usually expressed as a percentage.

packet switching

A technique in which a message is broken into smaller units called packets, which may be individually addressed and routed through the network, possibly using several different routes. The receiving-end node ascertains that all packets are received and in the proper sequence before forwarding the complete message to the addressee.

Packet Transfer Mode (see PTM)

PACS (picture archiving and communication system)

The hardware and software required for the storage and management of digital images, as well as the management of workflow within the radiology department. It has added functionality, compared to teleradiology, as it can provide historical context for current radiographs (old studies) and is an enabling technology that allows for the creation of the digital department.

PAD (packet assembler/disassembler)

An interface device that buffers data sent to and from character mode devices, and assembles and disassembles the packets needed for X.25 operation.

page description language (see PDL)

page, nonimpact, plain paper (see PNPP)

PAL (Phase Alternate Line)

A color television broadcasting system developed in West Germany and the United Kingdom that uses 650 picture lines and a 50-hertz (Hz) field frequency. See NTSC (National Television System Committee) and SECAM (Sequential Couleur a Memoire).

PAL (programmable array logic)

Palm

This brand name refers to Palm Inc.'s line of PDA devices, as well as its operating-system platform, the Palm OS. Palm once dominated the PDA market with an

estimated 80 percent market share, but competitive platforms such as Symbian and Windows CE emerged in 2000. (See PDA, Symbian and Windows CE.)

PAM (pulse amplitude modulation)

The encoding of information in a signal based on the fluctuation of carrier waves. The amplitude of the pulse carrier is varied in accordance with successive samples of the modulating signal.

PAMS (parallel application management system)

PAN (personal-area network)

A network in which personal devices or applications dynamically locate and interact with one another. Bluetooth and Jini are examples of emerging PAN technologies. Gartner believes the lifestyle impact of PANs — together with other emerging technologies, such as flexible displays and speech recognition — will be enormous within the next 10 years, leading to a fundamental rethinking of how personal communications and information technology are used in people's everyday lives.

PAP (Password Authentication Protocol)

A security protocol without encryption that uses password authentication to allow access to a network or host. See Challenge Handshake Authentication Protocol.

Parallel Enterprise Server (see PES)

parallel processing

The solution of a single problem across more than one processor. Little parallel processing is done today outside of research laboratories, because it is difficult to decompose tasks into independent parts, and the compiler technology does not yet exist that will extensively parallelize applications code.

Parallel Query Server (see PQS)

Parallel Reduced Instruction Set Multiprocessor (see PRISM)

Parallel Sysplex

The cornerstone of IBM's large-systems operating-system strategy. It is designed to allow multiple central electronics complexes (CECs) to share a common database so a transaction can execute on one of several processors that share a single view of the data. This implementation substantially improves parallelism, and therefore overall system throughput, at a multitasking level.

Parallel Sysplex License Charge (see PSLC)

parallel systems

Hardware configurations having nine or fewer central processing units (CPUs). Compare with highly parallel systems (HPS) and massively parallel systems.

Parallel Transaction Server (see PTS)

parallel transmission

Simultaneous transmission of the bits making up a character or byte, either over separate channels or on different carrier frequencies on one channel. See serial transmission.

PARC (Palo Alto Research Center)

parcel manifesting system (see PMS)

PA-RISC (Precision Architecture Reduced Instruction Set Computer)

Hewlett-Packard's RISC-based processor architecture, the basis for its desktop-to-mainframe processor products.

parity bit

A check bit appended to an array of binary digits to make the sum of all the binary digits, including the check bit, always odd or always even.

parity check

The addition of noninformation bits to data to make the number of ones in a grouping of bits either always even or always odd. This procedure enables detection of bit groupings that contain single errors. It can be applied to characters, blocks or any specific bit grouping.

partial document encryption (see PDE)

partitioned data set (see PDS)

partitioning code

Applications can be broken into three logical parts — presentation, logic and data. These are areas in which the program can be separated to facilitate execution of each logical piece on a different machine. Each segment is known as a partition. For example, the thin-client Web model requires that interface presentation be handled by the browser, application logic by the Web server and other application servers, and data by a database server. Developers are responsible for determining where the separation occurs.

partner interface process (see PIP)

partner relationship management (see PRM)

PAS (productivity application service)

Passive Optical Network (see PON)

password

A code used to restrict access to database and other sources of computerized information. The use of passwords is a commonly used security measure, but it has limitations, because a computer can only verify the accuracy of the password, not the authenticity of the person using the password.

With the explosion of the Internet and extranets, passwords are becoming the security measure of choice to guard against unauthorized access to protected areas. Unfortunately, the same reasons that make passwords a popular security measure — easy distribution and administration — also make them undependable. Other procedures, such as certification processes used in combination with passwords, provide better security for critical interactions.

Password Authentication Protocol (see PAP)

patch

1. A temporary electrical connection.
2. To make an improvised modification; to change a software routine in a rough or expedient way.

patient care management (see PCM)

patient financial system (see PFS)

path overhead (see POH)

PATX (private automatic telex exchange)

PAX (private automatic exchange)

payload pointer

Indicates the beginning of a synchronous payload envelope (a SONET term).

PBA (production business application)

PBM (pharmacy benefit management)

An entity that manages the delivery of pharmacy and drug-related benefits for members of a managed care or payer plans.

PBN (policy-based networking)

The application of business rules to govern network performance.

PBX (private branch exchange)

A telephone switch located on a customer's premises that primarily establishes voice-grade circuits (over tie lines to a telephone company central office) between individual users and the public switched telephone network. The PBX also provides switching within the customer premises local area, and usually offers numerous enhanced features, including least-cost routing and call detail recording.

PC (personal computer)

A microcomputer designed primarily for individual use. Sharing resources with another computer is optional. The first PCs were introduced in the late 1970s.

PCA (planning, control and administration)

PCB (printed circuit board)

A flat board made of fiberglass or plastic, on which chips and electronic components are interconnected via copper pathways. The primary PCB in a system is called a system board or motherboard, while smaller ones that plug into the slots in the main board are called riser boards or cards.

PC Card

A standard of the Personal Computer Memory Card International Association (see PCMCIA).

PC-DOS

The original operating system that IBM supplied with its personal computers, licensed from Microsoft.

PCI (Peripheral Component Interconnect)

The industry standard system bus architecture that is used in most desktop and server models. PCI provides faster communications between the processor and components (such as memory, disk and video) than the original IBM bus, now called Industry Standard Architecture (ISA).

PCL (Printer Control Language)

The set of commands used by Hewlett-Packard and compatible printers that governs how a document will be printed. Similar to, but less powerful and complex than, PostScript.

PCM (patient care management)

A system that enrolls or assigns patients to interventions across the continuum of health and illness. It includes wellness exams and routine screenings, utilization reviews, event focus, short-term case management, and the management of long-term chronic conditions.

PCM (plug-compatible manufacturer)

A hardware vendor whose products are direct replacements for the products of a computer system manufacturer. Initially PCMs provided peripherals that were “plug-for-plug” compatible with IBM peripherals, but the PCM industry has since grown to include IBM-compatible mainframes and peripherals for other vendors’ systems.

PCM (pulse code modulation)

A digital technique that involves sampling an analog signal at regular intervals and coding the measured amplitude into a series of binary values, which are transmitted by modulation of a pulsed, or intermittent, carrier. It is the standard technique in telecommunications transmission.

PCMCIA (Personal Computer Memory Card International Association)

A nonprofit trade association founded in 1989 to standardize the PC card. The PCMCIA standard has since been renamed the PC Card standard. The standard cards are removable modules that can hold memory, fax/modems, radio transceivers, network adapters, solid-state disks or hard disks in personal digital assistants (PDAs) and other lightweight or portable systems.

PCN (personal communications network)

A defunct term that was originally used for Global System for Mobile Communications (GSM) 1800 networks in Europe.

PCO (physician contracting organization)

A legal entity representing multiple physicians, practices and clinics that contracts with other entities to provide healthcare services.

P-code

An intermediate representation of a program that is file format independent and is executed by a virtual machine interpreter.

PCR (peak cell rate)

An asynchronous transfer mode (ATM) parameter used to determine the maximum number of cells permitted in a specified time period.

PCR (preventive cyclic retransmission)

The cyclic retransmission of all unacknowledged frames on an idle link. The term pertains to satellite and high-propagation delay links.

PCS (packet-controlled switching)

PCS (personal communications services)

A lower-powered, higher-frequency digital cellular technology. It operates in the 1.5MHz to 1.8MHz range. In the United States, PCS also operates at 1.9GHz.

PCS (Personal Conferencing Specification)

PC/SC (Personal Computer/Smart Card)

Microsoft's specification for using smart cards with PCs. To help forge standards and develop the integration of smart cards and PCs, Microsoft founded the PC/SC Workgroup with other smart card and technology industry companies, including Schlumberger, Hewlett-Packard and Groupe Bull.

PCTE (Portable Common Tool Environment)

PCX (private communications exchange)

A high-capacity evolution of the digital private branch exchange (PBX), which incorporates a cell-based switching fabric or asynchronous (ATM) interfaces, an open client/server architecture and a high dependence on computer telephony integration (CTI) to deliver advanced functionality while retaining enterprise investment in PBX line cards and terminal equipment.

PD (predictive dialer)

PDA (personal digital assistant)

A handheld computer that serves as an organizer, electronic book or note taker. It typically uses a stylus or pen-shaped device for data entry and navigation. Types of PDA include clamshell (a computer system that weighs less than 3 pounds and opens lengthwise to expose a keyboard and screen), and tablet (a computer system that weighs less than 4 pounds and that is operated by direct screen contact via a pen or touch interface).

PDC (Personal Data Cartridge)

PDC (personal digital cellular)

A Japanese standard for mobile telephony in the 800MHz to 1,500MHz spectrum.

PDC (Primary Domain Controller)

PDE (partial document encryption)

Encryption and delivery of only those pages requested or those that the requester is allowed to see. A certificate server only releases the required certificate when an authorized user makes a request. Using PDE, each page of the document is

encrypted and associated with a unique security certificate, allowing for selective retrieval.

PDES (Product Data Exchange Specification)

A standard format for the exchange of data between advanced computer-aided design (CAD) and computer-aided manufacturing (CAM) programs. It describes a complete product, including the geometric aspects of the images, manufacturing features, tolerance specifications, material properties and finish specifications.

PDF (Portable Document Format)

Adobe's PDF is based on its PostScript page description language, with some major changes. PostScript allows operators to describe a page layout independent of computer platform, device and resolution. PDF has substantially all of PostScript's cross-platform graphic capabilities, but treats each element in a job as a separate object. Modification and proofing of print publication data can be done more easily with PDF than with PostScript. Raster image processor outputs can easily be generated one page at a time via PDF conversions to PostScript. In addition, PDF is emerging as a de facto industry standard for a wide range of context-sensitive document management and electronic-form applications. PDF files must be viewed through Adobe Acrobat installed on a client device.

PDH (Plesiochronous Digital Hierarchy)

The initial structure for organizing the hierarchy or possible bit rates in synchronous time-division multiplexing. It is now superseded by Synchronous Digital Hierarchy (SDH). PDH is defined by the International Telecommunication Union (ITU) in the G.703 standard.

PDL (page description language)

The programming language used to specify the way a document will be printed or displayed.

PDM (product data management)

PDM technologies and products have historically been positioned as the primary application backbone for managing and controlling the flow of design intent across the three major design stages: concept design, detail design and production. But in practice, PDM has served as a complementary application tower to computer-aided design (CAD) and enterprise resource planning (ERP) systems by providing the main repository for production-approved engineering data (i.e., vaulting) and managing the changes to production - approved data (e.g., engineering change orders and configuration management). PDM is a key enabler for constructing a concurrent art-to-product environment (CAPE) for enterprisewide design and production. A well-constructed PDM enables all participants involved with the capture, communication and maintenance of design intent to freely share and disseminate all heterogeneous data related to the product. PDM evolved from the need to better manage paper, electronic documents, engineering change orders, and bills of materials during the product development process.

PDM (pulse duration modulation)

See PWM.

PDM II (product development management)

A new strategy for managing the interenterprise product development process from concept through production. The goal of a PDM II strategy is to provide the proper

combination of people, process and technologies to support the management of the entire interenterprise product development commercialization process to enhance competitiveness. The PDM II vision provides a framework for rationalizing the roles of product data management (PDM), virtual PDM (VPDM) and enterprise resource planning (ERP) technologies.

PDN (public data network)

A shared-use X.25 packet-switched network operated by a carrier. PDNs may offer value-added services at a reduced cost because of communications resource sharing and usually provide increased reliability due to built-in redundancy.

PDO (Portable Distributed Objects)

PDQ (Parallel Data Query)

PDS (partitioned data set)

A data set that is divided into physically distinct units, or members, which contain data or partial or complete programs.

PDU (Protocol Data Unit)

An Open Systems Interconnection (OSI) term for the data to be transmitted with the headers and trailers attached by each layer.

PDX (private digital exchange)

PDXI (Process Data Exchange Institute)

The organization standardizing the representation of process data for exchange among companies involved in the design and construction of process manufacturing facilities.

PE (phase encoding)

PEA (purchased enterprise applications)

Large, enterprisewide applications software purchased as a standardized packaged offering and typically customized by third-party or in-house developers. Examples include PeopleSoft and SAP R/3.

peak cell rate (see PCR)

peering agreement

A reciprocal arrangement between Internet providers that allows subscribers to access newsgroups and File Transfer Protocol (FTP) sites particular to a provider.

peer-to-peer (see P2P)

peer-to-peer remote copy (see PPRC)

PEM (privacy-enhanced mail)

A method of interenterprise e-mail messaging that works with existing e-mail systems to ensure the authenticity, integrity and optional confidentiality, through encryption, of messages.

performance monitoring and tuning tools

Facilities that enable users to monitor midrange-system performance, identify system and application performance bottlenecks, and set optimal performance-tuning parameters.

Performance Optimized With Enhanced RISC (see POWER)

Peripheral Component Interconnect (see PCI)

peripheral device

A piece of hardware that is connected to a computer. Examples are:

- Monitor
- Keyboard
- CD-ROM drive
- Modem
- Floppy disk drive

Peripheral Interchange Program (see PIP)

PERL (Practical Extraction Report Language)

A Unix scripting language for high-level system control, often used to manage Web servers.

permanent virtual circuit (see PVC)

personal-area network (see PAN)

personal communications network (see PCN)

personal communications services (see PCS)

personal computer (see PC)

Personal Computer Memory Card International Association (see PCMCIA)

Personal Computer/Smart Card (see PC/SC)

personal digital assistant (see PDA)

personal digital cellular (see PDC)

Personal HandyPhone System (see PHS)

personal health record (see PHR)

personal identification number (see PIN)

personal information manager (see PIM)

personalization

Use of continually adjusted user profiles to match content and services to individuals. This includes determining users' interests based on their preferences or behavior, constructing business rules to select relevant content based on those preferences or behaviors, and presenting the content to users in an integrated, cohesive format.

PERT (Program Evaluation and Review Technique)

A management tool for graphically displaying projected tasks, milestones, schedules and discrepancies.

PES (Parallel Enterprise Server)

A label used by IBM for its CMOS-based mainframes.

PFM (personal financial manager)

PFS (patient financial system)

Applications that collect service charges for healthcare services rendered to produce a patient bill and medical claim.

PFS (process flow scheduling)

Adds network process schedule planning and optimization functionality to a process-control system.

PGA (pin grid array)

An interconnection system for getting electrical signals into and out of a large integrated circuit (IC) such as a microprocessor.

PGML (Precision Graphics Markup Language)

An application proposed to the World Wide Web Consortium (W3C) to define 2-D graphics in Extensible Markup Language (XML).

PGP (Pretty Good Privacy)

Public-key encryption software sold by Network Associates (NA). PGP began as an open standard for message encryption. Add-ons are available for a number of desktop products. The underlying protocol has been designated a military weapon by the United States and some other countries, in an effort to regulate its distribution, but similar software is widely available on the Internet.

PGS (proposal generation system)

pharmacy benefit management (see PBM)

Phase Alternate Line (see PAL)

phase jitter

In telephony, the measurement, in degrees out of phase, that an analog signal deviates from the referenced phase of the main data-carrying signal. It is often caused by alternating current components in a network.

phase modulation

One of three ways of modifying a sine wave signal to make it carry information. The sine wave or “carrier” has its phase changed in accordance with the information to be transmitted. Also known as Phase-Shift Keying (PSK) or Differential Phase-Shift Keying (DPSK).

phase shift

Errors that occur when the clock and data signals become unsynchronized.

phase-shift keying (see PSK)

PHIGS (Programmers’ Hierarchical Interactive Graphics Standard)

A graphics system and language used to create 2-D and 3-D images that interface between the application program and the device-independent graphics subsystem. Graphics objects are managed in a hierarchical manner so that a complete assembly can be specified with all of its subassemblies. High-performance workstations and host processing are required to handle the standard.

PHO (physician hospital organization)

A legal entity representing joint contractual service arrangements between hospitals or integrated delivery systems and physicians.

PHR (personal health record)

A concept for capturing, managing and sharing a consumer’s healthcare profile. PHRs contain the following components: a consumer profile, a healthcare providers’ section, a healthcare encounter section, a family history section, an emergency profile and public-key infrastructure (PKI) security.

PHS (Personal HandyPhone System)

A Japanese standard for digital cellular service. It provides low-mobility or fixed wireless access in the 1,900-megahertz radio band.

PHY (Physical Layer Protocol)

One of two sublayers of the Fiber Distributed Data Interface (FDDI) physical layer. See PMD (Physical Medium Dependent).

physical asymmetric (see P/A)

physical layer

Within the Open Systems Interconnection (OSI) model, the lowest level of network processing concerned with the electrical, mechanical and handshaking procedures over the interface that connects a device to a transmission medium.

Physical Layer Convergence Procedure (see PLCP)

Physical Layer Protocol (see PHY)

Physical Medium Dependent (see PMD)

Physical Unit (see PU)

physician contracting organization (see PCO)

physician hospital organization (see PHO)

PIC (programmable interrupt controller)

PICS (Platform for Internet Content Selection)

A set of technical specifications from the World Wide Web Consortium. PICS is designed to enable users to find appropriate content easily, and to avoid content they consider undesirable or inappropriate, either for themselves or their children.

picture archiving and communication system (see PACS)

picture element (see pixel)

P&ID (piping and instrumentation drawing)

A style of drawing used in process manufacturing which depicts key process and control elements and is often used as a basis for construction, operation and maintenance of a facility.

PIDS (Patient Identification Services)

PIE (Pocket Internet Explorer)

PIM (personal information manager)

Software that organizes and manages random information for fast retrieval on a daily basis. It provides a combination of features, including telephone list with automatic dialing, calendar and scheduler.

PIMS (production information management system)

Also known as a "process information management system," a PIMS is a client/server application for the acquisition, display, archiving and reporting of information from a wide variety of control, plant and business systems. A critical component in a manufacturing enterprise's application architecture for creating a common repository of plant information that can be effectively leveraged in enterprise and supply chain management applications.

PIN (personal identification number)

A numeric code — typed on an automated teller machine or telephone keypad, or a computer keyboard — used to gain access to personal funds or information.

PIN (Processor Independent NetWare)

ping

A Transmission Control Protocol/Internet Protocol (TCP/IP) procedure that uses the Internet Control Message Protocol (ICMP) echo to check whether a network device is active.

PIP (partner interface process)

A feature of the RosettaNet Implementation Framework (RNIF). PIPs define business processes between trading partners. See RosettaNet and RNIF.

PIP (Peripheral Interchange Program)

A Control Program for Microcomputers (CP/M) utility that allows files to be transferred between devices.

pipng and instrumentation drawing (see P&ID)

PIR (protocol-independent routing)

A method in which routing decisions are made with no reference to the protocol in use.

PIS (product information system)

PIT (programmable interval timer)

PITC (point-in-time copy)

PIX (private Internet exchange)

pixel (picture element)

The smallest resolvable dot in an image display.

PKCS (Public-Key Cryptography Standard)

PKI (public-key infrastructure)

The techniques necessary to manage public-key cryptography (see separate entry), as well as the various systems for authentication, nonrepudiation and integrity that can be built on top of a public-key system.

PKIX (Public-Key Infrastructure Exchange)

An initiative, led by the Internet Engineering Task Force (IETF) Public-Key Infrastructure Exchange (PKIX) subcommittee, addressing interoperability and cross-certification among certificate authorities.

PL/1 (Programming Language 1)

An early scientific programming language, now rarely used.

PLA (programmable logic array)

Plant Maintenance (see PM)

platform

An individual hardware or software architecture or an operating system.

Platform for Internet Content Selection (see PICS)

Platform for Privacy Preferences (see P3P)

platform-independent

Software that can run on a variety of hardware platforms or software architectures. Platform-independent software can be used in many different environments, requiring less planning and translation across an enterprise. For example, the Java programming language was designed to run on multiple types of hardware and multiple operating systems. If Java platform-independence becomes a reality, organizations with multiple types of computers will be able to write a specialized application once and have it be used by virtually everyone, rather than having to write, distribute and maintain multiple versions of the same program.

PLC (programmable logic controller)

The fundamental building block of factory and process automation. A specialty purpose computer, including input/output processing and serial communications, used for executing control programs, especially control logic and complex interlock sequences. PLCs can be embedded in machines or process equipment by OEMs, used stand-alone in local control environments or networked in system configurations.

PLCP (Physical Layer Convergence Procedure)

The specification for mapping asynchronous transfer mode (ATM) cells onto the physical media.

PLD (programmable logic device)

Plesiochronous Digital Hierarchy (see PDH)

Plesiochronous network

A network that makes use of more than one clock source. It consists of different portions of the network where the signal may have the same nominal digit rate but does not stem from the same clock.

PLL (phase-locked loop)

PLM (Product Lifecycle Management)

A product development and management product from SAP.

PLP (Packet Level Protocol)

Network Layer protocol in the X.25 protocol stack.

PL/SQL (Programming Language/Structured Query Language)

One of the two deliverables of Oracle Systems' Transaction Processing Option (TPO). It is a 3GL-level extension of Structured Query Language (SQL) that groups SQL statements for transmission across the tools/database management system (DBMS) interface or across a network. PL/SQL must be used to take advantage of the Oracle version 6 performance enhancements.

plug-compatible manufacturer (see PCM)

plug-in

A program that uses a particular browser's API. Each plug-in is browser- and platform-specific. Plug-ins are stored locally, on the same machine as the browser. The best-known plug-ins are those that allow the display or playback of special file types (such as animation, audio or video) directly in the browser window.

PM (Plant Maintenance)

An SAP product for manufacturing enterprises, based on its R/3 enterprise resource planning system.

PM (practice management)

PM (predictive modeling)

Data mining that uses pattern recognition technologies and statistical and mathematical techniques on large amounts of data to support decision-making.

PM (Presentation Manager)

The graphical user interface offered in IBM's OS/2 operating system. It was the first manifestation of IBM's User Interface Architecture, consistent with the Common User Access (CUA) component of Systems Application Architecture (SAA).

PM (preventive maintenance)

A practice to reduce hardware or software downtime by replacement of components prior to failure via:

- A set of periodic inspections to highlight and repair functions prior to failure.
- Pre-emptive replacement based on mean time between failure (MTBF) statistics.

PM (project management)

PMBX (private manual branch exchange)

PMCC (project management center of competency)

PMD (Physical Medium Dependent)

One of two sublayers of the Fiber Distributed Data Interface (FDDI) physical layer, PMD interfaces directly with the media. See PHY.

PMD (programmable multilevel logic device)

PMH (product marshaling and handling)

An application suite that provides tight integration of warehouse and transportation operations and planning.

PMI (Project Management Institute)

PMLCD (passive-matrix liquid crystal display)

PMMU (paged memory management unit)

PMO (program management office)

PMOS (p-channel metal-oxide semiconductor)

PMP (Project Management Professional)

An information technology (IT) certification by the Project Management Institute (PMI), requiring two years of work experience and passing eight 50-minute tests (on project scope, quality, time, cost, risk, human resources, procurement and communications management).

PMR (private mobile radio)

PMR (projection microradiography)

PMS (parcel manifesting system)

A system that provides the ability to weigh and rate small parcel shipments. The PMS contains the rate tables of the various parcel carriers such as UPS, RPS and Federal Express. The PMS also provides a detailed manifest of all parcel shipments.

PNNI (Private Network-to-Network Interface)

A standard that maps out the network from an asynchronous transfer mode (ATM) standpoint and, through multiple phases of the specification, will become the router-to-router protocol implementation in ATM switches.

PNPP (page, nonimpact, plain paper)

A laser- or LCD-type printer.

POC (Policy Oversight Committee)

Formerly the Internet International Ad Hoc Committee (IAHC), the POC is carrying out and expanding the policies outlined in the Genetic Top-Level Domain-Memorandum of Understanding (gTLD-MoU). The POC is composed of members from Council of Registrars (CORE), Internet Architecture Board (IAB), Internet Assigned Numbers Authority (IANA), Federal Networking Council (FNC), International Trademark Association (INTA), Internet Society (ISOC), International Telecommunication Union (ITU) and World Intellectual Property Organization (WIPO).

Pocket PC

A personal digital assistant (PDA) platform launched by Microsoft and several partner hardware vendors in April 2000. Pocket PCs, which work with Microsoft's Windows CE operating system, represent a major initiative to gain a major share of the PDA market, once dominated by the Palm-compliant devices, which run the Palm OS operating system. See PDA, Windows CE and Palm.

POCSAG

A U.K. standard for radio paging formulated by the Post Office Coding Standards Advisory Group in 1982, now adopted by the Conference on European Posts and Telecommunications (CEPT) as a Pan-European standard.

POD (print on demand)

A technique that allows an enterprise to print a quantity of documents electronically as needed. Typically, the documents are forms, brochures, booklets or similar documents previously printed in large quantities, in monochrome or color, and stored in a warehouse.

POE (physician order entry system)

POE (point of entry)

POEMS (Platinum Open Enterprise Management System)

POH (path overhead)

Used for payload transport functions in SONET. The POH is assigned and transported with a payload until it is demultiplexed.

point of presence (see POP)

point of sale (see POS)

point of service (see POS)

point to point

Describes a circuit that connects two points directly, where there are generally no intermediate processing nodes, although there could be switching facilities. See multipoint.

Point-to-Point Protocol (see PPP)

Point-to-Point Tunneling Protocol (see PPTP)

polarization

A property of an electromagnetic wave characterized by the direction of the electric field.

polar keying

A technique of current loop signaling in which current flow direction establishes the two-level binary code.

policy-based networking (see PBN)

policy-based routing

The forwarding of packets to specified destinations based on user configurations.

Policy Oversight Committee (see POC)

polling

A means of controlling communications lines. When many stations are connected to the same circuit, polling from the center is used to ensure an orderly flow of data to the central location. Polling is an alternative to contention that makes sure no terminal is kept waiting for a long time.

POLS (private online service)

polymorphism

The capability of an operation to accept arguments of different or unknown types. Parametric polymorphism executes the same operation on different types. Overloading polymorphism selects appropriate operations according to the type.

POMS (Process Operations Management System)

Originally an IBM-and customer-supported initiative to link manufacturing resource planning (MRP) applications running on AS/400 minicomputers and shop floor data collection systems running on PS/2 computers. POMS was later reorganized as an independent software company.

PON (Passive Optical Network)

A fiber-based access system to customer premises for the delivery of video services such as video-on-demand that allows multiplexed traffic to be distributed to a number of users in such a fashion as to limit the number of optical terminations required.

POP (point of presence)

1. Since divestiture, the physical access location within a local access and transportation area (LATA) of a long-distance and/or interLATA common carrier.

2. The point to which the local telephone company terminates subscribers' circuits for long-distance, dial-up, or leased-line communications.
3. An Internet provider's node that allows subscribers to dial in using modems and voice lines.

POP (Post Office Protocol)

An application protocol that provides a mailbox-retrieval service for Internet PC users. It is being replaced by Internet Message Access Protocol (IMAP).

port

The entrance or physical access point to a computer, multiplexer, device or network, where signals may be supplied, extracted or observed.

portability

The ability to move software easily to computers from multiple vendors.

Portable Document Format (see PDF)

portal

A high-traffic Web site with a wide range of content, services and vendor links. It acts as a value-added middleman by selecting the content sources and assembling them in a simple-to-navigate and customize interface for presentation to the end user. Portals typically offer such services as Web searching, news, reference tools, access to online shopping venues, and communications capabilities including e-mail and chat rooms.

portal-enabling middleware

The toolset for a portal including the platform middleware and the integration and context management tools.

ported proprietary

Refers to a system that is a compliant implementation of an evolving set of proprietary or vendor-neutral specifications (or both) for interfaces, services, protocols and formats. These specifications let a user configure and operate a system, its applications and its components only with other equally compliant implementations available from the same vendor or with licensed implementations from different vendors.

portfolio management

A shift from the practice of using a single integrated application for the support of business requirements to using a collection of applications, technologies and services to create a system that addresses the unique requirements of an organization and leverages best-of-breed opportunities.

portfolio of technology solutions (see POTS)

porting

Modifying code that runs on one hardware platform or operating system so that it will properly execute on another hardware platform or operating system.

POS (point of sale)

POS systems use personal computers or specialized terminals in combination with cash registers, optical scanners or magnetic-stripe readers to capture and record data at the time of transaction. POS systems are usually online to a central computer for credit checking and inventory updating. Alternatively, they may be independent systems that store daily transactions until they can be transmitted to the central system for processing.

POS (point of service)

A type of health maintenance organization (HMO) plan that offers limited coverage for care received outside the HMO's network.

POSI (Promoting Conference for OSI)

A Japanese vendor organization for standards promotion.

Posix

A Unix-based standard developed by the Institute of Electrical and Electronics Engineers (IEEE) Posix committee. Posix has been adopted as a Federal Information Processing Standard (FIPS), and is included in the Open Group's X/Open specifications.

Posix 1003.1 compliance

Operating-system compliance with the Institute of Electrical and Electronics Engineers (IEEE) Posix 1003.1 system services standard.

POST (power-on self test)

Diagnostic tools that run automatically when a hardware device is turned on.

postal, telegraph, and telephone (see PTT)

Post Office Protocol (see POP)

PostScript printer support

Support for the printing of document information, including "rich text," on PostScript printers.

POTS (plain old telephone service)

POTS (portfolio of technology solutions)

A business model wherein vendors interface with major customers through solution-focused groups. These groups deliver repeatable solutions that address specific processes in specific industry sectors and are composed of consulting and implementation services based on leading-edge technologies and market-leading applications. The solution groups are supported by resources such as corporate services, technology centers of expertise, and industry and process centers of expertise. The vendor can treat these solution groups as profit-and-loss centers that are then easily valued for investment, incentive, and divestiture purposes.

POWER (Performance Optimized With Enhanced RISC)

IBM's proprietary reduced instruction set computer (RISC) processor technology used in the RS/6000 and selectively licensed to other suppliers, including Motorola.

power unit (see PU)

power-fail restart

Automatic restart initiation upon restoration of power after a power failure.

power-on self test (see POST)

PPA (personal productivity application)

PPC (palm-sized PC)

PPC (plain paper copier)

ppm (pages per minute)

A measure of document output speed.

PPM (project portfolio management)

A strategy for oversight of a portfolio of related or interdependent projects, with the intent of limiting duplicate work efforts and leveraging decision making and skills across projects. PPM takes a holistic view of projects and their relationships, and focuses on the potential for project benefits to be leveraged across the enterprise.

PPM tools, which support this strategy, are high-end, multiproject management tools with:

- Resource profiling and allocation capability
- Integrated time reporting
- Executive information reporting (such as "red light" and "green light" status)
- Project accounting interfaces

PPMS (physician practice management system)

PPO (preferred provider organization)

PPP (Point-to-Point Protocol)

A proposed Internet standard specified in Request for Comments (RFC) 1331.

PP-PI (Production Planning for Process Industries)

A module intended to provide vertical integration between the enterprise resource planning (ERP) elements of SAP R/3 and the production floor in batch/continuous manufacturing.

PPRC (peer-to-peer remote copy)

A synchronous controller-to-controller remote-copy capability. It uses standard Enterprise Systems Connection (ESCON) links to communicate between control units.

PPS (pulses per second)

PPTP (Point-to-Point Tunneling Protocol)

A Microsoft extension to Point-to-Point Protocol (PPP), used to create multiprotocol virtual private networks (VPNs) via the Internet.

PQS (Parallel Query Server)

A product announced by IBM in 1994 that used complementary metal-oxide semiconductor (CMOS) technology to provide parallelism to Information Management System/Database Manager (IMS/DM) applications.

Practical Extraction Report Language (see PERL)

Preboot Execution Environment (see PXE)

Precision Architecture Reduced Instruction Set Computer (see PA-RISC)

Precision Graphics Markup Language (see PGML)

predicate

A clause within a Structured Query Language statement (query) that specifies a constraint that the selected data (rows) must satisfy.

predicate transitive closure

The practice of adding semantically redundant predicates to a Structured Query Language (SQL) statement (query) to help the optimizer perform better. This can be done by application programmers or by the optimizer itself.

predictive behavior analysis

The use of techniques such as data mining, data visualization, algorithm clustering, and neural networking to find patterns or trends in data. These patterns or trends are used to forecast future behavior based on current or past behavior. Uses of predictive behavior analysis include identifying customers likely to drop out or default; identifying products customers are likely to buy next; developing customer segments or groups; and product development.

predictive modeling (see PM)

preprocessor

A program that converts data from the format of an emulated system to the format accepted by an emulator.

presentation layer

In the Open Systems Interconnection (OSI) model, the layer of processing that provides services to the application layer, enabling it to interpret the data exchanged, as well as to structure data messages to be transmitted in a specific display and control format.

Presentation Manager (see PM)

Pretty Good Privacy (see PGP)

preventive cyclic retransmission (see PCR)

preventive maintenance (see PM)

PRI (Primary Rate Interface)

An Integrated Services Digital Network (ISDN) interface that connects private branch exchanges (PBXs) to the network at the North American 1.544 megabits per second (Mbps) T1 rate. It is electrically identical to the ISDN Basic Rate Interface (BRI).

North American standards for this interface specify 23 64-kilobit-per-second (Kbps) B, or bearer, channels, plus one 64-Kbps D, or delta, channel. Thus, it is also referred to as 23B+D. In Europe, where it is known as Primary Rate Access (PRA), standards call for 30B+D.

Primary Rate Access (PRA)

See PRI.

Primary Rate Interface (see PRI)

primary station

The data terminal in a network that selects and transmits information to a secondary terminal and has the responsibility to ensure information transfer.

PRINCE (Projects in Controlled Environments)

A public domain standard for IT developers for the U.K. government. PRINCE was widely used in the United Kingdom for a range of projects in commercial and government organizations. The U.K. government, through the Central Computer and Telecommunications Agency (CCTA), owns the PRINCE name and methodology. PRINCE provides a process model to define what is involved in setting up, managing and closing down each stage of a project. The model specifies eight components for creating a project management environment: organization, planning, control, risk, stages, quality, change control and configuration management.

print on demand (see POD)

printed circuit board (see PCB)

Printer Control Language (see PCL)

print server

A network device allowing sharing of printers on a network without using the file server.

priority indicator

In data communications, a group of symbols that specifies the relative urgency of a message and its order of transmission.

PRISM (Parallel Reduced Instruction Set Multiprocessor)

A reduced instruction set computer (RISC) architecture developed by Apollo for its high-end workstation products.

privacy-enhanced mail (see PEM)

privacy seal (see OPSP)

private automatic branch exchange (see PABX)

private branch exchange (see PBX)

private communications exchange (see PCX)

private exchange (see PX)

private key

The confidential half of the asymmetric key pair used in public-key cryptography. Unlike the “secret key” used in symmetric-key cryptography — a single key known by both the sender and the receiver — a private key is known only by the recipient. See public-key cryptography and secret-key cryptography.

Private Network-to-Network Interface (see PNNI)

PRM (partner relationship management)

The customer relationship management (CRM) element that extends sales, marketing, customer service and other enterprise business functions to partners to foster more-collaborative channel partner relationships.

problem management

The core function of a customer service and support (CSS) application used by call centers. It coordinates a multitier, multiowner service and support environment, enables pattern analysis, provides management reports, and facilitates requesting additional service and support resources by providing hard numbers on the service workload and its changing nature. Because PM tools can also track service-level agreements (SLAs), they are valuable for monitoring compliance.

process

An interconnected network of activities that begins at the specification of a requirement for a product or service and terminates with the acceptance of consideration for providing the product or service.

process control

The regulation of variables that influence or control the conduct of a process so that a specified quality and quantity of product are obtained.

Process Data Exchange Institute (see PDXI)

process flow scheduling (see PFS)

process information management system (see PIMS)

process manufacturing

Manufacturing that adds value by performing chemical reactions or physical actions to transform materials, or by extracting, mixing, separating or forming materials in batch or continuous production modes.

processing, batch

A method of computer operation in which a number of similar input items are accumulated and sorted for processing.

processing, line

The processing of transactions as they occur, with no preliminary editing or sorting before they enter the system.

process management

A management concept that describes the goal of increasing intraenterprise coordination of separate business functions. Business process re-engineering is a form of process management that focuses on replacing traditional functions (e.g.,

sales, marketing and service) with functionally integrated processes (e.g., customer relationship management). The growth in the demand for enterprise software reflects the need for increased integration and sharing of business information throughout an enterprise.

Process Operations Management System (see POMS)

Processor Resources/Systems Manager (see PR/SM)

procurement

All activities concerned with the acquisition of goods and services, including ordering, negotiations and delivery.

procurement handbook

A handbook published by organizations such as Government OSI Procurement (GOSIP) to aid the buying process of Open Systems Interconnection (OSI) systems.

Product Data Exchange Specification (see PDES)

product data management (see PDM)

product development management (see PDM II)

Product Lifecycle Management (see PLM)

product marshaling and handling (see PMH)

production information management systems (see PIMS)

Production Planning for Process Industries (see PP-PI)

PR/OEE (Platinum Repository/Open Enterprise Edition)

Professional Office System (see PROFS)

professional services administration (see PSA)

profile

A definition of customer preferences, behaviors or demographics.

PROFS (Professional Office System)

An IBM office system for the Virtual Machine (VM) environment.

program

An organized list of instructions that, when executed, causes a computer to behave in a predetermined way. Without programs, computers are useless. A program is like a recipe that contains a combination of two things:

1. A list of ingredients (called variables). Variables represent numeric data, text or graphical images.

2. A list of directions (called statements). Statements provide the instructions for what to do with the variables. There are many programming languages that are used to create this combination of variables and statements.

Program Evaluation and Review Technique (see PERT)

program library

A collection of computer instruction packages and routines.

programmable logic controller (see PLC)

Programmers' Hierarchical Interactive Graphics Standard (see PHIGS)

programming language

A defined group of commands and syntax that a program developer uses to write software. Languages range from primitive (similar to machine code) to high-level (closer to standard written language and somewhat automated) and have been classified into generations, the fifth being the most recent. C, C++, Pascal, BASIC, FORTRAN, COBOL, LISP and Java are examples of programming languages.

Programming Language 1 (see PL/1)

Programming Language/Structured Query Language (see PL/SQL)

program sharing

In a network, the ability for several users or computers to use a group of instructions stored at another connection point.

project accounting (see PA)

Project Management Professional (see PMP)

project portfolio management (see PPM)

Projects in Controlled Environments (see PRINCE)

PROM (programmable read-only memory)

Promoting Conference for OSI (see POSI)

prompts

Messages from a computer that give instructions to the user.

propagation delay

The period between when a signal is placed on a circuit and when it is recognized and acknowledged at the other end. Propagation delay is of great importance in satellite channels because of the great distances involved.

proprietary software

Software that is owned by an organization or an individual, as opposed to "public-domain software," which is freely distributed. The explosion in the use of the Internet has expanded the reach of public-domain software since it is now much easier to transmit these programs. While many commercial software developers have

developed software that has become the de facto standard (e.g., Microsoft's Windows programs), proprietary software that is based on proprietary protocols, or standards, can create obstacles for application development and usage.

protected mode

A mode of the Intel microprocessors with hardware memory protection.

protocol

A set of procedures in telecommunications connections that the terminals or nodes use to send signals back and forth. Transmission Control Protocol/Internet Protocol (TCP/IP) is the standard protocol for the Internet and related networks such as intranets and extranets. Local-area networks (LANs) often rely on a different protocol. Networks and systems cannot communicate unless they use the same protocol or make use of a gateway.

protocol conversion

The process of translating the protocol native to an end-user device (e.g., a terminal) into a different protocol (e.g., ASCII to BSC), allowing that end-user device to communicate with another device (e.g., a computer) with which it would otherwise be incompatible. Protocol conversion can be performed by a dedicated device (a protocol converter); by a software package loaded onto an existing system, such as a general-purpose computer, front-end processor, or private branch exchange (PBX) system; or by a value-added network, such as Telenet.

Protocol Data Unit (see PDU)

protocol-independent routing (see PIR)

protocol stack

A defined protocol with options applicable for specific functions that can be implemented as a product. Also called a functional standard or functional profile.

protocol-transparent

Refers to a device's capability to perform its function independent of the communications protocol.

proxy agent

A network management agent that comes between an unmanaged device and a management system, allowing management by proxy, i.e., on behalf of the device.

proxy servers

Devices that process and filter all Internet Protocol (IP) packets that are directed to them and decide which protocols and services can be served out of their caches. Proxy servers tend to offer the greatest range of protocol and caching support since they cache Hypertext Transport Protocol (HTTP), Secure Hypertext Transport Protocol (HTTPS), File Transfer Protocol (FTP) and, in some cases, streaming content such as RealAudio and PointCast. Each workstation addresses the proxy server directly by setting specific parameters in each browser on each workstation.

PR/SM (Processor Resources/Systems Manager)

IBM's logical-partitioning facility that allows the computer architecture to be separated from its physical implementation to provide a logical organization of

resources that allows multiple operating systems (or copies of the same system) to run on one physical central processing unit (CPU) or complex.

PSA (professional services administration)

The integration of planning, resource management, project management and project accounting for service organizations. PSA vendors base their businesses on software licensing or subscription, mainly to external service providers (ESPs), and often offer resource exchange capability in conjunction with their software. PSA applications offer opportunity management, time-and-expense, invoicing and other features targeted to professional-service firms. Service process optimization (SPO) is a related term, but it encompasses internal service departments as well as ESPs. See SPO.

PSDN (public switched data network)

PSF (print service facility)

PSK (phase-shift keying)

Phase modulation that uses discrete, present changes of phase.

PSLC (Parallel Sysplex License Charge)

A software licensing arrangement available to vendors if the pieces making up their Parallel Sysplex complex have been qualified by IBM. Charges are based on the total machine capacity.

PSM (purchasing and supply management)

PSN (private switching network)

PSO (professional services organization)

PSTN (public switched telephone network)

A worldwide voice telephone network.

PTA (post and telecommunications administration)

PTAT (private transatlantic fiber-optic cables)

PTB (post and telecommunications bureau)

PTF (program temporary fix)

PTM (Packet Transfer Mode)

A "variable-length cell" transfer mode employed in IBM broadband networks built using IBM's broadband and asynchronous transfer mode (ATM) switching technology.

PTO (public telecommunications operator)

PTS (Parallel Transaction Server)

A product announced by IBM in 1994 that introduced CMOS-based parallelism to Information Management System/Database Manager (IMS/DB) applications.

PTT (postal, telegraph and telephone)

A term used to refer to the agency responsible for postal, telegraph and telephone services at a governmental level.

PU (Physical Unit)

A physical device used in IBM Systems Network Architecture (SNA) communications, including the software in an SNA node controlling the node's communications hardware. PU is synonymous with "node type" within SNA. PU 2 devices are intelligent cluster controllers/terminals. A PU 2.1 node enables local user ports to communicate without going through a host node's services.

PU (power unit)

A licensing metric used by Oracle defined as 1MHz of power in an Intel-compatible or reduced instruction set computer (RISC) processor in any computer of the designated systems on which the programs are installed and operating.

public data network (see PDN)

public key

The public half of the asymmetric key pair used in public-key cryptography (see separate entry).

public-key cryptography

An encryption technique developed to overcome the limitations of secret-key cryptography (see separate entry). Public key (also called "asymmetric key") cryptography uses two mathematically related keys: A public key to encrypt messages, and a private key to decrypt them. In a public-key system, you communicate privately by encrypting your message using the public key of your intended recipient. Although everyone else knows the recipient's public key, it is useless for decrypting a message encrypted with it. Only the corresponding private key, known only to the recipient, can decrypt the message.

public-key infrastructure (see PKI)

Public-Key Infrastructure Exchange (see PKIX)

public switched telephone network (see PSTN)

publish and subscribe

A communication pattern in which information sources "publish" (i.e., send) information to a somewhat intelligent middleware infrastructure, and information consumers "subscribe" by specifying what kind of information they want to receive from the infrastructure. The middleware must be able to physically transport messages from one or more publishers to one or more subscribers. It also must be smart enough to find the proper destinations by matching each message to subscription criteria. This model naturally supports one-to-many or many-to-many communication — in contrast to either message passing or message queuing, both of which mostly (but not entirely) aim at one-to-one communication.

publish/subscribe architecture

Processes that acquire or receive messages from source applications and publish these messages on a systemwide bulletin board or, in alternative terminology, put them on a software bus where all other processes can "see" them. Application

processes generally indicate which messages they are interested in by supplying one or more rules to the integration broker (IB). Also described as a software bus architecture.

PUC (Public Utilities Commission)

pulse

A brief change of current or voltage produced in a circuit to operate a switch or relay or which can be detected by a logic circuit.

pulse amplitude modulation (see PAM)

pulse carrier

A series of identical pulses intended for modulation.

pulse code modulation (see PCM)

pulse duration modulation (PDM)

See PWM.

pulse radio (see UWB)

pulse width modulation (see PWM)

push technology

Software that automates the delivery of information to users. In contrast, the Web is a "pull" environment that requires a user to seek information.

In a "push" environment, information is sent to a person proactively, through a Web browser, e-mail, or even voice mail or a pager. In business, push technology can be used for the conveyance of time-sensitive information, like changes in commodity pricing or the introduction of promotional programs to a sales force. Enterprises can employ push technology to communicate externally with their clients or internally with their employees over a network.

PV (present value)

PVC (permanent virtual circuit)

A virtual circuit established through manual or protocol interaction rather than via a call establishment sequence.

PVCS

A software configuration management product from Intersolv. PVCS was originally called Polytron Version Control System.

PWM (pulse width modulation)

A method of encoding information based on variations of the duration of carrier pulses. Also called pulse duration modulation (PDM).

PX (private exchange)

An internal telephone exchange serving a single organization and having no connection with a public exchange. Also private automatic exchange (PAX). Other types of private exchange include:

1. Private automatic branch exchange (PABX): A private automatic telephone exchange that provides for the transmission of calls internally and to and from the public telephone network. Also private branch exchange (PBX).
2. Private automatic telex exchange (PATX): Provides telex service within and without an enterprise (largely obsolete).
3. Private manual branch exchange (PMBX): Private, manually operated telephone exchange that provides private telephone service to an organization and allows calls to be transmitted to or from the public telephone network.
4. Private digital exchange (PDX): Private exchange employing digital transmission techniques.

PXE (Preboot Execution Environment)

A technology that allows a PC to be controlled in a pre-boot execution state. Part of the Wired for Management specification, it is generally used for pristine operating system loads.

Px64

A videoconferencing standard for two-way, full-motion video.

Q

Q.2931

The broadband out-of-band signaling standard for call management. Also known as Q.93 broadband.

QA (quality assurance)

The historic focus of the National Committee on Quality Assurance (NCQA), fundamentally based on measures of compliance to standards or processes.

QAM (quadrature amplitude modulation)

A combination of amplitude and phase modulation (and shift-keying) techniques used to transmit 9,600 bits per second (bps) over a 2,400-baud line.

QAS (quasi-associated signaling)

In QAS, an ISDN term, signaling links are separated from the transmission route but are switched via the signal transfer point.

QBE (query by example)

A query method (also known as "find similar") that enables the user of an information retrieval system to obtain search results by using examples — such as text strings or document names — that are characteristic of the information sought.

QCIF (Quarter Common Intermediate Format)

Part of the International Telecommunication Union Telecommunications Standards Sector's (ITU) H.261 video format standard, centered on Common Intermediate Format (CIF), which is a compromise between National Television Standards Committee (NTSC) and phase alteration line (PAL). In the H.261 standard, only QCIF is specified (176x144 pixel resolution).

QCTC (Quad Channel I/O Controller)**QEMM (Quarterdeck Expanded Memory Management)****QFA (quick file access)****QI (quality improvement)****QIC (quarter-inch cartridge)****QM (Quality Module)**

A module of SAP's R/3 enterprise resource planning system.

QMF (Query Management Facility)

IBM's strategic query and report-writing product for relational database management systems (RDBMSs). Its main objective is to widen the RDBMS user base, i.e., to allow anyone to use relational data. QMF has basic analysis and graphics features.

QOS (quality of service)

A negotiated contract between a user and a network provider that renders some degree of reliable capacity in the shared network.

QphD

A "query per hour" metric used in the Transaction Processing Performance Council's TPC-D benchmark.

QPM (QoS Policy Manager)

A Cisco Systems offering.

QppD

A power metric used in the Transaction Processing Performance Council's TPC-D benchmark.

QPSX (Queued Packet Synchronous Exchange)

A metropolitan-area network based on the Institute of Electrical and Electronics Engineers' 802.6 Distributed Queue Dual Bus standard for speeds up to 150 megabits per second.

QR (quick response)

QSAM (Queued Sequential Access Method)

An extended version of the Basic Sequential Access Method (BSAM). QSAM queues up input data blocks pending processing and output data blocks pending transfer.

Q-SIG

A PBX networking protocol produced by the European Computer Manufacturer's Association in line with the International Telecommunication Union's ISDN recommendations.

QthD

A throughput metric used in the Transaction Processing Performance Council's TPC-D benchmark.

quad

A cable of four separately insulated conductors, twisted together in such a way as to provide two pairs.

quadrature amplitude modulation (see QAM)

quality assurance (see QA)

quality of service (see QOS)

quantization noise

Signal errors caused by the process of digitizing a continuously variable slope.

Quarter Common Intermediate Format (see QCIF)

quartet signaling

The part of the 100BASE-VG proposed standard that orders transmission of data over four wires simultaneously, enabling the increase in speed to 100 megabits per second.

quasi-associated signaling (see QAS)

Quel (Query Language)

A relational data manipulation language that was a rival to Structured Query Language (SQL) but has now fallen into general disuse.

query

A request for information placed to a computer system or database. Queries may be performed by human beings (for example, a Web user entering a query into a search engine), but are also commonly performed by computers themselves (for example, a program placing an automated query to a database).

Query Language (see Quel)

Query Management Facility (see QMF)

query/report tools

Basic interactive and batch facilities that allow users to query files and databases and create terminal and printed-report output.

queue

1. In telephony, a series of telephone calls arranged in sequence, the two ends being the "head" and "tail." New calls are added to the tail. Calls can be removed either from the head or tail.
2. Any series of data streams, such as a print queue, waiting to access a device.

Queued Packet Synchronous Exchange (see QPSX)

Queued Sequential Access Method (see QSAM)

Q.x series

Any of several International Telecommunication Union (ITU) specifications for data communication standards, which include:

1. Q.93B: Specification for establishing, maintaining, and clearing Broadband Integrated Digital Services Network (B-ISDN) network connections.
2. Q.920/Q.921: Specifications for the ISDN user-to-network interface data link layer.
3. Q.922A: Specification for frame relay encapsulation.
4. Q.931: Specification for establishing, maintaining, and clearing ISDN network connections.

R

R/2

SAP's mainframe enterprise management suite, the precursor to R/3.

R/3

SAP's integrated enterprise resource planning (ERP) system for client/server and distributed open systems. It was ported (with enhancements) from R/2.

RA (registration authority)

Firmly proves a user's identity before passing the user's credentials to the certification authority for certificate creation. A certification authority creates, signs and revokes digital certificates that bind public keys to user identities. A repository or directory stores digital certificates and CRLs to allow users to obtain the public keys of other users and determine revocation status. Typically, the repository is a traditional X.500 directory or a database that supports LDAP access.

RAAD (rapid architected application development)

An approach to large-scale application development that includes the following phases, which are executed by up to 10 teams of 10 people each during a period of no more than 18 months:

- Phase 1: Business and technical architecture reconciliation.
- Phase 2: Functional requirements gathering and specification.
- Phase 3: Initial architecture design, building and implementation.
- Phase 4: First building and testing of application, including user interface, data access and business logic.
- Phase 5: Initial installation of application.
- Phase 6: Concurrent engineering for subsequent builds.
- Phase 7: Rapid release plan.

RACE (R&D for Advanced Communications in Europe)

RACF (Resource Access Control Facility)

The IBM product offering back-end security in the System/370 Multiple Virtual Storage (MVS) and Virtual Machine (VM) environments. RACF had been difficult to use, but release 1.7 achieved a level of flexibility and usability close to what is offered by leading-edge, third-party software. Computer Associates' Top Secret and Access Control Facility-2 (ACF2) are the leading independent software vendor (ISV) offerings.

rack

A framework or structure that holds computer servers or networking equipment, usually by means of shelves or mounting plates. The height of computer equipment is expressed in rack units (U), which equal the distance between shelf increments in a standard rack (see rack unit).

rack unit (U)

A standard increment used to express the height of a piece of rack-mounted computer or networking equipment, abbreviated as "U" and equal to 1.75 inches. For example, a server with a height of 4U takes up seven inches of vertical rack space. The most common dimensions for an industry-standard rack are 42U (73.5 inches) high and 19 inches wide.

RAD (rapid application development)

An application development approach that includes small teams (typically two to six people, but never more than 10) using joint application development (JAD) and iterative-prototyping techniques to construct interactive systems of low to medium complexity within a time frame of 60 to 120 days.

RADIO (Reliable Architecture for Distributed I/O)

radio channel

A frequency band allocated to a service provider or transmitter.

radio circuit

A physical circuit consisting of two unidirectional radio links and connections to terminal exchanges.

radio communications

Any telecommunications by means of radio waves.

radio determination

The use of radio waves to determine the position or velocity of a remote object. Also used to collect information.

radio frequency (see RF)

radio frequency identification (see RFID)

radio frequency interference (see RFI)

radiology information system (see RIS)

RADIUS (Remote Authentication Dial-In User Service)

A security protocol used to transport passwords between the access device and the authentication server.

RAID (redundant array of independent disks)

A method of mirroring or striping data on clusters of low-end disk drives; data is copied onto multiple drives for faster throughput, error correction, fault tolerance and improved mean time between failures.

With the exception of RAID 0, all RAID levels provide automated recovery of data in the event of a disk failure. The RAID levels and their key features are:

- RAID 0 — provides disk striping without parity information; data is written by segment across multiple disks sequentially until the end of the array is reached, and then writing starts at the beginning again. Provides greater logical disk capacity with faster access time on reads (multiple segments read simultaneously). However, RAID 0 provides no data redundancy — if one drive fails, the entire disk array subsystem is unavailable.
- RAID 1 — provides fault tolerance by using disk mirroring (also called shadowing). Each byte of data on a disk is duplicated on another physical drive, providing 100 percent data redundancy. RAID 1 provides immediate access to data when either the primary or secondary drive fails, but it has the highest cost of all RAID types, since duplicate hardware is required.
- RAID 0+1 and 1+0 — combinations of RAID 0 and RAID 1 that provide the benefits of striping and fault tolerance (disk mirroring). RAID 0+1 (also known as RAID 01 or RAID 0/1) uses a mirrored configuration of two striped disk sets; RAID 1+0 (also known as RAID 10 or RAID 1/0) is a stripe across a number of mirrored sets.
- RAID 2 — eliminates the 100 percent redundancy overhead of RAID 1 by using a powerful error detection and correction code (Hamming), with bits of the data pattern written across multiple disks.
- RAID 3 — similar to RAID 2, but uses a single check disk per group that contains the bit parity of the data disks; data is interleaved across all disks. Because disk reads are performed across the entire array and all data is transferred to the controller in parallel, RAID 3 is well suited for applications that require high data read/write transfer rates for large sequential files.
- RAID 4 — instead of interleaving blocks of data across all drives, writes the first block on drive 1, the second block on drive 2, and so on. This technique dramatically improves read time, since many reads are single block (single drive), freeing other drives for additional read requests.
- RAID 5 — eliminates the dedicated parity drive by writing parity with the data across all drives in the array. Consequently, the single-write restriction and some performance degradation of RAID 1 through RAID 4 are eliminated. If a drive fails, the controller can rebuild the data from the parity and data on the remaining drives.
- RAID 6 — provides two-disk parity and one spare, so that two simultaneous disk failures per array of disks can be tolerated. With the occurrence of a failure, a spare is brought online and transparent reconstruction begins automatically in the background with negligible impact on performance.

Raike Public Key (see RPK)

RAM (random-access memory)

The primary memory in a computer. Memory that can be overwritten with new information. The "random access" part of the name comes from the fact that the next bit of information in RAM can be located — no matter where it is — in an equal amount of time.

RAMAC

Remote-copy technology from IBM.

RAMAC Virtual Array (see RVA)

Rambus Dynamic Random Access Memory (see RDRAM)

RAMP-C

An IBM proprietary benchmark constructed to measure performance on online transaction processing (OLTP) workloads. IBM has used RAMP-C both to measure the relative performance of its processors and architectures and to compare its processors to the competition. It has not released the code, so independent judgments on the relevance of the benchmark are impossible.

random access

A storage device that allows information or blocks of information to be read in any order, e.g., a disk or solid-state memory.

random-access memory (see RAM)

rapid application development (see RAD)

rapid architected application development (see RAAD)

rapid education curricula (see REC)

RARP (Reverse Address Resolution Protocol)

A Transmission Control Protocol/Internet Protocol (TCP/IP) protocol for translating a Data Link Control address to an Internet Protocol (IP) address. RARP can be used to obtain a logical IP address. Responding to a RARP broadcast from a workstation, a RARP server can send back the IP address.

RAS (reliability, availability and serviceability)

A reference to a product's quality, availability of optional features, and ease of diagnosis and repair.

RAS (Remote Access Server)

A remote-node access feature of Windows NT.

raster

Originally a German word naming the screen used in photoengraving; now, the array of scan lines used to cover a planar area to read or depict image information on that area. The electron beam of a TV picture tube writes a raster on the phosphor screen.

raster image processor (see RIP)

rate averaging

A telephone company's method for establishing uniform toll rates based on distance, rather than on the relative cost or volume of telephone traffic on a given route.

rate center

A defined geographic point used by telephone companies in determining distance measurements for interLATA mileage rates.

RAW (read after write)

RBISO (role-based IS organization)

A distributed type of IS organization that represents an association of multiple IS organizational styles and sources fulfill different roles for the enterprise. Instead of a single, integrated IT function within the enterprise, this newer RBISO encompasses consultants, business liaisons, resource pools and services.

RBOC (regional Bell operating company)

A local or regional telephone company in one of seven regions of the United States formed by the divestiture of AT&T. Also known as Bell operating company (see BOC) or, colloquially, as a "baby Bell."

RBRS (resource-based relative value scale)

RC2, RC4, RC5 (Rivest Code 2, 4 and 5)

Alternatives to Data Encryption Standard (DES). RC2 (Rivest Cipher 2) is an exportable Rivest, Shamir Adelman (RSA) proprietary cipher for bulk encryption.

RCC (radio common carrier)

RCF (remote call forwarding)

RCM (reliability-centered maintenance)

RCP (Remote Copy Protocol)

Allows file copies to and from a remote host or server on a network; uses Transmission Control Protocol (TCP) to ensure the safe delivery of data.

RCS (Revision Control System)

A Unix utility that provides version control. The majority of Unix developers rely on basic utilities such as RCS for version control.

R&D (research and development)

RDA (Remote Data Access)

A standard (ISO/JCTI/SC21 WG3) to address the interconnection of applications and databases. The standard originally attempted to cover any kind of data access and concerned itself only with effective dialog management, but inherent complexity has encouraged it to focus more on Structured Query Language (SQL).

Rdb

A database management system from Hewlett-Packard (originally from Digital Equipment, and later Compaq Computer). Rdb is integrated with the other Virtual Address Extension (VAX) Information Architecture products, and is designed to provide true multiplatform, distributed-database capabilities.

RDBMS (relational database management system)

A database management system (DBMS) that incorporates the relational-data model, normally including a Structured Query Language (SQL) application programming interface. It is a DBMS in which the database is organized and accessed according to the relationships between data items. In a relational database, relationships between data items are expressed by means of tables. Interdependencies among these tables

are expressed by data values rather than by pointers. This allows a high degree of data independence.

RDD (rigid disk drive)

RDF (Resource Description Framework)

A proposed World Wide Web Consortium (W3C) standard for metadata descriptions.

RDLAP (Radio Data Link Access Protocol)

RDM (see remote data management)

RDP (Remote Desktop Protocol)

Microsoft's protocol for remote presentation.

RDRAM (Rambus Dynamic Random-Access Memory)

A memory chip design, from Rambus Inc. (Los Altos, California), that speeds up computer systems by accelerating the exchange of data between memory and surrounding logic chips.

RDS (report distribution system)

RDSS (Radio Determination Satellite Service)

RDТ (recall dial tone)

read-only memory (see ROM)

real time

The description for an operating system that responds to an external event within a short and predictable time frame. Unlike a batch or time-sharing operating system, a real-time operating system provides services or control to independent ongoing physical processes. It typically has interrupt capabilities (so that a less important task can be put aside) and a priority-scheduling management scheme.

real-time clock (see RTC)

Real-Time Data Intelligence (see REDI)

real-time document sharing (see RTDS)

real-time variable bit rate (see rtVBR)

reasonableness checks

Tests made on information reaching a real-time system or being transmitted from it to ensure that the data in question lie within a given range.

REC (rapid education curricula)

Short-term courses, offered through 12-month continuing education programs and at technical schools and community colleges, that typically target basic Structured Query Language and database administrator skills, as well as minimal essential programming concepts, language syntax and specific application development toolsets.

received line signal detector (see RLSD)

A signal indicating that a connection has been made. Also known as "carrier detect."

receive only (see RO)

record

Any information-bearing media generated or received by an organization — e.g., documents, spreadsheets, images, Web pages and e-mail.

record communication

Communication that produces a hard-copy record of the transmission, such as facsimile.

recording density

The number of bits that can be written on a specific area of magnetic media; generally measured in bits per inch (bpi).

Record Management System (see RMS)

record management

The systematic organization and managed storage of diverse information sources through the end of their life cycles. Record management can be viewed as a subset of integrated document management (IDM) and is one of the five most important IDM library service functions. The others are check-in/check-out, version control, document-level security and attribute, and full-text index search and retrieval.

record management application (see RMA)

recovery from fallback

When the system has switched to a fallback mode of operation and the cause of the fallback has been removed. This is the process that restores the system to its former condition.

Recovery Manager (see RMAN)

recovery point objective (see RPO)

recovery time objective (see RTO)

red book

Informal term for the CCITT (now ITU) standards approved in 1984.

red, green, blue monitor (see RGB)

REDI (Real-Time Data Intelligence)

A storage architecture from XIOTech (Eden Prairie, Minn.).

reduced instruction set computer (see RISC)

redundancy

1. Portion of the total information contained in a message that can be eliminated without loss of essential information.

2. Provision of duplicate, backup equipment or links that immediately take over the function of equipment or transmission lines that fail.

redundancy check

Automatic or programmed check based on the systematic insertion of components or characters used for checking purposes.

redundant array of independent disks (see RAID)

redundant processor

A computer that duplicates or partially duplicates the operation of another computer to substantially reduce the possibility of system failure.

redundant routing

By using an algorithm for redundant routing, a network management system automatically uses spare capacity to route by alternative paths in case of line failure.

reed switch

A special type of relay consisting of five moving, reed-like contacts controlled by an electromagnet where the reeds themselves are part of the magnetic as well as the electrical circuit being controlled.

Reference Information Model (see RIM)

referential integrity

The underlying rules defining the semantic interdependencies among items in a database. The goal is to define those items declaratively (i.e., as business-oriented assertions) rather than through procedural logic. A relational database management system (RDBMS) should enforce referential integrity as defined by Addendum 1 (X3.135.1-1989) to the American National Standards Institute (ANSI) Structured Query Language (SQL) standard.

Reflective Memory Technology (see RMT)

refresh rate

The speed at which an image is painted onto a monitor screen.

regenerative repeater

1. Repeater used in telegraph applications to retiming and retransmit the received signal impulses and restore them to their original strength. These repeaters are speed- and code-sensitive and are used with standard telegraph speeds and codes.
2. Repeater used in pulse code modulation (PCM) or digital circuits which detects, re-times and reconstructs the bits transmitted.

regenerator

Equipment that takes a digital signal that has been distorted by transmission and produces a new signal in which the shape, timing, and amplitude of the pulses are the same as those of the original before distortion.

regional Bell operating company (see RBOC)

regional center

Class of exchange.

regional (computer) network

A communications pathway with connections limited to a defined geographic area.

regional holding company (see RHC)

ReGIS (Remote Graphics Instruction Sets)

A graphics file transfer protocol.

register

First unit in the assembly of common control equipment in an automatic exchange. The register receives address information in the form of dial pulses or dual-tone multifrequency (DTMF) signals and stores it for possible conversion or translation.

registered jack (see RJ)

registration authority (see RA)

reinitiation time

The time required for a device or system to restart (usually after a power outage).

relational database management system (see RDBMS)

relational online analytical processing (see ROLAP)

relationship manager (see RM)

relay

An electronically operated device that causes abrupt changes in an electrical circuit, such as breaking the circuit, changing the circuit connection, or varying the circuit characteristics.

reliability, availability and serviceability (see RAS)

remote access

The ability to log on to a network from a distant location. Generally, this requires a computer, a modem and remote-access software to allow the computer to dial into the network over a telephone line and connect.

Remote Access Server (see RAS)

remote-access software

Sometimes called remote-control software, this type of program is a superset of the asynchronous communications software market. Remote-access software allows a PC to have complete control over another PC at a different site.

remote-access VPN

A virtual private network (VPN) that creates encrypted tunnels between remote users and a campus network.

Remote Authentication Dial-In User Service (see RADIUS)

remote batch

Transmission of bulk information stored by a distant terminal or computer.

remote computing

Using a distant processor to manipulate data.

remote console operation

A remote-operator console facility that enables the remote operator to perform normal operational tasks, including storage backup, on a running system.

remote control method

Application resides at the central location and the remote worker executes it via presentation software running on the remote device. The concept describes legacy applications as accessed over a 3270 terminal and PC applications using remote-control products.

Remote Copy Protocol (see RCP)

Remote Data Access (see RDA)

remote data concentration

Communications processors used for multiplexing data from low-speed lines or terminals onto one or more high-speed lines. See multiplexing.

remote data management

A form of client/server computing in which the database management system or file management system executes on a different node than the rest of the application logic. See client/server.

Remote Desktop Protocol (see RDP)

remote environmental monitoring

Hardware and software facilities to permit monitoring of computer room environment from a remote site, and to notify remote operators and sound alarms on conditions outside of selectable parameters.

Remote File Sharing (see RFS)

Remote Graphics Instruction Sets (see ReGIS)

remote initial program load

A remote-operator console facility that enables the remote operator to initiate as well as continue systems operation after power-on or abnormal system failure.

remote job entry (see RJE)

Remote Job Entry — Binary Synchronous Communication (see RJE-BSC)

Remote Login (see Rlogin)

remote log-on

A facility that allows an interactive user on one midrange system to log on to any other midrange system of the same supplier in the network.

Remote Method Invocation (see RMI)

Remote Monitoring (see RMON)

Remote Monitoring Management Information Base (see RMON MIB)

remote-node method

Users connect remotely, but have the status of a full node on the local-area network (LAN). In theory, these remote users may run the same applications that are available to LAN-based users, albeit at a much lower speed.

remote operation

An action started by one application entity and performed by another.

remote peripheral access

The ability to attach peripherals logically on other midrange systems in the network to a user's local process or program.

remote power-off

A system command facility that when executed results in a full power-off of the midrange system. The command could be invoked by local or remote terminal access to the system or could be time-released as an unattended processing task.

remote presentation

A form of client/server computing in which the presentation executes on a different computer than the rest of the application logic and data management. See client/server.

remote procedure call (see RPC)

remote request (see RR)

remote software maintenance system

A facility available to a user support group providing for distribution to selected remote midrange-system nodes of both new versions and "fixes" to midrange system software (e.g., operating system, database and communications).

remote SQL database calls

The availability of Structured Query Language (SQL) database calls on a PC that provide cross-network access to a relational database on a midrange system. The calls most likely would be syntactically and semantically equivalent to similar calls available on the midrange system.

remote storage device management

A function that reduces an enterprise's dependency on local operational support for storage by providing the ability to configure, monitor and administer remote storage devices over a network connection.

remote system monitoring

Software facilities to enable monitoring of systems operation from a remote site, and to notify remote operators on conditions outside of selectable parameters.

Removable Storage Management (see RSM)

REN (ringer equivalence number)

A measurement of the load on a telephone circuit; if overloaded, not all handsets on the circuit will ring.

REOS (Real-Time Electronic Online System)

repeatable solutions

Replicable, integrated solutions to a specific process improvement or application requirement. A repeatable solution typically exhibits the following characteristics:

- Fixed pricing, sometimes with a shared business-benefit upside.
- Fixed delivery schedule with rapid implementation.
- Specified performance.
- Initial pilot solution with rapid prototype, followed by roll-out and incremental additions.
- A prime contractor with one point of contact for the client.
- Long-standing and battle-proven partner relationships.
- Joint research and development and joint marketing among the partners.
- Opportunity for residual values in the form of royalties, licensing, etc.

Repeatable solutions must be targeted at specific processes, technologies, subindustries or topical issues. They require high-end professional services such as business process re-engineering, system and network integration, and software customization.

repeater

1. In digital transmission, this equipment receives a pulse train, amplifies it, re-times it and reconstructs the signal for retransmission.
2. In fiber optics, a device that decodes a low-power light signal, converts it to electrical energy, and retransmits it via a light-emitting diode (LED) or laser light source, often including some form of signal amplification.
3. A device on a local-area network (LAN) that extends the range by re-amplifying and re-timing the signal. See regenerative repeater.

replicated-site applications

Identical versions of applications that are distributed to remote sites (e.g., sales offices, bank branches, field service offices, insurance agencies, supermarkets and pharmacies) to perform identical roles.

repository

A facility for storing descriptions and behaviors of objects in an enterprise, including requirements, policies, processes, data, software libraries, projects, platforms and personnel, with the potential of supporting both software development and operations management. As a single point of definition for all system resources, it should stimulate both program and installation management productivity. A system repository would include configuration definitions, tuning parameters and performance goals, while an application repository would include data definitions.

requester/server

A type of relationship between two independent programs in which:

- Two programs interoperate to carry out the processing for a single unit of application work.
- One directs the other; i.e., the requester program specifies a function to be performed and the server program fulfills that service.

request for comment (see RFC)

request for information (see RFI)

request for proposal (see RFP)

request for quotation (see RFQ)

resale carrier

A company that redistributes the services of another common carrier and retails the services to the public.

Research in Motion (see RIM)

residual error rate, undetected error rate

The ratio of the number of bits, unit elements, characters, or blocks incorrectly received but undetected or uncorrected by the error-control equipment to the total number of bits, unit elements, characters or blocks sent.

resolution

The measure of the capability of a visual system, e.g., television, facsimile, etc., to reproduce detail. Usually given as resolution along the scanning lines and parallel to the scanning lines, as these two may differ.

resource

Any means available to system users, including computational power, programs, data files, and storage capacity.

Resource Access Control Facility (see RACF)

resource accounting and chargeback

Mechanisms for recording the use of resources such as processor time, physical memory, transactions, logical or physical input/output (I/O) operations, disk storage and licensed software, and reporting the utilization charged against users, projects, applications or cost centers.

Resource Description Framework (see RDF)

resource exchange

A Web site that enables enterprises in need of specific talent (for example, database administration) to select from many suppliers. Resource exchanges focus on putting the buyer in control. They usually aggregate content from multiple suppliers of staff and services, and provide decision support tools to help buyers make informed decisions.

resource requirements planning

The process of converting the production plan or the master production schedule into the impact on key resources, e.g., man hours, machine hours, storage, standard cost dollars, shipping dollars and inventory levels.

Resource Reservation Protocol (see RSVP)

response time

The time period between a terminal operator's completion of an inquiry and the receipt of a response. Response time includes the time taken to transmit the inquiry, process it by the computer, and transmit the response back to the terminal. Response time is frequently used as a measure of the performance of an interactive system.

Restructured Extended Executor (see REXX)

return on equity (see ROE)

return on information technology (see ROIT)

return on competitiveness (see ROC)

return on investment (see ROI)

reuse

An application development methodology that catalogs and makes available application components so that they may be incorporated into other applications.

Reverse Address Resolution Protocol (see RARP)

reverse-battery signaling

A type of loop signaling in which battery and ground are reversed on the tip and ring of the loop to give an off-hook signal when the called party answers.

reverse channel

A simultaneous data path in the reverse direction over a half-duplex facility. Normally, it is used for positive/negative acknowledgments of previously received data blocks. See half duplex.

Revised IP Security Option (see RIPS0)

Revision Control System (see RCS)

REXX (Restructured Extended Executor)

A structured, interpretive language used to create operator-level command streams to manage and link applications. Introduced in the Virtual Machine (VM) environment.

RF (radio frequency)

The electromagnetic frequency range used for radio communication, i.e., frequencies between approximately 20 kilohertz and 3 gigahertz.

RFC 2613 (see SMON)

RFC (request for comment)

A document submitted for comment and put through a review process under the auspices of the Internet Engineering Task Force (IETF). When accepted, it has the weight of a standard in the Internet community. Each RFC is given a tracking number. For example, RFC 822 describes the address format and data definitions for addressing electronic messages over the Internet, while RFC 1490 is a standard specification for encapsulating multiple protocols over a wide-area frame relay network.

RFI (radio frequency interference)

Signal interference generated at or near a received wavelength. The Federal Communications Commission (FCC) establishes RFI standards to reduce RFI.

RFI (request for information)

Notification of an intended acquisition sent to potential suppliers to determine interest and gather general descriptive product materials, though not prices.

RFID (radio frequency identification)

An analog-to-digital conversion technology that uses radio frequency (RF) waves to transfer data between a movable item and a reader for identification, tracking or location purposes. It does not require physical contact or a line of sight between the reader or scanner and the tagged item. This is one advantage over a bar code system, while another is that RFID tags can be read over a longer range — 100 feet or more. A typical RFID system has three components:

- An antenna
- RFID tags (sometimes called transponders or e-tags), which are electronically programmed with unique information
- An RF module with a decoder (transceiver)

RFP (request for proposal)

An invitation for vendors to bid on supplying goods and services.

RFQ (request for quotation)

Solicitation for pricing for a specific component software product, service or system.

RFS (Remote File Sharing)

Permits multivendor access to data on remote machines.

RFT (request for technology)

RGB (red, green, blue monitor)

A color display screen for computers.

RHC (regional holding company)

RHCs are the companies that were created from the AT&T divestiture (originally, there were seven; now there are five). Each RHC acts as a holding company within a region of the United States for one or more of the divested Bell operating companies, as well as for their unregulated subsidiaries. The RHCs are: Ameritech, Verizon (formerly Bell Atlantic), BellSouth, SBC Communications and US West.

RHIN (regional healthcare information network)

Rich Text Format (see RTF)

RIF (routing information field)

A token ring header field used by a source-routing bridge to determine through which segments it must send a packet.

RIM (Reference Information Model)

A comprehensive object-oriented information model for clinical healthcare that serves as a semantic point of reference for all the standards development efforts of Health Level Seven (HL7).

RIM (Research in Motion)

A vendor of wireless communication products based in Waterloo, Ontario. Its products include the popular BlackBerry handheld device (see BlackBerry).

ring

1. Ring-shaped contact of a plug usually positioned between, but insulated from, the tip and sleeve of a connector.
2. Audible alerting signal on a telephone line.
3. A network topology in which stations are connected to one another in a closed logical circle, with access to the medium passing sequentially from one station to the next by means of polling from a master station or by passing an access token from one station to another. Also called a loop.

ringer equivalence number (see REN)

ring in/ring out (see RI/RO)

ring network

A network topology in which each node is connected to two adjacent nodes to form a continuous ring configuration. Used particularly in the context of local-area networks.

RIP (raster image processor)

A device that converts computer graphics into a raster image — grids of small dots — from which the printing device will recreate them on paper.

RIP (Routing Information Protocol)

An internal router protocol used for informing network computers of changes in configuration.

RIPSO (Revised IP Security Option)

Internet Engineering Task Force Request for Comment (IETF RFC) 1108 specifies an optional Internet Protocol (IP) header field that contains a security classification and handling label.

RI/RO (ring in/ring out)

A method of joining token rings.

RIS (radiology information system)

A system that supports the information-processing and business requirements of radiology departments and freestanding image centers.

RISC (reduced instruction set computer)

A processor architecture that shifts the analytical process of a computational task from the execution or runtime to the preparation or compile time. By using less hardware or logic, the system can operate at higher speeds. RISC cuts down on the number and complexity of instructions, on the theory that each one can be accessed and executed faster, and that less semiconductor real estate is required to process them. The result is that for any given semiconductor technology, a more powerful microprocessor can be produced with RISC than with complex instruction set computer (CISC) architectures.

This simplification of computer instruction sets gains processing efficiencies. That theme works because all computers and programs execute mostly simple instructions. RISC has five design principles:

- Single-cycle execution — In most traditional central processing unit (CPU) designs, the peak possible execution rate is one instruction per basic machine cycle, and for a given technology, the cycle time has some fixed lower limit. Even on complex CPUs, most compiler-generated instructions are simple. RISC designs emphasize single-cycle execution, even at the expense of synthesizing multi-instruction sequences for some less-frequent operations.
- Hard-wired control, little or no microcode — Microcode adds a layer of interpretive overhead, raising the number of cycles per instruction, so even the simplest instructions can require several cycles.
- Simple instructions, few addressing modes — Complex instructions and addressing modes, which entail microcode or multicycle instructions, are avoided.
- Load and store, register-register design — Only loads and stores access memory; all others perform register-register operations. This tends to follow from the previous three principles.
- Efficient, deep pipelining — To make convenient use of hardware parallelism without the complexities of horizontal microcode, fast CPUs use pipelining. An n-stage pipeline keeps up to “n” instructions active at once, ideally finishing one (and starting another) every cycle. The instruction set must be carefully tuned to support pipelining.

RISC System 6000 (see RS/6000)

Rivest Code 2, 4 and 5 (see RC2, RC4, RC5)

Rivest-Shamir-Adelman (see RSA)

RJ (registered jack)

A connector type used for telephone and network connections. See RJx series.

RJE (remote job entry)

Refers to a set of communications protocols for input and output of information to a mainframe in a batch mode (as opposed to interactive).

RJE-BSC (Remote Job Entry — Binary Synchronous Communication)

Emulation in the midrange system of IBM bisynchronous 2780 or 3780 protocols allowing terminal operators to exchange job control and job output with the mainframe processing queues.

RJx series

From “registered jack”; any of several physical specifications for telephone and other modular connectors. This includes:

1. RJ11: Wiring jack commonly used in U.S. telephone systems and modems worldwide.
2. RJ12: Wiring jack for six-wire modular connectors.
3. RJ14: Similar to RJ11; used for connection of two telephone lines.
4. RJ45: Wiring jack for eight-wire modular connectors; used in data-and telecommunications.
5. RJ48C: Wiring jack with eight-wire modular connections as used in the DSX-1 interface.

RLL (run length limited)

Rlogin (Remote Login)

An Internet service similar to Telnet, developed for use between Berkeley Unix-based systems attached to the same local-area network (LAN). It was developed when Telnet did not support all the services that users required. Berkeley’s Unix development program intends to phase out this protocol.

RLSD (received line signal detector)

Signal indicating connection has been made. Also known as carrier detect (CD).

RM (relationship manager)

A staff member who acts as an organizational liaison, typically between an information services (IS) department and business-unit or function. The relationship manager may fulfill a combination of roles that correspond to the degree of trust and authority granted to the person in that role by his/her customer. The job content of the position (in order of increasing business trust and responsibility) is classified in four levels:

- Level 1: Inform and Communicate
- Level 2: Advise and Influence
- Level 3: Coordinate and Integrate
- Level 4: Manage and Oversee

Relationship management is essentially a matrixed position, with reporting requirements to at least two managers (typically a chief information officer — CIO — and business function or division manager). In practice, however, the position is usually linked more strongly to one side than another, and is most often aligned to the business management side.

RMA (record management application)

An application that enables enterprises to manage electronic and paper-based records throughout the life cycles of those records.

RMAN (Recovery Manager)

An Oracle database utility for managing data backup and recovery.

RMF (Resource Measurement Facility)

RMI (Remote Method Invocation)

A specific type of application component that allows one application process to invoke services existing in another remote application environment that relies on specific Java code.

RMON (Remote Monitoring)

A network management protocol that remotely manages networks providing multivendor interoperability between monitoring devices and management stations.

RMON1

The RMON1 specification builds on the functionality of Simple Network Management Protocol (SNMP) by extending the definition of the SNMP Management Information Base (MIB). The extended RMON1 MIB definitions enable network managers to monitor sub network devices (via the SNMP MIB). In addition, RMON1 enables the local collection of device data therefore helping network administrators address the bandwidth constraints imposed by SNMP's device polling design.

RMON2

Version 2 of the Remote Monitoring extension to the Simple Network Management Protocol (SNMP). It is used by management applications to collect statistics from network analysis probes. Statistics include traffic levels by application, across data circuits and between node pairs. RMON-1 products can assist in tasks involving load balancing, capacity planning, troubleshooting, security and accounting.

RMON MIB (Remote Monitoring Management Information Base)

An extension of the Simple Network Management Protocol (SNMP) Management Information Base II (MIB II) which provides a standard-based method for tracking, storing, and analyzing remote network management information. Developed by the Internet Engineering Task Force (IETF).

RMS (Record Management System)

The basic data management system under Hewlett-Packard's (originally Digital Equipment's, and later Compaq's) Virtual Memory System (VMS) operating system.

RMT (Reflective Memory Technology)

A high-speed memory-to-memory bus with very little overhead.

RNC (rugged notebook computer)

Required in applications where environmental concerns threaten reliable operation and drive higher tolerances in some aspect of design. Such concerns include temperature, humidity, drop specifications, vibration, altitude, electrostatic discharge. These requirements are addressed through a variety of methods, including hard disk shock mounting, sealed ports and keyboards, metal caseworks, internal construction and padding. Two categories of RNCs have emerged. Standard

clamshell designs have permanently attached keyboards and screens, like standard notebooks. Tablet-based designs offer the logic and storage inside the screen housing, providing keyboard input via a separate tray or a port replicator.

RNI (Raw Native Interface)

RNIF (RosettaNet Implementation Framework)

A framework that provides generic specifications for sending Extensible Markup Language (XML) messages securely over the Internet using Hypertext Transport Protocol (HTTP) or Secure HTTP (S-HTTP). RNIF is defined by the RosettaNet consortium. See RosettaNet.

RO (receive only)

1. A printer terminal without a keyboard for data entry.
2. A satellite earth station capable of receiving but not transmitting a signal.

roaming

A search for available network carriers performed by Global System for Communications (GSM) phones. GSM phones automatically roam a list of cellular operators so that, for example, when one arrives in a foreign country, the phone will automatically select a local carrier. If the home carrier has negotiated roaming agreements with multiple GSM operators in a foreign country, the user also has the possibility of manually selecting a local carrier.

roaming user profile (see RUP)

ROC (return on competitiveness)

A nomenclature and philosophy that goes beyond the traditional return on investment (ROI) concept by focusing the metric or calculation on how competitiveness is affected by investment. A good way to visualize and quantify the overall return on IT competitiveness is to build a spider diagram that identifies and maps all the relevant dimensions of competitiveness within a given industry sector, and then measure the changes in the overall mapping area during a given time interval (e.g., a year).

ROE (return on equity)

A measure of a company's financial performance (net income divided by the value of the stockholders' equity, and expressed in percent).

ROI (return on investment)

Financial gain expressed as a percentage of funds invested to generate that gain.

ROIT (return on information technology)

Financial gain expressed as a function of an enterprise's investment in information technology.

ROLAP (relational online analytical processing)

A relational approach to multidimensionality, offering multidimensional operations (initially read only) on top of relational data (heavily denormalized).

role-based IS organization (see RBISO)

rollback

A programmed return to a prior checkpoint or rerun point.

rollback recovery

"Before" and "after" images of a transaction are maintained in a log file so that if the system fails, the unsuccessful updates can be backed out, or rolled back.

ROM (read-only memory)

Data stored in computer memory that can be accessed and read by the user but not modified. ROM is often permanent, and stores system-control software.

root bridge

Within a spanning tree domain, the bridge designated to provide topology information to other bridges.

RosettaNet

A consortium that provides XML-based application standards to align supply chain business processes. Its standards, which include the RosettaNet Implementation Framework (RNIF), are widely respected for their approach to standardizing messages for multistep processes. See RNIF.

RosettaNet Implementation Framework (see RNIF)

round-robin lookups

A feature in the Domain Naming System (DNS) — post-BIND 4.8.3 — implemented by many enterprises in order for Web servers to share the client connection load. Essentially, the DNS tables contain multiple addresses (or "A" records) for a single name, such as www.athena.com, that correspond to multiple Web servers with identical content. When a client makes a uniform resource locator (URL) request, the DNS response shuffles the list of addresses returned to the client, redirecting the connection to successively different machines each and every time.

router

A device that connects two networks, performing a function similar to a local or remote bridge. Routers receive packets of information from computers or other routers on the network; they then send these packets to their destinations based on "addresses" at the beginning of the packets and a road map of the other computers and peripherals on the network.

routing

A sequence of operations or steps involved in the manufacture of a product.

routing code

The address or group of characters in the heading of a message defining the final circuit or terminal to which the message has to be delivered. Also called routing indicator.

routing indicator (see routing code)

routing information field (see RIF)

Routing Information Protocol (see RIP)

routing metric

How a routing algorithm determines the best route for a packet. The resulting information is stored in routing tables.

routing table

A table associated with a network node that states for each message (or packet) destination the preferred outgoing link that the message should use.

RPC (remote procedure call)

A mechanism that extends the notion of a local (i.e., contained in a single address space) procedure call to a distributed computing environment, enabling an application to be distributed among multiple systems in a way that is highly transparent to the application-level code.

RPG

A procedural programming language most commonly used on the AS/400 family of computers.

RPK (Raiké Public Key)

An algorithm being forwarded by a New Zealand-based company operating in the United States as RPK, Inc. Based on the Diffie-Hellman public-key algorithm, it uses patented "Mixture Generator" technology, a combination of long linear feedback shift registers to do high-speed exponentiation using the keys to synchronize the RPK "Encryptonite" engine to a randomly generated state. It then uses the high-speed key stream output of the same engine to encrypt the data.

RPM (relative performance measurement)

RPO (recovery point objective)

The point in time until which data can be recovered using backups, journals or transaction logs. It defines what is an acceptable loss of data.

RPS (remittance processing system)

RR (remote request)

A single read-only request to a single data source.

RS/6000 (RISC System/6000)

IBM's 32-bit technical workstation introduced in 1990. It is based on reduced instruction set computer (RISC) architecture and runs the Unix-based Advanced Interactive Executive (AIX) operating system.

RSA (RAMAC Scalable Array)

RSA (Rivest-Shamir-Adelman)

1. An encryption software company owned by Security Dynamics.
2. The public-key algorithm invented by Rivest, Shamir, and Adelman. It gets its security from the difficulty of factoring large numbers. RSA is slower than Data Encryption Standard (DES) but widely deployed in commercial products, often in conjunction with DES.

RSA (rural service area)

A U.S. term used by the Federal Communications Commission (FCC). For licensing purposes, the FCC divided the U.S. into metropolitan statistical areas (MSAs) and Rural Service Area (RSAs). There are 428 designated rural markets.

RSM (Removable Storage Management)

Provides access serialization for libraries and drives, library slot management and removable media catalog services. It also implements the tape and library device driver abstraction layer that simplifies vendor development and application certification.

RSM (route switch module)

RSP (remote store procedure)

RSVP (Resource Reservation Protocol)

A quality-enhancing protocol for improving the performance of a network.

RS-x series

Any of several interface specifications administered by the Electronic Industries Association (EIA). It includes:

1. RS-232-C: Physical interface, with associated electrical signaling, between Data Circuit-terminating Equipment (DCE) and Data Terminal Equipment (DTE); the most commonly employed interface between computers and modems. Now known as EIA/TIA-232.
2. RS-366: An EIA standard for providing dialing commands to network access equipment that uses RS-232 electrical specifications but different connector pinouts and signal functions.
3. RS-422-A: Specification for electrical characteristics of balanced-voltage digital interface circuits. Now known as EIA-530.
4. RS-423-A: Specification for electrical characteristics of unbalanced-voltage digital interface circuits. Now known as EIA-530.
5. RS-443: A standard describing electrical characteristics for unbalanced-voltage digital interface circuits.
6. RS-449: Specification for general-purpose, 37-position and 9-position interface for Data Terminal Equipment (DTE) and Data Circuit-terminating Equipment (DCE) employing serial binary data interchange. RS-449-1 includes Addendum 1. Now known as EIA/TIA-449.

RTC (real-time clock)

A hardware clock system powered by battery when the PC is switched off. This clock is paced by a quartz crystal and operates in a similar fashion to a digital watch, maintaining information for year, month, day, hour, minute and second.

RTCP (Real-Time Control Protocol)

RTDS (real-time document sharing)

Real-time document-sharing applications enable two or more individuals to share documents. RTDS functionality includes application sharing, file transfer, whiteboard

and chat. Some of these applications also include audio- and videoconferencing, but real-time document sharing is the central functionality.

RTE (runtime environment)

RTF (Rich Text Format)

A file format that encodes documents so their messages include boldface, italics and other limited text stylings across platforms and applications. Differences exist between implementations by Lotus and Microsoft.

RTMP (Routing Table Maintenance Protocol)

RTO (recovery time objective)

Often called the recovery window, RTO defines how quickly information systems, services and processes must be operational following some kind of incident such as fire, flood, or hardware failure.

RTOS (real-time operating system)

RTP (Real-Time Transport Protocol)

RTR (Reliable Transaction Router)

RTS (Reliable Transfer Service)

RTS (runtime system)

rtVBR (real-time variable bit rate)

An asynchronous transfer mode (ATM) service category, defined by the ATM Forum, that provides service guarantees for traffic with timing constraints (usually for compressed voice or video).

rugged notebook computer (see RNC)

rule-based systems

Systems that apply a collection of rules to recorded facts to deduce or infer new facts, using an interface engine. Applications include diagnosis, underwriting, regulatory compliance and customer service.

rule-based technology

Technology that enables workflow and knowledge management, allowing users or developers to more easily define, manage and update their knowledge in terms of workflow routing, decision thresholds, alert notifications, necessary actions and more. Knowledge management also includes capturing tacit knowledge from knowledge workers and distributing it to all workers.

run-length encoding

A lossless compression technique where sequences of repetitive characters are compressed. Suitable for still and motion video.

runtime component

A dynamically bindable package of one or more programs managed as a unit and accessed through documented interfaces that can be discovered at runtime.

RUOW (remote unit of work)

A single transaction that updates a single data source.

RUP (roaming user profile)

A feature offered by Windows NT/2000 Server, along with Windows NT Workstation and Windows 9x clients, that enables users to log on at any PC and be presented with their own customized "Start" menu and desktop settings.

rural service area (see RSA)

RV (residual value)

RVA (RAMAC Virtual Array)

An IBM disk storage system.

RWR (read while write)

S

S3 (stand-alone shared storage)

S.100

Defines a client/server model and a set of services for voice/call processing. It is derived from Signal Computing System Architecture (SCSA), and controlled by the Enterprise Computer Telephony Forum (ECTF).

S/370 (see System/370)

S/390 (see System/390)

SA (see server appliance)

SA (System Architect)

An application development (AD) tool suite from Popkin Software.

SAA (Systems Application Architecture)

IBM's name for the architecture once intended to unify its disparate computer architectures.

SA/DA (Source Address/Destination Address)

The combination of source and destination Media Access Control (MAC) addresses which determine a flow over shared media such as Ethernet.

SAE (Society of Automotive Engineers)

SAG (Software AG)

SAIC (Science Applications International Corp.)

An IT and systems integration company, founded in 1969 and headquartered in San Diego.

sales

The domain of CRM that is based on driving shorter sales cycles, sales order configuration, opportunity management, account management/contact management and the integration of sales channels. Technology-enabled selling uses integrated technology to provide continuous improvement in sales effectiveness, as well as the ability to balance and optimize each enterprise sales channel.

sales and operations planning (see S&OP)

sales force automation (see SFA)

SAM (Sequential Access Method)

SAM (software asset management)

A process for making software acquisition and disposal decisions. It includes strategies that identify and eliminate unused or infrequently used software, consolidating software licenses or moving toward new licensing models.

SAM (systems administration management)

SAM tools (systems administration management tools)

A flexible set of application maintenance utilities and application administration tools. The key value of such tools is derived from how well they manage changes. A well-designed tool permits an administrator to describe either a database or a form field change, ensures the integrity of all of the application components tied to the change and propagates the change to all affected users. SAM tools for central administration are used by system administrators and thus provide a level of technical competency, whereas administrative functionality for managers (e.g., sales managers, marketing content managers and call center supervisors) must be limited by the capabilities of the end user.

SAN (storage area network)

A SAN consists of two tiers: The first tier — the storage plumbing tier — provides connectivity between nodes in a network and transports device-oriented commands and status. At least one storage node must be connected to this network. The second tier — the software tier — uses software to provide value-added services that operate over the first tier.

SANS (System Administration, Networking and Security)

The organization that has established System Network Assurance Program (SNAP), a security industry certification program.

SAP

An enterprise resource planning (ERP) vendor based in Walldorf, Germany. The letters "SAP" stand for "Systeme, Anwendungen, Produkte in der Datenverarbeitung," German for "Systems, Applications and Products in Data Processing."

SAP (Service Advertising Protocol)

In routing, broadcasts a message used to identify which servers and end stations are available to which end stations.

SAP (System Assist Processor)

SAPI (server application programming interface)

SAPI (Speech Application Programming Interface)

SAP R/2 (see R/2)

SAP R/3 (see R/3)

SAS (sales analysis system)

SAS (shared application services)

SAS (Statistical Analysis Software)

SASD (server-attached storage device)

SA/SD (structured analysis/structured design)

SASI (Shugart Associates Standard Interface)

SATAN (Security Analysis Tool for Auditing Networks)

satellite communications

The use of geostationary orbiting satellites to relay transmissions from one earth station to another or to multiple earth stations.

satellite computer

A programmable machine that relieves a primary processor of such time-consuming operations as compiling, editing, and controlling input/output devices.

SATS (semiconductor assembly and test services)

SAX (Simple API for XML)

A public domain alternative to Document Object Model (DOM) that defines an event-oriented interface for Java applications.

SBA (strategic business application)

SBM (skills-based management)

A program that objectively defines what skills an enterprise possesses, what skills it will need in the future, when it will need those skills, what strategic value it will place on those skills and how the information technology (IT) employees' competency levels match the value of the strategically significant skills. It is, in effect, a road map for developing competency improvement programs and other methods to fill any skill gaps.

SBS (satellite business systems)

SBU (strategic business unit)

SBus (Sun Bus)

SCADA (supervisory control and data acquisition)

A system used in manufacturing for acquiring measurements of process variables and machine states, and for performing regulatory or machine control across a process area or work cell.

scalability

The measure of a system's ability to increase or decrease in performance and cost in response to changes in application and system processing demands. Examples would include how well a hardware system performs when the number of users is increased, how well a database withstands growing numbers of queries, or how well an operating system performs on different classes of hardware. Enterprises that are

growing rapidly should pay special attention to scalability when evaluating hardware and software.

Scalable Processor Architecture (see SPARC)

Scalable Vector Graphics interface (see SVG interface)

scanner

A device that resolves a two-dimensional object, such as a business document, into a stream of bits by raster scanning and quantization.

scattering

A cause of light wave signal loss in optical fiber transmission. Diffusion of a light beam caused by microscopic variations in the material density of the transmission medium.

SCC (shipping container code)

SCC (specialized common carrier)

SCC (supply chain collaboration)

SCCS (Source Code Control System)

A Unix software configuration management utility used for source code version control.

SCE (supply chain execution)

A subset of supply chain management, this is a framework of execution-oriented applications that enables the efficient procurement and supply of goods, services and information across enterprise boundaries to meet customer-specific demand. In its broadest sense, SCE includes the manufacturing execution system (MES), warehouse management system and other execution systems within the enterprise, as well as throughout the supply chain. The logistics-oriented elements of SCE include the transportation management system, warehouse management system, international trade systems (ITSs), real-time decision support systems (e.g., dynamic routing and dynamic sourcing systems), and supply chain inventory visibility systems.

SCE (System Control Element)

The SCE's major function is to provide paths and controls for the communications between the major functional units and central storage within the IBM 308, 3090 and ES/9000 processor families.

schematic

A diagram that details the electrical elements of a circuit or system.

Science Applications International Corp. (see SAIC)

SCIV (supply chain inventory visibility)

Applications that allow enterprises to monitor and manage events across the supply chain to plan their activities more effectively and pre-empt problems. SCIV systems enable enterprises not only to track and trace inventory globally on a line-item level, but also submit plans and receive alerts when events deviate from expectations. This

visibility into orders and shipments on a real-time basis gives enterprises reliable advance knowledge of when goods will arrive.

SCM (Select Component Manager)

SCM (semiconductor contract manufacturer)

SCM (software configuration management)

Also known as "software change management," SCM is a methodology for software problem/change request initiation and tracking; change impact analysis; version control; security administration of software assets; software promotion; quality reviews; and software distribution.

SCM (supply chain management)

The process of optimizing the delivery of goods, services and information from supplier to customer. SCM is a set of business processes that engages a trading-partner community in the common goal of satisfying the end customer. Functionally, SCM encompasses enterprise resource planning; manufacturing, warehouse and transportation management systems; and business intelligence and analysis.

SCO (The Santa Cruz Operation)

SCOPE (Supply Chain Optimization, Planning and Execution)

SCOR (Supply Chain Operations Reference Model)

SCP (service control point)

The database machine at the apex of the Signaling System 7 (SS7) network, containing information on how to process, route and bill calls.

SCP (Supply Chain Planner)

A supply-chain-planning tool from i2 Technologies (Irving, Texas).

SCP (supply chain planning)

A subset of supply chain management (SCM), SCP is the process of coordinating assets to optimize the delivery of goods, services and information from supplier to customer, balancing supply and demand. An SCP suite sits on top of a transactional system to provide planning, what-if scenario analysis capabilities and real-time demand commitments. Typical modules include network planning, capacity planning, demand planning, manufacturing planning and scheduling, distribution and deployment planning, and transportation planning and scheduling.

SCPC (single channel per carrier)

A transmission system in which a physical channel is allocated solely to one carrier for the duration of the transmission.

SCR (sustainable cell rate)

In asynchronous transfer mode (ATM), the average cell rate a source is allowed to maintain.

scrambler

A coding device applied to a digital channel that produces an apparently random bit sequence. A corresponding device is used to decode the channel, i.e., the coding is reversible.

screening

Prevention of electric, magnetic, or electromagnetic fields from escaping or entering an enclosed area by means of a barrier. Also called shielding.

screen popping

Populating a call center agent's screen with just-in-time customer information.

screen sharing

Similar to application sharing, but not all parties can update the document simultaneously.

scrolling

The vertical movement (up or down) of lines of data displayed on a cathode ray tube (CRT) screen.

SCS (sales configuration system)

SCSA (Signal Computing System Architecture)

A hardware and software architecture, defined by Dialogic, that specifies interfaces between components of a call-handling system, including media control. Media management is a missing component of Computer Supported Telephony Architecture (CSTA), Telephony Application Programming Interface (TAPI) and Telephony Services Application Programming Interface (TSAPI). Applications must make assumptions about the type of call (i.e., voice, data, fax or image) or interface with another piece of hardware to determine the type of call. SCSA can mesh with CSTA, TAPI and TSAPI. SCSA can provide computer-telephony integration (CTI) applications with the interfaces needed to identify and manage different call types.

SCSI (Small Computer Systems Interface)

A peripheral interface for up to seven devices. SCSI provides a high-speed, parallel data transfer of up to 40 megabits per second (Mbps) and has the advantage of connecting multiple peripherals while taking up only one slot in the computer.

SCSI Over IP (see iSCSI)

SCSL (Sun Community Source License)

The Sun Microsystems license model for Java. The new agreement leverages a community-based development model similar to "free software" initiatives that have existed in several forms for many years.

SCV (supply chain visibility)

SD (software distribution)

SDDI (Shielded Distributed Data Interface)

A proposal by IBM, and backed by several other vendors, for transmitting Fiber Distributed Data Interface (FDDI) signals over copper media. SDDI competes with

Copper Distributed Data Interface (CDDI), which runs on both shielded and unshielded twisted-pair media.

SDH (Synchronous Digital Hierarchy)

The European equivalent of the Synchronous Optical Network (SONET) standard, with transmission speeds and network management suited to the European standards.

SDML (Signed Document Markup Language)

SDK (software development kit)

A set of development utilities for writing software applications, usually associated with specific environments (e.g., the Windows SDK).

SDLC (Synchronous Data Link Control)

IBM's subset of High-Level Data Link Control (HDLC). The protocol is used for serial (by bit) information transfer over a data communications channel. Concerned with the lower two layers of the Open Systems Interconnection (OSI) model (physical and link control), SDLC was half of the Systems Network Architecture (SNA) attack on the diverse link controls, access methods and terminal types that existed in the pre-SNA era.

SDLC (systems development life cycle)

SDLT (SuperDLT-Quantum)

A storage product architecture that extends the capacity and transfer rate of digital linear tape (DLT) drives beyond the DLT 7000 drive.

SDM (System Data Mover)

The software component of extended remote copy (XRC) that runs under Multiple Virtual Storage (MVS).

SDMI (Secure Digital Music Initiative)

A framework to address new secure digital music technologies that will protect copyrighted material through all delivery channels. Announced on 15 December 1998 by the Recording Industry Association of America (RIAA) and endorsed by the major industry record labels and the technology community, the SDMI plans to create a digital music security specification. The SDMI's objectives are to answer consumer demand for online content access to quality recordings, while ensuring copyright protection for artists' work.

SDMS (Symmetrix Data Migration Service)

SDN (software-defined network)

SDRAM (synchronous dynamic random-access memory)

SDSAF (Switched Digital Services Applications Forum)

A consortium of equipment vendors, service vendors and users with the goal of advancing the state of switched digital services.

SDSL (symmetric digital subscriber line)

A digital subscriber line (DSL) technology that operates over voice-grade lines at 1.5 megabits per second one way, or 768 kilobits per second both ways, over an 8,000-foot distance.

SDV (switched digital video)

SEA (Software Engineers of America)

SEAL (Software-Optimized Encryption Algorithm)

A cipher designed and patented by IBM scientists for 32-bit processors.

Sealink

A version of xmodem.

SEA-ME-WE 2 (Southeast Asia-Middle East-Western Europe 2)

A fiber-optic submarine system operating on two fiber pairs at 560 megabits per second (Mbps), providing 16,000 nominal voice grade telephone circuits between Singapore and France.

search engine

A tool that enables Web pages to be found and retrieved based on the words they contain. Search engines use a searchable index of Web pages that is automatically updated by spiders or Web crawlers and housed on a central server connected to the Internet. Examples include Google, Yahoo and AltaVista.

search index

A machine-searchable matrix of documents used by a search engine.

SEAS (Signaling Engineering and Administration System)

Part of the Operations and Maintenance Application Part (OMAP) of Integrated Services Digital Network (ISDN).

SECAM (Sequential Couleur a Memoire)

A color television broadcasting system using 625 picture lines and a 50-hertz (Hz) field frequency, in which the two color-difference signals are transmitted sequentially instead of simultaneously. Developed and used in France and also used in the former Soviet Union. See NTSC and PAL.

SECC (Single Edge Contact Cartridge)

secondary channel

A low-speed channel established on a four-wire circuit over which diagnostic or control information is passed. User data is passed on the primary, high-speed channels of the circuit.

secondary station

On a communications pathway, a terminal device that has been selected to operate under the control of another terminal device.

Second-generation office systems

The class of multiuser time-sharing integrated office systems introduced around 1983, including Digital Equipment's ALL-IN-1, IBM's PROFS and Data General's CEO.

These products were largely superseded by client/server third-generation products of the 1990s.

secret key

The symmetric key used in secret-key cryptography. It is a secret shared between communicating parties, but is not truly private. See secret-key cryptography, private key and public-key cryptography.

secret-key cryptography

In this cryptography method (also known as symmetric-key cryptography), the single key needed to encrypt and decrypt messages is a shared secret between the communicating parties. The biggest problem with this method is that the secret key must be communicated through an external mechanism separate from the communication channel over which the encrypted text flows. In addition, secret-key systems do not support digital signatures. These limitations are addressed in public-key cryptography (see separate entry).

SECS (Semiconductor Equipment Communications Standards)

The oldest of the standards activities within the realm of manufacturing execution systems. Specific to the semiconductor manufacturing industry, the SECS collection of standards includes SECS I, SECS II and the SECS Messaging Standard.

Secure Digital Music Initiative (see SDMI)

Secure Electronic Network for Travelers' Rapid Inspection (see SENTRI)

Secure Electronic Transaction (see SET)

Secure Hash Algorithm (see SHA)

Secure Hypertext Transport Protocol (see S-HTTP)

Secure MIME (see S/MIME)

Secure Network Communications (see SNC)

Secure Sockets Layer (see SSL)

Secure Store and Forward (see SSF)

SED (stream editor)

A Unix text parser.

segment

A physical section of a network bounded by internetworking devices.

SEI (Software Engineering Institute)

A recognized source of research, process models and appropriate practices for application development. Based at Carnegie Mellon University, the SEI uses its Capability Maturity Model (CMM) as a framework for examining the adequacy of an enterprise's processes and improving those processes. It also provides hands-on experience to practitioners through groups such as software process improvement networks. See CMM.

selector

The last octet of an asynchronous transfer mode (ATM) address.

self-describing messages

A message that contains data as well as the metadata that describes the format and the meaning (i.e., the syntax and the semantics) of that data. For example, Extensible Markup Language (XML) is a self-describing message format that consists of tag/value pairs.

self-relocating program

A group of instructions capable of assigning blocks of memory as they are needed.

self-test and fault isolation

Most systems include a processor-check capability that allows the controlling computer to test itself and the rest of the system. If a fault is found, an alarm light is lit and a message is given on the system printer teletype, if one is provided. This feature also expedites service since the computer can pinpoint faulty equipment, saving diagnostic time.

sell side

Processes for companies to sell their products, including catalogs, transaction processors, payment processors, and supply chain management methods and tools.

SEM (Strategic Enterprise Management)

A suite of SAP analytical applications that supports integrated planning, decision making and performance monitoring. SEM uses multidimensional online analytical processing (OLAP) functionality to process data held in SAP's Business Information Warehouse (BIW).

semantic data model

A method of organizing data that reflects the basic meaning of data items and the relationships among them. This organization makes it easier to develop application programs and to maintain the consistency of data when it is updated.

Semantic Nets

A type of knowledge representation that uses nodes (representing objects or events) and links (representing relationships between those objects or events).

Semiconductor Equipment Communications Standards (see SECS)

SENTRI (Secure Electronic Network for Travelers' Rapid Inspection)

A face recognition system developed by Visionics under contract to the U.S. Department of Transportation (DOT) and the Immigration and Naturalization Service (INS) for U.S. ports of entry to Mexico and the Caribbean.

Sequenced Packet Exchange (see SPX)

Sequenced Packet Protocol (see SPP)

Sequential Couleur a Memoire (see SECAM)

sequential function chart (see SFC)

serial

One bit following another over a single pathway.

serial, impact, dot matrix (see SIDM)

serial, impact, fully formed (see SIFF)

serial, nonimpact, direct thermal (see SNDT)

serial, nonimpact ink jet (see SNIJ)

serial interface

An interconnection that transmits information bit by bit rather than a whole character at a time. It is much slower and cheaper than a parallel interface.

Serial Storage Architecture (see SSA)

serial transmission

A method whereby the bits of a character are sent sequentially on a single transmission channel. See parallel transmission.

server

A system or a program that receives requests from one or more client systems or programs to perform activities that allow the client to accomplish certain tasks. The term is usually applied to computers that provide specific services to other computers on a network. Routing servers connect subnetworks of like architecture; gateway servers connect networks of different architectures by performing protocol conversions; and terminal, printer and file servers provide interfaces between peripheral devices and systems on the network.

server appliance

A type of computing appliance that creates, manipulates or provides information to other network-connected computing devices. Unlike storage appliances, server appliances use an application context for the creation, manipulation or provision of information. Examples include Sun Microsystems' Cobalt Qube, Oracle's 8i Appliance and Dell Computer's PowerApp.web. Potential server appliance functions include database, e-mail, file storage and directory services. See computing appliance and storage appliance.

Service Advertising Protocol (see SAP)

service bureau

A company that processes various types of data for a client for a fee. Station message detail reporting (SMDR) and call-costing reports are typically provided.

service control point (see SCP)

service desk

A help desk that is equipped with the resources for resolving service requests and problem calls. It gives the customer service representative or end user the ability to efficiently diagnose, troubleshoot and correct technical-support problems, rather than being a "pass through."

service-level agreement (see SLA)

service-level management (see SLM)

service-level objective (see SLO)

Service-Level Reporter (see SLR)

service management agreement (see SMA)

service organization

Any company that contracts to provide for and operate computers not owned or leased by the company.

service-oriented architecture (see SOA)

Service Profile Identifier (see SPID)

service process optimization (see SPO)

service switching point (see SSP)

Services Tracking, Analysis and Reporting System (see STARS)

serving area

1. The region surrounding a broadcasting station where signal strength is at or above a stated minimum.
2. The geographic area handled by a telephone exchange, generally equivalent to a local access and transport area (LATA).

servlet

A form of server-based Java that operates in conjunction with a Web server and offers an alternative to using Common Gateway Interface (CGI) and server application programming interfaces (SAPIs) to communicate with Web server processes. In addition, servlets are independent of a given type of Web server, as the most prominent Web servers support servlets.

SES (severely errored second)

Any second with more than 1,544 bit errors.

SESAME (Secure European System for Applications in a Multivendor Environment)

session

1. A period of time in which an end user engages in dialog with an interactive computer system.
2. Layer 5 of the International Organization for Standardization's (ISO's) Open Systems Interconnection (OSI) reference model for network architectures.

Session Initiation Protocol (see SIP)

SET (Secure Electronic Transaction)

A standard conceived in 1995 by Visa and MasterCard to ensure that all Internet-based payment transaction details are encrypted, the parties authenticated, acknowledgments fully recorded and the customer payment details made available only to the bank. Due to its complexity and security flaws, SET failed to gain widespread acceptance. Meanwhile, Visa and MasterCard have pushed ahead with their own, separate standards: Verified by Visa and MasterCard Secure Payment Application.

severely errored second (see SES)

SF (single frequency)

SF (status field)

SF-36 (Short Form 36)

SFA (sales force automation)

Using technology to automate the sales process. Some technologies that can be used for sales automation are:

- Laptop computers
- Personal digital assistants
- Contact databases
- Interactive selling systems

SFC (sequential function chart)

A programming language for industrial logic controllers.

SFC (shop floor control)

A system of computers and/or controllers tools used to schedule, dispatch and track the progress of work orders through manufacturing based on defined routings. SFCs typically calculate work in process based on a percentage of completion for each order and operation that are useful in inventory valuations and materials planning.

SFDR (software failure detection and recovery)

Should support the concept of a transaction, including atomicity (either all changes take place or none take effect) to enable operating-system or application data recovery mechanisms to be implemented.

SFF-LR (small-form-factor, legacy-reduced)

Small footprint PC that has reduced support for legacy technologies such as mouse and keyboard ports, PCI slots, serial and parallel ports.

SFL (S/MIME Freeware Library)

SFM (store-and-forward manager)

A component that handles interapplication, asynchronous messaging for data consistency and multistep process coordination.

SFS (Structured File Server)

SG&A (selling, general and administrative)

SGML (Standard Generalized Markup Language)

An International Organization for Standardization (ISO) standard adopted in 1986 for the markup of electronic documents, identifying the basic structural elements of a text document. SGML addresses the structure of a document, not its format or presentation.

SGRAM (synchronous graphics random-access memory)

SHA (Secure Hash Algorithm)

Designed by the National Institute of Standards and Technology (NIST) and National Security Agency for use with the digital signature standard. Similar to Message Digest hash function MD5.

shadow/mirror databases

A system-level facility to enable shadowing or mirroring (i.e., duplication) of selected databases to a separate disk or disk set. The purpose is to minimize the space required for backup data while providing for the continuation of critical processing in the event of the loss of a disk containing related databases.

shared services

The centralization and consolidation of departments or multiple functions into a single organization. Shared services reduce personnel costs and build expertise. Consolidating to one central location or process lowers cost and increases efficiency.

shared-solution center (see SSC)

Shared Virtual Array (see SVA)

ShareNET

Formed by a number of clearinghouse participants in September 1997, ShareNET is a not-for-profit consortium promoting the creation of a virtual private network for healthcare electronic data interchange (EDI) administrative transactions. Initially, the strategy was to create direct links between participant networks. However, in February 1998, ShareNET augmented its initial connectivity strategy by selecting Electronic Data Systems (EDS) to develop a hub communications infrastructure.

shell

A cross-application operating environment accessed through a single user interface. A network connectivity shell is the environment used to access network resources — e.g., a suite with a single user interface for host, Transmission Control Protocol/Internet Protocol (TCP/IP) and Web access.

SHF (super-high frequency)

Denotes frequencies from 3 gigahertz (GHz) to 30 GHz.

Shielded Distributed Data Interface (see SDDI)

shielded pair

A pair of conductors in a cable that are wrapped with metallic foil designed to insulate the pair from interference.

shielded twisted pair (see STP)

shim

A software modification inserted into an operating system or application, to intercept the normal data flow and provide additional functionality. Often used by third party vendors to provide enhanced networking features.

Short Message Service (see SMS)

shrink-wrapped

1. A term used to refer to packaged software applications (from the shrink-wrapped packaging typical of such products).
2. Term initially used to describe an unsigned software license agreement that is deemed accepted when the user breaks a shrink-wrapped seal or opens an enclosed sealed envelope in the package containing the software media, such as a floppy disk or CD. Use of the term has been expanded to refer to software license agreements that are accepted electronically. See click-wrapped.

shop floor control (see SFC)

Shortest Path First (see SPF)

SHT (short holding time)

S-HTTP (Secure Hypertext Transport Protocol)

Also known as HTTPS, this is an extension of Hypertext Transport Protocol (HTTP) that provides security services for transaction confidentiality, authenticity and integrity between HTTP servers and clients. For the purposes of Internet browsers, S-HTTP is a competitive alternative to the more widely used Secure Sockets Layer (SSL) standard. Developed by Enterprise Integration Technologies (which was acquired by Verifone in 1995), S-HTTP was designed for use in browser applications, so it cannot be used to secure non-browser-based Internet applications.

SI (see system integration and system integrator)

SIA (Semiconductor Industry Association)

SIAC (Securities Industry Automation Corporation)

SIB (Securities and Investment Board)

SIC (Standard Industrial Code)

sideband

A frequency band on either the upper or lower side of the carrier frequency within which fall the frequencies produced by the process of modulation.

SIDM (serial, impact, dot matrix)

A type of printer, suitable for making carbon copies, but now largely obsolete.

SIFF (serial, impact, fully formed)

A type of printer using a character wheel similar to a typewriter. Suitable for making carbon copies, but now largely obsolete.

SIG (special-interest group)

signal

A physical, time-dependent energy value used for the purpose of conveying information through a transmission line.

Signal Computing System Architecture (see SCSA)

signal filtering

See "filtering."

signaling

The process by which a caller or equipment on the transmitting end of a line informs a particular party or equipment at the receiving end that a message is to be communicated.

Signaling, Engineering and Administration System (see SEAS)

Signaling, Protocols, and Switching (see SPS)

Signaling System 7 (see SS7)

signal-terminating equipment (see STE)

signal-to-noise ratio (see SNR)

signature

Any mark or symbol accepted by both parties to show intent, approval of, or responsibility for, a document. In e-business, for example, a "from" line on an e-mail, a mouse click of acceptance of terms, an e-mail closing, a biometric and electronic signatures of several types are accepted as signatures. Some laws specify a written signature to be legally acceptable. New laws give the same status to electronic signatures as written signatures.

signature verification

A biometric technique that uses characteristics of a person's signature (including pressure, pen lifts, speed and direction of pen strokes) to authenticate identity. Signature is less accurate than some other biometrics (e.g., fingerprints, iris) but is popular in document authentication applications that have traditionally used written signatures. Some growth may be fueled by the adoption of pen-based devices (e.g., the Palm Pilot) that can double as a signature input tablet.

significant digit

The integer of a number that represents the largest percentage.

sign off

Synonymous with "log off."

sign on

Synonymous with "log on."

SIIA (Software & Information Industry Association)

silicon on sapphire (see SOS)

SIM (software installation and maintenance)

SIM (subscriber identity module)

A small smart card containing a cellular service subscriber's identity key. SIMs enable users to swap phones without changing their subscriptions. The SIM is the key to security on Global System for Mobile Communications (GSM) networks. See GSM.

SIMAC

A Center for Facilitation of Procedures and Practices, Administration, Commerce and Transport (CEFACT) working group focused on simplifying electronic data interchange (EDI) for use by small to midsize businesses.

SIMD (single instruction, multiple data)

A design for parallel computers characterized by instructions that can directly trigger a large number (in parallel) of data operations on different data. Vector processors fall into this category.

SIMM (single in-line memory module)

A small printed circuit board that plugs into a socket on a personal computer and increases the available random-access memory (RAM).

Simple API for XML (see SAX)

Simple Internet Protocol Plus (see SIPP)

Simple Mail Transfer Protocol (see SMTP)

Simple Multicast Routing Protocol (see SMRP)

Simple Network Management Protocol (see SNMP)

Simple Object Access Protocol (see SOAP)

Simple Workflow Access Protocol (see SWAP)

simplex

Pertaining to the capability to transmit in one direction only. See half duplex and full duplex.

simplex circuit

A circuit permitting the transmission of signals in one specified direction only. Also known as single duplex.

SIMS (Sun Internet Mail Server)

simulation

The use of a mathematical or computer representation of a physical system for the purpose of studying constraint effects.

simulation routines

Various routines using historical information to simulate future alternatives for supply chain operations design. Usually strategically focused for use in future operations, these may then be optimized or prioritized.

SIN (status indication normal)

An Integrated Services Digital Network (ISDN) term for link test feedback indicating normal operation.

single channel per carrier (see SCPC)

single-image mode

A mode of operation in which multiple physical central processing units (CPUs) within a complex logically appear as one system running under control of a single copy of the operating system.

single in-line memory module (see SIMM)

single instruction, multiple data (see SIMD)

single large expensive disk (see SLED)

single-mode fiber

A fiber with a small core diameter allowing the propagation of a single light path.

single sideband (see SSB)

single sign-on (see SSO)

single threading

A group of instructions that completes the processing of one message before starting another. See multithreading.

Sinix

Siemens' version of Unix.

sink

The terminal connection that collects overflow transmissions on a communications pathway.

S-Interface

In Integrated Services Digital Network (ISDN), the standard four-wire connection between the terminal adapter and the network.

SIO (serial input/output)

SIO (strategic information office)

A business-unit-neutral information office whose role is to propagate the importance of enterprise information management to all business units, generate excitement for

these initiatives, negotiate organizational and technological issues across the enterprise, and enforce implementation of and compliance with standards at all levels of the enterprise. Members of the SIO should have in-depth knowledge of, and experience in, both business and technology, with at least one representative from senior management.

SIOS (status indication out of service)

An Integrated Services Digital Network (ISDN) term for an error condition in a tested circuit.

SIP (Session Initiation Protocol)

A protocol used to initiate interactive communication sessions of various types — including voice, video, chat, interactive games and virtual reality — between Internet users. A proposed standard of the Internet Engineering Task Force (IETF), the protocol is administered under the IETF's SIP Working Group.

SIP (SMDS Interface Protocol)

A frame-based protocol for access to a Switched Multimegabit Data Service (SMDS) by customer premises equipment based on Distributed Queue Dual Bus (DQDB) technology.

SIPP (Simple Internet Protocol Plus)

A proposed revision to the Internet Protocol (IP). IP must be modified to support an address threshold capable of supporting the exponential growth of the Internet. SIPP would increase the address space to eight bytes.

SIPRnet (Secret IP Router Network)

SIS (student information system)

A system used in academic environments to enroll and register students, and to track student information such as prior transcripts, courses taken, grades received and progress toward a degree.

SITA (Societe Internationale de Telecommunications Aeronautiques)

SITE (strategy, iteration, testing, enablement)

Information systems (IS) departments have a history in application development of following a design process of planning, prototyping, testing and development, resulting in justifiable design decisions. SITE is a Gartner concept that seeks to build on this long-established IS department practice by creating a structured framework within which both business unit (BU) managers and IS staff see clearly identified roles maximize their talents and skills. As a result, issues such as staffing, success factors, return on investment (ROI) guidelines, request for proposals (RFPs), interface design and graphic design are addressed in an environment where decisions are not arbitrary. Whether final execution is in-house or outsourced, adhering to SITE principles provides a project blueprint meaningful to multiple audiences.

Six Sigma

A management philosophy that strives to reduce out-of-specification product or service delivery to the Six Sigma (i.e., less than one-in-a-million) level. While ostensibly a quality program, Six Sigma is process-centric, as opposed to traditional statistical process control (SPC) activities, which are product-centric. In Six Sigma,

all business practices in manufacturing, not just the shop floor, are examined and steps are taken to reduce or eliminate variation. Common SPC tools, such as X-bar and R charts, are used, but when searching for assignable causes when failure rates are so low, extremely large volumes of data must be processed.

S/KEY

A one-time password authentication technology.

skill mining

A knowledge management (KM) functionality that automatically identifies the skills of knowledge workers by analyzing past behavior. This behavior may be implicit (e.g., looking for recurring concepts in documents that the worker has produced), or explicit (e.g., a worker's willingness and ability to answer a question in the past). Skill mining enables users to identify who in their enterprise has the expertise to address specific questions or problems.

skills-based management (see SBM)

SKIP (Simple Key Management for Internet Protocols)

SKU (stock-keeping unit)

A unique identification number that defines an item at the identifiable inventory level; for example, in retail applications, the SKU may designate style, size and color. A more detailed level would be at the serial number or unique identifier level.

SLA (service-level agreement)

An agreement that sets the expectations between the service provider and the customer and describes the products or services to be delivered, the single point of contact for end-user problems and the metrics by which the effectiveness of the process is monitored and approved.

slave

A terminal or computer controlled by another computer.

SLC (subscriber line charge)

An access charge paid by all telecommunications customers, developed by the Federal Communications Commission (FCC) to help local-exchange carriers (LECs) pay for their installed network infrastructure and for new network developments after the split-up of AT&T in 1984.

SLED (single large expensive disk)

Traditional disk drive. That is, a conventional large-system disk system that has, on average, almost twice the diameter of a lower-cost redundant array of independent disks (RAID) system and, as its name implies, is significantly more expensive to manufacture.

SLI (system-level integration)

SLIC (Subscriber Line Interface Circuit)

SLIP (Serial Line Internet Protocol)

SLM (service-level management)

The ongoing process of using service-level agreements (SLAs) to maintain high quality in the provision of services — and to ensure that service-level objectives (SLOs) and performance meet the changing needs of the recipient's business — through continuous improvement of service activities, functions and processes. See SLA and SLO.

SLM (software license management)

A mechanism for systematically ensuring compliance with vendors' software license restrictions (for example, maximum users or maximum nodes).

SLO (service-level objective)

Within service-level agreements (SLAs), SLOs are the objectives that must be achieved — for each service activity, function and process — to provide the best opportunity for service recipient success (see SLA).

SLOC (source lines of code)

SLR (Service-Level Reporter)

An IBM product that provides direct-access storage device (DASD) administrators with predefined storage report formats.

SLR (Single-Channel Linear Recording)

SLS (Signaling Link Selection)

SMA (service management agreement)

A tool for monitoring, measuring, managing and demonstrating the contribution of IT to the enterprise.

SMA (Systems Management Architecture)

small and midsize businesses (see SMBs)

Small Computer Systems Interface (see SCSI)

small-form-factor, legacy-reduced (see SFF-LR)

small office/home office (see SOHO)

small-systems consultant (see SSC)

Small Value Payments Co. (see SVPCo)

SMART (Self-Monitoring Analysis and Reporting Technology)

smart card

A plastic card that contains a microprocessor and a memory chip or just a memory chip. The microprocessor card has the ability to add, delete and manipulate information on the card. A memory-chip card, such as a phone card, can only add information. By maintaining all necessary functions and information on the card, smart cards do not require access to remote databases. Growing interest in smart cards crosses barriers of geography, industry and business functions. However, the

variety of smart card applications belies the fact that smart cards are capable of only three functions:

1. **Information Storage and Management:** Storing data off-line is an important feature of the smart card. It can maintain, for example, all of an individual's health-related information. Data can be stored and updated continuously either in a write-once format (e.g., a patient's blood type or known allergies) or in a read/write memory (e.g., a patient's last visit to the doctor). Memory can store information that does not change, build a history of transactions, or maintain a current data value (as in a phone card). The ability to add and subtract data value from the card makes it particularly useful in financial transactions that use electronic cash or credit and debit. The basic technique is the same whether it is applied to monetary values or frequent-flyer miles. Because the records are always in the user's possession, privacy is enhanced since duplicate information may be stored at a secure, off-line site in case the card is lost or damaged.
2. **Authentication:** Smart cards can manipulate the data they contain to enable real-time transactions. Authentication is the process that determines whether the card will allow a transaction.
3. **Encryption/Decryption:** Encryption changes data into a form that can be read only by the intended receiver. To decipher the message, the receiver of the encrypted data must have the proper decryption key. By its very nature, data encryption and decryption must take place in a local, secure environment. The capability of smart cards to perform this function within the embedded chip means that secure data does not have to be transferred from the card to a reader or other device. This reduces the chance that information can be stolen from the card.

smart phone

A large-screen, voice-centric handheld device designed to offer complete phone functions while simultaneously functioning as a personal digital assistant (PDA). See enhanced phone and PDA.

smart terminal

A display terminal that can operate in either conversational or block mode and can support a full range of local editing capabilities.

SMB (Server Message Block)

SMBs (small and midsize businesses)

Businesses which, due to their size, have different IT requirements — and often face different IT challenges — than large enterprises, and whose IT resources (usually budget and staff) are often highly constrained. For the purposes of its research, Gartner defines SMBs using the following approximate size categories:

- *Small business:* In the United States, less than \$50 million in annual revenue and up to 100 employees; in Europe, less than \$10 million in annual revenue and up to 75 employees
- *Midsize business:* In the United States, between \$50 million and \$300 million in annual revenue and between 100 and 1,000 employees; in Europe, between \$10 million and \$150 million in annual revenue and between 75 and 300 employees

SMC (Select Component Manager)

SMDR (station message detail recording)

See CDR (call detail recording)

SMDS (Switched Multimegabit Data Service)

A Bell Communications Research standard providing DS-1 and DS-3 access to a high-speed metropolitan-area network (MAN). It is based on the Institute of Electrical and Electronics Engineers (IEEE) 802.6 standard. SMDS is based on the data connectivity portion of the 802.6 MAN standard (but does not provide isochronous circuits). It is a high-volume, high-speed, switched digital service. Supported speeds range from 1.544 megabits per second (Mbps) to 45 Mbps (with 155 Mbps possible in the future).

Several local-exchange carriers (LECs) and interexchange carriers (IXCs) are now offering or testing SMDS services. Additionally, the service description for public networking includes an addressing capability, enabling users to interoperate with other SMDS users. Asynchronous transfer mode (ATM) is the underlying protocol of Broadband Integrated Services Digital Network (B-ISDN). It is dependent on the B-ISDN addressing scheme and must rationalize addressing schemes for non-B-ISDN implementations.

SMDS Interface Protocol (see SIP)

SME (small-to-midsize enterprise)

Another name for an SMB — see SMBs (small and midsize businesses).

SMF (Standard Message Format)

A file format protocol for NetWare Message Handling Service (MHS).

SMI (server management interface)

SMI (supplier-managed inventory)

See VMI (vendor-managed inventory).

SMIL (Synchronized Multimedia Integration Language)

One of the first four Extensible Markup Language (XML) specifications, approved by the Worldwide Web Consortium (W3C) in 1998, along with XML, Document Object Model (DOM) and Mathematical Markup Language (MathML). MIL Boston, under development in the W3C, adds functions, including support for television broadcasts, animation and interactivity as well as integration with Web standards. The primary goal of SMIL is to decrease the bandwidth required to transmit television-like multimedia content through the Web. If adopted, SMIL will greatly enhance the ability of commercial content providers to bring television-like consumer content to the Web.

S/MIME or SMIME (Secure MIME)

The Secure Multipurpose Internet Mail Extensions standard for secure messaging. Promoted by RSA Data Security, the S/MIME method for protecting and digitally signing e-mail encapsulates a message within a MIME envelope and uses Rivest-Shamir-Adelman's (RSA's) PKCS #7 to encrypt it. S/MIME specifies that X.509 certificates validate the correspondent's public keys and electronic signatures. These

certificates can come from the entity (i.e., self-certifying keys) or from a third-party certification authority service such as VeriSign.

SMLI (stateful multilayer inspection)

SMON (RFC 2613)

Officially called "Remote Network Monitoring MIB Extensions for Switched Networks Version 1.0," the Internet Engineering Task Force's (IETF's) request for comment (RFC) 2613 (SMON) continues the architecture created in the Remote Monitoring Management Information Base (RMON MIB) by providing RMON analysis for switched networks. Since current RMON and RMON2 standards were designed with shared media networks in mind and face limitations in today's switched networks, vendors of switching equipment have implemented proprietary monitoring mechanisms to overcome these limitations. RFC 2613 is an effort to standardize these mechanisms, and it addresses issues such as "copy port" and virtual LAN monitoring.

SMP (symmetric multiprocessing)

A multiprocessor architecture in which all processors are identical, share memory and execute both user code and operating-system code.

SMR (specialized mobile radio)

A wireless communications technology in competition with analog cellular services. In an SMR system, the base station equipment supplier is the licensee of the transmitters. Users have access to the multiple channels of the network rather than the limited number of channels of a private mobile radio network. Many users share all of the available channels. Sharing is accomplished on a first-come, first-served basis. When users want to initiate a call, they activate the push-to-talk button on the handset. Assuming the portable unit (and dispatcher or other portable unit) is tuned to an available channel, a communication path is established. If channels that the sender and receiver can use are not available, the call cannot be completed, and the operator must wait for another opportunity to try again.

SMRP (Simple Multicast Routing Protocol)

Routes multicast data streams in conjunction with multicast extensions to AppleTalk.

SMS (Shared Medical Systems)

SMS (Short Message Service)

A bidirectional paging function that is built into Global System for Telecommunications (GSM) systems. Each message can be up to 160 ASCII characters long. The network stores messages for up to several days (typically a maximum of 72 hours) and attempts to deliver the messages whenever the portable phone is switched on. Confirmation of receipt is available as an option in some networks.

SMS (Storage Management Subsystem)

A component of IBM's Multiple Virtual Storage/Data Facility Product (MVS/DFP) that is used to automate and centralize the management of storage by providing the storage administrator with control over data class, storage class, management class, storage group and ACS routine definitions.

SMS (system-managed storage)

The term used for the conceptualization of an architecture for attachment, management and reconfiguration of secondary storage. Among the SMS basic design goals is the separation of logical-device management from physical-device management.

SMS (Systems Management Server)

Systems management software that runs under Microsoft's Windows 2000 Server. It is used for distributing software, monitoring and analyzing network usage, and performing a variety of other network administration functions.

SMT (surface mount technology)

An integrated-circuit (IC) mounting system that uses flat pads instead of pins, eliminating the need to drill boards and making automated mounting easier.

SMTF (Symmetrix Multihost Transfer Facility)

SMTP (Simple Mail Transfer Protocol)

A messaging protocol governing electronic-mail transmission in Transmission Control Protocol/Internet Protocol (TCP/IP) networks. It is used to transfer e-mail between computers. It is a server-to-server protocol. SMTP supports only text and cannot handle attachments. It supports negative delivery notifications, not the positive notifications required by electronic data interchange (EDI).

SNA (Systems Network Architecture)

An IBM-developed layered network architecture. The layers isolate applications from system network services, enabling users to write applications independent of the lower networking software layer.

SNA Distribution Services (see SNADS)

SNADS (SNA Distribution Services)

An architecture for interchanging data through a Systems Network Architecture (SNA) network in a store-and-forward fashion. SNADS is an SNA extension that provides an asynchronous interface. It is mostly appropriate for applications without strong time requirements, such as document distribution, file transfer and e-mail. SNADS requires that Logical Unit (LU) 6.2, and the software that implements SNADS, reside in the same processor as the LU 6.2 support.

SNAP (System Network Assurance Program)

A security industry certification program established by the System Administration, Networking, and Security (SANS) organization. It consists of a standard series of briefings, courses and tasks used to demonstrate a more-detailed technical body of knowledge.

SNC (Secure Network Communications)

A SAP initiative intended to identify products that will protect the communication links between the distributed components of R/3 software. R/3 supports application-independent security products that adhere to the Generic Security Services Application Programming Interface (GSS-API) v.2 standard.

SNDT (serial, nonimpact, direct thermal)

A printer technology.

SNIA (Storage Networking Industry Association)

sniffer

A network management tool that monitors data packets on a network to help administrators ensure message integrity and service quality.

SNIJ (serial, nonimpact ink jet)

A printer technology.

SNMP (Simple Network Management Protocol)

A Transmission Control Protocol/Internet Protocol (TCP/IP)-derived protocol governing network management and the monitoring of network devices. Strictly speaking, SNMP is the Management Information Base (MIB) described in the SNMP standard; extensions to this MIB proposed by the Electronic Messaging Association permit the monitoring and reporting of all conforming messaging components through standard SNMP management tools for network components.

SNOMED (Systematized Nomenclature of Medicine)

A nomenclature created by the College of American Pathologists (CAP) for use in pathology. SNOMED has gradually been extended to cover other domains of medicine. It contains over 150,000 items and includes coverage for numerous medical specialties. CAP is also developing SNOMED RT for reference terminology and SNOMED CT for combined terminology.

SNR (signal-to-noise ratio)

The power of a signal relative to channel noise.

SNTT (serial, nonimpact, thermal transfer)

A printer technology.

SOA (service-oriented architecture)

A model for application construction that decouples server business functionality from client logic. The business functionality is organized as a collection of modules (services) each of which has a clear business identity and strict formal interfaces. The identity allows a requester to map a function to a service, and the formal interface enables any qualified requester to invoke the service without any information regarding its internal design. Behind the interface, the implementation of the service may be a single program or a flow of multiple programs and may include calls to a message broker.

SOAP (Simple Object Access Protocol)

A protocol introduced by Microsoft in conjunction with some small vendors. Designed to be simple, it creates transparent mapping of the Distributed Component Object Model (DCOM) interface definition language and Extensible Markup Language (XML) definitions. It provides the key transport technology for Web services, the next-generation paradigm for delivering applications as a set of Internet-enabled services (see Web services).

SOAP (subjective, objective, assessment and plan)

A process used by clinicians to identify and assess a patient's health status, and the subsequent treatments or course of therapy to improve this status.

SOC (software oversight committee)

A healthcare committee charged with ensuring that patients are not placed at risk by implemented software solutions. SOC's contain representative from all aspects of the healthcare enterprise. It has been suggested that SOC's be formed along the lines of the Joint Commission on Accreditation of Healthcare Organizations' (JCAHO's) institutional review boards (IRBs). IRBs typically are composed of risk management lawyers, medical ethicists, physicians and other interested parties within the care delivery organization (CDO). Because of the technical nature of the task, an SOC must have information systems (IS) representation. An SOC should meet on a regular basis to review all current and planned medical software implementations. In addition, it should form, and possibly enforce, a policy designed to ensure patient safety where medical software is concerned. Specific SOC responsibilities include validating that applications perform reliably and as expected, for example, by creating and running trial scenarios looking at both system and clinical functionality. A SOC should ensure that all decision support applications are based on accurate data, that they are implemented correctly and that the rule base is kept current. It should also examine how the various packages operate in tandem, paying close attention to combinations of functions that could potentially lead to errors that place patients at risk.

Society for Worldwide Interbank Financial Telecommunication (see SWIFT)

sockets

The Berkeley interprocess communications model. A socket specifies the end points of a two-way communications channel which connects two processes together so they can exchange information.

SOCKS

An NEC virtual private network (VPN) open protocol and Internet Engineering Task Force (IETF) standard.

SOD (statement of direction)

A vendor's announcement of future plans.

SODA (services-oriented development of applications)

softswitch

A type of network-switching technology. Softswitches are software-based products used to control communications networks.

software

Programs that control computer hardware. The two primary categories are system software (which governs the workings of the computer itself, such as the operating system and utilities) and application software (which performs specific tasks for the user, such as word processing, spreadsheets and accounts payable).

software asset management (see SAM)

Software Asset Management Interest Group (see SWAMI)

software configuration management (see SCM)

software development

Project management, specifications, design, programming, testing, installation and training associated with a specific application development project of any size.

software development kit (see SDK)

Software Engineering Institute (see SEI)

software failure detection and recovery (see SFDR)

software license management (see SLM)

software maintenance

Updating software, adding new functions, fixing bugs and solving problems. Technology vendors often sell a maintenance contract with their software. This contract is usually calculated as an annual fee based on some percentage of the total software cost. It generally provides for overall support and maintenance of a software product, including applications. Support may include telephone assistance time as well.

Software-Optimized Encryption Algorithm (see SEAL)

software oversight committee (see SOC)

Software Process Improvement Network (see SPIN)

software token

A software component that restricts the use of a specific ID/password combination to the machines on which the token is installed. Software tokens can be installed over the Internet on additional machines for a single session, or semipermanently, by providing the token server with the answers to a randomly ordered set of questions defined by the user at the time he or she initially enrolled in the authentication system. Such questions might include, "What is your mother's maiden name?" or "What was your roommate's nickname in your first year of college?"

SOGA (SNA Open Gateway Architecture)

SOHO (small office/home office)

A market segment for office equipment or computing peripherals (e.g., printers or copiers). Products targeted to the SOHO market tend to be lower in price and functionality than those designed to support large, corporate office environments.

SOI (silicon on insulator)

SOI (statement of intent)

solid-state device

Electronic pathways made of solid materials, e.g., chips.

solid-state disk (see SSD)

SOM (System Object Model)

IBM's object model designed to provide users with scalability, portability and interoperability of distributed applications across Hewlett-Packard and IBM platforms.

SONET (Synchronous Optical Network)

A International Telecommunications Union Telecommunications standard for synchronous transmission up to multigigabit speeds. The standard includes multivendor interoperability, improved troubleshooting and network survivability. As a Layer 1 standard, it is a foundation for Broadband Integrated Services Digital Network (B-ISDN) services.

S&OP (sales and operations planning)

A formal planning process used to determine the sales and operations strategy that best meets all aspects of the enterprise's objectives. The SOP process mediates sales, marketing, finance, operations and logistics objectives to formulate a single, achievable plan for the near-term — usually under one year.

SOP 98-1

A standard used to account for the cost of developing and maintaining software in-house. For financial statements prepared for fiscal years beginning after 15 December 1998, the American Institute of Certified Public Accountants (AICPA) early in 1998 approved the Statement of Position (SOP) 98-1. Under the guideline, maintaining legacy software is an internal maintenance expense — but if a company significantly enhances a system's functionality, under SOP 98-1 that work may be capitalized. Thus, portions of euro compliance efforts, for example, may be capitalized, depending on how the upgrade is done.

SOS (silicon on sapphire)

A semiconductor manufacturing term.

Source Address/Destination Address (see SA/DA)

Source Code Control System (see SCCS)

source encoding

A compression technique that takes into account the nature of the information to be compressed. See entropy encoding.

source routing

A technique used in local-area networks (LANs) in which the source of the frame specifies the route that the frame has to follow; the source furnishes a routing information field that designates the entire route to the destination.

source traffic descriptor

In asynchronous transfer mode (ATM), a set of parameters used during the setting up of a connection.

SOW (statement of work)

The statement of work includes an objectives section allowing the customer to emphasize the desired end state or performance metric to be achieved. It also mandates the assessment of past performance, technical approach and cost for each task order. The customer determines the relative importance of each criterion.

SOX (Schema for Object-Oriented XML)

SP (service pack)

A minor revision to a software product that contains feature updates or bug fixes, but not enough new code to warrant a new version number.

SP (subportal)

Provides highly targeted aggregate content and interactive capabilities and focuses on a specific vertical-market segment, as opposed to the broader market spectrum of portals (e.g., American Online, Excite or Microsoft Network). SPs are hybrid environments in that they can be deployed as intranet or Internet solutions, or a combination of both.

SP (system processor)

space diversity reception

To reduce the effects of fading and attenuation, a radio signal is received at more than one site, the sites being separated by a few wavelengths. The signals are then combined or selected.

space-division switching

A method for switching circuits in which each connection through the switch takes a physically separate path.

space segment

In a satellite circuit, the satellite itself, and necessary tracking, monitoring, and control functions. Excludes any ground equipment.

spam

Usenet messages flooded to many newsgroups indiscriminately. The term is also loosely applied to junk mail.

spamdexing

A slang term for the practice of hiding a slew of words or phrases in a Web page to increase the number of hits the page will register in Internet searches.

Spanning Tree

The Institute of Electrical and Electronics Engineers IEEE 802.1D committee standard for bridging local-area networks (LANs) without loops. The algorithm ensures only one path connects any pair of stations, selecting one bridge as the root bridge from which all paths are considered to radiate.

SPARC (Scalable Processor Architecture)

A reduced instruction set computer (RISC) architecture developed by Sun Microsystems and used in the Sun workstation family. SPARC chips are available from several semiconductor manufacturers, as Sun positions SPARC as a de facto standard.

spare/service parts planning (see SPP)

SPC (statistical process control)

Maintaining or improving process capability by employing statistical techniques to analyze process outputs and provide feedback for process control loops.

SPE (Sequenced Packet Exchange)

SPE (synchronous payload envelope)

See STS (synchronous transport signal).

speakerphone

A telephone device that has a speaker-microphone unit, allowing hands-free conversation. Also called a speakerset.

Speaker Recognition API (see SRAPI)

speaker verification

A biometrics-based alternative to typed passwords or personal identification numbers. A computer matches a person's spoken voice to an electronically captured voice sample (analogous to a unique fingerprint).

Speaker Verification API (see SVAPI)

SPEC (System Performance Evaluation Cooperative)

A vendor consortium that selects and standardizes benchmark programs submitted by members or others, for the purpose of rating and comparing the performance of processors.

SPECfp (SPEC floating point)

A test of floating-point computations, established by the System Performance Evaluation Cooperative (SPEC).

specialized carrier

A company that provides value-added or limited communications facilities.

specialized mobile radio (see SMR)

specialty center

A specialty center, or center of excellence, is a group of people dedicated to identifying best practices in an area of expertise and to building an internal service organization around that expertise. A center of excellence is a central clearinghouse for knowledge capital that is used across all business transformation projects. Specialty centers are an excellent means to leverage specialized skill sets, and are typically staffed to address a known demand.

specification, tracking, evaluation, production (see STEP)

SPECint (SPEC integer)

A System Performance Evaluation Cooperative (SPEC) benchmark to measure the integer performance of a processor.

spectrum

A continuous range of frequencies, usually wide in extent, within which waves have some specific common characteristics.

speech circuit

A circuit designed for the transmission of speech, either analog or encoded, but which can also be used for data transmission or telegraphy.

speech recognition

A computer's ability to convert spoken input to text. Special software takes vocal sounds, digitizes them and then compares them to a library of sound patterns. When matches are found, the computer can use those words just as if they had been typed on a keyboard. Speech recognition can improve personal productivity in the workplace, increase the effectiveness of interactive voice response (IVR) systems, and bring computing to untraditional environments (e.g., a hospital emergency room). The term "voice recognition" is also commonly used to refer to the ability of a machine or program to recognize and carry out voice commands or take dictation.

Speech recognition software generally falls into three categories along a continuum:

- Command systems were the earliest and simplest form; the computer learns a small number of voice commands like "open file" or "print document," freeing the user from having to use a keyboard or mouse to perform those tasks.
- Discrete speech recognition, the second stage in this evolution, can be used for dictation and other natural speaking conditions, but pauses are required between words.
- Continuous speech recognition software is emerging today. These systems understand natural speech without pauses, and their vocabularies and accuracy will continue to expand and improve.

speed dialing

A feature that enables a private branch exchange (PBX) or PBX station to store certain telephone numbers and dial them automatically when a short (one-, two- or three-digit) code is entered. Also called speed calling.

SPF (Shortest Path First)

A link-state protocol that uses a set of user-defined parameters to find the optimum route between two points. See RIP.

SPI (software process improvement)**SPID (Service Profile Identifier)**

In Integrated Services Digital Network (ISDN), a number assigned to an ISDN line by the provider.

spider

A piece of software (also called a Web crawler) designed to follow hyperlinks to their completion, and return information on Internet addresses passed.

SPIN (Software Process Improvement Network)

A best-practices group run by the Software Engineering Institute that provides hands-on experience to IT professionals.

splitter

A simple device that takes one input and sends it to several identical ports without amplification; commonly used residentially for sharing telephone lines between handsets and modems or for using two televisions on one antenna or cable point.

SPMD (single program, multiple data)

SPO (service process optimization)

Software designed to track and allocate the major resources of service companies or departments — people, intellectual capital and time. Most SPO applications consists of six core pieces of functionality:

- Project initiation (opportunity management)
- Engagement structuring
- Engagement execution (project management)
- Resource management
- Knowledge management
- A business-to-business exchange for procuring resources and collaborating with clients and colleagues

Professional services administration (PSA) is a related term, but one that applies only to external service providers (ESPs). SPO applies to internal service departments as well as ESPs. See PSA.

SPOC (single point of contact)

spoofing

1. A process whereby a router responds to keep alive messages from a host rather than passing them on the remote client, thus saving call charges. Used mainly in Integrated Services Digital Network (ISDN).
2. Of a packet, falsely claiming to be from an address different from that from which it actually originated.

spool

Simultaneous online peripherals operations; software that manages access to peripherals, such as printers.

spooling

Temporary storage of batch input and output until the processor is ready.

spot beam

A satellite signal that is concentrated on a small geographic area.

SPP (Sequenced Packet Protocol)

Xerox's connection-based, flow-controlled packet transmission for client processes; part of the XNX suite.

SPP (spare/service parts planning)

A specific aspect of supply chain planning dealing with parts used to repair or maintain products.

SPS (Signaling, Protocols and Switching)

One of the Conference on European Posts and Telecommunications' (CEPT's) working groups.

SPS (Standard Procurement System)

A procurement solution, based on a commercial package, to replace the U.S. Department of Defense's legacy systems with a single application capable of supporting the diverse computing needs of nearly 50,000 users across 900 contracting sites worldwide.

SPS/QCS (statistical process systems/quality control systems)

SPSS (statistical package for the social sciences)

SPX (Sequenced Packet Exchange)

A Novell NetWare session-based communications protocol used to govern the transport of messages across a network. It is designed to provide reliable end-to-end data transport, including error detection between two end-user devices.

SQA (software quality assurance)

SQC (statistical quality control)

SQL (Structured Query Language)

A relational data language that provides a consistent, English keyword-oriented set of facilities for query, data definition, data manipulation and data control. It is a programmed interface to relational database management systems (RDBMSs). IBM introduced SQL as the main external interface to its experimental RDBMS, System R, which it developed in the 1970s. SQL statements include:

- Data manipulation language statements: SELECT, INSERT, UPDATE and DELETE
- Data definition language statements, including the CREATE and DROP statements for tables and indexes. Statements that control data consistency, and grant and revoke authority.
- SQL statements are called "dynamic" when they are not completely specified until the program is executed. They are called "static" when they are completely specified when the program is compiled. SQL is precise, because it is based on predicate logic, but is difficult for average users to deal with, and its most fruitful position is as a protocol for software-to-software connectivity, rather than for human-to-software access.

SQL/400

The IBM relational database management system (RDBMS) for the AS/400.

SQL Access Group

A consortium of vendors set up in November 1989 to accelerate the development of Remote Data Access (RDA) standards, deliver protocols for interconnectivity among multiple SQL-based products, and define the application programming interfaces (APIs) necessary for the interoperability and portability of different SQL systems and applications.

SQL DB Access (SQL Database Access)

Availability of a relational interface to a midrange system database that is compliant with Level 1 of the American National Standards Institute (ANSI) X3.135-1986 or International Organization for Standardization (ISO) 9075 SQL standard. Both a

command (interpretive) and programming (compiled) implementation of the interface should be available.

SQL/DS (SQL/Data System)

SQLJ (Structured Query Language for Java)

The larger database management system (DBMS) vendors (e.g., Oracle and IBM) have proposed SQLJ as the standard way of using embedded Structured Query Language (SQL) within Java applications.

SQLNet

Middleware automatically bundled into Oracle6 and Oracle7 to enable fat-client, two-tier applications.

SRAM (static RAM)

SRAPI (Speaker Recognition API)

A biometric standardization group focused on speech biometrics.

SRDF (Symmetrix Remote Data Facility)

Implements controller-to-controller remote copy. It can operate in synchronous, semi-synchronous or adaptive mode.

SRM (storage resource management)

SRM (supplier relationship management)

A systematic approach to supplier evaluation, selection and ongoing relationship management, with the goal of cutting the costs of goods and services and boosting profits. SRM spans functional areas and enterprise boundaries, providing important decision support processes and functions to purchasing departments and key decision makers within the supply chain. It uses an extended-enterprise approach to squeeze waste out of the supply chain and to re-engineer the processes that link buyers and suppliers together.

SRM (System Resource Manager)

SRPI (server/requester programming interface)

SRT (Source Routing Transparent)

SS7 (Signaling System 7)

An out-of-band signaling scheme, typically with a 64-kilobit-per-second (Kbps) transmission rate. It is designed for digital networks, and is key to Integrated Services Digital Network (ISDN) implementations. Also called Common Channel Signaling No. 7 (CCS7).

SSA (Serial Storage Architecture)

IBM's disk storage technology and product line of disk subsystems. For most purposes, it is a single-sourced proprietary interface, especially useful for users who have standardized on RS/6000 systems with applications that require very high data rates that are not particularly cache-friendly.

SSA (System Software Associates)

SSAP (Statements of Standard Accounting Practice)

SSB (single sideband)

To make efficient use of the frequency band available, the carrier and the unwanted sideband of an amplitude-modulated wave can be filtered out so that only the sideband that contains all the information is transmitted. This is known as SSB transmission.

SSC (shared-solution center)

In Stage 1, an SSC is primarily a "showcase" for software demos and training. In Stage 2, the SSC can be used for "jump starts," while the client builds its own infrastructure. In Stage 3, certain project phases such as report development are done at an SSC. In Stage 4, more project phases including testing are done at the SSC, and the SSC reaches "scale" — at any given time 20 or more clients are being serviced from an SSC. Finally, in Stage 5, there are phase-based "factories" where testing or data conversion specialists and tools can be leveraged across multiple projects, whether on SAP, Oracle or other software.

SSC (small-systems consultant)

The SSC facilitates the transfer of skills needed by business unit (BU) developers from part-time developers to end users. The role of an SSC is to provide and enthusiastically promote easy-access, user-friendly application development (AD) consulting on a rapid, just-in-time basis. Although the SSC may be involved in the actual project work, his or her focus is to help the BU developers plan projects, model applications and "learn by doing."

SSCP (System Services Control Points)

Focal points of a Systems Network Architecture (SNA) network that contain the ability to manage configuration, coordinate network requests, and provide services.

SSD (solid-state disk)

A disk drive with no moving parts. It typically includes one or more circuit boards loaded with memory chips, and a controller that presents a standard disk drive interface to the computer system. Since it is on the peripheral side of the input/output (I/O) boundary, an SSD suffers the performance impacts of command overhead — typically less than 0.1 millisecond (ms) — and of I/O bus bandwidth. However, because it uses memory chips instead of mechanical access arms and rotating disk media, SSD eliminates the mechanical seek-and-rotation times that add 10 ms or more to the average access time of a hard-disk drive (HDD). Thus, while an HDD might deliver 100 I/Os per second, an SSD can deliver 4,000 I/Os per second. To emulate the permanence of magnetic disk storage, an SSD generally contains a battery backup power supply and an HDD that can save the contents of the memory in the event of a power failure.

SSEM (System and Server Evaluation Model)

A Gartner model that focuses on the six major categories of differentiation that exist between server platforms: performance and scalability, high availability, software vendor enthusiasm, platform architectural longevity, systems and network management software, and maintenance.

SSF (Secure Store and Forward)

A SAP initiative to certify third-party products. Security functions supported by SSF include application-level end-to-end security, smart-card authentication and single

sign-on. R/3 supports application-independent security products that adhere to the Generic Security Services Application Programming Interface (GSS-API) v.2 standard. SSF provides support for products that protect data and documents using such mechanisms as digital signatures and digital envelopes.

SSI (single-system image)

SSI (subsystem interface)

SSID (subsystem identification)

SSL (Secure Sockets Layer)

An Internet security standard developed by Netscape Communications. SSL offers session-level security; that is, after a secure session has been initiated, all information transmitted over the Internet during that session is encrypted. SSL also offers features such as server and client authentication as well as message integrity.

SSN (switched service network)

A telephone network consisting of terminals, transmission links, and at least one exchange, on which any user can communicate with any other user at any time.

SSO (single sign-on)

A term describing the function of consolidating the number of log-on processes for end users among various information technology (IT) resources.

SSP (service switching point)

The telecommunications switch in an intelligent network which handles a call with reference to supplementary routing and database information contained at a service control point (SCP).

SSP (storage service provider)

A company that provides computer storage capacity and management services. In addition to the storage itself, SSPs typically offer periodic backup and archiving, and some offer to consolidate data from multiple enterprise locations so that all locations can share the data effectively.

SSPC (structure, scripting, populating and channel outputs)

A framework for creating documents that requires four core mechanisms for defining the structure, scripting, populating (filling) and output channels of documents.

1. Structure definition consists of tools to define components (e.g., title, author, introduction, conclusion) and their sequence (e.g., "introductions" precede "conclusions"). The defined sequence cannot be modified or removed through editing.
2. Scripting is procedural logic that defines how to derive (i.e., calculate) content, and how to access or create queries that test and conditionally incorporate content from external sources (e.g., databases), other applications and content repositories.
3. Populating executes scripts against the structural definition to create an instance of the content model expressed in the document definition. This includes executing data and content retrieval, queries and computations to generate a

fleshed-out document (an “instance”) from the document skeleton defined by the structural and scripting elements.

4. Output channels control the content so that the same content can be formatted and composed for multiple media, such as print, Web or CD-ROM delivery.

SSR (smart switch router)

SSSWG (Storage System Standards Working Group)

A group that models, and develops standards for, generic mass storage systems.

ST (synchronous transport)

stack

An area in memory for the temporary storage of information. Information stored here is not retrieved by address, but rather in chronological order, last in-first out (LIFO).

stackable

Network devices that may be stacked one on another but where the stack is managed as one, as opposed to daisy-chaining, where each device is viewed separately by the network management system. Types of devices available in this format are hubs, switches, and routers.

staging area

An architectural database construct used to move data from one style of database to another (e.g., from an operational data store to a data warehouse or data mart). No user updating or analysis is performed on data in the staging area; rather, its contents are accessed only through programmatic means (e.g., by data transformation or cleansing utilities). The staging area may be fed by one or multiple sources, and its contents are normally refreshed and subsequently transformed or cleansed on a scheduled basis.

STAIRS (Storage and Information Retrieval System)

IBM's mainframe contextual search and library package (STAIRS/VS for MVS and STAIRS/CMS for VM).

stalking horses

Conceptual models used to test out new concepts and stimulate dialog about them.

standard

A document that recommends a protocol, interface, type of wiring, or some other aspect of a system. It may even recommend something as general as a conceptual framework or model (e.g., a communications architecture). De jure standards are developed by internationally or nationally recognized standards bodies or vendors. De facto standards are widely used vendor-developed protocols or architectures.

Standard for the Exchange of Product Model Data (see STEP)

Standard Generalized Markup Language (see SGML)

Standard Message Format (see SMF)

Standard Procurement System (see SPS)

standards

Specifications or styles that are widely accepted by users and adopted by several vendors. Standards are critical to the compatibility of hardware, software, and everything in between. Industry standards enable the essential elements of a computer and related infrastructure to work together. Standards provide specifications to hardware manufacturers and software developers that allow them to create products that will work together. Deviation from standards could result in the following problems:

- A plug on a keyboard does not fit into the related outlet on a computer.
- A piece of software does not work with a particular operating system.
- An Internet browser cannot read a certain page on the Web because the page is not formatted according the browser requirements.
- Proprietary software does not work on the Internet.

standby central control

A second control computer that can be provided to direct private branch exchange (PBX) operations if the primary one fails.

Starlan

A local network design and specification within IEEE 802.3 standards subcommittee, characterized by 1 megabit per second (Mbps) baseband data transmission over two-pair, twisted-pair wiring.

star network

A network topology in which each station is connected only to a central station by a point-to-point link and communicates with all other stations through the central station.

STARS (Services Tracking, Analysis and Reporting System)

A fraud and abuse detection, analysis and tracking system product of VIPS Healthcare Information Solutions.

start-stop (signaling)

Signaling in which each group of code elements corresponding to an alphabetical signal is preceded by a start signal (which serves to prepare the receiving mechanism for the reception and registration of character) and is followed by a stop signal (which serves to bring the receiving mechanism to rest in preparation for the reception of the next character).

statement of direction (see SOD)

statement of work (see SOW)

station

One of the input or output points of a communications system, e.g., a telephone set in the telephone system or the point where a business machine interfaces to a network, also called a node.

station message detail recording ((SMDR)

See CDR.

statistical multiplexing

A time-division multiplexing technique in which time slots are dynamically allocated on the basis of need (i.e., slots are allocated to equipment with data to be transmitted).

statistical process control (see SPC)

statistical quality control/statistical process control

A set of techniques based on statistical principles and methods used to regulate the quality of products and processes.

statistical time-division multiplexing (see STDM)

status indication normal (see SIN)

status indication out of service (see SIOS)

STDM (statistical time-division multiplexing)

Time shared dynamically between active channels on a multiplexer.

STE (signal-terminating equipment)

A network node used in the interconnection of public data networks.

STEP (specification, tracking, evaluation, production)

The four stages on which essential activities in the successful deployment of advanced technology are based. They are:

- Specification — which aligns corporate strategy and technology focus
- Tracking — which assesses individual technologies for maturity and business impact
- Evaluation — which involves prototyping and other in-depth evaluation activities to further gauge the readiness and relevance of the technology
- Production — which sees the technology being piloted and, if successful, rolled out into full deployment.

STEP (Standard for the Exchange of Product Model Data)

An international standards effort for defining the exchange of complete, unambiguous product model data. The goal of STEP is not only neutral file exchange, but also to serve as the basis for implementing, sharing and archiving entire product databases.

STI (self-timed interface)

stickiness

The general term applied to Web site qualities that attract and hold visitors. A sticky Web site is assumed to be offering higher value than one that is not sticky.

S/T interface

The electrical interface between a network terminator (NT1) device and one or more Integrated Services Digital Network (ISDN) communications devices that do not contain their own NT1s. A U.S. term.

STM (synchronous transfer mode)

A technique for multiplexing several circuits over transmission links and switches whereby time is divided into slots or buckets, and circuits are given time units whether or not they have any data to transmit. See asynchronous transfer mode.

STN (super-twisted nematic)

An LCD display-screen technology

stock-keeping unit (see SKU)

stop bit

The last bit of a character in asynchronous serial transmission, used to ensure recognition of the next start element.

stop element

In asynchronous transmission, the quiescent state following the transmission of a character. It is usually required to be at least 1, 1.42, 1.5 or 2 bit times long.

stop word

A word ignored by a search engine. Different document bases and search engines might employ different stop words, although most stop-word lists include common conjunctions, articles and prepositions (such as "and," "or," "the," "a," and "of" in English-language documents). See "search engine."

store-and-forward manager (see SFM)

Storage and Information Retrieval System (see STAIRS)

storage appliance

A type of computing appliance that provides data to, or manages data for, other network-connected computing devices. Unlike server appliances, storage appliances provide or manage data without an application context. This category of computing devices includes network-attached storage (NAS) and storage-area network (SAN) devices. See computing appliance, server appliance, NAS and SAN.

storage area network (see SAN)

Storage Management Subsystem (see SMS)

storage service provider (see SSP)

store and forward

A type of packet switching where the whole frame is processed before further transmission.

stored procedures

A limited form of remote procedure call (RPC) that enables procedural code (e.g., referential integrity constraints) to be stored and, thus, shared in the kernel of the

database management system (DBMS). The DBMS provides a facility for storing procedural code associated with the database, and enforces its use during any database operation.

STP (shielded twisted pair)

A pair of twisted transmission wires with a metallic shield to minimize interference.

STP (signal transfer point)

STP (straight-through processing)

A popular strategy for financial institutions that is also pursued by some utility companies under the name “flow-through processing.” Its fundamental premise is that a transaction, such as a payment, a trade, or a change to a residential phone service, is entered only once. Thereafter, it proceeds in an automated fashion through the rest of its life cycle, which may include dozens of steps in different application systems in various locations. The goal is to avoid re-keying information, thereby reducing input errors and shortening the time to completion.

straight-through cable

A cable with wires that have terminating ends with the same wire assignments. See crossover cable.

straight-through processing (see STP)

Strategic Enterprise Management (see SEM)

strategic information office (see SIO)

strategic sourcing

The ongoing analysis of the sourcing of products and services by an enterprise. Strategic sourcing focuses on determining the right mix of products and services — and the appropriate suppliers, terms and conditions — that will balance cost, quality and risk for the enterprise.

strategy, iteration, testing, enablement (see SITE)

stream editor (see SED)

streaming

One-way (either point-to-point or broadcast to multiple receivers) transmission of video and audio contents over the Internet or advanced wideband wireless networks.

structure, scripting, populating and channel outputs (see SSPC)

Structured Query Language (see SQL)

STS (Synchronous Transport Signal)

The following terms are associated with STS signaling:

1. STS-1: The basic SONET logical building block signal, with a transmission rate of 51.84 megabits per second (Mbps).

2. STS-3c: SONET format specifying the frame structure for the lines used to carry asynchronous transfer mode (ATM) cells.
3. STS-n: Signal created with byte interleaving of "N" STS-1 signals together (SONET term). The transmission rate is "N" times the STS-1 rate of 51.84 Mbps. See STS-1.
4. STS-Nc: Concatenated synchronous transport signal level N. An STS-N line layer signal combining the STS envelope capabilities from the "N" STS-1s forming an STS-Nc synchronous payload envelope, which is transported as a single entity.
5. STS envelope capacity: The total bandwidth within the STS frame carrying the STS synchronous payload envelope (SPE). Total bandwidth is the combination of "N" STS-1s required to carry an STS-Nc SPE.
6. STS payload capacity: The maximum bandwidth of an STS SPE.
7. STS Path Overhead (POH): Starting with the first byte of an STS SPE, nine evenly distributed bytes per 125 microseconds. The STS POH provides the communication from the creation of an STS SPE to its disassembly.
8. STS Path Terminating Equipment (PTE): Network devices that multiplex and demultiplex the STS payload. These devices can originate, terminate, and modify the STS POH as required to transport the STS payload.
9. STS synchronous payload envelope (SPE): A generic term referring to both STS-1 STS-Nc SPEs, which represent a 125-microsecond frame structure composed of STS POH and STS payload capacity.

STTL (Schottky transistor-transistor logic)

student information system (see SIS)

subjective, objective, assessment and plan (see SOAP)

subnet

A portion of network that may be physically independent of another network portion, but both portions of the network share the same network address, and the portion is distinguished by a subnet number.

subnet mask

The bits of an Internet Protocol (IP) address used for a subnetwork.

subnet number

The portion of the Internet address that designates a subnet. It is used for intranet routing but is ignored for Internet-routing purposes.

subscriber identity module (see SIM)

subscriber line charge (see SLC)

sub-voice-grade channel

A channel with a bandwidth narrower than that of voice-grade channels. Such channels are usually subchannels of a voice-grade line.

suite

A group of programs designed to work together, and which often use the same underlying architecture.

Summit

Also referred to as FS/Summit. Summit is the name used to denote the IBM high-end processor series that follows the 3090. The FS/Summit processors are the ES/9000 Models 820 and 900.

Sun Community Source License (see SCSL)

SunOS (Sun Microsystems Operating System)

Based on Berkeley 4.2; merged with AT&T's SVR4 Unix operating system.

SuperDLT-Quantum (see SDLT)

Superframe

In T1 lines, a set of 12 frames for synchronous transmission.

supergroup

An assembly of five 12-channel groups occupying adjacent bands in the spectrum for the purpose of simultaneous modulation or demodulation, i.e., 60 voice channels.

super-high frequency (see SHF)

super-rate signal

A signal carried only by an STS-Nc (SONET term). See STS-Nc.

superserver

Multiprocessor machines designed as network servers, combining input/output (I/O) capabilities of minicomputers with the capability to run industry-standard network operating systems.

Super Video Graphics Array (see SVGA)

supervisor

The part of an operating system that coordinates the use of resources and maintains the flow of central processing unit (CPU) operations.

supervisory control

Characters or signals that automatically actuate equipment or indicators at a remote terminal.

supervisory control and data acquisition (see SCADA)

supervisory programs

Computer programs designed to coordinate, service, and augment the machine components of the system and coordinate and service application programs. They handle work scheduling, input/output operations, error actions, and other functions. Also known as operating systems.

supplier-managed inventory (SMI)

See vendor-managed inventory (VMI).

supplier relationship management (see SRM)

supply chain execution (see SCE)

supply chain inventory visibility (see SCIV)

supply chain management (see SCM)

Supply Chain Planner (see SCP)

supply chain planning (see SCP)

Supra

Cincom Systems' overall environment that includes productivity tools and database managers.

Supranet

A Gartner term describing the emerging, ubiquitous network infrastructure that links the "e-world" (i.e., the world of electronic devices such as computers, phones and televisions) and the "p-world" (i.e., the physical world of paper, houses, people, vehicles and other objects) within natural human interactions.

The Supranet is enabled by four key phenomena:

- Embedded computers in many everyday objects
- Next-generation wireless networking, providing global indoor and outdoor connectivity to the Internet
- Interfacing technologies that enable bidirectional communication between p-world and e-world components (e.g., bar code scanning, speech recognition and electronic identification)
- The design of applications that satisfy user needs in a natural way with combinations of media and devices

surface mount technology (see SMT)

sustainable cell rate (see SCR)

SVA (Shared Virtual Array)

A direct-access storage device (DASD) product introduced by Storage Technology (StorageTek) in 1999. SVA is based on the same technology as the RAMAC Virtual Array, an IBM-branded storage system for which StorageTek served as the original equipment manufacturer.

SVAPI (Speaker Verification API)

A committee that is an offspring of Speaker Recognition API (SRAPI) founded and spearheaded by Novell. This committee has prepared a Human Authentication Application Programming Interface (HA-API) translation dealing with low-level interfaces. The specifications of the committee are aimed at the voice verification industry and are not vendor-specific.

SVC (switched virtual circuit)

A virtual connection with a call establishment and tear-down procedure to allow its temporary use.

SVDR (simultaneous voice and data recording)

SVG interface (Scalable Vector Graphics interface)

An imaging application and language written in XML, supported by key electronic publishing vendors. The SVG interface offers a solution to the problem of sharing many sophisticated, Web-based images. Vector graphics are more compact than bitmapped images and can potentially be altered by client devices to best suit display parameters.

SVGA (Super Video Graphics Array)

A Video Electronics Standards Association (VESA) display standard that provides higher resolution than the 640 x 480 Video Graphics Array (VGA) standard. It can support as many as 16 million colors, depending on the computer system and amount of available memory.

SVID (System V Interface Definition)

The specifications for Unix System V, set by AT&T. The fear that SVID might become a de facto standard, though not vendor-independent, was one reason for the creation of the Open Software Foundation (OSF).

SVPCo (Small Value Payments Co.)

A banking consortium, formed in July 1998, that is governed by 12 large U.S. banks and that plans to extend its board to 20 members. SVPCo is concentrating on Electronic Check Presentment (ECP), which requires cooperation from many banks in order to succeed.

SVR4 (System V Release 4)

SW (see software)

SWAMI (Software Asset Management Interest Group)

A worldwide network of influential users who are helping to shape more equitable software business practices and value-based pricing and licensing models. Working together, SWAMI members strive to understand and affect software licensing terms that provide long-term price protection and facilitate best-in-class deals. SWAMI holds general sessions annually in the United States and Europe.

SWAP (Simple Workflow Access Protocol)

An Internet-based protocol designed to provide a Hypertext Transport Protocol (HTTP)-based way to access a generic workflow service or a workflow enabled process or to interoperate with it. SWAP, which has been endorsed by the Workflow Management Coalition (WfMC), will offer a set of standard definitions for the type and structure of messages passed between cooperating workflow systems. SWAP will allow a workflow system to start, monitor, exchange data with and control workflow instances on a different workflow system. It will also provide a way to integrate workflow systems with other Web-based services.

SWIFT (Society for Worldwide Interbank Financial Telecommunication)

A self-describing messaging format that is used within the banking and finance industry to support electronic funds transfer. SWIFT is also the name of the format's creator, an organization that provides messaging and transaction-processing services to member banks and other financial organizations, including brokers, securities depositories, clearing organizations and stock exchanges.

switch

A device that makes, breaks, or changes the connections in an electrical circuit; to shift to another electrical circuit by means of a switch. In the telecommunications industry, the term is often used as a synonym for private branch exchange (PBX) or central office (CO) switch.

Switched 56

A U.S. circuit-switched service offering transmission at 56 kilobits per second.

Switched Digital Services Applications Forum (see SDSAF)

switched line

One of a series of lines that can be interconnected through a switching center; a line on the public telephone network.

Switched Multimegabit Data Service (see SMDS)

switched network

A multipoint communications pathway with circuit-switching capabilities, e.g., the telephone network.

switched service network (see SSN)

switched virtual circuit (see SVC)

switching

The establishment of a transmission path from a particular inlet to a particular outlet, within a group of such inlets and outlets.

switching center

A location that terminates multiple circuits and is capable of interconnecting circuits or transferring traffic between circuits.

switching matrix (see matrix)

switchover

When a failure occurs in the equipment, a switch to an alternative component can occur. Also called failover.

SWV (sampled write verify)

Symbian

A joint-venture company started in 1998 by Psion, Ericsson, Nokia and Motorola to develop Psion's EPOC 32 operating system into a real-time operating system for handheld phones and PDAs. Matsushita, NTT DoCoMo, Siemens and Sun

Microsystems have since joined. Its first products were released in 1999. Symbian has released reference designs and a software developer's kit.

symmetric cryptography

See secret-key cryptography.

symmetric digital subscriber line (see SDSL)

symmetric-key encryption

See secret-key cryptography.

symmetric multiprocessing (see SMP)

Symmetrix Remote Data Facility (see SRDF)

Synchronized Multimedia Integration Language (see SMIL)

synchronization

Establishment of common timing between sending and receiving equipment.

Synchronization Markup Language (see SyncML)

synchronous

Having a constant time interval between successive bits, characters or events. Synchronous transmission uses no redundant information to identify the beginning and end of characters, and is faster and more efficient than asynchronous transmission, which uses start and stop bits. The timing is achieved by transmitting sync characters prior to data; usually synchronization can be achieved in two-or three-character times.

synchronous communications

High-speed transmission of contiguous groups of characters. The stream of monitored and read bits uses a clock rate.

Synchronous Data Link Control (see SDLC)

Synchronous Digital Hierarchy (see SDH)

synchronous network

A network in which all the communications links are synchronized to a common clock.

Synchronous Optical Network (see SONET)

synchronous transfer mode (see STM)

synchronous transmission

Timing is achieved by transmitting sync characters prior to data. It does not use such redundant information as the start and stop bits in asynchronous transmission to identify the beginning and end of characters and is thus faster and more efficient.

Synchronous Transport Signal (see STS)

SyncML (Synchronization Markup Language)

The SyncML initiative aims to develop a uniform synchronization protocol — operating on any device, over any network — among various vendors' products. The protocol will provide support for a wide range of transports and media types. For the growing requirement of Web-based synchronization, SyncML will support both Hypertext Transport Protocol (see HTTP) and Extensible Markup Language (see XML). Support will also be provided for Wireless Session Protocol (see WSP), which is used widely in today's wireless phones. Short-range wireless technologies, such as Bluetooth and infrared, will also be supported through the Object Exchange protocol. In addition, support for several e-mail protocols and data types will be provided.

syntax

Message format or grammar (e.g., field lengths and delineators, headers, footers and optional fields).

Sysplex

A system complex of closely coupled Multiple Virtual Storage (MVS) processors sharing status information over a channel-to-channel link using the MVS Cross-System Coupling Facility (XCF) component. Sysplex became available with the first release of MVS/ESA SP 4.

System/370

A generic term for IBM's older mainframe architecture. Families with 370 architecture include 370/xxx, 9370, 43xx, 303x and 3090.

System/390

The generic term for IBM's processor family announced in September 1990, which replaced the S/370 family. S/390 consists of the ES/9000 mainframes.

System Administration, Networking and Security (see SANS)

System and Server Evaluation Model (see SSEM)

Systems Application Architecture (see SAA)

System Architect (see SA)

System Control Element (see SCE)

System Data Mover (see SDM)

system-managed storage (see SMS)

system management

Any of a number of "housekeeping" activities intended to preserve, maintain or correct the operation of a computer system. Included are such routine but critical processes as hardware diagnostics, software distribution, backup and recovery, file and disk integrity checking, and virus scanning.

System Network Assurance Program (see SNAP)

System Object Model (see SOM)

System Performance Evaluation Cooperative (see SPEC)

System R

Prototype version of the relational model developed by IBM, of which the important result was the precursor of Structured Query Language (SQL).

System Services Control Points (see SSCP)

Systematized Nomenclature of Medicine (see SNOMED)

systems administration management tools (see SAM tools)

system integration

The process of creating a complex information system that may include designing or building a customized architecture or application, integrating it with new or existing hardware, packaged and custom software, and communications. Most enterprises rely on an external contractor for program management of most or all phases of system development (see system integrator).

system integrator

An organization or an individual that performs system integration. Major system integration projects often require the assistance of a specialty firm that has the resources and expertise to manage a project plan that could last over several months or even years. This external vendor generally also assumes a high degree of the project's risks.

Systems Management Server (see SMS)

Systems Network Architecture (see SNA)

System V

AT&T's Unix operating system, introduced in 1983.

SystemView

An IBM network and systems management product.

System V Interface Definition (see SVID)

T

T1/T1C/T2/T3/T4

A digital carrier system introduced in the United States by the Bell System in the 1960s. It was the first to successfully support DS-1 formatted digitized voice transmission that connected directly to a network. T1 and T3 lines are often used in Internet service provider (ISP) connections to the Internet

T-series speeds include:

1. T1: 1.544 megabits per second (Mbps)
2. T1C: 3.152 Mbps
3. T2: 6.312 Mbps
4. T3: 44.746 Mbps
5. T4: 274.176 Mbps

T.120

A key standard for real-time document sharing ratified by the International Telecommunication Union (ITU).

TA (telecommunications administration)

Also called a PTT (postal, telephone and telegraph), a TA is an agency in a European or Asian country that is usually a government-run monopoly. It is responsible for all postal, telegraph and telephone services. It also has certain regulatory functions for telecommunications, such as providing specifications for modems and certifying that equipment may connect to the telephone network. In some countries, the TA also provides its own line of equipment, e.g., modems. TAs are undergoing varying amounts of structural change.

TA (terminal adapter)

An interfacing device for an Integrated Services Digital Network (ISDN) environment. It enables non-ISDN terminals to connect at the physical layer.

T&A (time and attendance)

Business applications used for the collection and tracking of hours worked, scheduled time and nonproductive work time. These applications support the payroll process, project accounting, workforce planning and other business processes.

TAB (tape-automated bonding)

table driven

1. A logical computer process, widespread in the operation of communications devices and networks, where a user-entered variable is matched against an array of predefined values.
2. A frequently used logical process in network routing, access security and modem operation.

tablet

A computing device that weighs less than 4 pounds and is operated by direct screen contact via a pen or touch interface.

TACACS (Terminal Access Controller Access Control System)

An authentication protocol from Digital Data Network (DDN) providing remote access security and related services.

tacit knowledge

The personal knowledge resident within the mind, behavior and perceptions of individuals. Tacit knowledge includes skills, experiences, insight, intuition and judgment. Tacit knowledge is typically shared through discussion, stories, analogies and person-to-person interaction and is, therefore, difficult to capture or represent in explicit form. Because individuals continually add personal knowledge, which changes behavior and perceptions, tacit knowledge is, by definition, uncaptured.

TACS (Total Access Communications System)

A derivative of the AT&T-developed analog cellular radio standard AMPS (Advanced Mobile Phone Service) adopted by U.K. Cellnet and Racal-Vodaphone, operating at 900 megahertz (MHz).

T&E (travel and expense)

The management of employee expense report processing.

Tag Control Information (see TCI)

Tag Image File Format (see TIFF)

Tag Protocol Identifier (see TPID)

tail

In satellite networks, any terrestrial extension used to connect end users to an earth station.

TAM (technical account manager)

tandem

The connection of networks or circuits in series, i.e., the connection of the output of one circuit to the input of another.

tandem data circuit

A data circuit that contains two or more sets of data circuit-terminating equipment in series.

tandem tie trunk network (see TTTN)

Tap1

Ethernet connection for coaxial wiring.

Tap2

Connection that can be in series or parallel to the main transmission media.

Tape Cartridge License

A format-specific Linear Tape Open (LTO) license package for media manufacturers.

Tape Mechanism License

A format-specific Linear Tape Open (LTO) license package for manufacturers that want to produce the drive.

TAPI (Telephony Application Programming Interface)

An interface standard from Microsoft and Intel that brings a well-defined computer/telephone application programming interface (API) to the desktop. The desktop linkage for TAPI can be accomplished via serial RS-232 interfaces. Additionally, it also allows for a board associated with the telephone to plug into the desktop computer's bus. The telephone station does not disappear from the desktop for several reasons (e.g., computer and application uptime, and handset requirement). The stable APIs that exist between Windows and the Nx64 Kbps circuit-switching systems are not limited to voice; they can accommodate data, images and video.

tariff

The formal process whereby services and rates are established by and for communications common carriers. Carriers submit tariffs to the appropriate regulatory agencies, which then review, amend and approve or disallow them.

TAT (transatlantic telephone cable)

TAXI (transparent asynchronous transmitter/receiver interface)

A 100-megabit-per-second (Mbps) multimode fiber interface.

TB (terabyte)

One trillion bytes.

TB/admin (terabytes-per-administrator ratio)

A metric used in data center storage management. The ratio is not a measurement of efficiency. It is merely a way of relating labor costs to the size of an organization's direct-access storage device (DASD) installed base.

T&C (terms and conditions)

TCA (Telecommunications Association)

TCAM (Telecommunications Access Method)

An IBM macro language for creating communication application programs and message control.

TCAP (Transaction Capabilities Application Part)

In Integrated Services Digital Network (ISDN) feature for transaction-type information transfer.

T-carrier services

High-speed, private-line digital transmission services that provide large-volume transmission between two stations. T1, the most common T-carrier service, operates at an aggregate data rate of 1.544 megabits per second (Mbps) and consists of 24 separate channels. (T2 and T3 are rated at 6.312 Mbps and 44 Mbps, respectively.) T1 channels are 6 to 30 times more cost-effective on a per-bit basis than voice-grade lines, but users must be able to fill the 1.544 Mbps channel to take full advantage of savings. Fractional T services, however, enable businesses to lease a fraction of a T line. Only T1 and T3 are widely supported. In Europe, e-carrier services are provided, although the corresponding data rates are not identical.

TCB (transaction control block)

TCF (transparent computing facility)

TCI (Tag Control Information)

Part of the virtual local-area network (VLAN) Tag Header in the Institute of Electrical and Electronics Engineers (IEEE) 802.1q draft standard.

TCM (Thermal Conduction Module or Thermo-Conductive Module)

The basic building block technology of the IBM 308X, 3090 and ES/9000. The TCM is a sealed container enclosing a multiple-chip carrying substrate amid a chip interconnection structure.

TCMP (tightly coupled multiprocessing)

A configuration in which all processors share a single pool of memory.

TCO (total cost of ownership)

A comprehensive assessment of information technology (IT) or other costs across enterprise boundaries over time. For IT, TCO includes hardware and software acquisition, management and support, communications, end-user expenses and the opportunity cost of downtime, training and other productivity losses.

TCP (Transmission Control Protocol)

A communications protocol based on the U.S. Department of Defense's standards for reliable internetwork delivery of data.

TCP/IP (Transmission Control Protocol/Internet Protocol)

A set of protocols covering (approximately) the network and transport layers of the seven-layer Open Systems Interconnection (OSI) network model. TCP/IP was developed during a 15-year period under the auspices of the U.S. Department of Defense. It has achieved de facto standard status, particularly as higher-level layers over Ethernet.

TCP/UDP (Transmission Control Protocol/User Datagram Protocol)

TDI (technical data interchange)

TDM (time division multiplexing)

A data, voice and video communications technique that interleaves several low-speed signals into one high-speed transmission channel.

TDMA (time division multiple access)

A technology for digital transmission of radio signals between, for example, a mobile phone and a radio base station. TDMA breaks signals into sequential pieces of defined length, places each piece into an information conduit at specific intervals and then reconstructs the pieces at the end of the conduit.

TDMF (Transparent Data Migration Facility)

An Amdahl-developed Multiple Virtual Storage (MVS) application that migrates data between geometrically similar mainframes from different vendors.

TDOA (time difference of arrival)

Based on triangulation, a method of processing cellular phone signals, TDOA allows the physical position of switched on phones to be located. The position is determined by comparing the time difference of the arrival of the existing reverse control channel at various cell sites.

TDR (time domain reflectometry)

A technique used to track faults in networks, such as cable breaks or loose connections. A pulse of a known shape is transmitted over the network and an echo is created when the pulse hits an obstacle or cable end. The time lapsed between the sending of the pulse and receiving the echo can be used to locate the origin of the echo and therefore the break. Sophisticated testing equipment is available to perform this test and to analyze the results.

TDS (telecommunications data systems)

A frequency allocation technique based on allotting discrete time slots to users, permitting many simultaneous transmissions.

TE (Terminal Equipment)

One of the Conference on European Posts and Telecommunications' (CEPT's) working groups.

tearing

The effect of removing a smart card too quickly from a reader during a transaction: the transaction fails and the reader may stall.

TEB (technology-enabled buying)

An integrated set of applications that improves both Web-buyer effectiveness and sales management efficiency. TEB solutions employ a mix of software applications, databases, middleware, operating systems and hardware environments. TEB can help enterprises to integrate their Web-buying sites with other distribution channels; to use the Web as a medium to build longer-term customer relationships; and to use Web-based buying to influence customers, sellers and suppliers.

TEC (technology-enabled content)

The integration of content with information technology (IT) to create value-added information that directly supports a business process.

TEC (Tivoli Enterprise Console)

technical components

Application building blocks that adhere to a specific component object model (e.g., COM or Enterprise JavaBeans). Technical components generally are delivered as

visual widgets and screen elements such as grid controls, message dispatchers and embedded Hypertext Markup Language (HTML) rendering engines. These components are gaining in complexity and nonvisual representation and are making up the bulk of components sold to applications developers.

technical license management (see TLM)

technology-enabled buying (see TEB)

technology-enabled content (see TEC)

technology-enabled marketing (see TEM)

technology-enabled relationship management (see TERM)

technology-enabled selling (see TES)

technology exchange option (see TEO)

telco

A contraction of the term "telephone company." It generally refers to the local-exchange carrier (LEC).

telecommunications

Any process that permits the passage of information from a sender to one or more receivers in any usable form (e.g., printed copy, fixed or moving pictures, and visible or audible signals) by means of any electromagnetic system (electrical transmission by wire, radio, optical transmission, waveguides, etc.). It includes telegraphy, telephony, video-telephony and data transmission.

Telecommunications Access Method (see TCAM)

Telecommunications Act of 1996

Besides opening up the local and long-distance markets to competition, the Telecommunications Act of 1996 was intended to fix many of the flaws in universal service and access charges. The new telecommunications law required that the Federal Communications Commission (FCC) separate access charges from universal service and that part of access charges be based on actual costs. The Telecommunications Act also made provisions to include Internet access for U.S. schools, libraries and hospitals as part of universal service.

telecommunications administration (see TA)

telecommunications data systems (see TDS)

Telecommunications Industry Association (see TIA)

telecommunications lines

Telephone and other communication pathways that are used to transmit information from one location to another.

Telecommunications Management Network (see TMN)

telecommuter

A work-at-home computer user who connects to the corporate local-area network (LAN) backbone using remote access technologies.

telecommuting

Any significant portion of working hours spent at a remote site (e.g., home) using communications lines to send and receive information, interact with customers and peers, and deliver work projects.

teleconference

Bidirectional audio on either a one-to-many or a many-to-many basis.

Telcordia Technologies

A telecommunications infrastructure technology firm, which changed its name from Bellcore in 1999. See Bellcore.

telecottage

A well-equipped, well-connected small office that may be used by people working remotely for the same or different companies.

TeleManagement Forum (see TMF)

telephone channel

A transmission path designed for the transmission of signals representing human speech or other telephone communication (e.g., facsimile) requiring the same bandwidth. The bandwidth allotted is usually 64 kilobits per second (Kbps), but can be reduced to 32 Kbps or even 16 Kbps with multiplexing techniques.

telephone circuit

An electrical connection permitting the establishment of telephone communication in both directions between two points. Also called telephone line.

telephone company (see telco)

telephone exchange

A switching center for the interconnection of lines that terminate therein. Also called central office (CO).

telephone frequency

Any frequency within the part of the audio-frequency range that is essential for the transmission of speech of commercial quality, i.e., 300 hertz (Hz) to 3,000 Hz. Also called voice frequency (VF).

telephony

A generic term for voice telecommunications.

Telephony Application Objects

The software model for Signal Computing System Architecture (SCSA), introduced by Dialogic. It includes the Enterprise Computer Telephony Forum (ECTF).

Telephony Application Programming Interface (see TAPI)

Telephony Services Application Programming Interface (see TSAPI)

Telephony User Part (see TUP)

telepoint

A means of accessing the public switched telephone network (PSTN) via low-cost, low-power radio base stations and compact, portable telephones. Access is obtained at up to 200 meters from base stations located in public places such as railway stations, airports and shopping centers.

teleprocessing

The processing of information that is received from or sent to remote locations by way of telecommunications lines. Such systems are necessary to hook up remote terminals or connect geographically separated computers. See telecommunications.

teletext

Generically, one-way data transmission designed for widespread broadcasting of graphics and textual information for display on subscriber televisions or low-cost terminals.

telink

An enhanced version of xmodem, mainly used on Fido bulletin boards.

Telnet

The Internet standard protocol for remote terminal connection. It allows a user at one site to interact with a remote device or system that expects terminal-mode traffic.

TEM (technology-enabled marketing)

Also known as marketing automation, TEM involves analyzing and automating the marketing process. Because the role of technology in all aspects of business is growing, marketing departments must make it a strategic imperative to use information and IT to build competitive differentiation. TEM includes a proactive strategy for using information and IT in marketing, with the ultimate goal of TEM is to allocate marketing resources to the activities, channels and media with the best potential return and impact on profitable customer relationships. The new metrics of customer profitability, customer lifetime value and share of customer will be needed to supplement the traditional metrics of market share and penetration. The components of TEM include:

- **Data Cleansing:** This involves the use of tools for data support (e.g., cleansing, manipulation and reconciliation) to produce quality data and data consistency.
- **Data Analysis:** Also called business intelligence, this involves the use of software for ad hoc query, reporting and analysis capabilities, supporting strategic decision-making processes with a data warehouse or data mart. A data warehouse is a consolidated database that stores all — or significant portions of — the data collected by an enterprise's multiple business systems. Data from online transaction-processing applications and other sources is selectively collected, extracted, integrated, transformed and cleaned. A data mart contains a subset of the data typically found in a data warehouse and is designed to support the unique business intelligence requirements of a specific business process/application requirement.
- **Content Management Systems:** Also known as marketing content management

(MCM) systems or marketing encyclopedia systems (MES), this category of applications allow enterprises to view and access marketing content.

- **Campaign Management System (CMS):** A CMS is a database management (DBM) tool used by marketers to design single-channel or multichannel marketing campaigns and track the effects of those campaigns by customer segment over time. CMS applications are also used by sales organizations to execute sales campaigns, such as achieving a specific market share with a particular product by a certain date.

Tempest

The U.S. military standard for electronic products that can withstand electromagnetic pulses.

TEN (total effect of networking)

TEO (technology exchange option)

A guaranteed residual value (RV) for equipment at 12-, 18-, 24- and 30-month intervals that allows the user to use the RV toward the purchase of additional capacity, features or functionality, or toward trade-in for newer technology. TEOs guarantee that future acquisitions will be at a lower original discount rate or current average selling price.

terabyte (see TB)

terabytes-per-administrator ratio (see TB/admin)

teraFLOPS (see TFLOPS)

TERM (technology-enabled relationship management)

The concept of forming one enterprisewide view of the customer across all customer contact channels (i.e., sales, marketing, and customer service and support). It is a complex area, requiring complex solutions to problems of integration, data flow, data access and marketing strategy. A critical component is the database that serves as the customer information repository.

terminal

A device, combining keyboard and display screen, that communicates with a computer. Terminals are divided into different classes depending on whether they are able to process data on their own.

- **Dumb terminals** — display monitor or simple input/output (I/O) devices that send and accept data from a network server or mainframe. They have no built-in processing capabilities. Workers enter data and commands, which are sent to a computer located elsewhere.
- **Smart terminals** — monitors that process limited amounts of information.
- **Intelligent terminals** — devices that contain main memory and a central processing unit (CPU) to perform special display functions. Examples include an information kiosk and AT&T Display Phones.

- 3270 terminals — IBM display stations used to communicate with mainframes made by IBM and other manufacturers. They are in widespread use, and are widely copied.

Terminal Access Controller Access Control System (see TACACS)

terminal adapter (see TA)

terminal emulation

Imitation of a specific terminal (VT100, for example) by a device, such as a PC, through software. PCs often use terminal emulation methods to connect to specific hosts, such as Digital Virtual Address Extensions (VAXs) or IBM mainframes, with which they would otherwise be incapable of communicating.

Terminal Equipment (see TE)

terminal job

In systems with time sharing, the processing done on behalf of one terminal user from log-on to log-off. See time sharing.

terminal server

A local-area network (LAN) device that allows asynchronous dumb terminals to communicate with a host computer also attached to the LAN. It is useful in minimizing the amount of cabling where several terminals need to be attached to a mainframe.

Terminal Services (see TS)

terminal user

In systems with time sharing, anyone who is eligible to log on. See time sharing.

Terrestrial Trunked Radio (see TETRA)

TES (technology-enabled selling)

Also known as “sales automation” or “technology-enabled sales,” this refers to the application of technology to enable selling through all desired sales channels, including field/mobile sales, inside sales/telesales, selling partners (i.e., e-partners), Web selling (i.e., e-sales) and retail sales. The goal of TES is to integrate technology with optimal processes to provide continuous improvement in sales team effectiveness, as well as balance and optimize each enterprise sales channel. The components of TES include:

- Field sales: Also known as mobile sales or sales force automation (SFA), this includes applications for salespeople who most often work outside the boundaries of the enterprise and without the benefit of continuous LAN or high-speed WAN connections.
- Inside sales: Also known as telesales or inside selling, this involves applications for salespeople who most often work inside the boundaries of the enterprise and with the benefit of continuous LAN or high-speed WAN connections. To perform their jobs, they spend a majority of their time using the phone, the Web or e-mail.

- **e-partner:** Also known as extended selling enterprise (ESE), this includes applications and technologies provided by the enterprise to assist third-party selling channel partners (e.g., brokers, agents, distributors and value-added resellers) in achieving selling objectives.
- **e-sales:** Also known as technology-enabled buying (TEB), unassisted selling or Web selling, this component of TES involves customer-direct, business or consumer Web-selling applications. These are customer-facing technologies and applications that allow consumers and businesses to “sell themselves” and conduct transactions without the assistance of a salesperson.
- **Retail sales:** Includes applications that enable retailers to sell their products to consumers through traditional brick-and-mortar outlets (such as department stores, specialty shops and outlet malls) or via new options such as home shopping, the Internet and warehouse clubs. Merchandising, relationship marketing and e-retailing are typical examples of retail sales applications.

test data generator

Communications instructions for forming files containing sets of information developed specifically to ensure the adequacy of a computer run or system.

tethered remote access

A terrestrial or nonmobile connection of individual users or small workgroups to a data source or network interface beyond the boundary defined by a building or campus. The connection is generally via a wide-area network (WAN), which, in most cases, is outside the sphere of ownership and management of the enterprise making the remote connection.

TETRA (Terrestrial Trunked Radio)

TETRA is a European Telecommunications Standards Institute (ETSI) standard for the digital private mobile radio (PMR) and public access mobile radio (PAMR) technology for police, ambulance and fire services, security services, utilities, the military, public access services, fleet management, transport services, closed-user groups, factory site services, mining and other uses. TETRA uses TDMA technology with four user channels on one radio carrier and 25kHz spacing between carriers.

text

The information portion of a transmitted message, as contrasted with the header, check characters, and end-of-text characters. Sometimes called the body of the message.

text editing

Manipulation of the information portion of a transmitted message by changing or rearranging symbols, words, sentences and paragraphs.

text information management system (see TIMS)

text mining

The process of extracting information from collections of textual data and utilizing it for business objectives.

text retrieval

Software used for finding units of textual information such as documents by matching a user's search terms to those in a full-text index derived from the collection of textual units.

TFA (trusted financial advisor)

TFAs provide or facilitate access for their clients to consumer banking functions, budgeting and expense-tracking services and financial advice. They also serve as middlemen between individuals and prequalified best-of-breed financial-services vendors, actively seeking out "better deals" for their subscribers. It is not necessary for a TFA to be an existing financial-services provider because TFA functions can be fulfilled by a wide range of firms or by low-cost existing or developmental personal software products. All other functionality offered by the TFA can be outsourced.

TFLOPS (teraFLOPS)

Trillion floating-point operations per second.

TFT (thin film transistor)

TFTP (Trivial File Transfer Protocol)

A protocol used for basic file transfers, as well as booting systems that communicate with the Transmission Control Protocol/Internet Protocol (TCP/IP) suite using the TFTP boot protocol. TFTP boot is similar to bootp, but contains an additional file transfer capability. This protocol is commonly used to provide configurations to remote routers, allowing network managers to centrally generate the remote router configuration file and then download that configuration once the router is powered up and connected to the wide area network.

THA (time horizon to action)

A measurement used in system compliance efforts, such as euro currency conversion. The THA is the time period between the present and the point at which at which conversion efforts must begin if the compliance deadline is to be met.

THC (time horizon to compliance)

A measurement used in system compliance efforts. The THC is the time period between the present and the point at which all systems must be compliant with a new format or requirement (e.g., euro currency support).

THD (time horizon to decision)

A measurement used in system compliance efforts, such as euro currency conversion. The THD is the time period between the present and the point at which a decision must be reached as to how compliance efforts will proceed (e.g., which systems must be changed, replaced or retired).

Thermal Conduction Module or Thermo-Conductive Module (see TCM)

THF (time horizon to failure)

Defines the date on which a system will cease to function properly if corrective action is not taken.

thick client

See fat client.

Thick Ethernet

10Base-5 coaxial Ethernet cable.

thin client

Term used to describe a type of client/server computing where applications are run, and data is stored, on the server rather than on the client. Because the applications are executed on the server, they do not require client-resident installation, although the graphical user interface and some application logic may be rendered to the client. Thin-client computing is dominated by Microsoft Windows Terminal Servers with Citrix MetaFrame. Other examples of thin-client technology include The Santa Cruz Operation's Tarantella and Sun Microsystems' Sun Ray.

A common misperception is that a thin-client application requires the use of a thin-client device (i.e., a stripped-down desktop machine that costs less to buy and maintain than a regular PC). However, while thin-client applications enable such devices to be used, they do not require it. In fact, more than 85 percent of devices used to display thin-client Windows applications are regular PCs, typically configured with both "fat-client" applications and access to thin-client ones.

See client/server and fat client.

Thin Ethernet

Flexible 10Base-2 coaxial cable useful in connecting PCs, workstations and systems arranged in a workgroup. It supports segment distances up to 185 meters.

thin server (see server appliance)**third generation (see 3G)****third-party logistics (see 3PL)****third-party maintenance**

Maintenance performed by a service organization contracted by the owner of the system.

three-schema architecture

A framework for managing access to data that involves three layers or schemas: the external or programming view, the conceptual or data administration view, and the internal or database administration view. Such ideas were developed by an American National Standards Institute/Scalable Processor Architecture subcommittee in 1971 but received little practical implementation by database management system (DBMS) vendors. The principle is that the conceptual schema consists of business rules derived from a semantic data model, which provides independence between programs and data structures. The emphasis has since shifted to computer-aided software engineering (CASE) tools and "repository" standards.

throughput

A computer term for the volume of work or information flowing through a system. Particularly meaningful in information storage and retrieval systems, in which throughput is measured in units such as accesses per hour.

thunking

The process by which legacy 16-bit code is converted to 32-bit code for execution.

TIA (Telecommunications Industry Association)

An U.S. association that develops telecommunications standards. TIA represents the communications sector of the Electronic Industries Alliance (see EIA).

TickIT

An initiative sponsored by the U.K. government. It is a scheme that adapts International Organization for Standardization (ISO) 9000 to the special needs of software development. TickIT is designed to make it easier for systems developers to apply Quality standards. TickIT is not a standard in itself; rather, it is a way of understanding and applying ISO 9000. Enterprises and information systems (IS) departments that undertake software development in the United Kingdom can apply for certification under TickIT. This certifies that their software development procedures conform to Quality standards. It does not imply any approval of other processes in the department or the enterprise.

tie line (see TL)

TIFF (Tag Image File Format)

A de facto standard format for bit-mapped image files.

tightly coupled multiprocessing (see TCMP)

TIM (transaction incident management)

The art and practice of ensuring that business transactions enable users, consumers and suppliers to perform business transaction tasks securely and in privacy according to business specifications and service-level agreements. The objectives of TIM are to detect any kind of abnormal incidents in real time and to resolve them rapidly, depending on their levels of business criticality.

time and attendance (see T&A)

time difference of arrival (see TDOA)

time division multiple access (see TDMA)

time division multiplexing (see TDM)

time-division signaling

Signaling over a time-division multiplex system in which all voice channels share a common signaling channel, with time division providing the separation between signaling channels.

time-division switching

The switching method for a time-division multiplexing (TDM) channel requiring the shifting of data from one slot to another in the TDM frame. The slot in question can carry a bit, a byte, or, in principle, any other unit of data.

time domain reflectometry (see TDR)

time horizon to action (see THA)

time horizon to compliance (see THC)

time horizon to decision (see THD)

time horizon to failure (see THF)

time of arrival (see TOA)

time out

The set time period before a terminal system performs some action. Typical uses include a poll release (when a terminal is disconnected if the time-out period elapses before keying resumes) or an access time out (when a terminal on a local area network is prevented from transmitting for a specified time period).

time sharing

Use of a computer facility by many users for different purposes at (apparently) the same time. Although the computer services the users sequentially, it allocates segments of time so that each user appears to have the dedicated resources of the machine. In the 1970s and 1980s, time sharing was common. With the introduction of more powerful, less expensive computers, it has become less common.

Time-Sharing Option (see TSO)

time slice

An interval on the central processing unit (CPU) allocated to performing a specific task. Once the interval has expired, CPU time is allocated to another task. See time sharing.

TIMS (text information management systems)

The genre of products that combines content-based retrieval and document database functions.

T-Interface

In Integrated Services Digital Network (ISDN), four-wire Basic Rate Interface (BRI) circuit connection.

TI RPC (Transport Independent Remote Procedure Call)

Tivoli Service Desk (see TSD)

TL (tie line)

A private-line communications channel of the type provided by communications common carriers for linking two or more points together, typically private branch exchanges (PBXs). Also called tie trunk.

TLF (trunk link frame)

See crossbar switch.

TLM (technical license management)

The management and control of software via a platform-independent automated facility that:

1. Ensures that access and use are in alignment with associated licensing agreements.

2. Provides the basis for determining enterprise use requirements.
3. Integrates with systems and network management tools.

TLMS (tape library management system)

TLP (transmission-level point)

Any point in a transmission system at which the power level of the signal is measured.

TLS (transport layer security)

Internet-based transaction security provided by the Secure Sockets Layer (SSL) protocol.

T&M (time and materials)

Tmap

Developed by Nortel Networks with support from Intel, Microsoft and Novell, Tmap maps Telephony Application Programming Interface (TAPI) programming calls to Telephony Services Application Programming Interface (TSAPI), enabling TAPI applications to run on networks that support TSAPI.

TMF (TeleManagement Forum)

A consortium of more than 100 equipment vendors and carriers that are developing implementation specifications for Open Systems Interconnection (OSI)-based network management. Founded in 1988, TMF (formerly known as the Network Management Forum) seeks solutions to business and technology challenges resulting from global telecommunications deregulation. The group provides a forum for industry leaders to address interoperability and process automation issues that they could not address as cost-effectively or comprehensively on their own.

TMM (tape mount management)

TMN (Telecommunications Management Network)

Model defined in the International Telecommunication Union (ITU) Recommendation M30. TMN will form the standard basis for network management in such advanced networks as synchronous digital hierarchy (SDH) and Groupe Speciale Mobile (GSM). It defines management functions with standard interfaces using a management network which is separate from the transmission network, and uses standard protocols such as Common Management Information Protocol (CMIP).

TMS (transportation management system)

System used to plan freight movements, do freight rating and shopping across all modes, select the appropriate route and carrier, and manage freight bills and payments.

TOA (technical office automation)

TOA (time of arrival)

Based on triangulation, a method of processing cellular phone signals, TOA allows the physical position of switched-on phones to be located. The position is determined by comparing the TOA of the existing reverse control channel at various cell sites.

token

1. A small piece of hardware or software used to authenticate user access to an information system. See "hardware token" and "software token."
2. A special bit pattern (also called a "supervisory frame") that grants access to a station seeking to transmit data over a local-area network. Stations wishing to transmit must receive the token before doing so. See "token bus," "token passing" and "token ring."

token bus

A local network access mechanism and topology in which all stations actively attached to the bus listen for a broadcast token or supervisory frame. Stations wishing to transmit must receive the token before doing so; however, the next physical station to transmit is not necessarily the next physical station on the bus. Bus access is controlled by preassigned priority algorithms.

token passing

A LAN access technique in which participating stations circulate a special bit pattern that grants access to the communications pathway to any station that holds the sequence. It is often used in networks with a ring topology.

token ring

A LAN access mechanism and topology in which a supervisory frame or token is passed from station to station in sequential order. Stations wishing to gain access to the network must wait for the token to arrive before transmitting data. In a token ring, the next logical station receiving the token is also the next physical station on the ring. The ring may be a logical, rather than physical, ring in the case of switched token ring.

TOP (Technical Office Protocol)

topology

The logical or physical arrangement of stations on a network in relation to one another. See bus, ring, star network and tree.

TORP (task order requirements package)

Total Access Communications System (see TACS)

total cost of ownership (see TCO)

total quality control (see TQC)

Total Solutions Lease (see TSL)

touchpoint

A contact point between an enterprise and its customers. Touchpoints may occur in any channel (e.g., via phone, the Web or direct contact with a salesperson).

touch-sensitive

Refers to the technology that enables a system to identify a point of contact on the screen by coordinates and transmit that information to a program.

Touch-tone

The registered AT&T trademark for push-button dialing. See dual-tone multifrequency signaling (DTMF).

TP (transaction processing)

A mode of processing characterized by short transactions recording business events that normally requires high availability and consistent, short response times. A category of application that requires a request for service to be answered within a predictable period that approaches real time, and a transaction that transforms this model from one business state to another. Tasks such as making reservations, scheduling and inventory control are especially complex; all the information must be current.

TPA (third-party administrator)

TPAML (Trading Partner Agreement Markup Language)

TPC (Transaction Processing Performance Council)

An organization that has developed several standardized transaction processing (TP) benchmarks, among which are TPC-A, TPC-B, and TPC-C. TPC prohibits testing systems that are specially optimized for benchmarking or lack real-world applicability.

TPC-A (Transaction Processing Performance Council Benchmark A)

A revised and superior version of the debit/credit online transaction processing (OLTP) benchmark. Ratified in late 1989, it came into widespread use in 1990. The major improvements in TPC-A were the requirements for full disclosure and the inclusion of the front-end network and terminals. TPC-A is intended to replace debit/credit as the only industrywide measure for OLTP performance and price/performance. It is a good test since it measures end-to-end performance, but it still is only one test reflecting a single type of transaction.

TPC-C (Transaction Processing Performance Council Benchmark C)

The Transaction Processing Performance Council (TPC) benchmark model for complex transaction-processing workloads. It is a well-defined set of programs and testing protocols that are used to derived consistent, repeatable benchmark results for online transaction processing. The TPC-C benchmark is most often used to define the maximum scalability supported by a database management system (DBMS) or an application server platform.

TPC-D (Transaction Processing Performance Council Benchmark D)

The Transaction Processing Performance Council (TPC) benchmark model for decision support workloads.

TPF (Transaction Processing Facility)

A high-availability IBM system designed to support transaction-driven applications.

TPID (Tag Protocol Identifier)

Part of the virtual local-area network (VLAN) Tag Header in the Institute of Electrical and Electronics Engineers (IEEE) 802.1q draft standard.

tpm (transactions per minute)

TPM (third-party maintenance)

TPM (transaction-processing monitor)

A process whereby the computer handles each instruction or query individually as it is entered.

tpmC (transactions per minute of TPC-C)

TPP (third-party participant)

TP-PMD (Twisted Pair-Physical Media Dependent)

The American National Standards Institute (ANSI) standard for Fiber Distributed Data Interface (FDDI) over copper, replacing Shielded Distributed Data Interface (SDDI), which runs on shielded cable and Copper Distributed Data Interface (CDDI), which runs on unshielded cable.

tps (transactions per second)

TQC (total quality control)

A theory of quality whereby the maker of the part has responsibility for the quality of that part.

TQM (total quality management)

TR (see token ring)

trace packet

In packet switching, a special kind of packet that functions as a normal packet but causes a report of each stage of its progress to be sent to the network control center.

track

A storage channel on a disk or tape that is made up of a series of sectors.

tracking, telemetry, control and monitoring (see TTC&M)

Tradacoms

A European electronic data interchange (EDI) standard published and maintained by the Article Numbering Association (ANA).

traditional processing

The availability of traditional and basic data processing facilities at the midrange system. Examples include sequential and keyed files, compilers such as COBOL, FORTRAN, PL/I, BASIC and C, interactive access, batch processing, multiprocessing and multitasking.

traffic

1. Messages sent and received over a communications channel.
2. Quantitative measurement of the total messages and their length, expressed in hundred call seconds (CCS) or other units.

traffic flow

The measure of the density of traffic on a telephone network expressed in Erlangs.

traffic matrix

Matrix of which the X, Y element contains the amount of traffic originated at node X and destined for node Y. The unit of measurement may be telephone calls or data packets per second, depending on the kind of network.

train time

The initialization time for full-duplex operation on a modem.

transaction

A logical update that takes a database from one consistent state to another.

Transaction Capabilities Application Part (see TCAP)

transaction incident management (see TIM)

transaction logging

A concept in which a detailed record is kept of all operations in a transaction; in case of a failure, the transaction could be backed out and the former state reconstructed.

transaction monitor

A subsystem that ensures that all transactions against a database leave it in a consistent state or, in case of a transaction failure, returns the database to its pre-transaction state.

transaction processing (see TP)

Transaction Processing Facility (see TPF)

transaction-processing monitor (see TPM)

Transaction Processing Performance Council (see TPC)

transceiver

A device that can transmit and receive traffic. It is used to connect nodes to Ethernet (LAN).

transducer

A device for converting signals from one form to another, such as a microphone or a receiver.

transfer rate

The speed at which information can be sent across a bus or communications link.

transit VLAN starvation

The condition a switch suffers in a non-ATM (asynchronous transfer mode) network when it becomes congested and cells or cell streams representing Layer 3 virtual local-area network (VLAN) packets are missed or dropped, leading to the VLAN becoming unstable. Also known as beatdown.

translator

A device that converts information from one system of representation into equivalent information in another system of representation.

transmission

Sending information in the form of electrical signals over electric wires, waveguides, or radio.

Transmission Control Protocol (see TCP)

Transmission Control Protocol/Internet Protocol (see TCP/IP)

Transmission Convergence Sublayer

In asynchronous transfer mode (ATM), a protocol for preparing cells for transmission.

transmission-level point (see TLP)

transmission speed

The rate at which information is passed through communications lines, generally measured in bits per second (bps).

transmit

To send information from one location to another.

transparency

If a signal passes through a network or facility unchanged, that network or facility is said to be transparent to it.

transparent asynchronous transmitter/receiver interface (see TAXI)

Transparent Data Migration Facility (see TDMF)

transponder

The receiver-transmitter combination that retransmits the received signal greatly amplified and at a different frequency. Communications satellites usually contain several transponders.

transport layer

In the Open Systems Interconnection (OSI) model, the network processing entity responsible — in conjunction with the underlying network, data link and physical layers — for the end-to-end control of transmitted data and the optimized use of network resources.

transport layer security (see TLS)

transport overhead

Overhead to the synchronous transport signal synchronous payload envelope (STS SPE) for transmission purposes (SONET term). See STS.

transportation management system (see TMS)

transportation planning and scheduling

Specifies how, when and where to transport goods. Transportation planning and scheduling applications may provide weight/size restrictions, merge-in-transit, continuous move, mode or carrier selection, and truckload planning functionality.

travel and expense (see T&E)

traveling wave tube amplifier (see TWTA)

tree

A type of bus network topology in which the medium branches at certain points along its length connect stations or clusters of stations. Also called a branching bus.

Trellis Coding

A method of modulation that combines both amplitude and phase.

Triple DES (Triple Data Encryption Standard)

A variant of DES. It uses three different DES keys, in sequence, to encrypt a file.

Triple X

A colloquial name given to the three International Telecommunication Union (ITU) protocols (X.3, X.28 and X.29) connecting character mode terminals to packet switched networks using the X.25 packet definition.

Trivial File Transfer Protocol (see TFTP)

TRN (token ring network)

See token ring.

Trojan horse

A form of malicious code that may be deliberately planted to perform a destructive act on a computer. It is effective because it is not what it appears to be. That is, the execution of a Trojan horse may have an undesirable and unexpected effect on the user's work environment, but it is the user who initiates the execution of the code (e.g., by clicking on a button in a graphical user interface that appears harmless). Unlike a computer virus, a Trojan horse is unable to replicate and is not parasitic.

trouble ticket

A record of a customer complaint or problem, usually created in a call or contact center. The ticket remains active until the issue has been resolved.

Tru64

Compaq's (now Hewlett-Packard's) 64-bit RISC Unix for Alpha (previously called Digital Unix).

TRUSTe (Trusted Universal Standards in Electronic Transactions)

An online privacy-seal program (OPSP) that identifies Internet companies utilizing appropriate processes and operations to protect the confidentiality of consumer information. License fees are based on a company's annual revenue. Vendors must reapply for licensing each year, and during this process their operations are re-evaluated to ensure they meet all requirements for displaying the program emblem.

trusted financial advisor (see TFA)

Trusted Universal Standards in Electronic Transactions (see TRUSTe)

TS (Terminal Services)

TS (transaction server)

TSAF (Transparent Services Access Facility)

TSAPI (Telephony Services Application Programming Interface)

A computer-telephony integration (CTI) application programming interface (API) developed by Novell and AT&T that provides a client/server implementation and supports first-and third-party call control. The major benefit of third-party call control is that, via the command link to the switch, it is possible to control calls between third parties (much like an operator does today). This permits CTI applications to provide multiparty services, call routing based on automatic number identification and other advanced features. However, TSAPI requires a Novell NetWare networking environment. TSAPI does not require additional hardware in each PC, but does require a server CTI connection to the private branch exchange (PBX) equipment.

TSD (Tivoli Service Desk)

A consolidated service desk product from Tivoli that provides help desk, change management, problem management, asset management and decision support functionality.

TSE (Terminal Server Edition)

TSL (Total Solutions Lease)

An IBM leasing program.

TSO (Time-Sharing Option)

The name originally given to IBM's compatible extension of Multiple Virtual Storage (MVS) background facilities into foreground execution. The name has stuck, although TSO is no longer an option in MVS.

TSO/E (Time-Sharing Option/Extended)

TSR (terminate and stay resident)

TSTN (triple-supertwist nematic)

TT (trouble ticket)

TTC&M (tracking, telemetry, control and monitoring)

Specialized ground stations used to track and control satellites and to monitor their performance.

TTL (transistor-transistor logic)

TTP (trusted third party)

TTTN (tandem tie trunk network)

A private network making use of tandem switching.

TUA (Telecommunications Users Association)

tuning

The process of adjusting computer system control variables to make a system divide its resources most efficiently for a workload.

tunneling

Provides services on a point-to-point basis without the necessity to change the data to accommodate differing network types or protocols.

TUP (Telephony User Part)

An Integrated Services Digital Network (ISDN) feature for call setup and control.

turnaround time

The time required to reverse the direction of transmission, e.g., to change from receive mode to transmit mode in order to acknowledge on a half-duplex line when individual blocks are acknowledged, as is required in certain protocols. The turnaround time has a major effect on throughput, particularly if the propagation delay is lengthy, such as on a satellite channel.

turnkey system

A complete communications system, including hardware and software, assembled and installed by a vendor and sold as a total package.

TVML (TV Markup Language)

An open standard promoted by WebTV and Microsoft as an extension to open Hypertext Markup Language (HTML) standards.

twinax

Coaxial cable with two center conductors at each connection point.

twisted pair

Two insulated wires twisted together, which can be shielded (STP) or unshielded (UTP).

Twisted Pair-Physical Media Dependent (see TP-PMD)

two-phase commit (see 2PC)

two-wire circuit

A circuit formed by two conductors insulated from each other, providing "send" and "return" channels in the same frequency.

TWTA (traveling wave tube amplifier)

A satellite component used to amplify an incoming signal before transmitting to receiving earth station.

Type A, B and C enterprises

A Gartner framework that classifies enterprises or their subdivisions according to a technology adoption profile. Classification is based not only on an enterprise's current technology adoption strategy, but also on whether the strategy is supported by top management and is adequately funded.

The Gartner Glossary of Information Technology Acronyms and Terms

- Type A enterprises are typically technically aggressive and well-funded, and use information technology (IT) to gain a competitive advantage.
- Type B enterprises, which are in the majority, are mainstream IT users with adequate funding that use IT for productivity.
- Type C enterprises are technologically conservative and risk-averse, and seek to control IT costs.

Recognizing an enterprise's type offers company strategists a meaningful way to compare an enterprise's use of technology against that of competitors, and to make decisions about when, how and where to adopt new technologies.

U

U (see rack unit)

UA (Upgrade Advantage)

A Microsoft software license program.

UART (universal asynchronous receiver/transmitter)

An integrated circuit attached to the parallel bus in a computer that conducts serial communications efficiently.

UAWG (Universal ADSL Working Group)

A vendor consortium promoting adoption of asymmetric digital subscriber line (ADSL) as a remote access method, with an emphasis on the adoption of International Telecommunication Union (ITU) standards.

UB (uniform billing code)

UBR (unspecified bit rate)

An asynchronous transfer mode (ATM) service category, defined by the ATM Forum, that provides no service guarantees and therefore is a "best effort" service equivalent to a shared-media LAN.

UCB (unit control block)

UCC (Uniform Code Council)

The U.S.-based, not-for-profit membership organization, founded in 1972, that administers the EAN-UCC retail bar code system together with EAN International (its European counterpart, founded in 1974). Through its UCCnet subsidiary, UCC is currently focused on developing standards and specifications for an integration marketplace for a range of industries. See EAN and UCCnet.

UCCnet

A subsidiary of the Uniform Code Council focused on developing an integration marketplace for some two dozen industries. Its initial focus is on item synchronization standards in grocery and apparel markets. Some 75 companies (primarily consumer product and retail companies in North America) subscribe to its services, while more than 100 others are working on the development of the integration marketplace through contributions to UCCnet Implementation Action Groups (monthly sessions to develop the standards).

UCITA (Uniform Computer Information Transactions Act)

UCR (usual and customary rates)

UDB (Universal Database)

A IBM brand name applied to several of its DB2 database management system products (e.g., DB2 UDB for S/390, and DB2 UDB for Unix and Windows).

UDDI (Universal Description, Discovery and Integration)

A specification for business-to-business (B2B) e-commerce integration introduced by Microsoft, IBM and Ariba in August 2000. UDDI provides a standard format with which enterprises can describe themselves and their method of conducting e-business transactions within an Internet-based business registry. UDDI is designed to jump-start B2B integration and e-commerce by enabling businesses to publish and locate available Web services, their characteristics and interfaces. More than 200 companies have agreed to support the specification, which is described on the UDDI Web site at www.uddi.org. See Web services.

UDF (user-defined functions)

UDP (User Datagram Protocol)

A Transmission Control Protocol/Internet Protocol (TCP/IP) technology that enables an application to send a message to one of several applications running in a destination machine. UDP is stateless — it differentiates sources and destinations within hosts and provides no other services. Often services do not use predefined port numbers, so filtering on the basis of “well known ports” will not work.

UFDL (Universal Form Definition Language)

A technology developed by UWI.com (now PureEdge Solutions) for e-form management. UFDL defined a high-level language used to construct, display and use e-forms. It was eventually rewritten in Extensible Markup Language (XML) syntax and renamed Extensible Forms Description Language (XFDL).

UHF (ultra-high frequency)

UI (user interface)

The connection between the user and a computer’s hardware or software that permits the user to work productively with a system or a program. User interface design requires significant skill and attention and has become a recognized specialty.

U Interface

In Integrated Services Digital Network (ISDN), a two-wire loop of twisted-pair cables.

ultraportable

A computer system that meets all the criteria for a notebook PC but is lighter and may not have an internal floppy disk drive. It typically weighs 4 pounds or less with the battery and weight-saver modules. The keyboard and screen are often compromised to meet weight targets and the unit must be augmented with a standard keyboard and mouse for long-term use.

UltraSCSI

A high-speed version of the Small Computer Systems Interface (SCSI) bus. It supports 40-megabyte-per-second transfers.

Ultra Wideband (see UWB)

Ultrix

A version of Unix from Compaq (formerly Digital Equipment), now merged with Hewlett-Packard.

UM (unified messaging)

The convergence of a variety of communications media (e.g., e-mail, fax and voice) into a single queue of messages, providing senders and recipients the freedom to choose media and access devices.

UMB (upper memory block)

UML (Unified Modeling Language)

A language for specifying, visualizing, constructing and documenting the artifacts of software systems.

UMLS (Unified Medical Language System)

UMTS (Universal Mobile Telecommunications System)

The first of the third-generation (3G) cellular networks, UMTS is being designed to offer speeds of at least 144 Kbps to fast-moving (e.g., vehicle-based) mobile devices, and offer an initial 2 Mbps to campus sites — designers expect to increase this to 10 Mbps by 2005.

unattended operation

Operation of a communications or data-processing system transmission that is controlled automatically and does not require a human operator. Also called "lights-out operation."

unbalanced-to-ground

In a two-wire circuit, the condition in which the impedance-to-ground on one wire is measurably different from that on the other. See balanced-to-ground.

UN/EDIFACT standard message (see UNSM)

UNI (user network interface)

In asynchronous transfer mode (ATM) environments, the point at which the user joins the network.

unified messaging (see UM)

Unified Modeling Language (see UML)

Unified Network Management Architecture (see UNMA)

unified point of service (see UPOS)

Uniform Code Council (see UCC)

uniform resource locator (see URL)

uniform-spectrum random noise

Noise distributed over the spectrum in such a way that the power-per-unit bandwidth is constant. Also called white noise.

uninterruptible power supply (see UPS)

Universal ADSL Working Group (see UAWG)

universal asynchronous receiver/transmitter (see UART)

Universal Database (see UDB)

Universal Description, Discovery and Integration (see UDDI)

Universal Form Definition Language (see UFDL)

Universal Mobile Telecommunications System (see UMTS)

Universal Networked Object (see UNO)

universal power unit (see UPU)

universal queue

A process (and technology) whereby all contact channels and media — e.g., phone, interactive voice response, fax, Web and e-mail — are integrated into the same queue to standardize processing and handling.

Universal Serial Bus (see USB)

Unix

An operating system originally designed by Bell Laboratories, Unix has proven to be adaptable to a variety of platforms. Developed in 1969, Unix has gone through a multitude of upgrades and enhancements. It has become a dominant operating system for servers and high-end workstations because of its scalability and support of complex processing. However, Microsoft's Windows 2000 (formerly Windows NT) has been gaining ground in this market segment.

Unix-to-Unix Copy Program (see UUCP)

Unix-to-Unix encoding (see uuencode)

UN/JEDI (UN Joint Electronic Data Interchange)

A working group set up by the United Nations (UN) Economic Commission for Europe (ECE), which established the Electronic Data Interchange for Administration, Commerce and Transportation (EDIFACT).

UNMA (Unified Network Management Architecture)

An AT&T architecture for integrated network management.

UNO (Universal Networked Object)

A proposal approved by the Object Management Group (OMG) in 1994, UNO mandated that CORBA2-compliant object request brokers (ORBs) be able to interoperate with other, heterogeneous ORBs using General Inter-ORB Protocol (GIOP).

unshielded twisted pair (see UTP)

UNSM (UN/EDIFACT standard message)

An Electronic Data Interchange for Administration, Commerce and Transportation (EDIFACT) generic message.

unspecified bit rate (see UBR)

Unstructured Supplementary Services Data (see USSD)

uplink

The portion of a satellite circuit extending from an earth station to the satellite. See downlink.

UPOS (unified point of service)

An international initiative to produce an architectural specification that is both operating-system-independent and language-neutral for retail point of sale (POS) systems, standardizing the POS application programming interfaces (APIs). While UPOS specifies the APIs for peripheral devices, such as scanners, printers and cash drawers, JavaPOS and OPOS outline the implementations, mapping the architecture to a retailer's specific hardware and software computing environment.

upper-CASE tool

A computer-aided software engineering (CASE) tool for modeling application requirements, system analysis and data structure.

UPS (uninterruptible power supply)

A device that provides temporary power upon failure of the main power source.

UPU (universal power unit)

A licensing metric offered by Oracle to license a number of its products. A UPU is one unit of platform-dependent power. The number of UPU's is calculated using an equation that involves the number of processors on which the software is running by the MHz rating of each processor, and a hardware platform index multiplier that varies by platform.

UR (utilization review)

UREP (Universal Repository)

URI (uniform resource identifier)

URL (uniform resource locator)

The character string that identifies an Internet document's exact name and location.

USART (universal synchronous/asynchronous receiver/transmitter)

USB (Universal Serial Bus)

A commercial desktop standard input/output (I/O) bus that provides a single peripheral connection and vastly increases bus speed. It simplifies peripheral connections via a "daisy chaining" scheme whereby the desktop system has only one I/O port to which all peripherals are connected in a series. Up to 120 peripherals can be connected to a single system.

Usenet

Usenet — also known as "Netnews" — is a part of the Internet that is made up of discussion forums or "newsgroups." The hierarchy is divided into seven high-level categories — e.g., computing (comp), recreational (rec) and alternative (alt) — which are in turn divided into numerous subgroups.

user

An individual who interacts with a computing device through applications. Technical personnel are not considered to be users when they are programming or operating the computer.

User Datagram Protocol (see UDP)

user-friendly

Any system, software or device that is easy to learn and use. Graphical user interfaces (GUIs) are designed to be user-friendly.

user interface (see UI)

username

The unique identifier for each user on a network.

user network interface (see UNI)

USLI (ultra-large-scale integration)

More than 1 million transistors in a chip.

USOC (Uniform Service Order Code)

USS (Unix System Services)

USSD (Unstructured Supplementary Services Data)

A Global Systems for Mobile Telecommunications (GSM) bearer service, similar to Short Message Service (SMS). However, unlike SMS messages, USSD messages are not stored and forwarded. USSD acts more like a transactional environment in which a USSD message generates a reply in near real-time. USSD is likely to find increased application with Wireless Application Protocol (WAP) services because it eliminates the delay of SMS store and forward.

USTA (United States Telephone Association)

UTP (unshielded twisted pair)

Wiring with one or more pairs of twisted insulated conductors housed in a single plastic sheath. The wires are twisted around each other to minimize interference from other twisted pairs in a cable bundle. UTP has no coaxial shielding.

UTRA (UMTS Terrestrial Radio Access)

UUCP (Unix-to-Unix Copy Program)

A protocol used for communication between geographically dispersed Unix systems.

uuencode (Unix-to-Unix encoding)

Converts files from binary to American Standard Code for Information Interchange (ASCII) format for transmission onto the Internet as e-mail messages. Uuencode has been largely superceded by Multipurpose Internet Messaging Extensions (MIME), but still exists among older products.

UWB (Ultra Wideband)

Also known as "pulse radio," an emerging wireless technology that uses pulsed radio techniques to transmit data. The transmitter sends a low-power broadband signal, with each channel from 10 to 40 million pulses per second. The correlator, which knows the timing code of the transmitter, listens at these intervals and decodes the signal. Time Domain's Larry Fullerton invented the concept. Time Domain's impulse transmitters emit ultra-short Gaussian monocycle pulses with pulse widths of between 1.5 nanoseconds and 2.0 nanoseconds. UWB also has applications in radar systems, including systems that can detect people through walls or rubble.

V

VA (volt ampere)

V AC or VAC (volts alternating current)

VACC (virtual access call center)

Whereas call centers are predominantly focused on voice-based contact with customers, the VACC will manage voice, e-mail, voice mail, fax, Web sites and scanned paper mail images for customer interactions.

VAD (value-added distributor or dealer)

VAG (VisualAge Generator)

VAISP (value-added Internet service provider)

VAJ (VisualAge for Java)

value-added carrier

A vendors that enhances a public phone network and resells the service.

value-added common carrier

A company that sells the services of a value-added network. It can be a postal, telegraph and telephone services (PTT) agent or subsidiary, or an independent company.

value-added reseller (see VAR)

value-added service (see VAS)

Value Chain Initiative (see VCI)

VAM (vendors against Microsoft)

A descriptive term applied to various vendor alliances that have formed over the years to offer alternatives to Microsoft technologies and platforms. No actual group exists by this name.

vampire tap

Connection into an Ethernet cable by piercing.

VAN (value-added network)

A private network through which value-added carriers provide special data transmission services.

VAR (value-added reseller)

An organization that buys equipment from a vendor at a discount, adds value (such as application software packaged and sold with underlying system software) and remarkets it.

variable bit rate (see VBR)

variable-measure trade item

An item always produced in the same predefined version (e.g., type, design and packaging) that may be sold at any point in the supply chain. However, it may vary in weight and size by its nature, or it may be traded without a predefined weight/size/length.

VAS (value-added service)

A communications facility using communications common carrier networks for transmission and providing enhanced extra data features with separate equipment. Such extra features, including store-and-forward message switching, terminal interfacing and host interfacing, are common.

VAS (VisualAge for Smalltalk)

VAX (Virtual Address Extension)

The Digital Equipment architecture that was the company's principal product line prior to the Alpha. VAX was enormously successful in Digital's traditional engineering and scientific customer base and, more significantly, enabled Digital to penetrate the commercial data-processing and office-automation markets. In 1999, Compaq Computer (which acquired Digital Equipment) announced plans to phase out the VAX platform in favor of Alpha system products.

VB (Visual Basic)

A high-level programming language from Microsoft.

VBA (Visual Basic for Applications)

A version of Microsoft's Visual Basic used to create basic and customized programs.

VBR (variable bit rate)

An asynchronous transfer mode (ATM) quality of service category, defined by the ATM Forum. Both real-time variable bit rate (rtVBR) and non real-time variable bit rate (nrtVBR) are defined. Apart from the traffic parameters for peak cell rate (PCR) and sustainable cell rate (which defines the average bit rate required by the application), additional quality-of-service parameters such as maximum cell transfer delay, cell delay variation and maximum burst size must be agreed upon. The typical application for which it is used is compressed voice and videoconferencing for rtVBR, and response-time-sensitive data such as Systems Network Architecture (SNA) for nrtVBR.

VBScript

A Microsoft proprietary language derived from Visual Basic (VB). Like JavaScript, VBScript is intended for use as a browser-based language, a server-side (Active Server Pages — ASP) language, and an administrative (Windows Scripting Host — WSH) language. Unlike JavaScript, support for it in browsers is limited to Microsoft's Internet Explorer.

VBUG (Voice Browser Usability Group)

A group within the World Wide Web Consortium (W3C) developing voice portal usability guidelines and best practices. See voice browser, voice portal and W3C.

VBX (Visual Basic Extension)

A Visual Basic add-on that Microsoft is converting to an Object Linking and Embedding (OLE) Custom Controls (OCX) infrastructure.

VC (virtual channel)

In asynchronous transfer mode (ATM), a communications track between two nodes giving the bandwidth needed for a virtual connection across the network.

VC (virtual circuit)

In packet switching, network facilities that appear to users to be an end-to-end circuit, but are in fact a dynamically variable network connection in which sequential user data packets may be routed differently during the course of a "virtual connection." Transmission facilities may be shared by many virtual circuits simultaneously.

VCI (Value Chain Initiative)

An effort by Microsoft and participating partners to define broadly accepted program interfaces to support the integration of applications across enterprises in specific vertical industries, such as banking, insurance and healthcare.

VCP (voice and call processing)

VCR (videocassette recorder)

VDB (virtual database)

VDSL (very high-speed digital subscriber line)

Extremely high-speed digital subscriber line (DSL) connections over short distances.

VDI (video display terminal)

VDU (visual display unit)

vector facility (see VF)

vector quantization

A source-encoding compression technique based on the division of the data stream into blocks called vectors, which are compared to vectors in a code book, the references of which are transmitted.

Vendor Independent Messaging (see VIM)

Vector Markup Language (see VML)

vendor-managed inventory (see VMI)

vendor-neutral

A state in which no one vendor can control the definition, revision or distribution of a specification. Vendor-neutral specifications encourage the development of competing yet compatible implementations, freeing the purchaser to choose from a multitude of vendors without suffering a loss of functionality. Vendor-neutral specifications must be comprehensive, consistent, and either publicly available or licensed at a nominal fee. Additionally, they must be defined by a multilateral association that is

representative of a broad cross-section of the computer industry, open to new members, publishes the rules of membership and operates according to democratic principles. Preferably, a vendor-neutral specification is supplemented with at least one reference implementation. This reference would be available in a format that allows re-creation — that format would be source code for software implementations — and a set of conformance tests that sufficiently ensure the implementation's integrity under all reasonable conditions of projected use.

vendors against Microsoft (see VAM)

vendor-unique master enterprise identifier (see VUMEI)

VEP (vertical enterprise portal)

A portal that typically includes a business application and focuses on a specific business function (e.g., investment or customer service).

Veronica

A former Internet search tool that used descriptive text; a precursor to today's Web search engines.

vertical enterprise portal (see VEP)

vertical portal

A portal that provides content aggregation relevant to a specific industry, links to related sites, collaboration capabilities and e-commerce services for relevant products and services.

very high-speed digital subscriber line (see VDSL)

very large database (see VLDB)

very large-scale integration (see VLSI)

very long instruction word (see VLIW)

very small aperture terminal (see VSAT)

VESA (Video Electronics Standards Association)

VESA local bus (see VL bus)

VF (vector facility)

An attachment to a processor that enables the processor to run programs that issue vector instructions, which are particularly useful in scientific calculations, but are not particularly useful for database operations.

VF (voice frequency)

VFast

The precursor to V.34 from Rockwell International, originally endorsed by many modem manufacturers.

VGA (Video Graphics Array)

A hardware display and software resolution standard for personal computers.

VHDFDD (very high-density flexible disk drive)

VHDL (VLSI Hardware Description Language)

An industry standard format for describing integrated-circuit logic and behavior.

VHDL Initiative Towards Standardization of ASIC Libraries (see VITAL)

VHE (virtual home environment)

A concept used to indicate the network infrastructure needed to support mobile users. VHE is part of IMT-2000 and UMTS.

VHF (very high frequency)

VHS (video home system)

A videocassette format.

VHSIC (very high speed integrated circuit)

VICS (Voluntary Interindustry Commerce Standards)

A committee dedicated to the adoption of bar coding and electronic data interchange (EDI) in the department store and mass merchandise industries. It has established collaborative planning, forecasting and replenishment (CPFR) standards for the consumer goods industry that are published by the Uniform Code Council (UCC).

videoconferencing

Communication by individuals or groups using systems that support image, voice and data transfer over digital networks or telephone circuits. Videoconferencing systems can take the form of large, dedicated units for group meetings or can be integrated with desktop personal computers.

Video Graphics Array (see VGA)

video on demand (see VOD)

video random-access memory (see VRAM)

video server

A server that delivers streams of digital video and audio.

video signal

A signal comprising frequencies normally required to transmit pictorial information (1 to 6 MHz).

videotex

An interactive data communications application broadcast over the public switched telephone network and received on adapted television receivers.

VIE (virtually integrated enterprise)

A business framework hypothesized to emerge as the predominant competitive model for global enterprises in the next decade. The virtual enterprise has evolved

from the traditional “vertically integrated enterprise” of the industrial era to the “virtually integrated enterprise” (VIE) of the rapidly developing “connected” economy. The VIE is a logically, not physically, defined combination of independent enterprises required to fulfill a defined customer need. The model is highly adaptive, speed-based and agile. It is aimed at instant response, and trades speed and agility for precision control.

VIM (Vendor Independent Messaging)

An application programming interface (API) developed to support the exchange of electronic mail among programs from different vendors

VIM (virtually interoperable manufacturing)

A way for manufacturers to gain competitive advantage by integrating business processes across core disciplines and the supply chain, while customizing and integrating the underlying business application products.

Vines

A network operating system (NOS) developed by Banyan Systems (now ePresence).

virtual access call center (see VACC)

Virtual Address Extension (see VAX)

virtual channel (see VC)

virtual circuit (see VC)

virtual classroom

An online location where a course can be experienced. An instructor may lead or facilitate the learning event, while students can merely participate or collaborate with one another. These activities may occur synchronously or asynchronously. Virtual-classroom functionality is similar to that used for conferencing, chat rooms, and other forums in which participants exchange comments or engage in other online interactions.

virtual company

A company integrating several ideals: outsourced noncore competencies; a focus on core business strengths; little or no physical presence or infrastructure; a network of business alliances; the exploitation of intellectual capital; and a heavy reliance on telecommunications. Virtual companies have outsourced the physical processes and administrative attributes of traditional business, and have expanded and combined intellectual activities (e.g., problem solving) with standard business processes such as marketing.

virtual home environment (see VHE)

virtual LAN (see VLAN)

virtual mall

A site on the Internet created and maintained by third parties. Service providers (e.g., insurance companies) can contract services with the parties owning and operating each site to present their products and services online. Consumers can

visit these sites to obtain research on products and generate rate quotes or purchase products from multiple carriers.

virtually interoperable manufacturing (see VIM)

Virtual Machine (see VM)

virtual machine (see VM)

Virtual Memory System (see VMS)

virtual path (see VP)

virtual private data network (see VPDN)

virtual private network (see VPN)

virtual product development management (see VPDM)

virtual reality (see VR)

Virtual Reality Modeling Language (see VRML)

Virtual Router Redundancy Protocol (see VRRP)

Virtual Server Facility (see VSF)

virtual storage

A computer that appears to have a much larger memory than its real memory. This is accomplished by software that moves pages rapidly in and out of a high-speed, random-access storage device, usually a disk.

Virtual Storage Access Method (see VSAM)

Virtual Storage Extended (see VSE)

Virtual Source/Virtual Destinations (see VS/VD)

virtual tape subsystem (see VTS)

Virtual Telecommunications Access Method (see VTAM)

Virtual Terminal (see VT)

Virtual Terminal Protocol (see VTP)

Virtual Tributary (see VT)

virtually integrated enterprise (see VIE)

virus

Software used to infect a computer. After the virus code is written, it is buried within an existing program. Once that program is executed, the virus code is activated and

attaches copies of itself to other programs in the system. Infected programs copy the virus to other programs.

VIS (voice information services)

Visual Basic (see VB)

Visual Basic Extension (see VBX)

Visual Basic for Applications (see VBA)

visualization

To illustrate information objects and their relationships on a display. Strategic visualization graphically illustrates the strength of relationships by the proximity of objects on the display. Advanced technology can make a significant difference in users' ability to interface to large knowledge repositories. These advances use the distance between objects on the display to reflect the similarity of meaning, similarity of content or other relationships (e.g., association with a group).

Visual J++

A Microsoft variant of the Java language.

Visual Studio (see VS)

VITAL (VHDL Initiative Towards Standardization of ASIC Libraries)

A standard for defining the format of integrated-circuit simulation libraries.

VLA (Volume License Agreement)

A software licensing option offered under the Novell "Customer Connections" program, with an entry point that opens discounts to smaller organizations.

VLAN (virtual LAN)

A virtual LAN (local-area network) is a set of systems that, regardless of higher-layer addressing or location, is designated as a logical LAN and treated as a set of contiguous systems on a single LAN segment. Virtual LANs can be proprietary or standardized using the Institute of Electrical and Electronics Engineers (IEEE) 802.1Q. Typical grouping parameters for VLANs include the port number of the hub, switch or router, the higher-layer protocol such as Internet Protocol (IP) or Internetwork Packet Exchange (IPX), the Media Access Control (MAC) address, and the traditional subnet. The goal of VLANs is to provide simpler administration, simple moves/adds/changes to network devices, and partitioning at the MAC layer.

VL bus (VESA local bus)

A local bus architecture introduced by the Video Electronic Standards Association (VESA) to work with the Intel 486.

VLDB (very large database)

A database greater than 100 gigabytes.

VLF (very low frequency)

VLIW (very long instruction word)

A processor architecture in which each data word contains multiple instructions. Used in Intel's next-generation IA-64 ("Merced") products.

VLM (very large memory)

VLM (Virtual Loadable Module)

VLSI (very large-scale integration)

A technology that makes it possible to place the equivalent of between 100,000 and 1 million transistors on a chip.

VLSIC (very large-scale integrated circuit)

VLSI Hardware Description Language (see VHDL)

VM (virtual machine)

A software implementation of a hardware-like architecture, which executes predefined instructions in a fashion similar to a physical central processing unit (CPU). A VM can be used to create a cross-platform computing environment that loads and runs on computers independently of their underlying CPUs and operating systems. A notable example is the Java Virtual Machine, the environment created on a host computer to run Java applets. Although VMs have existed longer than Java, Java has made VMs highly visible (see Java).

VM (Virtual Machine)

The IBM Virtual Machine (VM) set of operating system products (VM/SP, VM/HPO, VM/XA, VM Migration and VM/ESA) manages a system so that all its resources — processors, storage and input/output (I/O) devices — are available to many users at the same time. Each user has at his disposal the functional equivalent of a real dedicated system. Because this functional equivalent is simulated by VM and does not really exist, it is called a "virtual" machine.

VMA (Voice Mail Association)

VM/CP (voice messaging/call processing)

VMEbus

A high-speed parallel backplane system bus originally developed by Motorola, and now widely used on technical workstations and small multiuser systems.

VMI (vendor-managed inventory)

In the VMI process, the vendor assumes responsibility for managing the replenishment of stock. Rather than a customer submitting orders, the vendor will replenish stock as needed. This is sometimes referred to as supplier-managed inventory (SMI) or co-managed inventory.

VML (Vector Markup Language)

An application proposed to the World Wide Web Consortium (W3C) for defining 2-D graphics in Extensible Markup Language (XML).

VML (Veritas Media Librarian)

VMS (Virtual Memory System)

A mainframe operating system developed by Digital Equipment (later acquired by Compaq Computer, which recently merged with Hewlett-Packard).

VOD (video on demand)

A multimedia application with which a user can access motion or still video, which may be available as public or private services.

VoD (voice over data)

Transmission of voice communication over a data network.

VoFR (voice over frame relay)

Transmission of voice communications over a frame relay data network.

voice band

A voice-grade channel.

voice browser

A system that allows telephone access to voice portal sites. It prepares and presents information to callers, interprets their commands and enables them to navigate the site. Architectures and implementations vary, but many will use VoiceXML or a similar protocol to access the portal application. See voice portal and VoiceXML.

Voice Browser Usability Group (see VBUG)

voice digitization

Conversion of an analog voice signal into digital form for storage or transmission.

voice encryption

A function that enables a device to transport digitized voice signals.

Voice Extensible Markup Language (see VoiceXML)

voice-grade channel

A channel with bandwidth equivalent to a telephone line obtained through the public telephone network. The maximum potential bandwidth of a voice-grade channel is approximately 20 kilohertz (KHz); however, most voice grade channels in a transmission facility are usually spaced 4,000 hertz (Hz) apart, and not all of that bandwidth is generally available to a user due to the presence of noise-limiting loading coils. The telephone network itself is usually defined in terms of channels, with frequencies from 300 to 3,400 Hz.

voice mail

A service providing digitized voice message handling.

Voice Markup Language (see VoxML)

voice-messaging system

Also called voice mail and, incorrectly, voice-processing systems. Voice-messaging systems are hardware and software products that operate with most private branch exchange (PBX), hybrid, or key telephone systems, enabling users to send, receive, and redirect voice messages through office telephone systems and computers. Voice-

messaging hardware includes a central processor, analog-to-digital converters, disk storage and input/output ports.

voice-operated device

A piece of equipment that can be controlled by spoken commands.

Voice Over Data (see VoD)

voice over frame relay (see VoFR)

voice over IP (see VoIP)

voice portal

A system that uses advanced speech recognition technology to provide phone access to information on the Internet. Key components of most voice portals include:

- Text-to-speech translation
- Information aggregation
- Categorization software
- Telephony and Internet interfaces
- Administrative interfaces

Optional components include software to support context-sensitive, personalized assistance and support for VoiceXML. See voice browser and VoiceXML.

voice-processing system

A hardware and/or software product that encompass voice messaging, voice recognition, voice response and any other applications that deal with the processing of voice communications.

voice recognition

A hardware and software system that translates human speech into binary text. Also called speech recognition.

voice response system (see VRS)

voice response unit (see VRU)

voice store-and-forward (see VSF)

VoiceXML (Voice Extensible Markup Language)

A voice-technology-based version of Extensible Markup Language (XML) supported by more than 200 companies. It was founded by AT&T, IBM, Lucent and Motorola. See voice browser, voice portal and XML.

VoIP (voice over IP)

Transmission of voice communications over Internet Protocol (IP) data networks, such as IP-based LANs, intranets or the Internet. Many carriers offer integrated services such as voice and data over a single "pipe." However, VoIP still poses several concerns such as voice quality, traffic congestion, slow acceptance of standards, regulatory ambiguity and potential lack of future demand.

Volume License Agreement (see VLA)

volume purchase agreement (see VPA)

volume source cardinality (see VSC)

volume table of contents (see VTOC)

Voluntary Interindustry Commerce Standards (see VICS)

VoxML (Voice Markup Language)

A markup language that encodes Web-site information so that it can be read over the telephone. VoxML was introduced in September 1998 by Motorola's Internet and connectivity services division. VoxML allows for an assortment of speech technologies to be incorporated so Web information on the Web can be relayed through any phone. Sites can convert the text of Web pages to computerized speech, play pre-recorded audio content or construct sentences out of pre-recorded words and phrases.

VP (virtual path)

In asynchronous transfer mode (ATM), the bandwidth between two points on a network used by one or more virtual channels (VCs).

VPA (volume purchase agreement)

An agreement between a computer vendor and a customer under which the vendor grants discounted prices in return for the customer's commitment to purchase a minimum quantity of products.

VPDM (virtual product development management)

A vision that describes a new application realm combining characteristics of computer-aided design (CAD), product data management (PDM) and Web technologies focused primarily on the front end of the design process, where the greatest competitive gains for design innovation occur. While PDM and enterprise resource planning (ERP) technologies have proven themselves in the detail and production stages of product design, they are not yet effective in the conceptual product design stage, which has a much different process model and functional demands, particularly when interenterprise collaboration on large design projects is required.

VPDN (virtual private data network)

A data networking service offered by the provider in a virtually private mode, using virtual private network (VPN) technologies.

VPIM (Voice Profile for Internet Messaging)

VPN (virtual private network)

A system that delivers private communications services on a shared, public-network infrastructure, and provides customized operating characteristics uniformly and universally across an enterprise. The term "VPN" is generically used to refer to voice VPNs. To avoid confusion, those used for data, rather than voice, communications are more properly referred to as "data VPNs" or "IP VPNs." VPN service providers define a VPN as a wide-area network of permanent virtual circuits, generally using asynchronous transfer mode (ATM) or frame relay to transport IP. VPN technology

providers often define "virtual private networking" as the use of encryption software or hardware to bring privacy to communications over a public or untrusted data network.

VPO (Volume Purchase Option)

VR (virtual reality)

A computerized process, usually including special equipment, that projects the user into a simulated three-dimensional space. It gives the user the sensation of being in the simulated environment and the ability to respond to the simulation.

VRAM (video random-access memory)

A type of dynamic random access memory (DRAM) used in high-speed processing of visual data.

VRML (Virtual Reality Modeling Language)

A means of rendering 3-D worlds from mathematical equations or descriptions. A VRML browser can create shapes and text within a navigable 3-D context. The v.2.0 specifications further enhance the immersive experience, allowing for such real-world events as interaction between "visitors" and collision detection when a user "bumps into" an object or other users.

VRRP (Virtual Router Redundancy Protocol)

A network protocol that enables the same virtual Internet Protocol (IP) address to be used by several routers in an IP network — one of which acts as the master router, with the others acting as backups in case the master fails. Specifications for the protocol are defined and developed by the VRRP Working Group of the Internet Engineering Task Force.

VRS (voice response system)

Specialized technologies designed for providing callers with verbal and faxed answers to inquiries without assistance from a person. They provide account information, fulfill requests for mailable items, pre-screen callers for script customization, interact with host systems (read and write) and produce reports.

VRU (voice response unit)

An automated telephone answering system consisting of hardware and software that allows the caller to navigate through a series of prerecorded messages and use a menu of options through the buttons on a touch-tone telephone or through voice recognition.

VS (Visual Studio)

A Microsoft package of several application development (AD) tools with complementary, albeit overlapping, focuses, including Visual Basic, Visual C++ and Visual J++.

VSAM (Virtual Storage Access Method)

IBM's chosen access method for direct-access files. It is optimized for a virtual storage environment. Multiple Virtual Storage (MVS), Data Facility Product (DFP), Information Management System (IMS), Customer Information Control System (CICS), and DB2 all use VSAM.

VSAT (very small aperture terminal)

A satellite-based networking service that has a low per site cost (\$175 per month per site or less). VSATs achieved their low cost by using shared, high-power satellite capacity, asymmetric data flows and a star topology — enabling remote “earth stations” at the sites to be as inexpensive as \$5,000 (including installation). VSAT economies improve as the number of remote or branch sites increase; a VSAT network may connect hundreds or thousands of branch sites to a central site. VSATs have been heavily deployed in retail outlets.

VSC (volume source cardinality)

A model for the deployment of document scanners based on the volume of documents, the number of distinct document types (cardinality), and the source of the document (internal or external).

VSE (Virtual Storage Extended)

A multitasking, 370-architected IBM operating system akin to Multiple Virtual Storage (MVS). VSE work runs in partitions rather than address spaces, and uses Performance Optimized With Enhanced RISC (POWER) for input/output (I/O) rather than Job Entry Subsystem (JES), but is largely similar to MVS.

VSF (voice store-and-forward)

A processor-controlled system that enables voice messages to be created, edited, sent, stored and forwarded. Users access and operate the system by means of any 12-button dial pad in response to voice prompts from the system.

VSF (Virtual Server Facility)

A feature of second-generation Advanced CMOS-ECL (ACE) technology that effectively allows a user to physically partition a system into multiple systems, all within the cabinetry of the bigger system.

VSM (Virtual Storage Manager)

VSS (Versatile Storage Server)

VS/VD (Virtual Source/Virtual Destinations)

One of three types of feedback an asynchronous transfer mode (ATM) switch can provide.

VT (Virtual Terminal)

An application allowing interoperability between heterogeneous terminals (OSI term).

VT (Virtual Tributary)

A structure that transports and switches sub-synchronous transport signal-1 (STS-1) payloads (a SONET term). A VT can be described by size, such as VT_x, with x representing 1.5, 2, 3 or 6.

VTAM (Virtual Telecommunications Access Method)

The main Systems Network Architecture (SNA) subsystem resident in an IBM mainframe that manages session establishment and data flow between terminals and application programs, or between application programs.

VTOA (Voice and Telephony over ATM)

VTOC (volume table of contents)

The data structure on a direct-access volume that describes each data set on that volume.

VTP (Virtual Terminal Protocol)

The protocol driving the Virtual Terminal (VT) application.

VTR (videotape recorder)

VTs (Virtual Tape Server)

VTs (virtual tape subsystem)

Tape library hardware and software extensions that utilize direct-access storage device (DASD) buffers to multiply the tape device count, throughput and storage density of tape library systems.

VT superframe

An organized 500-microsecond superframe structure overlaid on a 125-microsecond synchronous transport signal-1 synchronous payload envelope (STS-1 SPE — a SONET term). Within this structure are the Virtual Tributary (VT) payload pointer and the Virtual Tributary synchronous payload envelope (VT SPE).

VUMEI (vendor-unique MEI)

A master enterprise identifier (MEI) that is generated in such a way that it is guaranteed to be unique across all of a vendor's enterprise master person index (EMPI) systems.

V.x series

A series of communications specifications administered by the International Telecommunication Union (ITU). It includes:

1. V.10: recommendation for unbalanced high-speed communications.
2. V.11: recommendation for balanced high-speed communications.
3. V.17: fax recommendation using TCM modulation.
4. V.21: recommendation for full-duplex asynchronous modems up to 300 bits per second (bps) on dialup lines.
5. V.22: recommendation for full-duplex asynchronous modems of 600 or 1,200 bps on dialup lines.
6. V.22bis: recommendation for full-duplex modems of 2,400 bps on dialup or two-wire leased lines.
7. V.23: recommendation for half-duplex asynchronous modems of 0-600 or 0-1,200 bps on dialup lines.
8. V.24: definitions of interchange circuits between terminal and communications equipment (e.g., between a PC and a modem).
9. V.25bis: definition of call procedures for call setup and tear-down over a data terminal equipment/data circuit-terminating equipment (DTE/DCE) interface.

10. V.26: recommendation for full-duplex synchronous modems of 2,400 bps on four-wire leased lines.
11. V.26bis: recommendation for full-duplex synchronous modems of 1,200 or 2,400 bps on dialup lines.
12. V.26ter: recommendation for full-duplex modems of 2,400 bps on dialup and two-wire leased lines.
13. V.27: recommendation for full-duplex synchronous modems of 4,800 bps on dialup lines.
14. V.27bis: recommendation for full-duplex synchronous modems of 2,400 or 4,800 bps on four-wire leased lines.
15. V.27ter: recommendation for half-duplex synchronous modems of 2,400 or 4,800 bps on dialup lines.
16. V.28: definition of the functions of the RS232 interface.
17. V.29: recommendation for synchronous modems of 4,800, 7,200, or 9,600 bps on four-wire leased lines.
18. V.32: serial line protocol for 4.8 kilobits per second (Kbps) or 9.6 Kbps bidirectional transmission.
19. V.32bis: recommendation that extends V.32 to speeds of 14.4 Kbps.
20. V.33: recommendation for full-duplex synchronous modems of 12,000 or 14,400 bps on four-wire leased lines.
21. V.34: recommendation for communications between a network access device and a packet network; a synchronous protocol driving speeds up to 48 Kbps. See Vfast.
22. V.34+: recommendation that improves V.34 to achieve speeds up to 33.6 Kbps over dialup lines.
23. V.35: recommendation for the transmission of data over the public switched telephone network.
24. V.36: recommendation for four-wire communications at speeds greater than 48 Kbps.
25. V.42: recommendation for error correction using Link Access Procedure for Modems (LAPM).
26. V.42bis: recommendation for modem data compression using a ratio of 4:1.
27. V.52: recommendation for Bit Error Rate Test (BERT) procedures on modem links.
28. V.54: recommendation for loopback tests on modems.
29. V.110: recommendation for supporting DTE equipment on Integrated Services Digital Network (ISDN).
30. V.120: recommendation for asynchronous rate adaptation up to 64 Kbps over ISDN bearer channels.

W

W2P (Windows 2000 Professional)**W2S (Windows 2000 Server)****W3C (World Wide Web Consortium)**

An organization formed in 1994 to promote the Web's evolution as a platform for information, commerce and communication. The W3C, which has more than 500 member organizations, has developed numerous commonly used Web standards — notably Extensible Markup Language (XML) and a variety of XML-related specifications.

W-4GL (workgroup fourth-generation language)**WABI (Windows application binary interface)****WAC (Web authorization and control)**

An access-control technology that works at the Web server, enabling an administrator to define generic user roles and authorize the user access to Web-based data and resources across multiple Web servers, generally at the sub-uniform-resource-locator (URL) level. Solutions are likely to include a central server where a user presents authentication credentials and then receives an encrypted "cookie" or embedded information in the URL. The URL or the cookie is then acknowledged by each of the Web servers in the systems running the WAC agent.

WAE (WAP application environment)

The components used to develop applications for WAP devices. It essentially consists of WML, WMLScript and WTAI specifications.

wafer

A thin, flat piece of semiconductor material used in integrated circuits.

WAG (wireless application gateway)

A server-based gateway that provides wireless access to enterprise applications. WAGs plug into the enterprise's application infrastructure, separating the data from the presentation layer and avoiding redundant development efforts. Leading WAGs provide secure access to any data source and the ability to render the data to any device (e.g., PDA, wireless telephone, pager or desktop). A WAG server can be deployed either as an internal platform installed within the enterprise, or as an outsourced platform hosted by a third party operating as a service bureau.

WAM (Web authorization management)

A building block of an extranet. In addition to authorization and management features being offered as part of an e-commerce system, there are security software

products that work with Web servers and e-commerce systems, allowing administrators to define generic user roles and authorize user access to Web-based data and resources across multiple applications (generally at the sub-URL level).

WAN (wide-area network)

A communications network that connects computing devices over geographically dispersed locations. While a local-area network (LAN) typically services a single building or location, a WAN covers a much larger area such as a city, state or country. WANs can use either phone lines or dedicated communication lines.

WAN performance monitor

A tool or toolset (hardware and software) to allow monitoring of wide-area network (WAN) traffic and problems.

WAN replacement

Wide-area network (WAN) replacement uses Internet-based virtual private networks (VPNs) or managed VPN services to connect branch offices.

WAP (Wireless Application Protocol)

WAP is a set of specifications developed by the WAP Forum (see separate entry) for efficient communication of data over wireless networks to small devices, such as personal digital assistants and cell phones. WAP specifications are based on Internet standards, with extensions to reflect the wireless device environment.

Specifications in the WAP architecture are arranged in a protocol stack consisting of application, session, transaction, security and transport layers. See WAE, WSP, WTP, WTLS and WDP.

WAP application environment (see WAE)

WAP browser

A microbrowser used to locate and display information on WAP-enabled devices. WAP browsers perform client-side functions required to render Web content to a WAP device.

WAP Forum

Founded in 1997 by Ericsson, Motorola, Nokia and Unwired Planet (now Phone.com), the WAP Forum is responsible for publishing and developing Wireless Application Protocol (WAP) specifications. The WAP Forum works closely with the World Wide Web Consortium (W3C) and the Telecommunications Industry Association (TIA). The goal of the WAP Forum is to provide an industry standard to foster interoperability among wireless devices.

WAP Identity Module (see WIM)

WAPI (Workflow API)

The Workflow Management Coalition's (WfMC's) application program interface (API), designed to enable workflow application interoperability in a heterogeneous workflow environment in five key areas between: business process re-engineering (BPR)/modeling tools and workflow systems; workflow systems and workflow-client applications; workflow systems and desktop applications; workflow engines; and common workflow administration and monitoring services.

war dialer

A tool used to scan phone lines to find modem connections. Often used by hackers, these tools dial a series of phone numbers at high speed until a modem connection is detected.

Warehouse Management (see WM)

warehouse management system (see WMS)

warehouse simulation

A detailed mathematical or computer model of all relevant activities of a distribution center. The simulation recreates the center's activities, determines how long the activities take and considers any physical constraints. Warehouse simulations typically focus on capacity planning, throughput calculations, inventory analysis, and the overall performance and cost impacts of selected logistics strategies.

WAS (Windows 2000 Advanced Server)

WASP (wireless application service provider)

A vendor that provides hosted wireless applications so that companies will not have to build their own wireless infrastructures.

WATS (wide-area telephone service)

A telephone company service providing reduced costs for certain telephone call arrangements. It may be IN-WATS or 800-number service, for which calls can be placed to a location from anywhere at no cost to the calling party, or OUT-WATS, for which calls are placed from a central location. The cost is based on hourly usage per WATS circuit and on distance-based zones, or bands, to which (or from which) calls are placed.

wave division multiplexing (see DWDM)

waveguide

A transmission path in which a system of boundaries guides electromagnetic energy. The most common of these are hollow metallic conducting tubes (microwave communications) or rods of dielectric material. See fiber-optic waveguides.

wavelength

The distance between the crests of a wave in a radio signal, measured as the speed of light divided by the frequency in hertz (Hz).

WBEM (Web-Based Enterprise Management)

An initiative started by BMC software, Compaq, Cisco, Intel and Microsoft in 1996 and accepted by the Desktop Management Task Force (DMTF) in June 1998. Its objective is to define a management environment where all managing systems and applications can access, control and share management information with each other and with any agent or managed device. The two important parts of WBEM are the Common Information Model (CIM) specification — which defines the WBEM implementation requirements and the CIM schema — and a standard management schema to hold object definitions for servers, applications, databases and network.

W-CDMA (wideband code division multiple access)

A technology for wideband digital radio communications in the Internet, multimedia, video and other capacity-demanding applications. W-CDMA utilizes the radio spectrum to provide a maximum data rate of 2 Mbps.

WCL (Windows Communication Library)

WCS (warehouse control system)

WDC (wireless data communication)

A form of communication that uses the radio spectrum rather than a physical medium. It may carry analog or digital signals and may be used on LANs or WANs in one- or two-way networks.

WDCS (Windows 2000 Datacenter Server)

WDM (wave division multiplexing)

See DWDM.

WDP (Wireless Datagram Protocol)

This allows Wireless Application Protocol (WAP) to be bearer-independent by adapting the transport layer of the underlying bearer service and presenting a consistent data format to the higher layer of the WAP protocol stack.

wearable computers

A new category of computing technology representing the far end of the portability spectrum. A typical wearable computer might consist of the following components — all of which have been created in at least prototype systems: a belt-mounted unit that contains the main processor, storage and batteries; use of wireless technology — a pair of glasses with built-in speakers, microphone and miniature video camera — to communicate with the I/O unit; and computer displays that are projected using the glasses to generate a transparent holographic display, which is superimposed over the user's normal field of vision. Commands would be entered using speech or a small handheld device that acts as a keyboard/mouse equivalent. The system might also include biosensors in a wristwatch-like device. The most visible development of wearable computers is occurring as part of the Massachusetts Institute of Technology (MIT) Media Laboratory's Wearable Computing Group.

Weather Observation Markup Format (see OMF)

Web

The Web (short for World Wide Web) is a hypertext-based global information system that was originally developed at the European Laboratory for Particle Physics in Geneva. It is a subset of the Internet, technically defined as the community on the Internet where all documents and resources are formatted using Hypertext Markup Language (HTML). HTML, and the related Hypertext Transport Protocol (HTTP), make it easy to find and view data and documents stored on computers connected to the Internet. HTML creates the links ("hyperlinks") that enable the user to move among many Web documents with the click of a mouse.

Web analytics

The use of a range of quantitative analyses to understand Web site performance and visitor experience. These analyses include usage levels and patterns on an individual

and aggregate level. Data sources may include clickstream data from the Web server log, Web transaction data, submitted data from input fields on the Web site and data in the Internet customer repository. The results may be used to improve site performance (from a technical and content perspective), enhance visitor experience (and thus loyalty), contribute to overall understanding of customers and channels, and identify opportunities and risks.

Web ATM

An umbrella term for several different versions of technology that link the Internet and an automated teller machine (ATM) networks.

Web authorization and control (see WAC)

Web authorization management (see WAM)

Web-Based Enterprise Management (see WBEM)

Web browser

A program used to access and browse the Web.

Web clipping

The wireless browser technology used in Palm PDAs.

Web crawler

A piece of software (also called a spider) designed to follow hyperlinks to their completion and to return to previously visited Internet addresses.

WebDAV (Web Distributed Authoring and Versioning)

An integrated document management (IDM) protocol proposed as an extension of Hypertext Transport Protocol (HTTP) 1.1 to provide library services for Web server resources. In a WebDAV IDM implementation, library services are executed on a resource basis rather than through a traditional relational database. This architecture enables shared locking, partial updates and reservations.

Web Distributed Authoring and Versioning (see WebDAV)

Web e-mail

An e-mail option that requires only a browser. A user can walk up to any Internet-connected device (e.g., a PC or airport kiosk), launch a browser, connect to a Web mail server, enter a user name and password and check e-mail.

Web-enabled

Refers to any application or document that uses the Internet as a communication backbone while exploiting Hypertext Transport Protocol (HTTP) as a means to link to other applications or content.

Web hosting

A service in which a vendor offers the housing of business-to-business (B2B) or business-to-consumer (B2C) e-commerce Web sites via vendor-owned shared or dedicated servers and applications for enterprises at the provider-controlled facilities. The vendor is responsible for all day-to-day operations and maintenance of the Web site. The customer is responsible for the content.

Web integration servers

Web servers that directly support Hypertext Transport Protocol (HTTP); execute a proprietary, high-level 4GL or scripting language; and include one or more adapters for databases, legacy systems and packaged applications. Extensible Markup Language (XML)-based Web integration servers are similar; however, they also use XML data internally within the server and externally with clients and other applications.

Weblog

A form of online publishing that uses a daily or frequently updated, log-type format. Personal weblogs (also known as "blogs") have become popular among individuals as a means of sharing their thoughts or creating informal forums for discussion. They typically take the form of a daily record of a person's thoughts, observations or opinions, posted together with links to related sites. Postings to the weblog may be limited to the individual who manages it, or others may be invited to participate. Enterprises may also use personal-publishing formats and technologies to create corporate weblogs, as a means of communicating, sharing knowledge or fostering discussions to further enterprise goals. In this respect, weblogs are a precursor to the trend of personal knowledge management becoming an important part of the enterprise environment.

Webmaster

The person who maintains a Web site.

Web page

A Web document — usually based on Hypertext Markup Language (HTML) — that may contain text, graphics, online audio, video, Java or ActiveX objects. These pages are linked to other pages using hypertext to form a Web site.

Web phone

A cell phones equipped with a microbrowser and network data capability through Wireless Application Protocol (WAP) or other Web integration technologies. These devices differ from smart phones (see separate entry) in that the latter are more data-centric, offering network-independent (offline) applications such as contact management and expense reporting.

Web server

The central location that hosts Web pages or a Web site and enables a remote "client" (system or program) to access the material held.

Web services

A software concept and infrastructure — supported by several major computing vendors (notably Microsoft and IBM) — for program-to-program communication and application component delivery. The Web services concept treats software as a set of services accessible over ubiquitous networks using Web-based standards and protocols.

Specifically, a Web service is a software component that can be accessed by another application (such as a client, a server or another Web service) through the use of generally available, ubiquitous protocols and transports, such as Hypertext Transport Protocol (HTTP). Joint efforts between IBM and Microsoft, with the support of other vendors such as Ariba and Iona Technologies, have produced agreement on a basic

set of XML-based standards for Web service interface definition, discovery and remote calling. They include:

- Web Services Description Language (WSDL) for describing Web service interfaces
- Universal Description, Discovery and Integration (UDDI) as the means for users to publish and locate available Web services, their characteristics and interfaces
- Simple Object Access Protocol (SOAP), which enables an application to call a Web service

See HTTP, .NET, SOAP, UDDI, WSDL and XML.

Web Services Description Language (see WSDL)

Web site

A collection of files accessed through a Web address, covering a particular theme or subject, and managed by a particular person or organization. Its opening page is called a home page. A Web site resides on servers connected to the Web network and is able to format and send information requested by worldwide users 24 hours a day, seven days a week. Web sites typically use HTML to format and present information and to provide navigational facilities that make it easy for the user to move within the site and around the Web.

WebTrust

The American Institute of Certified Public Accountants' online privacy seal program (OPSP) to identify Internet companies utilizing appropriate processes and operations to protect the confidentiality of consumer information.

Web TV

Web TV services enable a user to access the Web on a television set using a special remote control and a decoder that sits on top of the TV. Services are offered by various types of provider — e.g., TV broadcasters, satellite operators, and telecom operators. The connection can be provided over various media — analog or digital telephone lines, cable network or satellite links — depending on local infrastructure.

WECA (Wireless Ethernet Compatibility Alliance)

A group formed in 1999 to promote global use of the IEEE's 802.11b wireless LAN standard (which WECA calls "Wi-Fi"), and to certify product compliance with the standard. See Wi-Fi.

WEDI (Workgroup for Electronic Data Interchange)

WEP (Wired Equivalent Privacy)

A feature used to encrypt and decrypt data signals transmitted between Wireless LAN (WLAN) devices. An optional IEEE 802.11 feature, WEP provides data confidentiality equivalent to that of a wired LAN that does not employ advanced cryptographic techniques to enhance privacy. WEP makes WLAN links as secure as wired links.

WFC (Windows Foundation Classes)

Microsoft's object-oriented Java application framework and component model based on J/Direct Java Virtual Machine (JVM). The Java applications leveraging WFC and J/Direct can bypass much of the overhead of the Java runtime environment by directly accessing the Win32 application programming interface (API).

WfM (Wired for Management)

A blueprint, developed by Intel in partnership with and supported by leading information technology (IT) vendors, for making PC-based systems — including desktops, mobile systems and servers — universally manageable. The initiative encompasses advances in hardware and software that enable system management software applications that provide management to the desktop, mobile system or server, and the integration with systems management tools and frameworks. The innovations in hardware and software collectively enterprises should understand, evaluate and incorporate into all PC acquisitions.

WFM (workforce management)

1. Workflow and messaging associated with moving a candidate through the recruiting and new-hire process, as well as managing daily employee business activities, such as time and attendance, project status, expense reporting, invoicing and payment, and detailed reporting.
3. In call centers, WFM systems maximize the use of agent labor by projecting incoming call volumes and scheduling staff to meet needs exactly, by time of the day, day of the week. WFM systems use historical calling records, which are collected from the automatic call distribution system, to project future calling patterns and volumes for specified time frames. Features include:
 - Call volume forecasting
 - Calculation of the required number of agents, based on the desired average speed of answer
 - Agent scheduling
 - Meeting and vacation planning
 - Reporting
 - "What if" analysis

WfMC (Workflow Management Coalition)

The Workflow Management Coalition, founded in 1993, is a nonprofit international organization of workflow vendors, users, analysts and university research groups. WfMC's mission is to promote and develop the use of workflow technology through the establishment of standards for software terminology, interoperability and connectivity between workflow products.

WfMC API (see WAPI)

WfMC Interface 1 (Workflow Management Coalition Interface 1)

The WfMC's interface between modeling and definition tools and runtime workflow management software. It consists of an interchange format and applications programming interface (API) calls, which can support the exchange of process definition information over a variety of physical or electronic interchange media. The interface may support the exchange of a complete process definition or a subset —

e.g., a set of process definition changes or the attributes of a particular activity within the process definition.

WfMC Interface 2 (Workflow Management Coalition Interface 2)

The WfMC's interface designed to provide access from a workflow application to the workflow engine and work list, irrespective of the nature of actual product implementation.

WfMC Interface 3 (Workflow Management Coalition Interface 3)

The WfMC's interface intended for application agents and (longer-term) applications that have been designed to be "workflow enabled" (i.e., to interact directly with a workflow engine). This interface has been amalgamated into Interface 2.

WfMC Interface 4 (Workflow Management Coalition Interface 4)

The WfMC's interface designed to enable two or more workflow engines to communicate and work together to coordinate work. Interface 4 (i.e., WAPI4-WFMC-TC-1012) establishes the definition of all the requests and responses that two workflow engines must exchange to support processes that interoperate across a network. The exact coding used to transport those requests and responses is released in a separated binding specification. This separation allows the specification of different bindings using different network and encoding protocols. The first defined binding was released in March 1997, and is called WAPI4/MIME (WFMC-TC-1018). WAPI4/MIME uses Internet mail as the transport mean and Multipurpose Internet Mail Extensions (MIME) and Common Gateway Interface (CGI) for content encoding. See JFlow.

WfMC Interface 5 (Workflow Management Coalition Interface 5)

The WfMC's interface designed to let an independent management application interact with different workflow domains, although alternative implementation scenarios are also feasible. For example, the management application may be an integral part of one enactment service, although capable of managing various functions across additional (heterogeneous) workflow domains.

WGS (workgroup systems)

A future-oriented, vendor-independent model for office information systems (OIS). Key WGS proposals include a complete, cohesive architecture, distributed-logic client/server technology and interenterprise-capable, enterprise-class platforms for communications, collaboration, word and knowledge reuse.

what you see is what you get (see WYSIWYG)

WI (Windows Installer)

WID (wireless interactive device)

An emerging form of mobile computing and communications device — interactive and "always on" in nature — which Gartner expects to evolve over the course of the next 10 years. Driven by the coalescence of a number of technologies, WIDs will evolve from personal digital assistants, Wireless Application Protocol (WAP) telephones and other form factors. Gartner expects that increasing price/performance capabilities of WIDs will combine with improvements in wireless access and user interface technologies (e.g., speech recognition and wearable

displays) to provide inexpensive, lightweight devices that will be broadly adopted and frequently used. Features such as location sensing and tiny, low-cost cameras will further enhance the range of capabilities and services that users will experience with WIDs.

wide-area network (see WAN)

wide-area telephone service (see WATS)

wideband (see broadband)

wideband channel

A channel wider in bandwidth than a voice-grade channel.

wideband code division multiple access (see W-CDMA)

WIDL (Web Interface Definition Language)

Wi-Fi (Wireless Fidelity)

The Wireless Ethernet Compatibility Alliance's (WECA's) name for the IEEE's 802.11b standard for wireless LANs operating at 2.4 gigahertz (GHz). WECA promotes the standard's use for wireless products, and performs interoperability certification on products submitted by member companies for testing. The Wi-Fi standard is supported by more than 40 wireless technology vendors. See 802.11b.

Wi-Fi5

The certification mark issued by the WECA to certify that a product conforms to the 802.11a standard for wireless LANs operating at 5GHz. See 802.11a.

WIM (WAP Identity Module)

A WIM is used in performing wireless transport layer security (WTLS) and application-level security functions. It can be used to process and store user identification and authorization information. It can also be used to store encryption and authentication keys, as well as perform encryption and digital signature functions on the module. A WIM can be a hardware device, such as a smart card or Subscriber Identity Module (SIM).

WIMPS (windows, icons, menus, pointers, scroll bars)

A style of graphical user interface (GUI) originally developed by Xerox, popularized by the Apple Macintosh, and adopted by Microsoft for Windows.

Win16

An abbreviation for "16-bit Windows" — i.e., versions of Microsoft Windows (e.g., Windows 3.1) that predate Windows 95.

Win2000

Abbreviation for "Windows 2000" (see separate entry).

Win32

Abbreviation for "32-bit Windows" — i.e., Windows 95 and later versions of Microsoft Windows.

Win95

Abbreviation for "Windows 95" (see separate entry).

Win98

Abbreviation for "Windows 98" (see separate entry).

windowing

A display technique that uses multiple screen segments to display different items of information. The display can take two forms: tiling (breaking up the screen into discrete segments) and overlapping (producing a three-dimensional effect by having a screen segment partially or fully obscure another segment).

Windows

The operating system developed by Microsoft for PCs. The first version of Windows was introduced in 1985, and although it presented a graphical user interface (GUI) rather than its text-based predecessor, MS-DOS, it was clumsy compared to Apple's Macintosh operating system. However, an overwhelming base of MS-DOS users combined with superior marketing tactics have since made Microsoft's Windows the dominant operating system for PCs.

Windows 2000

Microsoft's family of high-end operating system (OS) products for client/server computing, formerly known as Windows NT. Launched in February 2000, the Windows 2000 product line includes Windows 2000 Professional (replacing Windows NT Workstation) as the client-side OS, and Windows 2000 Server (replacing Windows NT Server) on the server side.

Windows 95

A 32-bit operating system introduced by Microsoft in 1995. It offered significant enhancements over prior releases of Windows, with features like memory protection, multithreading, integrated networking and pre-emptive multitasking.

Windows 98

Microsoft's follow-up to Windows 95, launched in June 1998. Enhancements over Windows 95 included improved self-maintenance and update features, and native support for Universal Serial Bus (USB).

Windows CE

Microsoft's lightweight, micro-kernel-based operating system for Pocket PC computing devices, such as personal digital assistants (PDAs), smart phones, set-top boxes, retail point-of-sale devices and displays. Windows embedded operating systems now shipping include Windows CE.NET and Windows XP Embedded. Currently in development is "Mira," which is the code name for a set of Windows CE.NET-based technologies designed to extend the Windows XP experience throughout the home in "smart" wireless devices.

Windows Foundation Classes (see WFC)

windows, icons, menus, pointers, scroll bars (see WIMPS)

Windows Internet Naming System (see WINS)

Windows Management Initiative (see WMI)

Windows Me

Microsoft's desktop operating focused on the consumer market, released in September 2000. Enhancements over Windows 98 include new digital-media, home-networking and the online features designed to be easy for home PC users to use and understand.

Windows NT

Former name for Microsoft's high-end 32-bit operating system for client/server computing, now known as Windows 2000.

Windows Open Services Architecture (see WOSA)

Windows Scripting Host (see WSH)

Windows Query Language (see WQL)

WINS (Windows Internet Naming System)

A Windows NT network utility that resolves named nodes to their IP addresses.

Wintel

Refers to the personal computer (PC) environment composed of a Microsoft Windows operating system running on an Intel microprocessor. This environment dominates the PC market, and therefore Microsoft and Intel together control many aspects of the PC industry. The widely accepted standards of this environment have encouraged significant software development for this de facto standard platform, allowing software, files and data to be more easily shared among users and enhancing productivity. However, some have claimed that the dominance of Microsoft and Intel in the PC marketplace has reduced innovation and allowed Microsoft and Intel to set guidelines that enhance their corporate profits rather than benefit users.

WIP (Windows Interface Protocol)

WIP (work in progress)

WIPO (World Intellectual Property Organization)

Wired Equivalent Privacy (see WEP)

Wired for Management (see WfM)

wireless application gateway (see WAG)

Wireless Application Protocol (see WAP)

wireless application service provider (see WASP)

wireless certificate services

The initial focus of wireless transport layer security (WTLS) will be the use of digital certificates at the Wireless Application Protocol (WAP) gateway to provide strong authentication to the cellular phone that is connected to a legitimate server. This will involve cellular phones being preloaded with root certificates signed by certificate authorities with which the wireless device manufacturer has entered into trust relationships.

wireless credit card terminal

A wireless device that can be used to process purchases such as taxi rides or delivered pizzas.

wireless data communication (see WDC)

Wireless Datagram Protocol (see WDP)

Wireless Ethernet Compatibility Alliance (see WECA)

Wireless Fidelity (see Wi-Fi)

wireless instant messaging

Instant messaging available from a wireless device. This is an evolution of two-way Short Message Service (SMS) and paging technologies. Instant messaging not only offers Internet compatibility but, with its concept of "buddy lists," permits autodiscovery of addressable recipients — improvements that could lure many committed SMS and paging users to move to the new level of functionality. See instant messaging and SMS.

wireless interactive device (see WID)

wireless Internet service provider (see WISP)

wireless LAN (see WLAN)

Wireless Markup Language (see WML)

wireless office system (see WOS)

Wireless Public Key Infrastructure (see WPKI)

Wireless Session Protocol (see WSP)

wireless telephony application interface (see WTAI)

Wireless Transaction Protocol (see WTP)

Wireless Transport Layer Security (see WTLS)

wireline cellular company

A U.S. term referring to the Block B carrier. Under the Federal Communications Commission's (FCC's) initial licensing procedures, the Block B carrier is the local telephone company. Wireline or Block B systems operate on the frequencies ranging from 869 megahertz (MHz) to 894 MHz.

wiring closet

A dedicated room or enclosure serving as a central point for a data communications or a telecommunications network's equipment.

WISP (wireless Internet service provider)

A wireless gateway service that connects the wired Internet to one or more wireless bearer services.

WLAN (wireless LAN)

A LAN communication technology in which radio, microwave or infrared links take the place of physical cables. Three physical media types of WLAN are available. The first two — direct-sequence spread spectrum (DSSS) and frequency-hopping spread spectrum (FHSS) — are based on radio technologies that are not interoperable. The third is based on infrared, a nonradio technology based on lightwaves. Infrared can coexist with DSSS and FHSS radio-based systems in one enterprise network. However, Internet working issues between access points prevent an enterprise from mixing and matching WLAN devices from multiple vendors. WLAN standards include IEEE 802.11 and HIPERLAN/2 (see separate entries). Currently, two versions of the IEEE standard prevail: 802.11b (Wi-Fi), offering up to 11 Mbps, and 802.11a (Wi-Fi5), offering up to 54 Mbps. Although WLANs can be found in a corporate environment, a number of service providers are offering commercial services in "hot spots," such as airline lounges and coffee bars.

WLL (wireless local loop)

A wireless connection of a telephone in a home or office to a fixed telephone network.

WM (Warehouse Management)

An SAP R/3 module.

WMF (Workflow Management Facility)

See JFlow.

WMI (Windows Management Initiative)

Microsoft's technology implementation of the Desktop Management Task Force's (DTMF's) Web-Based Enterprise Management (WBEM) architecture. WMI extends the DMTF's Common Information Model (CIM) to represent management objects in Windows management environments, thus supporting system and application management based on CIM.

WML (Wireless Markup Language)

A programming language similar to XML, used to create pages that can be displayed in a Wireless Application Protocol (WAP) browser (e.g., a WAP phone).

WMLScript

A scripting language similar to JavaScript.

WMS (warehouse management system)

A software application that manages the operations of a warehouse or distribution center. Application functionality includes receiving, inventory management, cycle counting, task interleaving, wave planning, order allocation, order picking, replenishment, packing, shipping, labor management and automated material-handling equipment interfaces. The use of radio frequency technology in conjunction with bar codes provides the foundation of a WMS, delivering accurate information in real time.

WORA (write once, run anywhere)

workflow

The automation of work among users where the system is intelligent enough to act based on the definition of work types, users and tasks, and the recognition of dynamic processing conditions.

Workflow API (see WAPI)

workflow management

There are two types of workflow management:

1. Internal and external process integration — a workflow approach that allows for the definition of business processes that span applications, including those that come from different vendors. This usually requires a standards-based commercial workflow development environment
2. Automated events or processes — a workflow approach that enables automated tasks (e.g., the automation of steps in a marketing campaign or a sales process) to be performed

Workflow Management Coalition (see WfMC)

workforce analytics

An advanced set of data analysis tools and metrics for comprehensive workforce performance measurement and improvement. It analyzes recruitment, staffing, training and development, personnel, and compensation and benefits, as well as standard ratios that consist of time to fill, cost per hire, accession rate, retention rate, add rate, replacement rate, time to start and offer acceptance rate.

workforce management (see WFM)

workgroup systems (see WGS)

work management

A set of software products and services that apply workflow structure to the movement of information as well as to the interaction of business processes and human worker processes that generate the information. Work management streamlines and transforms crucial business processes and thus can improve results and performance.

workstation

A physical location where work is performed.

World Wide Web (see Web)

World Wide Web Consortium (see W3C)

worm

A malicious software program able to distribute multiple copies of itself within monolithic or distributed systems. Unlike a computer virus, it does not have a separate executable payload. Therefore, the primary impact of this type of threat is to system availability.

WORM (write once, read many)

A digital, optical storage medium on which information can be recorded once and read many times. Provides for extremely compact storage of data at relatively low prices compared to traditional magnetic storage.

WOS (wireless office system)

A technology that allows the user to transfer calls to a mobile telephone.

WOSA (Windows Open Services Architecture)

An architecture and set of application programming interfaces (APIs) that position Windows as a universal client. It standardizes the interfaces that developers can use in accessing underlying network services.

WP (word processing)

WPBX (wireless private branch exchange)

WPKI (Wireless Public Key Infrastructure)

A proposed method for handling public and private keys and digital certificates on the wireless-client (handset) side. The standards for WPKI and Wireless Transport Layer Security (WTLS) are complementary. See WTLS.

wpm (words per minute)

WQL (Windows Query Language)

A service included in Microsoft's Windows Management Initiative (WMI) to query the information model.

WRAM (Windows Random-Access Memory)

wrap

Automatic action taken by a Fiber Distributed Data Interface (FDDI) network in case of failure. Stations on each side of a break create a single logical ring out of the primary and secondary rings so that the network can still run.

wrapper

A layer of software that provides a new interface to the program around which it is wrapped. The purpose of a wrapper is to make the underlying program accessible to an otherwise incompatible external requesting program. A wrapper is a custom proxy program that performs either or both of the following two functions:

- Protocol bridging, such as from an object request broker (ORB) to another ORB, a transaction processing (TP) monitor, a file or a messaging system.
- Message content transformation, e.g., American Standard Code for Information Interchange (ASCII) to Extended Binary-Coded Decimal Interchange Code (EBCDIC), data type translation or truncating, padding or dropping fields.

Sometimes the wrapper must reconcile dissimilar communication patterns. This might involve talking to a wrapped legacy application via conversational terminal

emulation while displaying a message-queuing or ORB-style interface to external requesters.

wrapping

A general label for techniques that interpose conversion software between the source and user of data, and perform currency conversion as data is read or written through the security wrapper. Wrappers generally are used to access legacy function from an object-oriented environment, although they can also be developed the other way around — by having legacy applications access object-oriented functions on different systems. Wrappers can be developed to operate bidirectionally.

write once, read many (see WORM)

WSDL (Web Services Description Language)

A language that provides a document format and an Extensible Markup Language (XML) grammar for working with Web services. It is used to describe Web services interfaces for publication in a public registry based on Universal Description, Discovery and Integration (UDDI). The WSDL specification was introduced by Ariba, IBM and Microsoft in September 2000, and later submitted to the World Wide Web consortium (W3C) with a request that a W3C working group be formed to oversee its development. See Web services and UDDI.

WSH (Windows Scripting Host)

A Microsoft script execution facility that enables the use of scripts for noninteractive administrative tasks (e.g., bulk addition of users, log-on scripts). By default, WSH supports two Microsoft-supplied scripting language engines: JScript and VBScript. Scripting languages are implemented by plugging scripting engines into the WSH framework, which includes an object model. The object model provides script-independent access to Microsoft ActiveX objects, e.g., Active Directory Services Interface (ADSI), ActiveX Data Objects (ADO), Windows Management Interface (WMI) and Microsoft Office components. Implementing ActiveX object access inside WSH excuses scripting languages from that chore.

WSI (wafer scale integration)

WSP (Wireless Session Protocol)

A layer that links the Wireless Application Environment (WAE) to two session services: a connection-oriented service operating above the Wireless Transaction Protocol (WTP) and a connectionless service operating above the Wireless Datagram Protocol (WDP).

WSTS (World Semiconductor Trade Statistics)

WTAI (Wireless Telephony Application Interface)

Specifies how Wireless Application Protocol (WAP) applications can access mobile phone functionality, e.g., initiate a call or send a Short Message Service (SMS) message.

WTLS (Wireless Transport Layer Security)

Within the Wireless Application Protocol (WAP) framework for cellular-phone interface services, WTLS provides security functions similar to those of the Secure Sockets Layer (SSL) protocol used on the Web. See WAP.

WTP (Wireless Transaction Protocol)

A WAP protocol that runs on top of a datagram service such as Wireless Datagram Protocol (WDP) to provide a simplified protocol suitable for low bandwidth mobile applications. WTP offers three classes of transaction service: unreliable one-way request, reliable one-way request and reliable two-way request respond.

WWW (World Wide Web)

See Web.

WYSIWYG (what you see is what you get)

A user interface that presents a screen image that closely matches the appearance of the printed document.

X

X12

A electronic data interchange standard from the American National Standards Institute.

X.25

A wide-area network protocol for system and network interconnection.

X.400

An International Organization for Standardization (ISO) and International Telecommunication Union (ITU) messaging protocol that allows e-mail to move between different mail systems.

X.400 Application Programming Interface (see XAPI)**X.500**

The International Telecommunication Union (ITU) standard for directory services.

X.509

The certification authority standard administered by the International Telecommunication Union (ITU).

XA

A component of the Open Group's X/Open distributed transaction processing standards, which defines the interface between the Resource Manager and the Transaction Manager.

XA (Extended Architecture)

System/370 XA was the second major extension of IBM's System/360 architecture. It was delivered in 1983 on the 308x processors, and was later made available on the 3090 and 4381.

XAPI (X.400 Application Programming Interface)

APIs developed by the X.400 API Association (XAPIA) to sit on top of the X.400 protocols, simplifying the task of submitting and fetching X.400-compliant messages. The message submitted is transported by an X.400-compliant message transfer agent.

XAPIA (X.400 API Association)**xCBL (XML Common Business Library)**

The specification for the XML-centric transactional infrastructure used in Commerce One's e-marketplace platform. xCBL was originally developed by Veo Systems, an XML developer acquired by Commerce One in January 1999. xCBL is more ambitious

than Ariba's cXML, in that its usage is not designed to be restricted to the marketplace environment.

Xcert Universal Database (see XUDA)

XDR (Extended Data Reduced)

A proprietary set of Extensible Markup Language (XML) tags required to establish valid XML schema data types (e.g., "integer") in Microsoft's BizTalk Framework.

XDS

An interface that connects to X.500 directory services, designed by the X.400 API Association (XAPIA).

xDSL

Any of several variants of digital subscriber line (DSL). See ADSL and SDSL.

Xenix

A multiuser operating system developed by Microsoft. A subset of Unix, it is no longer sold.

Xeon

Intel's high-performance workstation processor, which is based on the same "P6" core as the Pentium II and Pentium III chips.

Xerox Network Systems (see XNS)

XFDL (Extensible Form Definition Language)

UWI.Com's Universal Form Definition Language (UFDL) version rewritten in Extensible Markup Language (XML) syntax. See UDFL.

XFN (X/Open Federated Naming)

A federated-naming service specification, similar to Messaging Application Programming Interface (MAPI) and Apple Open Collaborative Environment (AOCE), for the Unix environment. XFN, which first shipped with Sun's Solaris operating system in 1994, is now a technical standard of the Open Group.

XGA (Extended Graphics Array)

A high-resolution video display mode that provides screen pixel resolution of 1,024 by 768 in 256 colors, or 640 by 480 in high (16-bit) color.

XHTML (Extensible HTML)

XIE (X-Windows Image Extensions)

Extensions that support client compression/decompression.

Xjack

A miniature RJ11 modem jack for the Personal Computer Memory Card Industry Association's PCMCIA/PC Card modems.

XLink (XML Linking Language)

The language used to create and manage links among content components written in Extensible Markup Language (XML). XLink uses XML syntax to create constructs that can be inserted into XML documents to create links of varying complexity — from the

simple, unidirectional hyperlinks of Hypertext Markup Language to more sophisticated, multidirectional links. Specifications for the language are defined in the World Wide Web Consortium's proposed XLink Recommendation. See XML.

XMI (XML Metadata Interchange Format)

An Extensible Markup Language (XML) grammar for exchanging metadata between repositories.

XML (Extensible Markup Language)

A metalanguage approved as a World Wide Web Consortium (W3C) recommendation in February 1998. A simplified version of Standard Generalized Markup Language (SGML), XML captures SGML's key advantages (e.g., extensibility) without its more obscure features. Because it is a metalanguage (a language to define languages), it intrinsically offers Hypertext Markup Language (HTML) capabilities and can be used to create HTML documents.

A family of XML-related standards (formally called "recommendations") have been under development by the W3C since 1997. These include XML Linking Language (XLink), XML Path Language (XPath), XML Pointer Language (XPointer), Extensible Stylesheet Language (XSL) and XSL Transformations (XSLT). Together, they form a critical foundation for today's Web-based computing and e-commerce infrastructures. (See XLink, XPath, XPointer, XSL and XSLT.)

XML Common Business Library (see xCBL)

XML/EDI (Extensible Markup Language/Electronic Data Interchange)

Specifically refers to the effort being made by the Data Interchange Standards Association (DISA), CommerceNet and the XML/EDI group, not only to use the XML standards and technologies for business data interchanges, but also to leverage the use of the existing X12 data dictionaries, code lists, semantics, and general business domain knowledge encapsulated in the 4010 version of the X12 EDI standards.

The XML/EDI effort is being driven by the X12C Ad Hoc Task Group on the use of XML with X12 EDI. The group consists primarily of members from CommerceNet, DISA and the XML/EDI group.

XML Linking Language (see XLink)

XML Metadata Interchange Format (see XMI)

XML Namespace

An XML specification for qualifying the tag names used in XML documents by associating them with their source.

XML Path Language (see XPath)

XML Pointer Language (see XPointer)

XML-QL (XML Query Language)

XML RPC (Extensible Markup Language remote procedure call)

Specifications that define how an Extensible Markup Language (XML) document can be used to remotely invoke methods. XML tags within the document are used to define the method name, input and output parameters, and exceptions.

Xmodem

An 8-bit, public-domain error-checking protocol developed in the late 1970s. The file transfer protocol uses a 128-byte data block and cyclic redundancy check (CRC) or checksum error checking.

XMP (X/Open Management Protocol)

XNS (Xerox Network Systems)

An Ethernet-based data communications protocol developed by Xerox.

X-on/X-off

A method of controlling data communications. It essentially enables a terminal to activate a line when it is ready to receive information and to suspend line activity when the terminal is overloaded. See asynchronous.

X/Open

Originally, an international consortium of computer vendors working to advance the deployment of Unix. X/Open merged with the Open Software Foundation to form the Open Group in 1996, and the term "X/Open" is now used as an Open Group branding program.

X/Open Distributed Transaction Processing

An online transaction processing (OLTP) monitor compliant with the Open Group's XA distributed-transaction-processing specification.

X/Open Federated Naming (see XFN)

X/Open ISAM

Support, compliant with the Open Group's X/Open Portability Guide (XPG) volume 5, for Indexed Sequential Access Method (ISAM) files.

X/Open NLS

National language support (NLS) compliant with the Open Group's X/Open XPG3 NLS specification.

X/Open Transport Interface (see XTI)

XPAF (Xerox Printer Access Facility)

XPath (XML Path Language)

A language used to create addresses mapping the various parts of Extensible Markup Language (XML) document. XPath gets its name from its use of a path notation (as used in Internet URLs) for navigating through the hierarchical structure of an XML document. XPath addresses are used by Extensible Stylesheet Language Transformations (XSLT) and XML Pointer Language (XPointer). Specifications for the language are defined in the World Wide Web Consortium's XPath Recommendation. See XML, XPointer and XSLT.

XPE (Cross Platform Extension)

XPG (X/Open Portability Guide)

XPG3 (X/Open Portability Guide release 3)

XPG4 (X/Open Portability Guide release 4)

XPointer (XML Pointer Language)

XPointer provides the method for referencing Extensible Markup Language (XML) content. It is used to reference content components or anchors, the constructs that support addressing in the internal structures of XML documents. In particular, it provides for specific reference to elements, character strings and other parts of XML documents, whether or not they bear an explicit ID attribute. Specifications for the language are defined in the World Wide Web Consortium's XPointer Recommendation. See XML.

XPS (Cross Platform Server)

XRC (extended remote copy)

An asynchronous remote-copy capability that works with IBM's MVS operating system.

XRF (Extended Recovery Facility)

A reliability enhancement for the Information Management System/Virtual Storage (IMS/VS) database/data communications environment. It provides the capability of creating a "mirror" IMS system and synchronizing it to the primary subsystem. The mirror IMS/VS system takes over the workload from the primary subsystem in the case of some, but not all, disruptive events, thereby improving availability. XRF also facilitates testing and installation of new software releases.

XSA (Extended Subsystem Architecture)

XSL (Extensible Stylesheet Language)

XSL associates presentation characteristics (e.g., layout) with the markup used in Extensible Markup Language (XML). One of the XML family of languages developed by the World Wide Web Consortium, XSL is used to create XML "stylesheets," which describe how XML documents are presented. XSL provides independent control of presentation from content and can describe output of the same content to different formats (e.g., audio or print).

XSL is made up of three components:

- XSL Transformations (see XSLT)
- XML Path Language (see XPath)
- XSL Formatting Objects, an XML vocabulary for specifying formatting semantics

XSLT (Extensible Stylesheet Language Transformations)

A component of Extensible Stylesheet Language (see XSL), XSLT controls views of Extensible Markup Language (XML) documents and the ordering of XML elements. It is used to create new content structures from existing structures or subsets, based on interest, access privileges or security, and it transforms XML structure to

Hypertext Markup Language (HTML). Specifications for the language are defined in the World Wide Web Consortium's XSLT Recommendation, published in November 1999.

XSL Transformations (see XSLT)

X Terminal

A workstation operating under X-Windows.

XTI (X/Open Transport Interface)

Support provided for applications that use the Open Group's X/Open XTI programming interface for peer-to-peer communications.

XUDA (Xcert Universal Database)

A programmer's interface technology product from Xcert. It is a Secured Socket Layer (SSL)-secured Lightweight Directory Access Protocol (LDAP) compliant toolkit for developers, used for accessing distributed, multiplatform encryption modules.

X-Windows

The software system written for managing windows under Unix. A graphics architecture, application programming interface (API) and prototype implementation developed by the Massachusetts Institute of Technology, X-Windows defines a client/server relationship between the applications program and the workstation. It is not, however, a complete graphical user interface (GUI), but rather the basis upon which one can be built.

X-Windows Image Extensions (see XIE)

X.x series

Any of a number of technical specifications codified or administered by the International Telecommunication Union (ITU). It includes:

1. X.3: Specification for the packet assembler/disassembler (PAD) facility in a public data network.
2. X.21: Describes the interface used in the ITU X.25 packet-switching protocol and in some types of circuit switched data transmissions.
3. X.21bis: Defines the physical layer protocol for data terminal equipment/circuit-terminating equipment (DTE/DCE) communication in an X.25 network.
4. X.25: Specifies the interface between user DTE and packet-switching DCE.
5. X.26: Recommendation for unbalanced high-speed data communications.
6. X.28: DTE/DCE interface of a start/stop mode DTE accessing the PAD on a PDN situated in the same country.
7. X.29: Procedures for exchange of control information and user data between a packet mode DTE and a PAD facility.
8. X.32: Defines how X.25 connects to a packet network via a switched telephone system.
9. X.75: A standard for connecting X.25 networks.

- 10. X.121: International public data network numbering scheme.
- 11. X.400: Recommendation covering the handling of messages, such storing and forwarding e-mail, over digital circuits.
- 12. X.435: Recommendation covering the transmission of electronic data exchange messages by X.400 e-mail systems.
- 13. X.500: Recommendation covering the implementation of addressing databases for devices attached to a network.

Y

Y2K (year 2000)

The Y2K problem was the result of an industry practice that for more than 35 years had represented years in dates as only two digits, assuming in application logic that the first two digits of the year are "19." A variety of errors were introduced once such applications had to deal with dates beyond 31 December 1999.

yard management system (see YMS)**year 2000 compliance**

Confirmation that applications are free from the year 2000 problem, i.e., that they do not abnormally end or produce erroneous results as the result of incorrectly interpreting the year 2000 as "00" due to the use of only two digits to represent the year.

Ymodem

A packet-based error-checking protocol using 128-kilobyte or 1024-kilobyte blocks.

YMS (yard management system)

A logistic management system that provides the ability to monitor and track the movement and activities of vehicles entering and departing trailer yards. This capability allows enterprises to know the contents and location of all trailers within their facilities. A YMS also controls and directs the movement of trailers within a yard and between yard locations and dock doors.

YTD (year-to-date)

Z

ZAI (Zero Administration Initiative)

Microsoft's initiative to lower the total cost of ownership (TCO) necessary to manage its technology. ZAI consists of many components, including Zero Administration Kit, Systems Management Server (SMS), IntelliMirror and Common Information Model (CIM). Most of ZAI's functionality is included with Windows 2000.

ZAK (Zero Administration Kit)

A Microsoft product designed to reduce the management complexity associated with the Windows platform. Initially announced as an offering for Windows NT 4.0, ZAK is a component of Microsoft's Zero Administration Windows (ZAW) initiative.

ZAW (Zero Administration Windows)

A Microsoft initiative to lower the total cost of ownership (TCO) associated with the Windows platform by reducing costs associated with administration. More extensive and comprehensive than the Zero Administration Kit, ZAW will be deployed across a variety of Windows platforms over a period of years.

Zero Administration Initiative (see ZAI)**Zero Administration Kit (see ZAK)****Zero Administration Windows (see ZAW)****zero insertion force (see ZIF)****zero latency**

A goal to keep moving goods or information in the supply chain to provide near-real-time information management and reduce in-transit inventory costs.

zero-latency enterprise (see ZLE)**zero suppression**

The elimination of nonsignificant zeroes in a numeral.

ZIF (zero insertion force)

A type of chip socket that allows the chip to be removed and reinserted without damage.

ZigBee

The ZigBee Alliance's name for the proposed IEEE 802.15.4 standard. The technology defined by the standard is designed for "command and control"; therefore, it will not support audio or video, but it can be used to send text messages and voice. A ZigBee network will be able to control lights, fire or smoke detectors,

thermostats, or home security systems. It could also be used as a cable replacement technology. See 802.15.4.

ZigBee Alliance

A vendor initiative started by Philips Semiconductor, Honeywell and Invensys Metering Systems and now including more than 20 companies. It is responsible for developing applications — as well as a certification process, program, logo and marketing strategy — for technology based on the IEEE 802.15.4 standard, which the alliance calls "ZigBee." See 802.15.4 and ZigBee.

zip

The principle form of file-by-file data compression in the Windows environment.

ZIP (Zone Information Protocol)

A protocol that maps network numbers to zone names in an AppleTalk network.

ZLE (zero-latency enterprise)

Any strategy that exploits the immediate exchange of information across technical and organizational boundaries to achieve business benefit. For example, technical boundaries exist between different operating systems, database management systems and programming languages. "Immediate" implies being fast enough to bring all of the business benefits that simultaneous knowledge can potentially achieve. Latency cannot literally be zero in any real system because computers need time to "think."

Zmodem

An error-checking protocol that enables files to be transferred in one batch. If transmission is interrupted, file transfer is resumed from the point of interruption.

Zone Information Protocol (see ZIP)